



FOUNDATIONS OF CLASSICAL ELECTRODYNAMICS

– Electric charge and magnetic flux conservation
and the spacetime relation –

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Preface

In this book we display the fundamental structure underlying classical electrodynamics, i.e., the phenomenological theory of electric and magnetic effects. The book can be used as a textbook for an advanced course in theoretical electrodynamics for physics and mathematics students and, perhaps, for some highly motivated electrical engineering students.

We expect from our readers that they know elementary electrodynamics in the conventional $(1 + 3)$ -dimensional form including Maxwell's equations. Moreover, they should be familiar with linear algebra and elementary analysis, including vector analysis. Some knowledge of differential geometry would help.

Our approach rests on the metric-free integral formulation of the conservation laws of electrodynamics in the tradition of F. Kottler (1922), É. Cartan (1923), and D. van Dantzig (1934), and we stress, in particular, the axiomatic point of view. In this manner we are led to an understanding of why the Maxwell equations have their specific form. We hope that our book can be seen in the classical tradition of the book by E. J. Post (1962) on the *Formal Structure of Electromagnetics* and of the chapter "Charge and Magnetic Flux" of the encyclopedia article on classical field theories by C. Truesdell and R. A. Toupin (1960), including R. A. Toupin's Bressanone lectures (1965); for the exact references see the end of the introduction on page 9.

The manner in which electrodynamics is conventionally presented in physics courses à la R. Feynman (1962), J. D. Jackson (1999), and L. D. Landau & E. M. Lifshitz (1962) is distinctly different, since it is based on a flat spacetime manifold, i.e., on the (rigid) Poincaré group, and on H. A. Lorentz's approach (1916) to Maxwell's theory by means of his theory of electrons. We believe that the approach of this book is appropriate and, in our opinion, even superior for a good understanding of the structure of electrodynamics as a *classical field theory*. In particular, if gravity cannot be neglected, our framework allows for a smooth and trivial transition to the curved (and contorted) spacetime of general relativistic field theories. This is by no means a minor merit when one has to treat magnetic fields of the order of 10^9 tesla in the neighborhood of a neutron star where spacetime is appreciably curved.

Mathematically, integrands in the conservation laws are represented by exterior differential forms. Therefore exterior calculus is the appropriate language in which electrodynamics should be spelled out. Accordingly, we exclusively use this formalism (even in our computer algebra programs which we introduce in Sec. A.1.12). In Part A, and later in Part C, we try to motivate and to supply the necessary mathematical framework. Readers who are familiar with this formalism may want to skip these parts. They could start right away with the physics in Part B and then turn to Part D and Part E.

In Part B four axioms of classical electrodynamics are formulated and the consequences derived. This general framework has to be completed by a specific *electromagnetic spacetime relation* as a fifth axiom. This is done in Part D. The Maxwell–Lorentz theory is then recovered under specific conditions. In Part E,

we apply electrodynamics to moving continua, inter alia, which requires a sixth axiom on the formulation of electrodynamics inside matter.

This book grew out of a scientific collaboration with the late *Dermott McCrea* (University College Dublin). Mainly in Part A and Part C, Dermott's handwriting can still be seen in numerous places. There are also some contributions to "our" mathematics from *Wojtek Kopczyński* (Warsaw University). At Cologne University in the summer term of 1991, *Martin Zirnbauer* started to teach the theoretical electrodynamics course by using the calculus of exterior differential forms, and he wrote up successively improved notes to his course. One of the authors (FWH) also taught this course three times, partly based on Zirnbauer's notes. This influenced our way of presenting electrodynamics (and, we believe, also his way). We are very grateful to him for many discussions.

There are many colleagues and friends who helped us in critically reading parts of our book and who made suggestions for improvement or who communicated to us their own ideas on electrodynamics. We are very grateful to all of them: Carl Brans (New Orleans), Jeff Flowers (Teddington), David Hartley (Adelaide), Christian Heinicke (Cologne), Yakov Itin (Jerusalem), Martin Janssen (Cologne), Gerry Kaiser (Glen Allen, Virginia), R. M. Kiehn (formerly Houston), Attay Kovetz (Tel Aviv), Claus Lämmerzahl (Konstanz/Bremen), Bahram Mashhoon (Columbia, Missouri), Eckehard Mielke (Mexico City), Wei-Tou Ni (Hsin-chu), E. Jan Post (Los Angeles), Dirk Pützfeld (Cologne), Guillermo Rubilar (Cologne/Concepción), Yasha Shnir (Cologne), Andrzej Trautman (Warsaw), Arkady Tseytlin (Columbus, Ohio), Wolfgang Weller (Leipzig), and others.

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Please convey critical remarks to our approach or the discovery of mistakes by surface or electronic mail (hehl@thp.uni-koeln.de, yo@thp.uni-koeln.de) or by fax +49-221-470-5159.

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Introduction

Five plus one axioms

In this book we display the structure underlying classical electrodynamics. For this purpose we formulate six axioms: conservation of electric *charge* (first axiom), existence of the Lorentz *force* (second axiom), conservation of magnetic *flux* (third axiom), local *energy-momentum* distribution (fourth axiom), existence of an electromagnetic *spacetime* relation (fifth axiom), and finally, the splitting of the electric current into material and external pieces (sixth axiom).

The axioms expressing the conservation of electric charge and magnetic flux are formulated as integral laws, whereas the axiom for the Lorentz force is represented by a *local* expression basically defining the electromagnetic field strength $F = (E, B)$ as force per unit charge and thereby linking electrodynamics to mechanics; here E is the electric and B the magnetic field strength. Also the energy-momentum distribution is specified as a local law. The fifth axiom, the Maxwell–Lorentz spacetime relation is not as unquestionable as the first four axioms and extensions encompassing dilaton, skewon, and axion fields are discussed and nonlocal and nonlinear alternatives mentioned.

We want to stress the fundamental nature of the *first* axiom. Electric charge conservation is experimentally firmly established. It is valid for single elementary particle processes (like β -decay, $n \rightarrow p^+ + e^- + \bar{\nu}$, for instance, with n as neutron, p as proton, e as electron, and $\bar{\nu}$ as electron antineutrino). In other words, it is a *microscopic* law valid without any known exception.

Accordingly, it is basic to electrodynamics to assume a new type of entity called *electric charge*, carrying a positive or negative sign, with its own physical dimension, independent of the classical fundamental variables mass, length, and time. Furthermore, electric charge is conserved. In an age in which single electrons and (anti)protons are *counted* and caught in traps, this law is so deeply ingrained in our thinking that its explicit formulation as a fundamental law (and not only as a consequence of Maxwell’s equations) is often forgotten. We show that this *first axiom* yields the inhomogeneous Maxwell equation together with a definition of the electromagnetic excitation $H = (\mathcal{H}, \mathcal{D})$; here $\mathcal{H}$ is the magnetic excitation (“magnetic field”) and $\mathcal{D}$ the electric excitation (“electric displacement”). The excitation H is a microscopic field of an analogous quality as the field strength F . There exist operational definitions of the excitations $\mathcal{D}$ and $\mathcal{H}$ (via Maxwellian double plates or a compensating superconducting wire, respectively).

The *second* axiom for the Lorentz force, as mentioned above, leads to the notion of the field strength and is thereby exhausted. Thus we need further axioms. The only conservation law that can be naturally formulated in terms of the field strength is the conservation of magnetic flux (lines). This *third* axiom has the homogeneous Maxwell equation — that is, Faraday’s induction law and the vanishing divergence of the magnetic field strength — as a consequence.

Moreover, with the help of these first three axioms we are led, although not completely uniquely, to the electromagnetic *energy-momentum* current (fourth axiom), which subsumes the energy and momentum densities of the electromagnetic field and their corresponding fluxes, and to the *action* of the electromagnetic field. In this way, the basic structure of electrodynamics is set up, including the complete set of Maxwell's equations. To make this set of electrodynamic equations well determined, we still have to add the fifth axiom.

Magnetic monopoles are alien to the structure of the axiomatics we are using. In our axiomatic framework, a clear *asymmetry* is built in between electricity and magnetism in the sense of Oersted and Ampère wherein magnetic effects are created by moving electric charges. This asymmetry is characteristic for and intrinsic to Maxwell's theory. Therefore the conservation of magnetic *flux* and not that of magnetic charge is postulated as the third axiom.

The existence of a magnetic charge in violation of our third axiom would have far-reaching consequences: First of all, the electromagnetic potential A would not exist. Accordingly, in Hamiltonian mechanics, we would have to give up the coupling of a charged particle to the electromagnetic field via $\Pi = p - e A$. Moreover, the second axiom on the Lorentz force could be invalidated since one would have to supplement it with a term carrying the magnetic charge density. By implication, an extension of the fifth axiom on the energy-momentum current would be necessary.

In other words, if ever a *magnetic* monopole¹ were found, our axiomatics would lose its coherence, its compactness, and its plausibility. Or, to formulate it more positively: Not long ago, He [22], Abbott et al. [1], and Kalbfleisch et al. [32] determined experimentally new improved limits for the nonexistence of (Abelian or Dirac) magnetic monopoles. This ever increasing accuracy in the exclusion of magnetic monopoles speaks in favor of the axiomatic approach in Part B.

Topological approach

Since the notion of metric is a complicated one, which requires measurements with clocks and scales, generally with rigid bodies, which themselves are systems of great complexity, it seems undesirable to take metric as fundamental, particularly for phenomena which are simpler and actually independent of it.

E. Whittaker (1953)

The distinctive feature of this type of axiomatic approach is that one only needs minimal assumptions about the structure of the spacetime in which these axioms are formulated. For the first four axioms, a 4-dimensional differentiable

¹Our arguments refer only to Abelian gauge theory. In *non*-Abelian gauge theories the situation is different. There monopoles seem to be a must, at least if a Higgs field is present.

manifold is required that allows for a *foliation* into 3-dimensional hypersurfaces. Thus no *connection* and no *metric* are explicitly introduced in Parts A and B. The Poincaré and the Lorentz groups are totally ignored. Nevertheless, we recover Maxwell's equations already in Part B. This shows that electrodynamics is not as closely related to special relativity theory as is usually supposed.

This minimalistic *topological type* of approach may appear contrived at first look. We should recognize, however, that the metric of spacetime in the framework of general relativity theory represents the gravitational potential and, similarly, the connection of spacetime (in the viable Einstein–Cartan theory of gravity, for example) is intimately linked to gravitational properties of matter. We know that we really live in a curved and, perhaps, contorted spacetime. Consequently our desire should be to formulate the foundations of electrodynamics such that the metric and the connection don't interfere or interfere only in the least possible way. Since we know that the gravitational field permeates all the laboratories in which we make experiments with electricity, we should take care that this ever present field doesn't enter the formulation of the first principles of electrodynamics. In other words, a *clear separation* between pure electrodynamic effects and gravitational effects is desirable and can indeed be achieved by means of the axiomatic approach to be presented in Part B. Eventually, in the spacetime relation (see Part D), the metric does enter.

The power of the topological approach is also clearly indicated by its ability to describe the *phenomenology* (at low frequencies and large distances) of the quantum Hall effect successfully (not, however, its quantization). Insofar as the macroscopic aspects of the quantum Hall effect can be approximately understood in terms of a 2-dimensional electron gas, we can start with $(1 + 2)$ -dimensional electrodynamics, the formulation of which is straightforward in our axiomatics. It is then a mere finger exercise to show that in this specific case of $1 + 2$ dimensions there exists a *linear* constitutive law that doesn't require a metric. As a consequence the action is metric-free too. Thus the formulation of the quantum Hall effect by means of a *topological* (Chern–Simons) Lagrangian is imminent in our way of looking at electrodynamics.

Electromagnetic spacetime relation as a fifth axiom

Let us now turn to that domain where the metric does enter the 4-dimensional electrodynamical formalism. When the Maxwellian structure, including the Lorentz force and the action, is set up, it does not represent a concrete physical theory yet. What is missing is the electromagnetic spacetime relation linking the excitation to the field strength, i.e., $\mathcal{D} = \mathcal{D}(E, B)$, $\mathcal{H} = \mathcal{H}(B, E)$, or written 4-dimensionally, $H = H(F)$.

Trying the simplest approach, we assume *locality* and *linearity* between the excitation H and the field strength F , that is, $H = \kappa(F)$ with the linear operator κ . Together with two more “technical” assumptions, namely that $H = \kappa(F)$ is electric/magnetic *reciprocal* and κ *symmetric* (these properties will be discussed in detail in Part D), we are able to *derive* the metric of spacetime from

$H = \kappa(F)$ up to an arbitrary (conformal) factor. Accordingly, the light cone structure of spacetime is a consequence of a linear electromagnetic spacetime relation with the additional properties of reciprocity and symmetry. In this sense, the light cones are derived from electrodynamics.

Electrodynamics doesn't live in a preformed rigid Minkowski spacetime. Rather it has an arbitrary $(1 + 3)$ -dimensional spacetime manifold as its habitat which, as soon as a linear spacetime relation with reciprocity and symmetry is supplied, is equipped with local light cones everywhere. With the light cones it is possible to define the Hodge star operator $*$ that maps p -forms to $(4-p)$ -forms and, in particular, H to F according to $H \sim *F$.

Thus, in the end, that property of spacetime that describes its local “constitutive” structure, namely the metric, enters the formalism of electrodynamics and makes it into a complete theory. One merit of our approach is that it doesn't matter whether it is the rigid, i.e., flat, *Minkowski* metric of special relativity or the “flexible” *Riemannian* metric field of general relativity that changes from point to point according to Einstein's field equation. In this way, the traditional discussion of how to translate electrodynamics from special to general relativity loses its sense: The Maxwell equations remain the same, i.e., the exterior derivatives (the “commas” in coordinate language) are kept and are *not* substituted by something “covariant” (the “semicolons”), and the spacetime relation $H = \lambda_0 *F$ looks the same (λ_0 is a suitable factor). However, the Hodge star “feels” the difference in referring either to a constant or to a spacetime-dependent metric, respectively (see [51, 23]).

Our formalism can accommodate generalizations of classical electrodynamics, including those violating Lorentz invariance, simply by suitably modifying the fifth axiom while keeping the first four axioms as indispensable. Then a scalar *dilaton* field $\lambda(x)$, a pseudoscalar *axion* field $\alpha(x)$, and/or a tensorial traceless *skewon* field $\mathcal{S}_i{}^j$ can come up in a most straightforward way. Also the Heisenberg–Euler and the Born–Infeld electrodynamics are prime examples of such possible modifications. In the latter cases, the spacetime relation becomes effectively *nonlinear*, but it still remains a local expression.

Electrodynamics in matter and the sixth axiom

Eventually, we have to face the problem of formulating electrodynamics inside matter. We codify our corresponding approach in the sixth axiom. The total electric current, entering as the source in the inhomogeneous Maxwell equation, is split into bound charge and free charge. In this way, following Truesdell & Toupin [63] (see also the textbook of Kovetz [34]), we can develop a consistent theory of electrodynamics in matter. For simple cases, we can amend the axioms by a *linear constitutive law*. Since in our approach $(\mathcal{H}, \mathcal{D})$ are microscopic fields, like (E, B) , we believe that the conventional theory of electrodynamics inside matter needs to be redesigned.

In order to demonstrate the effectiveness of our formalism, we apply it to the electrodynamics of moving matter, thereby returning to the post-Maxwellian era of the 1880s when a relativistic version of Maxwell's theory had gained momentum. In this context, we discuss and analyse the experiments of Walker & Walker and James and those of Röntgen–Eichenwald and Wilson & Wilson.

List of axioms

1. Conservation of electric charge: (B.1.17).
2. Lorentz force density: (B.2.8).
3. Conservation of magnetic flux: (B.3.1).
4. Localization of energy-momentum: (B.5.7).
5. Maxwell–Lorentz spacetime relation: (D.6.13).
6. Splitting of the electric current in a conserved matter piece and an external piece: (E.3.1) and (E.3.2).

A reminder: Electrodynamics in 3-dimensional Euclidean vector calculus

Before we start to develop electrodynamics in 4-dimensional spacetime in the framework of the calculus of exterior differential forms, it may be useful to remind ourselves of electrodynamics in terms of conventional 3-dimensional Euclidean vector calculus. We begin with the laws obeyed by electric charge and current.

If $\vec{\mathcal{D}} = (\mathcal{D}_x, \mathcal{D}_y, \mathcal{D}_z)$ denotes the *electric excitation* field (historically “electric displacement”) and ρ the electric charge density, then the integral version of the *Gauss law*, ‘flux of $\mathcal{D}$ through any closed surface’ equals ‘net charge inside’ reads

$$\int_{\partial V} \vec{\mathcal{D}} \cdot d\vec{S} = \int_V \rho dV, \quad (\text{I.1})$$

with $d\vec{S}$ as area and dV as volume element. The *Oersted–Ampère law* with the *magnetic excitation* field $\vec{\mathcal{H}} = (\mathcal{H}_x, \mathcal{H}_y, \mathcal{H}_z)$ (historically “magnetic field”) and the electric current density $\vec{j} = (j_x, j_y, j_z)$ is a bit more involved because of the presence of the Maxwellian electric excitation current: The ‘circulation of $\vec{\mathcal{H}}$ around any closed contour’ equals ‘ $\frac{d}{dt}$ (flux of $\vec{\mathcal{D}}$ through surface spanned by contour)’ plus ‘flux of $\vec{j}$ through surface’ (t = time):

$$\oint_{\partial S} \vec{\mathcal{H}} \cdot d\vec{r} = \frac{d}{dt} \left(\int_S \vec{\mathcal{D}} \cdot d\vec{S} \right) + \int_S \vec{j} \cdot d\vec{S}. \quad (\text{I.2})$$

Here $\vec{dr}$ is the vectorial line element. Here the dot $\cdot$ always denotes the 3-dimensional metric-dependent scalar product, S denotes a 2-dimensional spatial surface, V a 3-dimensional spatial volume, and ∂S and ∂V the respective boundaries. Later we recognize that both, (I.1) and (I.2), can be derived from the *charge conservation law*.

The *homogeneous* Maxwell equations are formulated in terms of the electric field strength $\vec{E} = (E_x, E_y, E_z)$ and the magnetic field strength $\vec{B} = (B_x, B_y, B_z)$. They are defined operationally via the expression of the Lorentz force $\vec{F}$. An electrically charged particle with charge q and velocity $\vec{v}$ experiences the force

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}). \quad (\text{I.3})$$

Here the cross $\times$ denotes the 3-dimensional vector product. Then *Faraday's induction law* in its integral version, namely ‘circulation of $\vec{E}$ around any closed contour’ equals ‘minus $\frac{d}{dt}$ (flux of $\vec{B}$ through surface spanned by contour)’ reads:

$$\oint_{\partial S} \vec{E} \cdot \vec{dr} = -\frac{d}{dt} \left(\int_S \vec{B} \cdot d\vec{S} \right). \quad (\text{I.4})$$

Note the minus sign on its right-hand side, which is chosen according to the Lenz rule (following from energy conservation). Finally, the ‘flux of $\vec{B}$ through any closed surface’ equals ‘zero’, that is,

$$\int_{\partial V} \vec{B} \cdot d\vec{S} = 0. \quad (\text{I.5})$$

The laws (I.4) and (I.5) are inherently related. Later we formulate the law of *magnetic flux conservation* and (I.4) and (I.5) just turn out to be consequences of it.

Applying the Gauss and the Stokes theorems, the integral form of the Maxwell equations (I.1), (I.2) and (I.4), (I.5) can be transformed into their differential versions:

$$\text{div } \vec{\mathcal{D}} = \rho, \quad \text{curl } \vec{\mathcal{H}} - \dot{\vec{\mathcal{D}}} = \vec{j}, \quad (\text{I.6})$$

$$\text{div } \vec{B} = 0, \quad \text{curl } \vec{E} + \dot{\vec{B}} = 0. \quad (\text{I.7})$$

Additionally, we have to specify the spacetime relations $\vec{\mathcal{D}} = \varepsilon_0 \vec{E}$, $\vec{B} = \mu_0 \vec{\mathcal{H}}$, and if matter is considered, the constitutive laws.

This formulation of electrodynamics by means of 3-dimensional Euclidean vector calculus represents only a preliminary version since the 3-dimensional metric enters the scalar and the vector products and, in particular, the differential operators $\text{div} \equiv \vec{\nabla} \cdot$ and $\text{curl} \equiv \vec{\nabla} \times$, with $\vec{\nabla}$ as the nabla operator. In the Gauss law (I.1) or (I.6)₁, for instance,² only *counting procedures* enter,

²The subscript 1 refers to the first equation in (I.6).

namely counting of elementary charges inside V (taking care of their sign, of course) and counting of flux lines piercing through a closed surface ∂V . No length or time measurements and thus no metric are involved in such processes, as described in more detail below. Since similar arguments apply also to (I.5) or (I.7)₁, respectively, it should be possible to remove the metric from the Maxwell equations altogether.

On the literature

Basically not too much is new in our book. Probably Part D and Part E are the most original. Most of the material can be found somewhere in the literature. What we do claim, however, is some originality in the *completeness* and in the appropriate *arrangement* of the material, which is fundamental to the structure electrodynamics is based on. Moreover, we try to stress the *phenomena* underlying the axioms chosen and the *operational interpretation* of the quantities introduced. The *explicit derivation* in Part D of the metric of space-time from pre-metric electrodynamics by means of linearity, reciprocity, and symmetry, although considered earlier mainly by Toupin [61], Schönberg [53], and Jadczyk [28], is new and rests on recent results of Fukui, Gross, Rubilar, and the authors [42, 24, 41, 21, 52]. Also the generalization encompassing the dilaton, the axion, and/or the skewon field opens a new perspective. In Part E the electrodynamics of moving bodies, including the discussions of some classic experiments, contains much new material.

Our *main sources* are the works of Post [46, 47, 48, 49, 50], of Truesdell & Toupin [63], and of Toupin [61]. Historically, the metric-free approach to electrodynamics, based on integral conservation laws, was pioneered by Kottler [33], É. Cartan [10], and van Dantzig [64]. The article of Einstein [16] and the books of Mie [39], Weyl [65], and Sommerfeld [58] should also be consulted on these matters (see as well the recent textbook of Kovetz [34]). A description of the corresponding historical development, with references to the original papers, can be found in Whittaker [66] and, up to about 1900, in the penetrating account of Darrigol [12]. The driving forces and the results of Maxwell in his research on electrodynamics are vividly presented in Everitt's [17] concise biography of Maxwell.

In our book, we consistently use exterior calculus,³ including de Rham's *odd* (or twisted) differential forms. *Textbooks* on electrodynamics using exterior calculus are scarce. In English, we know only of Ingarden & Jamiołkowski [26], in German of Meetz & Engl [38] and Zirnbauer [67], and in Polish, of Jancewicz [31] (see also [30]). However, as a discipline of mathematical physics, corresponding presentations can be found in Bamberg & Sternberg [4], in Thirring

³Baylis [6] also advocates a geometric approach, using Clifford algebras (see also Jancewicz [29]). In such a framework, however, at least the way Baylis does it, the metric of 3-dimensional space is introduced right from the beginning. In this sense, Baylis' Clifford algebra approach is complementary to our metric-free electrodynamics.

[60], and as a short sketch, in Piron [44] (see also [5, 45]). Bamberg & Sternberg are particularly easy to follow and present electrodynamics in a very transparent way. That electrodynamics in the framework of exterior calculus is also in the scope of electrical engineers can be seen from Deschamps [14], Bossavit [8], and Baldomir & Hammond [3].

Presentations of exterior calculus, partly together with applications in physics and electrodynamics, were given amongst many others by Burke [9], Choquet-Bruhat et al. [11], Edelen [15], Flanders [19], Frankel [20], Parrott [43], and Ślebodziński [57]. For differential geometry we refer to the classics of de Rham [13] and Schouten [54, 55] and to Trautman [62].

What else influenced the writing of our book? The *axiomatics* of Bopp [7] is different but related to ours. In the more microphysical axiomatic attempt of Lämmerzahl et al. Maxwell's equations [35] (and the Dirac equation [2]) are deduced from direct experience with electromagnetic (and matter) waves, inter alia. The clear separation of *differential*, *affine*, and *metric* structures of space-time is nowhere more pronounced than in Schrödinger's [56] *Space-time structure*. A further presentation of electrodynamics in this spirit, somewhat similar to that of Post, has been given by Stachel [59]. Our $(1 + 3)$ -*decomposition* of spacetime is based on the paper by Mielke & Wallner [40]. More recently, Hirst [25] has shown, mainly based on experience with neutron scattering on magnetic structures in solids, that *magnetization* $\mathcal{M}$ is a *microscopic* quantity. This is in accord with our axiomatics which yields the magnetic excitation $\mathcal{H}$ as a microscopic quantity, quite analogously to the field strength B , whereas in conventional texts $\mathcal{M}$ is only defined as a macroscopic *average* over microscopically fluctuating magnetic fields. Clearly, $\mathcal{H}$ and also the electric excitation $\mathcal{D}$, i.e., the electromagnetic excitation $H = (\mathcal{H}, \mathcal{D})$ altogether, ought to be a microscopic field.

- Sections and subsections of the book that can be skipped at a first reading are marked by the symbol $\otimes$.

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Part A

Mathematics: Some exterior calculus



Why exterior differential forms?

In Part A and later in Part C, we are concerned with assembling the geometric concepts that are needed to formulate a classical field theory like *electrodynamics* and/or the theory of *gravitation* in the language of differential forms. The basic geometric structure underlying such a theory is that of a spacetime continuum or, in mathematical terms, a 4-dimensional *differentiable manifold* X_4 . The characteristics of the gravitational field will be determined by the nature of the additional geometric structures that are superimposed on this “bare manifold” X_4 . For instance, in Einstein’s general relativity theory (GR), the manifold is endowed with a *metric* together with a torsion-free, metric-compatible connection: It is a 4-dimensional Riemannian spacetime V_4 . The meanings of these terms are explained in detail in what follows.

In Maxwell’s theory of electrodynamics, under most circumstances, gravity can be safely neglected. Then the Riemannian spacetime becomes *flat*, i.e., its curvature vanishes, and we have the (rigid) Minkowskian spacetime M_4 of special relativity theory (SR). Its spatial part is the ordinary 3-dimensional Euclidean space R_3 . However, and this is one of the messages of the book, for the fundamental axioms of electrodynamics we don’t need to take into account the metric structure of spacetime and, even more so, we should not take it into account. This helps to keep electrodynamical structures cleanly separated from gravitational ones.

This separation is particularly decisive for a proper understanding of the emergence of the *light cone*. On the one side, by its very definition, it is an electrodynamical concept in that it determines the front of a propagating electromagnetic disturbance; on the other hand it constitutes the main (conformally invariant) part of the metric tensor of spacetime and is as such part of the grav-

itational potential of GR. This complicated interrelationship we try to untangle in Part D.

A central role in the formulation of classical electrodynamics adopted in the present work is played by the conservation laws of electric charge and magnetic flux. We start from their integral formulation. Accordingly, there is a necessity for an adequate understanding of the concepts involved when one writes down an *integral* over some domain on a differentiable manifold.

Specifically, in the Euclidean space R_3 in Cartesian coordinates, one encounters integrals like the electric tension (voltage)

$$\int_C (E_x dx + E_y dy + E_z dz) \quad (\text{A.1})$$

evaluated along a (1-dimensional) curve C , the magnetic flux

$$\int_S (B_x dy dz + B_y dz dx + B_z dx dy) \quad (\text{A.2})$$

over a (2-dimensional) surface S , and the (total) charge

$$\int_V \rho dx dy dz \quad (\text{A.3})$$

integrated over a (3-dimensional) volume V .

The fundamental result of classical integral calculus is Stokes' theorem which relates an integral over the boundary of a region to one taken over the region itself. Familiar examples of this theorem are provided by the expressions

$$\begin{aligned} \int_{\partial S} (E_x dx + E_y dy + E_z dz) &= \int_S \left[\left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right) dy dz \right. \\ &\quad \left. + \left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \right) dz dx + \left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \right) dx dy \right], \quad (\text{A.4}) \end{aligned}$$

and

$$\begin{aligned} \int_{\partial V} (B_x dy dz + B_y dz dx + B_z dx dy) &= \\ &= \int_V \left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} \right) dx dy dz, \quad (\text{A.5}) \end{aligned}$$

where ∂S and ∂V are the boundaries of S and V , respectively. The right-hand sides of these equations correspond to $\int_S \text{curl} \vec{E} \cdot d\vec{f}$ and $\int_V \text{div} \vec{B} dV$, respectively.

Consider the integral

$$\int \rho(x, y, z) dx dy dz \quad (\text{A.6})$$

and make a change of variables:

$$x = x(u, v, w), \quad y = y(u, v, w), \quad z = z(u, v, w). \quad (\text{A.7})$$

For simplicity and only for the present purpose, let us suppose that the Jacobian determinant

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} := \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} \quad (\text{A.8})$$

is *positive*. We obtain

$$\begin{aligned} \int \rho(x, y, z) dx dy dz = \\ \int \rho[x(u, v, w), y(u, v, w), z(u, v, w)] \frac{\partial(x, y, z)}{\partial(u, v, w)} du dv dw. \end{aligned} \quad (\text{A.9})$$

This suggests that we should write

$$dx dy dz = \frac{\partial(x, y, z)}{\partial(u, v, w)} du dv dw = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} du dv dw. \quad (\text{A.10})$$

If we set $x = y$ or $x = z$ or $y = z$ the determinant has equal rows and hence vanishes. Also, an odd permutation of x, y, z changes the sign of the determinant while an even permutation leaves it unchanged. Hence, we have

$$dx dx = 0, \quad dy dy = 0, \quad dz dz = 0, \quad (\text{A.11})$$

$$dx dy dz = dy dz dx = dz dx dy = -dy dx dz = -dx dz dy = -dz dy dx. \quad (\text{A.12})$$

It is this *alternating algebraic structure of integrands* that gave rise to the development of exterior algebra and calculus which is becoming more and more recognized as a powerful tool in mathematical physics. In general, an exterior p -form is an expression

$$\omega = \frac{1}{p!} \omega_{i_1 \dots i_p} dx^{i_1} \dots dx^{i_p}, \quad (\text{A.13})$$

where the components $\omega_{i_1 \dots i_p}$ are completely antisymmetric in the indices and $i_m = 1, 2, 3$. Furthermore, summation from 1 to 3 is understood over repeated indices. Then, when translating (A.5) in exterior form calculus, we recognize B as a 2-form

$$\int_{\partial V} B = \int_{\partial V} \frac{1}{2!} B_{[ij]} dx^i dx^j = \int_V \frac{1}{3!} \partial_{[k} B_{ij]} dx^k dx^i dx^j = \int_V dB. \quad (\text{A.14})$$

Note that $B_{[ij]} := (B_{ij} - B_{ji})/2$, $C_{(ij)} := (C_{ij} + C_{ji})/2$, etc. Accordingly, in R_3 , we have the *magnetic* field B as a *2-form* and, from (A.1), the *electric* field as a *1-form*. The *charge* ρ in (A.3) turns out to be a *3-form*.

In the 4-dimensional *Minkowski* space M_4 , the electric current J , like ρ in R_3 , is represented by a 3-form. Since the action functional of the electromagnetic field is defined in terms of a 4-dimensional integral, the integrand, the Lagrangian L , is a 4-form. The coupling term in L of the current J to the potential A , constructed as the exterior product $\sim J \wedge A$, identifies A as a 1-form. In the inhomogeneous Maxwell equation $J = dH$, the 3-form character of J requires the excitation H to be a 2-form. If, eventually, we execute a gauge transformation $A \rightarrow A + df$, $F \rightarrow F$, we will encounter a 0-form f . Consequently, a (gauge) field theory, starting from a conserved current 3-form (here the electric current J), generates in a straightforward way forms of all ranks $p \leq 4$.

We know from classical calculus that if the Jacobian determinant (A.8) above has negative values, i.e., the two coordinate systems do not have the same orientation, then, in (A.9) and (A.10), the Jacobian determinant must be replaced by its absolute value. In particular, instead of (A.10), we get the general formula

$$dx dy dz = \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| du dv dw. \quad (\text{A.15})$$

This behavior under a change of coordinates is typical of what is known as a *density*. We shall see that, if we wish to drop the requirement that all our coordinate systems should have the same orientation, then densities become important and these latter are closely related to the *twisted differential forms* that one has to introduce, in addition to the ordinary differential forms, since the electric current, for example, is of such a twisted type.

In Chapter A.1, we first consider a vector space and its dual and study the algebraic aspect of tensors and of geometrical quantities of a more general type. Then we turn our attention to exterior forms and their algebra and to a corresponding computer algebra program.

Since the tangent space at every point of a differentiable manifold is a linear vector space, we can associate an exterior algebra with each point and define differentiable *fields* of exterior forms or, more concisely, *differential forms* on the manifold. This is done in Chapter A.2, while Chapter A.3 deals with integration on a manifold.

It is important to note that in Part A, we are dealing with the “bare manifold”. The linear connection and metric will be introduced in Part C, after the basic axiomatics of electrodynamics are laid down in Part B.

A.1

Algebra

A.1.1 A real vector space and its dual

Our considerations are based on an n -dimensional real vector space V . One-forms are the elements of the dual vector space V^ defined as linear maps of the vector space V into the real numbers. The dual bases of V and V^* transform reciprocally to each other with respect to the action of the linear group. The vectors and 1-forms can be alternatively defined by their components with a specified transformation law.*

Let V be an n -dimensional real vector space. We can depict a vector $v \in V$ by an arrow. If we multiply v by a factor f , the vector has an f -fold size (see Fig. A.1.1). Vectors are added according to the parallelogram rule. A linear map $\omega : V \rightarrow \mathbb{R}$ is called a *1-form* on V . The set of all 1-forms on V can be given the structure of a vector space by defining the sum of two arbitrary 1-forms ω and φ ,

$$(\omega + \varphi)(v) = \omega(v) + \varphi(v), \quad v \in V, \quad (\text{A.1.1})$$

and the product of ω by a real number $\lambda \in \mathbb{R}$,

$$(\lambda\omega)(v) = \lambda(\omega(v)), \quad v \in V. \quad (\text{A.1.2})$$

This vector space is denoted V^* and called the *dual* of V . The dimension of V^* is equal to the dimension of V . The identification $V^{**} = V$ holds for finite dimensional spaces.

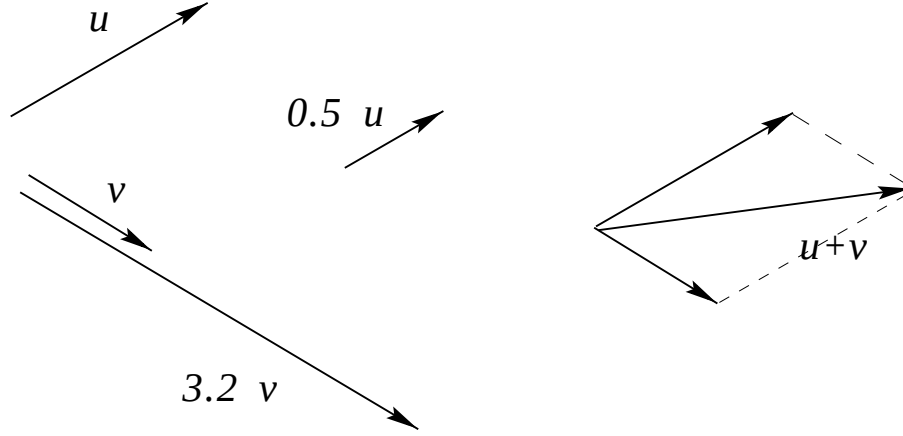


Figure A.1.1: Vectors as arrows: their multiplication by a factor, their addition (by the parallelogram rule).

In accordance with (A.1.1), (A.1.2), a 1-form can be represented by a pair of ordered hyperplanes, namely $(n-1)$ -dimensional subspaces (see Fig. A.1.2). The nearer the hyperplanes are to each other, the stronger is the 1-form. In Fig. A.1.3, the action of a 1-form on a vector is depicted.

Denote by $e_\alpha = \{e_1, \dots, e_n\}$ a (vector) *basis* in V . An arbitrary vector v can be decomposed with respect to such a basis: $v = v^\alpha e_\alpha$. Summation from 1 to n is understood over repeated indices (Einstein's summation convention). The n real numbers v^α , $\alpha = 1, \dots, n$, are called components with respect to the given basis. With a basis e_α of V we can associate its dual 1-form, or covector basis, the so-called *cobasis* $\vartheta^\alpha = \{\vartheta^1, \dots, \vartheta^n\}$ of V^* . It is determined by the relation

$$\vartheta^\alpha(e_\beta) = \delta_\beta^\alpha. \quad (\text{A.1.3})$$

Here δ_β^α is the Kronecker symbol with $\delta_\beta^\alpha = 1$ for $\alpha = \beta$ and $\delta_\beta^\alpha = 0$ for $\alpha \neq \beta$.

The components ω_α of a 1-form ω with respect to the cobasis ϑ^α are then given by

$$\omega = \omega_\alpha \vartheta^\alpha \quad \implies \quad \omega_\alpha = \omega(e_\alpha). \quad (\text{A.1.4})$$

A transformation from a basis e_α of V to another one ("alpha-prime" basis) $e_{\alpha'} = \{e'_1, \dots, e'_n\}$ is described by a matrix $L := (L_{\alpha'}^\alpha) \in GL(n, \mathbb{R})$ (general linear real n -dimensional group):

$$e_{\alpha'} = L_{\alpha'}^\alpha e_\alpha. \quad (\text{A.1.5})$$

The corresponding cobases are thus connected by

$$\vartheta^{\alpha'} = L_{\alpha'}^{\alpha} \vartheta^\alpha, \quad (\text{A.1.6})$$

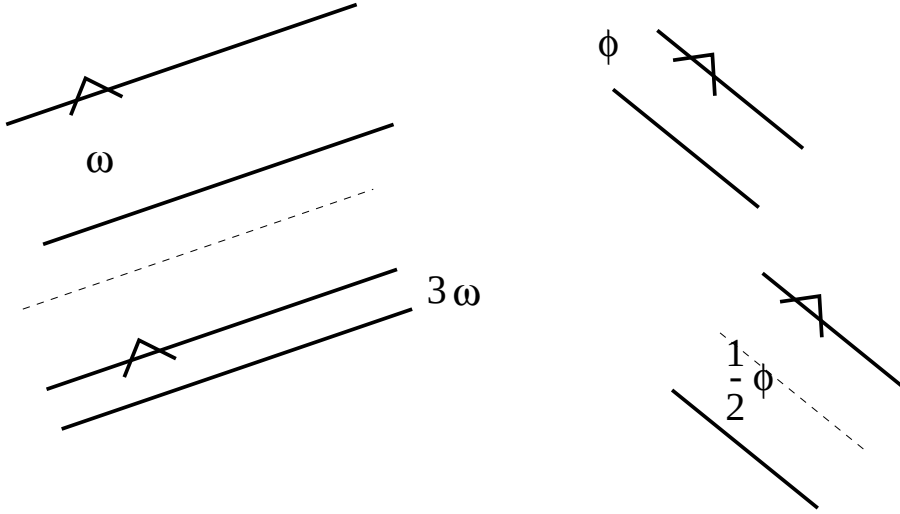


Figure A.1.2: One-forms are represented as two parallel hyperplanes (straight lines in $n = 2$) with a direction; their multiplication by a factor.

where $(L_{\alpha}^{\alpha'})$ is the inverse matrix to $(L_{\alpha'}^{\alpha})$, i.e., $L_{\alpha}^{\alpha'} L_{\alpha'}^{\beta} = \delta_{\alpha}^{\beta}$. Symbolically, we may also write $e' = L e$ and $\vartheta' = (L^T)^{-1} \vartheta$. Here T denotes the transpose of the matrix L .

Consequently, one can view a vector $v \in V$ as n ordered numbers v^{α} that transform under a change (A.1.5) of the basis as

$$v^{\alpha'} = L_{\alpha}^{\alpha'} v^{\alpha}, \quad (\text{A.1.7})$$

whereas a 1-form $\omega \in V^*$ is described by its components ω_{α} with the transformation law

$$\omega_{\alpha'} = L_{\alpha'}^{\alpha} \omega_{\alpha}. \quad (\text{A.1.8})$$

The similarity of (A.1.7) to (A.1.6) and of (A.1.8) to (A.1.5) and the fact that the two matrices in these formulas are contragradient (i.e., inverse and transposed) to each other explains the old-fashioned names for vectors and 1-forms (or covectors): *contravariant* and *covariant* vectors, respectively. Nevertheless, one should be careful: (A.1.7) represents the transformation of n components of one vector, whereas (A.1.6) encrypts the transformation of n different 1-forms.

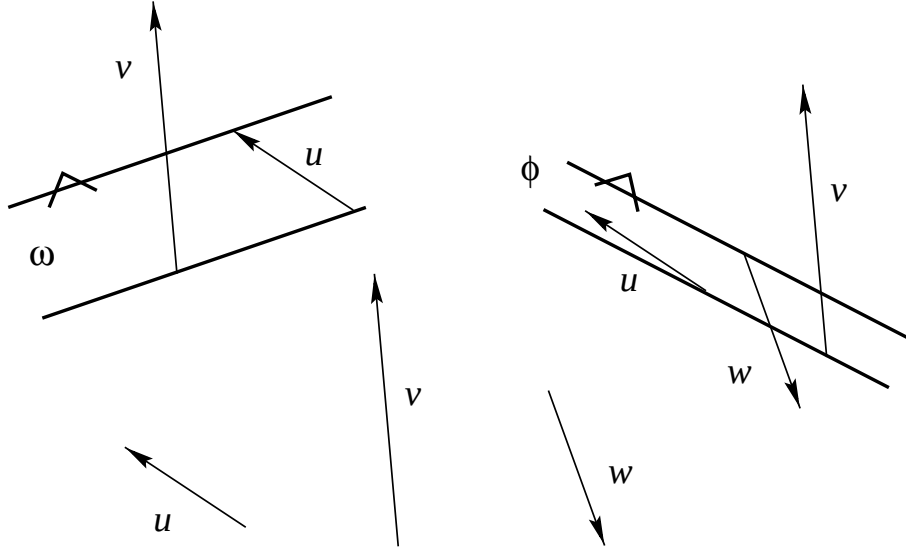


Figure A.1.3: A 1-form acts on a vector. Here we have $\omega(u) = 1$, $\omega(v) \approx 2.3$; $\phi(u) \approx 0.3$, $\phi(v) \approx 4.4$; $\phi(w) \approx -2.1$. A 1-form can be understood as a machine: You input a vector and the output is a number which can be read off from our images.

A.1.2 Tensors of type $\begin{bmatrix} p \\ q \end{bmatrix}$

A tensor is a multilinear map of a product of vector and dual vector spaces into the real numbers. An alternative definition of tensors specifies the transformation law of their components with respect to a change of the basis.

The related concepts of a vector and a 1-form can be generalized to objects of higher rank. The prototype of such an object is the stress *tensor* of continuum mechanics. A tensor T on V of type $\begin{bmatrix} p \\ q \end{bmatrix}$ is a multi-linear map

$$T : \underbrace{V^* \times \cdots \times V^*}_p \times \underbrace{V \times \cdots \times V}_q \rightarrow \mathbb{R}. \quad (\text{A.1.9})$$

It can be described as a geometrical quantity whose components with respect to the cobasis ϑ^α and the basis e_β are given by

$$T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q} = T(\vartheta^{\alpha_1}, \dots, \vartheta^{\alpha_p}; e_{\beta_1}, \dots, e_{\beta_q}). \quad (\text{A.1.10})$$

The transformation law for tensor components can be deduced from (A.1.5) and (A.1.6):

$$T^{\alpha'_1 \dots \alpha'_p}_{\beta'_1 \dots \beta'_q} = L_{\alpha_1}^{\alpha'_1} \dots L_{\alpha_p}^{\alpha'_p} L_{\beta'_1}^{\beta_1} \dots L_{\beta'_q}^{\beta_q} T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q}. \quad (\text{A.1.11})$$

If we have two tensors, T of type $\begin{bmatrix} p \\ q \end{bmatrix}$ and S of type $\begin{bmatrix} r \\ s \end{bmatrix}$, we can construct their tensor product, the tensor $T \otimes S$ of type $\begin{bmatrix} p+r \\ q+s \end{bmatrix}$, as follows

$$\begin{aligned} (T \otimes S)(\omega_1, \dots, \omega_{p+r}; v_1, \dots, v_{q+s}) \\ = T(\omega_1, \dots, \omega_p; v_1, \dots, v_q) S(\omega_{p+1}, \dots, \omega_{p+r}; v_{q+1}, \dots, v_{q+s}), \end{aligned}$$

for any 1-forms ω and any vectors v .

Tensors of type $\begin{bmatrix} p \\ q \end{bmatrix}$ that have the form $v_1 \otimes \dots \otimes v_p \otimes \omega_1 \otimes \dots \otimes \omega_q$ are called *decomposable*. Each tensor is a linear combination of decomposable tensors. More precisely, using the definition of the components of T according to (A.1.11), one can prove that

$$T = T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q} e_{\alpha_1} \otimes \dots \otimes e_{\alpha_p} \otimes \vartheta^{\beta_1} \otimes \dots \otimes \vartheta^{\beta_q}. \quad (\text{A.1.12})$$

Therefore tensor products of basis vectors e_α and of basis 1-forms ϑ^β constitute a basis of the vector space V_q^p of tensors of type $\begin{bmatrix} p \\ q \end{bmatrix}$ on V . Thus the dimension of this vector space is n^{p+q} .

Elementary examples of the tensor spaces are given by the original vector space and its dual, $V_0^1 = V$ and $V_1^0 = V^*$. The Kronecker symbol δ_β^α is a tensor of type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, as can be verified with the help of (A.1.11).

A.1.3 $\otimes$ A generalization of tensors: geometric quantities

A geometric quantity is defined by the action of the general linear group on a certain set of elements. Important examples are tensor-valued forms, the orientation, and twisted tensors.

In field theory, tensors are not the only objects needed for the description of nature. Twisted forms or vector-valued forms, for example, require a more general definition.

As we have seen above, there are two ways of dealing with tensors: either we can describe them as elements of the abstract tensor space V_q^p or as components, i.e., elements of $\mathbb{R}^{n^{p+q}}$, that have a prescribed transformation law. These observations can be generalized as follows.

Let W be a set, and let ρ be a left action of the group $GL(n, \mathbb{R})$ in the set W , i.e., to each element $L \in GL(n, \mathbb{R})$ we attach a map $\rho_L : W \rightarrow W$ in such a way that

$$\rho_{L_1} \circ \rho_{L_2} = \rho_{L_1 L_2}, \quad L_1, L_2 \in GL(n, \mathbb{R}). \quad (\text{A.1.13})$$

Denote by $P(V)$ the space of all bases of V and consider the Cartesian product $W \times P(V)$. The formula (A.1.5) provides us with a left action of $GL(n, \mathbb{R})$

in $P(V)$ which can be compactly written as $e' = Le$. Then $L_1(L_2e) = (L_1L_2)e$ holds. Thus we can define the left action of $GL(n, \mathbb{R})$ on the product $W \times P(V)$:

$$(w, e) \mapsto (\rho_L(w), Le). \quad (\text{A.1.14})$$

An orbit of this action is called a *geometric quantity* of type ρ on V . In other words, a geometric quantity of type ρ on V is an equivalence class $[(w, e)]$. Two pairs (w, e) and (w', e') are equivalent if and only if there exists a matrix $L \in GL(n, \mathbb{R})$ such that

$$w' = \rho_L(w) \quad \text{and} \quad e' = Le. \quad (\text{A.1.15})$$

In many physical applications, the set W is an N -dimensional vector space $\mathbb{R}^N$ and it is required that the maps ρ_L are linear. In other words, ρ_L is a representation $\rho_L : GL(n, \mathbb{R}) \rightarrow GL(N, \mathbb{R})$ of the linear group $GL(n, \mathbb{R})$ in the vector space W by $N \times N$ matrices $\rho_L = \rho_A^B(L) \in GL(N, \mathbb{R})$, with $A, B, \dots = 1, \dots, N$. Let us denote as e_A the basis of the vector space W . Then, we can represent the geometric quantity $w = w^A e_A$ by means of its components w^A with respect to the basis. The action of the group $GL(n, \mathbb{R})$ in W results in a linear transformation

$$e_A \longrightarrow e_{A'} = \rho_{A'}^B(L) e_B, \quad (\text{A.1.16})$$

and accordingly, the components of the geometric quantity transform as

$$w^A \longrightarrow w^{A'} = \rho_B^{A'}(L^{-1}) w^B. \quad (\text{A.1.17})$$

Examples:

- 1) We can take $W = V_q^p$ and $\rho_L = \text{id}_W$. The corresponding geometric quantity is then a tensor of type $\begin{bmatrix} p \\ q \end{bmatrix}$.
- 2) We can take $W = \mathbb{R}^{n^{p+q}}$ and choose ρ in such a way that (A.1.15) induces (A.1.11) together with (A.1.5). This type of geometric quantity is also a tensor of type $\begin{bmatrix} p \\ q \end{bmatrix}$. For instance, if we take $W = \mathbb{R}^n$ and either $\rho_L = L$ or $\rho_L = (L^T)^{-1}$, then from (A.1.16) and (A.1.17) we get vectors (A.1.7) or 1-forms (A.1.8), respectively.
- 3) The two examples above can be combined. We can take $W = \mathbb{R}^{n^{p+q}} \otimes V_s^r$ and ρ as in example 2). That means that we can consider objects with components $T^{\alpha_1 \dots \alpha_p}_{\beta_1 \dots \beta_q}$ belonging to V_s^r which transform according to the rule (A.1.11). This mixture of two approaches seems strange at first sight, but it appears productive if, instead of V_s^r , we take spaces of s -forms $\Lambda^s V$. Such *tensor-valued forms* turn out to be useful in differential geometry and physics.
- 4) Let $W = \{+1, -1\}$ and $\rho_L = \text{sgn}(\det L)$. This geometric quantity is an *orientation* in the vector space V . A frame $e \in P(V)$ is said to have a positive orientation if it forms a pair with $+1 \in W$. Each vector space has two different orientations.

- 5) Combine the examples 1) and 4). Let $W = V_q^p$ and $\rho_L = \text{sgn}(\det L) \text{id}_W$. This geometric quantity is called a *twisted* (or odd) tensor of type $\begin{bmatrix} p \\ q \end{bmatrix}$ on V . Particularly useful are twisted exterior forms since they can be integrated even on a manifold which is nonorientable.

A.1.4 Almost complex structure

An *even-dimensional* vector space V with $n = 2k$ can be equipped with an additional structure that has many interesting applications in electrodynamics and in other physical theories. We say that a real vector space V has an *almost complex structure*¹ if a tensor $\mathbf{J}$ of type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is defined on it that has the property

$$\mathbf{J}^2 = -1. \quad (\text{A.1.18})$$

With respect to a chosen frame, this tensor is represented by the components J_α^β and the above condition is then rewritten as

$$J_\alpha^\gamma J_\gamma^\beta = -\delta_\alpha^\beta. \quad (\text{A.1.19})$$

By means of a suitable choice of the basis e_α , the complex structure can be brought into the canonical form

$$J_\alpha^\beta = \begin{pmatrix} 0 & -\mathbf{I}_k \\ \mathbf{I}_k & 0 \end{pmatrix}. \quad (\text{A.1.20})$$

Here $\mathbf{I}_k$ is the k -dimensional unity matrix with $k = n/2$.

A.1.5 Exterior p -forms

Exterior forms are totally antisymmetric covariant tensors. Any tensor of type $\begin{bmatrix} 0 \\ p \end{bmatrix}$ defines an exterior p -form by means of the alternating map involving the generalized Kronecker.

As we saw at the beginning of Part A, exterior p -forms play a particular role as integrands in field theory. We now turn to their general definition.

Once again, let V be an n -dimensional linear vector space. An exterior p -form ω on V is a real-valued linear function

$$\omega : \underbrace{V \times V \times \cdots \times V}_{p \text{ factors}} \longrightarrow \mathbb{R} \quad (\text{A.1.21})$$

¹See Choquet-Bruhat et al. [4].

such that

$$\omega(v_1, \dots, v_\alpha, \dots, v_\beta, \dots, v_p) = -\omega(v_1, \dots, v_\beta, \dots, v_\alpha, \dots, v_p) \quad (\text{A.1.22})$$

for all $v_1, \dots, v_p \in V$ and for all $\alpha, \beta = 1, \dots, p$. In other words, ω is a completely antisymmetric tensor of type $\begin{bmatrix} 0 \\ p \end{bmatrix}$. In terms of a basis e_α of V and the cobasis ϑ^α of V^* , the linear function ω can be expressed as

$$\omega = \omega_{\alpha_1 \dots \alpha_p} \vartheta^{\alpha_1} \otimes \dots \otimes \vartheta^{\alpha_p}, \quad (\text{A.1.23})$$

where each coefficient $\omega_{\alpha_1 \dots \alpha_p} := \omega(e_{\alpha_1}, \dots, e_{\alpha_p})$ is completely antisymmetric in all its indices.

The space of real-valued p -linear functions on V was denoted by V_p^0 . Then, for any $\varphi \in V_p^0$ with

$$\varphi = \varphi_{\alpha_1 \dots \alpha_p} \vartheta^{\alpha_1} \otimes \dots \otimes \vartheta^{\alpha_p}, \quad (\text{A.1.24})$$

we can define a corresponding (alternating) exterior p -form $\text{Alt } \varphi$ by

$$\text{Alt } \varphi = \varphi_{[\alpha_1 \dots \alpha_p]} \vartheta^{\alpha_1} \otimes \dots \otimes \vartheta^{\alpha_p}. \quad (\text{A.1.25})$$

Here we have

$$\varphi_{[\alpha_1 \dots \alpha_p]} := \frac{1}{p!} \delta_{\alpha_1 \dots \alpha_p}^{\beta_1 \dots \beta_p} \varphi_{\beta_1 \dots \beta_p} \quad (\text{A.1.26})$$

with the generalized Kronecker delta

$$\delta_{\alpha_1 \dots \alpha_p}^{\beta_1 \dots \beta_p} := \begin{cases} +1 & \text{if } \beta_1, \dots, \beta_p \text{ is an even} \\ & \text{permutation of } \alpha_1, \dots, \alpha_p, \\ -1 & \text{if } \beta_1, \dots, \beta_p \text{ is an odd} \\ & \text{permutation of } \alpha_1, \dots, \alpha_p, \\ 0 & \text{otherwise,} \end{cases} \quad (\text{A.1.27})$$

where $\alpha_1, \dots, \alpha_p$ are p *different* numbers from the set $1, \dots, n$. Provided $\varphi_{\alpha_1 \dots \alpha_p}$ is already antisymmetric in all its indices, then

$$\varphi_{[\alpha_1 \dots \alpha_p]} = \varphi_{\alpha_1 \dots \alpha_p}. \quad (\text{A.1.28})$$

The set of exterior p -forms on V forms an $\binom{n}{p}$ -dimensional subspace of V_p^0 which we denote by $\Lambda^p V^*$. Here $\binom{n}{p}$ represent the binomial coefficients. In particular, for $p = 0$ and $p = 1$, we have

$$\Lambda^0 V^* = \mathbb{R}, \quad \Lambda^1 V^* = V^*. \quad (\text{A.1.29})$$

For $n = 4$, the dimensions of the spaces for p -forms are $\binom{4}{p} = \frac{4!}{p!(4-p)!}$ or

$$p = 0, 1, 2, 3, 4 \quad \sim \quad 1, 4, 6, 4, 1 \text{ dimensions}, \quad (\text{A.1.30})$$

respectively (see Table A.1.5).

Table A.1.1: Number of components of p -forms in 3 and 4 dimensions and examples from electrodynamics: ρ electric charge and j electric current density, $\mathcal{D}$ electric and $\mathcal{H}$ magnetic excitation, E electric and B magnetic field, $\mathcal{A}$ covector potential, φ scalar potential, and f gauge function. L is the Lagrangian, $J = (\rho, j)$, $H = (\mathcal{D}, \mathcal{H})$, $F = (E, B)$, $A = (\varphi, \mathcal{A})$. The forms J and H are twisted forms (see Sec. A.2.6).

p -form	$n = 3$	examples	$n = 4$	examples
0-form	1	φ, f	1	f
1-form	3	$\mathcal{H}, E, \mathcal{A}$	4	A
2-form	3	$j, \mathcal{D}, B$	6	H, F
3-form	1	ρ	4	J
4-form	0	–	1	L
5-form	0	–	0	–

A.1.6 Exterior multiplication

The exterior product defines a $(p+q)$ -form for every pair of p - and q -forms. The basis of the space of p -forms is then naturally constructed as the p -th exterior power of the 1-form basis. The exterior product converts the direct sum of all forms into an algebra.

In order to handle exterior forms, we have to define their multiplication. The exterior product of the p 1-forms $\omega^1, \dots, \omega^p \in V^*$, taken in that order, is a p -form defined by

$$\omega^1 \wedge \dots \wedge \omega^p := \delta_{\alpha_1 \dots \alpha_p}^{1 \dots p} \omega^{\alpha_1} \otimes \dots \otimes \omega^{\alpha_p}, \quad (\text{A.1.31})$$

spoken as “omega-one wedge... wedge omega- p ”. It follows that for any set of vectors $v_1, \dots, v_p \in V$,

$$(\omega^1 \wedge \dots \wedge \omega^p)(v_1, \dots, v_p) = \begin{vmatrix} \omega^1(v_1) & \omega^1(v_2) & \dots & \omega^1(v_p) \\ \omega^2(v_1) & \omega^2(v_2) & \dots & \omega^2(v_p) \\ \vdots & \vdots & \ddots & \vdots \\ \omega^p(v_1) & \omega^p(v_2) & \dots & \omega^p(v_p) \end{vmatrix}. \quad (\text{A.1.32})$$

Given $\omega \in \Lambda^p V^*$ so that

$$\omega = \omega_{\alpha_1 \dots \alpha_p} \vartheta^{\alpha_1} \otimes \dots \otimes \vartheta^{\alpha_p} \quad \text{with} \quad \omega_{[\alpha_1 \dots \alpha_p]} = \omega_{\alpha_1 \dots \alpha_p}, \quad (\text{A.1.33})$$

we have

$$\begin{aligned}\omega &= \omega_{[\alpha_1 \dots \alpha_p]} \vartheta^{\alpha_1} \otimes \dots \otimes \vartheta^{\alpha_p} \\ &= \frac{1}{p!} \delta_{\alpha_1 \dots \alpha_p}^{\beta_1 \dots \beta_p} \omega_{\beta_1 \dots \beta_p} \vartheta^{\alpha_1} \otimes \dots \otimes \vartheta^{\alpha_p},\end{aligned}\quad (\text{A.1.34})$$

and hence, cf. (A.13),

$$\boxed{\omega = \frac{1}{p!} \omega_{\beta_1 \dots \beta_p} \vartheta^{\beta_1} \wedge \dots \wedge \vartheta^{\beta_p}.} \quad (\text{A.1.35})$$

Since, in addition, the $\binom{n}{p}$ p -forms $\{\vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_p}, 1 \leq \alpha_1 < \alpha_2 < \dots < \alpha_p \leq n\}$ are linearly independent, it follows that they constitute a basis for $\Lambda^p V^*$. Equation (A.1.35) may also be written as

$$\omega = \sum_{\beta_1 < \beta_2 < \dots < \beta_p} \omega_{\beta_1 \dots \beta_p} \vartheta^{\beta_1} \wedge \dots \wedge \vartheta^{\beta_p}. \quad (\text{A.1.36})$$

The indices $\beta_1 < \beta_2 < \dots < \beta_p$ are called *strongly ordered*. Furthermore, it is clear from (A.1.35) that a p -form with $p > n$ is equal to zero.

The exterior product of two arbitrary forms is introduced as a map

$$\wedge : \Lambda^p V^* \times \Lambda^q V^* \longrightarrow \Lambda^{p+q} V^* \quad (\text{A.1.37})$$

as follows: Let $\psi \in \Lambda^p V^*$ and $\phi \in \Lambda^q V^*$. Then $\psi \wedge \phi \in \Lambda^{p+q} V^*$ is defined by

$$\psi \wedge \phi = \frac{(p+q)!}{p! q!} \text{Alt}(\psi \otimes \phi). \quad (\text{A.1.38})$$

In terms of a 1-form basis ϑ^α of V^* , we shall have

$$\psi = \frac{1}{p!} \psi_{\beta_1 \dots \beta_p} \vartheta^{\beta_1} \wedge \dots \wedge \vartheta^{\beta_p}, \quad (\text{A.1.39})$$

$$\phi = \frac{1}{q!} \phi_{\beta_1 \dots \beta_q} \vartheta^{\beta_1} \wedge \dots \wedge \vartheta^{\beta_q}, \quad (\text{A.1.40})$$

and their exterior product reads

$$\boxed{\psi \wedge \phi = \frac{1}{p! q!} \psi_{[\beta_1 \dots \beta_p} \phi_{\beta_{p+1} \dots \beta_{p+q}]} \vartheta^{\beta_1} \wedge \dots \wedge \vartheta^{\beta_{p+q}}.} \quad (\text{A.1.41})$$

From the definition (A.1.31), it is a straightforward matter to derive the following properties of exterior multiplication:

- 1) $(\lambda + \mu) \wedge \nu = \lambda \wedge \nu + \mu \wedge \nu$ [distributive law],
- 2) $(a\lambda) \wedge \nu = \lambda \wedge (a\nu) = a(\lambda \wedge \nu)$ [multiplicative law],

- 3) $(\lambda \wedge \nu) \wedge \omega = \lambda \wedge (\nu \wedge \omega)$ [associative law],
 4) $\lambda \wedge \nu = (-1)^{pq} (\nu \wedge \lambda)$ [(anti)commutative law],

where $\lambda, \mu \in \Lambda^p V^*$, $\nu \in \Lambda^q V^*$, $\omega \in \Lambda^r V^*$, and $a \in \mathbb{R}$.

With the exterior multiplication introduced, the direct sum of the spaces of all forms

$$\Lambda^* V^* := \bigoplus_{p=0}^n \Lambda^p V^* \quad (\text{A.1.42})$$

becomes an algebra over V^* . This is usually called the *exterior algebra*.

A.1.7 Interior multiplication of a vector with a form

The interior product decreases the rank of an exterior form by one.

By exterior multiplication, we *increase* the rank of a form. Besides this “constructive” operation, we need a “destructive” operation *decreasing* the rank of a form. Here interior multiplication comes in.

For $p > 0$, the interior product is a map

$$\lrcorner : V \times \Lambda^p V^* \longrightarrow \Lambda^{p-1} V^* \quad (\text{A.1.43})$$

which is introduced as follows: Let $v \in V$ and $\phi \in \Lambda^p V^*$. Then $(v \lrcorner \phi) \in \Lambda^{p-1} V^*$ is defined by

$$(v \lrcorner \phi)(u_1, \dots, u_{p-1}) := \phi(v, u_1, \dots, u_{p-1}), \quad (\text{A.1.44})$$

for all $u_1, \dots, u_{p-1} \in V$. We say “*v in ϕ* ”. In the literature sometimes the interior product of v and ϕ is alternatively abbreviated as $i_v \phi$. For $p = 0$,

$$v \lrcorner \phi := 0. \quad (\text{A.1.45})$$

Note that if $p = 1$, the definition (A.1.44) implies

$$v \lrcorner \phi = \phi(v). \quad (\text{A.1.46})$$

The following properties of interior multiplication follow immediately from the definitions (A.1.44), (A.1.45):

- 1) $v \lrcorner (\phi + \psi) = v \lrcorner \phi + v \lrcorner \psi$ [distributive law],
 2) $(v + u) \lrcorner \phi = v \lrcorner \phi + u \lrcorner \phi$ [linearity in a vector],
 3) $(av) \lrcorner \phi = a(v \lrcorner \phi)$ [multiplicative law],
 4) $v \lrcorner u \lrcorner \phi = -u \lrcorner v \lrcorner \phi$ [anticommutative law],

$$5) \quad v \lrcorner (\phi \wedge \omega) = (v \lrcorner \phi) \wedge \omega + (-1)^p \phi \wedge (v \lrcorner \omega) \quad [(\text{anti})\text{Leibniz rule}],$$

where $\phi, \psi \in \Lambda^p V^*, \omega \in \Lambda^q V^*, v, u \in V$, and $a \in \mathbb{R}$.

Let e_α be a basis of V and ϑ^α the cobasis of V^* . Then, by (A.1.46),

$$e_\alpha \lrcorner \vartheta^\beta = \vartheta^\beta(e_\alpha) = \delta_\alpha^\beta. \quad (\text{A.1.47})$$

Hence, if we apply the vector basis e_β to the p -form

$$\psi = \frac{1}{p!} \psi_{\alpha_1 \dots \alpha_p} \vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_p}, \quad (\text{A.1.48})$$

i.e., $e_\beta \lrcorner \psi$, then the properties listed above yield

$$e_\beta \lrcorner \psi = \frac{1}{(p-1)!} \psi_{\beta \alpha_2 \dots \alpha_p} \vartheta^{\alpha_2} \wedge \dots \wedge \vartheta^{\alpha_p}. \quad (\text{A.1.49})$$

If we multiply this formula by ϑ^β , we find the identity

$$\vartheta^\beta \wedge (e_\beta \lrcorner \psi) = p\psi. \quad (\text{A.1.50})$$

A.1.8 $\otimes$ Volume elements on a vector space, densities, orientation

A volume element is a form of maximal rank. Thus, it has one nontrivial component. Under the action of the linear group, this component is a density of weight +1. Orientation is an equivalence class of volume forms related by a positive real factor. The choice of an orientation is equivalent to the selection of similarly oriented bases in V .

The space $\Lambda^n V^*$ of exterior n -forms on an n -dimensional vector space V is 1-dimensional, and for $\omega \in \Lambda^n V^*$, we have

$$\omega = \frac{1}{n!} \omega_{\alpha_1 \dots \alpha_n} \vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_n} = \omega_{1\dots n} \vartheta^1 \wedge \dots \wedge \vartheta^n. \quad (\text{A.1.51})$$

The nonzero elements of $\Lambda^n V^*$ are called *volume elements*.

Consider a linear transformation (A.1.5) of the basis e_α of V . The corresponding transformation of the cobasis ϑ^α of V^* is given by (A.1.6).

Let $L := \det(L_{\alpha'}^\beta)$. Then

$$\begin{aligned} \vartheta^{1'} \wedge \dots \wedge \vartheta^{n'} &= L_{\alpha_1}^{1'} \dots L_{\alpha_n}^{n'} \vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_n} \\ &= (L_{\alpha_1}^{1'} \dots L_{\alpha_n}^{n'} \delta_{1\dots n}^{\alpha_1 \dots \alpha_n}) \vartheta^1 \wedge \dots \wedge \vartheta^n. \end{aligned}$$

Hence, for the volume element², we have:

$$\boxed{\vartheta^{1'} \wedge \cdots \wedge \vartheta^{n'} = \det(L_{\beta}^{\alpha'}) \vartheta^1 \wedge \cdots \wedge \vartheta^n.} \quad (\text{A.1.52})$$

Since, in terms of the basis $\vartheta^{\alpha'}$,

$$\omega = \omega_{1' \dots n'} \vartheta^{1'} \wedge \cdots \wedge \vartheta^{n'}, \quad (\text{A.1.53})$$

it follows from (A.1.51) and (A.1.52) that

$$\omega_{1 \dots n} = \det(L_{\beta}^{\alpha'}) \omega_{1' \dots n'} = L^{-1} \omega_{1' \dots n'} \quad (\text{A.1.54})$$

with $L^{-1} = \det(L_{\beta}^{\alpha'})$ and, conversely,

$$\omega_{1' \dots n'} = \det(L_{\alpha'}^{\beta}) \omega_{1 \dots n} = L \omega_{1 \dots n}. \quad (\text{A.1.55})$$

The geometric quantity with transformation law given by (A.1.55) is called a *scalar density*. It can easily be generalized. The geometric quantity $\mathcal{S}$ with transformation law

$$\mathcal{S}' = \det(L_{\alpha'}^{\beta})^w \mathcal{S} \quad (\text{A.1.56})$$

is called *scalar density of weight w* . The generalization to tensor densities of weight w in terms of components, see (A.1.11), reads

$$\mathcal{T}^{\alpha' \dots \beta' \dots}_{\beta' \dots \alpha' \dots} = \det(L_{\alpha'}^{\beta})^w L_{\alpha}^{\alpha'} \cdots L_{\beta'}^{\beta} \cdots \mathcal{T}^{\alpha \dots \beta \dots}_{\beta \dots \alpha \dots}, \quad (\text{A.1.57})$$

whereas *twisted* tensor densities of weight w on the right hand side pick up an extra factor $\text{sgn} \det(L_{\alpha'}^{\beta})$.

Let ω and φ be two arbitrary volume forms on V . We say that volumes are equivalent if a positive real number $a > 0$ exists such that $\varphi = a\omega$. This definition divides the space $\Lambda^n V^*$ into the two equivalence classes of n -forms. One calls each of the equivalence classes an *orientation* on V because a specification of a volume form uniquely determines a class of oriented bases on V and conversely. This can be demonstrated as follows.

On the intuitive level, in the case of a straight line, one speaks of an orientation by distinguishing between positive and negative directions, whereas in a plane one chooses positive and negative values of an angle, see Fig. A.1.4. The generalization of these intuitive ideas to an n -dimensional vector space is contained in the concept of *orientation*: one says that two bases e_{α} and h_{α} of V are *similarly oriented* if $h_{\alpha} = A_{\alpha}^{\beta} e_{\beta}$ with $\det(A_{\alpha}^{\beta}) > 0$. This is clearly an equivalence relation that divides the space of all bases of V into two classes. An *orientation of the vector space V* is an equivalence class of ordered bases, and V is called *oriented vector space* when a choice of orientation is made.

²For the definition of a volume element *without* the use of a metric, see also Synge and Schild [32], and in particular, Laurent [14].

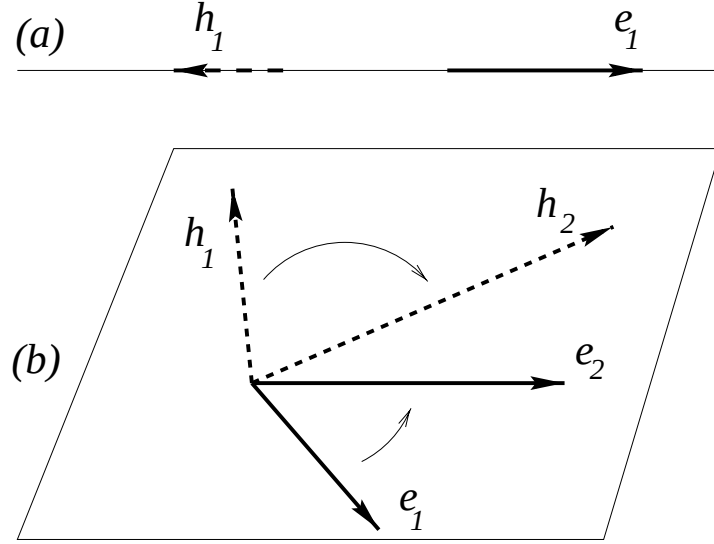


Figure A.1.4: Distinguishing orientation (a) on a line and (b) on a plane: The vector bases e_α and h_α are differently oriented.

Now we return to the volume elements in V . Given the volume ω we can define the function

$$o_\omega(e) := \text{sgn } \omega(e_1, \dots, e_n) \quad (\text{A.1.58})$$

on the set of all bases of V . It has only two values: $+1$ and -1 , and accordingly, we have a division of the set of all bases into the two subsets. One class is constituted of the bases for which $o_\omega(e) = 1$, and the other for which $o_\omega(e) = -1$. In each subset, the bases are similarly oriented.

In order to show this, let us assume that a volume form (A.1.51) is chosen and fixed, and let us take an arbitrary cobasis ϑ^α . The value of the volume form on the vectors of the basis e_α reads $\omega(e_1, \dots, e_n) = \omega_{1\dots n}(\vartheta^1 \wedge \dots \wedge \vartheta^n)(e_1, \dots, e_n) = \omega_{1\dots n}$. Suppose this number is positive; then in accordance with (A.1.58) we have $o_\omega(e) = +1$. For a different basis $h_\alpha = A_\alpha^\beta e_\beta$ we obtain $\omega(h_1, \dots, h_n) = \det(A_\alpha^\beta) \omega(e_1, \dots, e_n) = \det(A_\alpha^\beta) \omega_{1\dots n}$. Consequently, if the basis h_α is in the same subset as e_α , that is $o_\omega(h) = +1$, then $\det(A_\alpha^\beta) > 0$, which means that the bases e_α and h_α are similarly oriented. Conversely, assuming $\det(A_\alpha^\beta) > 0$ for any two bases $h_\alpha = A_\alpha^\beta e_\beta$, we find that (A.1.58) holds true for both bases.

Clearly, every volume form that is obtained by a “rescaling” $\omega_{1\dots n} \rightarrow \varphi_{1\dots n} = a \omega_{1\dots n}$ with a positive factor a will define the same orientation function (A.1.58): $o_\omega(e) = o_{a\omega}(e)$. This yields the whole class of equivalent volume forms that we introduced at the beginning of our discussion.

The *standard orientation* of V for an arbitrary basis e_α is determined by the volume form $\vartheta^1 \wedge \cdots \wedge \vartheta^n$ with cobasis ϑ^α . A simple reordering of the vectors (for example, an interchange of the first and the second leg) of a basis may change the orientation.

A.1.9 $\otimes$ Levi-Civita symbols and generalized Kronecker deltas

The Levi-Civita symbols are numerically invariant quantities and close relatives of the volume form. They can arise by applying the exterior product $\wedge$ or the interior product $\lrcorner$ n times, respectively. Levi-Civita symbols are totally antisymmetric tensor densities, and their products can be expressed in terms of the generalized Kronecker delta.

Volume forms provide a natural definition of very important tensor densities, the Levi-Civita symbols. In order to describe them, let us choose an arbitrary cobasis ϑ^α and consider the form of maximal rank

$$\hat{e} := \vartheta^1 \wedge \cdots \wedge \vartheta^n. \quad (\text{A.1.59})$$

We call this *an elementary volume*. Recall that the transformation law of this form is given by (A.1.52), which means that $\hat{e}$ is the *n -form density of weight -1* . By simple inspection it turns out that the wedge product

$$\vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_n} \quad (\text{A.1.60})$$

is either zero (when at least two of the wedge factors are the same) or equal to $\hat{e}$ up to a sign. The latter holds when all the wedge factors are different, and the sign is determined by the number of permutations that are needed for bringing the product (A.1.60) to the ordered form (A.1.59). This suggests a natural definition of the object $\epsilon^{\alpha_1 \cdots \alpha_n}$ that has similar symmetry properties. That is, we define it by the relation

$$\vartheta^{\alpha_1} \wedge \cdots \wedge \vartheta^{\alpha_n} =: \epsilon^{\alpha_1 \cdots \alpha_n} \hat{e}. \quad (\text{A.1.61})$$

As one can immediately check, the *Levi-Civita symbol* $\epsilon^{\alpha_1 \cdots \alpha_n}$ can be expressed in terms of the generalized Kronecker symbol (A.1.27)³:

$$\epsilon^{\alpha_1 \cdots \alpha_n} = \delta_{1 \cdots n}^{\alpha_1 \cdots \alpha_n}. \quad (\text{A.1.62})$$

In particular, we see that the only nontrivial component is $\epsilon^{1 \cdots n} = 1$. With respect to the change of basis, this quantity transforms as the $\begin{bmatrix} n \\ 0 \end{bmatrix}$ -valued 0-form density of weight $+1$:

$$\epsilon^{\alpha'_1 \cdots \alpha'_n} = \det(L_{\alpha'}^\beta) L_{\alpha_1}^{\alpha'_1} \cdots L_{\alpha_n}^{\alpha'_n} \epsilon^{\alpha_1 \cdots \alpha_n}. \quad (\text{A.1.63})$$

³See Sokolnikoff [29].

Recalling the definition of the determinant, we see that the components of the Levi-Civita symbol have *the same numerical values* with respect to all bases,

$$\epsilon^{\alpha'_1 \dots \alpha'_n} = \epsilon^{\alpha_1 \dots \alpha_n}. \quad (\text{A.1.64})$$

They are $+1$, -1 , or 0 .

Another fundamental antisymmetric object can be obtained from the elementary volume $\hat{e}$ with the help of the interior product operator. As we have learned from Sec. A.1.7, the interior product of a vector with a p -form generates a $(p-1)$ -form. Thus, starting with the elementary volume n -form and using a vector of basis e_α , we find an $(n-1)$ -form

$$\hat{e}_\alpha := e_\alpha \lrcorner \hat{e}. \quad (\text{A.1.65})$$

The transformation law of this object defines it as a *covector-valued* $(n-1)$ -form density of weight -1 :

$$\hat{e}_{\alpha'} = \det(L_{\alpha'}{}^\beta)^{-1} L_{\alpha'}{}^\alpha \hat{e}_\alpha. \quad (\text{A.1.66})$$

Applying once more the interior product of the basis to (A.1.65), one obtains an $(n-2)$ -form, and so on. Thus, we can construct chain of forms:

$$\hat{e}_{\alpha_1 \alpha_2} := e_{\alpha_2} \lrcorner \hat{e}_{\alpha_1}, \quad (\text{A.1.67})$$

$$\vdots$$

$$\hat{e}_{\alpha_1 \dots \alpha_n} := e_{\alpha_n} \lrcorner \dots \lrcorner e_{\alpha_1} \hat{e}. \quad (\text{A.1.68})$$

The last object is a 0-form. Property 4) of the interior product forces all these epsilons to be totally antisymmetric in all their indices. Similarly to (A.1.66), we can verify that for $p = 0, \dots, n$ the object $\hat{e}_{\alpha_1 \dots \alpha_p}$ is a $\begin{bmatrix} 0 \\ p \end{bmatrix}$ -valued $(n-p)$ -form density of the weight -1 . These forms $\{\hat{e}, \hat{e}_\alpha, \hat{e}_{\alpha_1 \alpha_2}, \dots, \hat{e}_{\alpha_1 \dots \alpha_n}\}$, alternatively to $\{\vartheta^\alpha, \vartheta^{\alpha_1} \wedge \vartheta^{\alpha_2}, \dots, \vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_n}\}$, can be used as a basis for arbitrary forms in the exterior algebra $\Lambda^* V$.

In particular, we find that (A.1.68) is the $\begin{bmatrix} 0 \\ n \end{bmatrix}$ -valued 0-form density of weight -1 . This quantity is also called the Levi-Civita symbol because of its evident similarity to (A.1.61). Analogously to (A.1.62) we can express (A.1.68) in terms of the generalized Kronecker symbol (A.1.27):

$$\hat{e}_{\alpha_1 \dots \alpha_n} = \delta_{\alpha_1 \dots \alpha_n}^{1 \dots n}. \quad (\text{A.1.69})$$

Thus we find again that the only nontrivial component is $\hat{e}_{1 \dots n} = +1$. Note that despite the deep similarity, we cannot identify the two Levi-Civita symbols in the absence of the metric; hence the different notation (with and without hat) is appropriate.

It is worthwhile to derive a useful identity for the product of the two Levi-Civita symbols:

$$\begin{aligned}
\epsilon^{\alpha_1 \dots \alpha_n} \hat{\epsilon}_{\beta_1 \dots \beta_n} &= \hat{\epsilon}^{\alpha_1 \dots \alpha_n} e_{\beta_n} \lrcorner \dots \lrcorner e_{\beta_1} \epsilon \\
&= e_{\beta_n} \lrcorner \dots \lrcorner e_{\beta_1} (\vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_n}) \\
&= (\vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_n}) (e_{\beta_1}, \dots, e_{\beta_n}) \\
&= \begin{vmatrix} \delta_{\beta_1}^{\alpha_1} & \dots & \delta_{\beta_n}^{\alpha_1} \\ \vdots & \ddots & \vdots \\ \delta_{\beta_1}^{\alpha_n} & \dots & \delta_{\beta_n}^{\alpha_n} \end{vmatrix} = \delta_{\beta_1 \dots \beta_n}^{\alpha_1 \dots \alpha_n}. \tag{A.1.70}
\end{aligned}$$

The whole derivation is based just on the use of the corresponding definitions. Namely, we use (A.1.68) in the first line, (A.1.61) in the second line, (A.1.44) in the third line, and (A.1.32) in the last line. This identity helps a lot in calculations of the different contractions of the Levi-Civita symbols. For example, we easily obtain from (A.1.70):

$$\epsilon^{\alpha_1 \dots \alpha_p \gamma_1 \dots \gamma_{n-p}} \hat{\epsilon}_{\beta_1 \dots \beta_p \gamma_1 \dots \gamma_{n-p}} = (n-p)! \delta_{\beta_1 \dots \beta_p}^{\alpha_1 \dots \alpha_p}. \tag{A.1.71}$$

Furthermore, let us take an integer $q < p$. The contraction of (A.1.70) over the $(n-q)$ indices yields the same result (A.1.71) with p replaced by q . Comparing the two contractions, we then deduce for the generalized Kroneckers:

$$\delta_{\beta_1 \dots \beta_q \gamma_{q+1} \dots \gamma_p}^{\alpha_1 \dots \alpha_q \gamma_{q+1} \dots \gamma_p} = \frac{(n-q)!}{(n-p)!} \delta_{\beta_1 \dots \beta_q}^{\alpha_1 \dots \alpha_q}. \tag{A.1.72}$$

In particular, we find

$$\delta_{\alpha_1 \dots \alpha_p}^{\alpha_1 \dots \alpha_p} = \frac{n!}{(n-p)!}. \tag{A.1.73}$$

Let us collect for a vector space of 4 dimensions the decisive formulas for going down the p -form ladder by starting from the 4-form density $\hat{\epsilon}$ and arriving at the 0-form $\hat{\epsilon}_{\alpha\beta\gamma\delta}$:

$$\begin{aligned}
\hat{\epsilon}_\alpha &= e_\alpha \lrcorner \hat{\epsilon} = \hat{\epsilon}_{\alpha\beta\gamma\delta} \vartheta^\beta \wedge \vartheta^\gamma \wedge \vartheta^\delta / 3!, \\
\hat{\epsilon}_{\alpha\beta} &= e_\beta \lrcorner \hat{\epsilon}_\alpha = \hat{\epsilon}_{\alpha\beta\gamma\delta} \vartheta^\gamma \wedge \vartheta^\delta / 2!, \\
\hat{\epsilon}_{\alpha\beta\gamma} &= e_\gamma \lrcorner \hat{\epsilon}_{\alpha\beta} = \hat{\epsilon}_{\alpha\beta\gamma\delta} \vartheta^\delta, \\
\hat{\epsilon}_{\alpha\beta\gamma\delta} &= e_\delta \lrcorner \hat{\epsilon}_{\alpha\beta\gamma}.
\end{aligned} \tag{A.1.74}$$

Going up the ladder yields:

$$\begin{aligned}
\vartheta^\alpha \wedge \hat{\epsilon}_{\beta\gamma\delta\mu} &= \delta_\mu^\alpha \hat{\epsilon}_{\beta\gamma\delta} - \delta_\delta^\alpha \hat{\epsilon}_{\beta\gamma\mu} + \delta_\gamma^\alpha \hat{\epsilon}_{\beta\delta\mu} - \delta_\beta^\alpha \hat{\epsilon}_{\gamma\delta\mu}, \\
\vartheta^\alpha \wedge \hat{\epsilon}_{\beta\gamma\delta} &= \delta_\delta^\alpha \hat{\epsilon}_{\beta\gamma} + \delta_\gamma^\alpha \hat{\epsilon}_{\delta\beta} + \delta_\beta^\alpha \hat{\epsilon}_{\gamma\delta}, \\
\vartheta^\alpha \wedge \hat{\epsilon}_{\beta\gamma} &= \delta_\gamma^\alpha \hat{\epsilon}_\beta - \delta_\beta^\alpha \hat{\epsilon}_\gamma, \\
\vartheta^\alpha \wedge \hat{\epsilon}_\beta &= \delta_\beta^\alpha \hat{\epsilon}.
\end{aligned} \tag{A.1.75}$$

One can, with respect to the $\hat{e}$ -system, define a (pre-metric) duality operator $\diamond$ that establishes an equivalence between p -forms and totally antisymmetric tensor densities of weight $+1$ and of type $\begin{bmatrix} n \\ 0 \end{bmatrix}^p$. In terms of the bases of the corresponding linear spaces, this operator is introduced as

$$\diamond(\hat{e}_{\alpha_1 \dots \alpha_p}) := \delta_{\alpha_1 \dots \alpha_p}^{\beta_1 \dots \beta_p} e_{\beta_1} \otimes \dots \otimes e_{\beta_p}. \quad (\text{A.1.76})$$

Consequently, given an arbitrary p -form ω expanded with respect to the $\hat{e}$ -basis as

$$\omega = \frac{1}{(n-p)!} \omega^{\alpha_1 \dots \alpha_{n-p}} \hat{e}_{\alpha_1 \dots \alpha_{n-p}}, \quad (\text{A.1.77})$$

the map $\diamond$ defines a tensor density by

$$\diamond \omega := \frac{1}{(n-p)!} \omega^{\alpha_1 \dots \alpha_{n-p}} e_{\alpha_1} \otimes \dots \otimes e_{\alpha_{n-p}}. \quad (\text{A.1.78})$$

For example, in $n = 4$ we have $\diamond \hat{e}_\alpha = e_\alpha$ and $\diamond \hat{e} = 1$. Thus, every 3-form $\varphi = \varphi^\alpha \hat{e}_\alpha$ is mapped into a vector density $\diamond \varphi = \varphi^\alpha e_\alpha$, whereas a 4-form ω yields a scalar density $\diamond \omega$.

A.1.10 The space M^6 of two-forms in four dimensions

Electromagnetic excitation and field strength are both 2-forms. On the 6-dimensional space of 2-forms, there exists a natural 6-metric, which is an important property of this space.

Let e_α be an arbitrary basis of V with $\alpha, \beta, \dots = 0, 1, 2, 3$. In later applications, the zeroth leg e_0 can be related to the time coordinate of spacetime, but this will not always be the case (for the null symmetric basis (C.2.14), for instance, all the e_α 's have the same status with respect to time). The three remaining legs will be denoted by e_a with $a, b, \dots = 1, 2, 3$. Accordingly, the dual basis of V^* is represented by $\vartheta^\alpha = (\vartheta^0, \vartheta^a)$.

In the linear space of 2-forms $\Lambda^2 V^*$, every element can be decomposed according to $\varphi = \frac{1}{2} \varphi_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta$. The basis $\vartheta^\alpha \wedge \vartheta^\beta$ consists of six simple 2-forms. This 6-plet can be alternatively numbered by a collective index. Accordingly, we enumerate the antisymmetric index pairs 01, 02, 03, 23, 31, 12 by uppercase letters $I, J, \dots$ from 1 to 6:

$$\begin{aligned} B^I &= \begin{pmatrix} B^1 \\ B^2 \\ B^3 \\ B^4 \\ B^5 \\ B^6 \end{pmatrix} = \begin{pmatrix} \vartheta^0 \wedge \vartheta^1 \\ \vartheta^0 \wedge \vartheta^2 \\ \vartheta^0 \wedge \vartheta^3 \\ \vartheta^2 \wedge \vartheta^3 \\ \vartheta^3 \wedge \vartheta^1 \\ \vartheta^1 \wedge \vartheta^2 \end{pmatrix} \\ &= \begin{pmatrix} \vartheta^0 \wedge \vartheta^a \\ \frac{1}{2} \hat{e}_{bcd} \vartheta^c \wedge \vartheta^d \end{pmatrix} = \begin{pmatrix} \beta^a \\ \hat{e}_b \end{pmatrix}. \end{aligned} \quad (\text{A.1.79})$$

With the B^I as basis (say “cyrillic B” or “Beh”), we can set up a 6-dimensional vector space $M^6 := \Lambda^2 V^*$. This vector space will play an important role in our considerations in Parts D and E. The extra decomposition with respect to β^a and $\hat{e}_b$ is convenient for recognizing where the electric and where the magnetic pieces of the field are located.

We denote the elementary volume 3-form by $\hat{e} = \vartheta^1 \wedge \vartheta^2 \wedge \vartheta^3$. Then $\hat{e}_a = e_a \lrcorner \hat{e}$ is the basis 2-form in the space spanned by the 3-coframe ϑ^a , see (A.1.65). This notation has been used in (A.1.79). Moreover, as usual, the 1-form basis can then be described by $\hat{e}_{ab} = e_b \lrcorner \hat{e}_a$. Some useful algebraic relations can be immediately derived:

$$\hat{e}_a \wedge \vartheta^b = \delta_a^b \hat{e}, \quad (\text{A.1.80})$$

$$\hat{e}_{ab} \wedge \vartheta^c = \delta_a^c \hat{e}_b - \delta_b^c \hat{e}_a, \quad (\text{A.1.81})$$

$$\hat{e}_{ab} \wedge \hat{e}_c = \hat{e} \hat{e}_{abc}. \quad (\text{A.1.82})$$

Correspondingly, taking into account that $\vartheta^0 \wedge \hat{e} =: \text{Vol}$ is the elementary 4-volume in V , we find

$$\beta^a \wedge \beta^b = 0, \quad (\text{A.1.83})$$

$$\hat{e}_a \wedge \hat{e}_b = 0, \quad (\text{A.1.84})$$

$$\hat{e}_a \wedge \beta^b = \delta_a^b \text{Vol}, \quad (\text{A.1.85})$$

$$\hat{e}_{ab} \wedge \beta^c \wedge \vartheta^d = (-\delta_a^c \delta_b^d + \delta_b^c \delta_a^d) \text{Vol}. \quad (\text{A.1.86})$$

Every 2-form, being an element of M^6 , can now be represented as $\varphi = \varphi_I B^I$ by its six components with respect to the basis (A.1.79). A 4-form ω , i.e., a form of the maximal rank in four dimensions, is expanded with respect to the wedge products of the B -basis as $\omega = \frac{1}{2} \omega_{IJ} B^I \wedge B^J$. The coefficients ω_{IJ} form a *symmetric* 6×6 matrix since the wedge product between 2-forms is evidently commutative.

A 4-form has only one component. This simple observation enables us to introduce a natural metric on the 6-dimensional space M^6 as the symmetric bilinear form

$$\varepsilon(\omega, \varphi) := (\omega \wedge \varphi)(e_0, e_1, e_2, e_3), \quad \omega, \varphi \in M^6, \quad (\text{A.1.87})$$

where e_α is a vector basis. Although the metric (A.1.87) apparently depends on the choice of basis, the linear transformation $e_{\alpha'} \rightarrow L_{\alpha'}^\alpha e_\alpha$ induces the pure rescaling $\varepsilon \rightarrow \det(L_{\alpha'}^\alpha) \varepsilon$. Using the expansion of the 2-forms with respect to the bivector basis B^I , the bilinear form (A.1.87) turns out to be

$$\varepsilon(\omega, \varphi) = \omega_I \varphi_J \varepsilon^{IJ}, \quad \text{where} \quad \varepsilon^{IJ} = (B^I \wedge B^J)(e_0, e_1, e_2, e_3). \quad (\text{A.1.88})$$

A direct inspection by using the definition (A.1.79) and the identity (A.1.85) shows that the 6-metric components read explicitly

$$\varepsilon^{IJ} = \begin{pmatrix} 0 & \mathbf{I}_3 \\ \mathbf{I}_3 & 0 \end{pmatrix}. \quad (\text{A.1.89})$$

Here $\mathbf{I}_3 := \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is the 3×3 unit matrix. Thus we see that the metric

(A.1.87) is always nondegenerate. Its signature is $(+, +, +, -, -, -)$. Indeed, the eigenvalues λ of the matrix (A.1.89) are defined by the characteristic equation $\det(\varepsilon^{IJ} - \lambda \delta^{IJ}) = (\lambda^2 - 1)^3 = 0$. The symmetry group that preserves the 6-metric (A.1.87) is isomorphic to $O(3, 3)$.

By construction, the elements of (A.1.89) numerically coincide with the components of the Levi-Civita symbol ϵ^{ijkl} , see (A.1.62):

$$\varepsilon^{IJ} = \epsilon^{IJ} := \begin{pmatrix} 0 & \mathbf{I}_3 \\ \mathbf{I}_3 & 0 \end{pmatrix}. \quad (\text{A.1.90})$$

Similarly, the covariant Levi-Civita symbol $\hat{\epsilon}_{mnpq}$, see (A.1.69), can be represented in 6D notation by the matrix

$$\hat{\epsilon}_{IK} = \hat{\epsilon}_{KI} := \begin{pmatrix} 0 & \mathbf{I}_3 \\ \mathbf{I}_3 & 0 \end{pmatrix}. \quad (\text{A.1.91})$$

One can immediately prove by multiplying the matrices (A.1.89) and (A.1.91) that their product is equal to 6D-unity, in complete agreement with (A.1.71). Thus, the Levi-Civita symbols can be consistently used for raising and lowering indices in M^6 .

⊗ Transformation of the M^6 -basis

What happens in M^6 when the basis in V is changed, i.e., $e_\alpha \rightarrow e_{\alpha'}$? As we know, such a change is described by the linear transformation (A.1.5). Then the cobasis transforms in accordance with (A.1.6):

$$\vartheta^\alpha = L_{\alpha'}{}^\alpha \vartheta^{\alpha'}. \quad (\text{A.1.92})$$

In the $(1 + 3)$ -matrix form, this can be written as

$$\begin{pmatrix} \vartheta^0 \\ \vartheta^a \end{pmatrix} = \begin{pmatrix} L_0^0 & L_b^0 \\ L_0^a & L_b^a \end{pmatrix} \begin{pmatrix} \vartheta'^0 \\ \vartheta'^b \end{pmatrix}. \quad (\text{A.1.93})$$

Correspondingly, the 2-form basis (A.1.79) transforms into a new bivector basis

$$B'^I = \begin{pmatrix} \beta'^a \\ \hat{\epsilon}'_b \end{pmatrix}, \quad (\text{A.1.94})$$

$$\beta'^a = \begin{pmatrix} \vartheta'^0 \wedge \vartheta'^1 \\ \vartheta'^0 \wedge \vartheta'^2 \\ \vartheta'^0 \wedge \vartheta'^3 \end{pmatrix}, \quad \hat{\epsilon}'_b = \begin{pmatrix} \vartheta'^2 \wedge \vartheta'^3 \\ \vartheta'^3 \wedge \vartheta'^1 \\ \vartheta'^1 \wedge \vartheta'^2 \end{pmatrix}.$$

Substituting (A.1.92) into (A.1.79), we find that the new and old 2-form bases are related by an induced linear transformation

$$\begin{pmatrix} \beta^a \\ \hat{\epsilon}_a \end{pmatrix} = \begin{pmatrix} P^a_b & W^{ab} \\ Z_{ab} & Q_a^b \end{pmatrix} \begin{pmatrix} \beta'^b \\ \hat{\epsilon}'_b \end{pmatrix}, \quad (\text{A.1.95})$$

where

$$P^a_b = L_0^0 L_b^a - L_0^a L_b^0, \quad Q_b^a = \det L_c^d (L^{-1})_b^a, \quad (\text{A.1.96})$$

$$W^{ab} = L_c^0 L_d^a \epsilon^{bcd}, \quad Z_{ab} = \hat{\epsilon}_{acd} L_0^c L_b^d. \quad (\text{A.1.97})$$

The 3×3 matrix $(L^{-1})_b^a$ is inverse to the 3×3 subblock L_a^b in (A.1.92). The inverse transformation is easily computed:

$$\begin{pmatrix} \beta'^a \\ \hat{\epsilon}'_a \end{pmatrix} = \frac{1}{\det L} \begin{pmatrix} Q_b^a & W^{ba} \\ Z_{ba} & P_a^b \end{pmatrix} \begin{pmatrix} \beta^b \\ \hat{\epsilon}_b \end{pmatrix}, \quad (\text{A.1.98})$$

where the determinant of the transformation matrix (A.1.92) reads

$$\det L := \det L_{\alpha'}^{\beta} = [L_0^0 - L_a^0 (L^{-1})_b^a L_0^b] \det L_c^d. \quad (\text{A.1.99})$$

One can write an arbitrary linear transformation $L \in GL(4, \mathbb{R})$ as a product

$$L = L_1 L_2 L_3 \quad (\text{A.1.100})$$

of three matrices of the form

$$L_1 = \begin{pmatrix} 1 & U_b \\ 0 & \delta_b^a \end{pmatrix}, \quad (\text{A.1.101})$$

$$L_2 = \begin{pmatrix} 1 & 0 \\ V^a & \delta_b^a \end{pmatrix}, \quad (\text{A.1.102})$$

$$L_3 = \begin{pmatrix} \Lambda_0^0 & 0 \\ 0 & \Lambda_b^a \end{pmatrix}. \quad (\text{A.1.103})$$

Here $V^a, U_b, \Lambda_0^0, \Lambda_b^a$, with $a, b = 1, 2, 3$, describe $3 + 3 + 1 + 9 = 16$ elements of an arbitrary linear transformation. The matrices $\{L_3\}$ form the group $\mathbb{R} \otimes GL(3, \mathbb{R})$ which is a subgroup of $GL(4, \mathbb{R})$, whereas the sets of unimodular matrices $\{L_1\}$ and $\{L_2\}$ evidently form two *Abelian* subgroups in $GL(4, \mathbb{R})$.

In the study of the covariance properties of various objects in M^6 , it is thus sufficient to consider the three separate cases (A.1.101)-(A.1.103). Using (A.1.96) and (A.1.97), we find for $L = L_1$

$$P^a_b = Q_b^a = \delta_b^a, \quad W^{ab} = \epsilon^{abc} U_c, \quad Z_{ab} = 0. \quad (\text{A.1.104})$$

Similarly, for $L = L_2$ we have

$$P^a_b = Q_b^a = \delta_b^a, \quad W^{ab} = 0, \quad Z_{ab} = \hat{\epsilon}_{abc} V^c, \quad (\text{A.1.105})$$

and for $L = L_3$

$$P^a_b = \Lambda_0^0 \Lambda_b^a, \quad Q_b^a = (\det \Lambda) (\Lambda^{-1})_b^a, \quad W^{ab} = Z_{ab} = 0. \quad (\text{A.1.106})$$

A.1.11 Almost complex structure on M^6

An almost complex structure on the space of 2-forms determines a splitting of the complexification of M^6 into two invariant 3-dimensional subspaces.

Let us introduce an almost complex structure $\mathbf{J}$ on M^6 . We recall that every tensor of type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ represents a linear operator on a vector space. Accordingly, if $\varphi \in M^6$, it is of type $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\mathbf{J}(\varphi)$ can be defined as a contraction. The result will also be an element of M^6 . By definition, $\mathbf{J}(\mathbf{J}(\varphi)) = -\mathbf{I}_6 \varphi$ or

$$\mathbf{J}^2 = -1, \quad (\text{A.1.107})$$

see (A.1.19).

As a tensor of type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, the operator $\mathbf{J}$ can be represented as a 6×6 matrix. Since the basis in M^6 is naturally split into $3 + 3$ parts in (A.1.79), we can write it in terms of the set of four 3×3 matrices,

$$J_I^K = \begin{pmatrix} C^a{}_b & A^{ab} \\ B_{ab} & D_a{}^b \end{pmatrix}. \quad (\text{A.1.108})$$

Because of (A.1.107), the 3×3 blocks A, B, C, D are constrained by

$$\begin{aligned} A^{ac} B_{cb} + C^a{}_c C^c{}_b &= -\delta_b^a, \\ C^a{}_c A^{cb} + A^{ac} D_c{}^b &= 0, \\ B_{ac} C^c{}_b + D_a{}^c B_{cb} &= 0, \\ B_{ac} A^{cb} + D_a{}^c D_c{}^b &= -\delta_a^b. \end{aligned} \quad (\text{A.1.109})$$

$\otimes$ Complexification of M^6

An almost complex structure on M^6 motivates a *complex* generalization of M^6 to the *complexified* linear space $M^6(\mathbb{C})$. The elements of $M^6(\mathbb{C})$ are the complex 2-forms $\omega \in M^6(\mathbb{C})$, i.e. their components ω_I in a decomposition $\omega = \omega_I B^I$ are complex. Alternatively, one can consider $M^6(\mathbb{C})$ as a real 12-dimensional linear space spanned by the basis (B^I, iB^I) , where i is the imaginary unit. We denote by $\overline{M^6(\mathbb{C})}$ the complex conjugate space.

The same symmetric bilinear form as in (A.1.87) also defines a natural metric in $M^6(\mathbb{C})$. Note however, that now an orthogonal (complex) basis can always be introduced in $M^6(\mathbb{C})$ so that $\varepsilon^{IJ} = \delta^{IJ}$ in that basis.

Incidentally, one can define another scalar product on a complex space $M^6(\mathbb{C})$ by

$$\varepsilon'(\omega, \varphi) := (\omega \wedge \overline{\varphi})(e_0, e_1, e_2, e_3), \quad \omega, \varphi \in M^6. \quad (\text{A.1.110})$$

The significant difference between these two metrics is that ε' assigns a real length to any complex vector, whereas ε defines complex vector lengths.

We will assume that the $\mathbf{J}$ operator is defined in $M^6(\mathbb{C})$ by the same formula as $\mathbf{J}(\omega)$ in M^6 . In other words, $\mathbf{J}$ remains a *real* linear operator in $M^6(\mathbb{C})$, i.e. for every complex 2-form $\omega \in M^6(\mathbb{C})$ one has $\overline{\mathbf{J}(\omega)} = \mathbf{J}(\overline{\omega})$. The eigenvalue problem for the operator $\mathbf{J}(\omega_\lambda) = \lambda \omega_\lambda$ is meaningful only in the complexified space $M^6(\mathbb{C})$ because, in view of the property (A.1.107), the eigenvalues are $\lambda = \pm i$. Each of these two eigenvalues has multiplicity three, which follows from the reality of $\mathbf{J}$. Note that the 6×6 matrix of the $\mathbf{J}$ operator has six eigenvectors, but the number of eigenvectors with eigenvalue $+i$ is equal to the number of eigenvectors with eigenvalue $-i$ because they are complex conjugate to each other. Indeed, if $\mathbf{J}(\omega) = i\omega$, then the conjugation yields $\overline{\mathbf{J}(\omega)} = \mathbf{J}(\overline{\omega}) = -i\overline{\omega}$.

Let us denote the 3-dimensional subspaces of $M^6(\mathbb{C})$ that correspond to the eigenvalues $+i$ and $-i$ by

$$\begin{aligned} M^{(s)} &:= \{ \omega \in M^6(\mathbb{C}) \mid \mathbf{J}(\omega) = i\omega \}, \\ M^{(a)} &:= \{ \varphi \in M^6(\mathbb{C}) \mid \mathbf{J}(\varphi) = -i\varphi \}, \end{aligned} \quad (\text{A.1.111})$$

respectively. Evidently, $\overline{M^{(s)}} = M^{(a)}$. Therefore, we can restrict our attention only to the self-dual subspace. This will be assumed in our derivations from now on.

Accordingly, every form ω can be decomposed into a self-dual and an anti-self-dual piece⁴,

$$\omega = \omega^{(s)} + \omega^{(a)}, \quad (\text{A.1.112})$$

with

$$\omega^{(s)} = \frac{1}{2} [\omega - i\mathbf{J}(\omega)], \quad \omega^{(a)} = \frac{1}{2} [\omega + i\mathbf{J}(\omega)]. \quad (\text{A.1.113})$$

It can be checked that $\mathbf{J}(\omega^{(s)}) = +i\omega^{(s)}$ and $\mathbf{J}(\omega^{(a)}) = -i\omega^{(a)}$.

A.1.12 Computer algebra

Also in electrodynamics, research usually requires the application of computers. Besides numerical methods and visualization techniques, the manipulation of formulas by means of “computer algebra” systems is nearly a must. By no means are these methods confined to pure algebra. Differentiations and integrations, for example, can also be executed with the help of computer algebra tools.

“If we do work on the *foundations* of classical electrodynamics, we can dispense with computer algebra,” some true fundamentalists will claim. Is this really true? Well later, in Chap. D.2, we analyze the Fresnel equation; we couldn’t have done it to the extent we did without using an efficient computer algebra

⁴A discussion of the use of self dual and anti-self dual 2-forms in general relativity can be found in Kopczyński and Trautman [13], e.g..

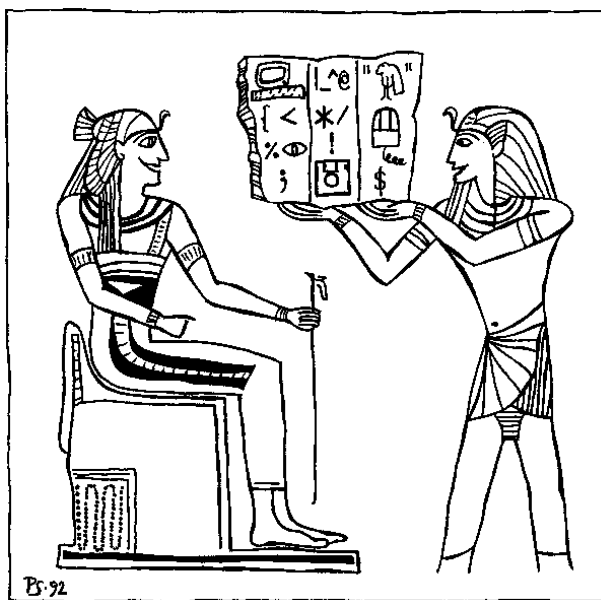


Figure A.1.5: “Here is the new Reduce-update on a hard disk.”

system. Thus, our fundamentalist is well advised to learn some computer algebra. Accordingly, in addition to introducing some mathematical tools in exterior calculus, we mention computer algebra systems like *Reduce*⁵, *Maple*⁶, and *Mathematica*⁷ and specifically explain how to apply the Reduce package *Excalc*⁸ to the exterior forms that occur in electrodynamics.

In practical work in solving problems by means of computer algebra, it is our experience that it is best to have access to different computer algebra systems. Even though in the course of time good features of one system “migrated” to other systems, still, for a certain specified purpose one system may be better suited than another one — and for different purposes these may be different systems. There does not exist as yet *the* optimal system for all purposes. There-

⁵Hearn [8] created this Lisp-based system. For introductions to Reduce, see Toussaint [36], Grozin [6], MacCallum and Wright [15], or Winkelmann and Hehl [38]; in the latter text you can learn how to get hold of a Reduce system for your computer. Reduce as applied to general-relativistic field theories is described, for example, by McCrea [16] and by Socorro et al. [28]. In our presentation, we partly follow the lectures of Toussaint [36].

⁶Maple, written in C, was created by a group at the University of Waterloo, Canada. A good introduction is given by Char et al. [3].

⁷Wolfram (see [39]) created the C-based Mathematica software package which is in very widespread use.

⁸Schrufer@Schrüfer [25, 26] is the creator of that package (cf. also [27]). Excalc is applied to Maxwell's theory by Puntigam et al. [22].

fore, it is not a rare occasion that we have to feed the results of a calculation by means of one system as input into another system.

For computations in electrodynamics, relativity, and gravitation, we keep the three general-purpose computer algebra systems: Reduce, Maple, and Mathematica. Other systems are available⁹. Our workhorse for corresponding calculations in exterior calculus is the Reduce package Excalc, but also in the *MathTensor* package¹⁰ of Mathematica exterior calculus is implemented. For the manipulation of tensors we use the following packages: In Reduce the library of McCrea¹¹ and *GRG*¹², in Maple *GRTensorII*¹³, and in Mathematica, besides MathTensor, the *Cartan* package¹⁴.

Computer algebra systems are almost exclusively interactive systems nowadays. If one is installed on your computer, you can usually call the system by typing its name or an abbreviation thereof, i.e., ‘`reduce`’, ‘`maple`’, or ‘`math`’, and then hitting the return key, or clicking on the corresponding icon. In the case of ‘`reduce`’, the system introduces itself and issues a ‘`1:`’. It waits for your first command. A command is a statement, usually some sort of expression, a part of a formula or a formula, followed by a terminator¹⁵. The latter, in Reduce, is a semicolon ; if you want to see the answer of the system, and otherwise a dollar sign \$. Reduce is case insensitive, i.e., the lowercase letter `a` is not distinguished from the uppercase letter `A`.

Formulating Reduce input

As an input statement to Reduce, we type in a certain legitimately formed expression. This means that, with the help of some operators, we compose formulas according to well-defined rules. Most of the built-in operators of Reduce, like the arithmetic operators `+` (plus), `-` (minus), `*` (times), `/` (divided by), `**` (to the power of)¹⁶ are self-explanatory. They are so-called *infix operators* since they are positioned *in* between their arguments. By means of them we can construct combined expressions of the type $(x + y)^2$ or $x^3 \sin x$, which in Reduce read `(x+y)**2` and `x**3*sin(x)`, respectively.

If the command

⁹In the review of Hartley [7] possible alternative systems are discussed (see also Heinicke et al. [11]).

¹⁰Parker and Christensen [21] created this package; for a simple application see Tsantilis et al. [37].

¹¹See McCrea’s lectures [16].

¹²The GRG system, created by Zhytnikov [40], and the GRG_{EC} system of Tertychniy [35, 34, 20] grew from the same root; for an application of GRG_{EC} to the Einstein–Maxwell equations, see [33].

¹³See the documentation of Musgrave et al. [19]. Maple applications to the Einstein–Maxwell system are covered in the lectures of McLenaghan [17].

¹⁴Soleng [30] is the creator of ‘Cartan’.

¹⁵Has nothing to do with Arnold Schwarzenegger!

¹⁶Usually one takes the circumflex for exponentiation. However, in the Excalc package this operator is redefined and used as the wedge symbol for exterior multiplication.

Switch	description if switch is on	example
* allfac	factorize simple factors	$2x + 2 \rightarrow 2(x + 1)$
div	divide by the denominator	$(x^2 + 2)/x \rightarrow x + 2/x$
* exp	expand all expressions	$(x + 1)(x - 1) \rightarrow x^2 - 1$
* mcd	make (common) denominator	$x + x^{-1} \rightarrow (x^2 + 1)/x$
* lcm	cancel least common multiples	
gcd	cancel greatest common divisor	
rat	display as polynomial in factor	$\frac{x+1}{x} \rightarrow 1 + x^{-1}$
* ratpri	display rationals as fraction	$1/x \rightarrow \frac{1}{x}$
* pri	dominates allfac,div,rat,revpri	
revpri	display polynom. in opposite order	$x^2 + x + 1 \rightarrow 1 + x + x^2$
rounded	calculate with floats	$1/3 \rightarrow 0.333333333333$
complex	simplify complex expressions	$1/i \rightarrow -i$
nero	don't display zero results	$0 \rightarrow$
* nat	display in Reduce input format	$\frac{x^2}{3} \rightarrow x**2/3$
msg	suppress messages	
fort	display in Fortran format	
tex	display in TeX format	

Table A.1.2: Switches for Reduce's reformulation rules. Those marked with * are turned on by default; the other ones are off.

```
(x+y)**2;
```

is executed, you will get the expanded form $x^2 + 2xy + y^2$. There is a so-called switch **exp** in Reduce that is usually switched on. You can switch it off by the command

```
off exp;
```

Type in again

```
(x+y)**2;
```

Now you will find that Reduce doesn't do anything and gives the expression back as it received it. With **on exp**; you can go back to the original status.

Using the switches is a typical way to influence Reduce's way of how to evaluate an expression. A partial list of switches is presented in a table on the next page.

Let us give some more examples of expressions with infix operators:

```
(u+v)*(y-x)/8
(a>b) and (c<d)
```

Here we have the logical and relational operators **and**, **>** (greater than), **<** (less than). Widely used are also the infix operators:

```
neq  >=  <=  or  not  :=
```

The operator **neq** means not equal. The assignment operator **:=** assigns the value of the expression on its right-hand side (its second argument) to the identifier on its left-hand side (its first argument). In Reduce, logical (or Boolean) expressions have only the truth values **t** (true) or **nil** (false). They are only allowed within certain statements (namely in **if**, **while**, **repeat**, and **let** statements) and in so-called rule lists.

A *prefix operator* stands in front of its argument(s). The arguments are enclosed by parentheses and separated by commas:

```
cos(x)
int(cos(x),x)
factorial(8)
```

In ordinary notation, the second statement reads $\int \cos x \, dx$.

The following mathematical functions are built-in as prefix operators:

sin	sind	cos	cosd	tan	tand
cot	cotd	sec	secd	csc	cscd
asin	asind	acos	acosd	atan	atand
acot	acotd	asec	asecd	acsc	acscd
sinh	cosh	tanh	coth	sech	csch
asinh	acosh	atanh	acoth	asech	acsch
sqrt	exp	ln	log	log10	logb
dilog	erf	expint	cbrt	abs	hypot
factorial					

Identifiers ending with **d** indicate that this operator expects its argument to be expressed in degrees. **log** stands for the natural logarithm and is equivalent to **ln**. **logb** is the logarithm in base n ; accordingly n must be specified as a second argument of **logb**. **hypot** calculates the hypotenuse according to $\text{hypot}(x, y) = \sqrt{x^2 + y^2}$. **csc** is the cosecant, **dilog** the Euler dilogarithm with $\text{dilog}(z) = -\int_0^z \log(1-\zeta)/\zeta \, d\zeta$, **erf** the Gaussian error function with $\text{erf}(x) = 2/\sqrt{\pi} \int_0^x e^{-t^2} dt$, **abs** the absolute value function, **expint** the exponential

integral with $\text{expint}(x) = \int_{-\infty}^x e^t/t \, dt$, and finally, `cbrt` the operator for the cubic root.

Reduce only knows a few elementary rules for these operators. In addition to these built-in rules and operators, the Reduce user may want to define her or his own rules and operators (i.e., functions) by calling the command `operator`. No arguments are specified in the declaration statement. After the declaration, the specified operators may be used with arguments, just like `sin`, `cos`, etc.:

```
clear f,k,m,n$
operator f$
f(m);
f(n):=n**4+h**3+p**2+u;
f(4,k):=g;
```

If an operator is given with a certain argument, say `f(n)`, and an expression (here `n**4+h**3+p**2+u`, which contains the argument `n` of the operator) is assigned to the operator `f(n)`, this is a specific assignment only. There is *no* general functional relationship established between the argument of the operator and the same identifier that appears in the assigned expression. Such a relationship can only be created by self-defined rules. Let us demonstrate this somewhat difficult point as follows:

```
f(n):=n**4+h**3+p**2+u$
f(k);                                % does not evaluate to the value
                                     % k**4+h**3+p**2+u, but only to f(k)
f(n);                                % again yields n**4+h**3+p**2+u
```

A newly created operator, which has no previously assigned value, carries as its value its own name with the arguments (in contrast to the elements of an array which are initialized with value zero and which can never have as values the array name with their indices!). These operators have no properties unless `let` rules are specified.

All operators may

- have values assigned to them, as in

```
log(u):=12$
cos(2*k*pi):=1$
```

- have properties declared for some collections of arguments (for example, the value of `sin(integer*pi)` is always 0),
- be fully defined, either by the user or by Reduce, as is the case for the operator `df` for differentiation.

With the operators defined so far, we can construct Reduce expressions by combining variables and operators in such a way that they represent our mathematical formulas. Reduce distinguishes between three kinds of expressions: integer, scalar, and Boolean.

Integer expressions evaluate to whole numbers, as in:

```
2
9-6
5**7+9*(6-j)*(k+h)
```

provided the variables j, k, h evaluate to integers.

Scalar expressions consist of (syntactically correct) sequences of numbers, variables, operators, left and right parentheses, and commas and are the usual representation of mathematical expressions in Reduce:

```
sin(8*y**4)+h(u)-(a+b)**7
df(u,x,8)*pi
b(y)+factorial(9)
a
```

The minimal scalar expressions that are known to Reduce are variables or numbers. The following rules are applied on evaluation of scalar expressions:

- Variables and operators with a number of arguments have the algebraic value they were last assigned or, if never assigned, stand for themselves.
- Nevertheless, some special expressions, such as elements of arrays (indexed variables), initially have the value 0.
- Operators act according to the rules that are defined for them. If there is no matching rule, the operator with its argument stands for itself (`cos 0`, for example, evaluates to 1, but `cos x` won't get evaluated as long as x is an unbound variable). Note that an (inappropriate) assignment such as `cos(0):=7` will have the same effect as a rule that defines `cos(0)` to be 7.
- Procedures of expressions are evaluated with the values of their actual parameters used in the procedure call.
- The algebraic evaluation of expressions (also called simplification) is controlled by the switches, which may be turned on or off by the Reduce user.
- In any case, the standard rules of algebra apply. Parentheses are allowed. Expressions may be combined with legal operators to build new expressions. Those new expressions take on the new value built from the values of the subexpressions via the operators, taking into account the control switches.

Examples:

```

clear a,b$           % a and b are declared to be unbound
a*b;
pol;                 % still not assigned
pol:=(a+b)**3$       % now assigned
pol;
on gcd$              % greatest common divisor switch on
off exp$             % expansion switch off
pol;
f:=g*m*m/r**2;
on div$              % removes identical factors in
                     % numerator and denominator
f;
off gcd,div$ on exp$ % reset switches

```

We didn't give the output. You should try to get this yourself on your computer.

Boolean expressions use the well-known Boolean algebra and have truth values `t` for true and `nil` for false. For handling Boolean expressions we have already mentioned the Boolean infix operators. Boolean expressions are only allowed within `if`-, `while`-, or `repeat`-statements. Examples of typical Boolean expressions are

```

j neq 2
a=b and (d or g)
(a+7) > 18           % if a evaluates to an integer

```

If you want to display the truth value of a Boolean expression, use the `if`-statement, as in the following example:

```

if 2**28 < 10**7 then write "less"
    else write "greater or equal";

```

Rudiments of evaluation

A Reduce program is a follow-up of commands. And the evaluation of the commands may be conditioned by switches that we switch on or off (also by a command). Let us look into the evaluation process a bit closer. After a command has been sent to the computer by hitting the return key, the whole command is evaluated. Each expression is evaluated from left to right, and the values obtained are combined with the operators specified. Substatements or subexpressions existing within other expressions, as in

Now `a` has the value of $(7+h)^{**3}$ rather than $(g+h)^{**3}$.

Sometimes it is necessary to remove the assigned value from a variable or an expression. This can be achieved by using the operator `clear` as in

```
clear g,h$
a:=(g+h)**3$
g:=1$
a;
clear g$
a;
```

or by overwriting the old value by means of a new assignment statement:

```
clear b,u,v$
a:=(u+v)**2$
a:=a-v**2$
a;
b:=b+1$
b;
```

The evaluation of `a;` results in the value $u*(u+2*v)$ since $(u+v)^{**2}$ had been assigned to `a` and $a-v^{**2}$ (i.e., $(u+v)^{**2}-v^{**2}$) was reassigned to `a`. The assignment `b:=b+1;` will, however, lead to a difficulty: Since no value was previously assigned to `b`, the assignment replaces `b` *literally* with `b+1` (whereas the previous `a:=a-v**2` statement produces the *evaluation* `a:=(u+v)**2-v**2`). The last evaluation `b;` will lead to an error or will even hang up the system because `b+1` is assigned to `b`. As soon as `b` is evaluated, Reduce returns `b+1`, whereby `b` still has the value `b+1`, and so on. Therefore the evaluation process leads to an infinite loop. Hence we should avoid such recursions.

Incidentally, if you want to finish a Reduce session, just type in `bye;`

After these glimpses of Reduce, we will turn to the real object of our interest.

Loading Excalc

We load the Excalc package by

```
load_package excalc$
```

The system will tell us that the operator $\wedge$ is redefined since it became the new wedge operator.

Excalc is designed such that the input to the computer is the same as what would have been written down for a hand calculation. For example, the statement $f \wedge x \wedge y + u \wedge (y \wedge z \wedge x)$ would be a legitimately built Excalc expression with $\wedge$ denoting the exterior and $\wedge$ (underline followed by a vertical bar) the interior product sign. Note that *before* the interior product sign $\wedge$ (spoken *in*) there must be a blank; the other blanks are optional.

However, before Excalc can understand our intentions, we better declare u to be a (tangential) vector

```
tvector u;
```

f to be a scalar (i.e., a 0-form), and x, y, z to be 1-forms:

```
pform f=0, x=1, y=1, z=1;
```

A variable that is not declared to be a vector or a form is treated as a constant; thus 0-forms must also be declared. After our declarations, we can input our command

```
f*x^y+u |(y^z^x);
```

Of course, the system cannot do much with this expression, but it expands the interior product. It also knows, of course, that

```
u |f;
```

vanishes, that $y \wedge x = -x \wedge y$, or that $x \wedge x = 0$. If we want to check the rank of an expression, we can use

```
exdegree(x^y);
```

This yields 2 for our example.

Quite generally, Excalc can handle scalar-valued exterior forms, vectors and operations between them, as well as nonscalar valued forms (indexed forms). Simple examples of indexed forms are the Kronecker delta δ_α^β or the connection 1-form Γ_α^β of Sec. C.1.2. Their declaration reads

```
pform delta(a,b)=0, gamma1(a,b)=1;
```

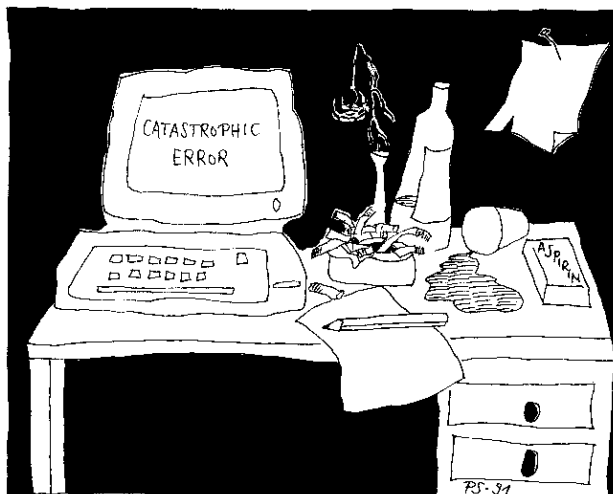


Figure A.1.6: “Catastrophic error,” a Reduce error message.

The names of the indices are arbitrary. Subsequently, in the program a lower index is marked by a minus sign and an upper index with a plus (or with nothing), i.e., δ_1^1 translates into `delta(-1,1)`, and so on.

Excalc is a good tool for studying differential equations, for making calculations in field theory and general relativity, and for such simple things as calculating the Laplacian of a tensor field for an arbitrarily given frame. Excalc is completely embedded in Reduce. Thus, all features and facilities of Reduce are available in a calculation.

If we declare the dimension of the underlying space by

```
spacedim 4;
```

then

```
pform a=2,b=3;  a^b;
```

yields 0.

These are the fundamental commands of Excalc for exterior algebra. As soon as we have introduced exterior calculus with frames and coframes, vector fields and fields of forms, not to forget exterior and Lie differentiation, we will come back to Excalc and better appreciate its real power.

A.2

Exterior calculus

Having developed the concepts involved in the exterior *algebra* associated with an n -dimensional linear vector space V , we now look at how this structure can be “lifted” onto an n -dimensional *differentiable manifold* X_n or, for short, onto X . The procedure for doing this is the same as for the transition from tensor algebra to tensor calculus. At each point x of X there is an n -dimensional vector space X_x , the tangent vector space at x . We identify the space X_x with the vector space V considered in the previous chapter. Then, at each point x , the exterior algebra of forms is determined on $V = X_x$. However, in differential geometry, one is concerned not so much with objects defined at isolated points as with *fields* over the manifold X or over open sets $U \subset X$. A field ω of p -forms on X is defined by assigning a p -form to each point x of X and, if this assignment is performed in a smooth manner, we shall call the resulting field of p -forms an *exterior differential p -form*. For simplicity we shall take “smooth” to mean C^∞ , although in physical applications the degree of differentiability may be less.

A.2.1 $\otimes$ Differentiable manifolds

A topological space becomes a differentiable manifold when an atlas of coordinate charts is introduced in it. Coordinate transformations are smooth in the intersections of the charts. The atlas is oriented when in all intersections the Jacobians of the coordinate transformations are positive.

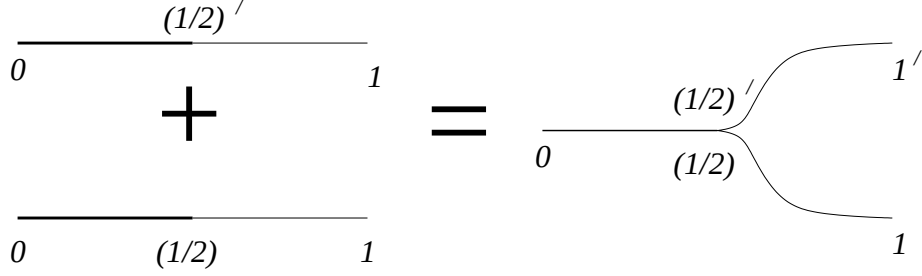


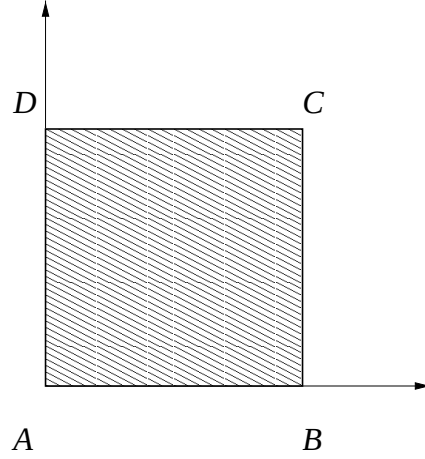
Figure A.2.1: *Non-Hausdorff* manifold: Take two copies of the line segment $\{0, 1\}$ and identify (paste together) their left halves *excluding* the points $(1/2)$ and $(1/2)'$. In the resulting manifold, the Hausdorff axiom is violated for the pair of points $(1/2)$ and $(1/2)'$.

In order to describe more rigorously how fields are introduced on X , we have to recall some basic facts about manifolds. At the start, one needs a *topological structure*. To be specific, we will normally assume that X is a connected, Hausdorff, and paracompact topological space. A *topology* on X is introduced by the collection of *open sets* $\mathcal{T} = \{U_\alpha \subset X \mid \alpha \in I\}$ which, by definition, satisfy three conditions: (i) both the empty set $\emptyset$ and the manifold itself X belong to that collection, $\emptyset, X \in \mathcal{T}$, (ii) any *union* of open sets is again open, i.e., $\bigcup_{\alpha \in J} U_\alpha \in \mathcal{T}$ for any subset $J \in I$, and (iii) any *intersection of a finite* number of open sets is open, i.e., $\bigcap_{\alpha \in K} U_\alpha \in \mathcal{T}$ for any finite subset $K \in I$.

A topological space X is *connected* if one cannot represent it by the sum $X = X_1 \cup X_2$ with open $X_{1,2}$ and $X_1 \cap X_2 = \emptyset$. Usually for a spacetime manifold, one further requires a *linear connectedness*, which means that any two points of X can be connected by a continuous path. A topological space X is *Hausdorff* when for any two points $p_1 \neq p_2 \in X$ one can find open sets $U_1, U_2 \subset X$ with $p_1 \in U_1$ and $p_2 \in U_2$ such that $U_1 \cap U_2 = \emptyset$. Hausdorff's axiom forbids the “branched” manifolds of the sort depicted in Fig. A.2.1. A connected Hausdorff manifold is *paracompact* when X can be covered by a countable number of open sets, i.e., $X = \bigcup_{\alpha \in K} U_\alpha$ for a countable subset $K \in I$. Finally, a manifold X is *compact* when it can be covered by a finite number of open sets selected from its arbitrary covering.

A collection of functions $\{\rho_\alpha : X \rightarrow \mathbb{R}\}$ is called a *partition of unity subordinate* to a covering $\{U_\alpha\}$ if support of $\rho_\alpha \subset U_\alpha$ and $0 \leq \rho_\alpha(p) \leq 1$, $\sum_\alpha \rho_\alpha(p) = 1$ for all points $p \in X$. The partition of unity always exists for paracompact manifolds, and it is a standard tool that helps to derive global constructions from the local ones.

A *differentiable manifold* is a topological space X plus a *differentiable structure* on it. The latter is defined as follows: A *coordinate chart* on X is a pair (U, ϕ) where $U \in \mathcal{T}$ is an open set and the map $\phi : U \rightarrow \mathbb{R}^n$ is a homeomorphism (i.e., continuous with a continuous inverse map) of U onto an open subset of the arithmetic space of n -tuples $\mathbb{R}^n$. This map assigns n labels or coordinates

Figure A.2.2: Rectangle $ABCD$ in $\mathbb{R}^2$.

$\phi(p) = \{x^1(p), \dots, x^n(p)\}$ to any point $p \in U \subset X$. Given any two intersecting charts, (U_α, ϕ_α) and (U_β, ϕ_β) with $U_\alpha \cap U_\beta \neq \emptyset$, the map

$$f_{\alpha\beta} := \phi_\alpha \circ \phi_\beta^{-1} : \mathbb{R}^n \longrightarrow \mathbb{R}^n \quad (\text{A.2.1})$$

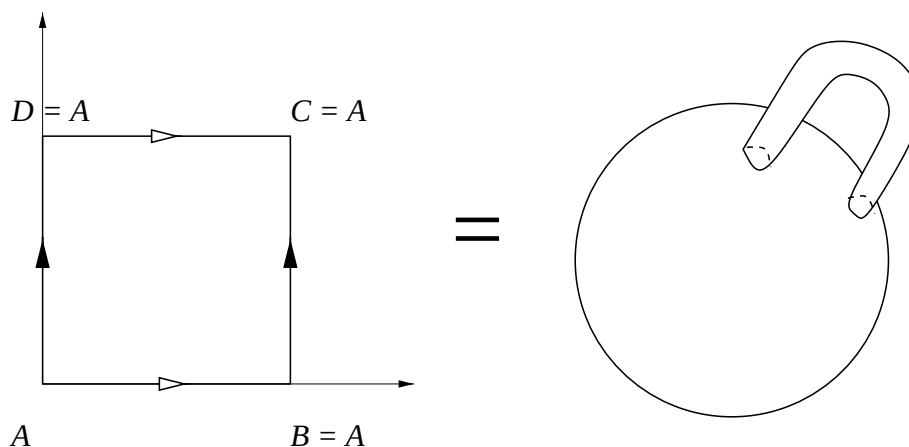
is C^∞ . The latter gives a coordinate transformation in the intersection of the charts. The whole collection of the charts $\{(U_\alpha, \phi_\alpha) \mid \alpha \in I\}$ is called an *atlas* for every open covering of $X = \bigcup_{\alpha \in I} U_\alpha$. The two atlases $\{(U_\alpha, \phi_\alpha)\}$ and $\{(V_\alpha, \psi_\alpha)\}$ are said to be compatible if their union is again an atlas. Finally, the *differentiable structure* on X is a maximal atlas $\mathcal{A}(X)$ in the sense that its union with any atlas gives again $\mathcal{A}(X)$.

The atlas $\{(U_\alpha, \phi_\alpha)\}$ on X is said to be *oriented* if all the transition functions (A.2.1) are orientation preserving, i.e., the corresponding Jacobian determinants are everywhere positive,

$$J(f_{\alpha\beta}) = \det \left(\frac{\partial x^i}{\partial y^j} \right) > 0, \quad (\text{A.2.2})$$

where $\phi_\alpha = \{x^i\}$ and $\phi_\beta = \{y^j\}$, with $i, j = 1, \dots, n$. Then $f_{\alpha\beta} = (x^1(y^1, \dots, y^n), \dots, x^n(y^1, \dots, y^n))$.

The differentiable manifold X is *orientable* if it has an oriented atlas. The notions of orientability and orientation on a manifold will prove to be very important in the theory of integration of differential forms. It is straightforward to provide examples of orientable and nonorientable manifolds. The following *2-dimensional* manifolds can be easily constructed with the help of the cut and paste techniques.

Figure A.2.3: Torus $\mathbb{T}^2$.

Consider an $\mathbb{R}^2$ and cut out a rectangle $ABCD$, as shown in Fig. A.2.2. As such, this is a compact two-dimensional manifold with boundary that is topologically equivalent to a disc.

However, after gluing together the sides of this rectangle, one can construct a number of compact manifolds *without a boundary*. The first example is obtained when we identify the opposite sides without twisting them, as shown in Fig. A.2.3. The resulting manifold is a 2-dimensional torus $\mathbb{T}^2$ that is topologically equivalent to a sphere with one handle. This 2-dimensional compact manifold is orientable.

Another possibility is to glue the rectangle $ABCD$ together after *twisting both pairs of opposite sides*. This is shown in Fig. A.2.4. As a result, one obtains a *real projective plane* $\mathbb{P}^2$, which is represented by a sphere with a disc removed and the resulting hole is closed up by a “cross-cap”, i.e., by identifying its diametrically opposite points (a more spectacular way is to say that the hole is closed by a Möbius strip). This 2-dimensional manifold is also compact, but it is *nonorientable*.

Finally, we can glue the rectangle $ABCD$ by *twisting one pair of opposite sides* while matching the two other sides untwisted, as shown in Fig. A.2.5. The resulting compact 2-dimensional manifold is a famous *Klein bottle* $\mathbb{K}^2$. The Klein bottle cannot be drawn in $\mathbb{R}^3$ without self-intersections. However, it is possible to understand it as a sphere with *two* discs removed and the holes closed up with two “cross-caps” (the Möbius strips). The Klein bottle is also a compact but *nonorientable* manifold.

In general, one can prove that any connected compact 2-dimensional manifold can be realized as a sphere with a certain number of small discs removed and a finite number of either handles or “cross-caps” attached in order to close the holes.

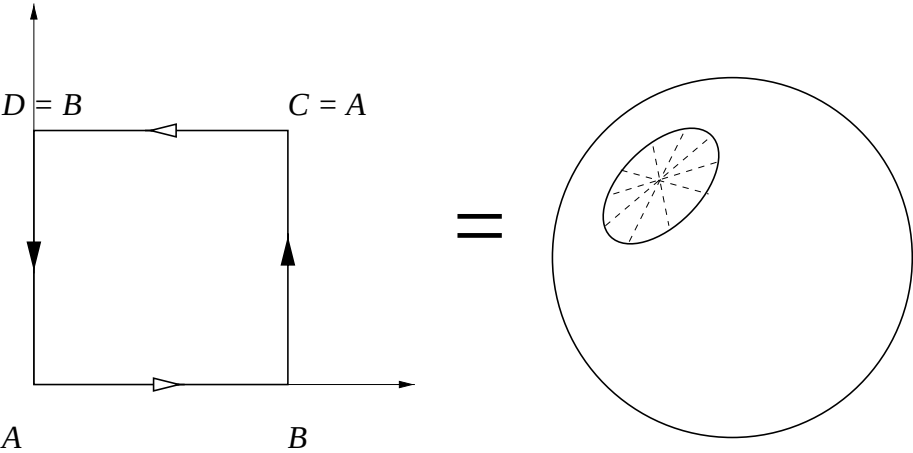


Figure A.2.4: Real projective plane $\mathbb{P}^2$.

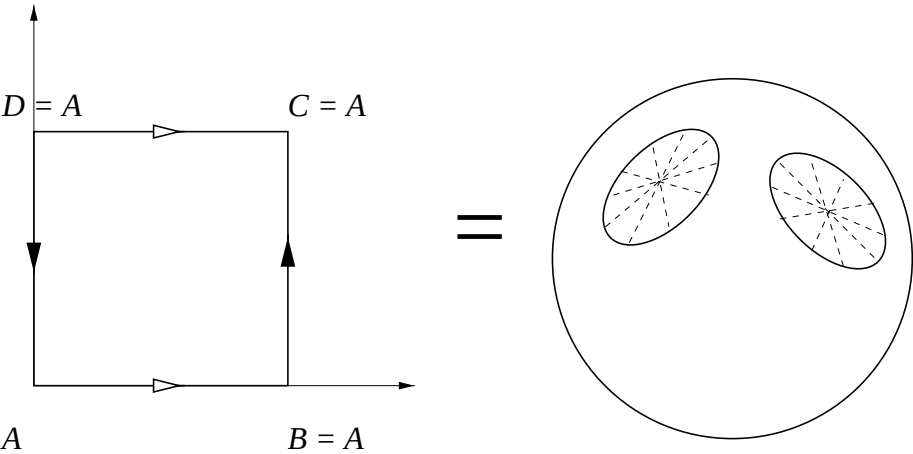


Figure A.2.5: Klein bottle $\mathbb{K}^2$.

A.2.2 Vector fields

Vector fields smoothly assign to each point of a manifold an element of the tangent space. The commutator of two vector fields is a new vector field.

Let us denote by $C(X)$ the algebra of differentiable functions on X . A *tangent vector* u at a point $x \in X$ is defined as an operator which maps $C(X)$ into $\mathbb{R}$ and satisfies the condition

$$u(fg) = f(x)u(g) + g(x)u(f), \quad \forall f, g \in C(X). \quad (\text{A.2.3})$$

A physical motivation comes from the notion of velocity. Indeed, let us consider a smooth curve $x(t)$ such that $0 \leq t \leq 1$ and $x(0) = x$. Then the directional derivative of a function $f \in C(X)$ along $x(t)$ at x ,

$$v(f) = \left. \frac{df(x(t))}{dt} \right|_{t=0}, \quad (\text{A.2.4})$$

is a linear mapping $v : C(X) \rightarrow \mathbb{R}$ satisfying (A.2.3). Choose a local coordinate system $\{x^i\}$ ($i = 1, \dots, n$) on a coordinate neighborhood $U \ni x$. Then the differential operator $\partial_i := \partial/\partial x^i$, for each i , satisfies (A.2.3). It can be demonstrated that the set of vectors $\{\partial_i\}$, $i = 1, \dots, n$, provides a basis for the tangent space X_x at $x \in U$. We will use Latin letters to label coordinate indices.

A mapping u that assigns a tangent vector $u_x \in X_x$ to each point x is called a *vector field* on the manifold X . If we consider a smooth function $f(x)$ on X , then $u(f) := u_x(f)$ is a function on X . A vector field is called differentiable when a function $u(f)$ is differentiable for any $f \in C(X)$. In local coordinates $\{x^i\}$, a vector field is described $u = u^i(x)\partial_i$ by its components $u^i(x)$, which are smooth functions of coordinates.

For every two vector fields u and v a *commutator* $[u, v]$ is naturally defined by

$$[u, v](f) := u(v(f)) - v(u(f)). \quad (\text{A.2.5})$$

This is again a vector field. Please check that the condition (A.2.3) is satisfied. In local coordinates, because of $u = u^i(x)\partial_i$ and $v = v^i(x)\partial_i$, we find the components of the commutator $[u, v] = [u, v]^i(x)\partial_i$ as

$$[u, v]^i = u^j v^i_{,j} - v^j u^i_{,j}. \quad (\text{A.2.6})$$

A.2.3 One-form fields, differential p -forms

One-form fields assign to each point of a manifold an element of the dual tangent space. Differential p -forms are then defined pointwise as the exterior products of 1-form fields.

The dual vector space X_x^* is called *cotangent space* at x . The elements of X_x^* are 1-forms ω that map X_x into $\mathbb{R}$. An *exterior differential 1-form* ω is defined on X if a 1-form $\omega_x \in X_x^*$ is assigned to each point x . A natural example is provided by the differential df of a function f , which is defined by

$$(df_x(u)) := u_x(f), \quad \forall u \in X_x, \quad (\text{A.2.7})$$

or in local coordinates $\{x^i\}$,

$$df = f_{,i} dx^i. \quad (\text{A.2.8})$$

Obviously, the coordinate differentials dx^i describe fields of 1-forms that provide a basis of X_x^* at each point. The bases $\{\partial_i\}$ and $\{dx^i\}$ are dual to each other, i.e., $dx^i(\partial_j) = \delta_j^i$. Repeating pointwise the constructions of Sec. A.2.2, we find that a differential p -form ω on $U \subset X$ is expressible in the form

$$\omega = \frac{1}{p!} \omega_{i_1 \dots i_p}(x) dx^{i_1} \wedge \dots \wedge dx^{i_p}, \quad (\text{A.2.9})$$

where $\omega_{i_1 \dots i_p}(x) = \omega_{[i_1 \dots i_p]}(x)$ are differentiable functions of the coordinates. The space of differential p -forms on X will be denoted by $\Lambda^p(X)$. In this context we may write $\Lambda^0(X) = C(X)$ for the set of differentiable functions on X .

A.2.4 Pictures of vectors and one-forms

The picture of a vector field (an arrow) arises from the velocity of a point moving along an arbitrary curve. The prototype of a picture of a 1-form (a pair of ordered hyperplanes) emerges from the differential of a function.

It seems worthwhile to provide simple pictures for differential 1-form fields and vector fields. The physical prototype of a tangent vector v is the *velocity* of a particle moving along a given curve. Therefore a vector can pictorially be represented by an *arrow* (see Fig. A.2.6).

The prototype of a 1-form ω can be represented by the differential df of a function f ; its components represent the gradient. Therefore, a suitable picture for a 1-form is given by *two parallel hyperplanes* (see Fig. A.2.6), that describe surfaces of constant value of f . An arrowhead indicates in which direction the value of f is increasing. The “stronger” the 1-form is, the closer those two planes are. In physics, a generic 1-form is the *wave* 1-form, defined as the gradient of the phase (think of a de Broglie wave!), or the *momentum* 1-form, defined as the gradient of the action function.

To give a specific example, let us consider a 3-dimensional manifold with local coordinates (x^1, x^2, x^3) . A hyperplane is then simply a 2-dimensional plane. A local basis of *vectors* is given by $e_i = \partial_i$, with $i = 1, 2, 3$, which are tangent to the coordinate lines x^i , respectively. Similarly, a local basis of *1-forms* $\vartheta^i = dx^i$

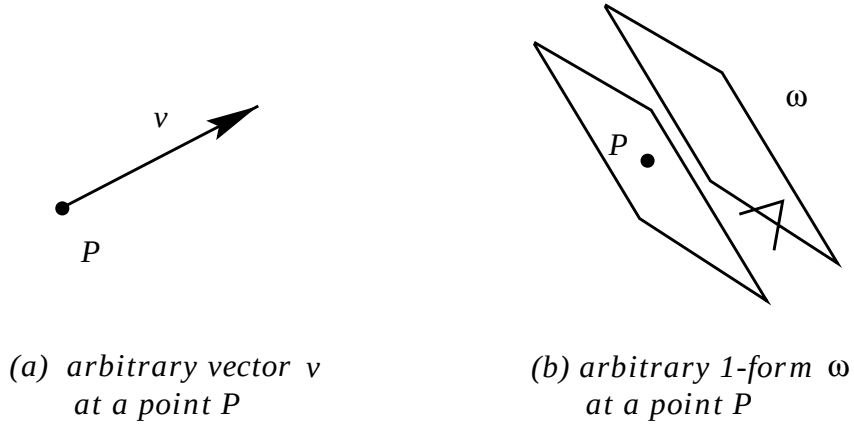


Figure A.2.6: (a) Picture of a vector (“contravariant vector”) at a point P . (b) Picture of a 1-form (“covector” or “covariant vector”) at a point P .

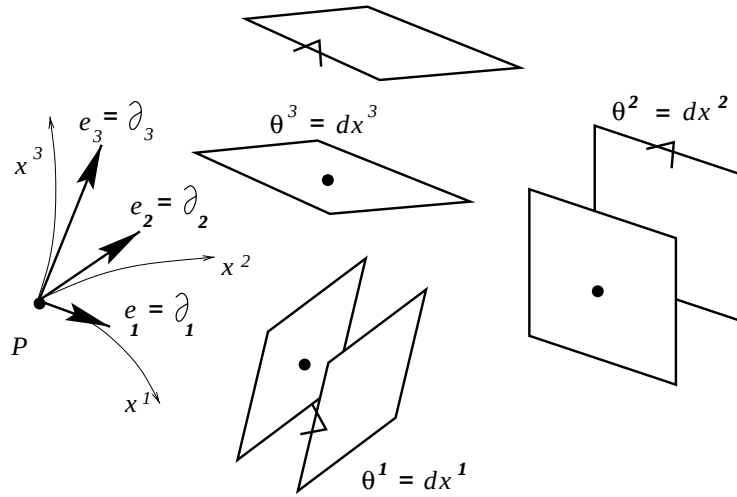


Figure A.2.7: Local coordinates (x^1, x^2, x^3) at a point P of a 3-dimensional manifold and the basis vectors (e_1, e_2, e_3) . The basis 1-forms $\vartheta^i = dx^i, i = 1, 2, 3$, are also supposed to be at P . Note that $\vartheta^1(e_1) = 1, \vartheta^1(e_2) = 0, \vartheta^1(e_3) = 0, \dots$, i.e., ϑ^i is dual to e_j according to $\vartheta^i(e_j) = \delta_j^i$.

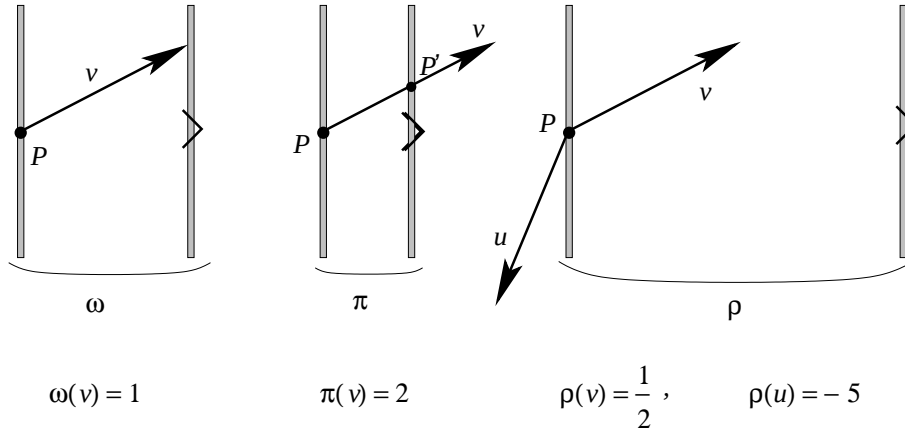


Figure A.2.8: Two-dimensional pictures of 1-forms ω , π , ρ of different strengths at P . They are applied to a certain vector v at the same point P and, in the case of ρ , also to the vector u .

is represented by the local planes, which depict surfaces of constant value of the relevant coordinate x^i (see Fig. A.2.7).

A 1-form is defined in such a way that, if applied to a vector, a number (scalar) pops out. The pictorial representation of such a number is straightforward (see Fig. A.2.8): A straight line starting at P in the direction of the vector v dissects the second hypersurface at P' . The number $\omega(v)$ is then the size of the vector v measured in terms of the segment PP' used as a unit.¹

A.2.5 $\otimes$ Volume forms and orientability

An everywhere nonvanishing n -form on an n -dimensional manifold is called a volume form.

As in the case of a vector space V , differential forms of maximal rank on a manifold X are closely related to volume and orientation. Any *everywhere nonvanishing* n -form ω is called a *volume form* on X . Obviously, it is determined by a single component in every local coordinate chart (U_α, ϕ_α) :

$$\omega = \frac{1}{n!} \omega_{i_1 \dots i_n}(x) dx^{i_1} \wedge \dots \wedge dx^{i_n} = \omega_{1 \dots n}(x) dx^1 \wedge \dots \wedge dx^n. \quad (\text{A.2.10})$$

¹Pictures of vectors and 1-forms, and of many other geometrical quantities, can be found in Schouten [24]; see, e.g., p. 55. Also easily accessible is Misner et al. [18], where in Chapter 4 a number of corresponding worked out examples and nice pictures are displayed. More recent books with beautiful pictures of forms and their manipulation include those of Burke [2] and Jancewicz [12].

A manifold X is orientable (for the definition, see Sec. A.2.1) *if and only if* it has a global volume form. Indeed, given a volume form ω which, by definition, does not vanish for any x , the function $\omega_{1\dots n}(x)$ is either everywhere positive or everywhere negative in U_α . If it is negative, we can simply replace the local coordinate x^1 by $x^{1'} = -x^1$. Then $\omega_{1'2\dots n}(x) > 0$. Thus, without any restriction, we have positive coefficient functions $\omega_{1\dots n}(x)$ in all charts of the atlas $\{(U_\alpha, \phi_\alpha)\}$. In an intersection of any two charts $U_\alpha \cap U_\beta$ with local coordinates $\phi_\alpha = \{x^i\}$ and $\phi_\beta = \{y^i\}$, we have

$$\begin{aligned}\omega &= \omega_{1\dots n}(y) dy^1 \wedge \dots \wedge dy^n = \omega_{1\dots n}(x) dx^1 \wedge \dots \wedge dx^n \\ &= \omega_{1\dots n}(x) J(f_{\alpha\beta}) dy^1 \wedge \dots \wedge dy^n.\end{aligned}\tag{A.2.11}$$

Thus $\omega_{1\dots n}(y) = \omega_{1\dots n}(x) J(f_{\alpha\beta})$. Since both $\omega_{1\dots n}(x)$ and $\omega_{1\dots n}(y)$ are positive, we conclude that the Jacobian determinant $J(f_{\alpha\beta})$ is also positive for all intersecting charts, cf. (A.2.2). Hence the atlas $\{(U_\alpha, \phi_\alpha)\}$ is oriented.

Conversely, let the atlas $\{(U_\alpha, \phi_\alpha)\}$ be oriented; i.e., (A.2.2) holds true. Then in each chart (U_α, ϕ_α) we have an evidently nonvanishing n -form $\omega^{(\alpha)} := dx^1 \wedge \dots \wedge dx^n$. In the intersections $U_\alpha \cap U_\beta$ one finds $\omega^{(\alpha)} = J(f_{\alpha\beta}) \omega^{(\beta)}$. Let $\{\rho_\alpha\}$ be a partition of unity subordinate to the covering $\{U_\alpha\}$ of X . Then we define a global n -form $\omega = \sum_\alpha \rho_\alpha \omega^{(\alpha)}$. Since in the overlapping charts all nontrivial forms $\omega^{(\alpha)}$ are positive multiples of each other and $\rho_\alpha(p) \geq 0$, $\sum_\alpha \rho_\alpha(p) = 1$ (i.e., all ρ_α cannot vanish at any point p), we conclude that ω is a volume form.

A.2.6 $\otimes$ Twisted forms

The differential forms that can be defined on a nonorientable manifold are called twisted differential forms. They are orientation-valued in terms of the conventional differential forms of Sec. A.2.3.

“Since the integral of a differential form on $\mathbb{R}^n$ is not invariant under the whole group of diffeomorphisms of $\mathbb{R}^n$, but only under the subgroup of orientation-preserving diffeomorphisms, a differential form cannot be integrated over a nonorientable manifold. However, by modifying a differential form we obtain something called a *density*, which can be integrated over any manifold, orientable or not.”²

Besides the conventional differential forms, one can define slightly different objects called *twisted forms*³ (or, sometimes, “odd forms”, “impar forms”,

²Bott & Tu [1], p. 79.

³“Twisted tensors were introduced by Hermann Weyl ... and de Rham ... called them tensors of odd kind ... We could make out a good case that the usual differential forms are actually the twisted ones, but the language is forced on us by history. Twisted differential forms are the natural representations for densities, and sometimes are actually called densi-

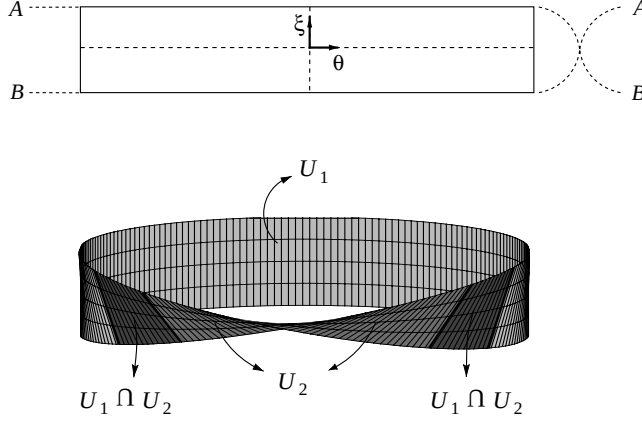


Figure A.2.9: Möbius strip.

“densities”, or “pseudoforms”). The twisted forms are necessary for an appropriate representation of certain physical quantities, such as the electric current density. Moreover, they are indispensable when one considers the integration theory on manifolds and, in particular, on nonorientable manifolds.

In Sec. A.1.3, see Examples 4) and 5), a twisted form was defined on a vector space V as a geometric quantity. Intuitively, a twisted form on the manifold X can be defined as an “orientation-valued” conventional exterior form. Given an atlas $\{(U_\alpha, \phi_\alpha)\}$, a *twisted p -form* is represented by a family of differential p -forms $\{\omega^{(\alpha)}\}$ such that in the intersections $U_\alpha \cap U_\beta$,

$$\omega^{(\alpha)} = \text{sgn} J(f_{\alpha\beta}) \omega^{(\beta)}, \quad (\text{A.2.12})$$

where, as in the previous section, $J(f_{\alpha\beta})$ is the Jacobian of the transition function $f_{\alpha\beta} := \phi_\alpha \circ \phi_\beta^{-1}$.

Example: Consider the Möbius strip, a nonorientable 2-dimensional compact manifold with boundary (see Fig. A.2.9). It can be easily realized by taking a rectangle $\{(\theta, \xi) \in \mathbb{R}^2 | 0 < \theta < 2\pi, -1 < \xi < 1\}$ and gluing it together with one twist along vertical sides. The simplest atlas for the resulting manifold consists of two charts (U_1, ϕ_1) and (U_2, ϕ_2) . The open domains $U_{1,2}$ are rectangles and they can be chosen as shown in Fig. A.2.9 with the evidently defined local coordinate maps $\phi_1 = (x^1, x^2)$ and $\phi_2 = (y^1, y^2)$, where the first coordinate runs along the rectangles and the second one across them. The intersection $U_1 \cap U_2$ is comprised of two open sets, $(U_1 \cap U_2)_{\text{left}}$ and $(U_1 \cap U_2)_{\text{right}}$. The transition functions $f_{12} = \phi_1 \circ \phi_2^{-1}$ are $f_{12} = \{x^1 = y^1, x^2 = y^2\}$ in $(U_1 \cap U_2)_{\text{left}}$ and $f_{12} = \{x^1 = y^1, x^2 = -y^2\}$ in $(U_1 \cap U_2)_{\text{right}}$, so that $J(f_{12}) = \pm 1$ in these

ties, which would be an ideal name were it not already in use in tensor analysis. I agonized over a notation for twisted tensors, say, a different typeface. In the end I decided against it ... ,” William L. Burke [2], p. 183.

domains, respectively. The 1-form $\omega = \{\omega^{(1)} = dx^2, \omega^{(2)} = dy^2\}$ is a twisted form on the Möbius strip.

In general, given a chart (U_α, ϕ_α) , both a usual and a twisted p -form is given by its components $\omega_{i_1 \dots i_p}(x)$, see (A.2.9). With a change of coordinates, the components of a twisted form, via (A.2.12), are transformed as

$$\omega_{i_1 \dots i_p}(x) = \left(\text{sgn det} \frac{\partial x^i}{\partial y^j} \right) \frac{\partial y^{j_1}}{\partial x^{i_1}} \dots \frac{\partial y^{j_p}}{\partial x^{i_p}} \omega_{j_1 \dots j_p}(y). \quad (\text{A.2.13})$$

For a conventional p -form, the first factor on the right-hand side, the sign of the Jacobian, is absent.

Normally, in gravity and in field theory one works on orientable manifolds with an oriented atlas chosen. Then the difference between ordinary and twisted objects disappears because of (A.2.2). However, twisted forms are very important on nonorientable manifolds on which the usual forms cannot be integrated.

A.2.7 Exterior derivative

The exterior derivative maps a p -form into a $(p+1)$ -form. Its crucial property is nilpotency, $d^2 = 0$.

Denote the set of vector fields on X by X_0^1 . For 0-forms $f \in \Lambda^0(X)$, the differential 1-form df is defined by (A.2.7), (A.2.8), i.e., by $df = f_{,i} dx^i$. We wish to extend this map $d : \Lambda^0(X) \rightarrow \Lambda^1(X)$ to a map $d : \Lambda^p(X) \rightarrow \Lambda^{p+1}(X)$. Ideally this should be performed in a coordinate-free way and we shall give such a definition at the end of this section. However, the definition of exterior derivative of a p -form in terms of a coordinate basis is very transparent. Furthermore, it is a simple matter to prove that it is, in fact, independent of the local coordinate system that is used. Starting with the expression (A.2.9) for a $\omega \in \Lambda^p(X)$, we define $d\omega \in \Lambda^{p+1}(X)$ by

$$d\omega := \frac{1}{p!} d\omega_{i_1 \dots i_p} \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}. \quad (\text{A.2.14})$$

By (A.2.7), (A.2.8), the right-hand side of (A.2.14) is

$$(1/p!) \omega_{i_1 \dots i_p, j} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}. \quad (\text{A.2.15})$$

Hence, because of the antisymmetry of the exterior product, we may write

$$d\omega = \frac{1}{p!} \omega_{[i_1 \dots i_p, j]} dx^j \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}. \quad (\text{A.2.16})$$

Under a coordinate transformation $\{x^i\} \rightarrow \{x^{i'}\}$, it is found that

$$\omega_{[i' \dots i'_p, j']} = \frac{\partial x^{i_1}}{\partial x^{i'_1}} \dots \frac{\partial x^{i_{p+1}}}{\partial x^{j'}} \omega_{[i_1 \dots i_p, i_{p+1}]}. \quad (\text{A.2.17})$$

Hence,

$$\begin{aligned} \omega_{[i'_1 \dots i'_p, j']} dx^{j'} \wedge dx^{i'_1} \wedge \dots \wedge dx^{i'_p} &= \\ &= \omega_{[i_1 \dots i_p, i_{p+1}]} dx^{i_{p+1}} \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}, \end{aligned} \quad (\text{A.2.18})$$

so that the exterior derivative, as defined by (A.2.16), is independent of the coordinate system chosen.

Proposition 1: The exterior derivative, as defined by (A.2.16), is a map

$$d : \Lambda^p(X) \longrightarrow \Lambda^{p+1}(X) \quad (\text{A.2.19})$$

with the following properties:

- 1) $d(\omega + \lambda) = d\omega + d\lambda$ [linearity],
- 2) $d(\omega \wedge \phi) = d\omega \wedge \phi + (-1)^p \omega \wedge d\phi$ [(anti)Leibniz rule],
- 3) $df(u) = u(f)$ [partial derivative for functions],
- 4) $d(d\omega) = 0$ [nilpotency].

Here, $\omega, \lambda \in \Lambda^p(X), \phi \in \Lambda^q(X), f \in \Lambda^0(X), u \in X_0^1(X)$.

Proof. 1) and 3) are obvious from the definition. Because of 1) and the distributive property of the exterior multiplication, it is sufficient to prove 2) for ω and ϕ of the ‘monomial’ form:

$$\omega = f dx^{i_1} \wedge \dots \wedge dx^{i_p}, \quad \phi = h dx^{j_1} \wedge \dots \wedge dx^{j_q}. \quad (\text{A.2.20})$$

Thus

$$\omega \wedge \phi = fh dx^{i_1} \wedge \dots \wedge dx^{i_p} \wedge dx^{j_1} \wedge \dots \wedge dx^{j_q}. \quad (\text{A.2.21})$$

Then

$$\begin{aligned} d(\omega \wedge \phi) &= d(fh) \wedge dx^{i_1} \wedge \dots \wedge dx^{j_q} \\ &= (f dh + h df) \wedge dx^{i_1} \wedge \dots \wedge dx^{j_q} \\ &= (df \wedge dx^{i_1} \wedge \dots \wedge dx^{i_p}) \wedge (h dx^{j_1} \wedge \dots \wedge dx^{j_q}) \\ &\quad + (-1)^p (f dx^{i_1} \wedge \dots \wedge dx^{i_p}) \wedge (dh \wedge dx^{j_1} \wedge \dots \wedge dx^{j_q}) \\ &= d\omega \wedge \phi + (-1)^p \omega \wedge d\phi. \end{aligned} \quad (\text{A.2.22})$$

To prove 4), we first of all note that, for a function $f \in \Lambda^0(X)$,

$$d(df) = f_{,[ij]} dx^j \wedge dx^i = 0, \quad (\text{A.2.23})$$

since partial derivatives commute. For a p -form it is sufficient to consider a monomial

$$\omega = f dx^{i_1} \wedge \dots \wedge dx^{i_p}. \quad (\text{A.2.24})$$

Then

$$d\omega = df \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_p}, \quad (\text{A.2.25})$$

and repeated application of property 2) and (A.2.23) yields the desired result $d(d\omega) = 0$. By linearity, this may be extended to a general p -form which is a linear combination of terms like (A.2.24).

Proposition 2: Invariant expression for the exterior derivative.

For $\omega \in \Lambda^p(X)$, we can express $d\omega$ in a coordinate-free manner as follows:

$$\begin{aligned} d\omega(u_0, u_1, \dots, u_p) &= \sum_{j=0}^p (-1)^j u_j (\omega(u_0, \dots, \widehat{u}_j, \dots, u_p)) \\ &+ \sum_{0 \leq i < j \leq p} (-1)^{i+j} \omega([u_i, u_j], u_0, \dots, \widehat{u}_i, \dots, \widehat{u}_j, \dots, u_p), \end{aligned} \quad (\text{A.2.26})$$

where $u_0, u_1, \dots, u_p$ are arbitrary vector fields and $\widehat{u}$ indicates that the field u is *omitted* as an argument. It is a straightforward matter to verify that (A.2.26) is consistent with (A.2.16). We shall make particular use of the case in which ω is a 1-form and (A.2.9) becomes

$$d\omega(u, v) = u(\omega(v)) - v(\omega(u)) - \omega([u, v]). \quad (\text{A.2.27})$$

A.2.8 Frame and coframe

A natural frame and natural coframe are defined at every local coordinate patch by ∂_i and dx^i , respectively. An arbitrary frame e_α and coframe ϑ^α are constructed by a linear transformation therefrom. The object of anholonomy measures how much a coframe differs from a natural one.

A *local* frame on an n -dimensional differentiable manifold X is a set e_α , $\alpha = 0, 1, \dots, n$, of n vector fields that are *linearly independent* at each point of an open subset U of X . They thus form a basis of the tangent (vector) space X_x at every point $x \in U$. There exist quite ordinary manifolds, the 2-dimensional sphere for example, where no continuous frame field can be introduced globally, i.e., at each point of the manifold X . Therefore, speaking of frames on X , we will always have in mind local frames. If e_α is a frame, then the corresponding *coframe* is the set ϑ^α of n different 1-forms such that

$$\vartheta^\alpha(e_\beta) = \delta^\alpha_\beta \quad (\text{A.2.28})$$

is valid at each point of X . In other words $\vartheta^\alpha|_x$ at each point $x \in X$ is the dual basis of 1-forms for X_x^* . We note in particular that, as a consequence of (A.2.28), every vector field $u \in X_0^1$ can be decomposed according to

$$u = u^\alpha e_\alpha, \quad \text{where} \quad u^\alpha = \vartheta^\alpha(u) = u \lrcorner \vartheta^\alpha. \quad (\text{A.2.29})$$

A local *coordinate* system defines a coordinate frame ∂_i on the open neighborhood U . Thus an arbitrary frame e_α may be expressed on U in terms of ∂_i in the form of

$$e_\alpha = e^i_\alpha \partial_i, \quad (\text{A.2.30})$$

where e^i_α are differentiable functions of the coordinates. For the corresponding coframe ϑ^α we have

$$\vartheta^\alpha = e_i^\alpha dx^i, \quad (\text{A.2.31})$$

where by (A.2.28),

$$e_i^\alpha e^i_\beta = \delta^\alpha_\beta. \quad (\text{A.2.32})$$

If a coframe ϑ^α has the property that

$$d\vartheta^\alpha = 0, \quad (\text{A.2.33})$$

it is said to be *natural* or *holonomic*. In this case, in the neighborhood of each point, there exists a coordinate system $\{x^i\}$ such that

$$\vartheta^\alpha = \delta_i^\alpha dx^i. \quad (\text{A.2.34})$$

Under these circumstances, the frame e_α is also natural or holonomic with $e_\alpha = \delta_\alpha^i \partial_i$. The 2-form

$$C^\alpha := d\vartheta^\alpha = \frac{1}{2} C_{ij}^\alpha dx^i \wedge dx^j = \frac{1}{2} C_{\beta\gamma}^\alpha \vartheta^\beta \wedge \vartheta^\gamma, \quad (\text{A.2.35})$$

with $C_{(\beta\gamma)}^\alpha \equiv 0$, is the object of *anholonomy* with its 24 independent components. It measures how much a given coframe ϑ^α fails to be holonomic. There is also a version of (A.2.35) in terms of the frame e_α . With the help of (A.2.27), it can be rewritten as

$$[e_\alpha, e_\beta] = -C_{\alpha\beta}^\gamma e_\gamma. \quad (\text{A.2.36})$$

The object of anholonomy has a nontensorial transformation behavior.

Example: On a 2-dimensional manifold with local coordinates $\{x, y\}$, the 1-forms $\{\vartheta^1 = x dy, \vartheta^2 = y dx\}$ are linearly independent. Such a coframe is anholonomic with $d\vartheta^1 = dx \wedge dy$ and $d\vartheta^2 = -dx \wedge dy$, i.e., $C^1 = -C^2 = dx \wedge dy$.

A.2.9 $\otimes$ Maps of manifolds: push-forward and pull-back

Pull-back φ^ and push-forward φ_* maps are the companions of every diffeomorphism φ of the manifold X . They relate the corresponding cotangent and tangent spaces at points x and $\varphi(x)$. The map φ^* commutes with the exterior differential.*

If a differentiable map $\varphi : X \rightarrow Y$ is given, various geometric objects can be transported either from X to Y (pushed forward) or from Y to X (pulled back). A push-forward is denoted by φ_* and a pull-back by φ^* .

Given a tangent vector u at a point $x \in X$, we can define its push-forward $\varphi_* u \in Y_{\varphi(x)}$ (which is also called the *differential*) by determining its action on a function $f \in C(Y)$ as

$$(\varphi_* u)(f) = u(f \circ \varphi). \quad (\text{A.2.37})$$

However, if u is not merely a tangent vector, but a vector field over X , it is in general not possible to define its push-forward to Y . There might be two reasons for that. Firstly, if φ is not injective and $\varphi(x_1) = \varphi(x_2)$ for $x_1 \neq x_2$, then the vectors pushed from X_{x_1} and X_{x_2} are different in general. Secondly, if φ is not surjective, the pushed forward vector field is not, in general, determined all over Y . It is always possible to define $\varphi_* v$ of a vector field if φ is a *diffeomorphism* (which can only be considered when $\dim X = \dim Y$).

Using the rule

$$\varphi_*(v_1 \otimes v_2) := \varphi_* v_1 \otimes \varphi_* v_2, \quad (\text{A.2.38})$$

we can define the push-forward of an arbitrary contravariant tensor at $x \in X$ to the space of tensors of the same type at $\varphi(x) \in Y$. So φ_* becomes a homomorphism of the algebras of contravariant tensors at $x \in X$ and $\varphi(x) \in Y$.

In a diagram we can depict the push-forward map φ_* of tangent vectors u and the pull-back map φ^* for 1-forms ω :

$$\begin{array}{ccc} \varphi^* \omega \in X_x^* & \xleftarrow{\varphi^*} & \omega \in Y_{\varphi(x)}^* \\ \uparrow & & \uparrow \\ x \in X & \xrightarrow{\varphi} & y = \varphi(x) \in Y \\ \downarrow & & \downarrow \\ u \in X_x & \xrightarrow{\varphi_*} & \varphi_* u \in Y_{\varphi(x)} \end{array}$$

Let $\{x^i\}$ be local coordinates in X and $\{y^j\}$ local coordinates in Y (with the ranges of indices i and j defined by the dimensionality of X and Y , respectively). Then the map φ is described by a set of smooth functions $y^j(x^i)$, and the push-forward map for tensors of type $\begin{bmatrix} p \\ 0 \end{bmatrix}$ in components reads

$$(\varphi_* T)^{j_1 \dots j_p} = \frac{\partial y^{j_1}}{\partial x^{i_1}} \dots \frac{\partial y^{j_p}}{\partial x^{i_p}} \bigg|_x T^{i_1 \dots i_p}. \quad (\text{A.2.39})$$

Comparing with (A.2.8) for the case when $Y = \mathbb{R}$, it becomes clear why φ_* is also called a differential map.

For a p -form $\omega \in \Lambda_{y=\varphi(x)}^p(Y)$, we can determine its pull-back $\varphi^*\omega \in \Lambda_x^p(X)$ by

$$(\varphi^*\omega)(v_1, \dots, v_p) = \omega(\varphi_*v_1, \dots, \varphi_*v_p). \quad (\text{A.2.40})$$

This definition can be straightforwardly extended to a homomorphism of the algebras of covariant type $\begin{bmatrix} 0 \\ p \end{bmatrix}$ tensors. In local coordinates it reads, analogously to (A.2.39),

$$(\varphi_*T)_{i_1 \dots i_p} = \frac{\partial y^{j_1}}{\partial x^{i_1}} \dots \frac{\partial y^{j_p}}{\partial x^{i_p}} \bigg|_x T_{j_1 \dots j_p}. \quad (\text{A.2.41})$$

Let ω be an exterior p -form (i.e., a p -form *field*) on Y . In order to determine its pull-back $\varphi^*\omega$ to X by (A.2.40), it is sufficient to have $\varphi_*v_1, \dots, \varphi_*v_p$ on the right hand side of (A.2.40) defined as vectors (i.e., not necessarily as vector fields). Therefore, the pull-back of exterior forms (and, in general, of covariant tensors) is determined for an arbitrary map φ . In exterior calculus, an important property is the commutativity of pull-back and exterior differentiation for any p -form ω :

$$\boxed{d(\varphi^*\omega) = \varphi^*(d\omega)}. \quad (\text{A.2.42})$$

If φ is a *diffeomorphism*, or at least a local diffeomorphism, we shall use the pull-back φ^* of arbitrary tensor fields. For contravariant tensors, it can be defined as

$$\varphi^* = (\varphi^{-1})_* = (\varphi_*)^{-1}. \quad (\text{A.2.43})$$

To define it for an arbitrary tensor of type $\begin{bmatrix} p \\ q \end{bmatrix}$, we have to require only that φ^* is an algebra isomorphism. Technically, in local coordinates, this amounts to the invertibility of the square matrices $\partial y^j / \partial x^i$.

When φ is a (local) diffeomorphism, we can also pull-back (or push-forward) geometric quantities constructed on the tangent space. Let $[(w, e)]$ be a geometric quantity; here $e = (e_1, \dots, e_n)$ is a frame in the tangent space Y_y and w belongs to the set W in which there is the left action ρ of $GL(n, \mathbb{R})$. As for vectors, we define $\varphi^*e = (\varphi^*e_1, \dots, \varphi^*e_n)$ and

$$\varphi^*[(w, e)] = [(w, \varphi^*e)]; \quad (\text{A.2.44})$$

i.e., the transported object has the same components as the initial object with respect to the transported frame. Certainly, this definition of the pull-back is consistent with that given earlier for tensors.

A.2.10 $\otimes$ Lie derivative

A vector field generates a group of diffeomorphisms on a manifold. Making use of this group action, the Lie derivative enables us to compare tensors and geometric quantities at different points.

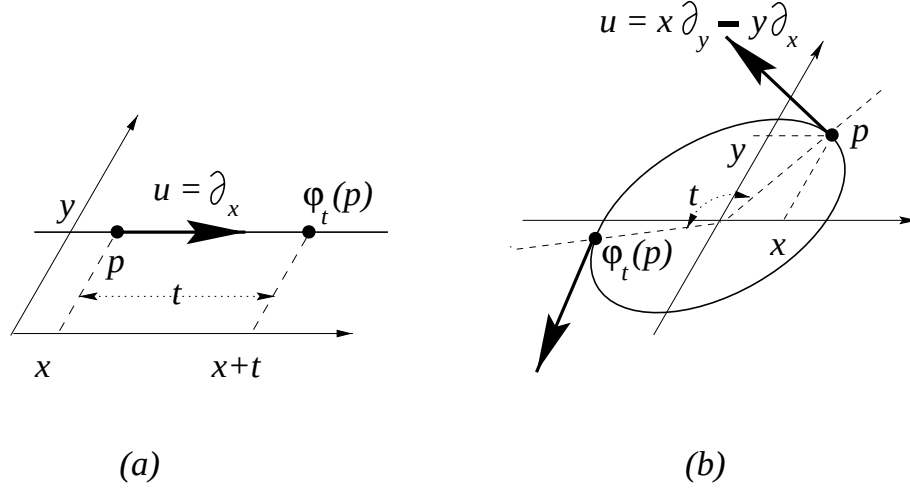


Figure A.2.10: Translations (a) and circular motion (b) generated on $\mathbb{R}^2$.

The main result of the present section will be equation (A.2.51), the Lie derivative of a differential form. However, we shall first explain the concept of the Lie derivative of a general geometrical quantity. Note that for a Lie derivative *no metric and no connection is required*; it can be defined on each differentiable manifold.

For each point $p \in X$, a vector field u with $u(p) \neq 0$ determines a unique curve $\sigma_p(t)$, $t \geq 0$, such that $\sigma_p(0) = p$ with u as the tangent vector field to the curve. The family of curves defined in this way is called the *congruence* of curves generated by the vector field u . Let $\{x^i\}$ be a local coordinate system with x_p^i as the coordinates of p and decompose u according to $u = u^i(x^1, \dots, x^n) \partial_i$. Then the curve $\sigma_p(t)$ is found by solving the system of ordinary differential equations

$$\frac{dx^i}{dt} = u^i(x^1(t), \dots, x^n(t)), \quad (\text{A.2.45})$$

with initial values $x^i(0) = x_p^i$. The congruence of curves obtained in this way defines (at least locally) a 1-parameter group of diffeomorphisms φ_t on X given by

$$\varphi_t(p) = \sigma_p(t), \quad \forall p \in X, \quad (\text{A.2.46})$$

with the properties that (a) $\varphi_t^{-1} = \varphi_{-t}$, (b) $\varphi_t \circ \varphi_s = \varphi_{t+s}$, and (c) φ_0 is the identity map. The integral curves of the congruence are called the *trajectories* of the group. Furthermore, the equations (A.2.45) are equivalent to

$$u(f)(p) = \lim_{t \rightarrow 0} \frac{f(\varphi_t(p)) - f(p)}{t} = \left. \frac{d}{dt} f(\varphi_t(p)) \right|_{t=0}, \quad (\text{A.2.47})$$

for all $p \in X$ and all differentiable functions f .

Examples in $\mathbb{R}^2$:

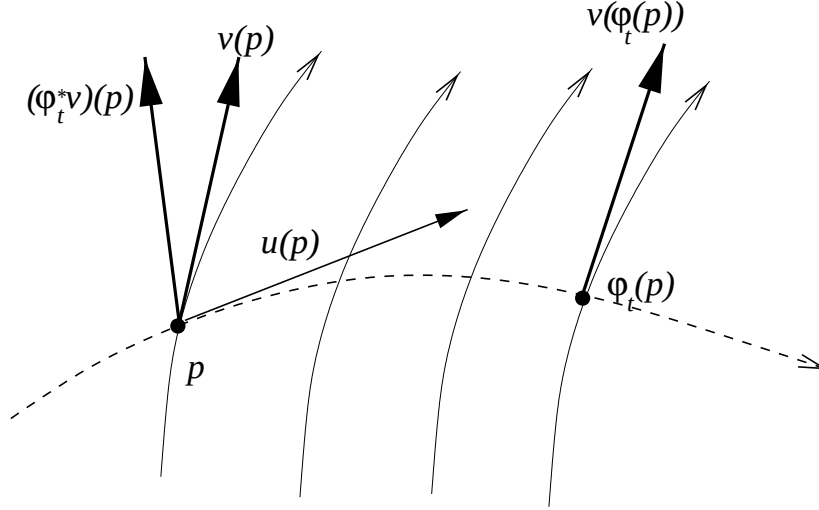


Figure A.2.11: The definition of the Lie derivative $\mathcal{L}_u v$ with respect to a vector u : The 1-parameter group φ_t , generated by the vector field u , is used in order to transfer the vector $v(\varphi_t(p))$ back to the initial point and to compare it with $v(p)$.

- 1) The vector field $u = \partial/\partial x$ generates translations $\varphi_t(x, y) = (x + t, y)$, $-\infty < t < +\infty$. The trajectories are the lines $y = \text{const}$. See Fig. A.2.10a.
- 2) The vector field $u = (x \partial/\partial y - y \partial/\partial x)$ generates the circular motion $\varphi_t(x, y) = (x \cos t - y \sin t, x \sin t + y \cos t)$, $0 \leq t < 2\pi$. The trajectories are concentric closed curves around the origin, see Fig. A.2.10b.

In general, if we take a coordinate patch U of a differentiable manifold with coordinates $\{x^i\}$, then φ_t is defined in terms of x^i by

$$\varphi_t(x^i) = y^i := f^i(t; x^j), \quad (\text{A.2.48})$$

where $f^i(t; x^j)$ are differentiable functions of (t, x^j) . Property (a) states that $x^i = x^i(t; y^j) = f^i(-t; y^j)$. By property (b) we have $f^i(t; f^j(s; x^k)) = f^i(t + s; x^j)$, while (c) means that $f^i(0; x^j) = x^i$.

For every value of t in a certain interval, the diffeomorphism φ_t induces corresponding pull-backs φ_t^* on functions, vectors, exterior forms, and general tensor fields of type $\begin{bmatrix} p \\ q \end{bmatrix}$. Accordingly, the *Lie derivative* of a tensor T with respect to a vector field u is defined by

$$\mathcal{L}_u T := \lim_{t \rightarrow 0} \frac{\varphi_t^* T - T}{t}. \quad (\text{A.2.49})$$

It is sufficient to have explicit expressions for the Lie derivatives of functions, vectors, and 1-forms in order to be in a position to do the same for a general tensor. The two most important cases are as follows.

For vectors $v \in X_0^1$,

$$\boxed{\mathcal{L}_u v = [u, v];} \quad (\text{A.2.50})$$

see (A.2.6) for a component version. For p -forms $\omega \in \Lambda^p(X)$ and $p \geq 0$, we find the *main theorem* for the Lie derivative of an exterior form:

$$\boxed{\mathcal{L}_u \omega = u \lrcorner (d\omega) + d(u \lrcorner \omega).} \quad (\text{A.2.51})$$

An alternative coordinate-free general formula for this Lie derivative reads:

$$(\mathcal{L}_u \omega)(v_1, \dots, v_p) = u(\omega(v_1, \dots, v_p)) - \sum_{i=1}^p \omega(v_1, \dots, [u, v_i], \dots, v_p). \quad (\text{A.2.52})$$

The Lie derivative for the functions $f \in C(X)$ is obtained as a particular case of (A.2.51) for $p = 0$:

$$\mathcal{L}_u f = u(f) = u \lrcorner df. \quad (\text{A.2.53})$$

The last formula is straightforwardly checked by a direct calculation,

$$\begin{aligned} \mathcal{L}_u f(p) &= \lim_{t \rightarrow 0} \frac{(\varphi_t^* f)(p) - f(p)}{t} \\ &= \lim_{t \rightarrow 0} \frac{f(\varphi_t(p)) - f(p)}{t} = u(f)(p), \end{aligned} \quad (\text{A.2.54})$$

by use of (A.2.47). The proof of (A.2.50) and (A.2.51) is left as an exercise to the readers. As a hint, we mention that the formula (A.2.51) follows from the Lie derivative of a 1-form ω ,

$$(\mathcal{L}_u \omega)(v) = u(\omega(v)) - \omega(\mathcal{L}_u v), \quad (\text{A.2.55})$$

where u, v are arbitrary vector fields.

The most important properties of the *Lie derivatives of exterior forms* may be summarized as follows:

- 1) $\mathcal{L}_u d\omega = d\mathcal{L}_u \omega$ [$\mathcal{L}_u$ and d commute],
- 2) $\mathcal{L}_u(\omega \wedge \varphi) = (\mathcal{L}_u \omega) \wedge \varphi + \omega \wedge \mathcal{L}_u \varphi$ [Leibniz rule],
- 3) $\mathcal{L}_{f u} \omega = f \mathcal{L}_u \omega + df \wedge (u \lrcorner \omega)$ [rescaled vector],
- 4) $\mathcal{L}_v \mathcal{L}_u \omega - \mathcal{L}_u \mathcal{L}_v \omega = \mathcal{L}_{[u, v]} \omega$ [noncommutativity],
- 5) $\mathcal{L}_v(u \lrcorner \omega) - u \lrcorner \mathcal{L}_v \omega = [v, u] \lrcorner \omega$ [$\mathcal{L}_v, u \lrcorner$ do not commute].

The formulas above contain all the necessary information about the Lie derivative for arbitrary tensors of type $\begin{bmatrix} p \\ q \end{bmatrix}$. In particular, by construction we have that $\mathcal{L}_u$ is type preserving, i.e., if $T \in T_q^p(X)$ then $(\mathcal{L}_u T) \in T_q^p(X)$. Moreover, it is clear that for any two tensor fields T and S of the same type,

$$\mathcal{L}_u(T + S) = \mathcal{L}_u T + \mathcal{L}_u S \quad (\text{A.2.56})$$

and

$$\mathcal{L}_{u+v}T = \mathcal{L}_uT + \mathcal{L}_vT. \quad (\text{A.2.57})$$

Finally, for $T \in T_s^r(X)$, $S \in T_q^m(X)$,

$$\mathcal{L}_u(T \otimes S) = (\mathcal{L}_uT) \otimes S + T \otimes (\mathcal{L}_uS). \quad (\text{A.2.58})$$

The proof follows straightforwardly from

$$\varphi_t^*(T \otimes S) = \varphi_t^*T \otimes \varphi_t^*S. \quad (\text{A.2.59})$$

These properties enable us to express the Lie derivative of a general tensor in terms of a local coordinate basis. Consider a tensor field of type $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, for example. In terms of a local coordinate system $\{x^i\}$,

$$T = T^{ij}{}_k(x) \partial_i \otimes \partial_j \otimes dx^k, \quad (\text{A.2.60})$$

and we easily find for $u = \xi^i(x) \partial_i$,

$$\mathcal{L}_uT = (T^{ij}{}_{k,r} \xi^r - T^{rj}{}_k \xi^i{}_{,r} - T^{ir}{}_k \xi^j{}_{,r} + T^{ij}{}_r \xi^r{}_{,k}) \partial_i \otimes \partial_j \otimes dx^k. \quad (\text{A.2.61})$$

For completeness, let us consider the Lie derivative of a *geometric quantity*. For this purpose we note the following: If $e_\alpha(x)$ is a frame taken at a given point $p \in X$, then $\varphi_t^*(e_\alpha(\varphi_t(x)))$ can be decomposed with respect to this frame with some t -dependent coefficients:

$$\varphi_t^*(e_\alpha(\varphi_t(x))) = \Phi_\alpha{}^\beta(-t, x) e_\beta(x). \quad (\text{A.2.62})$$

Differentiating this equation with respect to t at $t = 0$, we get

$$[u, e_\alpha] = - \left. \frac{d}{dt} \Phi_\alpha{}^\beta(t, x) \right|_{t=0} e_\beta(x), \quad (\text{A.2.63})$$

and thus formally we find the matrix

$$\psi_\alpha{}^\beta := \left. \frac{d}{dt} \Phi_\alpha{}^\beta(t, x) \right|_{t=0} = e_\alpha(u^\beta)(x) + u^\mu(x) C_{\mu\alpha}{}^\beta(x). \quad (\text{A.2.64})$$

Let us consider a field of a geometric quantity $[(w(x), e(x))]$ of type ρ , for short, $w(x)$. According to the definition of the pull-back of such objects, we have

$$\begin{aligned} (\varphi_t^*[(w, e)])(x) &= [(w(\varphi_t(x)), \varphi_t^*(e(\varphi_t(x))))] \\ &= [(w(\varphi_t(x)), \Phi(-t, x) e(x))] \\ &= [(\rho(\Phi(t, x)) w(\varphi_t(x)), e(x))]. \end{aligned}$$

We differentiate at $t = 0$ and find:

$$\begin{aligned} (\mathcal{L}_u w)(x) &= \left. \frac{d}{dt} (\rho(\Phi(t, x)) w(\varphi_t(x))) \right|_{t=0} \\ &= (uw)(x) + \rho_* \psi(x) w(x). \end{aligned} \quad (\text{A.2.65})$$

In all practical applications, a geometric quantity is described as a smooth field on X that takes values in the vector space $W = \mathbb{R}^N$ of a ρ -representation of the group $GL(n, \mathbb{R})$ of local linear frame transformations. In simple terms, the geometric quantity $w = w^A e_A$ is represented by its components w^A with respect to the basis e_A of the vector space $W = \mathbb{R}^N$. Hereafter $A, B, \dots = 1, \dots, N$. Thus, recalling (A.1.16), $\rho(L) = \rho_A{}^B(L) \in GL(N, \mathbb{R})$, and ρ_* maps the tangent space $End(n, \mathbb{R})$ of the group $GL(n, \mathbb{R})$ into the tangent space $End(N, \mathbb{R})$ of the group $GL(N, \mathbb{R})$ by means of the matrix

$$\rho_A{}^{B\alpha}{}_\beta := \left. \frac{\partial \rho_A{}^B(L)}{\partial L_{\alpha}{}^\beta} \right|_{L_{\alpha}{}^\beta = \delta_\alpha^\beta}. \quad (\text{A.2.66})$$

Therefore, taking into account (A.1.17), Eq.(A.2.65) reads for the components of the geometric quantity:

$$\mathcal{L}_u w^A = u(w^A) - \rho_B{}^{A\alpha}{}_\beta \psi_\alpha{}^\beta w^B, \quad (\text{A.2.67})$$

where $\psi_\alpha{}^\beta$ is defined by (A.2.64).

For a tensor field $T^\alpha{}_\beta$ of type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, for instance, we have

$$\mathcal{L}_u T^\alpha{}_\beta = u(T^\alpha{}_\beta) + \psi_\beta{}^\mu T^\alpha{}_\mu - \psi_\mu{}^\alpha T^\mu{}_\beta. \quad (\text{A.2.68})$$

As an interesting exercise, we propose to the reader to calculate the Lie derivative of a scalar density $\mathcal{S}$ of weight w , see (A.1.56):

$$\mathcal{L}_u \mathcal{S} = u(\mathcal{S}) + w \mathcal{S} \psi_\alpha{}^\alpha. \quad (\text{A.2.69})$$

It is a simple matter to generalize the relation for the Lie derivative of a p -form of type ρ . If ω is such a form, then

$$\mathcal{L}_u \omega = d(u \lrcorner \omega) + u \lrcorner d\omega + \rho(\psi) \omega. \quad (\text{A.2.70})$$

For a vector-valued p -form, for example, we have

$$\mathcal{L}_u \omega^\alpha = d(u \lrcorner \omega^\alpha) + u \lrcorner d\omega^\alpha - \omega^\beta \psi_\beta{}^\alpha. \quad (\text{A.2.71})$$

A.2.11 Excalc, a Reduce package

In Chap. A.2, after specifying an n -dimensional manifold, we introduced successively fields of 1-forms, p -forms, and vectors. Their declarations by means of `pform` and `tvector` are already known to us. Then the exterior derivative was specified. In Excalc, not surprisingly, the letter `d` is reserved for this operator. Partial differentiation is denoted by the operator `@`. Thus, `@(sin x, x);` yields `cos(x)`. We collect the different Excalc operators in Table A.2.1.

Unary means that there is one, binary that there are two, and “nary” that there is any number of arguments.

Let us load again Excalc by `load_package excalc$`. By means of a declaration with `fdomain`, an identifier can be declared to be a function of certain variables. With

Math.	Excalc	Operator	Operator Type
$\wedge$	$\wedge$	exterior product	nary infix
$\lrcorner$	$\lrcorner$	interior product	binary infix
∂	$@$	partial derivative	nary prefix
d	d	exterior derivative	unary prefix
$\mathcal{L}$	$ _$	Lie derivative	binary infix
$*$	$\#$	Hodge star operator	unary prefix

Table A.2.1: Translation of mathematical symbols into Excalc. The Hodge star operator will not be defined before Sec. C.2.8.

```
fdomain f=f(x,y),h=h(x);
@(x*f,x);    @(h,y);
```

i.e., $\partial_x(xf)$, $\partial_y h$, we find, respectively,

```
f + x*@ f x
0
```

i.e., $f + x\partial_x f$ and 0. The partial derivative symbol can also be an operator with a single argument, as in $@(z)$. Then it represents the leg ∂_z of a natural frame.

Coming back to the exterior derivative, the following example is now self-explanatory:

```
pform x=0,y=0,z=0,f=0;
fdomain f=f(x,y);
d f;

@ f*d x + @ f*d y
x          y
```

i.e., df evaluates to $(\partial_x f)dx + (\partial_y f)dy$. Products are normally differentiated out, i.e.,

```
pform x=0,y=p,z=q; d(x*y^z);

(- 1)^p *x*y^d z + x*d y^z + d x^y^z
```

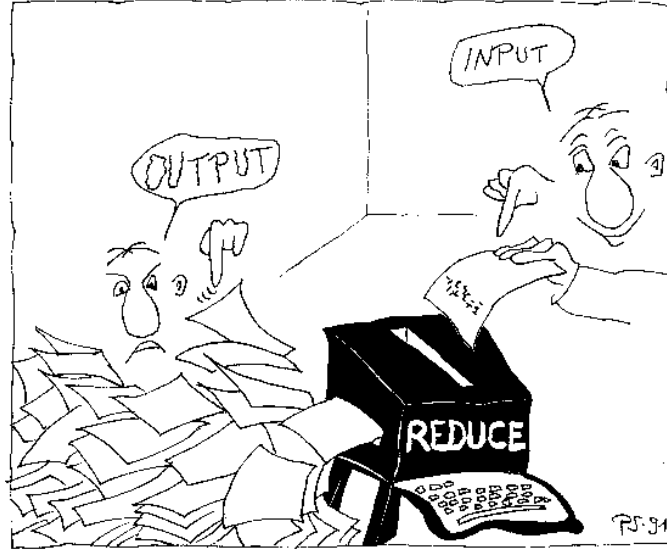


Figure A.2.12: The perennial computer algebra problem.

In an ordinary formula, we have $d(xy \wedge z) = (-1)^p xy \wedge dz + x dy \wedge z + dx \wedge y \wedge z$. This expansion can be suppressed by the command `noxpnd d;`. Expansion is performed again when the command `xpnd d;` is executed.

The Excalc operator `d` knows all the rules for the exterior derivative as specified in Proposition 1 in the context of (A.2.19). Let us declare the corresponding ranks of the forms in order to check the first two rules (note that *lambda* is a reserved identifier in Reduce and cannot be used):

```
pform omega=p, lam=p, phi=q;
```

Then we give the commands

```
d(omega+lam); d(omega^phi);
```

i.e., $d(\omega + \lambda)$ and $d(\omega \wedge \phi)$, and find, respectively,

```
d lam + d omega
```

```

      P
(- 1) *omega^d phi + d omega^phi

```

i.e., $d\omega + d\lambda$ and $(-1)^p \omega \wedge d\phi + d\omega \wedge \phi$.

The second to last entry in our table is the Lie derivative $\mathcal{L}$. In Excalc, it can be applied to an exterior form with respect to a vector or to a vector again with respect to a vector. It is represented by the infix operator $|_{\underline{\quad}}$ (vertical bar followed by an underline). If the Lie derivative is applied to a form, Excalc remembers the main theorem of Lie derivatives, namely (A.2.51). Thus,

```
pform z=k;   tvector u;   u|_z;
```

i.e., $\mathcal{L}_u z$, yields

```
d(u |_ z) + u |_ d z
```

i.e., $d(u \lrcorner z) + u \lrcorner dz$. The operator of the Lie derivatives fulfills the rules displayed after (A.2.55). We will check the rule for the rescaled vector as an example. Already above, the form ω has been declared to be a p -form, f to be a scalar, and u to be a vector. Hence we can type in directly

```
(f*u)|_omega;
```

i.e., $\mathcal{L}_f \omega$, and find

```
d(u |_ omega)*f + u |_ d omega*f + d f^u |_ omega
```

i.e., $d(u \lrcorner \omega)f + (u \lrcorner d\omega)f + df \wedge u \lrcorner \omega$. The rule is verified, but Excalc substituted (A.2.51) immediately. Anyway, we also see that $|_{\underline{\quad}}$ does exactly what we expect from it.

In Sec. A.2.8, we introduced the frame e_α and the coframe ϑ^α as bases of the tangent and the cotangent space, respectively. In Excalc we use the symbols $\mathbf{e}(-\mathbf{a})$ and $\mathbf{o}(\mathbf{a})$, respectively. In Excalc a coframe can only be specified *provided a metric is given at the same time*. This feature of Excalc is not ideal for our purposes. Nevertheless, even if we introduce the metric only in Part C, we have to use it in the Excalc program already here in order to make the programs of Part B executable.

As we saw already in Sec. A.1.12, we can introduce Excalc to the dimension of a manifold via `spacedim 4;`. This can also be done with the `coframe` statement, since we specify thereby the underlying four 1-forms of the coframe and, if the coframe is orthonormal, the signature of the metric. For a Minkowski spacetime with time coordinate t and spherical spatial coordinates r, θ, φ , we state

```
coframe   o(t)      =                d t,
          o(r)      =                d r,
```

```

o(theta) = r *          d theta,
o(phi)   = r * sin(theta) * d phi
with signature (1,-1,-1,-1);
frame e;

```

With `frame e;`, we assigned the identifier e to the name of the frame. In ordinary mathematical language, the coframe statement would read

$$\begin{aligned} \vartheta^t &= dt, & \vartheta^r &= dr, & \vartheta^\theta &= r d\theta, & \vartheta^\phi &= r \sin\theta d\phi, \\ g &= \vartheta^t \otimes \vartheta^t - \vartheta^r \otimes \vartheta^r - \vartheta^\theta \otimes \vartheta^\theta - \vartheta^\phi \otimes \vartheta^\phi. \end{aligned} \quad (\text{A.2.72})$$

Of course, the frame `e(-a)` and the coframe `o(a)` are inverse to each other, i.e., the command `e(-a) |o(b)`; will yield the Kronecker delta (if you switch on `nero`; then only the components which nonvanishing values will be printed out).

The coframe statement is very fundamental for Excalc. All quantities will be evaluated with respect to this coframe. This yields the anholonomic (or physical) components of an object.

The coframe statement of a corresponding spherically symmetric Riemannian metric with unknown function $\psi(r)$ reads:

```

load_package excalc$
pform psi=0$ fdomain psi=psi(r)$
coframe o(t)      = psi *          d t,
o(r)             = (1/psi) *       d r,
o(theta)         = r *            d theta,
o(phi)           = r * sin(theta) * d phi
with signature (1,-1,-1,-1)$

displayframe;      % displays the coframe o(a), input can be checked
frame e$

```

If a command is terminated by a dollar sign `$`, the output will be suppressed. Consequently, if we input the program segment into Reduce, only the coframe ϑ^α will be displayed:

```

t
o = d t*psi

r      d r
o = ----
psi

theta
o      = d theta*r

```

$$\frac{\phi}{r} = d\phi \sin(\theta) \star r$$

Perhaps we should remind ourselves that $\psi^2 = 1 - 2m/r$ represents the Schwarzschild solution of general relativity.

A.2.12 $\otimes$ Closed and exact forms, de Rham cohomology groups

Closed forms are not exact in general. Two closed forms belong to the same cohomology class when they differ by an exact form. Groups of cohomologies are topological invariants.

Let us consider the exterior algebra $\Lambda^*(X) = \bigoplus_{p=0}^n \Lambda^p(X)$ together with the exterior derivative defined in (A.2.19).

A p -form ω is called *closed* if $d\omega = 0$. The space of all closed p -forms

$$Z^p(X) := \{\omega \in \Lambda^p(X) \mid d\omega = 0\}, \quad p = 0, \dots, n, \quad (\text{A.2.73})$$

forms a (real) vector subspace of $\Lambda^p(X)$. A p -form ω is called *exact* if a $(p-1)$ -form φ exists such that $\omega = d\varphi$. The space of all exact p -forms

$$B^p(X) := \{\omega \in \Lambda^p(X) \mid \omega = d\varphi\}, \quad p = 1, \dots, n, \quad (\text{A.2.74})$$

is also a (real) vector space, and evidently $B^p(X) \subset Z^p(X)$ (each exact form is closed, since $dd \equiv 0$). One puts $B^0(X) = \emptyset$.

Obviously the exterior derivative defines an equivalence relation in the space of closed forms: two forms $\omega, \omega' \in Z^p(X)$ are said to be *cohomologically equivalent* if they differ by an exact form, i.e., $(\omega - \omega') \in B^p(X)$. The quotient space

$$H^p(X; \mathbb{R}) := Z^p(X)/B^p(X), \quad p = 0, \dots, n, \quad (\text{A.2.75})$$

consists of *cohomology classes* of p -forms. Each $H^p(X; \mathbb{R})$ is a vector space and, moreover, an Abelian group with an evident group action. The spaces $H^p(X; \mathbb{R})$ are named as *de Rham cohomology groups*.

Unlike the $\Lambda^p(X)$ which are infinite dimensional functional spaces, the de Rham groups, for compact manifolds X , are finite dimensional. The dimension

$$b^p(X) := \dim H^p(X; \mathbb{R}) \quad (\text{A.2.76})$$

is called the p -th *Betti number* of the manifold X .

Locally, an exterior derivative does not yield a difference between closed and exact forms. This fact is usually formulated as the *Poincaré lemma*: Locally, in

a given chart (U, ϕ) of X , every closed p -form ω with $d\omega = 0$ is exact, i.e., a $(p-1)$ -form φ exists in $U \subset X$ such that $\omega = d\varphi$. Let us illustrate this by an explicit example. Suppose we have a closed *one*-form ω . In local coordinates,

$$\begin{aligned}\omega &= \omega_i(x) dx^i, \\ d\omega &= 0 \iff \partial_i \omega_j(x) = \partial_j \omega_i(x).\end{aligned}\tag{A.2.77}$$

Then this form is locally exact, $\omega = d\varphi$, where the 0-form φ is given explicitly in the chart (U, ϕ) by

$$\varphi(x) = \int_0^1 \omega_i(tx) x^i dt.\tag{A.2.78}$$

Indeed, let us check directly by differentiation:

$$\begin{aligned}d\varphi &= \int_0^1 dt [(\partial_j \omega_i(tx)) t x^j dx^i + \omega_i(tx) dx^i] \\ &= \int_0^1 dt [t x^j \partial_j \omega_i(tx) + \omega_i(tx)] dx^i = \int_0^1 dt \frac{d[t \omega_i(tx)]}{dt} dx^i \\ &= \omega_i(x) dx^i = \omega.\end{aligned}\tag{A.2.79}$$

We used (A.2.77) when moving from the first line to the second one. The explicit construction (A.2.78) is certainly not unique but it is sufficient for demonstrating how the proof works. One can easily generalize (A.2.78) for the case when ω is a p -form, $p > 1$,

$$\varphi(x) = \int_0^1 t^{p-1} u \lrcorner \omega(tx) dt,\tag{A.2.80}$$

where the vector field u is *locally* defined by $u = x^i \partial_i$. Its integral lines evidently form a “star-like” structure with the center at the origin of the local coordinate system.

Globally, i.e., on the whole manifold X , however, not every closed form is exact: One usually states that topological obstructions exist. The importance of the de Rham groups is directly related to the fact that they present an example of *topological invariants* of a smooth manifold. Of course, the Betti numbers then also encode information about the topology of X . The zeroth number $b^0(X)$, for instance, simply counts the connected components of any manifold X . This follows from the fact that 0-forms are just functions of X , and hence, a closed form φ with $d\varphi = 0$ is a constant on every connected component. Since there are no exact 0-forms, $B^0(X) = \emptyset$, the elements of the group $H^0(X; \mathbb{R})$ are thus N -tuples of constants, with N equal to the number of connected components. Hence $b^0(X) = \dim H^0(X; \mathbb{R}) = N$.

Moreover, recall that in Sec. A.2.9 for any differentiable map $f : X \rightarrow Y$ we have described a pull-back map of exterior forms on a manifold Y to the forms on X . Since the pull-back commutes with the exterior derivative, see (A.2.42), we immediately find that any such map determines a map between the relevant cohomology groups:

$$f^* : H^p(Y; \mathbb{R}) \longrightarrow H^p(X; \mathbb{R}). \quad (\text{A.2.81})$$

With the help of this map one can prove a fundamental fact: If X and Y are homotopically equivalent manifolds, their de Rham cohomology groups are isomorphic. As a consequence, their Betti numbers are equal, $b^p(X) = b^p(Y)$. Homotopical equivalence essentially means that the manifolds X and Y can be “continuously deformed into one another”. An n -dimensional Euclidean space $\mathbb{E}^n$ is homotopically equivalent to an n -dimensional disk $D^n = \{(x^1, \dots, x^n) \in \mathbb{E}^n \mid \sqrt{\sum_{i=1}^n (x^i)^2} \leq 1\}$, for example, and both are homotopically equivalent to a point. Another example: A Euclidean plane $\mathbb{E}^2$ with one point (say, the origin) removed is homotopically equivalent to a circle S^1 . More rigorously, manifolds X and Y are homotopically equivalent, if there are two differentiable maps $f_1 : X \rightarrow Y$ and $f_2 : Y \rightarrow X$ such that $f_2 \circ f_1 : X \rightarrow X$ and $f_1 \circ f_2 : Y \rightarrow Y$ are homotopic to identity maps id_X and id_Y , respectively. Two maps are homotopic if they can be related by a smooth family of maps.

The alternating sum

$$\chi(X) := \sum_{p=0}^n (-1)^p b^p(X) \quad (\text{A.2.82})$$

is a topological invariant called the *Euler characteristic* of a manifold X . In two dimensions, every orientable closed (compact without a boundary) manifold is diffeomorphic to a sphere with a finite number of handles, $M_h^2 := S^2 + “h \text{ handles}”$, where $h = 0, 1, 2, \dots$ (for $h = 1$, we find a torus $M_1^2 = \mathbb{T}^2$ from Fig. A.2.3). Euler characteristics of these manifolds is $\chi(M_h^2) = 2 - 2h$. Analogously, for the nonorientable 2-dimensional manifolds $N_k^2 := S^2 + “k \text{ cross-caps}”$ (Figs. A.2.4, A.2.5 show $N_1^2 = \mathbb{P}^2$ and $N_2^2 = \mathbb{K}^2$, respectively), the Euler characteristic is equal $\chi(N_k^2) = 2 - k$.

A.3

Integration on a manifold

In this chapter we will describe the *integration of exterior forms on a manifold*. The calculus of differential forms provides us with a powerful technique. This occurs because one theorem, known as the Stokes or the Stokes–Poincaré theorem, replaces a number of different theorems known from 3-dimensional vector calculus. Both types of p -forms, ordinary and twisted ones, can be integrated over p -dimensional submanifolds, and in both cases one needs an additional structure, the orientation, in order to define them. For ordinary forms one needs the *inner* and for twisted forms the *outer* orientation. There are two exceptions: To integrate an ordinary 0-form or a twisted n -form, no orientation is necessary.

A.3.1 Integration of 0-forms and orientability of a manifold

The integral of a 0-form f over a 0-dimensional submanifold (set of points in X) is just a sum of values of f at these points.

Let f be a function on X , i.e., $f \in \Lambda^0(X)$, and let Ξ be a finite collection of points, $\Xi = (p_1, \dots, p_k)$. We can then define the integral of f over Ξ by

$$\int_{\Xi} f := \sum_{i=1}^k f(p_i). \quad (\text{A.3.1})$$

If f is, instead of being an ordinary function, a twisted function, then this definition is not satisfactory. Then the $f(p_i)$'s change their signs together with

the change of the orientation of the reference frames at each point p_i . If we fix one of the orientations at, say, the point p_1 , then we can try to propagate this orientation by continuity to all other points $p_2, \dots, p_k$. If this can be done unambiguously, then we say that the manifold is orientable, and we have just chosen an orientation for X . In such a case, the values of the function f , i.e., $f(p_1), \dots, f(p_k)$, can be taken with respect to any frame with positive orientation, and formula (A.3.1) defines unambiguously the integral $\int_{\Xi} f$ of a twisted 0-form.

A.3.2 Integration of n -forms

The integral of an n -form will be defined over an orientable n -dimensional manifold. In the case of a twisted n -form the orientation is not needed for that purpose.

A support for a p -form φ on X is defined as a set $\text{Supp}(\varphi) := \{p \in X \mid \varphi(p) \neq 0\}$. Let ω be an n -form on X , i.e., $\omega \in \Lambda^n(X)$. At first, let us consider the case when its support $\text{Supp}(\omega) \subset U$ is contained in *one* coordinate chart (U, ϕ) , $\phi = \{x^i\}$. Then in $U \subset X$ we have

$$\omega = f(x) dx^1 \wedge \dots \wedge dx^n, \quad (\text{A.3.2})$$

where we denoted the only component of the n -form as $f(x) = \omega_{1\dots n}(x)$, cf. (A.2.10). We can try to define

$$\int_X \omega := \int_{\phi(U)} f(x^1, \dots, x^n) dx^1 \dots dx^n, \quad (\text{A.3.3})$$

where $\phi(U)$ is the image of U in $\mathbb{R}^n$ for the coordinate map ϕ , and on the right-hand side of (A.3.3) we have a usual Riemann integral. The “definition” above is, however, ambiguous if one changes the coordinates in (A.3.2). Without touching U , one can consider, for example, an arbitrary diffeomorphism $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that will introduce a new local coordinate map $\phi' = A \circ \phi$ with coordinates $\{y^i\}$ in U . Under the change of variables $x^i = x^i(y^j)$, the right-hand side of (A.3.3) transforms into

$$\int_{\phi(U)} f(x) dx^1 \dots dx^n = \int_{\phi'(U)} f(y) |J(A^{-1})| dy^1 \dots dy^n, \quad (\text{A.3.4})$$

where $J(A^{-1}) = \det \partial x^i / \partial y^j$ is the Jacobian determinant of the variable change, cf. (A.2.2). But from (A.3.2), we have $\omega'_{1\dots n}(y) = f(y) J(A^{-1})$. Thus the transformed left-hand side of (A.3.3) is equal $\pm$ of the right-hand side, depending on the sign of the Jacobian determinant. A diffeomorphism A that changes the

orientation of the coordinate system leads to a change of sign of the integral (A.3.3). One has thus to *fix an orientation* in X in order to have a definite notion of the integral $\int_X \omega$.

If $\text{Supp}(\omega)$ is not contained in the domain of a single coordinate chart, the situation is more complicated. Then, for any atlas $\{(U_\alpha, \phi_\alpha)\}$, one has to use a partition of unity $\{\rho_\alpha\}$ subordinate to the covering $\{U_\alpha\}$ of X . In particular, since $\sum_\alpha \rho_\alpha = 1$, we can represent the n -form as a sum $\omega = \sum_\alpha \omega_\alpha$, where each $\omega_\alpha := \omega \rho_\alpha$ vanishes outside U_α , i.e., $\text{Supp}(\omega_\alpha) \subset U_\alpha$. For every ω_α we can then construct the integral via (A.3.3), and finally, an integral for an arbitrary n -form is defined by

$$\int_X \omega := \sum_\alpha \int_{\phi_\alpha(U_\alpha)} f_\alpha(x_\alpha^1, \dots, x_\alpha^n) dx_\alpha^1 \cdots dx_\alpha^n, \quad (\text{A.3.5})$$

where $f_\alpha(x_\alpha^i) = \rho_\alpha \omega_{1\dots n}(x_\alpha^i)$. The integral of an ordinary n -form can be defined unambiguously only in the case of an orientable manifold. Moreover, one can prove that it is *uniquely* defined over X if the orientation is prescribed. In particular, this definition is invariant under the change of an oriented atlas $\{(U_\alpha, \phi_\alpha)\}$ and/or the partition of unity $\{\rho_\alpha\}$.

The situation is quite different if, instead of an ordinary n -form, we consider a *twisted* n -form. If (A.3.2) holds for a twisted form ω in one coordinate system, then in another one we have

$$\omega' = f' dx^{1'} \wedge \cdots \wedge dx^{n'}, \quad (\text{A.3.6})$$

with

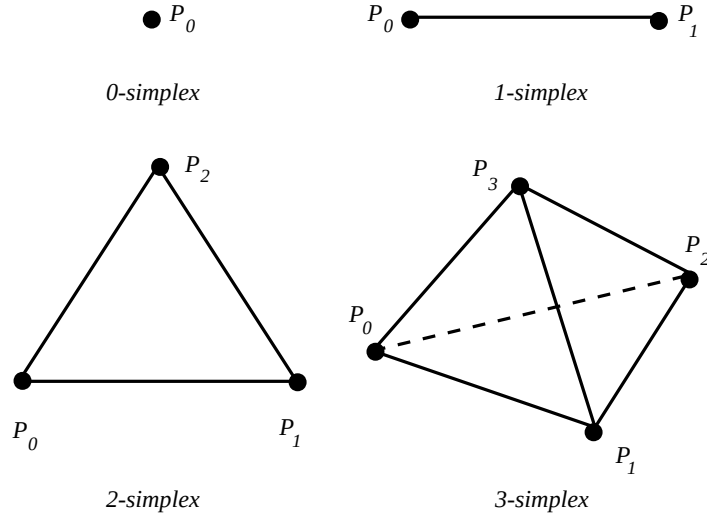
$$f' = \left| \det \left(\frac{\partial x^i}{\partial x^{i'}} \right) \right| f. \quad (\text{A.3.7})$$

Here the additional sign factor of (A.2.12) has been taken into account. We don't need any orientation in order to fix uniquely the meaning of (A.3.3). If many U_α 's cover $\text{Supp}(\omega)$, then we still need the partition of unity, but the consistency in the intersections $U_\alpha \cap U_\beta$ is automatically provided by (A.3.7). Summing up: Any *twisted* n -form ω with compact support can be integrated over an n -dimensional manifold, regardless of whether the latter is orientable or not.

A.3.3 $\otimes$ Integration of p -forms with $0 < p < n$

The integral of a p -form is defined for a singular p -simplex.

In order to define an integral of a p -form with $0 < p < n$ some preliminary constructions are needed that introduce suitable p -dimensional domains of X over which one can integrate.

Figure A.3.1: Simplices $\sigma^0, \sigma^1, \sigma^2$, and σ^3 .

As a first step, one considers p -simplices in $\mathbb{R}^n$. Take a set of $p + 1$ ordered points $P_0, P_1, \dots, P_p \in \mathbb{R}^n$ that are independent in the sense that the p vectors $(P_i - P_0)$ with $i = 1, \dots, p$ are linearly independent (recall that $\mathbb{R}^n$ is a linear vector space). One calls a p -dimensional simplex, or simply a p -simplex, the closed convex hull spanned by this set of points:

$$\sigma^p := (P_0, P_1, \dots, P_p). \quad (\text{A.3.8})$$

Geometrically, it is represented by

$$\sigma^p = \left\{ \sum_{i=0}^p t^i P_i \right\}, \quad \sum_{i=0}^p t^i = 1, \quad t^i \geq 0, \quad (\text{A.3.9})$$

where $t^0, \dots, t^p$ are real numbers.

Accordingly, every 0-simplex is simply one point (P_0); a 1-simplex is a directed line segment (P_0, P_1) ; a 2-simplex is a closed triangle with ordered vertices (P_0, P_1, P_2) ; a 3-simplex is a closed tetrahedron (P_0, P_1, P_2, P_3) , and so on (see Fig. A.3.1).

Each p -simplex σ^p has a natural $(p - 1)$ -dimensional boundary that is composed of faces. An i -th face $\sigma_{(i)}^{p-1}$ with $0 \leq i \leq p$ of a simplex $(P_0, P_1, \dots, P_p)$ is defined as a $(p - 1)$ -simplex $\sigma_{(i)}^{p-1} := (P_0, \dots, \widehat{P}_i, \dots, P_p)$ obtained from $(P_0, P_1, \dots, P_p)$ by removing the vertex P_i (as usual, the hat denotes that an element is omitted from the list). Defined in this way, a face lies opposite to the vertex P_i .

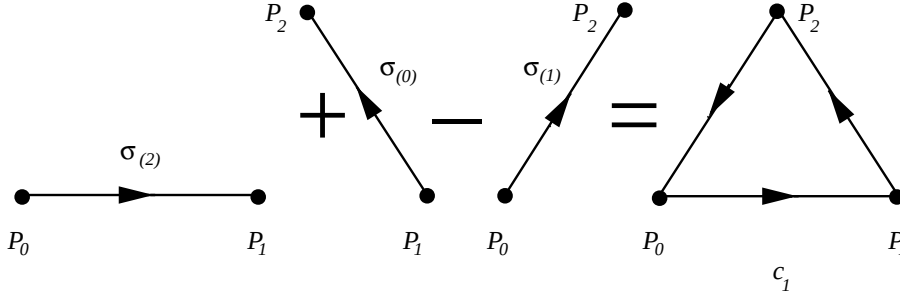


Figure A.3.2: A 1-chain $c_1 = \sigma_{(0)} - \sigma_{(1)} + \sigma_{(2)}$. With $\sigma_{(0)} = (P_1, P_2)$, $\sigma_{(1)} = (P_0, P_2)$, $\sigma_{(2)} = (P_0, P_1)$, the resulting chain is a *boundary* of a 2-simplex: $c_1 = \partial(P_0, P_1, P_2)$. Arrows show the ordering of the vertices.

For any p -simplex $\sigma^p = (P_0, P_1, \dots, P_p)$, with the help of the faces, one can define the formal sum of $(p-1)$ -simplices by

$$\partial\sigma^p := \sum_{i=0}^p (-1)^i \sigma_{(i)}^{p-1}. \quad (\text{A.3.10})$$

It is called the *boundary* of σ^p . More explicitly,

$$\partial(P_0, P_1, \dots, P_p) := \sum_{i=0}^p (-1)^i (P_0, \dots, \widehat{P}_i, \dots, P_p). \quad (\text{A.3.11})$$

The boundary of the 2-simplex (P_0, P_1, P_2) , for example, reads

$$\partial(P_0, P_1, P_2) = (P_0, P_1) - (P_0, P_2) + (P_1, P_2) \quad (\text{A.3.12})$$

(see Fig. A.3.2).

The *boundary of a boundary is zero*: for any p -simplex we have

$$\partial\partial(P_0, P_1, \dots, P_p) = 0. \quad (\text{A.3.13})$$

Let us check this for the 2-simplex (P_0, P_1, P_2) . From (A.3.12) and the definition (A.3.11), we find:

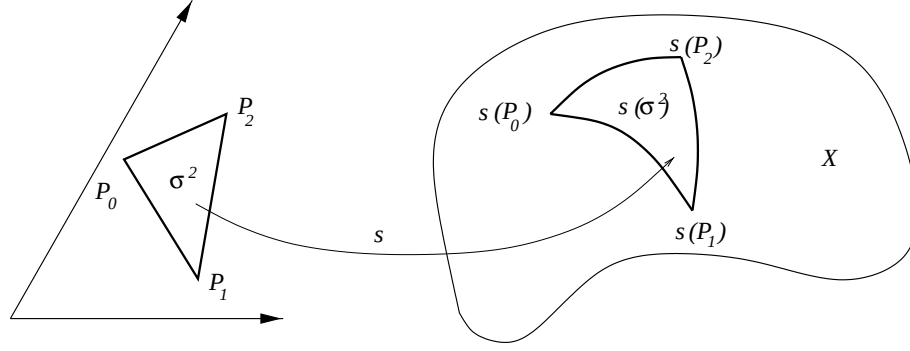
$$\partial\partial(P_0, P_1, P_2) = [(P_1) - (P_0)] - [(P_2) - (P_0)] + [(P_2) - (P_1)] = 0. \quad (\text{A.3.14})$$

From simplices one is able to construct chains. An arbitrary p -chain is a formal sum

$$c_p = \sum_i a_i \sigma_{(i)}^p, \quad (\text{A.3.15})$$

where the a_i are real coefficients and the $\sigma_{(i)}^p$ p -simplices. The boundary of a p -chain is a $(p-1)$ -chain defined by

$$\partial c_p = \sum_i a_i \partial\sigma_{(i)}^p. \quad (\text{A.3.16})$$

Figure A.3.3: Singular simplices on a smooth manifold X .

In Fig. A.3.2, we demonstrate the construction of a chain from simplices. In this particular case the chain turns out to be a boundary of a 2-simplex.

We are now in a position to define the integral of a p -form on the manifold X . For this we need to extend the concept of a p -simplex from $\mathbb{R}^n$ to the manifold X . A suitable domain of integration is given by singular simplex in X which is defined as follows. Given a p -simplex $\sigma^p \subset \mathbb{R}^p$, a *singular p -simplex* in the manifold X is defined as a differentiable map $s : \sigma^p \rightarrow X$. Every point $p \in X$ can evidently be treated as singular 0-simplex, and any smooth curve on X is just a singular 1-simplex. For a 2-simplex see, for example, Fig. A.3.3.

Now consider a p -form ω on the manifold X . Given a singular p -simplex $s : \sigma^p \rightarrow X$, the pull-back s^* maps ω to $\mathbb{R}^p$ and we define the integral of the form over the singular simplex by

$$\int_s \omega := \int_{\sigma^p} f(t^1, \dots, t^p) dt^1 \cdots dt^p, \quad (\text{A.3.17})$$

where $t^i = (t^1, \dots, t^p)$ are the standard coordinates in $\mathbb{R}^p$ and $f(t^1, \dots, t^p) := (s^*\omega)_{1\dots p}(t^i)$ is the single component of the form $s^*\omega$ on $\sigma^p \subset \mathbb{R}^p$. The right-hand side of (A.3.17) is understood in the usual sense of a Riemann integral.

As in the case of the integral for n -forms, the questions related to the orientation should be carefully studied separately first for ordinary and then for twisted p -forms.

Let ω now be an *ordinary* p -form on X . No orientation should be specified for X in the definition above. Instead, the preferred (and standard) orientation in $\mathbb{R}^p$ is used. This orientation, by means of the map s , is transported to $s(\sigma)$. In other words, the push-forward s_* maps the standard frame $(\frac{\partial}{\partial t^1}, \dots, \frac{\partial}{\partial t^p})$ to the frame $(s_* \frac{\partial}{\partial t^1}, \dots, s_* \frac{\partial}{\partial t^p})$ tangent to $s(\sigma) \subset X$. This frame determines an inner orientation all over $s(\sigma)$. The value of the integral (A.3.17) is not changed if we change s and σ^p in such a way (keeping ω untouched) that the orientation induced on $s(\sigma)$ is not changed. Such changes of s to $s' = (s \circ A)$ can be

induced by diffeomorphisms $A : \mathbb{R}^p \rightarrow \mathbb{R}^p$ that have positive determinant $J(A)$. The value of the integral (A.3.17) does not depend on s ; it depends, however, on the choice of *inner* orientation of the simplex σ^p . Therefore it must be assumed that σ^p is inner orientable.

Let us now turn to *twisted* p -forms. In this case, equation (A.3.17) is ambiguous since the choice between $+\omega$ and $-\omega$ depends on the orientation in X , and this fact is not taken care of properly. In order to overcome this ambiguity, we have to determine the *outer* orientation first of the tangent space Ξ_x and subsequently of the whole Ξ . Hereafter we denote a singular p -simplex by $\Xi := s(\sigma^p) \subset X$ and an arbitrary point by $x \in \Xi$. The tangent space of the singular simplex Ξ_x is a subspace of the tangent space X_x at this point. An outer orientation in the tangent space Ξ_x is an orientation in the complementary space X_x/Ξ_x . As usual, it is given by an equivalence class of frames in X_x/Ξ_x that are related by a matrix with positive determinant. In other words, the outer orientation is given by a sequence $(e_{p+1}, \dots, e_n)$ of linearly independent vectors in X_x that are *transversal* to Ξ . The submanifold is outer orientable if the outer orientations in the tangent spaces Ξ_x can be chosen continuously on the whole Ξ . Since Ξ is connected, there are only two orientations allowed. If the submanifold is outer oriented, we can require that the sign in front of ω on the right-hand side of (A.3.17) is consistent with the choice of the orientation for the frame

$$\left(s_* \frac{\partial}{\partial t^1}, \dots, s_* \frac{\partial}{\partial t^p}, e_{p+1}, \dots, e_n\right) \quad (\text{A.3.18})$$

all over the singular simplex $\Xi = s(\sigma)$. This finally removes, for twisted forms, the ambiguity inherent in (A.3.17). To put it differently, by fixing the outer orientation, the only allowed orientation-reversing coordinate transformations are induced by the orientation-changing diffeomorphisms of $\mathbb{R}^p$, and (A.3.17) is obviously invariant under such transformations.

A good example of the notions introduced so far is the Möbius strip, Fig. A.2.9, considered as a submersed submanifold (with boundary) of $\mathbb{R}^3$. In this case, neither an inner nor an outer orientation can be attached to the Möbius strip in a continuous way. Therefore neither ordinary nor twisted 2-forms, given on $\mathbb{R}^3$, can be integrated over the Möbius strip. However, one can embed the Möbius strip into a nonorientable 3-dimensional manifold that is defined as the direct product $X_3 = \mathbb{R} \times \text{Möbius strip}$. In this X_3 , the Möbius strip is a two-sided submanifold, and one can thus introduce the outer orientation on it. After fixing the outer orientation, any twisted 2-form on X_3 can be integrated on the Möbius strip.

A.3.4 Stokes' theorem

Stokes' theorem provides an n -dimensional generalization of the familiar 3-dimensional Gauss and 2-dimensional Stokes theorems.

The Stokes theorem is a far-reaching generalization of the fundamental integration theorem of calculus. Its importance for geometry and physics cannot be overestimated. There are several formulations of Stokes' theorem. Usually, that basic result is presented for an n -dimensional manifold X with a boundary ∂X . Let ω be an $(n-1)$ -form on X . Then

$$\boxed{\int_X d\omega = \int_{\partial X} \omega} . \quad (\text{A.3.19})$$

This theorem is true for an ordinary form on orientable manifolds as well as for a twisted form on nonorientable manifolds. One can show that, in the former case, a natural inner orientation, and in the latter case, a natural outer orientation is induced on the boundary ∂X . We will not give a rigorous proof here.¹

Another version of Stokes' theorem, the so-called combinatorial one, is related to the singular homology of a manifold (see Sec. A.3.5). For any singular p -simplex $s : \sigma^p \rightarrow X$ and a $(p-1)$ -form ω on the manifold X , the (combinatorial) Stokes theorem states that

$$\int_s d\omega = \int_{\partial s} \omega . \quad (\text{A.3.20})$$

Here, in accordance with the definition (A.3.10), the boundary of a singular simplex $s : \sigma^p \rightarrow X$ is defined by

$$\partial s(\sigma^p) := \sum_{i=0}^p (-1)^i s(\sigma_{(i)}^{p-1}) . \quad (\text{A.3.21})$$

Let us *demonstrate* this theorem for a 2-simplex. It is clear that, without loss of generality, one can always choose coordinates (t^1, t^2) in $\mathbb{R}^2 \supset \sigma^2$ in such a way that the vertices of the 2-simplex are the points $P_0 = (0, 0)$, $P_1 = (1, 0)$, and $P_2 = (0, 1)$. The simplex is then called *standard* with the canonical choice of coordinates. The standard 2-simplex is depicted on Fig. A.3.4. Incidentally, the generalization to higher-dimensional simplices in $\mathbb{R}^p$ is straightforward: A standard p -simplex $\sigma^p = (P_0, P_1, \dots, P_p)$ is defined by the points $P_0 = (0, \dots, 0)$, $P_1 = (1, 0, \dots, 0)$, $\dots$, $P_p = (0, \dots, 0, 1)$.

Given the parametrization of the standard 2-simplex, cf. (A.3.9),

$$\sigma^2 = \{(1 - t^1 - t^2) P_0 + t^1 P_1 + t^2 P_2\} , \quad 0 \leq t^i \leq 1, \quad (\text{A.3.22})$$

its boundary is described by the three 1-simplices (its faces):

$$\begin{aligned} \sigma_{(0)}^1 &= \{t^1 P_1 + t^2 P_2\} , & t^1 + t^2 &= 1, \\ \sigma_{(1)}^1 &= \{t^2 P_2 + (1 - t^2) P_0\} , & 0 \leq t^2 &\leq 1, \\ \sigma_{(2)}^1 &= \{(1 - t^1) P_0 + t^1 P_1\} , & 0 \leq t^1 &\leq 1. \end{aligned} \quad (\text{A.3.23})$$

¹A rigorous proof can be found in Choquet-Bruhat et al. [4].

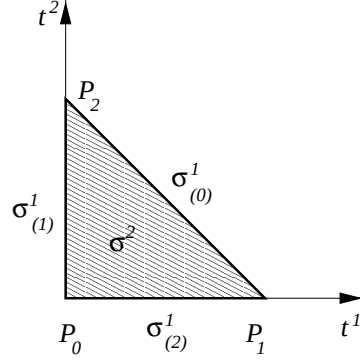


Figure A.3.4: Standard 2-simplex with canonical coordinates on it.

For a 1-form ω on X the pull-back on $\sigma^2 \subset \mathbb{R}^2$ is given by

$$(s^*\omega) = f_1(t^1, t^2) dt^1 + f_2(t^1, t^2) dt^2, \quad (\text{A.3.24})$$

with the two independent components $f_i(t^1, t^2) = (s^*\omega)_i$, $i = 1, 2$. Accordingly, the exterior derivative reads

$$(s^*d\omega) = \left(\frac{\partial f_2}{\partial t^1} - \frac{\partial f_1}{\partial t^2} \right) dt^1 \wedge dt^2. \quad (\text{A.3.25})$$

Now we have to apply the definition (A.3.17). For the left-hand side of Stokes' theorem we find, using the conventional rules for the multiple integrals,

$$\begin{aligned} \int_s d\omega &= \int_{\sigma^2} s^*d\omega = \iint_{(P_0, P_1, P_2)} \left(\frac{\partial f_2}{\partial t^1} - \frac{\partial f_1}{\partial t^2} \right) dt^1 dt^2 \\ &= \int_0^1 dt^2 \int_0^{1-t^2} dt^1 \frac{\partial f_2}{\partial t^1} - \int_0^1 dt^1 \int_0^{1-t^1} dt^2 \frac{\partial f_1}{\partial t^2} \\ &= \int_0^1 dt^2 [f_2(1-t^2, t^2) - f_2(0, t^2)] \\ &\quad - \int_0^1 dt^1 [f_1(t^1, 1-t^1) - f_1(t^1, 0)]. \end{aligned} \quad (\text{A.3.26})$$

The right-hand side of Stokes' theorem consists of the three integrals over the faces (A.3.23). Direct calculation of the corresponding line integrals yields:

$$\begin{aligned}
\int_{\sigma_{(0)}^1} s^* \omega &= \int_{P_1}^{P_2} f_i dt^i = - \int_0^1 dt^1 f_1(t^1, 1-t^1) + \int_0^1 dt^2 f_2(1-t^2, t^2), \\
\int_{\sigma_{(1)}^1} s^* \omega &= \int_{P_2}^{P_0} f_i dt^i = \int_0^1 dt^2 f_2(0, t^2), \\
\int_{\sigma_{(2)}^1} s^* \omega &= \int_{P_0}^{P_1} f_i dt^i = \int_0^1 dt^1 f_1(t^1, 0).
\end{aligned} \tag{A.3.27}$$

Taking into account that

$$\int_{\partial s} \omega = \sum_{i=0}^p (-1)^i \int_{\sigma_{(i)}^1} s^* \omega, \tag{A.3.28}$$

and recalling (A.3.21), we compare (A.3.26) and (A.3.27) to verify that (A.3.20) holds true for any 1-form and any singular 2-simplex on X .

A.3.5 $\otimes$ De Rham's theorems

The first theorem of de Rham states that a closed form is exact if and only if all of its periods vanish.

Recall that the de Rham cohomology groups, which were defined in Sec. A.2.12 with the help of the exterior derivative

$$d : \Lambda^p(X) \longrightarrow \Lambda^{p+1}(X) \tag{A.3.29}$$

in the algebra of differential forms $\Lambda^*(X)$, “feel” the topology of the manifold X . Likewise, singular simplices can also be used to study the topological properties of X . The relevant mathematical structure is represented by the *singular homology groups*. They are defined as follows: Like a chain constructed from simplices, see (A.3.15), a *singular p -chain* on a manifold X is defined as a formal sum

$$c_p = \sum_i a_i s_i^p, \tag{A.3.30}$$

with real coefficients a_i and singular p -simplices s_i^p . In the space $C_p(X)$ of all singular p -chains on X , a sum of chains and multiplication by a real constant are defined in an obvious way.

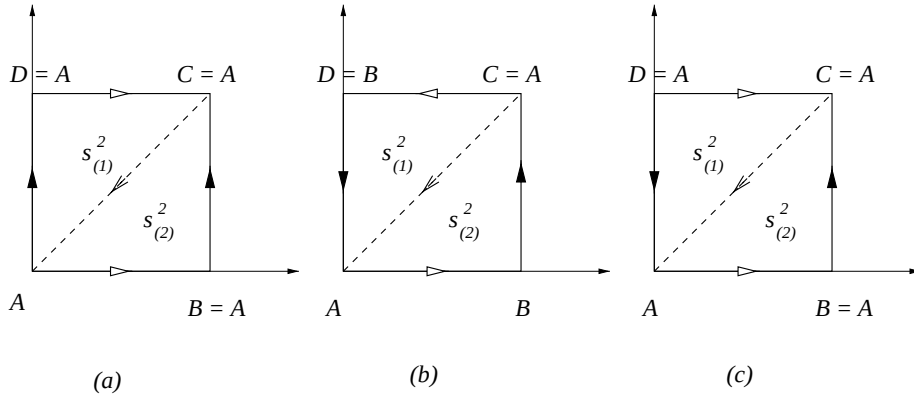


Figure A.3.5: Simplicial decomposition (triangulation) of (a) the torus $\mathbb{T}^2$, (b) the real projective plane $\mathbb{P}^2$, and (c) the Klein bottle $\mathbb{K}^2$.

The *boundary* map

$$\partial : C_p(X) \longrightarrow C_{p-1}(X) \quad (\text{A.3.31})$$

is introduced, in analogy to (A.3.16) and (A.3.21), by defining for every singular p -chain c_p a singular $(p-1)$ -chain:

$$\partial c_p = \sum_i a_i \partial s_i^p. \quad (\text{A.3.32})$$

In complete analogy with the *de Rham complex* $(\Lambda^*(X), d)$, a singular p -simplex z is called a *cycle* if $\partial z = 0$. The set of all p -cycles,

$$Z_p(X) := \{z \in C_p(X) \mid \partial z = 0\}, \quad p = 0, \dots, n, \quad (\text{A.3.33})$$

is a real vector subspace, $Z_p(X) \subseteq C_p(X)$.

A singular p -chain b is called a *boundary* if a $(p+1)$ -chain c exists such that $b = \partial c$. The space of p -boundaries

$$B_p(X) := \{b \in C_p(X) \mid b = \partial c\}, \quad p = 1, \dots, n, \quad (\text{A.3.34})$$

also forms a (real) vector space and $B_p(X) \subseteq Z_p(X)$, since $\partial \partial \equiv 0$. One sets $B_n(X) = \emptyset$.

Finally, the singular homology groups are defined as the quotient spaces

$$H_p(X; \mathbb{R}) := Z_p(X) / B_p(X), \quad p = 0, \dots, n. \quad (\text{A.3.35})$$

As an instructive example, let us briefly analyze the homological structure of the simplest compact 2-dimensional manifolds: The sphere S^2 , the torus $\mathbb{T}^2$ (these two are orientable), the real projective plain $\mathbb{P}^2$, and the Klein bottle

$\mathbb{K}^2$ (these are nonorientable). The three last manifolds are seen in Fig. A.2.3, Fig. A.2.4, and Fig. A.2.5, respectively. A standard approach to the calculation of homologies for a manifold X is to *triangulate* it, that is, to subdivide X into simplices in such a way that the resulting totality of simplices (called a simplicial complex) contains, together with each simplex, all of its faces. Every two simplices either do not have common points or they intersect over a common face of lower dimension. The triangulation of a sphere obviously reduces just to a collection of four 2-simplices that form the boundary of a 3-simplex, that is, the surface of a tetrahedron (see Fig. A.3.1). The triangulations of the torus, the projective plane, and the Klein bottle are depicted in Fig. A.3.5.

- 1) S^2 has as the only 2-cycle the manifold itself, $z^2 = S^2$. Direct inspection shows that there are no nontrivial 1-cycles (they are all boundaries of 2-dimensional chains). Finally, each vertex of the tetrahedron is trivially a 0-cycle, and they are all homological to each other because of the connectedness of S^2 . These facts are summarized by displaying the homology groups explicitly:

$$\begin{aligned} H_2(S^2; \mathbb{R}) &= \mathbb{R}, \\ H_1(S^2; \mathbb{R}) &= 0, \\ H_0(S^2; \mathbb{R}) &= \mathbb{R}. \end{aligned} \tag{A.3.36}$$

- 2) $\mathbb{T}^2$ is “composed” of two 2-simplices, $\mathbb{T}^2 = S_{(1)}^2 + S_{(2)}^2$, namely $S_{(1)}^2 = (A, C, D)$ and $S_{(2)}^2 = (A, B, C)$ with the corresponding identifications (gluing) of sides and points as shown in Fig. A.3.5(a). The direct calculation of the boundaries yields $\partial S_{(2)}^2 = (A, B) + (B, C) - (A, C)$ and $\partial S_{(1)}^2 = (C, D) - (A, D) + (A, C)$. Taking the identifications into account, we then find $\partial \mathbb{T}^2 = 0$; hence the torus itself is a 2-cycle. There are no other non-trivial 2-cycles. As for the 1-cycles, we find *two*: $z_{(1)}^1 = (A, B)|_{B=A}$ and $z_{(2)}^1 = (B, C)|_{C=B=A}$ (end points are identified). Geometrically, these cycles are just closed curves, one of which goes along and another across the handle. There are no other independent 1-cycles ($z_{(3)}^1 = (C, A)|_{C=A}$, for example, is homological to the sum of $z_{(1)}^1$ and $z_{(2)}^1$). Thus we have verified that the 1st homology group is two-dimensional. For 0-cycles the situation is exactly the same as for the sphere. In summary, we have for the torus:

$$\begin{aligned} H_2(\mathbb{T}^2; \mathbb{R}) &= \mathbb{R}, \\ H_1(\mathbb{T}^2; \mathbb{R}) &= \mathbb{R}^2, \\ H_0(\mathbb{T}^2; \mathbb{R}) &= \mathbb{R}. \end{aligned} \tag{A.3.37}$$

- 3) $\mathbb{P}^2 = S_{(1)}^2 + S_{(2)}^2$, where $S_{(1)}^2 = (A, C, D)$ and $S_{(2)}^2 = (A, B, C)$ with sides and points identified as shown in Fig. A.3.5b. Repeating the calculation

for the torus, we find $\partial\mathbb{P}^2 = 2c_{(1)}^1 + 2c_{(2)}^1$, where the 1-chains are $c_{(1)}^1 = (A, B)$ and $c_{(2)}^1 = (B, C)$. Thus, the projective plane itself is not a 2-cycle. Since there are no other homologically inequivalent 2-cycles, we conclude that the 2nd homology group is trivial. Moreover, we immediately verify $\partial c_{(1)}^1 = (B) - (A)$ and $\partial c_{(2)}^1 = (A) - (B)$, thus proving that $z^1 = c_{(1)}^1 + c_{(2)}^1$ is a 1-cycle. Moreover, it is a boundary because of $2z^1 = \partial\mathbb{P}^2$. No other 1-cycle exists in $\mathbb{P}^2$. Thus we conclude that the 1st homology group is also trivial. Because of connectedness, the final list reads:

$$\begin{aligned} H_2(\mathbb{P}^2; \mathbb{R}) &= 0, \\ H_1(\mathbb{P}^2; \mathbb{R}) &= 0, \\ H_0(\mathbb{P}^2; \mathbb{R}) &= \mathbb{R}. \end{aligned} \tag{A.3.38}$$

- 4) $\mathbb{K}^2 = S_{(1)}^2 + S_{(2)}^2$, where $S_{(1)}^2 = (A, C, D)$ and $S_{(2)}^2 = (A, B, C)$ with sides and points glued as shown in Fig. A.3.5c. By an analogous calculation, we find $\partial\mathbb{K}^2 = 2(B, C)$. There are no nontrivial 2-cycles on the Klein bottle. As for the torus, there are two independent 1-cycles, $z_{(1)}^1 = (A, B)|_{B=A}$ and $z_{(2)}^1 = (B, C)|_{C=B=A}$. However, the second one is a boundary $2z_{(2)}^1 = \partial\mathbb{K}^2$. Hence $z_{(1)}^1$ generates the only nontrivial homology class for the Klein bottle. Thus finally, the homology groups are:

$$\begin{aligned} H_2(\mathbb{K}^2; \mathbb{R}) &= 0, \\ H_1(\mathbb{K}^2; \mathbb{R}) &= \mathbb{R}, \\ H_0(\mathbb{K}^2; \mathbb{R}) &= \mathbb{R}. \end{aligned} \tag{A.3.39}$$

Like the de Rham cohomology groups $H^p(X; \mathbb{R})$ (see Sec. A.2.12) the singular homology groups $H_p(X; \mathbb{R})$ are *topological invariants* of a manifold. In particular, they do not change under a ‘smooth deformation’ of a manifold, i.e., they are homotopically invariant. Cohomologies and homologies are deeply related. In order to demonstrate this (although without rigorous proofs), we need the central notion of a period. For any *closed* p -form $\omega \in Z^p(X)$ and each p -cycle $z \in Z_p(X)$, a *period* of the form ω is the number

$$\text{per}_z(\omega) := \int_z \omega. \tag{A.3.40}$$

This real number is not merely a function of ω and z ; it rather depends on the whole cohomology class of the form $[\omega] \in H^p(X; \mathbb{R})$ and on the whole homology class of the cycle $[z] \in H_p(X; \mathbb{R})$. Stokes’ theorem underlies the proof: For any cohomologically equivalent p -form $\omega + d\varphi$ and for any homologically equivalent

p -cycle $z + \partial c$, we find

$$\begin{aligned} \text{per}_z(\omega + d\varphi) &= \int_z (\omega + d\varphi) = \int_z \omega + \int_{\partial z} \varphi = \int_z \omega = \text{per}_z(\omega), \\ \text{per}_{z+\partial c}(\omega) &= \int_{z+\partial c} \omega = \int_z \omega + \int_z d\omega = \int_z \omega = \text{per}_z(\omega), \end{aligned}$$

since $\partial z = d\omega = 0$. Therefore, in a strict sense, one has to write a period as $\text{per}_{[z]}([\omega])$. We recall the definition of a form as a linear map from a vector space V to the reals (see Sec. A.1.1). Accordingly, one can treat the period (A.3.40) as a 1-form on the space of cohomologies with $V = H^p(X; \mathbb{R})$, i.e., as an element of the dual space

$$\text{per}_{[z]} \in H_p(X; \mathbb{R})^*, \quad \text{for all } [\omega] \in H^p(X; \mathbb{R}). \quad (\text{A.3.41})$$

The linear map $\text{DR} : H^p(X; \mathbb{R}) \rightarrow H_p(X; \mathbb{R})^*$, defined via the equations (A.3.41) and (A.3.40) as $\text{DR}([\omega])([z]) := \text{per}_{[z]}([\omega])$, is called the *de Rham map*. A fundamental *theorem of de Rham* states that this map is an isomorphism. Sometimes, the proof of the de Rham theorem is subdivided into the two separate propositions known as the *first* and *second* de Rham theorems. The *first de Rham theorem* reads: *A closed form is exact if and only if all of its periods vanish*:

$$\left\{ \begin{array}{c} \omega \in Z^p(X) \\ \Downarrow \\ \omega \in B^p(X) \end{array} \right\} \iff \left\{ \begin{array}{c} \text{per}_z(\omega) = 0 \\ \text{for all} \\ z \in Z_p(X) \end{array} \right\}. \quad (\text{A.3.42})$$

In simple terms, the first theorem tells that $\text{DR}([\omega]) = 0 \Leftrightarrow [\omega] = 0$. A 1-form on a vector space V is determined by its components which give the values of that form with respect to a basis of V . Suppose we have chosen a basis $[z_i]$ of the p -th homology group $H_p(X; \mathbb{R})$, i.e., a complete set of homologically inequivalent singular p -cycles z_i . (For a compact manifolds this set is finite.) Denote as $\Theta^i \in V^* = H_p(X; \mathbb{R})^*$ the dual basis to $[z_i]$. Each 1-form on $V = H_p(X; \mathbb{R})$ is then an element $a_i \Theta^i$ specified by a set of real numbers $\{a_i\}$ with i running over the whole range of the basis z_i . The *second de Rham theorem* states that the de Rham map *is invertible*, that is, for every set of real numbers $\{a_i\}$ there exists a closed p -form ω on X such that

$$\text{DR}([\omega]) = a_i \Theta^i, \quad \text{i.e.,} \quad [\omega] = \text{DR}^{-1}(a_i \Theta^i). \quad (\text{A.3.43})$$

In combination with the second theorem, the first de Rham theorem clearly guarantees that the de Rham map is one-to-one: Suppose that for a given set $\{a_i\}$ one can find two 1-forms ω and ω' that both satisfy (A.3.43). Then we get $\text{DR}([\omega - \omega']) = 0$ and (A.3.42) yields $[\omega] = [\omega']$, i.e., ω and ω' differ by an exact form.

Incidentally, our earlier study of the homological structure of the 2-dimensional manifolds $S^2, \mathbb{T}^2, \mathbb{P}^2, \mathbb{K}^2$ gave explicit constructions of the bases $[z_i]$ of the homology groups. One can show that for an arbitrary *compact* manifold X both the cohomology and homology groups are finite-dimensional vector spaces. Then the de Rham map establishes the canonical isomorphism

$$H_p(X; \mathbb{R}) \stackrel{\text{DR}}{\cong} H^p(X; \mathbb{R}), \quad p = 0, \dots, n. \quad (\text{A.3.44})$$

In particular, $\dim H^p(X; \mathbb{R}) = \dim H_p(X; \mathbb{R})$. Then one can, for example, calculate the Euler characteristics (A.2.82) easily. Returning again to the 2-dimensional examples, we find: $\chi(S^2) = 1 - 0 + 1 = 2$, see (A.3.36); $\chi(\mathbb{T}^2) = 1 - 2 + 1 = 0$, see (A.3.37); $\chi(\mathbb{P}^2) = 0 - 0 + 1 = 1$, see (A.3.38); and $\chi(\mathbb{K}^2) = 0 - 1 + 1 = 0$, see (A.3.39).

Problem

Problem A.1.

Show that properties 1)–4) of Proposition 1 in Sec. A.2.7 lead uniquely to the formula (A.2.16), i.e., they provide also a definition of the exterior derivative.

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Part B

Axioms of classical electrodynamics

In Part B we put phenomenological classical electrodynamics into such a form that the underlying physical facts are clearly visible. We recognize that the conservation of *electric charge* and of *magnetic flux* are the two main experimentally well-founded axioms of electrodynamics. To formulate them we use exterior calculus because it is the appropriate mathematical framework for handling fields the integrals of which — here charge and flux — possess an invariant meaning.

The densities of the electric charge and electric current are assumed to be phenomenologically specified. These quantities will not be resolved any further and will be considered as fundamental for classical electrodynamics.

B.1

Electric charge conservation

... it is now discovered and demonstrated, both here and in Europe, that the Electrical Fire is a real Element, or Species of Matter, not created by the Friction, but collected only.

Benjamin Franklin (1747)¹

B.1.1 Counting charges. Absolute and relative dimension

Progress in semiconductor technology has enabled the fabrication of structures so small that they can contain just one mobile electron. By varying controllably the number of electrons in these ‘artificial atoms’ and measuring the energy required to add successive electrons, one can conduct atom physics exper-

¹See Heilbron [16], p. 330. On the same page Heilbron states: “Although Franklin did not ‘discover’ conservation, he was unquestionably the first to exploit the concept fruitfully. Its full utility appeared in his classic analysis of the condenser.” Note that the discovery of charge conservation preceded the discovery of the Coulomb law (1785) by more than 40 years. This historical sequence is reflected in our axiomatics. However, our reason is not an historical but a conceptual one: Charge conservation should come first and the Maxwell equations should be formulated so as to be compatible with that law.

iments in a regime that is inaccessible to experiments on real atoms.

R. C. Ashoori (1996)²

Phenomenologically speaking, electromagnetism has two types of sources: The electric charge density ρ and the electric current density $\vec{j}$. The electric current density can be understood, with respect to some reference system, as moving electric charge density.

In this section we give a *heuristic* discussion of charge conservation that will be used, in the next section, as a motivation for the formulation of a first axiom for electrodynamics.

Imagine a 3-dimensional simply connected region Ω_3 that is enclosed by the 2-dimensional boundary $\partial\Omega_3$ (see Fig. B.1.1). In the region Ω_3 , there are elementary particles with charge $\pm e$ (e = elementary electric charge) and quarks with charges $\pm\frac{1}{3}e$ and $\pm\frac{2}{3}e$, respectively, which all move with some velocity. It is assumed in electrodynamics that we can attribute to the 3-region Ω_3 at any time in a certain reference system a well-defined net charge Q with the *dimension*³ of charge q (in SI units, coulomb, abbreviated C):

$$Q = \int_{\Omega_3} \rho, \quad [Q] = q, \quad [\rho] = q. \quad (\text{B.1.1})$$

Here $[Q]$ should be read as “dimension of Q ” and, analogously, $[\rho]$ as “dimension of ρ .” We call $[Q]$ and $[\rho]$ also *absolute* dimension of Q and ρ , respectively.

The integrand ρ is called the electric charge density 3-form. It assigns to the volume 3-form in an arbitrary coordinate system $(a, b, \dots = 1, 2, 3)$

$$dx^a \wedge dx^b \wedge dx^c, \quad (\text{B.1.2})$$

a scalar-valued charge

$$\rho = \frac{1}{3!} \rho_{abc} dx^a \wedge dx^b \wedge dx^c, \quad \rho_{abc} = \rho_{[abc]}, \quad (\text{B.1.3})$$

which, as a scalar, can be added up in any coordinate system to yield Q . Thus, as already noted, even the charge *density* ρ carries the same absolute dimension as the net charge Q .

In spatial spherical coordinates $x^a = (r, \theta, \phi)$, for instance, the coordinates carry different dimensions: The x^1 or r -coordinate has the dimension of a length,

²See his article [1]. For a review on the counting of single electrons, see Devoret and Grabert [5] and, for more recent developments, Fujiwara and Takahashi [12].

³We base our dimensional analysis on the work of Dorgelo and Schouten (see Schouten [45], pp. 126–138 and Post [37], pp. 23–46). A guide of how to use the International System of Units (SI) has been provided by Nelson [34]; see also the references therein.

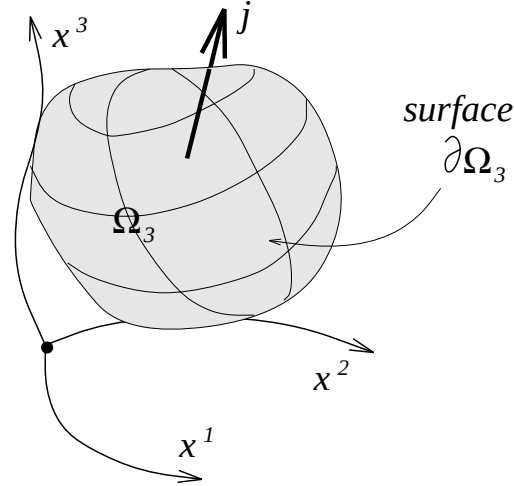


Figure B.1.1: Charge conservation in 3-dimensional space. The electric current density 2-form j flows out of the volume Ω_3 .

whereas $x^2 = \theta$ and $x^3 = \phi$ are dimensionless. Accordingly, the different coordinate components ρ_{abc} of ρ have different dimensions that cannot be measured straightforwardly in an experiment. Thus these components of ρ are unsuitable for a general definition of the relative (also called physical) dimension of a quantity.

Therefore we introduce an arbitrary 3-dimensional local coframe $\vartheta^{\hat{a}}$ with $\hat{a}, \hat{b}, \dots = \hat{1}, \hat{2}, \hat{3}$, and its dual frame $e_{\hat{b}}$ with $e_{\hat{b}} \lrcorner \vartheta^{\hat{a}} = \delta_{\hat{b}}^{\hat{a}}$. We mark the numbers $\hat{1}, \hat{2}, \hat{3}$ by a circumflex in order to be able to identify them as being related to *anholonomic* 3D indices. A physicist needs such a (co)frame to anchor his or her apparatus. We assign to each of the three 1-forms $\vartheta^{\hat{a}}$ the *dimension of a length* ℓ and to the corresponding vectors $e_{\hat{b}}$ the dimension of ℓ^{-1} :

$$[\vartheta^{\hat{a}}] = \ell, \quad \text{for } \hat{a} = \hat{1}, \hat{2}, \hat{3}, \quad (\text{B.1.4})$$

$$[e_{\hat{b}}] = \ell^{-1}, \quad \text{for } \hat{b} = \hat{1}, \hat{2}, \hat{3}. \quad (\text{B.1.5})$$

Length is here understood as a *segment*, that is, as part of a straight line between two points. Accordingly, length is synonymous with a 1-dimensional extension in affine geometry; it is not, however, the distance in the sense of Euclidean geometry. In other words, length as dimension does *not* presuppose the existence of a metric; it is a pre-metric concept.

Now we can decompose the charge density 3-form with respect to the coframe $\vartheta^{\hat{a}}$,

$$\rho = \frac{1}{3!} \rho_{\hat{a}\hat{b}\hat{c}} \vartheta^{\hat{a}} \wedge \vartheta^{\hat{b}} \wedge \vartheta^{\hat{c}}, \quad \text{rel. dim.} := [\rho_{\hat{a}\hat{b}\hat{c}}] = q \ell^{-3}. \quad (\text{B.1.6})$$

The dimensions of all *anholonomic* components $\rho_{\hat{a}\hat{b}\hat{c}}$ of the charge density ρ are the same. We call $[\rho_{\hat{a}\hat{b}\hat{c}}]$ the *relative dimension* of ρ . Accordingly, the absolute dimension of ρ is *charge*, i.e., $[\rho] = q$, whereas the relative dimension of ρ is *charge/length*³, i.e., $[\rho_{\hat{a}\hat{b}\hat{c}}] = q \ell^{-3}$.

In the *hypothesis of locality*⁴ it is assumed that the measuring apparatus in the coframe $\vartheta^{\hat{a}}$, even if the latter is accelerated, measures the anholonomic components of a physical quantity, such as the components $\rho_{\hat{a}\hat{b}\hat{c}}$, exactly as in a momentarily comoving inertial frame of reference. In the special case of the measurment of time, Einstein spoke about the clock hypothesis.

If we assume, as suggested by experience, that the electric charge Q has *no intrinsic screw-sense*, then the sign of Q does not depend on the orientation in space. Accordingly, the charge density ρ is represented by a *twisted* 3-form; for the definition of twisted quantities, see the end of Sec. A.1.3.

Provided the coordinates x^a are given in Ω_3 , we can determine dx^a and the volume 3-form (B.1.2). There is no need to use a metric nor a connection, the properties of a ‘bare’ differential manifold (continuum) are sufficient for the definition of (B.1.1). This can also be recognized as follows:

The net charge Q in (B.1.1) can be determined by counting the charged elementary particles inside $\partial\Omega_3$ and adding up their elementary electric charges. Nowadays one catches single electrons in traps. Thus the counting of electrons is an experimentally feasible procedure, not only a thought experiment devised by a theoretician. This consideration shows that it is not necessary to use a distance concept nor a length standard in Ω_3 for the determination of Q . Only ‘counting procedures’ are required and a way to delimit an arbitrary finite volume Ω_3 of 3-dimensional space by a boundary $\partial\Omega_3$ and to know what is inside $\partial\Omega_3$ and what outside. Accordingly, ρ is the *prototype* of a charge density with absolute dimension $[\rho] = q$ and relative dimension $[\rho_{abc}] = q \ell^{-3}$. It becomes the conventional charge density, that is charge per *scaled* unit volume, if a unit of distance (m in SI-units) is introduced additionally. Then, in SI-units, we have $[\rho_{\hat{a}\hat{b}\hat{c}}] = \text{C m}^{-3}$.

Out of the region Ω_3 , crossing its bounding surface $\partial\Omega_3$, there will, in general, flow a net electric current, see Fig.B.1.1,

$$\mathcal{J} = \oint_{\partial\Omega_3} j, \quad (\text{B.1.7})$$

with absolute dimension $q t^{-1}$ (t = time), which must not depend on the orientation of space either. The integrand j , the twisted electric current density

⁴The formulation of Mashhoon [30] reads: “...the hypothesis of locality — i.e., the presumed equivalence of an accelerated observer with a momentarily comoving inertial observer — underlies the standard relativistic formalism by relating the measurements of an accelerated observer to those of an inertial observer.” The two observers are assumed to be otherwise identical. That is, the two observers are copies of one observer: One copy is inertial and the other accelerated; this is the only difference between them. The limitations of the hypothesis of locality are discussed in [31].

2-form with the same absolute dimension $q t^{-1} \stackrel{\text{SI}}{=} \text{C s}^{-1} = \text{ampere} = \text{A}$, here s = second, assigns to the area element 2-form

$$dx^a \wedge dx^b \quad (\text{B.1.8})$$

a scalar-valued charge current

$$j = \frac{1}{2!} j_{ab} dx^a \wedge dx^b, \quad j_{ab} = j_{[ab]}. \quad (\text{B.1.9})$$

The postulate of electric charge conservation requires

$$\frac{dQ}{dt} = -\mathcal{J}, \quad (\text{B.1.10})$$

provided the area 2-form $dx^a \wedge dx^b$ is directed in such a way that the outflow is counted positively in (B.1.7). The time variable t , provisionally introduced here, does not need to possess a scale or a unit. It can be called a “smooth causal time” in the sense of parameterizing a future-directed curve in the spacetime manifold with t as a monotone increasing and sufficiently smooth variable. Substitution of (B.1.1) and (B.1.7) into (B.1.10) yields an integral form of charge conservation:

$$\frac{d}{dt} \int_{\Omega_3} \rho + \oint_{\partial\Omega_3} j = 0. \quad (\text{B.1.11})$$

By applying the 3-dimensional Stokes theorem, the differential version turns out to be

$$\frac{\partial \rho}{\partial t} + d j = 0. \quad (\text{B.1.12})$$

Let us put (B.1.11) into a 4-dimensional form. For this purpose we integrate (B.1.11) over a certain time interval from t_1 to t_2 (see Fig. B.1.2). Note that this figure depicts the same physical situation as in Fig. B.1.1:

$$\begin{aligned} & \int_{t_1}^{t_2} dt \frac{d}{dt} \int_{\Omega_3} \rho + \int_{t_1}^{t_2} dt \wedge \oint_{\partial\Omega_3} j \\ &= \int_{t=t_2, \Omega_3} \rho - \int_{t=t_1, \Omega_3} \rho + \int_{[t_1, t_2] \times \partial\Omega_3} dt \wedge j = 0. \end{aligned} \quad (\text{B.1.13})$$

Obviously we are integrating over a 3-dimensional boundary of a compact piece of the 4-dimensional spacetime. If we introduce in four dimensions the twisted 3-form

$$J := -j \wedge dt + \rho, \quad \text{with} \quad [J] = q, \quad (\text{B.1.14})$$

then the integral can be written as a 4-dimensional boundary integral,

$$\oint_{\partial\Omega_4} J = 0, \quad (\text{B.1.15})$$

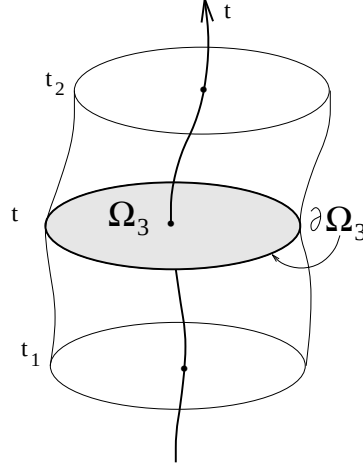


Figure B.1.2: Charge conservation in 4-dimensional spacetime. The 3-dimensional “cross section” Ω_3 sweeps out a 4-dimensional volume Ω_4 on its way from $t = t_1$ to $t = t_2$.

where $\Omega_4 = [t_1, t_2] \times \Omega_3$. The twisted 3-form J of the electric charge-current density with absolute dimension q plays the central role as source of the electromagnetic field.

B.1.2 Spacetime and the first axiom

Motivated by the integral form of charge conservation (B.1.15), we can now turn to an axiomatic approach of electrodynamics. First we will formulate a set of minimal assumptions that we shall need for defining an appropriate spacetime manifold.

Let spacetime be given as a 4-dimensional *connected*, *Hausdorff*, *orientable*, and *paracompact* differentiable manifold X_4 . This manifold is “bare,” that is, it carries neither a connection nor a metric so far. We assume, however, the conventional continuity and differentiability requirements of physics. To recall, a topological space X is Hausdorff when for any two points $p_1 \neq p_2 \in X$ one can find open sets $p_1 \in U_1 \subset X$, $p_2 \in U_2 \subset X$, such that $U_1 \cap U_2 = \emptyset$. An X is connected when any two points can be connected by a continuous curve. Finally, a connected Hausdorff manifold is paracompact when X can be covered by a countable number of coordinate charts. The (smooth) coordinates in arbitrary charts will be called x^i , with $i, j, k, \dots = 0, 1, 2, 3$. The vector basis (frame) of the tangent space will be called e_α and the 1-form basis (coframe) of the cotangent space ϑ^α with (anholonomic) indices $\alpha, \beta, \gamma, \dots = \hat{0}, \hat{1}, \hat{2}, \hat{3}$. On the X_4 we can

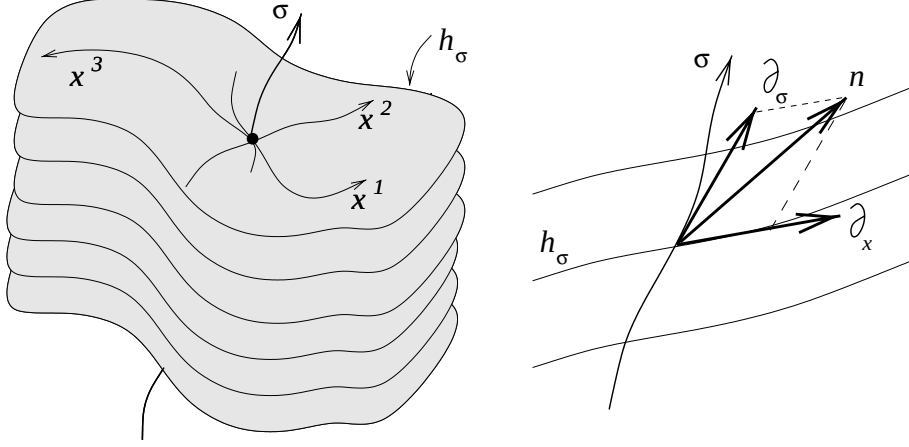


Figure B.1.3: Spacetime and its (1 + 3)-foliation.

define twisted and ordinary untwisted tensor-valued differential forms. In order to avoid possible violations of causality, we will, as usual, consider only *non-compact* spacetime manifolds X_4 .

The X_4 with the described topological properties is known to possess a (1+3)-foliation (see Fig. B.1.3); i.e., there exists a set of nonintersecting 3-dimensional hypersurfaces h_σ that can be parameterized by a monotone increasing (would-be time) variable σ with the dimension of time: $[\sigma] = t$. Although at this stage we do not introduce any metric on X_4 , it is well known that the existence of a (1 + 3)-foliation is closely related to the existence of pseudo-Riemannian structures.

Among the vector fields *transverse* to the foliation, we choose a vector field n (not to be confused with the dimension n of a manifold discussed in Part A) normalized by the condition

$$n \lrcorner d\sigma = \mathcal{L}_n \sigma = 1, \quad [n] = [\sigma]^{-1} = t^{-1}. \quad (\text{B.1.16})$$

Physically, the folia h_σ of constant σ represent a simple model of a “3-space,” while the function σ serves as a “time” variable. The vector field n is usually interpreted as a congruence of observer worldlines. In Sec. E.4.1, this rather formal mathematical construction becomes a full-fledged physical tool when the metric is introduced.

Now we are in a position to formulate our first axiom. We require the existence of a *twisted charge-current density 3-form* J with the absolute dimension of charge q , i.e., $[J] = q \stackrel{\text{SI}}{=} \text{C}$ which, if integrated over an arbitrary *closed* 3-dimensional submanifold $C_3 \subset X_4$, obeys

$$\oint_{C_3} J = 0, \quad \partial C_3 = 0 \quad (\text{first axiom}). \quad (\text{B.1.17})$$

We recall, a manifold is closed if it is compact and has no boundary. In particular, the 3-dimensional boundary $C_3 = \partial\Omega_4$ of an arbitrary 4-dimensional region Ω_4 is a closed manifold. However, in general, not every closed 3-manifold is a 3-boundary of some spacetime region.

This is the first axiom of electrodynamics. It has a firm phenomenological basis.

B.1.3 Electromagnetic excitation H

Since (B.1.17) holds for an arbitrary 3-dimensional boundary C_3 , we can choose $C_3 = \partial\Omega_4$. Then, by Stokes' theorem, one finds

$$\int_{\Omega_4} dJ = 0. \quad (\text{B.1.18})$$

Since Ω_4 can be chosen arbitrarily, the electric current turns out to be a closed form:

$$dJ = 0. \quad (\text{B.1.19})$$

This is, in four dimensions, the differential version of charge conservation.

Having proved that J is a closed 3-form, we now recognize (B.1.17) as the statement that all periods of the current J vanish. Then, by de Rham's first theorem (see (A.3.42)), the current is also an exact form:

$$J = dH. \quad (\text{B.1.20})$$

This is the inhomogeneous Maxwell equation. The twisted electromagnetic excitation 2-form H has the absolute dimension of charge q , i.e., $[H] = q$.

The excitation H in this setup appears as a potential of the electric current. It is determined only up to a closed 2-form

$$H \longrightarrow H + \psi, \quad d\psi = 0. \quad (\text{B.1.21})$$

In Sec. B.3.4, however, we discuss how a *unique* excitation field H is selected from the multitude of H 's occurring in (B.1.21) by a very weak assumption: If the field strength F , to be defined below via the second axiom in (B.2.8), vanishes, then the excitation H vanishes too. In this context we recognize that the electric and magnetic pieces of H can be *measured* by means of an ideal electric conductor and a superconductor of type II, respectively.

Charge conservation, as formulated in the first axiom, is experimentally verified in all microscopic experiments, in particular in those of high-energy elementary particle physics. Therefore the excitation H represents a *microscopic* field as well. Charge conservation is *not* only valid as a macroscopic average. Similarly, the excitation is not only a quantity that shows up in macrophysics, it rather is a microscopic field, too, analogous to the electromagnetic field strength F to be introduced below.

B.1.4 Time-space decomposition of the inhomogeneous Maxwell equation

Time is nature's way of keeping everything from happening at once.

Anonymous

Given the spacetime foliation, we can decompose any exterior form in “time” and “space” pieces. With respect to the fixed vector field n , normalized by (B.1.16), we define, for a p -form Ψ , the part *longitudinal* to the vector n by

$${}^\perp\Psi := d\sigma \wedge \Psi_\perp, \quad \Psi_\perp := n \lrcorner \Psi, \quad (\text{B.1.22})$$

and the part *transversal* to the vector n by

$$\underline{\Psi} := (1 - {}^\perp)\Psi = n \lrcorner (d\sigma \wedge \Psi), \quad n \lrcorner \underline{\Psi} \equiv 0. \quad (\text{B.1.23})$$

Then, we finally have

$$\Psi = {}^\perp\Psi + \underline{\Psi} = d\sigma \wedge \Psi_\perp + \underline{\Psi}, \quad (\text{B.1.24})$$

with the absolute dimensions

$$[\Psi_\perp] = [\Psi] t^{-1}, \quad [\underline{\Psi}] = [\Psi]. \quad (\text{B.1.25})$$

Thus the projection operators “ ${}^\perp$ ” and “ $\underline{}$ ” form a complete set. Furthermore, every 0-form is transversal, whereas every 4-form is longitudinal, that is, for $p = 0$, we find $\Psi = \underline{\Psi}$ and, for $p = 4$, $\Psi = {}^\perp\Psi$.

In order to apply this decomposition to field theory, the following rules for exterior multiplication can be derived from (B.1.22) and (B.1.23),

$${}^\perp(\Psi \wedge \Phi) = {}^\perp\Psi \wedge \underline{\Phi} + \underline{\Psi} \wedge {}^\perp\Phi = d\sigma \wedge (\Psi_\perp \wedge \underline{\Phi} + (-1)^p \underline{\Psi} \wedge \Phi_\perp), \quad (\text{B.1.26})$$

$$\underline{\Psi} \wedge \Phi = \underline{\Psi} \wedge \underline{\Phi}, \quad (\text{B.1.27})$$

if Ψ is a p -form. We introduce the 3-dimensional exterior derivative as the transversal part of d , that is, according to (B.1.23), as $\underline{d} := n \lrcorner (d\sigma \wedge d)$. Then the exterior derivative of a p -form decomposes as follows:

$${}^\perp(d\Psi) = d\sigma \wedge (\mathcal{L}_n \underline{\Psi} - \underline{d}\Psi_\perp), \quad \underline{d}\Psi = \underline{d} \underline{\Psi}. \quad (\text{B.1.28})$$

According to (A.2.51), the *Lie derivative* of a p -form along a vector field ξ can be written as

$$\mathcal{L}_\xi \Psi := \xi \lrcorner d\Psi + d(\xi \lrcorner \Psi). \quad (\text{B.1.29})$$

Notice that the Lie derivative along the foliation vector field n commutes with the projection operators as well as with the exterior derivative, i.e.,

$$\mathcal{L}_n {}^\perp\Psi = {}^\perp(\mathcal{L}_n \Psi), \quad \mathcal{L}_n \underline{\Psi} = \underline{\mathcal{L}_n \Psi}, \quad \mathcal{L}_n d\Psi = d(n \lrcorner d\Psi) = d(\mathcal{L}_n \Psi). \quad (\text{B.1.30})$$

These rules also imply that

$$\mathcal{L}_n \underline{d}\Psi = \underline{d} \mathcal{L}_n \Psi. \quad (\text{B.1.31})$$

The Lie derivative of the transversal piece $\underline{\Psi}$ of a form with respect to the vector n is abbreviated by a dot,

$$\dot{\underline{\Psi}} := \mathcal{L}_n \underline{\Psi}, \quad (\text{B.1.32})$$

since this turns out to be the time derivative of the corresponding quantity.

In order to make this decomposition formalism more transparent, it is instructive to consider natural (co)frames on a 3-dimensional hypersurface h_σ of constant σ (see Fig. B.1.3). Let x^a be local coordinates on h_σ with $a = 1, 2, 3$. The differentials dx^a are *not* transversal to n in general. Indeed, in terms of the local spacetime coordinates (σ, x^a) , the normalization (B.1.16) allows for a vector field of the structure $n = \partial_\sigma + n^a \partial_a$ with some (in general, nonvanishing) functions n^a . We use (B.1.22) and (B.1.23) and find a nontrivial longitudinal piece ${}^\perp(dx^a) = n^a d\sigma$, whereas the transversal piece reads $\underline{dx}^a = dx^a - n^a d\sigma$. Obviously $d\sigma$ is purely longitudinal. Hence it is convenient to choose, at an arbitrary point of spacetime, a basis of the cotangent space

$$(\vartheta^{\hat{0}}, \vartheta^a) = (d\sigma, \underline{dx}^a), \quad [\vartheta^{\hat{0}}] = t, \quad [\vartheta^a] = \ell, \quad (\text{B.1.33})$$

which is compatible with the foliation given. The coframe (B.1.33) manifestly spans the longitudinal and transversal subspaces of the cotangent space. The corresponding basis of the tangent space reads

$$(e_{\hat{0}}, e_a) = (n, \partial_a), \quad [e_{\hat{0}}] = t^{-1}, \quad [e_a] = \ell^{-1}. \quad (\text{B.1.34})$$

This coframe and this frame are anholonomic, in general.

The general decomposition scheme can be applied to the inhomogeneous Maxwell equation (B.1.20). First, we decompose its left-hand side. Then the current reads

$$J = {}^\perp J + \underline{J} = -j \wedge d\sigma + \rho =: (j, \rho), \quad (\text{B.1.35})$$

with

$$j := -J_\perp \quad \text{and} \quad \rho := \underline{J}. \quad (\text{B.1.36})$$

The minus sign is chosen in conformity with (B.1.14). Since $[J] = q$, we have for the absolute and the relative dimensions, respectively,

$$\begin{aligned} [j] &= q t^{-1} \stackrel{\text{SI}}{=} \text{C s}^{-1} = \text{A}, & [\rho] &= q \stackrel{\text{SI}}{=} \text{C}, \\ [j_{\hat{a}\hat{b}}] &= [j] \ell^{-2} \stackrel{\text{SI}}{=} \text{A m}^{-2}, & [\rho_{\hat{a}\hat{b}\hat{c}}] &= [\rho] \ell^{-3} \stackrel{\text{SI}}{=} \text{C m}^{-3}. \end{aligned} \quad (\text{B.1.37})$$

In three dimensions, we recover the twisted current density 2-form j (see (B.1.9)) and the twisted charge density 3-form ρ (see (B.1.3)). Now charge conservation (B.1.19) can easily be decomposed, too. We substitute (B.1.36)

into (B.1.28) for $\Psi = J$. Then, with the abbreviation (B.1.32), we recover (B.1.12):

$$\dot{\rho} + \underline{d}j = 0. \quad (\text{B.1.38})$$

This, at the same time, gives an exact meaning to the time derivative, which we treated somewhat sloppily in Sec. B.1.1. Note that $\underline{d}\rho = 0$, as a 4-form in three dimensions, represents an identity with no additional information.

Before we turn to the right-hand side of (B.1.20), we decompose the excitations according to⁵

$$H = {}^\perp H + \underline{H} = -\mathcal{H} \wedge d\sigma + \mathcal{D} =: (\mathcal{H}, \mathcal{D}) \quad (\text{B.1.39})$$

with the twisted magnetic excitation 1-form $\mathcal{H}$ and the twisted electric excitation 2-form $\mathcal{D}$:

$$\mathcal{H} := H_\perp \quad \text{and} \quad \mathcal{D} := \underline{H}. \quad (\text{B.1.40})$$

The signs in these two definitions are discussed in section B.4.2. The absolute and relative dimensions of $\mathcal{H}$ and $\mathcal{D}$ are, respectively,

$$\begin{aligned} [\mathcal{H}] &= qt^{-1} \stackrel{\text{SI}}{=} \text{A}, & [\mathcal{D}] &= q \stackrel{\text{SI}}{=} \text{C}, \\ [\mathcal{H}_{\hat{a}}] &= [\mathcal{H}] \ell^{-1} \stackrel{\text{SI}}{=} \text{A m}^{-1}, & [\mathcal{D}_{\hat{a}\hat{b}}] &= [\mathcal{D}] \ell^{-2} \stackrel{\text{SI}}{=} \text{C m}^{-2}. \end{aligned} \quad (\text{B.1.41})$$

Everything is now prepared: The longitudinal part of (B.1.20) reads

$${}^\perp J = d\sigma \wedge J_\perp = {}^\perp(dH) = d\sigma \wedge (\mathcal{L}_n \underline{H} - \underline{d}H_\perp) \quad (\text{B.1.42})$$

and the transversal part

$$\underline{J} = \underline{d}H = \underline{d} \underline{H}. \quad (\text{B.1.43})$$

By means of (B.1.35), (B.1.36), and (B.1.39), the last two equations can be rewritten as

$$\underline{d}\mathcal{H} - \dot{\mathcal{D}} = j \quad (\text{B.1.44})$$

and

$$\underline{d}\mathcal{D} = \rho, \quad (\text{B.1.45})$$

respectively. We find it remarkable that all we need to recover the inhomogeneous set of the Maxwell equations (B.1.44), (B.1.45) (see also their vector versions in (I.6) of the introduction) is electric charge conservation in the form (B.1.17) for arbitrary periods — and nothing more. Incidentally, the boundary conditions for $\mathcal{H}$ and $\mathcal{D}$, if required, can also be derived from (B.1.17). Because of the dot, formula (B.1.44) represents an evolution equation, whereas (B.1.45) constitutes a constraint on the initial distributions of $\mathcal{D}$ and ρ .

⁵The historical name of $\mathcal{H}$ is “magnetic field” and that of $\mathcal{D}$ “electric displacement.” The ISO names (International Organization for Standardization) are “magnetic field” for $\mathcal{H}$ and “electric flux density” for $\mathcal{D}$. As we can see, the ISO names are in conflict with a reasonable 4D interpretation; that is, the 4D quantity H remains without a name. The ISO names are based on the difference between a 1-form, here $\mathcal{H}$ (field), and a 2-form, here $\mathcal{D}$ (flux density).

B.2

Lorentz force density

B.2.1 Electromagnetic field strength F

By now we have exhausted the information contained in the axiom of charge conservation. We have to introduce new concepts in order to complete the fundamental structure of electrodynamics. Whereas the excitation $H = (\mathcal{H}, \mathcal{D})$ is linked to the charge current $J = (j, \rho)$, the electric and magnetic field strengths are usually introduced as forces acting on unit charges q at rest or in motion, respectively.

Let us start with a *heuristic* discussion. We turn first to classical mechanics. Therein the force $\mathcal{F}$ has to be a covector since this is the way it is defined in Lagrangian mechanics, namely $\mathcal{F}_a \sim \partial L / \partial x^a$, with the Lagrangian L . In the purely electric case, the force $\mathcal{F}$ acting on a test charge q at rest in spatial components reads:

$$\mathcal{F}_a \sim q E_a. \quad (\text{B.2.1})$$

Thereby we can *define* the electric field strength E in 3-dimensional space. The electric field strength E has three independent components exactly as the electric excitation $\mathcal{D}$.

In order to link up mechanics with the rudimentary electrodynamics of the first axiom, we consider a delta-function-like test charge current $J = (j, \rho)$ centered around some point with coordinates x^i . Generalizing (B.2.1), the simplest 4-dimensional ansatz for defining the electromagnetic field strength reads:

$$\text{force} \sim \text{field strength} \times \text{charge current}. \quad (\text{B.2.2})$$

Also in four dimensions, the *force* $\sim \partial L / \partial x^\alpha$ is represented by the components f_α of a *covector*. Accordingly, the ansatz (B.2.2) can be made more precise:

$$f_\alpha = F_\alpha \wedge J. \quad (\text{B.2.3})$$

The components f_α represent a twisted form since they change sign under spatial reflections. The Lagrangian 4-form has the absolute dimension of an action, h . Correspondingly the force, which is understood as the (variational) derivative of the action with respect to the spacetime coordinates, has the dimension

$$[f_\alpha] = h [\vartheta^\alpha]^{-1} = (h t^{-1}, h \ell^{-1}) = (\text{energy}, \text{momentum}). \quad (\text{B.2.4})$$

A force cannot possess more than four independent components. Since the current J in (B.2.3) is a twisted 3-form, the only possibility seems to be that f_α is a covector-valued 4-form (recall the general definition given in Sec. A.1.3, example 3). In other words, the covector index transforms under the 4-dimensional linear group as the frame does, namely $f_{\alpha'} = L_{\alpha'}^\alpha f_\alpha$ (see (A.1.5)). Then f_α has four components, indeed, and assigns to any 4-volume element the components of a covector:

$$f_\alpha = \frac{1}{4!} f_{ijkl\alpha} dx^i \wedge dx^j \wedge dx^k \wedge dx^l, \quad f_{ijkl\alpha} = f_{[ijkl]\alpha}. \quad (\text{B.2.5})$$

The anholonomic components of f_α , namely $f_{\mu\nu\rho\sigma\alpha}$, determine the relative dimension of f_α . Since $f_{\mu\nu\rho\sigma\alpha} = f_{[\mu\nu\rho\sigma]\alpha}$, one of the indices μ, ν, ρ, σ has to be a time index and the remaining three space indices. Therefore, we find (W = watt, N = newton),

$$\begin{aligned} [f_{\mu\nu\rho\sigma\alpha}] &= h [\vartheta^\alpha]^{-1} t^{-1} \ell^{-3} = (h t^{-2} \ell^{-3}, h t^{-1} \ell^{-4}) \\ &= (\text{power density}, \text{force density}) \\ &\stackrel{\text{SI}}{=} (\text{W m}^{-3}, \text{N m}^{-3}). \end{aligned} \quad (\text{B.2.6})$$

Thus we have identified the f_α in (B.2.3) indeed as 4-force density, the spatial components of which carry the relative dimension of $f \ell^{-3}$. This shows once more that the assignment of the dimensions in (B.2.4) was correct. As a consequence, the field F_α in (B.2.3) turns out to be an untwisted covector-valued 1-form $F_\alpha = F_{\alpha\beta} \vartheta^\beta$ with the coframe ϑ^β . The covector index of F_α transforms as $F_{\alpha'} = L_{\alpha'}^\alpha F_\alpha$, whereas, as a 1-form, we have $F_\alpha = F_{\alpha i} dx^i = F_{\alpha\beta} \vartheta^\beta$.

As such, $F_{\alpha\beta}$ would have 16 independent components in general. However, we know already that the electric field strength E has 3 independent components. If we expect the analogous to be true for the magnetic field strength B , then $F_{\alpha\beta}$ should carry 6 independent components at most. Then, it is near at hand to assume that $F_{\alpha\beta}$ is antisymmetric: $F_{\alpha\beta} = -F_{\beta\alpha}$. Accordingly, the electromagnetic field strength turns out to be a 2-form $F = \frac{1}{2} F_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta$ and $F_\alpha = e_\alpha \lrcorner F = F_{\alpha\beta} \vartheta^\beta$ with the frame e_α . For the absolute dimension of F this yields, if we use (B.2.3), (B.2.4), and $[J] = q$,

$$\begin{aligned} [F] &= [F_\alpha] [e_\beta]^{-1} = [f_\alpha] [J]^{-1} [e_\beta]^{-1} = h q^{-1} [\vartheta^\alpha]^{-1} [e_\beta]^{-1} \\ &= h q^{-1} \stackrel{\text{SI}}{=} \text{J s C}^{-1} = \text{V s} =: \text{Wb}. \end{aligned} \quad (\text{B.2.7})$$

In SI, we have the abbreviations J = joule (not to be mixed up with the current 3-form J), V = volt, and Wb = weber.

Finally, the force equation (B.2.3) for a test charge current reads $f_\alpha = (e_\alpha \lrcorner F) \wedge J$. Accordingly, we defined the field strength F as the incorporation of the possible forces acting in an electromagnetic field on test charges, thereby relating the mechanical notion of a force to the electromagnetic state around electric charges and currents. In the first axiom, we consider the *active* role of charge that creates the excitation field; here we study its *passive* role, namely which forces act on it in an electromagnetic environment. This concludes our heuristic considerations and we now turn to our new axiom.

B.2.2 Second axiom relating mechanics and electrodynamics

We have then

$$f_\alpha = (e_\alpha \lrcorner F) \wedge J \quad (\text{second axiom}). \quad (\text{B.2.8})$$

The *untwisted electromagnetic field strength 2-form* F carries the absolute dimension of a magnetic flux: $[F] = h q^{-1} \stackrel{\text{SI}}{=} \text{Wb}$. This equation for the Lorentz force density yields an *operational* definition of the electromagnetic field strength F and represents our second axiom of electrodynamics. Observe that (B.2.8), like (B.1.17), is an equation that is free from the metric and the connection; it is defined on any 4-dimensional differentiable manifold. Since $F \wedge J$ is a 5-form, (B.2.8) can alternatively be written as $f_\alpha = -F \wedge (e_\alpha \lrcorner J)$.

A decomposition of the 2-form F into “time” and “space” pieces according to

$$F = {}^\perp F + \underline{F} = E \wedge d\sigma + B =: (E, B), \quad (\text{B.2.9})$$

yields, in three dimensions, the untwisted electric field strength 1-form E and the untwisted magnetic field strength¹ 2-form B ,

$$E := -F_\perp \quad \text{and} \quad B := \underline{F}. \quad (\text{B.2.10})$$

The choices of the signs in these definitions are related to the Lenz rule, *inter alia*. They are discussed in section B.4.2. The absolute and relative dimensions

¹The historical names are “electric field” for E and “magnetic induction” for B . The ISO names “electric field” and “magnetic flux density,” respectively, again don’t merge to a 4D name. Note also that the ISO names don’t recognize the difference between a twisted and an untwisted form. The 1-forms $\mathcal{H}$ and E are both called “field” and $\mathcal{D}$ and B “flux density,” as if they were quantities of the same type. As a matter of fact, the four quantities $\mathcal{H}$, $\mathcal{D}$, E , B , constituting the electromagnetic field, obey four distinctly different transformation laws under diffeomorphisms (coordinate transformations).

of E and B are, respectively,

$$\begin{aligned} [E] &= hq^{-1}t^{-1} \stackrel{\text{SI}}{=} \text{V}, & [B] &= hq^{-1} \stackrel{\text{SI}}{=} \text{Wb}, \\ [E_{\hat{a}}] &= [E] \ell^{-1} \stackrel{\text{SI}}{=} \text{V m}^{-1}, & [B_{\hat{a}\hat{b}}] &= [B] \ell^{-2} \stackrel{\text{SI}}{=} \text{Wb m}^{-2} =: \text{T}, \end{aligned} \quad (\text{B.2.11})$$

where T abbreviates tesla. We can decompose the 1-form E and the 2-form B as follows: $E = E_a dx^a$ and $B = \frac{1}{2} B_{ab} dx^a \wedge dx^b = \mathcal{B}^c \hat{e}_c$. For the 2-form $\hat{e}_c$ in three dimensions, one should compare with (A.1.65). The components E_a and $\mathcal{B}^a = \frac{1}{2} \epsilon^{abc} B_{bc}$ correspond to the conventional Cartesian components of the electric and the magnetic field vectors $\vec{E}$ and $\vec{B}$, respectively.

Clearly then, the *electric line tension* (electromotive force or voltage) $\int_{\Omega_1} E$ and the *magnetic flux* $\int_{\Omega_2} B$ must play a decisive role in Maxwell's theory. Hence, as building blocks for laws governing the electric and magnetic field strengths, we have only the electric line tension and the magnetic flux at our disposal.

The Lorentz force density f_α , as a 4-form, that is, as a form of maximal rank ("top dimensional form"), is purely longitudinal with respect to the normal vector n . Thus only its longitudinal piece $(f_\alpha)_\perp$, a covector-valued 3-form, survives. It turns out to be

$$\begin{aligned} k_\alpha &:= (f_\alpha)_\perp = n \lrcorner [(e_\alpha \lrcorner F) \wedge J] = (e_\alpha \lrcorner E) \wedge J + (e_\alpha \lrcorner F) \wedge j \\ &= \rho \wedge (e_\alpha \lrcorner E) + j \wedge (e_\alpha \lrcorner B) - j \wedge E \wedge (e_\alpha \lrcorner d\sigma). \end{aligned} \quad (\text{B.2.12})$$

If we now display its time and space components, we find

$$k_{\hat{0}} = -j \wedge E, \quad [k_{\hat{0}}] = ht^{-2} \stackrel{\text{SI}}{=} \text{W}, \quad (\text{B.2.13})$$

$$k_a = \rho \wedge (e_a \lrcorner E) + j \wedge (e_a \lrcorner B), \quad [k_a] = h\ell^{-1}t^{-1} \stackrel{\text{SI}}{=} \text{N}. \quad (\text{B.2.14})$$

The zero in $k_{\hat{0}}$ carries a circumflex since it is an anholonomic index. The relative dimensions pick up an additional factor of $\ell^{-3} \stackrel{\text{SI}}{=} \text{m}^{-3}$. The time component $k_{\hat{0}}$ represents the *electric power density*, the space components k_a the 3-dimensional Lorentz force density.

Let us come back to the Lorentz force density f_α . We consider now a *point particle* with net electric charge q and the velocity vector $u = u^\alpha e_\alpha$. Its electric current J is a 3-form. Hence we define the flow 3-form of the particle as

$$U := u \lrcorner \hat{e} = u^\alpha e_\alpha \lrcorner \hat{e} = u^\alpha \hat{e}_\alpha \quad \text{or} \quad u = \diamond (\vartheta^\alpha \wedge U) e_\alpha, \quad (\text{B.2.15})$$

with the volume 4-form $\hat{e}$ and the 3-form $\hat{e}_\alpha$ of (A.1.74). Clearly, the flow 3-form U has the same number of independent components as the velocity vector u . Then the electric current of the point particle reads

$$J = qU = qu \lrcorner \hat{e}. \quad (\text{B.2.16})$$

We substitute this into the second axiom (B.2.8),

$$\begin{aligned} f_\alpha &= q(e_\alpha \lrcorner F) \wedge u \lrcorner \hat{e} \\ &= -q u \lrcorner [(e_\alpha \lrcorner F) \wedge \hat{e}] + q(u \lrcorner e_\alpha \lrcorner F) \hat{e}. \end{aligned} \quad (\text{B.2.17})$$

Since $\hat{e}$ is a 4-form, the expression in the square bracket is a 5-form and vanishes. Then the antisymmetry of the interior product yields:

$$f_\alpha = (-q e_\alpha \lrcorner u \lrcorner F) \hat{e} =: \mathcal{F}_\alpha \hat{e}. \quad (\text{B.2.18})$$

Consequently, for the components $\mathcal{F}_\alpha$ of the 1-form $\mathcal{F}$, we have

$$\mathcal{F}_\alpha = \diamond f_\alpha = -q u^\beta e_\alpha \lrcorner e_\beta \lrcorner F = -q u^\beta F_{\beta\alpha} = q F_{\alpha\beta} u^\beta, \quad (\text{B.2.19})$$

whereas the 1-form itself reads

$$\mathcal{F} = \mathcal{F}_\alpha \vartheta^\alpha = -q u \lrcorner F, \quad \text{with} \quad u \lrcorner \mathcal{F} = 0. \quad (\text{B.2.20})$$

This is the point particle version of the Lorentz force, which is in accord with our starting point in (B.2.1). In contrast, we chose as a second axiom (B.2.8) the field-theoretical version of the Lorentz force, namely its density, since only in that version does there appear the current 3-form J of the first axiom.

If there is an electromagnetic field configuration such that the Lorentz force density vanishes ($f_\alpha = 0$), we call it a *force-free electromagnetic* field:

$$(e_\alpha \lrcorner F) \wedge dH = 0. \quad (\text{B.2.21})$$

Here we have substituted the inhomogeneous Maxwell equation.

In plasma physics such configurations play a decisive role if restricted purely to the magnetic field. We (1 + 3)-decompose (B.2.21). A look at (B.2.13) and (B.2.14) shows that for the magnetic field only the space components (B.2.14) of the Lorentz force density matter:

$$k_a = \underline{d}\mathcal{D} \wedge (e_a \lrcorner E) - \dot{\mathcal{D}} \wedge (e_a \lrcorner B) + \underline{d}\mathcal{H} \wedge (e_a \lrcorner B). \quad (\text{B.2.22})$$

If the electric excitation and its time derivative vanish ($\mathcal{D} = 0, \dot{\mathcal{D}} = 0$) — clearly a frame dependent, i.e., noncovariant statement — then we find for the *force-free magnetic* field²

$$(e_a \lrcorner B) \wedge \underline{d}\mathcal{H} = 0 \quad (\text{B.2.23})$$

or, in components,

$$\mathcal{B}^b \partial_{[a} \mathcal{H}_{b]} = 0. \quad (\text{B.2.24})$$

²See Lundquist [26] and Lüst & Schlüter [27]. However, they as well as later authors put $B = \mu_0 {}^* \mathcal{H}$ right away and start with $\vec{\mathcal{H}} \times \nabla \times \mathcal{H} = 0$.

B.2.3 $\otimes$ The first three invariants of the electromagnetic field

The first axiom supplied us with the fields J (twisted 3-form) and H (twisted 2-form) and the second axiom, additionally, with F (untwisted 2-form). Algebraically we can construct therefrom the so-called first invariant of the electromagnetic field,

$$I_1 := F \wedge H, \quad [I_1] = h. \quad (\text{B.2.25})$$

It is a twisted 4-form or, equivalently, a scalar density of weight +1 with one independent component. Clearly, I_1 could qualify as a Lagrange 4-form: It is a twisted form and it has the appropriate dimension.

Furthermore, a second invariant can be assembled,

$$I_2 := F \wedge F, \quad [I_2] = (h/q)^2, \quad (\text{B.2.26})$$

an untwisted 4-form with the somewhat strange dimension of the square of a magnetic flux, and a third one,

$$I_3 := H \wedge H, \quad [I_3] = q^2, \quad (\text{B.2.27})$$

equally an untwisted form.

In order to get some insight into the meaning of these 4-forms, we substitute the $(1 + 3)$ -decompositions (B.1.39) and (B.2.9):

$$I_1 = F \wedge H = d\sigma \wedge (B \wedge \mathcal{H} - E \wedge \mathcal{D}), \quad (\text{B.2.28})$$

$$I_2 = F \wedge F = -2 d\sigma \wedge B \wedge E, \quad (\text{B.2.29})$$

$$I_3 = H \wedge H = 2 d\sigma \wedge \mathcal{H} \wedge \mathcal{D}. \quad (\text{B.2.30})$$

If for an electromagnetic field configuration the first invariant vanishes, then we find,

$$I_1 = 0 \quad \text{or} \quad \frac{1}{2}B \wedge \mathcal{H} = \frac{1}{2}E \wedge \mathcal{D}. \quad (\text{B.2.31})$$

As we will see in (B.5.61), this means that the magnetic energy density equals the electric energy density. Similarly, for the second invariant, we have

$$I_2 = 0 \quad \text{or} \quad B \wedge E = 0. \quad (\text{B.2.32})$$

Then the electric field strength 1-form can be called “parallel” to the magnetic field strength 2-form. Analogously, for the vanishing third invariant,

$$I_3 = 0 \quad \text{or} \quad \mathcal{H} \wedge \mathcal{D} = 0, \quad (\text{B.2.33})$$

the excitations are “parallel” to each other. It should be understood that the characterizations $I_1 = 0$ and so on are (diffeomorphism and frame) invariant statements about electromagnetic field configurations.

This is as far as we can go by algebraic manipulations. If we differentiate, we can construct dJ , dH , dF . The first two expressions are known, $dJ = 0$ and $dH = J$, whereas the last one, dF , is left open so far. We turn to it in the next chapter.

Let us collect our results in a table:

Table B.2.1: Invariants of the electromagnetic field in 4D.

Invariant	dimension	un-/twisted	name
$I_1 = F \wedge H$	h	twisted	$\sim$ Lagrangian
$I_2 = F \wedge F$	$(h/q)^2$	untwisted	Chern 4-form
$I_3 = H \wedge H$	q^2	untwisted	...
$I_4 = A \wedge J$	h	twisted	coupling term in Lagr.

The fourth invariant I_4 and the names are explained in Sec. B.3.3. The I_k 's are 4-forms, respectively, with one independent component each. We call them invariants. With the diamond operator $\diamond$, the dual with respect to the Levi-Civita epsilon (see (A.1.78) at the end of Sec. A.1.9), we can attach to each 4-form I_k a (metric-free) scalar *density* $\diamond I_k$.

B.3

Magnetic flux conservation

B.3.1 Third axiom

The spacetime manifold, which underlies our consideration and which has been defined at the beginning of Sec. B.1.2, is equipped with the property of an orientation. Then we can integrate the untwisted 2-form F in four dimensions over a 2-dimensional surface. Since F is a 2-form, the simplest invariant statement that comes to mind reads

$$\oint_{C_2} F = 0, \quad \partial C_2 = 0 \quad (\text{third axiom}), \quad (\text{B.3.1})$$

for any closed 2-dimensional submanifold $C_2 \subset X_4$. Indeed, this is the axiom we are looking for.

It is straightforward to find supporting evidence for (B.3.1) by using the decomposition (B.2.9). Faraday's induction law results from (B.3.1) if one chooses the 2-dimensional surface as $C'_2 = \partial\Omega'_3$ with $\Omega'_3 = [\sigma_0, \sigma] \times \Omega'_2$, where Ω'_2 is represented in Fig. B.3.1 by a line in h_{σ_0} ; i.e., it is transversal to the vector field n :

$$\oint_{\partial\Omega'_2} E + \frac{d}{d\sigma} \int_{\Omega'_2} B = 0. \quad (\text{B.3.2})$$

What is usually called the law of the absence of magnetic charge we find by again choosing the 2-dimensional surface C_2 in (B.3.1) as the boundary of a

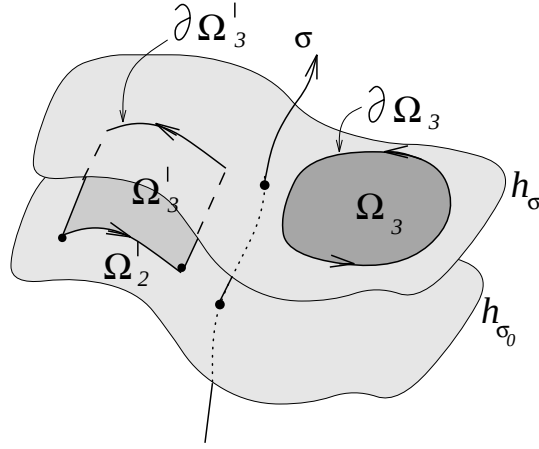


Figure B.3.1: Different 2-dimensional periods of the magnetic flux integral. The 3-dimensional integration region Ω'_3 lies transversal to the hypersurface h_σ whereas Ω_3 lies in it.

3-dimensional submanifold Ω_3 that lies in one of the folia h_σ (see Fig. B.3.1):

$$\oint_{\partial\Omega_3 \subset h_\sigma} B = 0. \quad (\text{B.3.3})$$

The proofs are analogous to those given in (B.1.13). They are given below on a differential level.

Magnetic flux $\int_{\Omega_2} B$ and its conservation are of central importance to electrodynamics. At low temperatures, certain materials can become superconducting; i.e., they lose their electrical resistance. At the same time, if they are exposed to an external (sufficiently weak) magnetic field, the magnetic field is expelled from their interior except for a thin layer at their surface (Meissner–Ochsenfeld effect). In the case of a superconductor of type II (niobium, for example), provided the external field is higher than a certain critical value, *quantized* magnetic flux lines carrying a flux quantum of¹ $\Phi_0 := h/(2e) \overset{\text{SI}}{\approx} 2.068 \times 10^{-15} \text{ Wb}$ can penetrate the surface of the superconductor and build up a triangular lattice, an Abrikosov lattice. A cross section of such a flux-line lattice is depicted in Fig. B.3.2; a schematic view is provided in Fig. B.3.3. In high-temperature superconductors half-integer flux quanta have also been observed.² What is important for us is that, at least under certain circumstances, single quantized flux lines can be *counted* and that they behave like a *conserved* quantity; i.e., they migrate but are not spontaneously created nor destroyed. The counting

¹Here h is Planck's constant and e the elementary charge.

²See the discussion of Tsuei and Kirtley [50].

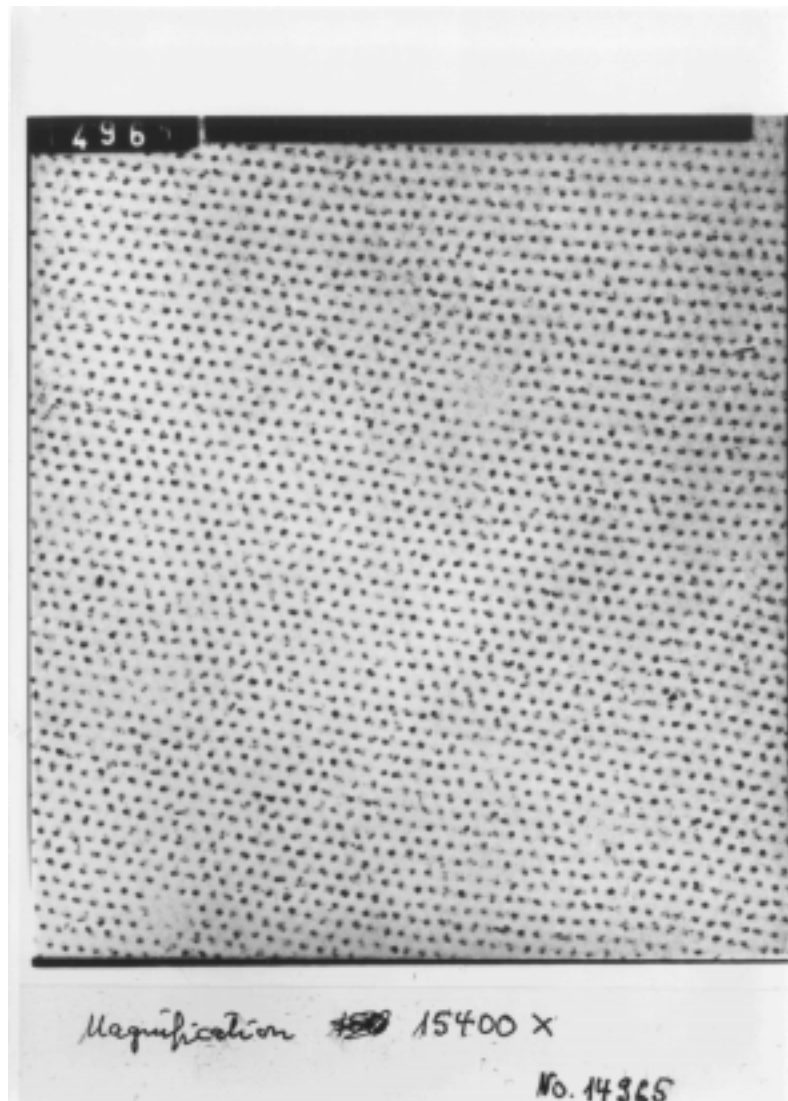


Figure B.3.2: Direct observation of individual flux lines in type II superconductors according to Essmann & Träuble [7, 8]. The image shown here belongs to a small superconducting Niobium disc (diameter 4 mm, thickness 1 mm) which, at a temperature of 1.2 K, was exposed to an external magnetic excitation of $\mathcal{H} = 78$ kA/m. At the surface of the disc the flux lines were decorated by small ferromagnetic particles that were fixed by a replica technique. Eventually the replica was observed by means of an electron microscope. The parameter of the flux-line lattice was 170 nm (courtesy of U. Essmann).

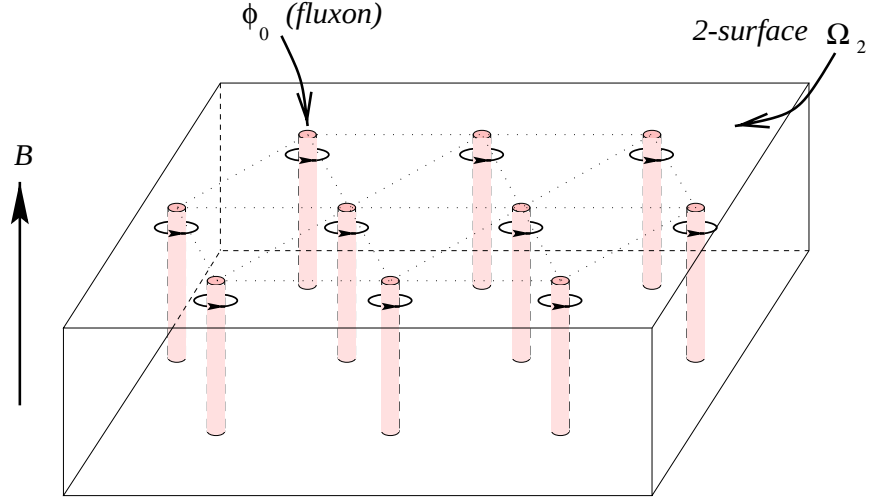


Figure B.3.3: Sketch of an Abrikosov lattice in a type II superconductor in 3-dimensional space.

argument supports the view that magnetic flux is determined in a metric-free way, the migration argument, even if this is somewhat indirect, that the flux is conserved. The computer simulation of the magnetic field of the Earth in Fig. B.3.4 may give one an intuitive feeling that the magnetic lines of force, which can be understood as unquantized magnetic flux lines, are controlled by the induction law and, at the same time, are close to our visual perception of magnetic field configurations.

B.3.2 Electromagnetic potential

In (B.3.1) we specialize to the case when C_2 is a 2-boundary $C_2 = \partial\Omega_3$ of an arbitrary 3-dimensional domain Ω_3 . Then, by Stokes' theorem, we find

$$\int_{\Omega_3} dF = 0. \quad (\text{B.3.4})$$

Since Ω_3 can be chosen arbitrarily, the electromagnetic field strength turns out to be a closed form:

$$dF = 0. \quad (\text{B.3.5})$$

This is the 4-dimensional version of the set of homogeneous Maxwell equations.

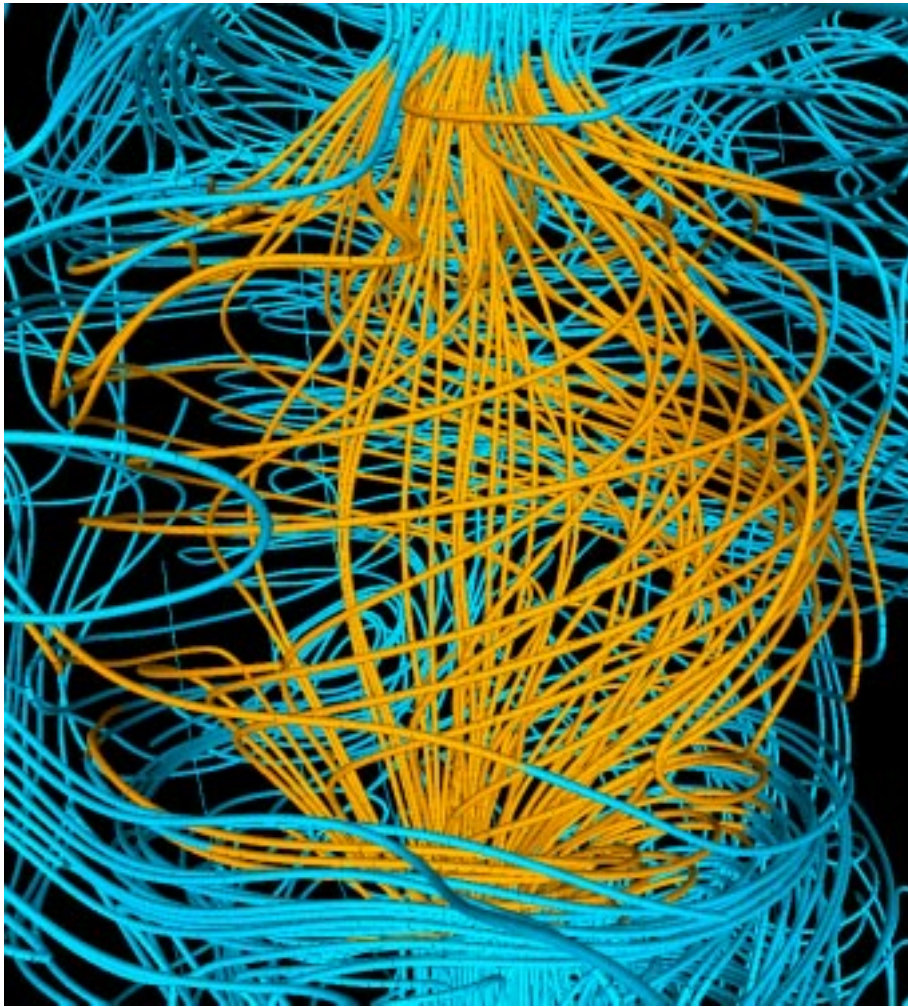


Figure B.3.4: Figure of G. A. Glatzmaier: Snapshot of magnetic lines of force in the core of our computer-simulated Earth. Lines are in gold (blue) where they are inside (outside) the inner core. The axis of rotation is vertical in this image. The field is directed inward at the inner core north pole (top) and outward at the south pole (bottom); the maximum magnetic field strength is about 30 mT (see [14] and [43]). In color you can look at such pictures at the website <http://www.es.ucsc.edu/~glatz/geodynamo.html>.

The axiom (B.3.1) now shows that all periods of F are zero. Consequently the field strength is an exact form

$$F = dA. \quad (\text{B.3.6})$$

Equation (B.3.5) is implied by (B.3.6) because of $dd = 0$. However, the inverse statement that (B.3.5) implies (B.3.6) does not hold in a global manner unless the conditions for the first de Rham theorem are met.

The untwisted electromagnetic potential 1-form A has the dimension of $\hbar q^{-1}$. Decomposed in “time” and “space” pieces, it reads

$$A = -\varphi d\sigma + \mathcal{A}, \quad (\text{B.3.7})$$

with

$$\varphi := -A_{\perp} \quad \text{and} \quad \mathcal{A} := \underline{A}. \quad (\text{B.3.8})$$

Here we recover the familiar 3-dimensional scalar and covector potentials φ and $\mathcal{A}$, respectively. The potential is only determined up to a closed 1-form

$$A \longrightarrow A + \chi, \quad d\chi = 0, \quad (\text{B.3.9})$$

a fact that has far-reaching consequences for the quantization of the electromagnetic field. We decompose (B.3.6) and find straightforwardly

$${}^{\perp}F = {}^{\perp}(dA) \quad \text{or} \quad E = -\underline{d}\varphi - \dot{\mathcal{A}} \quad (\text{B.3.10})$$

and

$$\underline{F} = \underline{dA} \quad \text{or} \quad B = \underline{d}\mathcal{A}. \quad (\text{B.3.11})$$

A decomposition of (B.3.5) by means of (B.1.28) and (B.2.10) yields the homogeneous set of Maxwell's equations,

$${}^{\perp}(dF) = d\sigma \wedge (\mathcal{L}_n \underline{F} - \underline{dF}_{\perp}) = 0, \quad \underline{dF} = 0, \quad (\text{B.3.12})$$

or in 3-dimensional notation,

$$\underline{d}E + \dot{B} = 0 \quad (\text{B.3.13})$$

and

$$\underline{d}B = 0, \quad (\text{B.3.14})$$

respectively. If we integrate these equations over a 2- or 3-dimensional volume and apply Stokes' theorem, we find (B.3.2) and (B.3.3). Again, by analogy with the inhomogeneous equations, (B.3.13) and (B.3.14) represent an evolution equation and a constraint, respectively.

B.3.3 $\otimes$ Abelian Chern–Simons and Kiehn 3-forms

For the electromagnetic theory, we now have the 1-form A as a new building block. This has an immediate consequence for the second invariant, namely

$$I_2 = (dA) \wedge F = d(A \wedge F) = d(A \wedge dA), \quad (\text{B.3.15})$$

since $dF = 0$. In other words, I_2 , the so-called Abelian *Chern* 4-form, is an *exact* form and, accordingly, cannot be used as a nontrivial Lagrangian. We read off from (B.3.15) the untwisted Abelian *Chern–Simons* 3-form

$$C_A := A \wedge F, \quad [C_A] = (h/q)^2, \quad (\text{B.3.16})$$

which has a topological meaning in that it is related to the winding number of the so-called $U(1)$ vector bundle of electromagnetism. Also in all other dimensions with $n \geq 3$, the Abelian Chern–Simons form is represented by a 3-form. If we $(1 + 3)$ -decompose C_A , we find

$$C_A = \mathcal{A} \wedge B + d\sigma \wedge (\varphi B + \mathcal{A} \wedge E), \quad (\text{B.3.17})$$

which includes the so-called magnetic helicity density³ $\mathcal{A} \wedge B$, which is also a 3-form (however in 3D) with one independent component. Obviously,

$$dC_A = F \wedge F. \quad (\text{B.3.18})$$

Consequently, even though C_A is not gauge invariant, its differential dC_A is gauge invariant.

What we just did to the second invariant, we can now implement in an analogous way for the first invariant:

$$I_1 = (dA) \wedge H = d(A \wedge H) + A \wedge J. \quad (\text{B.3.19})$$

We define the *twisted Kiehn* 3-form⁴

$$K := A \wedge H, \quad [K] = h. \quad (\text{B.3.20})$$

It carries the dimension of an action. In n dimensions, it would be an $(n - 1)$ -form — in sharp contrast to the Abelian Chern–Simons 3-form. If we decompose K into $1 + 3$, we find

$$K = \mathcal{A} \wedge \mathcal{D} + d\sigma \wedge (-\varphi \mathcal{D} + \mathcal{A} \wedge \mathcal{H}). \quad (\text{B.3.21})$$

The 2-form $\mathcal{A} \wedge \mathcal{H}$ is the purely magnetic piece of K in 3D. By differentiating K , or by rearranging (B.3.19), we find

$$dK = F \wedge H - A \wedge J. \quad (\text{B.3.22})$$

³The *magnetic helicity* is given by the integral $\int_V \mathcal{A} \wedge B$ (see Moffatt [33], Marsh [28, 29], and Rañada and Trueba[40, 41, 49]).

⁴See Kiehn and Pierce [24, 22, 23].

Even though the Kiehn 3-form changes under a gauge transformation $A \rightarrow A + d\psi$, its exterior derivative dK , provided we are in the free-field region with $J = 0$, is gauge *invariant*, as can be seen in (B.3.22).

From (B.3.22), we can read off the new twisted electromagnetic interaction 4-form

$$I_4 := A \wedge J, \quad [I_4] = h. \quad (\text{B.3.23})$$

It decomposes according to

$$A \wedge J = d\sigma \wedge (\varphi \rho + \mathcal{A} \wedge j). \quad (\text{B.3.24})$$

Because $dJ = 0$, it is gauge invariant up to an exact form:

$$I_4 \longrightarrow I_4 + (d\psi) \wedge J = I_4 + d(\psi \wedge J). \quad (\text{B.3.25})$$

Accordingly I_4 , like I_1 , also qualifies as a piece of an electrodynamic Lagrangian 4-form since it is twisted, of dimension h , and gauge invariant (up to an irrelevant exact form). Note, however, that dK in (B.3.22), as an exact form, cannot feature as a Lagrangian in 4D even though both pieces on the right-hand side of (B.3.22) independently have the correct behavior. In other words, the relative factor between I_1 and I_4 is inappropriate for a Lagrange 4-form.

Let us put our results together. If we start with the 2-form H and differentiate and multiply, then we find the sequence $(H, dH, H \wedge H)$. We cannot go any further since forms with a rank $p > 4$ will vanish identically. Similarly, for the 1-form A , we have $(A, dA, A \wedge dA, dA \wedge dA)$ and, if we mix both, the sequence $(A \wedge H, dA \wedge H, A \wedge dH)$. Note that $A \wedge A \equiv 0$ since A is a 1-form. In this way, we create the new 3-forms K (see (B.3.20)) and C_A (see (B.3.16)):

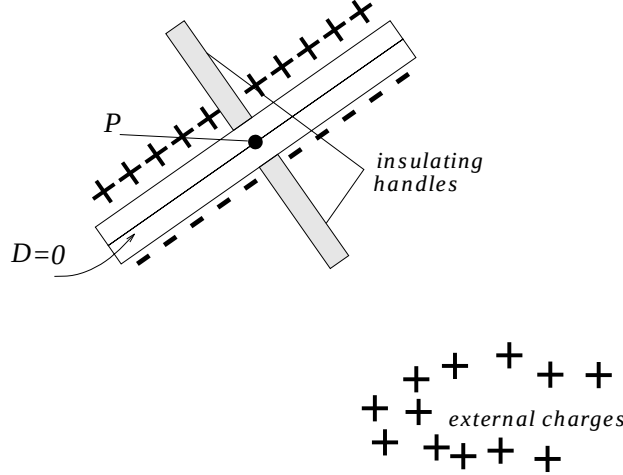
Table B.3.1: Three-forms of the electromagnetic field in 4D.

3-form	dimension	un-/twisted	name
$K = A \wedge H$	h	twisted	Kiehn
$C_A = A \wedge F$	$(h/q)^2$	untwisted	Chern–Simons
J	q	untwisted	electric current

By the same token, the four different invariants I_k of the electromagnetic field arise as 4-forms (see Table B.2.1, p. 123).

B.3.4 Measuring the excitation

The electric excitation $\mathcal{D}$ can be operationally defined using the Gauss law (B.1.45). Put at some point P in free space two small, thin, and electrically conducting (metal) plates of arbitrary shape with insulating handles (“Maxwellian

Figure B.3.5: Measurement of $\mathcal{D}$ at a point P .

double plates”) (see Fig. B.3.5). Suppose we choose local coordinates (x^1, x^2, x^3) in the neighborhood of P such that $P = (0, 0, 0)$. The vectors ∂_a , with $a = 1, 2, 3$, span the tangent space at P . Press the plates together at P and orient them in such a way that their common boundary is given by the equation $x^3 = 0$ (and thus (x^1, x^2) are the local coordinates on each plate’s surface). Separate the plates and measure the net charge $\mathcal{Q}$ and the area S of that plate for which the vector ∂_3 points outwards from its boundary surface. Determine experimentally $\lim_{S \rightarrow 0} \mathcal{Q}/S$ as well as possible. Then

$$\mathcal{D}_{12} = \lim_{S \rightarrow 0} \frac{\mathcal{Q}}{S}. \quad (\text{B.3.26})$$

Technically, one can realize the limiting process by constructing the plates in the form of parallelograms with the sides $a \partial_1$ and $a \partial_2$. Then (B.3.26) is replaced by

$$\mathcal{D}_{12} = \lim_{a \rightarrow 0} \frac{\mathcal{Q}}{a^2}. \quad (\text{B.3.27})$$

Repeat this measurement with the two other possible orientations of the plates (be careful to choose one of the two plates in accordance with the orientation prescription above). Then, similarly, one finds $\mathcal{D}_{23}$ and $\mathcal{D}_{31}$. Thus finally the electric excitation is measured:

$$\mathcal{D} = \frac{1}{2} \mathcal{D}_{ab} dx^a \wedge dx^b = \mathcal{D}_{12} dx^1 \wedge dx^2 + \mathcal{D}_{23} dx^2 \wedge dx^3 + \mathcal{D}_{31} dx^3 \wedge dx^1. \quad (\text{B.3.28})$$

The dimension of $\mathcal{D}$ is that of a *charge*, i.e. $[\mathcal{D}] = q$; for its components we have $[\mathcal{D}_{ab}] = q \ell^{-2}$.

Let us prove the correctness of this prescription. The double plates are assumed to be *ideal conductors*. If an external field is applied to an ideal conductor, a redistribution of the carriers of the electric charges (usually electrons) takes place in its interior by means of flowing electric currents such that the Lorentz force acting on each “electron” finally vanishes. This was found in experiments and later used to define an ideal conductor. In the end, no internal currents are flowing any more and, relative to the conductor, the charge distribution is at rest. Therefore the vanishing 3D Lorentz force density (B.2.14) implies that the electric field E inside the conductor has to vanish as well:

$$E|_{\text{inside}} = 0. \quad (\text{B.3.29})$$

The vanishing of E defines a unique electrodynamical state in the conductor. In particular, a certain electric charge distribution is specified. If we assume that (B.3.29) implies

$$\mathcal{D}|_{\text{inside}} = 0, \quad (\text{B.3.30})$$

then this charge distribution, as we will see in a moment, determines $\mathcal{D}$ uniquely: We can read off from the charge distribution the electric excitation.⁵ That (B.3.29) has (B.3.30) as a consequence is a rudiment of the spacetime relation that links F and H and that makes the excitation field $\mathcal{D}$ unique. For that reason we have postponed this discussion until now, since knowledge of the notion of the field strength E is necessary for it. By (B.3.30), we *selected* from the allowed class (B.1.21) of the excitations that $\mathcal{D}$ that is measured by the double plates.

In a more general setting, let us consider the electromagnetic excitation $\mathcal{D}$ near a 2-dimensional boundary surface $S \subset h_\sigma$ that separates two parts of space filled with two different types of matter. Choose a point $P \in S$. Introduce local coordinates (x^1, x^2, x^3) in the neighborhood of P in such a way that $P = (0, 0, 0)$ while (x^1, x^2) are the coordinates on S (see Fig. B.3.6). Let V be a 3-dimensional domain that is half (denoted V_1) in one medium and half (V_2) in another, with the two halves $V_{1,2}$ spanned by the triples of vectors $(\partial_1, \partial_2, a\partial_3)$ and $(\partial_1, \partial_2, -a\partial_3)$, respectively. For definiteness, we assume that ∂_3 points from medium 1 to medium 2. Now we consider the Gauss law (B.1.45) in this domain. Integrate (B.1.45) over V and take the limit $a \rightarrow 0$. The result is

$$\int_{\Delta S_{12}} \mathcal{D}_{(2)} - \int_{\Delta S_{12}} \mathcal{D}_{(1)} = \lim_{a \rightarrow 0} \int_V \rho, \quad (\text{B.3.31})$$

where $\mathcal{D}_{(1)}$ and $\mathcal{D}_{(2)}$ denote the values of the electric excitations in the medium 1 and 2, respectively, while ΔS_{12} is a piece of S spanned by (∂_1, ∂_2) .

Despite the fact that the volume V clearly goes to zero, the right-hand side of (B.3.31) is nontrivial when there is a *surface charge* exactly on the boundary

⁵The only really appropriate discussion of the definitions of $\mathcal{D}$ and E that we know of is given by Pohl [36].

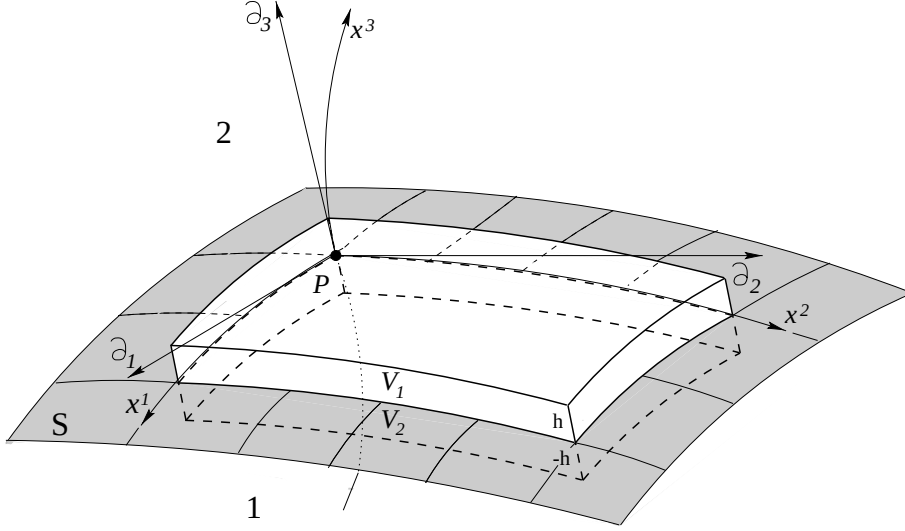


Figure B.3.6: Electric excitation on the boundary between two media.

S . Mathematically, in local coordinates (x^1, x^2, x^3) , this can be described by the δ -function structure of the charge density:

$$\rho = \rho_{123} dx^1 \wedge dx^2 \wedge dx^3 = \delta(x^3) \tilde{\rho}_{12} dx^1 \wedge dx^2 \wedge dx^3. \quad (\text{B.3.32})$$

Substituting (B.3.32) into (B.3.31), we then find

$$\int_{\Delta S_{12}} \mathcal{D}_{(2)} - \int_{\Delta S_{12}} \mathcal{D}_{(1)} = \int_{\Delta S_{12}} \tilde{\rho}, \quad (\text{B.3.33})$$

where $\tilde{\rho} = \tilde{\rho}_{12} dx^1 \wedge dx^2$ is the *2-form of the electric surface charge density*. Since P and ΔS_{12} are arbitrary, we conclude that on the separating surface S between two media, the electric excitation satisfies

$$\mathcal{D}_{(2)}|_S - \mathcal{D}_{(1)}|_S = \tilde{\rho}. \quad (\text{B.3.34})$$

Returning to the measurement process with plates, we have $\mathcal{D}_{(1)} = 0$ inside an ideal conductor, and (B.3.34) justifies the definition of the electric excitation as the charge density on the surface of the plate (B.3.26). Thereby we recognize that the electric excitation $\mathcal{D}$ is, by its very definition, the ability to separate charges on (ideally conducting) double plates.

A similar result holds for the magnetic excitation $\mathcal{H}$. Analogously to the small 3-dimensional domain V , let us consider a 2-dimensional domain Σ that is half (Σ_1) in one medium and half (Σ_2) in another one, with the two halves $\Sigma_{1,2}$

spanned by the vectors $(l, a\partial_3)$ and $(l, -a\partial_3)$, respectively. Here $l = l^1\partial_1 + l^2\partial_2$ is an arbitrary vector tangent to S at P . Integrating the Maxwell equation (B.1.44) over Σ , and taking the limit $a \rightarrow 0$, we find

$$\int_l \mathcal{H}_{(2)} - \int_l \mathcal{H}_{(1)} = \lim_{a \rightarrow 0} \int_{\Sigma} j. \quad (\text{B.3.35})$$

The second term on the left-hand side of (B.1.44) has a zero limit because $\dot{\mathcal{D}}$ is continuous and finite in the domain of a loop that contracts to zero. However, the right-hand side of (B.3.35) produces a nontrivial result when there are surface currents flowing exactly on the boundary surface S . Analogously to (B.3.32), this is described by

$$j = j_{13} dx^1 \wedge dx^3 + j_{23} dx^2 \wedge dx^3 = \delta(x^3) \tilde{j}_a dx^a \wedge dx^3. \quad (\text{B.3.36})$$

Substituting (B.3.36) into (B.3.35), we find

$$\int_l \mathcal{H}_{(2)} - \int_l \mathcal{H}_{(1)} = \int_l \tilde{j}, \quad (\text{B.3.37})$$

and, since l is arbitrary, eventually:

$$\mathcal{H}_{(2)}|_S - \mathcal{H}_{(1)}|_S = \tilde{j}. \quad (\text{B.3.38})$$

Thus we see that on the boundary surface between the two media the magnetic excitation is directly related to the *1-form of the electric surface current density* $\tilde{j}$.

To measure $\mathcal{H}$, following a suggestion by Zirnbauer,⁶ we can use the Meissner effect. Because of this effect, the magnetic field B is driven out of the superconductor. Take a thin superconducting wire (since the wire is pretty cold, perhaps around 10 K, “taking” is not to be understood too literally) and put it at the point P where you want to measure $\mathcal{H}$ (see Fig.B.3.7(a)). Because of the Meissner effect, the magnetic field B and, if we assume analogously to the electric case that $B = 0$ implies $\mathcal{H} = 0$, the excitation $\mathcal{H}$ is expelled from the superconducting region (apart from a thin surface layer of some 10 nm, where $\mathcal{H}$ can penetrate). According to the Oersted–Ampère law, (B.1.44) $\underline{d}\mathcal{H} = j$ (we assume quasi-stationarity in order to be permitted to forget about $\dot{\mathcal{D}}$). The compensation of $\mathcal{H}$ at P can be achieved by surface currents $\tilde{j}$ flowing around the superconducting wire. These induced surface currents $\tilde{j}$ are of such a type that they generate an $\mathcal{H}_{\text{ind}}$ that compensates the $\mathcal{H}$ to be measured: $\mathcal{H}_{\text{ind}} + \mathcal{H} = 0$. We have to change the angular orientation such that we eventually find the

⁶See his lecture notes [55]; compare also Ingarden and Jamiołkowski [17].

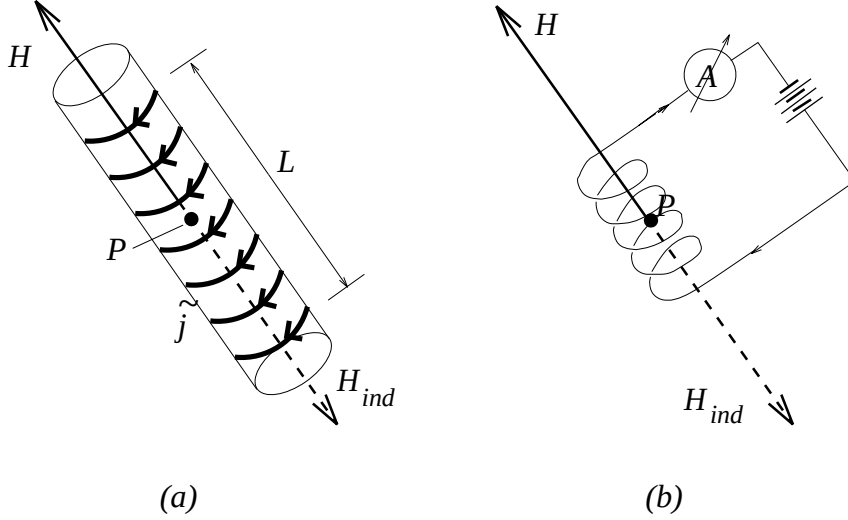


Figure B.3.7: Measurement of $\mathcal{H}$ at a point P . We denote the surface currents by $\tilde{j}$.

maximal current $\tilde{j}_{\max}$. Then, if the x^3 -coordinate axis is chosen tangentially to the “maximal orientation,” we have

$$\mathcal{H}_3 = \lim_{L \rightarrow 0} \frac{\tilde{j}_{\max}}{L}, \quad (\text{B.3.39})$$

where L is the length parallel to the wire axis transverse to which the current has been measured. Accordingly, we find

$$\mathcal{H} = \mathcal{H}_3 dx^3. \quad (\text{B.3.40})$$

Clearly the dimension of $\mathcal{H}$ is that of an *electric current*: $[\mathcal{H}] = q t^{-1} \stackrel{\text{SI}}{=} \text{A}$ and $[\mathcal{H}_{\hat{a}}] = q \ell^{-1} t^{-1} \stackrel{\text{SI}}{=} \text{A/m}$ (“ampere-turn per meter”).

This is admittedly a thought experiment so far. However, the surface currents are experimentally known to exist.⁷ Furthermore, with nanotechnological tools it should be possible to realize this experiment. It is conceivable that the surface currents can be measured by means of the Coulomb drag.⁸

Alternatively, one can take a real small *test coil* at P , orient it suitably, and read off the corresponding maximal current at a galvanometer as soon as the

⁷See the corresponding recent magneto-optical studies of Jooss et al. [21].

⁸The idea is the following (M. Zirnbauer, private communication): If in a first thin film a current flows (this represents the surface of the superconductor), then via the Coulomb interaction, it can drag along with it in a separate second conducting thin film the charge carriers which can then be measured (outside the superconductor). This Coulomb drag, which is known to exist (see [44]), requires, however, the breaking of translational invariance.

effective $\mathcal{H}$ vanishes (see Fig. B.3.7(b)). Multiply this current with the winding number of the coil and find $\tilde{j}_{\max}$. Divide by the length of the test coil and you are back to (B.3.39). Whether $\mathcal{H}$ is really compensated for, you can check with a magnetic needle which, in the field-free region, should be in an indifferent equilibrium state. Note that today one has magnetic “needles” of nanometer size.

In this way, we can build up the excitation $H = \mathcal{D} - \mathcal{H} \wedge d\sigma$ as a measurable electromagnetic quantity in its own right. The excitation H , together with the field strength F , we call the “electromagnetic field.”

B.4

Basic classical electrodynamics summarized, example

B.4.1 Integral version and Maxwell's equations

We are now in a position to summarize the fundamental structure of electrodynamics in a few lines. According to (B.1.17), (B.2.8), and (B.3.1), the three axioms on a connected, Hausdorff, paracompact, and oriented spacetime read, for any C_3 and C_2 with $\partial C_3 = 0$ and $\partial C_2 = 0$:

$$\oint_{C_3} J = 0, \quad f_\alpha = (e_\alpha \lrcorner F) \wedge J, \quad \oint_{C_2} F = 0. \quad (\text{B.4.1})$$

The first axiom governs matter and its conserved electric charge, the second axiom links the notion of that charge and the concept of a mechanical force to an operational definition of the electromagnetic field strength. The third axiom determines the flux of the field strength as sourcefree.

In Part C we learn that a metric of spacetime brings in the temporal and spatial distance concepts and a linear connection the inertial guidance field (and thereby parallel displacement). Since the metric of spacetime represents Einstein's *gravitational* potential (and the linear connection is also related to gravitational properties), the three axioms (B.4.1) of electrodynamics are not contaminated by gravitational properties, in contrast to what happens in the usual textbook approach to electrodynamics. A curved metric or a non-flat linear connection do not affect (B.4.1), since these geometric objects don't enter the axioms. Up to now, we could do without a metric and without a connection. Yet we do have the basic Maxwellian structure already at our disposal. Consequently, the structure of electrodynamics that has emerged so far has nothing

to do with Poincaré or Lorentz covariance. The transformations involved are diffeomorphisms and frame transformations alone.

If one desires to generalize special relativity to general relativity theory or to the Einstein–Cartan theory of gravity (a viable alternative to Einstein’s theory formulated in a non-Riemannian spacetime; see again Part C), then the Maxwellian structure in (B.4.1) is untouched by it. As long as a 4-dimensional connected, Hausdorff, paracompact and oriented differentiable manifold is used as the spacetime, the axioms in (B.4.1) stay covariant and remain the same. In particular, arbitrary frames, holonomic and anholonomic ones, can be used for the evaluation of (B.4.1).

For what reasons are charge conservation (B.4.1)₁ and flux conservation (B.4.1)₃ such “stable” natural laws? Apparently since the elementary “portions” of charge $e/3$ and of flux $\pi\hbar/e$ (with e = elementary charge and $\hbar$ = (reduced) Planck constant) are additive 4-dimensional scalars that are constant in time and space. In other words,¹

$$e = \text{const}, \quad \hbar = \text{const}, \quad e, \hbar \text{ are 4D-scalars}, \quad (\text{B.4.2})$$

guarantees that electric charge and magnetic flux are conserved quantities in classical electrodynamics. Therefore (B.4.2) secures the validity of the first and third axioms. A time dependence of e and/or $\hbar$, for example, would dismantle the Maxwellian framework.

According to (B.4.1), the more specialized differential version of electrodynamics (skipping the boundary conditions) reads as follows:

$$\begin{aligned} dJ &= 0, & f_\alpha &= (e_\alpha \lrcorner F) \wedge J, & dF &= 0, & (\text{B.4.3}) \\ J &= dH, & & & F &= dA. \end{aligned}$$

This is the structure that we defined by means of our axioms so far. But, in fact, we know a bit more: Because of the existence of conductors and superconductors, we can *measure* the excitation H . Thus, even if H emerges as a kind of potential for the electric current, it is more than that: It is measurable. This is in clear contrast to the potential A that is not measurable. Thus we have to take the gauge invariance under the substitution

$$A' = A + \chi, \quad \text{with} \quad d\chi = 0, \quad (\text{B.4.4})$$

very seriously, since measurable quantities must be gauge invariant.

The system (B.4.3) can be straightforwardly translated into the Excalc language: We denote the electromagnetic potential A by `pot1`. The “1” we wrote in order to remember better that the potential is a 1-form. The field strength F is written as `farad2`, i.e., as the Faraday 2-form. The excitation H is named `excit2`, and the left-hand sides of the homogeneous and the inhomogeneous Maxwell equations are called `maxhom3` and `maxinh3`, respectively. Then we need

¹See the corresponding discussion of Peres [35].

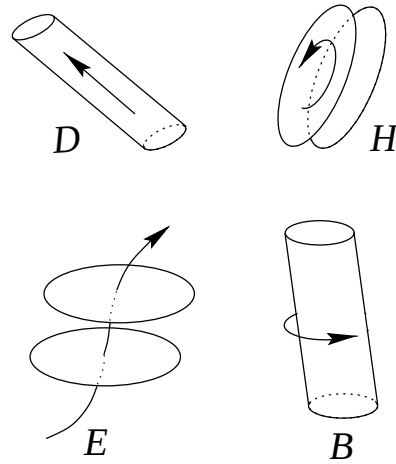


Figure B.4.1: Faraday–Schouten pictograms of the electromagnetic field in 3-dimensional space. The images of 1-forms are represented by two neighboring planes. The nearer the planes, the stronger the 1-form is. The 2-forms are pictured as flux tubes. The thinner the tubes, the stronger the flow. The difference between a twisted and an untwisted form accounts for the two different types of 1- and 2-forms, respectively.

the electric current density `curr3` and the left-hand side of the continuity equation `cont4`. For the first axiom, we have

```
pform cont4=4, curr3=3, maxinh3=3, excit2=2$
```

```
cont4      := d curr3;
maxinh3    := d excit2;
```

and for the second and third axiom,

```
pform force4(a)=4, maxhom3=3, farad2=2, pot1=1$
                                     % has to be preceded by a coframe statement
frame e$
farad2      := d pot1;
force4(-a)  := (e(-a) _|farad2)^curr3;
maxhom3     := d farad2;
```

These program bits and pieces, which look almost trivial, will be integrated into a complete and executable Maxwell sample program after we have learned about the energy-momentum distribution of the electromagnetic field and about its action.

The physical interpretation of equations (B.4.3) can be found via the $(1 + 3)$ -decomposition that we had derived earlier as

$$J = -j \wedge d\sigma + \rho =: (j, \rho), \quad (\text{B.4.5})$$

$$H = -\mathcal{H} \wedge d\sigma + \mathcal{D} =: (\mathcal{H}, \mathcal{D}), \quad (\text{B.4.6})$$

$$F = E \wedge d\sigma + B =: (E, B), \quad (\text{B.4.7})$$

$$A = -\varphi d\sigma + \mathcal{A} =: (\varphi, \mathcal{A}), \quad (\text{B.4.8})$$

(see (B.1.35), (B.1.39), (B.2.9), and (B.3.7), respectively). The signs in (B.4.5) to (B.4.8) will be discussed in the next section. A 3-dimensional visualization of the electromagnetic field is given in Fig. B.4.1.

We first concentrate on equations that contain only measurable quantities, namely the Maxwell equations. For their $(1 + 3)$ -decomposition we found (see (B.1.44), (B.1.45) and (B.3.13), (B.3.14)),

$$dH = J \quad \left\{ \begin{array}{ll} \underline{d}\mathcal{D} = \rho & (1 \text{ constraint eq.}), \\ \dot{\mathcal{D}} = \underline{d}\mathcal{H} - j & (3 \text{ time evol. eqs.}), \end{array} \right. \quad (\text{B.4.9})$$

$$dF = 0 \quad \left\{ \begin{array}{ll} \underline{d}B = 0 & (1 \text{ constraint eq.}), \\ \dot{B} = -\underline{d}E & (3 \text{ time evol. eqs.}). \end{array} \right. \quad (\text{B.4.10})$$

Accordingly, we have $2 \times 3 = 6$ time evolution equations for the $2 \times 6 = 12$ variables $(\mathcal{D}, B, \mathcal{H}, E)$ of the electromagnetic field. Thus the Maxwellian structure in (B.4.3) is underdetermined. We need, in addition, an electromagnetic spacetime relation that expresses the excitation $H = (\mathcal{H}, \mathcal{D})$ in terms of the field strength $F = (E, B)$, i.e., $H = H[F]$. For classical electrodynamics, this functional becomes the Maxwell–Lorentz spacetime relation that we discuss in Part D.

Whereas, on the unquantized level, the Maxwellian structure in (B.4.1) or (B.4.3) is believed to be of universal validity, the spacetime relation is more of an ad hoc nature and amenable to corrections. The “vacuum” can have different spacetime relations depending on whether we take it with or without vacuum polarization. We will come back to this question in Chapters D.6 and E.2.



@@@ Tables (put inside the **front cover of the book**)

Table II. SI units of the electromagnetic field and its source: C = coulomb, A = ampere, Wb = weber, V = volt, T = tesla, m = meter, s = second. The units oersted and gauss are phased out and do not exist any longer in SI. In the second column we find the units related to the absolute and in the third column those related to the relative dimensions.

Field	SI unit of field	SI unit of components of field
ρ	C	C/m^3
j	$\text{A} = \text{C}/\text{s}$	$\text{A}/\text{m}^2 = \text{C}/(\text{s m}^2)$
$\mathcal{D}$	C	C/m^2
$\mathcal{H}$	$\text{A} = \text{C}/\text{s}$	$\text{A}/\text{m} = \text{C}/(\text{s m}) \quad (\rightarrow 4\pi \times 10^{-3} \text{ oersted})$
E	$\text{Wb}/\text{s} = \text{V}$	$\text{V}/\text{m} = \text{Wb}/(\text{s m})$
B	Wb	$\text{Wb}/\text{m}^2 = \text{T} \quad (\rightarrow 10^4 \text{ gauss})$

@@@ Tables (put inside the front cover of the book)

Table I. The electromagnetic field and its source.

Field	name	mathematical object	independent components	related to	reflection	absolute dimension
ρ	electric charge	twisted 3-form	ρ_{123}	volume	$-\rho$	q = electric charge
j	electric current	twisted 2-form	j_{23}, j_{31}, j_{12}	area	$-j$	q/t
$\mathcal{D}$	electric excitation	twisted 2-form	$\mathcal{D}_{23}, \mathcal{D}_{31}, \mathcal{D}_{12}$	area	$-\mathcal{D}$	q
$\mathcal{H}$	magnetic excitation	twisted 1-form	$\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$	line	$-\mathcal{H}$	q/t
E	electric field strength	untwisted 1-form	E_1, E_2, E_3	line	E	Φ/t
B	magnetic field strength	untwisted 2-form	B_{23}, B_{31}, B_{12}	area	B	Φ = magnetic flux

B.4.2 $\otimes$ Lenz and anti-Lenz rule

In generally covariant form, the Maxwell equations read $dH = J$ and $dF = 0$. However, their physical interpretation requires, via (B.4.5)–(B.4.7), an $(1 + 3)$ -decomposition. We discussed the operational interpretation of the different quantities in ordinary 3-space, and now we turn to the signs involved.² Generally, we could write the 4-current as

$$J = i_T j \wedge d\sigma + i_S \rho, \quad (\text{B.4.11})$$

where we introduced the factors i_T and i_S with values from $\{+1, -1\}$. Any factor the absolute value of which differs from 1 is assumed to be absorbed in the corresponding form. Hence (B.4.11) is the most general decomposition relative to a given foliation.

We differentiate (B.4.11) and use $\underline{d}\rho = 0$. Then the first axiom yields

$$dJ = i_T dj \wedge d\sigma + i_S d\rho = d\sigma \wedge (-i_T \underline{d}j + i_S \dot{\rho}) = 0. \quad (\text{B.4.12})$$

We know that ρ is the charge density and j the current density. With the usual conventions on orientation, namely that the outward direction from an enclosed 3D volume is positive etc., both quantities fulfill the continuity equation (B.1.12). Thus, $i_S = -i_T$ and

$$J = i_T (j \wedge d\sigma - \rho). \quad (\text{B.4.13})$$

Let us now turn to the inhomogeneous Maxwell equation. For the excitation we have

$$H = h_T \mathcal{H} \wedge d\sigma + h_S \mathcal{D}, \quad (\text{B.4.14})$$

where we introduced again the sign factors h_T and h_S with values from $\{+1, -1\}$. We differentiate (B.4.14)

$$\begin{aligned} dH &= h_T \underline{d}\mathcal{H} \wedge d\sigma + h_S \underline{d}\mathcal{D} + h_S d\sigma \wedge \dot{\mathcal{D}} \\ &= d\sigma \wedge (h_T \underline{d}\mathcal{H} + h_S \dot{\mathcal{D}}) + h_S \underline{d}\mathcal{D}. \end{aligned} \quad (\text{B.4.15})$$

Accordingly, the inhomogeneous Maxwell equation $dH = J$ (see (B.4.9)) with the source (B.4.13) decomposes as

$$h_T \underline{d}\mathcal{H} + h_S \dot{\mathcal{D}} = i_T j, \quad h_S \underline{d}\mathcal{D} = -i_T \rho. \quad (\text{B.4.16})$$

The conventional definition of the sign of the electric excitation is that the (positive) electric charge ρ is its *source*: $\underline{d}\mathcal{D} = \rho$. Consequently, $h_S = -i_T$ and (B.4.16)₁, the Oersted-Ampère-Maxwell law, reads

$$h_T \underline{d}\mathcal{H} = -h_S (j + \dot{\mathcal{D}}). \quad (\text{B.4.17})$$

²We follow here the paper of Itin et al. [19], see also the references given there.

The contributions of j and $\dot{\mathcal{D}}$ add up correctly to the the Maxwell current $j + \dot{\mathcal{D}}$. In analogy to $\mathcal{D}$, the magnetic excitation $\mathcal{H}$ is defined such as to have $j + \dot{\mathcal{D}}$ as its source:

$$\underline{d}\mathcal{H} = j + \dot{\mathcal{D}}, \quad h_{\mathsf{T}} = -h_{\mathsf{S}} = i_{\mathsf{T}}. \quad (\text{B.4.18})$$

Accordingly,

$$H = i_{\mathsf{T}} (\mathcal{H} \wedge d\sigma - \mathcal{D}). \quad (\text{B.4.19})$$

We have not yet made a convention for the sign of the charge. If we manipulate with a cat's skin and a rod of amber, we can take the conventional choice. This amounts to $i_{\mathsf{T}} = -1$. Then the decompositions (B.4.13) and (B.4.19) coincide with (B.4.5) and (B.4.6), respectively.

Finally, we address the field strength:

$$F = f_{\mathsf{T}} E \wedge d\sigma + f_{\mathsf{S}} B = f_{\mathsf{T}} (E \wedge d\sigma + f_{\Lambda} B), \quad f_{\Lambda} := \frac{f_{\mathsf{S}}}{f_{\mathsf{T}}}. \quad (\text{B.4.20})$$

The Lenz factor f_{Λ} can be either $+1$ or -1 . The overall sign of F is again chosen by a convention in connection with the sign of the Lorentz force relative to a positive charge. Conventionally, $f_{\mathsf{T}} = 1$.

The homogeneous Maxwell equation $dF = 0$ (see (B.4.10)) decomposes according to

$$\underline{d}E + f_{\Lambda} \dot{B} = 0, \quad \underline{d}B = 0. \quad (\text{B.4.21})$$

Thus, the only sign left open is the one in Faraday's induction law (B.4.21)₁. The choice $f_{\Lambda} = +1$ is equivalent to the *Lenz rule*: The induced voltage $\int E$ is directed such that it acts opposite to the $\dot{B}$ that creates it. The (unphysical) choice $f_{\Lambda} = -1$ we call the anti-Lenz rule.

Our choice in (B.4.7) corresponds to the Lenz rule. We will justify this later. Therefore, we have derived all signs in (B.4.5)–(B.4.7) with the exception of that of the Lenz factor f_{Λ} . Since the Lenz rule follows from energy considerations, we have to postpone further discussion until we have an axiom about the electromagnetic energy distribution. At the end of section B.5.3, we will resume our discussion.

B.4.3 $\otimes$ Jump conditions for electromagnetic excitation and field strength

Equations (B.3.34) and (B.3.38) represent the so-called jump (or continuity) conditions for the components of the electromagnetic excitation. In this section we give a more convenient fomulation of these conditions. Namely, let us consider on a 3-dimensional slice h_{σ} of spacetime (see Fig. B.1.3) an arbitrary 2-dimensional surface S , the points of which are defined by the parametric equations

$$x^a = x^a(\xi^1, \xi^2), \quad a = 1, 2, 3. \quad (\text{B.4.22})$$

Here $\xi^A = (\xi^{\dot{1}}, \xi^{\dot{2}})$ are the two parameters specifying the position on S . We denote the corresponding indices $A, B, \dots = \dot{1}, \dot{2}$ by a dot in order to distinguish them from the other indices. We assume that this surface is not moving; i.e., its form and position are the same in every $\sigma = \text{const}$ hypersurface. We introduce the 1-form density ν normal to the surface S ,

$$\begin{aligned} \nu := & \left(\frac{\partial x^2}{\partial \xi^{\dot{1}}} \frac{\partial x^3}{\partial \xi^{\dot{2}}} - \frac{\partial x^2}{\partial \xi^{\dot{2}}} \frac{\partial x^3}{\partial \xi^{\dot{1}}} \right) dx^1 + \left(\frac{\partial x^3}{\partial \xi^{\dot{1}}} \frac{\partial x^1}{\partial \xi^{\dot{2}}} - \frac{\partial x^3}{\partial \xi^{\dot{2}}} \frac{\partial x^1}{\partial \xi^{\dot{1}}} \right) dx^2 \\ & + \left(\frac{\partial x^1}{\partial \xi^{\dot{1}}} \frac{\partial x^2}{\partial \xi^{\dot{2}}} - \frac{\partial x^1}{\partial \xi^{\dot{2}}} \frac{\partial x^2}{\partial \xi^{\dot{1}}} \right) dx^3, \end{aligned} \quad (\text{B.4.23})$$

and the two vectors *tangential* to S ,

$$\tau_A := \frac{\partial x^a}{\partial \xi^A} \partial_a, \quad A = \dot{1}, \dot{2}. \quad (\text{B.4.24})$$

Accordingly, we have the two conditions $\tau_A \lrcorner \nu = 0$. The surface S divides the whole slice h_σ into two halves. We denote them by the subscripts $_{(1)}$ and $_{(2)}$, respectively.

Then, by repeating the limiting process above for the integrals near S , we find, instead of (B.3.34) and (B.3.38), the jump conditions

$$(\mathcal{D}_{(2)} - \mathcal{D}_{(1)} - \tilde{\rho}) \Big|_S \wedge \nu = 0, \quad (\text{B.4.25})$$

$$\tau_A \lrcorner (\mathcal{H}_{(2)} - \mathcal{H}_{(1)} - \tilde{j}) \Big|_S = 0. \quad (\text{B.4.26})$$

Here, $\mathcal{D}_{(1)}$ and $\mathcal{H}_{(1)}$ are the excitation forms in the first half and $\mathcal{D}_{(2)}$ and $\mathcal{H}_{(2)}$ in the second half of the 3D space h_σ .

One can immediately verify that the analysis in Sec. B.3.4 of the operational determination of the electromagnetic excitation was carried out for the special case of a surface S defined by the equations $x^1 = \xi^{\dot{1}}$, $x^2 = \xi^{\dot{2}}$, $x^3 = 0$ (then $\nu = dx^3$ and $\tau_A = \partial_A$, $A = 1, 2$). It should be noted though that the formulas (B.3.34) and (B.3.38) are not less general than (B.4.25) and (B.4.26) since the local coordinates can always be chosen in such a way that (B.4.25), (B.4.26) reduces to (B.3.34), (B.3.38). However, in practical applications, the use of (B.4.25), (B.4.26) turns out to be more convenient.

In an analogous way, one can derive the jump conditions for the components of the electromagnetic field strength. Starting from the homogeneous Maxwell equations (B.3.13) and (B.3.14) and considering their integral form near S , we obtain

$$(B_{(2)} - B_{(1)}) \Big|_S \wedge \nu = 0, \quad (\text{B.4.27})$$

$$\tau_A \lrcorner (E_{(2)} - E_{(1)}) \Big|_S = 0. \quad (\text{B.4.28})$$

As above, $B_{(1)}$ and $E_{(1)}$ are the field strength forms in the first half and $B_{(2)}$ and $E_{(2)}$ in the second half of the 3D space h_σ .

The homogeneous Maxwell equation does not contain charge and current sources. Thus, equations (B.4.27), (B.4.28) describe the continuity of the tangential piece of the magnetic field and of the tangential part of the electric field across the boundary between the two domains of space. In (B.4.25) and (B.4.26), the components of the electromagnetic excitation are not continuous in general, with the surface charge and current densities $\tilde{\rho}$, $\tilde{j}$ defining the corresponding discontinuities.

B.4.4 Arbitrary local noninertial frame: Maxwell's equations in components

In (B.4.9) and (B.4.10), we displayed the Maxwell equations in terms of geometrical objects in a coordinate and frame invariant way. Sometimes it is necessary, however, to introduce locally arbitrary (co)frames of reference that are noninertial, that is, accelerated in general. Then the components of the electromagnetic field with respect to a coframe ϑ^α , the physical components emerge and the Maxwell equations can be expressed in terms of these physical components.

Let the current, the excitation, and the field strength be decomposed according to

$$J = \frac{1}{3!} J_{\alpha\beta\gamma} \vartheta^\alpha \wedge \vartheta^\beta \wedge \vartheta^\gamma, \quad (\text{B.4.29})$$

$$H = \frac{1}{2} H_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta, \quad F = \frac{1}{2} F_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta, \quad (\text{B.4.30})$$

respectively. We substitute these expressions into the Maxwell equations. Then the coframe needs to be differentiated. As a shorthand notation, we introduce the anholonomicity 2-form $C^\alpha := d\vartheta^\alpha = \frac{1}{2} C_{\mu\nu}^\alpha \vartheta^\mu \wedge \vartheta^\nu$ (see (A.2.35)). Then we straightforwardly find:

$$\partial_{[\alpha} H_{\beta\gamma]} - C_{[\alpha\beta}^\delta H_{\gamma]\delta} = \frac{1}{3} J_{\alpha\beta\gamma}, \quad \partial_{[\alpha} F_{\beta\gamma]} - C_{[\alpha\beta}^\delta F_{\gamma]\delta} = 0. \quad (\text{B.4.31})$$

If we use the (metric-free) Levi-Civita tensor density $\epsilon^{\alpha\beta\gamma\delta}$, then the excitation and the current,

$$\tilde{\mathcal{H}}^{\alpha\beta} := \frac{1}{2!} \epsilon^{\alpha\beta\gamma\delta} H_{\gamma\delta}, \quad \tilde{\mathcal{J}}^\alpha := \frac{1}{3!} \epsilon^{\alpha\beta\gamma\delta} J_{\beta\gamma\delta}, \quad (\text{B.4.32})$$

are represented as densities. Accordingly, Maxwell's equations (B.4.3) in components read alternatively

$$\partial_\beta \tilde{\mathcal{H}}^{\alpha\beta} + C_{\beta\gamma}^\alpha \tilde{\mathcal{H}}^{\beta\gamma} = \tilde{\mathcal{J}}^\alpha, \quad \partial_{[\alpha} F_{\beta\gamma]} - C_{[\alpha\beta}^\delta F_{\gamma]\delta} = 0. \quad (\text{B.4.33})$$

The terms with the C 's emerge if Maxwell's equations are referred to an arbitrary local (co)frame. Only if we restrict ourselves to natural (or coordinate) frames is $C = 0$, and Maxwell's equations display their conventional form.³

This representation of electrodynamics can be used in special or in general relativity. If one desires to employ a laboratory frame of reference, then this is the way to do it: The object of anholonomicity in the lab frame has to be calculated. By substituting it into (B.4.31) or (B.4.33), we find the Maxwell equations in terms of the components $F_{\alpha\beta}$ and so on of the electromagnetic field quantities with respect to the lab frame — and these are the quantities one observes in the laboratory. Therefore the $F_{\alpha\beta}$ and so on are called *physical components*. As soon as one starts from (B.4.3), the derivation of the sets (B.4.31) or (B.4.33) is an elementary exercise. Many discussions of the Maxwell equations within special relativity in noninertial frames could be appreciably shortened by using this formalism.

B.4.5 $\otimes$ Electrodynamics in flatland: 2-dimensional electron gas and quantum Hall effect

Our formulation of electrodynamics can be generalized straightforwardly to arbitrary dimensions n . If we assume again the charge conservation law as a first axiom, then the rank of the electric current must be $n-1$. The force density in mechanics, in accordance with its definition $\sim \partial L / \partial x^i$ within the Lagrange formalism, should remain a covector-valued n -form. Hence we keep the second axiom in its original form. Accordingly, the field strength F is again a 2-form:

$$\oint_{C_{n-1}} J = 0, \quad f_\alpha = (e_\alpha \lrcorner F) \wedge J, \quad \oint_{C_2} F = 0. \quad (\text{B.4.34})$$

This may seem like an academic exercise. However, at least for $n = 3$, there exists an application: Since the middle of the 1960s, experimentalists were able to create a *2-dimensional electron gas* (2DEG) in suitable transistors at sufficiently low temperatures and to position the 2DEG in a strong external transversal magnetic field. Under such circumstances, the electrons can only move in a plane transverse to B and one space dimension can be suppressed.

³To formulate electrodynamics in accelerated systems by means of tensor analysis, see J. Van Bladel [51]. New experiments in rotating frames (with ring lasers) can be found in Stedman [47].

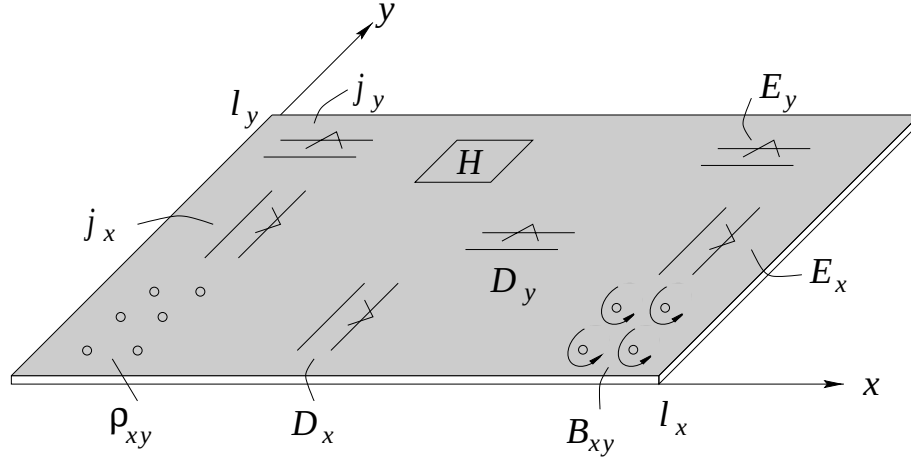


Figure B.4.2: Electrodynamics in $1 + 2$ dimensions: The arsenal of electromagnetic quantities in flatland. The sources are the charge density ρ_{xy} and the current density (j_x, j_y) (see (B.4.40)). The excitations $(\mathcal{D}_x, \mathcal{D}_y)$ and $\mathcal{H}$ (twisted scalar) are somewhat unusual (see (B.4.41)). The double plates for measuring $\mathcal{D}$, e.g., become double wires. The magnetic field has only one independent component B_{xy} , whereas the electric field has two, namely (E_x, E_y) (see (B.4.42)).

Electrodynamics in $1 + 2$ dimensions

In electrodynamics with one time and two space dimensions, we have from the first and third axioms (in arbitrary coordinates),

$$d^{(2)} J = 0, \quad J^{(2)} = d^{(1)} H, \quad (\text{B.4.35})$$

and

$$d^{(2)} F = 0, \quad F^{(2)} = d^{(1)} A, \quad (\text{B.4.36})$$

respectively, where we indicated the rank of the forms explicitly for better transparency. Here, the remarkable feature is that field strength F and current J carry the same rank; this is only possible for $n = 3$ spacetime dimensions. Moreover, the current J , the excitation H , and the field strength F all have the same number of independent components, namely three.

Now we $(1 + 2)$ -decompose the current and the electromagnetic field:

$$\text{twisted 2-form:} \quad J^{(2)} = -j \wedge d\sigma + \rho, \quad (\text{B.4.37})$$

$$\text{twisted 1-form:} \quad H^{(1)} = -\mathcal{H} d\sigma + \mathcal{D}, \quad (\text{B.4.38})$$

$$\text{untwisted 2-form:} \quad F^{(2)} = E \wedge d\sigma + B. \quad (\text{B.4.39})$$

Accordingly, in the space of the 2DEG, we have (again in arbitrary coordinates)

$$j = j_1 dx^1 + j_2 dx^2, \quad \rho = \rho_{12} dx^1 \wedge dx^2, \quad (\text{B.4.40})$$

$$\mathcal{H}, \quad \mathcal{D} = \mathcal{D}_1 dx^1 + \mathcal{D}_2 dx^2, \quad (\text{B.4.41})$$

$$E = E_1 dx^1 + E_2 dx^2, \quad B = B_{12} dx^1 \wedge dx^2. \quad (\text{B.4.42})$$

We recognize the rather degenerate nature of such a system. The magnetic field B , for example, has only one independent component B_{12} . Such a configuration is visualized in Fig. B.4.2 by using rectilinear coordinates; they need not be Cartesian coordinates.

Charge conservation (B.4.35) in decomposed form and in components reads,

$$\underline{d}j + \dot{\rho} = 0, \quad \partial_1 j_2 - \partial_2 j_1 + \dot{\rho}_{12} = 0. \quad (\text{B.4.43})$$

The $(1 + 2)$ -decomposed Maxwell equations look exactly as in (B.4.9) and (B.4.10). We also express them in components:

$$\underline{d}\mathcal{D} = \rho, \quad \partial_1 \mathcal{D}_2 - \partial_2 \mathcal{D}_1 = \rho_{12}, \quad (\text{B.4.44})$$

$$\underline{d}\mathcal{H} - \dot{\mathcal{D}} = j, \quad \partial_1 \mathcal{H} - \dot{\mathcal{D}}_1 = j_1, \quad (\text{B.4.45})$$

$$\partial_2 \mathcal{H} - \dot{\mathcal{D}}_2 = j_2, \quad (\text{B.4.46})$$

and

$$\underline{d}E + \dot{B} = 0, \quad \partial_1 E_2 - \partial_2 E_1 + \dot{B}_{12} = 0, \quad (\text{B.4.47})$$

$$\underline{d}B \equiv 0. \quad (\text{B.4.48})$$

We assume an infinite extension of flatland. If that cannot be assumed as a valid approximation, one has to allow for line currents at the boundary of flatland (“edge currents”) in order to fulfill the Maxwell equations.

In our formulation the Maxwell equations don’t depend on the metric. Thus, instead of the plane, as in Fig. B.4.2, we could have drawn an *arbitrary* 2-dimensional manifold, a surface of a cylinder or of a sphere, for example; the Maxwell equations (B.4.44) to (B.4.48) would still be valid since they had been derived for arbitrary (curvilinear) coordinates and arbitrary frames.

Before we can apply this formalism to the quantum Hall effect (QHE), we first remind ourselves of the classical Hall effect (of 1879).

Hall effect (*excerpt from the literature,⁴ conventional formalism*)

This subsection on the Hall effect and the next one on QHE are excerpts from the literature. Thus equations (B.4.49) to (B.4.56) are written in the conventional notation, and they refer to Cartesian coordinates and may contain the metric of 3-dimensional space. However, our subsequent presentation of QHE,

⁴See Landau & Lifshitz [25], pp. 96–98 or Raith [39], p. 502.

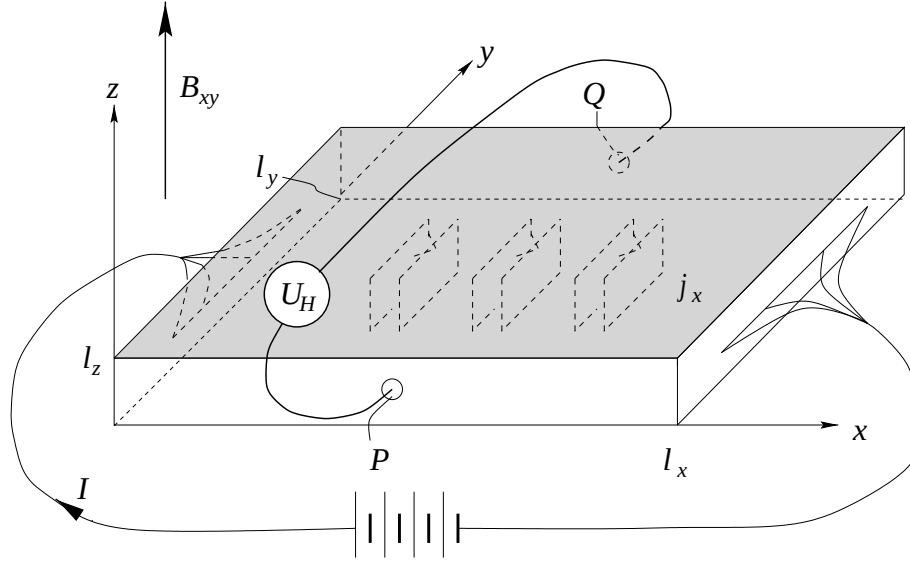


Figure B.4.3: Hall effect (schematic): The current density j in the conducting plate is affected by the external constant magnetic field B (in the figure only symbolized by one arrow) such as to create the Hall voltage U_H .

which starts with (B.4.57) and which contains all relevant features of a phenomenological description of QHE, is strictly metricfree.

We connect the two yz -faces of a (semi)conducting plate of volume $l_x \times l_y \times l_z$ with a battery (see Fig. B.4.3). A current I will flow and in the plate the current density is j_x . Transverse to the current, between the contacts P and Q , there exists no voltage. However, if we apply a constant magnetic field B along the z -axis, then the current j is deflected by the Lorentz force and the Hall voltage U_H occurs which, according to experiment, turns out to be

$$U_H = R_H I = A_H \frac{BI}{l_z}, \quad \text{with} \quad [A_H] = \frac{l^3}{q}. \quad (\text{B.4.49})$$

R_H is called the Hall resistance and A_H the Hall constant. We divide U_H by l_y . Because $E_y = U_H/l_y$, we find

$$E_y = A_H B_{xy} \frac{I_x}{l_y l_z} = A_H B_{xy} j_x. \quad (\text{B.4.50})$$

Let us stress that the classical Hall effect is a volume (or bulk) effect. It is to be described in the framework of ordinary $(1 + 3)$ -dimensional Maxwellian electrodynamics.

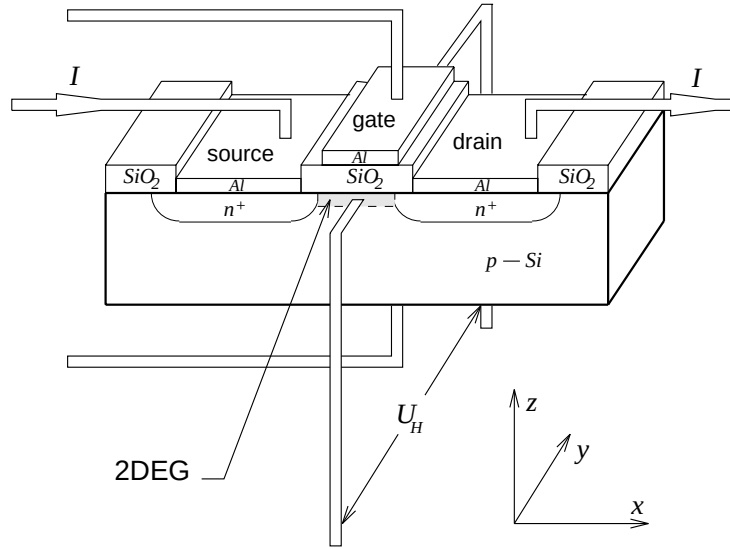


Figure B.4.4: A Mosfet with a 2-dimensional electron gas (2DEG) layer between a semiconductor (Si) and an insulator (SiO_2). Adapted from Braun [3]. In 1980, with such a transistor, von Klitzing et al. performed the original experiment on QHE.

Quantum Hall effect (*excerpt from the literature,⁵ conventional formalism*)

A prerequisite for the discovery in 1980 of QHE were the advances in transistor technology. Since the 1960s one was able to assemble 2-dimensional electron gas layers in certain types of transistors, such as in a **metal-oxide-semiconductor field effect transistor**⁶ (Mosfet) (see a schematic view of a Mosfet in Fig. B.4.4). The electron layer is only about 50 nanometers thick, whereas its lateral extension may go up to the millimeter region.

In the quantum Hall regime, we have very low temperatures (between 25 mK and 500 mK) and very high magnetic fields (between 5 T and 15 T). Then the conducting electrons of the specimen, because of a (quantum mechanical) excitation gap, cannot move in the z -direction, they are confined to the xy -plane. Thus an almost ideal *2-dimensional electron gas* (2DEG) is constituted.

The Hall conductance ($= 1/\text{resistance}$) exhibits very well-defined plateaus at integral (and, in the fractional QHE, at rational) multiples of the fundamental

⁵See, for example, von Klitzing [52], Braun [3], Chakraborty and Pietiläinen [4], Janssen et al. [20], Yoshioka [54], and references given there.

⁶A fairly detailed description can be found in Raith [39], pp. 579-582.

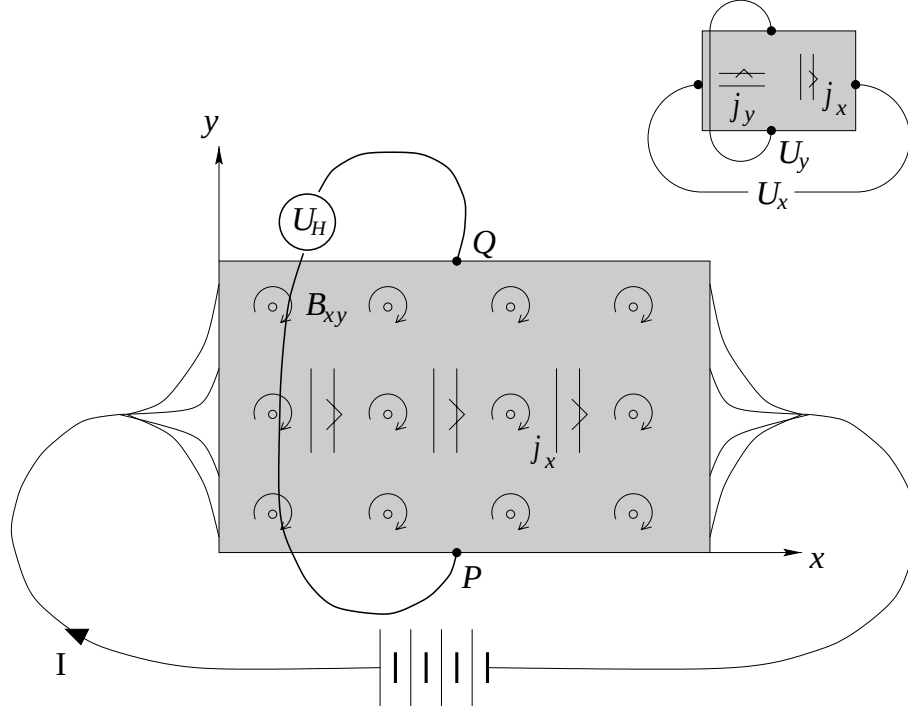


Figure B.4.5: Schematic view of a quantum Hall experiment with a 2-dimensional electron gas (2DEG). The current density j_x in the 2DEG is exposed to a strong transverse magnetic field B_{xy} . The Hall voltage U_H can be measured in the transverse direction to j_x between P and Q . In the inset we denoted the longitudinal voltage with U_x and the transversal one with $U_y (= U_H)$.

conductance⁷ of $e^2/h \equiv 1/(25\,812.807\,\Omega)$, where e is the elementary charge and h Planck's constant. Therefore this effect is instrumental in precision experiments for measuring, in conjunction with the Josephson effect, e and h very accurately. We concentrate here on the *integer* QHE.

Turning to Fig. B.4.5, we consider the rectangle in the xy -plane with side lengths l_x and l_y , respectively. The quantum Hall effect is observed in such a 2-dimensional system of electrons subject to a strong uniform transverse magnetic field $\vec{B}$. The configuration is similar to that of the classical Hall effect (see Fig. B.4.3), but the system is cooled down to a uniform temperature of about of 0.1 K. The *Hall resistance* R_H is defined by the ratio of the Hall voltage U_y and

⁷Recently, the conductance of a *single* H₂-molecule has been measured (see Smit et al. [46]). The value $2e^2/h$ was found, which is consistent with the fundamental conductance of QHE.

the electric current I_x in the x -direction: $R_H = U_y/I_x$. Longitudinally, we have the ordinary dissipative Ohm resistance $R_L = U_x/I_x$. For fixed values of the magnetic field B and area charge density of electrons $n_e e$, the Hall resistance R_H is a constant. Phenomenologically, QHE can be described by means of a *linear* tensorial Ohm–Hall law as a constitutive relation. Thus, for an *isotropic* material,

$$U_x = R_L I_x - R_H I_y, \quad (\text{B.4.51})$$

$$U_y = R_H I_x + R_L I_y. \quad (\text{B.4.52})$$

If we introduce the electric field $E_x = U_x/l_x$, $E_y = U_y/l_y$ and the 2D current densities $j_x = I_x/l_y$, $j_y = I_y/l_x$, we find

$$\vec{E} = \boldsymbol{\rho} \vec{j} \quad \text{or} \quad \vec{j} = \boldsymbol{\sigma} \vec{E}, \quad \text{with} \quad \boldsymbol{\rho} = \boldsymbol{\sigma}^{-1} = \begin{pmatrix} R_L \frac{l_y}{l_x} & -R_H \\ R_H & R_L \frac{l_x}{l_y} \end{pmatrix}, \quad (\text{B.4.53})$$

where $\boldsymbol{\rho}$, the specific resistivity, and $\boldsymbol{\sigma}$, the specific conductivity, are represented by second-rank tensors.

With a classical electron model for the conductivity, the Hall resistance can be calculated to be $R_H = B/(n_e e)$, where B is the only component of the magnetic field in 2-space. The magnetic flux quantum for an electron is $\hbar/e = 2\Phi_0$, with $\Phi_0 \stackrel{\text{SI}}{=} 2.07 \times 10^{-15}$ Wb. With the fundamental resistance we can rewrite the Hall resistance with the dimensionless *filling factor* ν as

$$R_H = \frac{B}{n_e e} = \frac{\hbar}{e^2} \nu^{-1}, \quad \text{where} \quad \nu := \frac{\hbar}{e} \frac{n_e}{B}. \quad (\text{B.4.54})$$

Observe that ν^{-1} measures the amount of magnetic flux (in flux units) per electron.

Thus classically, if we increase the magnetic field, keeping n_e (i.e., the gate voltage) fixed, we would expect a strictly linear increase of the Hall resistance. Surprisingly, however, we find the following (see Fig. B.4.6):

1. The Hall R_H has *plateaus* at rational heights. The plateaus at integer height occur with a high accuracy: $R_H = (\hbar/e^2)/i$, for $i = 1, 2, \dots$ (for holes we would get a negative sign).
2. When (ν, R_H) belongs to a plateau, the longitudinal resistance, R_L , *vanishes* to a good approximation.

In accordance with these results, we choose ν such that R_L is zero and σ_H constant at a plateau for one particular system, namely

$$\boldsymbol{\rho} = R_H \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{or} \quad \boldsymbol{\sigma} = \sigma_H \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (\text{B.4.55})$$

This yields

$$\vec{j} = \sigma_H \epsilon \vec{E} \quad \text{with} \quad \epsilon := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (\text{B.4.56})$$

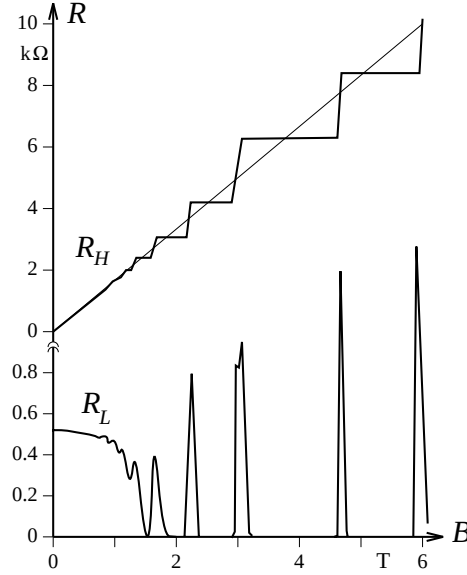


Figure B.4.6: The Hall resistance R_H at a temperature of 8 mK as a function of the magnetic field B (adapted from Ebert et al. [6]).

a phenomenological law valid at low frequencies and large distances. We will see in the next subsection that we can express this constitutive law in a generally covariant form. This shows that the laws governing QHE are entirely independent of the particular geometry (metric) under consideration.

Covariant description of the phenomenology at the plateaus

For a $(1 + 2)$ -dimensional covariant formulation,⁸ let us turn back to the first and the third axioms in (B.4.35) and (B.4.36), respectively. In (B.4.56), $\vec{j}$ is linearly related to $\vec{E}$. Accordingly, we will also assume *linearity* in $1+2$ dimensions between the 2-forms J and F ,

$$J_{ij} = -\sigma_{ij}{}^{kl} F_{kl}, \quad (\text{B.4.57})$$

with the Hall resistance $\sigma_{ij}{}^{kl}$. The minus sign is chosen because we want to recover (B.4.56) eventually. Since $J_{ij} = -J_{ji}$ and $F_{kl} = -F_{lk}$, we have

$$\sigma_{ij}{}^{kl} = -\sigma_{ji}{}^{kl} = -\sigma_{ij}{}^{lk}. \quad (\text{B.4.58})$$

⁸This can be found in the work of Fröhlich et al. [11, 9, 10] (see also Avron [2], Richter and Seiler [42], and references given there). An early article on QHE and topology is that of Post [38], which seems to have been largely overlooked.

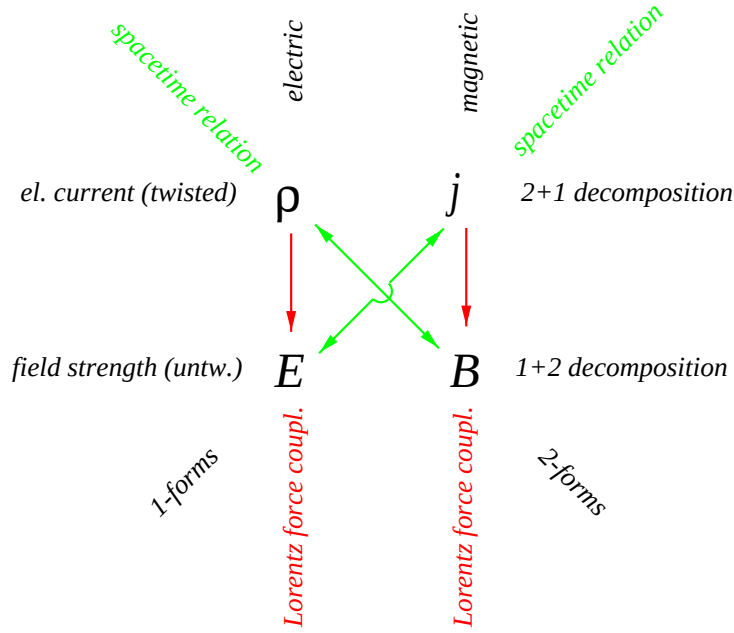


Figure B.4.7: Interrelationship between current and field strength in the Chern–Simons electrodynamics of QHE: In the *horizontal* direction, the 2-dimensional space part of a quantity is linked with its 1-dimensional time part to a 3-dimensional spacetime quantity. Lines in the *vertical* direction connect a pair of quantities that contribute to the Lorentz force density. And a *diagonal* line represents a spacetime relation.

Therefore the simplest $(1 + 2)$ -covariant ansatz is found by assuming additionally *isotropy* in the $1 + 2$ dimensions, that is,

$$\sigma_{ij}^{kl} = \sigma_H \delta_{ij}^{kl}, \quad (\text{B.4.59})$$

with the generalized Kronecker delta δ_{ij}^{kl} . Substituted into (B.4.57), this yields

$$\boxed{J = -\sigma_H F}, \quad [\sigma_H] = q^2/h = 1/\text{resistance}. \quad (\text{B.4.60})$$

The Hall resistance σ_H is a twisted scalar (or pseudoscalar). Of course, we can only hope that such a simple ansatz is valid provided the Mosfet is isotropic with respect to its Hall conductance in the plane of the Mosfet. Because of $(\text{B.4.35})_1$ and $(\text{B.4.36})_1$, we find quite generally that the Hall conductance must be *constant* in space and time:

$$\sigma_H = \text{const.} \quad (\text{B.4.61})$$

We should be aware that something remarkable happened here. We have *not* introduced a spacetime relation à la $H \sim F$, as is done in $(1 + 3)$ -dimensional

electrodynamics (see Part D). Since for $n = 3$, H and F have still the same number of independent components, this would be possible. However, it is the ansatz $J \sim F$ that leads to a successful description of the phenomenology of QHE. One can consider (B.4.60) as a spacetime relation for the $(1 + 2)$ -dimensional quantum Hall regime (see Fig. B.4.7). Of course, the integer or fractional numbers of σ_H cannot be derived from a classical theory. Nevertheless, classical $(1 + 2)$ -dimensional electrodynamics immediately suggests a relation of the type (B.4.60), a relation that is free of any metric. In other words, the $(1 + 2)$ -dimensional electrodynamics of QHE, as a classical theory, is metricfree. Equation (B.4.60) transforms the Maxwell equations (B.4.35)₂ and (B.4.36)₂ into a complete system of partial differential equations that can be integrated. At the same time, it also yields an explicit relation between H and F , namely

$$dH = -\sigma_H F \quad \text{or} \quad dH = -\sigma_H dA. \quad (\text{B.4.62})$$

The last equation can be integrated. We find

$$H = -\sigma_H A, \quad (\text{B.4.63})$$

since the potential 1-form A is only defined up to a gauge transformation anyway. The metricfree *differential* relation (B.4.62)₁ between excitation H and field strength F is certainly not what we would have expected from classical Maxwellian electrodynamics. It represents, for $n = 3$, a totally new type of (Chern–Simons) electrodynamics.

We compare (B.4.60) with (B.4.37) and (B.4.39) to find

$$j = \sigma_H E \quad \text{and} \quad \rho = -\sigma_H B. \quad (\text{B.4.64})$$

In (B.4.56), the current is represented as a vector. Therefore we introduce the vector density $\check{j} := \diamond j$ by means of the diamond operator of (A.1.78) or, in components, $\check{j}^a = \epsilon^{ab} j_b / 2$. Then $(a, b = 1, 2)$,

$$\check{j}^a = \frac{1}{2} \epsilon^{ab} j_b = (\sigma_H \epsilon^{ab}) E_a; \quad (\text{B.4.65})$$

that is, we recover (B.4.56), thereby verifying the ansatz (B.4.60). For the charge density $\check{\rho} := \diamond \rho = \epsilon^{ab} \rho_{ab} / 2$, we find

$$\check{\rho} = -\sigma_H B_{12}, \quad \text{with} \quad \partial_1 \check{j}^1 + \partial_2 \check{j}^2 + \dot{\check{\rho}} = 0. \quad (\text{B.4.66})$$

As mentioned, (B.4.57) and the subsequent equations in this subsection are all metricfree. In particular, this is true for (B.4.65). Nevertheless, we recover (B.4.56) from it. In our derivation the *crystal structure* does not enter. The assumptions of linearity and isotropy were sufficient. The (crystalline) material, in which QHE takes place, does define orthogonality by suitable crystal axes. However, this information is not needed for the derivation of (B.4.64).

Moreover, we really would need a theory of electrodynamics inside matter for the discussion of QHE. Such questions are treated in Part E. In spite of our naïve

ansatz (B.4.60) in the realm of $(1+2)$ -dimensional vacuum electrodynamics, we get the (long wavelength limit of the) phenomenology of QHE correct and we can post-dict the topological nature of QHE. This is not a minor achievement, and we consider it as a definite proof of the relevance of our way of formulating electrodynamics. Our guess is that the isotropy assumption in (B.4.59) is a sort of effective macroscopic average already. And probably for that reason the ansatz (B.4.60) works.

Preview: 3D Lagrangians

We introduce Lagrangians in Sec. B.5.4. Nevertheless, let us conclude this section on electrodynamics in flatland with a few remarks on the appropriate Lagrangian for QHE.

The Lagrangian in $(1+2)$ -dimensional electrodynamics has to be a twisted 3-form with the dimension h of an action. Obviously, the first invariant of (B.2.25) qualifies,

$$I_1 = F \wedge H = -\frac{1}{\sigma_H} J \wedge H = \frac{1}{\sigma_H} H \wedge dH. \quad (\text{B.4.67})$$

But also the Chern–Simons 3-form of (B.3.16), if multiplied by the twisted scalar σ_H , has the right characteristics,

$$\begin{aligned} \sigma_H C_A &= \sigma_H A \wedge F = \sigma_H A \wedge dA \\ &= -A \wedge dH = d(A \wedge H) - F \wedge H = -F \wedge H + dK, \end{aligned} \quad (\text{B.4.68})$$

with the Kiehn 3-form $K = A \wedge H$. Therefore, both candidate Lagrangians are intimately linked,

$$\sigma_H C_A = -I_1 + dK. \quad (\text{B.4.69})$$

In other words, they yield the same Lagrangian since they only differ by an irrelevant exact form.

Let us define then the Lagrange 3-form

$$L_{3D} := \frac{1}{2\sigma_H} H \wedge dH - A \wedge dH. \quad (\text{B.4.70})$$

By means of its Euler–Lagrange equation,

$$d\left(\frac{\partial L}{\partial(dH)}\right) + \frac{\partial L}{\partial H} = 0 \quad \text{or} \quad \frac{1}{\sigma_H} dH - dA = 0, \quad (\text{B.4.71})$$

we can recover the 3D spacetime relation (B.4.60) that we started from. Alternatively, we can consider J as the external field in the Lagrangian

$$L_{3D'} := -\frac{\sigma_H}{2} A \wedge dA - A \wedge J. \quad (\text{B.4.72})$$

A similar computation as in (B.4.71) yields directly (B.4.60).

B.5

Electromagnetic energy-momentum current and action

B.5.1 Fourth axiom: localization of energy-momentum

Minkowski's "greatest discovery was that at any point in the electromagnetic field in vacuo there exists a tensor of rank 2 of outstanding physical importance. . . . each component of the tensor $E_p{}^q$ has a physical interpretation, which in every case had been discovered many years before Minkowski showed that these 16 components constitute a tensor of rank 2. The tensor $E_p{}^q$ is called the energy tensor of the electromagnetic field."

Edmund Whittaker (1953)

Let us consider the Lorentz force density $f_\alpha = (e_\alpha \lrcorner F) \wedge J$ in (B.4.3). If we want to derive the energy-momentum law for electrodynamics, we have to try to express f_α as an *exact* form. Then energy-momentum is a kind of generalized potential for the Lorentz force density, namely $f_\alpha \sim d\Sigma_\alpha$. For that purpose, we start from f_α . We substitute $J = dH$ (inhomogeneous Maxwell equation) and subtract out a term with H and F exchanged and multiplied by a constant factor a :

$$f_\alpha = (e_\alpha \lrcorner F) \wedge dH - a(e_\alpha \lrcorner H) \wedge dF. \quad (\text{B.5.1})$$

Because of $dF = 0$ (homogeneous Maxwell equation), the subtracted term vanishes. The factor a will be left open for the moment. Note that we need a nonvanishing current $J \neq 0$ for our derivation to be sensible.

We partially integrate both terms in (B.5.1):

$$f_\alpha = d[a F \wedge (e_\alpha \lrcorner H) - H \wedge (e_\alpha \lrcorner F)] - a F \wedge d(e_\alpha \lrcorner H) + H \wedge d(e_\alpha \lrcorner F).$$

The first term already has the desired form. We recall the main formula for the Lie derivative of an arbitrary form Φ , namely $\mathcal{L}_{e_\alpha} \Phi = d(e_\alpha \lrcorner \Phi) + e_\alpha \lrcorner (d\Phi)$ (see (A.2.51)). This allows us to transform the second part of (B.5.2):

$$\begin{aligned} f_\alpha = & d[a F \wedge (e_\alpha \lrcorner H) - H \wedge (e_\alpha \lrcorner F)] \\ & - a F \wedge (\mathcal{L}_{e_\alpha} H) + H \wedge (\mathcal{L}_{e_\alpha} F) \\ & + a F \wedge e_\alpha \lrcorner (dH) - H \wedge e_\alpha \lrcorner (dF). \end{aligned} \quad (\text{B.5.2})$$

The last line can be rewritten as

$$+ a e_\alpha \lrcorner [F \wedge dH] - a (e_\alpha \lrcorner F) \wedge dH - e_\alpha \lrcorner [H \wedge dF] + (e_\alpha \lrcorner H) \wedge dF. \quad (\text{B.5.3})$$

As 5-forms, the expressions in the square brackets vanish. Two terms remain, and we find

$$\begin{aligned} f_\alpha = & d[a F \wedge (e_\alpha \lrcorner H) - H \wedge (e_\alpha \lrcorner F)] \\ & - a F \wedge (\mathcal{L}_{e_\alpha} H) + H \wedge (\mathcal{L}_{e_\alpha} F) \\ & - a (e_\alpha \lrcorner F) \wedge dH + (e_\alpha \lrcorner H) \wedge dF. \end{aligned} \quad (\text{B.5.4})$$

Because $dF = 0$, the third line adds up to $-af_\alpha$. Accordingly,

$$\begin{aligned} (1 + a) f_\alpha = & d[a F \wedge (e_\alpha \lrcorner H) - H \wedge (e_\alpha \lrcorner F)] \\ & - a F \wedge (\mathcal{L}_{e_\alpha} H) + H \wedge (\mathcal{L}_{e_\alpha} F). \end{aligned} \quad (\text{B.5.5})$$

Now we have to make up our minds about the choice of the factor a . With $a = -1$, the left-hand side vanishes and we find a mathematical identity. A real conservation law is only obtained when, eventually, the second line vanishes. In other words, here we need an a posteriori argument, i.e., we have to take some information from experience. For $a = 0$, the second line does not vanish. However, for $a = 1$, we can hope that the first term in the second line compensates the second term if somehow $H \sim F$. In fact, under “ordinary circumstances,” to be explored below, the two terms in the second line do compensate each other for $a = 1$. Therefore we postulate this choice and find

$$f_\alpha = (e_\alpha \lrcorner F) \wedge J = d^k \Sigma_\alpha + X_\alpha. \quad (\text{B.5.6})$$

Here the *kinematic energy-momentum* 3-form of the electromagnetic field, a central result of this section, reads

$${}^k \Sigma_\alpha := \frac{1}{2} [F \wedge (e_\alpha \lrcorner H) - H \wedge (e_\alpha \lrcorner F)] \quad (\text{fourth axiom}), \quad (\text{B.5.7})$$

and the remaining force density 4-form turns out to be

$$X_\alpha := -\frac{1}{2} (F \wedge \mathcal{L}_{e_\alpha} H - H \wedge \mathcal{L}_{e_\alpha} F). \quad (\text{B.5.8})$$

The absolute dimension of ${}^k\Sigma_\alpha$ as well as of X_α is h/ℓ .

Our derivation of (B.5.6) doesn't lead to a unique definition of ${}^k\Sigma_\alpha$. The addition of any closed 3-form would be possible,

$${}^k\Sigma'_\alpha := {}^k\Sigma_\alpha + Y_\alpha, \quad \text{with} \quad dY_\alpha = 0, \quad (\text{B.5.9})$$

such that

$$f_\alpha = d{}^k\Sigma'_\alpha + X_\alpha. \quad (\text{B.5.10})$$

In particular, Y_α could be exact: $Y_\alpha = dZ_\alpha$. The 2-form Z_α has the same dimension as ${}^k\Sigma_\alpha$. It seems impossible to build up Z_α exclusively in terms of the quantities e_α , H , F in an algebraic way. Therefore, $Y_\alpha = 0$ appears to be the most natural choice. Thus, by the fourth axiom we postulate that ${}^k\Sigma_\alpha$ in (B.5.7) represents the energy-momentum current that correctly localizes the energy-momentum distribution of the electromagnetic field in spacetime. We call it the *kinematic* energy-momentum current since we didn't find it by a dynamic principle, which we will not formulate before (B.5.89), but rather by means of some sort of kinematic arguments.

The current ${}^k\Sigma_\alpha$ can also be rewritten by applying the anti-Leibniz rule for $e_\alpha \lrcorner$ either in the first or second term on the right-hand side of (B.5.7). With the 4-form

$$\Lambda := -\frac{1}{2} F \wedge H, \quad (\text{B.5.11})$$

we find

$${}^k\Sigma_\alpha = e_\alpha \lrcorner \Lambda + F \wedge (e_\alpha \lrcorner H) = -e_\alpha \lrcorner \Lambda - H \wedge (e_\alpha \lrcorner F). \quad (\text{B.5.12})$$

Let us come back to (B.5.6). Of course, $d{}^k\Sigma_\alpha$ and ${}^k\Sigma_\alpha$ cannot be integrated since they are covector-valued forms. However, if we have a suitable vector field $\xi = \xi^\alpha e_\alpha$ available, we can transvect its components ξ^α with ${}^k\Sigma_\alpha$. Accordingly, we define the “charge” density

$$\mathcal{Q} := \xi^\alpha {}^k\Sigma_\alpha = \frac{1}{2} [F \wedge (\xi \lrcorner H) - H \wedge (\xi \lrcorner F)], \quad (\text{B.5.13})$$

which should *not* be mixed up with the notion of an *electric* charge. In flat spacetime, for example, with t as time coordinate, a suitable (Killing) vector would be $\xi = \partial_t$ and $\mathcal{Q} = \xi^\alpha {}^k\Sigma_\alpha = {}^k\Sigma_t$ would signify the energy density. Generally, the integrals $\int d\mathcal{Q}$ and $\int \mathcal{Q}$ are well defined.

Let us now compute $d\mathcal{Q}$. We transvect (B.5.6) with the vector components ξ^α . Then,

$$\begin{aligned} \xi^\alpha f_\alpha &= (\xi \lrcorner F) \wedge J = \xi^\alpha d{}^k\Sigma_\alpha + \xi^\alpha X_\alpha \\ &= d(\xi^\alpha {}^k\Sigma_\alpha) - {}^k\Sigma_\alpha \wedge d\xi^\alpha + \xi^\alpha X_\alpha \end{aligned} \quad (\text{B.5.14})$$

or

$$d\mathcal{Q} = (\xi \lrcorner F) \wedge J + {}^k\Sigma_\alpha \wedge d\xi^\alpha - \xi^\alpha X_\alpha. \quad (\text{B.5.15})$$

If we evaluate the last term by substituting (B.5.8), we have to remember the rule for the multiplication of a vector by a scalar in a Lie derivative (see Sec. A.2.10):

$$\mathcal{L}_{fu} \omega = f \mathcal{L}_u \omega + df \wedge (u \lrcorner \omega). \quad (\text{B.5.16})$$

We substitute $f = \xi^\alpha$, $u = e_\alpha$, $\omega = H$ and reorder:

$$\xi^\alpha \mathcal{L}_{e_\alpha} H = \mathcal{L}_\xi H - d\xi^\alpha \wedge (e_\alpha \lrcorner H). \quad (\text{B.5.17})$$

An analogous formula is valid for the 2-form F . Then (B.5.15) becomes

$$d\mathcal{Q} = (\xi \lrcorner F) \wedge J + \frac{1}{2} (F \wedge \mathcal{L}_\xi H - H \wedge \mathcal{L}_\xi F) \quad (\text{B.5.18})$$

or, in holonomic components, with $\check{Q}^i := \epsilon^{ijkl} Q_{jkl}/6$, $\check{J}^i := \epsilon^{ijkl} J_{jkl}/6$, and $\check{\mathcal{H}}^{ij} := \epsilon^{ijkl} H_{kl}/2$,

$$\partial_i \check{Q}^i = \xi^k F_{kl} \check{J}^l + \frac{1}{4} (F_{kl} \mathcal{L}_\xi \check{\mathcal{H}}^{kl} - \check{\mathcal{H}}^{kl} \mathcal{L}_\xi F_{kl}). \quad (\text{B.5.19})$$

Under which conditions is $\mathcal{Q}$ conserved, i.e., $d\mathcal{Q} = 0$? We have to assume vacuum, i.e., $J = 0$. Moreover, the conditions $\mathcal{L}_\xi H = \alpha(x)H$ and $\mathcal{L}_\xi F = \alpha(x)F$, with an arbitrary function $\alpha(x)$, are sufficient for the vanishing of the expression within the last two parentheses. We will come back to this type of question at the end of this section under *Preview*.

B.5.2 Properties of the energy-momentum current, electric/magnetic reciprocity

${}^k\Sigma_\alpha$ is trace free

The energy-momentum current ${}^k\Sigma_\alpha$ is a 3-form. We can blow it up to a 4-form according to $\vartheta^\beta \wedge {}^k\Sigma_\alpha$. Since it still has sixteen components, we haven't lost any information. If we recall that for any p -form Φ we have $\vartheta^\alpha \wedge (e_\alpha \lrcorner \Phi) = p\Phi$, we immediately recognize from (B.5.7) that

$$\vartheta^\alpha \wedge {}^k\Sigma_\alpha = 0, \quad (\text{B.5.20})$$

which amounts to *one* equation. This property — the vanishing of the “trace” of ${}^k\Sigma_\alpha$ — is connected with the fact that the electromagnetic field (the “photon”) carries no mass and the theory is thus invariant under dilations. Why we call it the trace of the energy-momentum will become clear below (see (B.5.38)).

${}^k\Sigma_\alpha$ is electric/magnetic reciprocal

Furthermore, we can observe another property of ${}^k\Sigma_\alpha$. It is remarkable how symmetric H and F enter (B.5.7). This was achieved by our choice of $a = 1$.

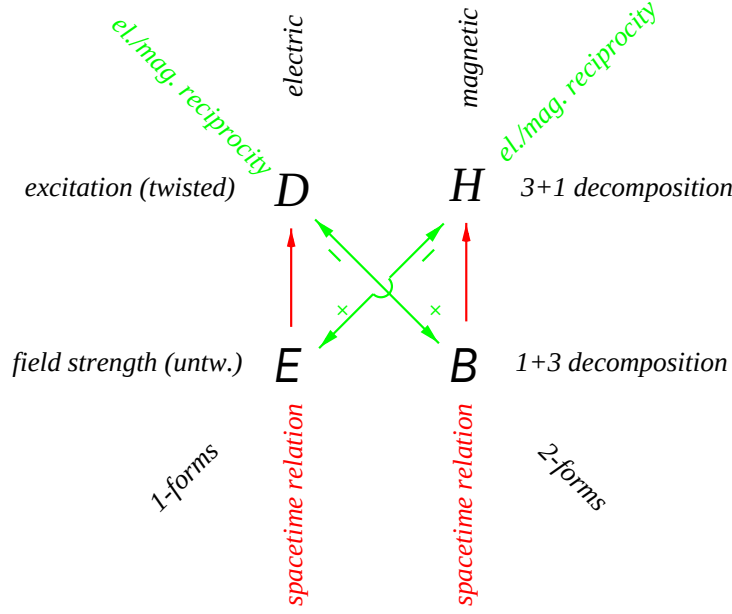


Figure B.5.1: Different aspects of the electromagnetic field: In the *horizontal* direction, the 3-dimensional space part of a quantity is linked with its 1-dimensional time part to a 4-dimensional spacetime quantity. The energy-momentum current remains invariant under the exchange of those quantities that are connected by a *diagonal* line. And a *vertical* line represents a space-time relation between quantities that are canonically related like momentum and velocity (see (B.5.84)₁).

The energy-momentum current is electric/magnetic reciprocal; i.e., it remains invariant under the transformation

$$H \rightarrow \zeta F, \quad F \rightarrow -\frac{1}{\zeta} H \quad \Longrightarrow \quad {}^k\Sigma_\alpha \rightarrow {}^k\Sigma_\alpha, \quad (\text{B.5.21})$$

with the twisted 0-form (pseudoscalar function) $\zeta = \zeta(x)$ of dimension $[\zeta] = [H]/[F] = q^2/h = 1/\text{resistance}$.

It should be stressed that in spite of ${}^k\Sigma_\alpha$ being electric/magnetic reciprocal, Maxwell's equations are *not*,

$$dH = J \quad \rightarrow \quad dF + F \wedge d\zeta/\zeta = J/\zeta, \quad (\text{B.5.22})$$

$$dF = 0 \quad \rightarrow \quad dH - H \wedge d\zeta/\zeta = 0, \quad (\text{B.5.23})$$

not even for $d\zeta = 0$, since we don't want to restrict ourselves to the free-field case with vanishing source $J = 0$.

For the invariants of the electromagnetic field, we find under a reciprocity transformation the following responses:

$$I_1 \rightarrow -I_1, \quad (\text{B.5.24})$$

$$I_2 \rightarrow \frac{1}{\zeta^2} I_3, \quad (\text{B.5.25})$$

$$I_3 \rightarrow \zeta^2 I_2. \quad (\text{B.5.26})$$

Thus, the products $\diamond I_1 \diamond I_1$ and $\diamond I_2 \diamond I_3$ are invariant under reciprocity transformations.

Equation (B.5.21) expresses a certain reciprocity between electric and magnetic effects with regard to their respective contributions to the energy-momentum current of the field. We call it electric/magnetic reciprocity.¹ That this naming is appropriate can be seen from a $(1 + 3)$ -decomposition. We recall the decompositions of H and F in (B.4.6) and (B.4.7), respectively. We substitute them into (B.5.21):

$$H \rightarrow \zeta F \quad \begin{cases} \mathcal{H} \rightarrow -\zeta E, \\ \mathcal{D} \rightarrow \zeta B, \end{cases} \quad (\text{B.5.27})$$

$$F \rightarrow -\frac{1}{\zeta} H \quad \begin{cases} E \rightarrow \frac{1}{\zeta} \mathcal{H}, \\ B \rightarrow -\frac{1}{\zeta} \mathcal{D}. \end{cases} \quad (\text{B.5.28})$$

Here it is clearly visible that a magnetic quantity is replaced by an electric one *and* an electric quantity by a magnetic one: *electric* $\longleftrightarrow$ *magnetic*. In this sense, we can speak of an electric/magnetic reciprocity in the expression for the energy-momentum current Σ_α . Alternatively we can say that Σ_α fulfills electric/magnetic reciprocity; it is electric/magnetic reciprocal.

Let us pause for a moment and wonder of how the notions “electric” and “magnetic” are attached to certain fields and whether there is a conventional element involved. By making experiments with a cat’s skin and a rod of amber, we can “liberate” what we call *electric* charges. In three dimensions, they are described by the charge density ρ . Set in motion, they produce an electric current j . The electric charge is conserved (first axiom) and is linked, via the Gauss law $\underline{d}\mathcal{D} = \rho$, to the *electric* excitation $\mathcal{D}$.

Returning to the Oersted experiment, it is clear that moving charges j induce magnetic effects in accordance with the Oersted–Ampère law $\underline{d}\mathcal{H} - \dot{\mathcal{D}} = j$, also a consequence of the first axiom. Hence we can unanimously attribute the term *magnetic* to the excitation $\mathcal{H}$. There is no room left for doubt about that.

¹...following Toupin [48] even if he introduced this notion in a somewhat more restricted context. Maxwell spoke of the mutual embrace of electricity and magnetism (see Wise [53]). In the case of a *prescribed metric*, discussions of the corresponding Rainich “duality rotation” were given by Gaillard & Zumino [13] and by Mielke [32], amongst others. Note, however, that our transformation (B.5.21) is metricfree and thus of a different type.

The second axiom links the electric charge density ρ to the field strength E according to $(e_a \lrcorner \rho) \wedge E$ and the electric current j to the field B according to $(e_a \lrcorner j) \wedge B$. Consequently, for the field strength F also, there can be no other way than to label E as *electric* and B as *magnetic* field strength.

These arguments imply that the substitutions $H \rightarrow \zeta F$ as well as $F \rightarrow -H/\zeta$ both substitute an electric by a magnetic field and a magnetic by an electric one (see (B.5.27) and (B.5.28)). Because of the minus sign (that is, because $a = 1$) that we found in (B.5.21) in analyzing the electromagnetic energy-momentum current Σ_α , we cannot speak of an equivalence of electric and magnetic fields; the expression reciprocity is much more appropriate. Fundamentally, electricity and magnetism enter into classical electrodynamics in an *asymmetric* way.

Let us try to explain the electric/magnetic reciprocity by means of a simple example. In (B.5.61) we will show that the electric energy density reads $u_{\text{el}} = \frac{1}{2} E \wedge \mathcal{D}$. If one wants to try to guess the corresponding expression for the magnetic energy density u_{mg} , one substitutes for an electric a corresponding magnetic quantity. However, the electric field strength is a 1-form. One cannot replace it by the magnetic field strength B since that is a 2-form. Therefore one has to switch to the magnetic excitation according to $E \rightarrow \frac{1}{\zeta} \mathcal{H}$, with the 1-form $\mathcal{H}$. The function ζ is needed because of the different dimensions of E and $\mathcal{H}$ and since E is an untwisted and $\mathcal{H}$ a twisted form. Analogously, one substitutes $\mathcal{D} \rightarrow \zeta B$, thereby finding $u_{\text{mg}} = \frac{1}{2} \mathcal{H} \wedge B$. This is the correct result, i.e., $u = \frac{1}{2}(E \wedge \mathcal{D} + B \wedge \mathcal{H})$, and we can be happy.

Naively, one would then postulate the invariance of u under the substitution $E \rightarrow \frac{1}{\zeta} \mathcal{H}$, $\mathcal{D} \rightarrow \zeta B$, $B \rightarrow \frac{1}{\zeta} \mathcal{D}$, $\mathcal{H} \rightarrow \zeta E$. But, as a look at (B.4.6) and (B.4.7) will show, this cannot be implemented in a covariant way. More generally, if we compare (B.4.14) with (B.4.20), then the mentioned invariance would require $h_\tau = f_\tau$ and $h_s = f_\Lambda f_\tau$. Consequently, $f_\Lambda = h_s/h_\tau$. However, according to (B.4.18), we have $h_s/h_\tau = -1$. Hence only the unphysical anti-Lenz rule with $f_\Lambda = -1$ could save the situation, an option that we don't take; we rather stick to the Lenz rule.

How are we going to save our rule of thumb for extracting the magnetic energy from the electric one? Well, if we turn to the substitutions (B.5.27) and (B.5.28), i.e., if we introduce two minus signs according to $E \rightarrow \frac{1}{\zeta} \mathcal{H}$, $\mathcal{D} \rightarrow \zeta B$, $B \rightarrow -\frac{1}{\zeta} \mathcal{D}$, $\mathcal{H} \rightarrow -\zeta E$, then u still remains invariant and we recover the covariant rule (B.5.21). In other words, the naive approach works *up to two minus signs*. Those we can supply by having insight into the covariant version of electrodynamics. Accordingly, the electric/magnetic reciprocity is the one that we knew all the time; we just have to be careful with the sign.

$\otimes^k \Sigma_\alpha$ expressed in terms of the complex electromagnetic field

We can understand the electric/magnetic reciprocity transformation as acting on the column vector consisting of H and ζF :

$$\begin{pmatrix} H' \\ \zeta F' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} H \\ \zeta F \end{pmatrix}. \quad (\text{B.5.29})$$

In order to compactify this formula, we introduce the *complex electromagnetic field* 2-form²

$$U := H + i \zeta F \quad \text{and} \quad U^* = H - i \zeta F, \quad (\text{B.5.30})$$

with $*$ denoting the complex conjugate. Now the electric/magnetic reciprocity (B.5.29) translates into

$$U' = -i U, \quad U^{*'} = i U^*. \quad (\text{B.5.31})$$

This corresponds, in the complex plane, where U lives, to a rotation by an angle of $-\pi/2$.

We can resolve (B.5.30) with respect to excitation and field strength:

$$H = \frac{1}{2} (U + U^*), \quad F = -\frac{i}{2\zeta} (U - U^*). \quad (\text{B.5.32})$$

We differentiate (B.5.30)₁. Then the Maxwell equation for the complex field turns out to be

$$dU + (U^* - U) \frac{d\zeta}{2\zeta} = J. \quad (\text{B.5.33})$$

Clearly, if we choose a constant ζ , i.e., $d\zeta = 0$, the second term on the left-hand side vanishes. The asymmetry between electric and magnetic fields finds its expression in the fact that the source term on the right-hand side of (B.5.33) is a *real* quantity.

If we substitute (B.5.32) into the energy-momentum current (B.5.7), we find, after some algebra,

$${}^k \Sigma_\alpha = \frac{i}{4\zeta} [U^* \wedge (e_\alpha \lrcorner U) - U \wedge (e_\alpha \lrcorner U^*)]. \quad (\text{B.5.34})$$

Now, according to (B.5.31), electric/magnetic reciprocity of the energy-momentum current is manifest.

²Even though we introduce the concept of a metric only in Part C, it is necessary to point out that the complex electromagnetic *field* U , subsuming excitation and field strength (see Fig. B.5.1), should be carefully distinguished from the complex electromagnetic field *strength* introduced conventionally: $\hat{F}^a := F^{0a} + i\epsilon^{abc} F_{bc}$, with $F^{0a} := g^{0i} g^{aj} F_{ij}$. This can only be defined *after* a metric has been introduced. Similarly, for the excitation we would then have $\hat{H}^a := -H^{0a} + i\epsilon^{abc} H_{bc}$, with $H^{0a} := g^{0i} g^{aj} H_{ij}$.

If we execute successively *two* electric/magnetic reciprocity transformations, namely $U \rightarrow U' \rightarrow U''$, then as can be seen from (B.5.31) or (B.5.21), we find a reflection (a rotation of $-\pi$), namely $U'' = -U$, i.e.,

$$U \rightarrow -U \quad \text{or} \quad (H \rightarrow -H, F \rightarrow -F). \quad (\text{B.5.35})$$

Only *four* electric/magnetic reciprocity transformations lead back to the identity. It should be stressed, however, that already *one* electric/magnetic reciprocity transformation leaves ${}^k\Sigma_\alpha$ invariant.

It is now straightforward to formally extend the electric/magnetic reciprocity transformation (B.5.31) to

$$U' = e^{+i\phi} U, \quad U^{*'} = e^{-i\phi} U^*, \quad (\text{B.5.36})$$

with $\phi = \phi(x)$ as an arbitrary “rotation” angle. The energy-momentum current ${}^k\Sigma_\alpha$ is still invariant under this extended transformation, but in later applications only the subcase of $\phi = -\pi/2$, treated above, will be of interest.

⊗ Energy-momentum tensor density ${}^k\mathcal{T}_\alpha{}^\beta$

Since ${}^k\Sigma_\alpha$ is a 3-form, we can decompose it either conventionally or with respect to the basis 3-form $\hat{e}_\beta = e_\beta \lrcorner \hat{e}$, with $\hat{e} = \vartheta^{\hat{0}} \wedge \vartheta^{\hat{1}} \wedge \vartheta^{\hat{2}} \wedge \vartheta^{\hat{3}}$ (see (A.1.74)):

$${}^k\Sigma_\alpha = \frac{1}{3!} {}^k\Sigma_{\lambda\mu\nu\alpha} \vartheta^\lambda \wedge \vartheta^\mu \wedge \vartheta^\nu =: {}^k\mathcal{T}_\alpha{}^\beta \hat{e}_\beta. \quad (\text{B.5.37})$$

The second-rank tensor density of weight 1, ${}^k\mathcal{T}_\alpha{}^\beta$, is the *Minkowski* energy tensor density. We can resolve this equation with respect to ${}^k\mathcal{T}_\alpha{}^\beta$ by exterior multiplication with ϑ^β . We recall $\vartheta^\beta \wedge \hat{e}_\gamma = \delta_\gamma^\beta \hat{e}$ and find

$${}^k\mathcal{T}_\alpha{}^\beta \hat{e} = \vartheta^\beta \wedge {}^k\Sigma_\alpha \quad (\text{B.5.38})$$

or with the new diamond operator $\diamond$ of (A.1.78),

$${}^k\mathcal{T}_\alpha{}^\beta = \diamond(\vartheta^\beta \wedge {}^k\Sigma_\alpha) = \frac{1}{3!} \epsilon^{\beta\lambda\mu\nu} {}^k\Sigma_{\lambda\mu\nu\alpha}. \quad (\text{B.5.39})$$

Thereby we recognize that $\vartheta^\alpha \wedge {}^k\Sigma_\alpha = 0$ (see (B.5.20)) is equivalent to the vanishing of the trace of the energy-momentum tensor density ${}^k\mathcal{T}_\alpha{}^\alpha = 0$. Thus ${}^k\mathcal{T}_\alpha{}^\beta$ as well as ${}^k\Sigma_\alpha$ have fifteen independent components at this stage. Both quantities are equivalent.

If we substitute (B.5.7) into (B.5.39), then we can express the energy-momentum tensor density in the components of H and F as follows:³

$${}^k\mathcal{T}_\alpha{}^\beta = \frac{1}{4} \epsilon^{\beta\mu\rho\sigma} (H_{\alpha\mu} F_{\rho\sigma} - F_{\alpha\mu} H_{\rho\sigma}). \quad (\text{B.5.40})$$

³We leave it to the readers to prove the formula ${}^k\mathcal{T}_\alpha{}^\gamma {}^k\mathcal{T}_\gamma{}^\beta = \frac{1}{4} \delta_\alpha^\beta {}^k\mathcal{T}^2$ which was derived first by Minkowski in 1907.

$\otimes {}^k \mathcal{T}_\alpha^\beta$ **alternatively derived by means of tensor calculus**

We start with the Maxwell equations (B.4.33) in holonomic coordinates, i.e., in the natural frame $e_\alpha = \delta_\alpha^i \partial_i$:

$$\partial_j \check{\mathcal{H}}^{ij} = \check{\mathcal{J}}^i, \quad \partial_{[i} F_{jk]} = 0. \quad (\text{B.5.41})$$

Here $\check{\mathcal{H}}$ is defined according to $\check{\mathcal{H}}^{kl} = \frac{1}{2} \epsilon^{klmn} H_{mn}$ (see (B.4.32)₁). We substitute the inhomogeneous Maxwell equation into the Lorentz force density and integrate partially:

$$f_i = F_{ij} \check{\mathcal{J}}^j = F_{ij} \partial_k \check{\mathcal{H}}^{jk} = \partial_k (F_{ij} \check{\mathcal{H}}^{jk}) - (\partial_k F_{ij}) \check{\mathcal{H}}^{jk}. \quad (\text{B.5.42})$$

The last term can be rewritten by means of the homogeneous Maxwell equation, i.e.,

$$-\partial_k F_{ij} = \partial_i F_{jk} + \partial_j F_{ki}, \quad (\text{B.5.43})$$

or

$$f_i = \partial_k (F_{ij} \check{\mathcal{H}}^{jk}) + (\partial_i F_{jk} + \partial_j F_{ki}) \check{\mathcal{H}}^{jk}. \quad (\text{B.5.44})$$

Again, we integrate partially. This time the two last terms are:

$$\begin{aligned} f_i &= \partial_k (F_{ij} \check{\mathcal{H}}^{jk}) + \partial_i (F_{jk} \check{\mathcal{H}}^{jk}) + \partial_j (F_{ki} \check{\mathcal{H}}^{jk}) \\ &\quad - F_{jk} \partial_i \check{\mathcal{H}}^{jk} - F_{ki} \partial_j \check{\mathcal{H}}^{jk}. \end{aligned} \quad (\text{B.5.45})$$

We collect the first three terms on the right-hand side and substitute the left-hand side of the inhomogeneous Maxwell equation into the last term:

$$f_i = \partial_k (\delta_i^k F_{jl} \check{\mathcal{H}}^{jl} + 2F_{ij} \check{\mathcal{H}}^{jk}) - F_{jk} \partial_i \check{\mathcal{H}}^{jk} - F_{ik} \check{\mathcal{J}}^k. \quad (\text{B.5.46})$$

The last term represents the negative of a Lorentz force density (see (B.5.42)). Thus we find

$$f_i = \partial_k \left(\frac{1}{2} \delta_i^k F_{jl} \check{\mathcal{H}}^{jl} + F_{ij} \check{\mathcal{H}}^{jk} \right) - \frac{1}{2} F_{jl} \partial_i \check{\mathcal{H}}^{jl}. \quad (\text{B.5.47})$$

The first and the third term on the right-hand side are of a related structure. We split the first term into two equal pieces and differentiate one piece:

$$f_i = \partial_k \left(\frac{1}{4} \delta_i^k F_{jl} \check{\mathcal{H}}^{jl} + F_{ij} \check{\mathcal{H}}^{jk} \right) + \frac{1}{4} \left[(\partial_i F_{jl}) \check{\mathcal{H}}^{jl} - F_{jl} \partial_i \check{\mathcal{H}}^{jl} \right]. \quad (\text{B.5.48})$$

Introducing the kinematic energy-momentum tensor density

$${}^k \mathcal{T}_i^j = \frac{1}{4} \delta_i^j F_{kl} \check{\mathcal{H}}^{kl} - F_{ik} \check{\mathcal{H}}^{jk} \quad (\text{B.5.49})$$

and the force density

$$\mathcal{X}_i = \frac{1}{4} \left[(\partial_i F_{jk}) \check{\mathcal{H}}^{jk} - F_{jk} \partial_i \check{\mathcal{H}}^{jk} \right], \quad (\text{B.5.50})$$

we finally have the desired result,

$$f_i = \partial_j {}^k \mathcal{T}_i^j + \mathcal{X}_i, \quad (\text{B.5.51})$$

which is the component version of (B.5.6). By means of (B.4.32)₁, it is possible to transform (B.5.49) into (B.5.40).

⊗ **Preview: Covariant conservation law and vanishing extra force density $\widehat{X}_\alpha$**

The Lorentz force density f_α in (B.5.6) and the energy-momentum current ${}^k\Sigma_\alpha$ in (B.5.7) are covariant with respect to frame and coordinate transformations. Nevertheless, both of the two terms on the right-hand side of (B.5.6), namely $d{}^k\Sigma_\alpha$ and X_α , are *not* covariant by themselves. What can we do?

For the first three axioms of electrodynamics, the spacetime arena is only required to be a $(1 + 3)$ -decomposable 4-dimensional manifold. We cannot be as economical as this in general. Ordinarily a *linear connection* Γ_α^β on that manifold is needed. The linear connection is the guiding field that transports a vector, for example, from one point of spacetime to a neighboring one.

The connection is only introduced in Part C. There, the covariant exterior differential is defined as $D = d + \Gamma_\alpha^\beta \rho(L_\beta^\alpha)$ (see (C.1.63)). With the help of this operator, a generally covariant expression $D{}^k\Sigma_\alpha$ can be constructed. Then (B.5.6) can be rewritten as

$$f_\alpha = D{}^k\Sigma_\alpha + \widehat{X}_\alpha, \quad (\text{B.5.52})$$

with the new supplementary force density

$$\widehat{X}_\alpha = \frac{1}{2} (H \wedge L_{e_\alpha} F - F \wedge L_{e_\alpha} H), \quad (\text{B.5.53})$$

which contains the covariant Lie derivative $L_\xi = D\xi \lrcorner + \xi \lrcorner D$ (see (C.1.71)). Note that the energy-momentum current ${}^k\Sigma_\alpha$ remains the same; only the force density X_α gets replaced by $\widehat{X}_\alpha$. It is remarkable that in (B.5.52) or in (B.5.6) the energy-momentum current can be defined even if (B.5.52) (as long as $\widehat{X}_\alpha \neq 0$) doesn't represent a genuine conservation law.

In this subsection only, and not in the rest of Part B, we use the linear connection and the *covariant* exterior derivative. Thus we are able to show that the fourth axiom is exactly what is needed for an appropriate and consistent derivation of the conservation law for energy-momentum. Let us exploit then, as far as possible, the arbitrary linear connection Γ_α^β introduced above. As auxiliary quantities attached to Γ_α^β , we need the *torsion* 2-form T^α and the *transposed connection* 1-form $\widehat{\Gamma}_\alpha^\beta := \Gamma_\alpha^\beta + e_\alpha \lrcorner T^\beta$, both to be introduced in Part C in (C.1.42) and (C.1.43), respectively.

Let us now go back to the extra force density $\widehat{X}_\alpha$ of (B.5.53). What we need is the gauge covariant Lie derivative of an arbitrary 2-form $\Psi = \Psi_{\mu\nu} \vartheta^\mu \wedge \vartheta^\nu / 2$ in terms of its components. Using the general formula (C.2.128) we have

$$L_{e_\alpha} \Psi = \frac{1}{2} \left(\widehat{D}_\alpha \Psi_{\mu\nu} \right) \vartheta^\mu \wedge \vartheta^\nu, \quad (\text{B.5.54})$$

where $\widehat{D}_\alpha := e_\alpha \lrcorner \widehat{D}$, with $\widehat{D}$ as the exterior covariant differential with respect to the transposed connection. Thus,

$$\widehat{X}_\alpha = \frac{1}{8} \left(H_{\rho\sigma} \widehat{D}_\alpha F_{\mu\nu} - F_{\rho\sigma} \widehat{D}_\alpha H_{\mu\nu} \right) \vartheta^\rho \wedge \vartheta^\sigma \wedge \vartheta^\mu \wedge \vartheta^\nu, \quad (\text{B.5.55})$$

or since $\vartheta^\rho \wedge \vartheta^\sigma \wedge \vartheta^\mu \wedge \vartheta^\nu = \epsilon^{\rho\sigma\mu\nu} \hat{e}$, we find as an alternative to (B.5.53),

$$\hat{X}_\alpha = \frac{\hat{e}}{8} \epsilon^{\rho\sigma\mu\nu} \left(H_{\rho\sigma} \widehat{D}_\alpha F_{\mu\nu} - F_{\rho\sigma} \widehat{D}_\alpha H_{\mu\nu} \right). \quad (\text{B.5.56})$$

This is as far as we can go with an arbitrary linear connection.

Now it becomes obvious how to achieve the vanishing of $\hat{X}_\alpha$. Our four axioms don't make electrodynamics a complete theory. What is missing is the *electromagnetic spacetime relation* between excitation H and field strength F . Such a fifth axiom is introduced in Chapter D.4. The starting point for arriving at such an axiom is the *linear ansatz*

$$\tilde{\mathcal{H}}^{\mu\nu} = \frac{1}{2} \chi^{\mu\nu\rho\sigma} F_{\rho\sigma}, \quad \text{with} \quad \tilde{\mathcal{H}}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} H_{\rho\sigma}. \quad (\text{B.5.57})$$

Substituting into (B.5.56), we have

$$\hat{X}_\alpha = -\frac{\hat{e}}{8} \left(\widehat{D}_\alpha \chi^{\rho\sigma\mu\nu} \right) F_{\rho\sigma} F_{\mu\nu}. \quad (\text{B.5.58})$$

Thus, the extra force density $\hat{X}_\alpha$ vanishes provided $\chi^{\rho\sigma\mu\nu}$ is covariantly constant with respect to the transposed connection of the underlying spacetime. We will come back to this discussion in Sec. E.1.4.

B.5.3 Time-space decomposition of the energy-momentum current and the Lenz rule

Another theory of electricity, which I prefer, denies action at a distance and attributes electric action to tensions and pressures in an all-pervading medium, these stresses being the same in kind with those familiar to engineers, and the medium being identical with that in which light is supposed to be propagated.

James Clerk Maxwell (1870)

If we 1 + 3 decompose the Lorentz force and the energy-momentum current, we arrive at the 3-dimensional version of the energy-momentum law of electrodynamics in a rather direct way. Recall that we work with a foliation-compatible frame e_α as specified in (B.1.34), i.e., with $e_{\hat{0}} = n$, $e_a = \partial_a$, together with the transversality condition $e_a \lrcorner d\sigma = 0$.

Consider the definition (B.5.7) of ${}^k\Sigma_\alpha$. Substitute into it the (1 + 3)-decompositions (B.4.6) and (B.4.7) of the excitation H and the field strength F , respectively. Then, we obtain

$${}^k\Sigma_{\hat{0}} = u - d\sigma \wedge s, \quad (\text{B.5.59})$$

$${}^k\Sigma_a = -p_a - d\sigma \wedge S_a, \quad (\text{B.5.60})$$

where we introduced the *energy* density 3-form

$$u := \frac{1}{2} (E \wedge \mathcal{D} + B \wedge \mathcal{H}) , \quad (\text{B.5.61})$$

the *energy flux* density (or Poynting) 2-form

$$s := E \wedge \mathcal{H} , \quad (\text{B.5.62})$$

the *momentum* density 3-form

$$p_a := -B \wedge (e_a \lrcorner \mathcal{D}) , \quad (\text{B.5.63})$$

and the Maxwell *stress* (or momentum flux density) 2-form of the electromagnetic field

$$\begin{aligned} S_a := & \frac{1}{2} [(e_a \lrcorner E) \wedge \mathcal{D} - (e_a \lrcorner \mathcal{D}) \wedge E \\ & + (e_a \lrcorner \mathcal{H}) \wedge B - (e_a \lrcorner B) \wedge \mathcal{H}] . \end{aligned} \quad (\text{B.5.64})$$

Accordingly, we can represent the scheme (B.5.59)–(B.5.60) in the form of a 4×4 matrix (for density we use the abbreviation d.):

$$({}^k \Sigma_\alpha) = \begin{pmatrix} \text{energy d.} & -\text{mom. d.} \\ \text{energy flux d.} & \text{mom. flux d.} \end{pmatrix} = \begin{pmatrix} u & -p_a \\ s & S_a \end{pmatrix} . \quad (\text{B.5.65})$$

The entries of the first row are 3-forms and those of the second row 2-forms.

The absolute dimensions of the quantities emerging in the 4×4 matrix can be determined from their respective definitions and the decompositions (B.4.6) and (B.4.7):

$$\begin{pmatrix} [u] & [-p_a] \\ [s] & [S_a] \end{pmatrix} = h \begin{pmatrix} t^{-1} & \ell^{-1} \\ t^{-2} & (t\ell)^{-1} \end{pmatrix} . \quad (\text{B.5.66})$$

The relative dimensions, that is, those of their respective components, read (here $i, j, k = 1, 2, 3$),

$$\begin{pmatrix} [u_{ijk}] & [-p_{ijk\ a}] \\ [s_{ij}] & [S_{ij\ a}] \end{pmatrix} = \frac{h}{t\ell^3} \begin{pmatrix} 1 & (\ell/t)^{-1} \\ \ell/t & 1 \end{pmatrix} . \quad (\text{B.5.67})$$

This coincides with the results from mechanics. A momentum flux density, e.g., should have the dimension $pv/\ell^3 = mv^2/\ell^3 = f/\ell^2 = \text{stress}$, in agreement with $[S_{ij\ a}] = h/(t\ell^3) = \text{energy}/\ell^3 = \text{stress} \stackrel{\text{SI}}{=} \text{Pascal}$. Note that the dimension of the energy flux density s_{ij} is the same as that of the momentum density $p_{ijk\ a}$ times the *square of a velocity* $(\ell/t)^2$.

Transvecting the Maxwell stress S_a , “familiar to engineers,” with ϑ^a , we find straightforwardly

$$\vartheta^a \wedge S_a = -u , \quad (\text{B.5.68})$$

which is the 3-dimensional version of (B.5.20). As soon as an electromagnetic spacetime relation is available, we can relate the energy flux density $s \wedge \vartheta^a$, which

has the same number of independent components as the 2-form s , namely three, to the momentum density p_a . In Sec. E.1.4, by means of the Maxwell–Lorentz spacetime relation, we prove the symmetry of the energy-momentum current in this way (see (E.1.30)).

The Lorentz force density is longitudinal with respect to n , i.e., $f_a = {}^\perp f_a$, whereas the forms u , s , p_a , and S_a are purely transversal. Equations (B.5.59)–(B.5.60) provide the decomposition of the energy-momentum 3-form into its “time” and “space” pieces. If we apply (B.1.28) to it, we find for the exterior differentials:

$$d^k \Sigma_{\hat{0}} = d\sigma \wedge (\dot{u} + \underline{d}s), \quad (\text{B.5.69})$$

$$d^k \Sigma_a = d\sigma \wedge (-\dot{p}_a + \underline{d}S_a). \quad (\text{B.5.70})$$

Combining all the results, we eventually obtain for the $(1 + 3)$ -decomposition of (B.5.6) the balance equations for the electromagnetic field energy and momentum:

$$k_{\hat{0}} = \dot{u} + \underline{d}s + (X_{\hat{0}})_\perp, \quad (\text{B.5.71})$$

$$k_a = -\dot{p}_a + \underline{d}S_a + (X_a)_\perp. \quad (\text{B.5.72})$$

Let us now come back to our discussion of the Lenz rule. We take ${}^k \Sigma_{\hat{0}}$ from the fourth axiom (B.5.7) and substitute the decompositions for excitation and field strength (B.4.14) and (B.4.20), respectively. After some algebra we find

$${}^k \Sigma_{\hat{0}} = f_{\text{T}} h_{\text{T}} \left(E \wedge \mathcal{H} \wedge d\sigma + \frac{h_{\text{S}}}{h_{\text{T}}} \frac{1}{2} \mathcal{D} \wedge E - f_{\Lambda} \frac{1}{2} B \wedge \mathcal{H} \right). \quad (\text{B.5.73})$$

In section B.4.2 we have shown that $h_{\text{T}} = -1$, $h_{\text{S}} = +1$, and $f_{\text{T}} = +1$. Thus,

$${}^k \Sigma_{\hat{0}} = -E \wedge \mathcal{H} \wedge d\sigma + \frac{1}{2} \mathcal{D} \wedge E + f_{\Lambda} \frac{1}{2} B \wedge \mathcal{H}. \quad (\text{B.5.74})$$

The electric energy density $u_{\text{el}} = \frac{1}{2} E \wedge \mathcal{D} = \frac{1}{2} \mathcal{D} \wedge E$ and the magnetic $u_{\text{mg}} = \frac{1}{2} B \wedge \mathcal{H} = \frac{1}{2} \mathcal{H} \wedge B$ add up for $f_{\Lambda} = +1$ (Lenz rule). In the opposite case, for $f_{\Lambda} = -1$ (anti-Lenz rule), they subtract. It is true, at the present stage we cannot make any statements about the signs of either u_{el} or u_{mg} . However, from the similarity of both expressions one would conclude that if u_{el} is the electric energy then u_{mg} has to be the magnetic one and similarly for $-u_{\text{el}}$ and $-u_{\text{mg}}$.

In any case, pre-metric electrodynamics clearly bifurcates into the Maxwell–Lenz and the Maxwell–anti-Lenz equations. Already in (B.4.7), we had opted for the Maxwell–Lenz case in anticipation of the adding up of electric and magnetic energy. For the rest of the book, we put $f_{\Lambda} = +1$.

Observe finally that all the formulas displayed in this section are independent of any metric and/or connection.

B.5.4 $\otimes$ Action

Why have we postponed the discussion of the Lagrange formalism for so long even though we know that this formalism helps so much in the effective organization of field-theoretical structures? We chose to base our axiomatics on the conservation laws of charge and flux, *inter alia*, which are amenable to direct experimental verification. And in the second axiom we used the concept of force from mechanics that also has the appeal of being able to be grasped directly. Accordingly, the *proximity to experiment* was one of our guiding principles in selecting the axioms.

Already via the second axiom the notion of a force density came in. We know that this concept, according to $f_i \sim \partial L / \partial x^i$, also has a place in the Lagrange formalism. When we “derived” the fourth axiom by trying to express the Lorentz force density f_α as an exact form $f_\alpha \sim d\Sigma_\alpha$, we obviously had already moved towards the Lagrange formalism. It became apparent in (B.5.11): the 4-form Λ is a possible Lagrangian. Still, we proposed the energy-momentum current without appealing to a Lagrangian. That seemed to be more secure because we could avoid all the fallacies related to a not directly observable quantity like L . We were led, practically in a unique fashion, to the fourth axiom (B.5.7).

In any case, having formulated the integral and the differential versions of electrodynamics including its energy-momentum distribution, we have enough understanding of its inner working to be able to reformulate it in a Lagrangian form in a very straightforward way.

As we discussed in Sec. B.4.1, for the completion of electrodynamics we need an electromagnetic spacetime relation $H = H[F]$. This could be a nonlocal and nonlinear functional in general, as we will discuss in Chapter E.2. The field variables in Maxwell’s equations are H and F . Therefore, the Lagrange 4-form of the electromagnetic field should depend on both of them:

$$V = V(H[F], F) = V[F]. \quad (\text{B.5.75})$$

From a dimensional point of view, it is quite obvious what type of action we would expect for the Maxwell field. For the excitation we have $[H] = q$ and for the field strength $[F] = h q^{-1}$. Accordingly, $\sim \int H \wedge F$ would qualify as action. And this is, indeed, what we will find out later.

We hasten to add that in Part D, when we introduce the hypothetical skewon field, a Lagrangian description turns out to be too narrow. We instead use the spacetime relation $H = H[F]$ directly without taking recourse to a Lagrangian.

Maxwell’s equations are first order in $H = H[F]$ and F , respectively. Since $F = dA$, the field strength F itself is first order in A . We take A to be a field variable. The Euler–Lagrange equations of a variational principle with a Lagrangian of differential order m are of differential order $2m$. The field equations are assumed at most of second differential order in the field variables A, ψ . Therefore, the Lagrangian is of first order in these fields.

Consider an electrically charged matter field ψ which, for the time being, is assumed to be a p -form. The total Lagrangian of the system, a twisted 4-form,

should consist of a free field part V of the electromagnetic field and a matter part L_{mat} , the latter of which describes the matter field ψ and its coupling to A :

$$L = V + L_{\text{mat}} = V(A, dA, \text{ST}) + L_{\text{mat}}(A, \psi, d\psi, \text{ST}). \quad (\text{B.5.76})$$

In L_{mat} , the potential A is needed for the coupling of matter to the electromagnetic field. However, dA is not assumed to arise: $\partial L_{\text{mat}}/\partial dA = 0$. This is dubbed *minimal coupling* since this is the most economical way of bringing a coupling about. Furthermore, ST are what we call the *structural* fields. Their presence, at this stage, is motivated by a technical concern: In order to *construct* a viable Lagrangian other than the trivial “topological” $F \wedge F$ Lagrangian, one should have supplementary fields, namely ST, that allow one to achieve this goal. We won’t specify their nature now. As soon as we have introduced a metric (and a connection) on spacetime, those fields will play the part of ST.

We require V to be *gauge invariant*, that is

$$\delta V = V(A + \delta A, d[A + \delta A], \text{ST}) - V(A, dA, \text{ST}) = 0, \quad (\text{B.5.77})$$

where $\delta A = d\omega$ represents an infinitesimal gauge transformation of the type (B.3.9) for $\chi \approx d\omega$. We obtain

$$\delta V = d\omega \wedge \frac{\partial V}{\partial A} = 0 \quad \Leftrightarrow \quad \frac{\partial V}{\partial A} = 0, \quad (\text{B.5.78})$$

or

$$V = V(dA, \text{ST}) = V(F, \text{ST}). \quad (\text{B.5.79})$$

Hence the free field Maxwell or *gauge* Lagrangian can depend on the potential A only via the field strength $F = dA$. The matter Lagrangian should also be gaugeinvariant.

The action reads

$$W = \int_{\Omega_4} L. \quad (\text{B.5.80})$$

The field equations for A are given by the stationary points of W under a variation δ of A , which commutes with the exterior derivative by definition, that is, $\delta d = d\delta$, and vanishes at the boundary, i.e., $\delta A|_{\partial\Omega_4} = 0$. Varying A yields

$$\begin{aligned} \delta_A W &= \int_{\Omega_4} \delta_A L = \int_{\Omega_4} \left[\delta A \wedge \frac{\partial L}{\partial A} + \delta dA \wedge \frac{\partial L}{\partial dA} \right] \\ &= \int_{\Omega_4} \left[\delta A \wedge \left\{ \frac{\partial L}{\partial A} - (-1)^1 d \frac{\partial L}{\partial dA} \right\} + d \left\{ \delta A \wedge \frac{\partial L}{\partial dA} \right\} \right] \\ &= \int_{\Omega_4} \delta A \wedge \frac{\delta L}{\delta A} + \int_{\partial\Omega_4} \delta A \wedge \frac{\partial L}{\partial dA}, \end{aligned} \quad (\text{B.5.81})$$

where the variational derivative of the 1-form A is defined according to

$$\frac{\delta L}{\delta A} := \frac{\partial L}{\partial A} + d \frac{\partial L}{\partial dA}. \quad (\text{B.5.82})$$

Stationarity of W leads to the *gauge field equation*

$$\frac{\delta L}{\delta A} = 0. \quad (\text{B.5.83})$$

Keeping in mind the inhomogeneous Maxwell field equation in (B.4.3), we define the excitation (“field momentum”) conjugated to A and the matter current by

$$H = -\frac{\partial V}{\partial dA} = -\frac{\partial V}{\partial F} \quad \text{and} \quad J = \frac{\delta L_{\text{mat}}}{\delta A} = \frac{\partial L_{\text{mat}}}{\partial A}, \quad (\text{B.5.84})$$

respectively. Then we recover, indeed:

$$dH = J. \quad (\text{B.5.85})$$

The homogeneous Maxwell equation is a consequence of working with the potential A , since $F = dA$ and $dF = ddA = 0$. In (B.5.85), we were also able to arrive at the inhomogeneous equation. The excitation H and the current J are, however, only implicitly given. As we can see from (B.5.84), only an explicit form of the Lagrangians V and L_{mat} promotes H and J to more than sheer placeholders. On the other hand, it is very satisfying to recover the structure of Maxwell’s theory already at such an implicit level. Equation (B.5.84)₁ represents the as yet unknown *spacetime relation* of Maxwell’s theory.

Let us turn to the variational of the matter field ψ . Its variational derivative reads

$$\frac{\delta L}{\delta \psi} := \frac{\partial L}{\partial \psi} - (-1)^p d \frac{\partial L}{\partial d\psi}. \quad (\text{B.5.86})$$

Since V does not depend on ψ , we find for the *matter field equation* simply

$$\frac{\delta L_{\text{mat}}}{\delta \psi} = 0. \quad (\text{B.5.87})$$

All that is left to do now in this context is to specify the spacetime relation (B.5.84)₁ and thereby to transform the Maxwell and the matter Lagrangian into an explicit form.

At this stage, the structural fields ST are considered nondynamical (“background”), so we do not have equations of motion for them.

B.5.5 $\otimes$ Coupling of the energy-momentum current to the coframe

In this section we show that the canonical definition of the energy-momentum current (as a Noether current corresponding to spacetime translations) coincides

with its *dynamic* definition as a source of the gravitational field that is represented by the coframe, and both are closely related to the kinematic current of our fourth axiom.

Let us assume that the interaction of the electromagnetic field with gravity is “switched on.” On the Lagrangian level, it means that (B.5.79) should be replaced by the Lagrangian

$$V = V(\vartheta^\alpha, F). \quad (\text{B.5.88})$$

From now on, the coframe assumes the role of the structural field ST. In a standard way, the *dynamic* (or Hilbert) energy-momentum current for the coframe field is defined by

$${}^d\Sigma_\alpha := -\frac{\delta V}{\delta \vartheta^\alpha} = -\left(\frac{\partial V}{\partial \vartheta^\alpha} + d\frac{\partial V}{\partial(d\vartheta^\alpha)}\right). \quad (\text{B.5.89})$$

The last term vanishes for the Lagrangian (B.5.88) under consideration. As before (cf. (B.5.84)), the electromagnetic field momentum is defined by

$$H = -\frac{\partial V}{\partial F}. \quad (\text{B.5.90})$$

The crucial point is the condition of general coordinate or diffeomorphism invariance of the Lagrangian (B.5.88) of the interacting electromagnetic and coframe fields. The general variation of the Lagrangian reads

$$\delta V = \delta \vartheta^\alpha \wedge \frac{\partial V}{\partial \vartheta^\alpha} + \delta F \wedge \frac{\partial V}{\partial F} = -\delta \vartheta^\alpha \wedge {}^d\Sigma_\alpha - \delta F \wedge H. \quad (\text{B.5.91})$$

If ξ is a vector field generating an arbitrary 1-parameter group $\mathcal{T}_t$ of diffeomorphisms on X , the variation in (B.5.91) is described by the Lie derivative, i.e., $\delta = \mathcal{L}_\xi = \xi \lrcorner d + d\xi \lrcorner$. Substituting this into the left-hand and right-hand sides of (B.5.91), we find, after some rearrangements, the identity

$$d\left[(\xi \lrcorner V) + (\xi \lrcorner \vartheta^\alpha) {}^d\Sigma_\alpha + (\xi \lrcorner F) \wedge H\right] \quad (\text{B.5.92})$$

$$- (\xi \lrcorner \vartheta^\alpha) d {}^d\Sigma_\alpha + (\xi \lrcorner d\vartheta^\alpha) \wedge {}^d\Sigma_\alpha - (\xi \lrcorner F) \wedge dH = 0.$$

Since ξ is arbitrary, the first and second lines vanish separately. Now, the final step is to put $\xi = e_\alpha$. Then (B.5.92) yields two identities.

From the first line of (B.5.92) we find that the dynamic current ${}^d\Sigma_\alpha$, defined in (B.5.89), can be identified with the *canonical* (or Noether) *energy-momentum* current, i.e.,

$${}^d\Sigma_\alpha \equiv \Sigma_\alpha, \quad \text{with} \quad \Sigma_\alpha := -e_\alpha \lrcorner V - (e_\alpha \lrcorner F) \wedge H. \quad (\text{B.5.93})$$

Thus we don't need to distinguish any longer between the *dynamic* and the *canonical* energy-momentum current. This fact matches very well with the structure of the *kinematic* energy-momentum current ${}^k\Sigma_\alpha$ of (B.5.7) or, better, of

(B.5.12). We compare (B.5.93) with (B.5.12). If we choose as our Lagrangian $V = -F \wedge H/2$, then ${}^k\Sigma_\alpha = \Sigma_\alpha$, and we can drop the k and the d from ${}^k\Sigma_\alpha$ and ${}^d\Sigma_\alpha$, respectively. However, this choice of Lagrangian is only legitimized by our fifth axiom in Chap. D.6.

The second line of (B.5.92) yields the *conservation law* of energy-momentum:

$$d\Sigma_\alpha = (e_\alpha \lrcorner d\vartheta^\beta) \wedge \Sigma_\beta + (e_\alpha \lrcorner F) \wedge J, \quad (\text{B.5.94})$$

where we have used the inhomogeneous Maxwell equation $dH = J$. The presence of the first term on the right-hand side guarantees the covariant character of that equation. It is easy to see that we can rewrite (B.5.94) in the equivalent form

$$\tilde{D}\Sigma_\alpha = (e_\alpha \lrcorner F) \wedge J \quad (\text{B.5.95})$$

by making use of the Riemannian covariant exterior derivative $\tilde{D}$ to be defined in Part C.

Our discussion can even be made more general,⁴ going beyond a pure electrodynamical theory. Namely, let us consider a theory of the coframe ϑ^α coupled to a *generalized matter* field Ψ . Note that the latter may not be just one function or an exterior form but an *arbitrary collection* of forms of all possible ranks and/or exterior forms of type ρ , i.e., $\Psi = (\psi^{(0)U}, \dots, \psi^{(p)A}, \dots)$. The ranges of indices for forms of different ranks are, in general, also different; that is, U and A run over different ranges, for example. We assume that the Lagrangian of such a theory depends very generally on frame, matter field, and its derivatives:

$$L = L(\vartheta^\alpha, d\vartheta^\alpha, \Psi, d\Psi). \quad (\text{B.5.96})$$

Normally, the matter Lagrangian does not depend on the derivatives of the coframe, but we include $d\vartheta^\alpha$ for greater generality. (It is important to realize that the Lagrangian (B.5.96) describes the most general theory: For example, the set Ψ can include not only the true matter fields described, say, by a p -form $\psi^{(p)A}$ but formally also other virtual gravitational potentials such as the metric 0-form $g_{\alpha\beta}$ and the connection 1-form $\Gamma_\alpha{}^\beta$ as soon as they are defined on X and are interacting with each other.)

The basic assumption about the Lagrangian (B.5.96) is that L is a *scalar-valued twisted n -form that is invariant under the spacetime diffeomorphisms*. This simple input has amazingly general consequences.

The dynamic energy-momentum current of matter is defined as in (B.5.89):

$${}^d\Sigma_\alpha := -\frac{\delta L}{\delta \vartheta^\alpha} = -\left(\frac{\partial L}{\partial \vartheta^\alpha} + d \frac{\partial L}{\partial (d\vartheta^\alpha)} \right). \quad (\text{B.5.97})$$

⁴See, for example, reference [15] and also [18].

Similarly, the variational derivative of the generalized matter reads

$$\frac{\delta L}{\delta \Psi} = \frac{\partial L}{\partial \Psi} - (-1)^p d \frac{\partial L}{\partial (d\Psi)}, \quad (\text{B.5.98})$$

where the sign factor $(-1)^p$ correlates with the relevant rank of a particular component in the set of fields Ψ . This is a simple generalization of (B.5.86).

According to the Noether theorem, the conservation identities of the matter system result from the postulated invariance of L under a local symmetry group. Actually, this is only true “weakly,” i.e., provided the Euler–Lagrange equation (B.5.98) for the matter fields is satisfied.

Here we consider the consequences of the invariance of L under the group diffeomorphisms on the spacetime manifold. Let ξ be a vector field generating an arbitrary 1-parameter group $\mathcal{T}_t$ of diffeomorphisms on X . In order to obtain a corresponding Noether identity from the invariance of L under a 1-parameter group of *local* translations $\mathcal{T}_t \subset \mathcal{T} \approx \text{Diff}(4, R)$, it is important to recall that infinitesimally the action of a 1-parameter group $\mathcal{T}_t$ on X is described by the conventional Lie derivative (A.2.49) with respect to a vector field ξ . Since we work with fields that are exterior forms of various ranks, the most appropriate formula for the Lie derivative is (A.2.51), i.e., $\mathcal{L}_\xi = \xi \lrcorner d + d\xi \lrcorner$. The general variation of the Lagrangian (B.5.96) reads:

$$\delta L = \delta \vartheta^\alpha \wedge \frac{\partial L}{\partial \vartheta^\alpha} + (\delta d\vartheta^\alpha) \wedge \frac{\partial L}{\partial d\vartheta^\alpha} + \delta \Psi \wedge \frac{\partial L}{\partial \Psi} + (\delta d\Psi) \wedge \frac{\partial L}{\partial d\Psi}. \quad (\text{B.5.99})$$

For the variations generated by a 1-parameter group of the vector field ξ we have to substitute $\delta = \mathcal{L}_\xi$ in (B.5.99). This is straightforward and, after performing some “partial integrations,” we find

$$\begin{aligned} d(\xi \lrcorner L) &= d \left[(\xi \lrcorner \vartheta^\alpha) \frac{\partial L}{\partial \vartheta^\alpha} + (\xi \lrcorner d\vartheta^\alpha) \wedge \frac{\partial L}{\partial d\vartheta^\alpha} \right. \\ &\quad \left. + (\xi \lrcorner \Psi) \wedge \frac{\partial L}{\partial \Psi} + (\xi \lrcorner d\Psi) \wedge \frac{\partial L}{\partial d\Psi} \right] \\ &\quad - (\xi \lrcorner \vartheta^\alpha) d \frac{\partial L}{\partial \vartheta^\alpha} + (\xi \lrcorner d\vartheta^\alpha) \wedge \frac{\partial L}{\partial \vartheta^\alpha} \\ &\quad + (\xi \lrcorner d\Psi) \wedge \frac{\partial L}{\partial \Psi} + (-1)^p (\xi \lrcorner \Psi) \wedge d \frac{\partial L}{\partial \Psi}. \end{aligned} \quad (\text{B.5.100})$$

Now we rearrange the equation above by collecting terms under the exterior derivative separately. Then (B.5.100) can be written as

$$A - dB = 0, \quad (\text{B.5.101})$$

where

$$\begin{aligned} A &:= -(\xi \lrcorner \vartheta^\alpha) d \frac{\delta L}{\delta \vartheta^\alpha} + (\xi \lrcorner d\vartheta^\alpha) \wedge \frac{\delta L}{\delta \vartheta^\alpha} \\ &\quad + (\xi \lrcorner d\Psi) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p (\xi \lrcorner \Psi) \wedge d \frac{\delta L}{\delta \Psi}, \end{aligned} \quad (\text{B.5.102})$$

$$\begin{aligned} B &:= \xi \lrcorner L - (\xi \lrcorner \vartheta^\alpha) \frac{\partial L}{\partial \vartheta^\alpha} - (\xi \lrcorner d\vartheta^\alpha) \wedge \frac{\partial L}{\partial d\vartheta^\alpha} \\ &\quad - (\xi \lrcorner \Psi) \wedge \frac{\partial L}{\partial \Psi} - (\xi \lrcorner d\Psi) \wedge \frac{\partial L}{\partial d\Psi}. \end{aligned} \quad (\text{B.5.103})$$

The functions A and B have the form

$$A = \xi^\alpha A_\alpha, \quad B = \xi^\alpha B_\alpha. \quad (\text{B.5.104})$$

Hence, by (B.5.101),

$$\xi^\alpha (A_\alpha - dB_\alpha) - d\xi^\alpha \wedge B_\alpha = 0, \quad (\text{B.5.105})$$

where both ξ^α and $d\xi^\alpha$ are *pointwise arbitrary*. Hence we can conclude that B_α as well as A_α vanish:

$$A = 0, \quad B = 0. \quad (\text{B.5.106})$$

Since the vector field ξ is arbitrary, it is sufficient to replace it via $\xi \rightarrow e_\alpha$ by the frame field. Then, for $B = 0$, we obtain from (B.5.103)

$$\begin{aligned} {}^d\Sigma_\alpha &= -e_\alpha \lrcorner L + (e_\alpha \lrcorner d\Psi) \wedge \frac{\partial L}{\partial d\Psi} + (e_\alpha \lrcorner \Psi) \wedge \frac{\partial L}{\partial \Psi} \\ &\quad - d \frac{\partial L}{\partial d\vartheta^\alpha} + (e_\alpha \lrcorner d\vartheta^\beta) \wedge \frac{\partial L}{\partial d\vartheta^\beta}. \end{aligned} \quad (\text{B.5.107})$$

For $A = 0$, equation (B.5.102) yields

$$d {}^d\Sigma_\alpha \equiv (e_\alpha \lrcorner d\vartheta^\beta) \wedge {}^d\Sigma_\beta + \mathcal{F}_\alpha, \quad (\text{B.5.108})$$

where

$$\mathcal{F}_\alpha := -(e_\alpha \lrcorner d\Psi) \wedge \frac{\delta L}{\delta \Psi} - (-1)^p (e_\alpha \lrcorner \Psi) \wedge d \frac{\delta L}{\delta \Psi}. \quad (\text{B.5.109})$$

In fact, when the matter Lagrangian is independent of the derivatives of the coframe field, i.e., $\partial L / \partial d\vartheta^\beta = 0$, equation (B.5.107) demonstrates the equality of the dynamic energy-momentum current that is coupled to the coframe with the canonical one, the Noether current of the translations.

B.5.6 Maxwell's equations and the energy-momentum current in Excalc

It is a merit of exterior calculus that electrodynamics and, in particular, Maxwell's equations can be formulated in a very succinct form. This translates into

an equally concise form of the corresponding computer programs in Excalc. The goal of better understanding the structure of electrodynamics leads us, hand in hand, to a more transparent and more effective way of computer programming.

In Excalc, as we mentioned in Sec. A.2.11, the electrodynamical quantities are evaluated with respect to the coframe that is specified in the program. If we put in an accelerating and rotating coframe ω^α , for example, then the electromagnetic field strength F in the program will be evaluated with respect to this frame: $F = F_{\alpha\beta} \omega^\alpha \wedge \omega^\beta / 2$. This is, of course, exactly what we discussed in Sec. B.4.4 when we introduced arbitrary noninertial coframes.

We cautioned our readers already in Sec. A.2.11 that we need to specify a coframe together with the metric. Thus, our Maxwell sample program proper, to be displayed below, is preceded by coframe and frame commands. We pick the spherical coordinate system of Sec. A.2.11 and require the spacetime to be Minkowskian, i.e., $\psi(r) = 1$.

Afterwards we specify the electromagnetic potential $A = A_\alpha \vartheta^\alpha$, namely `pot1`. Since we haven't defined a specific problem so far, we leave its components `aa0`, `aa1`, `aa2`, `aa3` open for the moment. Then we put in the pieces discussed after (B.4.3). In order to relate H and F , we have to make use of the fifth axiom, only to be pinned down in (D.6.13):

```
% file mustermx.exe, 2002-11-15

load_package excalc$

pform psi=0$ fdomain psi=psi(r)$
coframe o(0)      = psi          * d t,
              o(1)      = (1/psi)    * d r,
              o(2)      = r          * d theta,
              o(3)      = r * sin(theta) * d phi
              with signature (1,-1,-1,-1)$          % coframe defined
frame e$

psi:=1;                                % flat spacetime assumed

% start of Maxwell proper: unknown functions aa0, aa1, ...

pform {aa0,aa1,aa2,aa3}=0, pot1=1, {farad2,excit2}=2,
      {maxhom3,maxinh3}=3$
fdomain aa0=aa0(t,r,theta,phi),aa1=aa1(t,r,theta,phi),
      aa2=aa2(t,r,theta,phi),aa3=aa3(t,r,theta,phi)$

pot1    := aa0*o(0) + aa1*o(1) + aa2*o(2) + aa3*o(3)$
farad2  := d pot1;
maxhom3 := d farad2;
```

```

excit2 := lam * # farad2;           % spacetime relation, see
maxinh3 := d excit2;               % 5th axiom, Eq. (D.6.13)

% Maxwell Lagrangian and energy-momentum current assigned

pform lmax4=4, maxenergy3(a)=3$

lmax4      := -(1/2)*farad2^excit2;
maxenergy3(-a) := e(-a) _|lmax4 + (e(-a) _|farad2)^excit2;

% Use of a blank before the interior product sign is obligatory!
end;

```

If this sample program is written onto a file with name *mustermax.exi* (exi stands for excalc-input; the corresponding output file has the extension .exo), then this very file can be read into a Reduce session by the command `in"muster-max.exi"`;

As a trivial test, you can specify the potential of a point charge sitting at the origin of our coordinate system:

```

aa0 := -q/r;      aa1 := aa2 := aa3 := 0;

```

Determine its field strength by `farad2:=farad2`; its excitation by `excit2:=excit2`; and its energy-momentum distribution by `maxenergy3(-a):=maxenergy3(-a)`; You will find the results you are familiar with. And, of course, you want to convince yourself that Maxwell's equations are fulfilled by releasing the commands `maxhom3:=maxhom3`; and `maxinh3:=maxinh3`;

This sample program can be edited according to the needs one has. Prescribe a noninertial coframe, i.e., an accelerating and rotating coframe. Then you just have to edit the coframe command and can subsequently compute the corresponding physical components of an electromagnetic quantity with respect to that frame. Applications of this program include the Reissner–Nordström and the Kerr–Newman solutions of general relativity; they represent the electromagnetic and the gravitational fields of an electrically charged mass of spherical or axial symmetry, respectively. They are discussed in the outlook in the last part of the book.

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Part C

More mathematics

In Part A we outlined exterior algebra and exterior calculus on differential manifolds that are “bare” in the sense that no other geometrical structures are assumed on them. The corresponding mathematical machinery proved to be sufficient for the formulation in Part B of the general framework of classical electrodynamics.

In Part C we provide additional mathematical tools needed to complete electromagnetic theory: the connection and the metric. Quite generically, the connection and the metric can be introduced on a smooth manifold as two completely independent fields.

The (linear) connection defines parallel transport, which enables us to compare values of quantities at different points of the manifold. The existence of a connection in physical theories of spacetime is indispensable in that it naturally incorporates the inertial properties of spacetime. Chapter C.1 deals with the connection and its properties in detail.

The metric brings in the notions of distance and angles on a spacetime manifold, which are crucial for the operational definition of most physical measurements in space and time, namely those that are beyond mere counting procedures as in charge conservation and simple (Lorentz) force measurements. Clocks and rulers come into play, forming a firm basis for the description and an understanding of most physical observations. An important characteristic of the metric is its deep relation to certain duality operators. The relevant mathematical material, which is later heavily used in Part D, is displayed in Chapter C.2.

C.1

Linear connection

“...the essential achievement of general relativity, namely to overcome “rigid” space (i.e., the inertial frame), is *only indirectly* connected with the introduction of a Riemannian metric. The directly relevant conceptual element is the “displacement field” (Γ_{ik}^l), which expresses the infinitesimal displacement of vectors. It is this which replaces the parallelism of spatially arbitrarily separated vectors fixed by the inertial frame (i.e., the equality of corresponding components) by an infinitesimal operation. This makes it possible to construct tensors by differentiation and hence to dispense with the introduction of “rigid” space (the inertial frame). In the face of this, it seems to be of secondary importance in some sense that some particular Γ field can be deduced from a Riemannian metric ...”

A. Einstein (1955 April 04)¹

C.1.1 Covariant differentiation of tensor fields

The change of scalar functions f along a vector u is described by the directional derivative $\partial_u f$. The generalization of this notion from scalars f to tensors T is provided by the covariant differentiation $\nabla_u T$.

¹Translation by F. Gronwald, D. Hartley, and F.W. Hehl from the German original: See the preface in [14].

When calculating a *directional derivative* of a function $f(x)$ along a vector field u , one has to know the values of $f(x)$ at different points on the integral lines of u . With the standard definition, which involves taking a limit of the separation between points, the directional derivative reads

$$\begin{aligned}\partial_u f &:= u \lrcorner df = df(u) \\ &= u^i(x) \frac{\partial f(x)}{\partial x^i} \quad \text{in local coordinates } \{x^i\}.\end{aligned}\tag{C.1.1}$$

Obviously, (C.1.1) describes a map: $T_0^0(X) \times T_0^1(X) \rightarrow T_0^0(X)$ of scalar fields into scalar fields, i.e., tensors of type $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ are mapped again into $T_0^0(X) = C(X)$. This map has simple properties: 1) $\mathbb{R}$ -linearity, 2) $C(X)$ -linearity with respect to u , i.e., $\partial_{gu+hv} f = g\partial_u f + h\partial_v f$, 3) additivity $\partial_u(f+g) = \partial_u f + \partial_u g$.

In order to generalize the directional derivative to arbitrary tensor fields of type $\begin{bmatrix} p \\ q \end{bmatrix}$, one needs a recipe for comparing tensor quantities at two different points of a manifold. This is provided by an additional structure on X called the *linear connection*. The linear connection or, equivalently, the covariant differentiation is necessary in order to formulate differential equations for various physical fields such as the Einstein equation for the gravitational field or the Navier–Stokes equation of hydrodynamics.

In line with the directional derivative, a *covariant differentiation* ∇ is defined as a smooth $\mathbb{R}$ -linear map

$$\nabla : T_q^p(X) \times T_0^1(X) \longrightarrow T_q^p(X) \tag{C.1.2}$$

which to any vector field $u \in T_0^1(X)$ and to any tensor field $T \in T_q^p(X)$ of type $\begin{bmatrix} p \\ q \end{bmatrix}$ assigns a tensor field $\nabla_u T \in T_q^p(X)$ of type $\begin{bmatrix} p \\ q \end{bmatrix}$ that satisfies the following properties:

- 1) $C(X)$ -linearity with respect to u ,

$$\nabla_{fu+gv} T = f\nabla_u T + g\nabla_v T, \tag{C.1.3}$$

- 2) additivity with respect to T ,

$$\nabla_u(T+S) = \nabla_u T + \nabla_u S, \tag{C.1.4}$$

- 3) for a scalar field f a directional derivative is recovered,

$$\nabla_u f = u \lrcorner df, \tag{C.1.5}$$

- 4) the Leibniz rule with respect to the tensor product,

$$\nabla_u(T \otimes S) = \nabla_u T \otimes S + T \otimes \nabla_u S, \tag{C.1.6}$$

- 5) the Leibniz rule with respect to the interior product,

$$\nabla_u(v \lrcorner \omega) = (\nabla_u v) \lrcorner \omega + v \lrcorner \nabla_u \omega, \tag{C.1.7}$$

for all vector fields u, v , all functions f, g , all tensor fields T, S , and all forms ω .

The Lie derivative $\mathcal{L}_u$, defined in Sec. A.2.10, is also a map $T_q^p(X) \times T_0^1(X) \rightarrow T_q^p(X)$. The properties of the covariant differentiation 2)–5) are the same as those of the Lie derivative, cf. (C.1.4)–(C.1.7) with (A.2.56), (A.2.53), (A.2.58) and (A.2.55), respectively. The property (C.1.3) is, however, somewhat different from the corresponding property of the Lie derivative (A.2.57), which can be visualized, for example, by substituting $u \rightarrow fu$ in (A.2.57). This difference reflects the fact that the definition of $\mathcal{L}_u T$ at a given point $x \in X$ makes use of u in the neighborhood of this point, whereas in order to define $\nabla_u T$ at x one has to know only the value of u at x .

C.1.2 Linear connection 1-forms

The difference between covariant and partial differentiation of a tensor field is determined by the linear connection 1-forms Γ_i^j . They show up in the action of ∇_u on a frame ∂_i and supply a constructive realization of the covariant differentiation ∇_u of an arbitrary tensor field.

Consider the chart U_1 with the local coordinates x^i that contains a point $x \in U_1 \subset X$. Take the natural basis ∂_i at x . The covariant differentiation ∇_u of the vectors ∂_i with respect to an arbitrary vector field $u = u^k \partial_k$ reads

$$\nabla_u \partial_i = u^k \nabla_{\partial_k} \partial_i. \quad (\text{C.1.8})$$

Here we used the property (C.1.3). These n vector fields can be decomposed with respect to the coordinate frame ∂_i :

$$\nabla_u \partial_i := \Gamma_i^j(u) \partial_j. \quad (\text{C.1.9})$$

The connection 1-forms

$$\Gamma_i^j = \Gamma_{ki}^j dx^k \quad (\text{C.1.10})$$

can be read off with their components Γ_{ki}^j . Suppose a different chart U_2 with the local coordinates $x^{i'}$ intersects with U_1 , and the point $x \in U_1 \cap U_2$ belongs to an intersection of the two charts. Then the connection 1-forms satisfy the consistency condition

$$\Gamma_{i',j'}^{i'}(x') = \frac{\partial x^i}{\partial x^{i'}} \frac{\partial x^{j'}}{\partial x^j} \Gamma_i^j(x) + \frac{\partial x^{j'}}{\partial x^k} d \frac{\partial x^k}{\partial x^{i'}} \quad (\text{C.1.11})$$

everywhere in the intersection of the local charts $U_1 \cap U_2$.

Now, making use of properties 1)–5), one can describe a covariant differentiation of an arbitrary $\begin{bmatrix} p \\ q \end{bmatrix}$ tensor field

$$T = T^{i_1 \dots i_p}_{j_1 \dots j_q} \partial_{i_1} \otimes \dots \otimes \partial_{i_p} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_q} \quad (\text{C.1.12})$$

in terms of the connection 1-forms. Namely, we have explicitly the $\begin{bmatrix} p \\ q \end{bmatrix}$ tensor field

$$\nabla_u T = u \lrcorner (DT^{i_1 \dots i_p}_{j_1 \dots j_q}) \partial_{i_1} \otimes \dots \otimes \partial_{i_p} \otimes dx^{j_1} \otimes \dots \otimes dx^{j_q}, \quad (\text{C.1.13})$$

where the *covariant differential* of the natural tensor components is introduced by

$$\begin{aligned} DT^{i_1 \dots i_p}_{j_1 \dots j_q} &:= d(T^{i_1 \dots i_p}_{j_1 \dots j_q}) \\ &+ \Gamma_k^{i_1} T^{k i_2 \dots i_p}_{j_1 \dots j_q} + \dots + \Gamma_k^{i_p} T^{i_1 \dots i_{p-1} k}_{j_1 \dots j_q} \\ &- \Gamma_{j_1}^k T^{i_1 \dots i_p}_{k j_2 \dots j_q} - \dots - \Gamma_{j_q}^k T^{i_1 \dots i_p}_{j_1 \dots j_{q-1} k}. \end{aligned} \quad (\text{C.1.14})$$

The step in (C.1.9) can be generalized to an arbitrary frame e_α . Its covariant differentiation with respect to a vector field u reads

$$\nabla_u e_\alpha = \Gamma_\alpha^\beta(u) e_\beta, \quad (\text{C.1.15})$$

with the corresponding linear connection 1-forms Γ_α^β . In four dimensions, we have sixteen 1-forms Γ_α^β at our disposal. The components of the connection 1-forms with respect to the coframe ϑ^α are given by

$$\Gamma_\alpha^\beta = \Gamma_{\gamma\alpha}^\beta \vartheta^\gamma \quad \text{or} \quad \nabla_{e_\alpha} e_\beta = \Gamma_{\alpha\beta}^\gamma e_\gamma. \quad (\text{C.1.16})$$

In terms of a local coordinate system $\{x^i\}$,

$$\Gamma_\alpha^\beta = \Gamma_{i\alpha}^\beta dx^i \quad \text{where} \quad \Gamma_{i\alpha}^\beta = \Gamma_\alpha^\beta(\partial_i). \quad (\text{C.1.17})$$

The 1-forms Γ_α^β are not a new independent object: since an arbitrary frame may be decomposed with respect to the coordinate frame, we find, with the help of (A.2.30) and (A.2.31), the simple relation

$$\Gamma_\alpha^\beta = e_j^\beta \Gamma_i^j e_\alpha^i + e_i^\beta d e_\alpha^i. \quad (\text{C.1.18})$$

Under a change of frame that is described by a linear transformation

$$e_{\alpha'} = L_{\alpha'}^\alpha e_\alpha, \quad (\text{C.1.19})$$

the connection 1-forms transform in a nontensorial way,

$$\Gamma_{\alpha'}^{\beta'} = L_{\alpha'}^\alpha L_{\beta'}^{\beta} \Gamma_\alpha^\beta + L_{\gamma}^{\beta'} d L_{\alpha'}^\gamma. \quad (\text{C.1.20})$$

For an infinitesimal linear transformation, $L_\alpha^\beta = \delta_\alpha^\beta + \varepsilon_\alpha^\beta$,

$$\delta e_\alpha = e_{\alpha'} - e_\alpha = \varepsilon_\alpha^\beta e_\beta, \quad (\text{C.1.21})$$

the connection 1-form changes as

$$\delta \Gamma_\alpha^\beta = D \varepsilon_\alpha^\beta = d \varepsilon_\alpha^\beta + \Gamma_\gamma^\beta \varepsilon_\alpha^\gamma - \Gamma_\alpha^\gamma \varepsilon_\gamma^\beta. \quad (\text{C.1.22})$$

Although the transformation law (C.1.20) is inhomogeneous we cannot, on an open set U , achieve $\Gamma_{\alpha'}^{\beta'} = 0$ in general; it can only be done if the curvature (to be introduced later) vanishes in U . At a given point x_0 , however, we can always choose the first derivatives of $L_{\alpha'}^{\alpha}$ contained in $dL_{\alpha'}^{\alpha}$ in such a way that $\Gamma_{\alpha'}^{\beta'}(x_0) = 0$. A frame e_{α} such that

$$\Gamma_{\alpha}^{\beta}(x_0) = 0 \quad \text{at a given point } x_0 \quad (\text{C.1.23})$$

is called *normal* at x_0 . A normal frame is given up to transformations $L_{\alpha'}^{\alpha}$ whose first derivatives vanish at x_0 . This freedom may be used, and we can always choose a coordinate system $\{x^i\}$ such that the frame $e_{\alpha}(x)$ in (C.1.23) is also “trivialized”:

$$e_{\alpha} = \delta_{\alpha}^i \partial_i \quad \text{at a given point } x_0. \quad (\text{C.1.24})$$

Summing up², for a *trivialized frame* we have

$$(e_{\alpha}, \Gamma_{\alpha}^{\beta})^* = (\delta_{\alpha}^i \partial_i, 0) \quad \text{at a given point } x_0. \quad (\text{C.1.25})$$

Despite the fact that the normal frame looks like a coordinate frame (in the sense, e.g., that (C.1.24) shows apparently that the tangent vectors ∂_i of the coordinate frame coincide with the vectors of e_{α} basis at x_0), one cannot, in general, introduce new *local coordinates* in the neighborhood of x_0 in which (C.1.23) is fulfilled.

C.1.3 $\otimes$ Covariant differentiation of a general geometric quantity

What is the covariant differential of a tensor density, for example?

Let us now consider the covariant differentiation of a general geometric quantity that was introduced in Sec. A.1.3. As in Sec. A.2.10, we treat a geometric quantity w as a set of smooth fields w^A on X . These fields are the components of $w = w^A e_A$ with respect to a frame $e_A \in W = \mathbb{R}^N$ in the space of a ρ -representation of the group $GL(n, \mathbb{R})$ of local linear frame transformations (C.1.19). The transformation (C.1.19) of a frame of spacetime acts on the geometric quantity of type ρ by means of the local generalization of (A.1.17),

$$w^A \longrightarrow w^{A'} = \rho_B^{A'}(L^{-1}) w^B, \quad (\text{C.1.26})$$

or in the infinitesimal case (C.1.21),

$$\delta w^A = -\varepsilon_{\alpha}^{\beta} \rho_B^{A\alpha} w^B. \quad (\text{C.1.27})$$

²See von der Heyde [10], Hartley [9], and Iliev [11].

The generator matrix $\rho_B{}^A{}_\alpha{}^\beta$ was introduced in (A.2.66) when we discussed the Lie derivative of geometric quantities of type ρ .

A covariant differentiation for geometric quantities of type ρ is introduced as a natural generalization of the map (C.1.2) with all the properties 1)–5) preserved: The covariant differentiation for a W -frame reads

$$\nabla_u e_A = \rho_A{}^B{}_\alpha{}^\beta \Gamma_\alpha{}^\beta(u) e_B, \quad (\text{C.1.28})$$

whereas for an arbitrary geometric quantity $w = w^A e_A$ of type ρ , eqs. (C.1.13), (C.1.14) are replaced by

$$\nabla_u w = u \lrcorner (Dw^A) e_A, \quad \text{with} \quad Dw^A := dw^A + \rho_B{}^A{}_\alpha{}^\beta \Gamma_\alpha{}^\beta w^B. \quad (\text{C.1.29})$$

The general formula (C.1.29) is consistent with the covariant derivative of usual tensor fields when the latter are treated as a geometric quantity of a special kind; one can compare this with examples 2), 3) in Sec. A.1.3.

Two simple applications of this general technique are in order. First, we recall that the Levi-Civita symbols have the same values with respect to all frames, see (A.1.64). This means that they are geometric quantities of the type $\rho = \text{id}$ (identity transformation) or, plainly speaking, $\delta\epsilon^{\alpha_1 \dots \alpha_n} = 0$. Comparing this with (C.1.27) and (C.1.28), we conclude that

$$D\epsilon^{\alpha_1 \dots \alpha_n} = 0 \quad (\text{C.1.30})$$

for an arbitrary connection.

Second, let us take a scalar density $\mathcal{S}$ of weight w . This geometric quantity is described by the transformation law (A.1.56) or, in the infinitesimal form, by

$$\delta\mathcal{S} = w \varepsilon_\alpha{}^\alpha \mathcal{S}. \quad (\text{C.1.31})$$

Comparing with (C.1.27), we find $\rho^\alpha{}_\beta = -w \delta_\beta^\alpha$, and hence (C.1.29) yields

$$D\mathcal{S} = d\mathcal{S} - w \Gamma_\alpha{}^\alpha \mathcal{S}. \quad (\text{C.1.32})$$

If we generalize this to a tensor density $\mathcal{T}_{\alpha \dots}{}^{\beta \dots}$, we find

$$D\mathcal{T}_{\alpha \dots}{}^{\beta \dots} = d\mathcal{T}_{\alpha \dots}{}^{\beta \dots} - \Gamma_\alpha{}^\mu \mathcal{T}_{\mu \dots}{}^{\beta \dots} - \dots + \Gamma_\mu{}^\beta \mathcal{T}_{\alpha \dots}{}^{\mu \dots} + \dots - w \Gamma_\mu{}^\mu \mathcal{T}_{\alpha \dots}{}^{\beta \dots}. \quad (\text{C.1.33})$$

C.1.4 Parallel transport

By means of the covariant differentiation ∇_u , a tensor can be parallelly transported along a curve on a manifold. This provides a convenient tool for comparing values of the tensor field at different points of the manifold.

A connection enables us to define *parallel transport* of a tensor along a curve. A differentiable curve σ on X is a smooth map

$$\begin{aligned}\sigma : (a, b) &\rightarrow X \\ t &\mapsto x(t),\end{aligned}$$

where (a, b) is an interval in $\mathbb{R}$. In local coordinates $\{x^i\}$, the tangent vector to the curve $\sigma = \{x^i(t)\}$ is $u^i = dx^i/dt$. A tensor field T is said to be *parallelly transported* along σ if

$$\nabla_u T = 0 \quad (\text{C.1.34})$$

along σ . Taking into account (C.1.14), we get for T

$$\begin{aligned}\frac{d}{dt} T^{i_1 \dots i_p}_{j_1 \dots j_q}(x(t)) + \frac{dx^l}{dt} \left(\Gamma_{lk}^{i_1}(x(t)) T^{k i_2 \dots i_p}_{j_1 \dots j_q}(x(t)) + \dots \right. \\ \left. - \Gamma_{l j_q}^k(x(t)) T^{i_1 \dots i_p}_{j_1 \dots j_{q-1} k}(x(t)) \right) = 0. \quad (\text{C.1.35})\end{aligned}$$

If $0 \in (a, b)$ and $T(\sigma(0))$ is given, then there exists (at least locally) a unique solution $T(\sigma(t))$ of this linear ordinary differential equation for $t \in (a, b)$. Therefore, we have a linear map of tensors of type $\begin{bmatrix} p \\ q \end{bmatrix}$ at the point $\sigma(0)$ to tensors of type $\begin{bmatrix} p \\ q \end{bmatrix}$ at the point $\sigma(t)$.

Taking $T = u$, we obtain a differential equation for *autoparallels*:

$$\nabla_u u = 0 \quad \text{or} \quad \frac{d^2 x^i}{dt^2} + \Gamma_{jk}^i(x(t)) \frac{dx^j}{dt} \frac{dx^k}{dt} = 0. \quad (\text{C.1.36})$$

C.1.5 $\otimes$ Torsion and curvature

Torsion and curvature both measure the deviation from the flat spacetime geometry (of special relativity). When both of them are zero, one can globally define the trivialized frame (C.1.25) all over the spacetime manifold.

We now want to associate with a connection two tensor fields: torsion and curvature. As we have seen above, the connection form can always be transformed to zero at one given point. However, torsion and curvature will in general give a nonvanishing characterization of the connection at this point.

The *torsion* of a connection ∇ is a map T that assigns to each pair of vector fields u and v a vector field $T(u, v)$ by

$$T(u, v) := \nabla_u v - \nabla_v u - [u, v]. \quad (\text{C.1.37})$$

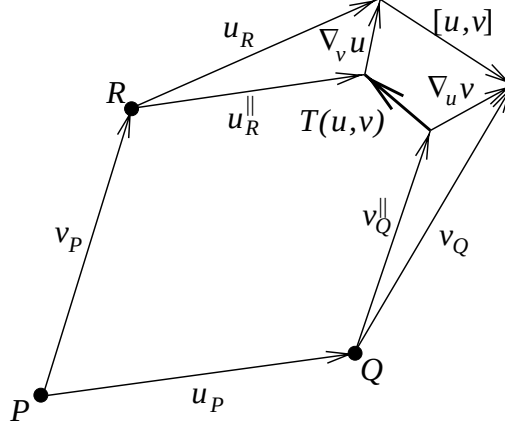


Figure C.1.1: On the geometrical interpretation of torsion: a closure failure of infinitesimal displacements. This is a schematic view. Note that R and Q are infinitesimally near to P .

The commutator $[u, v]$ has been defined in (A.2.5). One can straightforwardly verify the tensor character of $T(u, v)$ so that its value at any given point is determined by the values of u and v at that point.

Fig. C.1.1 illustrates schematically the geometrical meaning of torsion: Choose two vectors v and u at a point $P \in X$. Transport u parallelly along v to the point R and likewise v along u to the point Q . If the resulting parallelogram is *broken*, i.e., if it has a gap or a closure failure, then the connection carries a torsion. Such situations occur in the continuum theory of dislocations.³

Since the torsion $T(u, v)$ is a vector field, one can expand it with respect to a local frame,

$$T(u, v) = T^\alpha(u, v) e_\alpha. \quad (\text{C.1.38})$$

By construction, the coefficients $T^\alpha(u, v)$ of this expansion are 2-forms, i.e., functions that assign real numbers to every pair of the vector fields u, v . Taking the vectors of a frame,

$$T(e_\beta, e_\gamma) = T^\alpha(e_\beta, e_\gamma) e_\alpha = T_{\beta\gamma}^\alpha e_\alpha, \quad (\text{C.1.39})$$

we can read off the coefficients of this *vector-valued torsion 2-form*:

$$T^\alpha = \frac{1}{2} T_{\mu\nu}^\alpha \vartheta^\mu \wedge \vartheta^\nu = \frac{1}{2} T_{ij}^\alpha dx^i \wedge dx^j. \quad (\text{C.1.40})$$

³See Kröner [13] and references given there.

The explicit form of these coefficients is obtained directly from the definition (C.1.37) which we evaluate with respect to a frame e_α ,

$$T^\alpha(e_\beta, e_\gamma) e_\alpha = \nabla_{e_\beta} e_\gamma - \nabla_{e_\gamma} e_\beta - [e_\beta, e_\gamma]^\alpha e_\alpha. \quad (\text{C.1.41})$$

The first two terms on the right-hand side bring in the connection coefficients (C.1.16), whereas the commutator can be rewritten in terms of the object of anholonomy, see (A.2.36). Accordingly, we find for the components of the torsion

$$T_{\beta\gamma}{}^\alpha = \Gamma_{\beta\gamma}{}^\alpha - \Gamma_{\gamma\beta}{}^\alpha + C_{\beta\gamma}{}^\alpha. \quad (\text{C.1.42})$$

The torsion 2-form T^α allows one to define the 1-form $e_\alpha \lrcorner T^\beta$. If added to the connection 1-form $\Gamma_\alpha{}^\beta$, it is again a connection. We call it the *transposed connection*

$$\begin{aligned} \widehat{\Gamma}_\alpha{}^\beta &:= \Gamma_\alpha{}^\beta + e_\alpha \lrcorner T^\beta &= (\Gamma_{\gamma\alpha}{}^\beta + T_{\gamma\alpha}{}^\beta) \vartheta^\gamma \\ & &= (\Gamma_{\alpha\gamma}{}^\beta + C_{\alpha\gamma}{}^\beta) \vartheta^\gamma, \end{aligned} \quad (\text{C.1.43})$$

since in a natural frame, i.e. for $C_{\alpha\gamma}{}^\beta = 0$, the indices of the components of $\widehat{\Gamma}_\alpha{}^\beta$ are transposed with respect to that of $\Gamma_\alpha{}^\beta$.

The *curvature* of a connection ∇ is a map that assigns to each pair of vector fields u and v a linear transformation $R(u, v) : X_x \rightarrow X_x$ of the tangent space at an arbitrary point x by

$$R(u, v)w := \nabla_u \nabla_v w - \nabla_v \nabla_u w - \nabla_{[u, v]} w. \quad (\text{C.1.44})$$

One can verify the tensor character of the curvature so that the value of $R(u, v)w$ at any given point depends only on the values of vector fields u , v , and w at that point.

Analogously to torsion, we can expand the four vector fields $R(u, v)e_\alpha$ with respect to a local vector frame,

$$R(u, v) e_\alpha = R_\alpha{}^\beta(u, v) e_\beta. \quad (\text{C.1.45})$$

Thereby the curvature 2-form $R_\alpha{}^\beta$ is defined. Similarly to (C.1.40), we may express these 2-forms with respect to a coframe ϑ^α or a coordinate frame dx^i as follows:

$$R_\alpha{}^\beta = \frac{1}{2} R_{\mu\nu\alpha}{}^\beta \vartheta^\mu \wedge \vartheta^\nu = \frac{1}{2} R_{ij\alpha}{}^\beta dx^i \wedge dx^j. \quad (\text{C.1.46})$$

Taking (C.1.45) with respect to a frame, one finds the components of the curvature 2-form,

$$R(e_\mu, e_\nu) e_\alpha = R_{\mu\nu\alpha}{}^\beta e_\beta \quad \text{and} \quad R(\partial_i, \partial_j) e_\alpha = R_{ij\alpha}{}^\beta e_\beta. \quad (\text{C.1.47})$$

Thus, by (C.1.44),

$$R_{\mu\nu\alpha}{}^\beta = \partial_\mu \Gamma_{\nu\alpha}{}^\beta - \partial_\nu \Gamma_{\mu\alpha}{}^\beta + \Gamma_{\mu\sigma}{}^\beta \Gamma_{\nu\alpha}{}^\sigma - \Gamma_{\nu\sigma}{}^\beta \Gamma_{\mu\alpha}{}^\sigma + C_{\mu\nu}{}^\sigma \Gamma_{\sigma\alpha}{}^\beta \quad (\text{C.1.48})$$

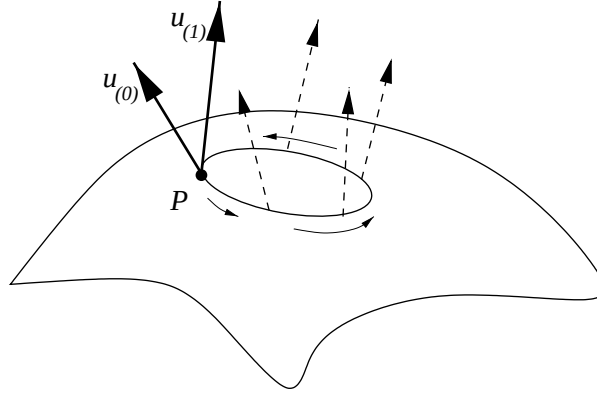


Figure C.1.2: On the geometrical interpretation of curvature: parallel transport of a vector around a closed loop.

and

$$R_{ij\alpha}{}^{\beta} = \partial_i \Gamma_{j\alpha}{}^{\beta} - \partial_j \Gamma_{i\alpha}{}^{\beta} + \Gamma_{i\sigma}{}^{\beta} \Gamma_{j\alpha}{}^{\sigma} - \Gamma_{j\sigma}{}^{\beta} \Gamma_{i\alpha}{}^{\sigma}. \quad (\text{C.1.49})$$

Here we introduced the abbreviation $\partial_{\mu} := e^i{}_{\mu} \partial_i$.

The curvature 2-form $R_{\alpha}{}^{\beta}$ can be contracted by means of the frame e_{β} . In this way we find the Ricci 1-form

$$\text{Ric}_{\alpha} := e_{\beta} \lrcorner R_{\alpha}{}^{\beta} = \text{Ric}_{\alpha\beta} \vartheta^{\beta}. \quad (\text{C.1.50})$$

Using (C.1.46), we immediately find

$$\text{Ric}_{\alpha\beta} = R_{\gamma\alpha\beta}{}^{\gamma}. \quad (\text{C.1.51})$$

The geometrical meaning of the curvature is revealed when we consider a parallel transport of a vector along a closed curve in X (see Fig. C.1.2). Let $\sigma : \{x^i(t)\}$, $0 \leq t \leq 1$, be a smooth curve that starts and ends at a point $P = x^i(0) = x^i(1)$ (in other words, σ is a 1-cycle). Taking a vector $u_{(0)}$ at P and transporting it parallelly along σ (which technically reduces to the solution of a differential equation (C.1.35)), one finds at the return point $x^i(1)$ a vector $u_{(1)}$ that differs from $u_{(0)}$. The difference is determined by the curvature,

$$\Delta u^{\alpha} = u_{(1)}^{\alpha} - u_{(0)}^{\alpha} = -\Delta S R_{\beta}{}^{\alpha} u^{\beta}, \quad (\text{C.1.52})$$

where ΔS is an infinitesimal 2-dimensional surface element that is bound by σ , i.e., $\sigma = \partial \Delta S$.

When the curvature is zero everywhere in X , we call such a manifold a *flat affine space with torsion* (or a teleparallelism spacetime). If the torsion vanishes additionally, we speak of a flat affine space.

Affine means that the connection ∇ is still there and it allows one to compare tensors at different points. Clearly, the curvature is vanishing for an everywhere

trivial connection form $\Gamma_\alpha^\beta = 0$. However, as we know, the components Γ_α^β depend on the frame field and, in general, if we have curvature, it is impossible to choose frames e_α in such a way that $\Gamma_\alpha^\beta = 0$ on the whole X . On a flat affine space, the components of a vector do not change after a parallel transport around a closed loop; in other words, parallel transport is integrable. Curvature is a measure of the deviation from the flat case.

C.1.6 $\otimes$ Cartan's geometric interpretation of torsion and curvature

On a manifold with a linear connection, the notion of a position vector can be defined along a curve. If we transport the position vector around an infinitesimal closed loop, it is subject to a translation and a linear transformation. The translation reveals the torsion and the linear transformation the curvature of the manifold.

Let us start, following Cartan, with the flat affine space in which a connection ∇ has zero torsion and zero curvature. In such a manifold, we may define an affine *position vector field* (or radius vector field) r as one that satisfies the equation

$$\nabla_v r = v \quad (\text{C.1.53})$$

for all vector fields v . With respect to the local coordinates $\{x^i\}$, $r = r^i \partial_i$, and (C.1.53) is a system of sixteen partial differential equations for the four functions $r^i(x)$, namely

$$D r^j = dx^j, \quad \text{or} \quad \partial_i r^j(x) + \Gamma_{ik}^j(x) r^k(x) = \delta_i^j. \quad (\text{C.1.54})$$

In flat affine space, a coordinate basis can always be chosen, at least within one chart, in such a way that $\Gamma_{ik}^j = 0$, and then the equation (C.1.53) or (C.1.54) is simply $\partial_j r^i = \delta_j^i$. The solution is $r^i = x^i + A^i$ where A^i a constant vector, so that r^i is, indeed, the position (or radius) vector of x^i with respect to an origin $x^i = -A^i$.

In an affine spacetime, the integrability condition for (C.1.53) is

$$\nabla_v \nabla_w r - \nabla_w \nabla_v r - \nabla_{[v,w]} r = \nabla_v w - \nabla_w v - [v, w], \quad (\text{C.1.55})$$

or

$$R(v, w)r - T(v, w) = 0, \quad (\text{C.1.56})$$

for all vector fields v, w . Hence a sufficient condition for the existence of global radius vector fields r is

$$R(v, w) = 0 \quad \text{and} \quad T(v, w) = 0 \quad (\text{C.1.57})$$

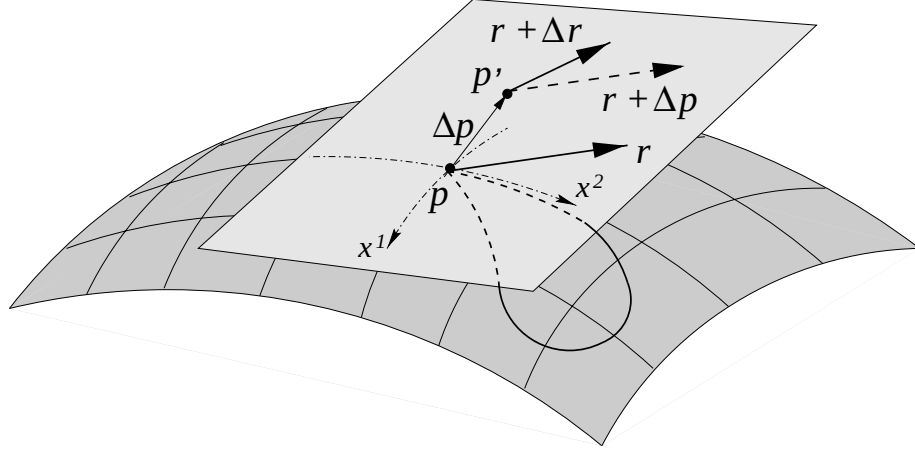


Figure C.1.3: Cartan's position vector on manifold with curvature and torsion: Affine transport of a vector r around an infinitesimal loop. Here p and p' denote the initial and final positions of the position vector in the affine tangent space.

for arbitrary v, w , i.e., vanishing curvature and torsion.

In a general manifold, when torsion and curvature are nonzero, the position vector field does not exist on X . Nevertheless, it is possible to define a position vector *along a curve*. This object turns out to be extremely useful for revealing the geometrical meaning of torsion and curvature.

Let $p \in X$ be an arbitrary point and $\sigma = \{x^i(t)\}$, $t \geq 0$, be a smooth curve that starts at p , i.e., $x^i(0) = x_p^i$. We now attach at p an affine tangent space or, to put it differently, we consider the tangent space X_p at p as an n -dimensional *affine vector space*. Recall that in an affine vector space, each element (vector) is given by its origin and its components with respect to some fixed basis. We construct Cartan's position vector as a map that assigns to each point of a curve σ a vector in the affine tangent space X_p . Geometrically such a construction can be conveniently understood as a generalized development map that is defined by "rolling" the tangent space along a given curve.

The defining equation of the position vector is again (C.1.53), but this time v is not an arbitrary vector field but tangent to the curve under consideration, i.e. $v = (dx^i/dt)\partial_i$. Substituting this into (C.1.53), we get

$$\frac{dr^\alpha(t)}{dt} = \left[e_i^\alpha(x(t)) - \Gamma_{i\beta}^\alpha(x(t)) r^\beta(t) \right] \frac{dx^i}{dt}. \quad (\text{C.1.58})$$

As t increases, one "moves" along the curve σ and the functions $r^\alpha(t)$ always describe the components of the position vector in the affine tangent space at the *fixed original* point p . Thus, for example, a displacement along the curve from x^i to $x^i + d\xi^i$ yields a change of the position vector

$$dr^\alpha = e_i^\alpha(x) d\xi^i - \Gamma_{i\beta}^\alpha(x) r^\beta(x) d\xi^i. \quad (\text{C.1.59})$$

Following Cartan, we may interpret this equation as telling us that the position vector map consists of a *translation* $e_i^\alpha d\xi^i$ and a *linear transformation* $-\Gamma_{i\beta}^\alpha d\xi^i r^\beta$ in the affine tangent space at p .

Let us now consider a closed curve σ such that $x^i(1) = x_p^i$ (see Fig. C.1.3). Then, upon integrating around σ , it is found that the total change in r^α is given by

$$\Delta r^\alpha = \Delta S (T^\alpha - R_\beta^\alpha r^\beta) \quad (\text{C.1.60})$$

where ΔS is the 2-dimensional infinitesimal surface enclosed by σ . Thus, in going around the infinitesimal closed loop σ , the position vector r in the affine tangent space at p undergoes a translation and a linear transformation of the same order of magnitude as the area of ΔS . The translation is determined by the torsion

$$\Delta p^\alpha = \Delta S T^\alpha, \quad (\text{C.1.61})$$

whereas the linear transformation is determined by the curvature (It is instructive to compare this with the change of a vector under the parallel transport (C.1.52).).

C.1.7 $\otimes$ Covariant exterior derivative

If an exterior form is generalized to a tensor-valued exterior form, then the usual definition of the exterior derivative can be naturally extended to the covariant exterior derivative. Covariant exterior derivatives of torsion and curvature are involved in the two Bianchi identities.

The torsion 2-form (C.1.38), (C.1.40) and the curvature 2-form (C.1.45), (C.1.46) are examples of tensor-valued p -forms, that is, of generalized geometric quantities. For such objects we need to introduce the notion of covariant exterior derivative which shares the properties of a covariant derivative of a geometric quantity and of an exterior derivative of a scalar-valued form.

Let φ^A be an arbitrary p -form of type ρ . It can be written as a sum of decomposable p -forms of type ρ , namely $\varphi^A = w^A \omega$ where w^A is a scalar of type ρ and ω a usual exterior p -form. For such a form we define

$$D\varphi^A := (\nabla w^A) \omega + w^A d\omega \quad (\text{C.1.62})$$

and extend this definition by $\mathbb{R}$ -linearity to arbitrary p -forms of type ρ . Using (C.1.29), it is straightforward to obtain the general formula

$$\boxed{D\varphi^A = d\varphi^A + \rho_B^{A\alpha} \Gamma_\alpha^\beta \wedge \varphi^B} \quad (\text{C.1.63})$$

and to prove that D satisfies the Leibniz rule

$$D(\varphi^A \wedge \psi^B) = D\varphi^A \wedge \psi^B + (-1)^p \varphi^A \wedge D\psi^B, \quad (\text{C.1.64})$$

where p is the degree of φ^A .

Unlike the usual exterior derivative, which satisfies $dd = 0$, the covariant exterior derivative is no longer nilpotent:

$$DD\varphi^A = \rho_B{}^A{}_\alpha{}_\beta R_\alpha{}^\beta \wedge \varphi^B. \quad (\text{C.1.65})$$

The simplest proof makes use of the normal frame (C.1.23) in which $D\varphi^A \stackrel{*}{=} d\varphi^A$, $R_\alpha{}^\beta \stackrel{*}{=} d\Gamma_\alpha{}^\beta$. We choose a normal frame and differentiate (C.1.63). Since the resulting formula is an equality of two $(p+2)$ -forms of type ρ , it holds in an arbitrary frame. The relation (C.1.65) is called the *Ricci identity*.

Now we can appreciably simplify all calculations involving frame, connection, curvature, and torsion. At first, noticing that the coframe ϑ^α is a 1-form of vector type, we recover the torsion 2-form (C.1.38), (C.1.40) as a covariant exterior derivative

$$T^\alpha = D\vartheta^\alpha = d\vartheta^\alpha + \Gamma_\beta{}^\alpha \wedge \vartheta^\beta. \quad (\text{C.1.66})$$

This equation is often called the first (Cartan) structure equation. Applying (C.1.65), we obtain the *1st Bianchi identity*:

$$\boxed{DT^\alpha \equiv R_\beta{}^\alpha \wedge \vartheta^\beta}. \quad (\text{C.1.67})$$

Analogously, after recognizing the curvature 2-form as a generalized 2-form of tensor type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, we immediately rewrite (C.1.45), (C.1.46) as

$$R_\alpha{}^\beta = d\Gamma_\alpha{}^\beta + \Gamma_\sigma{}^\beta \wedge \Gamma_\alpha{}^\sigma, \quad (\text{C.1.68})$$

which is called the second (Cartan) structure equation. Using again the trick with the normal frame, we obtain the *2nd Bianchi identity*:

$$\boxed{DR_\alpha{}^\beta \equiv 0}. \quad (\text{C.1.69})$$

Finally, we can link up the notions of Lie derivative and covariant derivative. For decomposable p -forms of type ρ , $\varphi^A = w^A \omega$, where w^A is a scalar of type ρ and ω a p -form, we define the *covariant* Lie derivative as

$$\mathbb{L}_u \varphi^A := (\nabla_u w^A) \omega + w^A \mathcal{L}_u \omega \quad (\text{C.1.70})$$

and extend this definition by $\mathbb{R}$ -linearity to arbitrary p -forms of type ρ . It is an interesting exercise to show that

$$\mathbb{L}_u \varphi^A = u \lrcorner D\varphi^A + D(u \lrcorner \varphi^A). \quad (\text{C.1.71})$$

C.1.8 $\otimes$ The forms `o(a)`, `conn1(a,b)`, `torsion2(a)`, `curv2(a,b)`

We come back to our Excalc programming. We put $n = 4$. On each differential manifold, we can specify an arbitrary coframe field ϑ^α , in Excalc `o(a)`. Excalc

is made familiar with $o(a)$ by means of the coframe statement as described in Sec. A.2.11. Moreover, since we introduced a linear connection 1-form Γ_α^β , we do the same in Excalc with `pform conn1(a,b)=1`; Then it is straightforward to implement the torsion and curvature 2-forms T^α and R_α^β by means of the structure equations (C.1.66) and (C.1.68), respectively:

```
pform torsion2(a)=2, curv2(a,b)=2;      % preceded by coframe command

torsion2(a):=d o(a) + conn1(-b,a)^o(b);
curv2(-a,b):=d conn1(-a,b) + conn1(-a,c)^conn1(-c,b);
```

In Excalc, the trace of the torsion $T := e_\alpha \lrcorner T^\alpha$ reads `e(-a) _| torsion2(a)`, and the corresponding trace part of the torsion $T^\alpha = \frac{1}{3} \vartheta^\alpha \wedge T$ becomes

```
pform trator2(a);
trator2(a) :=o(a)^(e(-b) _|torsion2(b));
```

The Ricci 1-form $Ric_\alpha := e_\beta \lrcorner R_\alpha^\beta$, see (C.1.50), is encoded as

```
pform ricci1(a)=1;
ricci1(-a) :=e(-b) _| curv2(-a,b);
```

Weyl's (purely non-Riemannian) segmental (or dilational) curvature 2-form $\frac{1}{4} \delta_\alpha^\beta R_\gamma^\gamma$ is the other generally covariant contraction of the curvature. We have

```
pform delta(a,b)=0, dilcurv2(a,b)=2;

delta(-0,0):=delta(-1,1):=delta(-3,3):=delta(-1,1):=1;
dilcurv2(-a,b):=delta(-a,b)*curv2(-c,c)/4;
```

These are the quantities that play a role in a 4-dimensional differential manifold with a prescribed connection. The corresponding Excalc expressions defined here can be put into an executable Excalc program. However, first we want to get access to a possible metric of this manifold.

C.2

Metric

Although in our axiomatic discussion of electrodynamics in Part B we adhered to the connection-free and metric-free point of view, the notions of connection and metric are unavoidable in the end. In the previous Chapter C.1, we gave the fundamentals of the geometry of manifolds equipped with a linear connection. Here we discuss the metric.

In Special Relativity theory (SR) and in the corresponding classical field theory in flat spacetime, the Lorentzian metric enters as a fundamental *absolute* element. In particular, all physical particles are defined in terms of representations of the Poincaré (or inhomogeneous Lorentz) group which has a metric built in from the very beginning.

In General Relativity theory (GR), the metric field is upgraded to the status of a *gravitational potential*. In particular, the Einstein field equation is formulated in terms of a Riemannian metric with Lorentz signature carrying on its right-hand side the symmetric (Hilbert or metric) energy-momentum tensor as a material source. The physical significance of the spacetime metric lies in the fact that it determines intervals ds^2 between events in spacetime and, furthermore, establishes the causal structure of spacetime.

It is important to realize that the two geometrical structures – the connection and the metric – are a priori absolutely independent from each other. Modern data convincingly demonstrate the validity of Riemannian geometry and Einstein's GR on macroscopic scales where mass (energy-momentum) of matter alone determines the structure of spacetime. However, at high energies, the properties of matter are significantly different, with additional spacetime-related characteristics, such as spin and scale charge coming into play. Corre-

spondingly, one can expect that the geometric structure of spacetime on small distances may deviate from Riemannian geometry.

“In the dilemma whether one should ascribe to the world primarily a metric or an affine structure, the best point of view may be the neutral one which treats the g ’s as well as the Γ ’s as independent state quantities. Then the two sets of equations, which link them together, become laws of nature without attributing a preferential status as definitions to one or the other half.”¹

C.2.1 Metric vector spaces

A metric tensor introduces the length of a vector and an angle between every two vectors. The components of the metric are defined by the values of the scalar products of the basis vectors.

Let us consider a linear vector space V . It is called a *metric vector space* if on V a scalar product is defined as a bilinear symmetric and nondegenerate map

$$\mathbf{g} : V \times V \longrightarrow \mathbb{R}. \quad (\text{C.2.1})$$

In other words, a scalar product is introduced by a *metric tensor* $\mathbf{g}$ of type $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$ that is symmetric, i.e., $\mathbf{g}(u, v) = \mathbf{g}(v, u)$ for all $u, v \in V$, and nondegenerate in the sense that $\mathbf{g}(u, v) = 0$ holds for all v if and only if $u = 0$. The real number

$$\mathbf{g}(u, u) \quad (\text{C.2.2})$$

is called a *length* of a vector u . The metric $\mathbf{g}$ defines a canonical isomorphism of the vector space and its dual,

$$\tilde{\mathbf{g}} : V \longrightarrow V^*, \quad (\text{C.2.3})$$

where the 1-form $\tilde{\mathbf{g}}(u)$, if applied to a vector v , yields

$$\tilde{\mathbf{g}}(u)(v) := \mathbf{g}(u, v), \quad \text{for all } v \in V. \quad (\text{C.2.4})$$

Alternatively, we may write $\tilde{\mathbf{g}}(u) = \mathbf{g}(u, \cdot)$.

In terms of a basis e_α of V and the dual basis ϑ^α of V^* ,

$$\mathbf{g} = g_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta, \quad \text{with } g_{\alpha\beta} := \mathbf{g}(e_\alpha, e_\beta) = g_{\beta\alpha}. \quad (\text{C.2.5})$$

¹“In dem Dilemma, ob man der Welt ursprünglich eine metrische oder eine affine Struktur zuschreiben soll, ist vielleicht der beste Standpunkt der neutrale, der sowohl die g wie die Γ als unabhängige Zustandsgrößen behandelt. Dann werden die beiden Sätze von Gleichungen, welche sie verbinden, zu Naturgesetzen ohne daß die eine oder andere Hälfte als Definitionen eine bevorzugte Stellung bekommen.” H. Weyl: *50 Jahre Relativitätstheorie* [18], our translation.

Thus the isomorphism (C.2.3) is technically reduced to the vertical motion of indices,

$$\tilde{\mathbf{g}}(e_\alpha)(e_\beta) = \mathbf{g}(e_\alpha, e_\beta) = g_{\alpha\beta} = g_{\alpha\gamma} \delta_\beta^\gamma = g_{\alpha\gamma} \vartheta^\gamma(e_\beta). \quad (\text{C.2.6})$$

Accordingly, the basis vectors define the 1-forms via

$$\tilde{\mathbf{g}}(e_\alpha) = g_{\alpha\gamma} \vartheta^\gamma =: \vartheta_\alpha. \quad (\text{C.2.7})$$

Under a change of the basis (A.1.5), the metric coefficients $g_{\alpha\beta}$ transform according to (A.1.11). Recall that a symmetric matrix can always be brought into a diagonal form by a linear transformation. A basis for which

$$g_{\alpha\beta} = \text{diag}(1, \dots, 1, -1, \dots, -1) \quad (\text{C.2.8})$$

is called *orthonormal*. We are mainly interested in 4-dimensional spacetime. Its tangent vector space at each event is Minkowskian. Therefore, from now on, let us take V to be a 4-dimensional Minkowskian vector space unless specified otherwise. The components of the metric tensor with respect to an orthonormal basis are then given by

$$g_{\alpha\beta} = o_{\alpha\beta} := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (\text{C.2.9})$$

C.2.2 $\otimes$ Orthonormal, half-null, and null frames, the coframe statement

A null vector has zero length. A set of null vectors is in many cases a convenient tool for the construction of a special basis in a Minkowski vector space.

The Minkowski (or Lorentz) metric has many interesting “faces” which we mention here only briefly.

Traditionally, in relativity theory the vectors of an *orthonormal* basis are labeled by 0, 1, 2, 3, thus underlining the fundamental difference between e_0 , which has a positive length $g_{00} = \mathbf{g}(e_0, e_0) = 1$, and e_a , $a = 1, 2, 3$, which have negative length $g_{aa} = \mathbf{g}(e_a, e_a) = -1$. In general, a vector $u \in V$ is called *timelike* if $\mathbf{g}(u, u) > 0$, *spacelike* if $\mathbf{g}(u, u) < 0$, and *null* if $\mathbf{g}(u, u) = 0$.

In Excalc one specifies the coframe as the primary quantity. If we use Cartesian coordinates, an orthonormal coframe and frame in Minkowski space read, respectively,

$$\text{coframe } o(0) = dt,$$

```

o(1) = d x ,
o(2) = d y ,
o(3) = d z with
metric g = o(0)*o(0)-o(1)*o(1)-o(2)*o(2)-o(3)*o(3);
frame e;

```

The blank between d and t and so on is necessary! Note that the phrase `with metric g=o(0)*o(0)-o(1)*o(1)-o(2)*o(2)-o(3)*o(3);` in the case of a diagonal metric can also be abbreviated by `with signature 1,-1,-1,-1;`

We recall that, in Excalc, a specific spherically symmetric coframe in a 4-dimensional Riemannian spacetime with Lorentzian signature has already been defined in Sec. B.5.6 in our Maxwell sample program. In general relativity, for the gravitational field of a mass m with an angular momentum per unit mass a , one has an *axially symmetric* metric with a *coframe* like

```

pform rr=0, delsqrt=0, ffsqrt=0$
fdomain rr=rr(rho,theta), delsqrt=delsqrt(rho), ffsqrt=ffsqrt(theta)$

coframe
o(0) = (delsqrt/rr)*(d t-(a0*sin(theta)**2)*d phi),
o(1) = (rr/delsqrt)*d rho,
o(2) = (rr/ffsqrt)*d theta,
o(3) = (ffsqrt/rr)*sin(theta)*(-a0*d t+(rho**2+a0**2)*d phi) with
signature 1,-1,-1,-1$

```

Here `rr`, `delsqrt`, `ffsqrt` are functions to be determined by the Einstein equation. This is an example of a coframe that is a bit more involved.

Starting from an orthonormal basis e_α with respect to which the metric has the standard form (C.2.9), we can build a new frame $e_{\alpha'} = (l, n, e_2, e_3)$ by the linear transformation:

$$l = \frac{1}{\sqrt{2}}(e_0 + e_1), \quad n = \frac{1}{\sqrt{2}}(e_0 - e_1), \quad (\text{C.2.10})$$

and $e_{2'} = e_2, e_{3'} = e_3$. The first two vectors of the new frame are null: $\mathbf{g}(l, l) = \mathbf{g}(n, n) = 0$. Correspondingly, the metric in this *half-null* basis reads

$$g_{\alpha'\beta'} = h_{\alpha'\beta'} := \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (\text{C.2.11})$$

In Excalc, again with Cartesian coordinates, we find

```

coframe h(0) = (d t+d x)/sqrt(2),

```

```

h(1) = (d t-d x)/sqrt(2),
h(2) = d y,
h(3) = d z with
metric hh = h(0)*h(1)+h(1)*h(0)-h(2)*h(2)-h(3)*h(3);

```

You can convince yourself by `displayframe`; and on `nero`; `hh(-a,-b)`; that all has been understood correctly by Excalc.

Following Newman & Penrose, we can further construct two more *null* vectors as the *complex* linear combinations of e_2 and e_3 :

$$m = \frac{1}{\sqrt{2}}(e_2 + i e_3), \quad \bar{m} = \frac{1}{\sqrt{2}}(e_2 - i e_3). \quad (\text{C.2.12})$$

Here i is the imaginary unit, and overbar means the complex conjugation. This transformation leads to the Lorentzian metric in a *null* (Newman–Penrose) basis $e_{\alpha''} = (l, n, m, \bar{m})$:

$$g_{\alpha''\beta''} = n_{\alpha''\beta''} := \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \quad (\text{C.2.13})$$

Such a basis is convenient for investigating the properties of gravitational and electromagnetic waves. In Excalc, we have

```

coframe n(0) = (d t + d x)/sqrt(2),
n(1) = (d t - d x)/sqrt(2),
n(2) = (d y + i*d z)/sqrt(2),
n(3) = (d y - i*d z)/sqrt(2) with
metric nn = n(0)*n(1)+n(1)*n(0)-n(2)*n(3)-n(3)*n(2);

```

In the Newman & Penrose frame, we have two real null legs, namely l and n , and two complex ones, m and $\bar{m}$.

It may be surprising to learn that it is also possible to define a *null symmetric* frame that consists of four *real* null vectors.² We start from an orthonormal basis e_α , with $g(e_\alpha, e_\beta) = o_{\alpha\beta}$, and define the new basis $f_{\tilde{\alpha}}$ according to

$$\begin{aligned} f_{\tilde{0}} &= (\sqrt{3} e_0 + e_1 + e_2 + e_3)/2, \\ f_{\tilde{1}} &= (\sqrt{3} e_0 + e_1 - e_2 - e_3)/2, \\ f_{\tilde{2}} &= (\sqrt{3} e_0 - e_1 + e_2 - e_3)/2, \\ f_{\tilde{3}} &= (\sqrt{3} e_0 - e_1 - e_2 + e_3)/2. \end{aligned} \quad (\text{C.2.14})$$

²Such completely symmetric real null frames had first been used by B. Coll and J.A. Morales [4] (see also Finkelstein and Gibbs [7]) and the corresponding coordinates even earlier by G.H. Derrick [5] and Coll [2]. Coll [3] also recognized the role of such frames in the framework of the Global Positioning System (GPS), see also Rovelli [15] and Blagojević et al.[1].

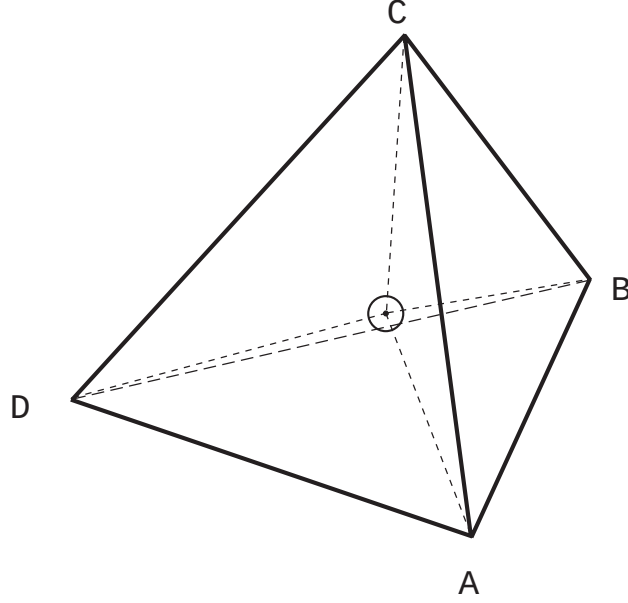


Figure C.2.1: A tetrahedron that represents the real symmetric null frame in 3-dimensional space. At time $t = 0$, light is emitted in O . It reaches the points A, B, C , and D at $t = \sqrt{3}/2$. The event $(0, O)$, together with the events $(\sqrt{3}/2, A)$, $(\sqrt{3}/2, B)$, etc., determine the four real null vectors.

Since $\mathbf{g}(f_{\tilde{\alpha}}, f_{\tilde{\alpha}}) = 0$ for all $\tilde{\alpha}$, the null symmetric frame consists solely of real nonorthogonal null vectors. The metric with respect to this frame reads

$$g_{\tilde{\alpha}\tilde{\beta}} = f_{\tilde{\alpha}\tilde{\beta}} := \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}. \quad (\text{C.2.15})$$

The metric (C.2.15) looks completely symmetric in all its components: Seemingly the time coordinate is not preferred in any sense. Nevertheless (C.2.15) is a truly Lorentzian metric. Its determinant is -3 and the eigenvalues are readily computed to be

$$3, \quad -1, \quad -1, \quad -1, \quad (\text{C.2.16})$$

which shows that the metric (C.2.15) has, indeed, the correct signature.

There is a beautiful geometrical interpretation of the four null legs of the null symmetric frame. In a Minkowski spacetime, let us consider the 3-dimensional spacelike hypersurface that is spanned by (e_1, e_2, e_3) . The four points that are defined by the spatial parts of the null symmetric basis vectors (C.2.14) with coordinates $A = (1, 1, 1)$, $B = (1, -1, -1)$, $C = (-1, 1, -1)$, and $D = (-1, -1, 1)$

form a perfect tetrahedron in the 3-subspace (see Figure C.2.1). The vertices A , B , C , and D lie at the same distances of $\sqrt{3}/2$ from the origin $O = (0, 0, 0)$, and correspondingly all sides of the tetrahedron are of equal length, namely $\sqrt{2}$. If we now send, at $t = 0$, a light pulse from the origin O , it reaches all four vertices of the tetrahedron at $t = \sqrt{3}/2$. Thus four light rays provide the operational definition for the *null* symmetric Coll–Morales basis (C.2.14). In a Riemannian spacetime, we can pick Riemannian normal coordinates at an arbitrary point. Then, provided the tetrahedron is sufficiently small, we have an analogous interpretation at the point chosen.

C.2.3 Metric volume 4-form

Given a metric, a corresponding orthonormal coframe determines a metric volume 4-form on every vector space.

Let ϑ^α be an orthonormal coframe in the 4-dimensional Minkowski vector space $(V^*, \mathbf{g})$. Let us define the product

$$\eta = \vartheta^0 \wedge \vartheta^1 \wedge \vartheta^2 \wedge \vartheta^3. \quad (\text{C.2.17})$$

If $\vartheta^{\alpha'}$ is another orthonormal coframe in $(V^*, \mathbf{g})$, then

$$\vartheta^{\alpha'} = L_\alpha^{\alpha'} \vartheta^\alpha, \quad (\text{C.2.18})$$

where the transformation matrix is (pseudo)orthogonal, i.e.,

$$o_{\alpha'\beta'} L_\alpha^{\alpha'} L_\beta^{\beta'} = o_{\alpha\beta} \quad \text{and} \quad L := \det(L_\alpha^{\alpha'}) = \pm 1. \quad (\text{C.2.19})$$

Therefore, under this change of the basis, we have

$$\vartheta^{0'} \wedge \vartheta^{1'} \wedge \vartheta^{2'} \wedge \vartheta^{3'} = L \vartheta^0 \wedge \vartheta^1 \wedge \vartheta^2 \wedge \vartheta^3. \quad (\text{C.2.20})$$

In other words, the definition (C.2.17) gives us a unique (i.e., basis independent) *twisted volume 4-form* of the Minkowski vector space. Alternatively, one may consider two untwisted volume 4-forms separately for each orientation.

If $\vartheta^{\alpha'}$ is an *arbitrary* (not necessarily orthonormal) frame, then we have $\vartheta^{\alpha'} = L_\alpha^{\alpha'} \vartheta^\alpha$ and, since η is twisted,

$$\eta = |L| \vartheta^{0'} \wedge \vartheta^{1'} \wedge \vartheta^{2'} \wedge \vartheta^{3'}. \quad (\text{C.2.21})$$

On the other hand $e_{\alpha'} = L_{\alpha'}^\alpha e_\alpha$. Thus, from the tensor transformation $g_{\alpha'\beta'} = L_{\alpha'}^\alpha L_{\beta'}^\beta o_{\alpha\beta}$, we obtain $|L|^2 = -\det(g_{\alpha'\beta'})$. Hence the twisted volume element with respect to the frame $e_{\alpha'}$ reads

$$\eta = \sqrt{-\det(g_{\alpha'\beta'})} \vartheta^{0'} \wedge \vartheta^{1'} \wedge \vartheta^{2'} \wedge \vartheta^{3'}. \quad (\text{C.2.22})$$

Dropping the primes in (C.2.22), we may write, for any basis ϑ^α , the metric volume 4-form as

$$\eta = \frac{1}{4!} \eta_{\alpha\beta\gamma\delta} \vartheta^\alpha \wedge \vartheta^\beta \wedge \vartheta^\gamma \wedge \vartheta^\delta, \quad (\text{C.2.23})$$

where the twisted antisymmetric tensor $\eta_{\alpha\beta\gamma\delta}$ of type $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$ is defined by

$$\eta_{\alpha\beta\gamma\delta} := \sqrt{-\det(g_{\mu\nu})} \hat{\epsilon}_{\alpha\beta\gamma\delta}, \quad (\text{C.2.24})$$

and $\hat{\epsilon}_{\alpha\beta\gamma\delta}$ is the Levi-Civita permutation symbol with $\hat{\epsilon}_{0123} = +1$. If we raise the indices in the usual way, we find the contravariant components as

$$\eta^{\alpha\beta\gamma\delta} = g^{\alpha\mu} g^{\beta\nu} g^{\gamma\rho} g^{\delta\sigma} \eta_{\mu\nu\rho\sigma} = - \frac{1}{\sqrt{-\det(g_{\mu\nu})}} \epsilon^{\alpha\beta\gamma\delta}. \quad (\text{C.2.25})$$

It follows from (C.2.25) that

$$\vartheta^\alpha \wedge \vartheta^\beta \wedge \vartheta^\gamma \wedge \vartheta^\delta = -\eta^{\alpha\beta\gamma\delta} \eta. \quad (\text{C.2.26})$$

C.2.4 Duality operator for 2-forms as a symmetric almost complex structure on M^6

The duality operator grows out of the almost complex structure $\mathbf{J}$ on M^6 when $\mathbf{J}$ is additionally self-adjoint with respect to the natural metric on M^6 .

Let us now turn again to the space M^6 of 2-forms in four dimensions, which we discussed in Sec. A.1.10.

When an almost complex structure $\mathbf{J}$ is introduced in M^6 in such a way that it is additionally *symmetric*, i.e., self-adjoint, with respect to the natural metric ϵ (A.1.87) on M^6 , then a linear operator on 2-forms

$$\# : \Lambda^2 V^* \longrightarrow \Lambda^2 V^*, \quad (\text{C.2.27})$$

defined by means of $\# = \mathbf{J}$, is called the *duality operator* on M^6 . In view of (A.1.107), the duality operator satisfies

$$\# \# = -1. \quad (\text{C.2.28})$$

This will be called the *closure relation* of the duality operator. The *self-adjointness* with respect to the 6-metric (A.1.87) means

$$\epsilon(\# \omega, \varphi) = \epsilon(\omega, \# \varphi) \quad (\text{C.2.29})$$

for all $\omega, \varphi \in M^6$.

The explicit action of the duality operator on the basis 2-forms reads

$$\# (\vartheta^\alpha \wedge \vartheta^\beta) = \frac{1}{2} \#_{\rho\sigma}{}^{\alpha\beta} (\vartheta^\rho \wedge \vartheta^\sigma), \quad (\text{C.2.30})$$

where the elements of the matrix $\#_{\rho\sigma}{}^{\alpha\beta}$ are, by definition, the components of the almost complex structure in M^6 . In the 6-dimensional notation introduced in Sec. A.1.10, the duality operator is thus described by

$$\# B^K = \#_I{}^K B^I. \quad (\text{C.2.31})$$

Expressed in terms of components, the metric reads $\varepsilon(\omega, \varphi) = \varepsilon^{IJ} \omega_I \varphi_J$, see (A.1.88). Therefore, the self-adjointness can also be written as $\varepsilon^{IJ} \#_I{}^K \omega_K \varphi_J = \varepsilon^{IJ} \#_J{}^K \omega_I \varphi_K$, or

$$\overset{\circ}{\chi}{}^{IJ} = \overset{\circ}{\chi}{}^{JI} \quad \text{for} \quad \overset{\circ}{\chi}{}^{IJ} := \varepsilon^{IK} \#_K{}^J. \quad (\text{C.2.32})$$

By construction, the components of the duality matrix can be expressed in terms of the 3×3 blocks imported from (A.1.108),

$$\#_I{}^K = \begin{pmatrix} C^a{}_b & A^{ab} \\ B_{ab} & C^b{}_a \end{pmatrix}. \quad (\text{C.2.33})$$

Here the components are constrained by the self-adjointness (C.2.29),

$$A^{ab} = A^{ba}, \quad B_{ab} = B_{ba}, \quad C^a{}_a = 0, \quad (\text{C.2.34})$$

as compared to the almost complex structure (in particular, the D -block is expressed in terms of C). Besides that, the algebraic condition (A.1.109) is replaced by the closure relation

$$A^{ac} B_{cb} + C^a{}_c C^c{}_b = -\delta_b^a, \quad C^{(a} A^{b)c} = 0, \quad C^c{}_{(a} B_{b)c} = 0. \quad (\text{C.2.35})$$

The existence of the duality operator has an immediate consequence for the complexified space $M^6(\mathbb{C})$ of 2-forms. As we saw in Sec. A.1.11, the almost complex structure provides for a splitting of the $M^6(\mathbb{C})$ into the two 3-dimensional subspaces corresponding to the $\pm i$ eigenvalues of $\mathbf{J}$. Now we can say even more: These two subspaces are orthogonal to each other in the sense of the natural 6-metric (A.1.87):

$$\varepsilon(\omega, \varphi) = 0, \quad \text{for all} \quad \omega \in \overset{(s)}{M}, \quad \varphi \in \overset{(a)}{M}. \quad (\text{C.2.36})$$

The proof is straightforward: $\varepsilon(\omega, \varphi) = i\varepsilon(\omega, \# \varphi) = i\varepsilon(\# \omega, \varphi) = -\varepsilon(\omega, \varphi)$, where we used the definitions (A.1.111) and the symmetry property (C.2.29).

The 6-metric ε induces a metric on the 3-dimensional subspaces (A.1.111), turning them into complex Euclidean 3-spaces. The symmetry group, which

preserves the induced 3-metric on $\overset{(s)}{M}$ (and on $\overset{(a)}{M}$), is $SO(3, \mathbb{C})$. This is a group-theoretic origin of the reconstruction of the spacetime metric from the duality operator: The Lorentz group, being isomorphic to $SO(3, \mathbb{C})$, is encoded in the structure of the self-dual (or, equivalently, anti-self-dual) complex 2-forms on X .

The significance of the duality operator will become clearer in the next sections where we explicitly demonstrate that $\#$ enables us to construct a Lorentzian metric on spacetime.

C.2.5 From the duality operator to a triplet of complex 2-forms

Every duality operator on M^6 determines a triplet of complex 2-forms that satisfy certain completeness conditions.

Suppose the duality operator (C.2.27) is defined in M^6 with the closure property (A.1.107) and self-adjointness (C.2.29). Its action on the basis 2-forms is given by (C.2.31), with the matrix (C.2.33), (C.2.34). Using the natural $3+3$ split of the 2-form basis (A.1.79), eq.(C.2.31) is rewritten, with the help of (C.2.33), as

$$\# \begin{pmatrix} \beta \\ \hat{\epsilon} \end{pmatrix} = \begin{pmatrix} C & A \\ B & C^T \end{pmatrix} \begin{pmatrix} \beta \\ \hat{\epsilon} \end{pmatrix}, \quad (\text{C.2.37})$$

or in terms of its matrix elements,

$$\begin{aligned} \# \beta^a &= C^a_b \beta^b + A^{ab} \hat{\epsilon}_b, \\ \# \hat{\epsilon}_a &= B_{ab} \beta^b + C^b_a \hat{\epsilon}_b. \end{aligned} \quad (\text{C.2.38})$$

By means of the duality operator $\#$ (as well as with $\mathbf{J}$ of the almost complex structure), one can decompose any 2-form into a *self-dual* and an *anti-self-dual* part. In terms of the 2-form basis, this reads,

$$B^I = \overset{(s)}{B}^I + \overset{(a)}{B}^I, \quad (\text{C.2.39})$$

where we define

$$\overset{(s)}{B}^I := \frac{1}{2}(B^I - i \# B^I), \quad (\text{C.2.40})$$

$$\overset{(a)}{B}^I := \frac{1}{2}(B^I + i \# B^I). \quad (\text{C.2.41})$$

Here i is the imaginary unit. One can check that

$$\begin{aligned} \# \overset{(s)}{B}^I &= +i \overset{(s)}{B}^I, \\ \# \overset{(a)}{B}^I &= -i \overset{(a)}{B}^I. \end{aligned} \quad (\text{C.2.42})$$

Thus the 6-dimensional space of complex 2-forms $M^6(\mathbb{C})$ decomposes into two 3-dimensional invariant subspaces that correspond to the two eigenvalues $\pm i$ of the duality operator.

In order to construct the bases of these subspaces, we have to inspect the (3+3)-representation (A.1.79). One has, using (A.1.79) in (C.2.40), the two sets of self-dual 2-forms,

$$\beta^{(s)a} = \frac{1}{2}(\beta^a - i \# \beta^a), \quad (\text{C.2.43})$$

$$\epsilon^{(s)}_a = \frac{1}{2}(\hat{\epsilon}_a - i \# \hat{\epsilon}_a), \quad (\text{C.2.44})$$

and similarly for the anti-self-dual forms. With the help of (C.2.38), we find explicitly:

$$\beta^{(s)a} = \frac{1}{2}[(\delta^a_b - i C^a_b) \beta^b - i A^{ab} \hat{\epsilon}_b], \quad (\text{C.2.45})$$

$$\epsilon^{(s)}_a = \frac{1}{2}[(\delta^b_a - i C^b_a) \hat{\epsilon}_b - i B_{ab} \beta^b]. \quad (\text{C.2.46})$$

Since the invariant subspace of the self-dual forms is 3-dimensional, these two triplets of self-dual forms cannot be independent from each other. Indeed, let us multiply (C.2.45) with B_{ca} and (C.2.46) with C^a_c . Then the sum of the resulting relation, with the help of the closure property (C.2.35), yields

$$B_{ac} \beta^{(s)a} = (i \delta^a_c - C^a_c) \epsilon^{(s)}_a. \quad (\text{C.2.47})$$

This shows that these two triplets are linearly dependent. For the nondegenerate B -matrix, one can express $\beta^{(s)a}$ in terms of $\epsilon^{(s)}_a$ explicitly.

Let us compute the exterior products of the triplets (C.2.45) and (C.2.46):

$$\beta^{(s)a} \wedge \beta^{(s)b} = -\frac{1}{2} \left(i A^{ab} + C^{(a}_c A^{b)c} \right) \text{Vol} = -\frac{i}{2} A^{ab} \text{Vol}, \quad (\text{C.2.48})$$

$$\epsilon^{(s)}_a \wedge \epsilon^{(s)}_b = -\frac{1}{2} \left(i B_{ab} + C^c_{(a} B_{b)c} \right) \text{Vol} = -\frac{i}{2} B_{ab} \text{Vol}. \quad (\text{C.2.49})$$

We used here the algebra (A.1.83)–(A.1.85) and the closure property (C.2.35). Furthermore, we have

$$\beta^{(s)a} \wedge \overline{\beta^{(s)b}} = \frac{1}{2} C^{(a}_c A^{b)c} \text{Vol} = 0, \quad (\text{C.2.50})$$

$$\epsilon^{(s)}_a \wedge \overline{\epsilon^{(s)}_b} = \frac{1}{2} C^c_{(a} B_{b)c} \text{Vol} = 0. \quad (\text{C.2.51})$$

Here the overbar denotes complex conjugate objects.

C.2.6 From the triplet of complex 2-forms to a duality operator

Every triplet of complex 2-forms with the completeness property determine a duality operator on M^6 .

Both (C.2.45) and (C.2.46) are particular representations of the following general structure: Given is a *triplet of self-dual complex 2-forms* $S^{(a)}$ such that

$$S^{(a)} \wedge S^{(b)} = -2i G^{ab} \text{Vol}, \quad (\text{C.2.52})$$

$$S^{(a)} \wedge \overline{S}^{(b)} = 0, \quad (\text{C.2.53})$$

where the matrix G^{ab} is *real and nondegenerate*. The overbar denotes complex conjugate objects. Equivalently, one can rewrite (C.2.52) as

$$S^{(a)} \wedge S^{(b)} = \frac{1}{3} G^{ab} G_{cd} S^{(c)} \wedge S^{(d)}, \quad (\text{C.2.54})$$

where G_{ab} denotes the matrix inverse to G^{ab} . We call (C.2.52), (C.2.53) the *completeness* conditions for a triplet of 2-forms.

In the previous section we saw that every duality operator defines a triplet of self-dual complex 2-forms. Here we show that the converse is also true: Let $S^{(a)}$ be an arbitrary triplet of complex 2-forms that satisfies the completeness conditions (C.2.52), (C.2.53). Then they determine a duality operator in M^6 .

Expanding the arbitrary 2-forms with respect to the basis (A.1.79), we can write

$$S^{(a)} = M^a_b \beta^b + N^{ab} \hat{e}_b. \quad (\text{C.2.55})$$

In view of (A.1.83)–(A.1.85), we find that (C.2.52) imposes an algebraic constraint on the matrix components,

$$M^a_c N^{bc} + M^b_c N^{ac} = -2G^{ab}. \quad (\text{C.2.56})$$

Introducing the real variables

$$M^a_b = V^a_b + i U^a_b, \quad N^{ab} = X^{ab} + i Y^{ab}, \quad (\text{C.2.57})$$

one can decompose (C.2.56) into the two real equations:

$$V^{(a}_c X^{b)c} - U^{(a}_c Y^{b)c} = 0, \quad (\text{C.2.58})$$

$$V^{(a}_c Y^{b)c} + U^{(a}_c X^{b)c} = -G^{ab}. \quad (\text{C.2.59})$$

Analogously, (C.2.53) yields another pair of real matrix equations:

$$V^{(a}_c X^{b)c} + U^{(a}_c Y^{b)c} = 0, \quad (\text{C.2.60})$$

$$V^{[a}_c Y^{b]c} - U^{[a}_c X^{b]c} = 0. \quad (\text{C.2.61})$$

Combining (C.2.58) and (C.2.60), we find

$$V^{(a}{}_c X^{b)c} = 0, \quad U^{(a}{}_c Y^{b)c} = 0, \quad (\text{C.2.62})$$

whereas the sum of (C.2.59) and (C.2.61) gives

$$V^a{}_c Y^{bc} + U^b{}_c X^{ac} = -G^{ab}. \quad (\text{C.2.63})$$

We can count the number of independent degrees of freedom. The total number of variables is 36 ($= 4 \times 9$ unknown components of the matrices V, U, X, Y). They are subject to 21 constraints ($= 6 + 6 + 9$) imposed by the equations (C.2.62) and (C.2.63). Thus, in general, 15 degrees are left over for the unknown matrices in (C.2.55).

In order to see how the triplet is related to the duality operator, let us define a new basis in M^6 by means of the linear transformation

$$\begin{pmatrix} \beta'^a \\ \hat{\epsilon}'_a \end{pmatrix} = \begin{pmatrix} V^a{}_b & X^{ab} \\ -U_{ab} & -Y_a{}^b \end{pmatrix} \begin{pmatrix} \beta^b \\ \hat{\epsilon}_b \end{pmatrix}, \quad (\text{C.2.64})$$

where $U_{ab} := G_{ac}U^c{}_b$ and $Y_a{}^b := G_{ac}Y^{cb}$. The new basis elements satisfy the same algebraic conditions

$$\beta'^a \wedge \beta'^b = 0, \quad \hat{\epsilon}'_a \wedge \hat{\epsilon}'_b = 0, \quad \hat{\epsilon}'_a \wedge \beta'^b = \delta_a^b \text{Vol}, \quad (\text{C.2.65})$$

as those in (A.1.83)–(A.1.85). The proof follows directly from (C.2.62) and (C.2.63). Interestingly, the transformation (C.2.64) is *always* invertible, with the inverse given by

$$\begin{pmatrix} \beta^a \\ \hat{\epsilon}_a \end{pmatrix} = \begin{pmatrix} -Y_b{}^a & X^{ba} \\ -U_{ba} & V^b{}_a \end{pmatrix} \begin{pmatrix} \beta'^b \\ \hat{\epsilon}'_b \end{pmatrix}. \quad (\text{C.2.66})$$

The direct check again involves only the completeness conditions (C.2.62) and (C.2.63).

The original triplet (C.2.55), with respect to the new basis (C.2.64), then reduces to

$$S^{(a)} = \beta'^a - iG^{ab}\hat{\epsilon}'_b. \quad (\text{C.2.67})$$

Now we are prepared to introduce the duality operator. We define it by simply postulating that its action on the triplet amounts to a mere multiplication by the imaginary unit, i.e.,

$$\# S^{(a)} = i S^{(a)} \quad \left(\text{hence} \quad \# \overline{S}^{(a)} = -i \overline{S}^{(a)} \right). \quad (\text{C.2.68})$$

From (C.2.67) we have $\beta'^a = (S^{(a)} + \overline{S}^{(a)})/2$ and $\hat{\epsilon}'_b = iG_{ab}(S^{(a)} - \overline{S}^{(a)})/2$. Then, immediately, we find the action of the duality operator on the new 2-form basis:

$$\# \begin{pmatrix} \beta'^a \\ \hat{\epsilon}'_a \end{pmatrix} = \begin{pmatrix} 0 & G^{ab} \\ -G_{ab} & 0 \end{pmatrix} \begin{pmatrix} \beta'^b \\ \hat{\epsilon}'_b \end{pmatrix}. \quad (\text{C.2.69})$$

We can now reconstruct the original basis in (C.2.37). We use (C.2.66) and (C.2.64) and find the 3×3 matrices

$$A^{ab} = G_{cd} (Y^{ac} Y^{bd} - X^{ac} X^{bd}), \quad (\text{C.2.70})$$

$$B_{ab} = G_{cd} (U^c{}_a U^d{}_b - V^c{}_a V^d{}_b), \quad (\text{C.2.71})$$

$$C^a{}_b = G_{cd} (U^c{}_b Y^{da} - V^c{}_b X^{da}). \quad (\text{C.2.72})$$

As a consequence, every triplet of complex 2-forms, which satisfy the completeness conditions (C.2.52) and (C.2.53), defines a duality operator with the closure and symmetry properties.

C.2.7 From a triplet of complex 2-forms to the metric: Schönberg–Urbantke formulas

The triplet of complex 2-forms is a building material for the metric of spacetime. The Lorentzian metric, up to a scale factor, can be constructed from the triplet of 2-forms by means of the Schönberg–Urbantke formulas.

The importance of the duality operator $\#$ and the corresponding triplet of 2-forms lies in the fact that they determine a Lorentzian metric on 4-space.

Let us formulate this result. Let a triplet of 2-forms $S^{(a)}$ that satisfy (C.2.52) be given on V with some symmetric regular matrix G . Then the Lorentzian metric of spacetime is recovered with the help of the Schönberg–Urbantke formulas:

$$\sqrt{\det g} g_{ij} = \frac{16}{3} \sqrt{\det G} \epsilon^{klmn} \hat{\epsilon}_{abc} S_{ik}^{(a)} S_{lm}^{(b)} S_{nj}^{(c)}, \quad (\text{C.2.73})$$

$$\sqrt{\det g} = \frac{1}{24} \epsilon^{klmn} G_{cd} S_{kl}^{(c)} S_{mn}^{(d)}. \quad (\text{C.2.74})$$

Here, ϵ^{klmn} is the Levi-Civita symbol, and the $S_{ij}^{(a)}$ are the components of the basis 2-forms with respect to the local coordinates $\{x^i\}$, i.e., $S^{(a)} = \frac{1}{2} S_{ij}^{(a)} dx^i \wedge dx^j$. Despite the appearance of the Levi-Civita symbols in (C.2.73)–(C.2.74), these expressions are tensorial.

A rigorous proof of the fact that on a 4-dimensional vector space V every three complex 2-forms $S^{(a)}$, which satisfy the completeness condition (C.2.52), define a (pseudo)Riemannian metric will not be presented here.³ The metric is, in general, complex; however it is real when (C.2.53) is fulfilled.

Note that the sign in the algebraic equations (A.1.107) and (C.2.35) is important. If instead of the minus there were a plus, then the resulting spacetime

³See, however, Schönberg [16], Urbantke [17], or Harnett [8].

metric would have a Riemannian (Euclidean) signature $(+1, +1, +1, +1)$ or a mixed one $(+1, +1, -1, -1)$.

As a comment to the Schönberg–Urbantke mechanism, we would like to mention the local isomorphism of the three following complex Lie groups (symplectic, special linear, and orthogonal):

$$Sp(1, \mathbb{C}) \approx SL(2, \mathbb{C}) \approx SO(3, \mathbb{C}). \quad (\text{C.2.75})$$

The easiest way to see this is to analyze the corresponding Lie algebras. At first, recall that the orthogonal algebra $so(3, \mathbb{C})$ consists of all skew-symmetric 3×3 matrices with complex elements:

$$a = \begin{pmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{pmatrix}. \quad (\text{C.2.76})$$

Here q_1, q_2, q_3 is a triplet of complex numbers.

Since the linear algebra $sl(2, \mathbb{C})$ consists of all traceless 2×2 complex matrices, its arbitrary element can be written as

$$\tilde{a} = \frac{1}{2} \begin{pmatrix} iq_3 & iq_1 + q_2 \\ iq_1 - q_2 & -iq_3 \end{pmatrix}. \quad (\text{C.2.77})$$

Assuming that the three complex parameters q_1, q_2, q_3 in (C.2.77) are the same as in (C.2.76), we obtain a map $so(3, \mathbb{C}) \rightarrow sl(2, \mathbb{C})$, which is obviously an isomorphism. It is straightforward to check, for example, that the commutator $[a, b]$ of any two matrices of the form (C.2.76) is mapped into the commutator $[\tilde{a}, \tilde{b}]$ of the corresponding matrices (C.2.77).

The symplectic algebra $sp(1, \mathbb{C})$ consists of all 2×2 matrices $\tilde{a}$ that satisfy $\tilde{a}s + s\tilde{a}^T = 0$, with $s := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. One can easily verify that any matrix (C.2.77) satisfies this relation, thus proving the isomorphism $sp(1, \mathbb{C}) = sl(2, \mathbb{C})$.

C.2.8 Hodge star and Excalc's

In a metric vector space, the Hodge operator establishes a map between p -forms and $(n - p)$ -forms. Besides the ϑ -basis for exterior forms we can define an η -basis that is the Hodge dual to the ϑ -basis.

Consider an n -dimensional vector space with metric $\mathbf{g}$. Usually (when an orientation is fixed), the Hodge star is defined as a linear map $\star : \Lambda^p V^* \rightarrow \Lambda^{n-p} V^*$, such that for an arbitrary p -form ω and for an arbitrary 1-form φ it satisfies

$$\star(\omega \wedge \varphi) = \tilde{\mathbf{g}}^{-1}(\varphi) \lrcorner \star\omega. \quad (\text{C.2.78})$$

The formula (C.2.78) reduces the definition of a Hodge dual for an arbitrary p -form to the definition of a 4-form dual to a number $*1$. Usually, $*1$ is taken as the ordinary (untwisted) volume form and such a procedure distinguishes a certain orientation in V . We change this convention and require, instead of the usual definition, that

$$\star : \Lambda^p V^* \longrightarrow \text{twisted } \Lambda^{n-p} V^*, \quad (\text{C.2.79})$$

and vice versa

$$\star : \text{twisted } \Lambda^p V^* \longrightarrow \Lambda^{n-p} V^*. \quad (\text{C.2.80})$$

Accordingly, we put $*1$ equal to the *twisted* volume form:

$$*1 = \eta. \quad (\text{C.2.81})$$

Let us now restrict our attention to the 4-dimensional Minkowski vector space. We can use (C.2.78) to define the Hodge dual for an arbitrary p -form. Take at first $\omega = 1$ and $\varphi = \vartheta^\alpha$ as a coframe 1-form. Then, with (C.2.4), (C.2.81), and (C.2.7), we find

$$\star \vartheta^\alpha = \tilde{\mathbf{g}}^{-1}(\vartheta^\alpha) \lrcorner *1 = g^{\alpha\mu} e_\mu \lrcorner \eta = \frac{1}{3!} \eta^\alpha{}_{\beta\gamma\delta} \vartheta^\beta \wedge \vartheta^\gamma \wedge \vartheta^\delta, \quad (\text{C.2.82})$$

where we used (A.1.49) in order to compute the interior product $e_\mu \lrcorner \eta$ for (C.2.23). We can now go on in a recursive way. Choosing in (C.2.78) $\omega = \vartheta^\alpha$ and again $\varphi = \vartheta^\beta$, we obtain:

$$\star(\vartheta^\alpha \wedge \vartheta^\beta) = e^\beta \lrcorner \star \vartheta^\alpha. \quad (\text{C.2.83})$$

Here $e^\alpha := g^{\alpha\beta} e_\beta$. Substituting (C.2.82) into (C.2.83) and again using (A.1.49) to evaluate the interior product, we find successively the formulas

$$\star \vartheta^\alpha = e^\alpha \lrcorner \eta = \frac{1}{3!} \eta^\alpha{}_{\beta\gamma\delta} \vartheta^\beta \wedge \vartheta^\gamma \wedge \vartheta^\delta =: \eta^\alpha, \quad (\text{C.2.84})$$

$$\star(\vartheta^\alpha \wedge \vartheta^\beta) = e^\beta \lrcorner (e^\alpha \lrcorner \eta) = \frac{1}{2!} \eta^{\alpha\beta}{}_{\gamma\delta} \vartheta^\gamma \wedge \vartheta^\delta =: \eta^{\alpha\beta}, \quad (\text{C.2.85})$$

$$\star(\vartheta^\alpha \wedge \vartheta^\beta \wedge \vartheta^\gamma) = e^\gamma \lrcorner (e^\beta \lrcorner (e^\alpha \lrcorner \eta)) = \eta^{\alpha\beta\gamma}{}_\delta \vartheta^\delta =: \eta^{\alpha\beta\gamma}, \quad (\text{C.2.86})$$

$$\star(\vartheta^\alpha \wedge \vartheta^\beta \wedge \vartheta^\gamma \wedge \vartheta^\delta) = \eta^{\alpha\beta\gamma\delta}. \quad (\text{C.2.87})$$

The newly defined η -system of p -forms, $p = 0, \dots, 4$,

$$\{\eta^{\alpha\beta\gamma\delta}, \eta^{\alpha\beta\gamma}, \eta^{\alpha\beta}, \eta^\alpha, \eta\} \quad (\text{C.2.88})$$

constitutes, along with the usual ϑ -system,

$$\{1, \vartheta^\alpha, \vartheta^\alpha \wedge \vartheta^\beta, \vartheta^\alpha \wedge \vartheta^\beta \wedge \vartheta^\gamma, \vartheta^\alpha \wedge \vartheta^\beta \wedge \vartheta^\gamma \wedge \vartheta^\delta\}, \quad (\text{C.2.89})$$

a new *metric dependent* basis for the exterior algebra over the Minkowski vector space. This construction can evidently be generalized to an n -dimensional case.

In the n -dimensional Minkowski space,

$$**\omega = (-1)^{p(n-p)+1}\omega, \quad \text{for } \omega \in \Lambda^p V^*. \quad (\text{C.2.90})$$

Thus, for $n = 4$, we have $**\omega = \omega$ for exterior forms of *odd* degree, $p = 1, 3$, and $**\omega = -\omega$ for forms of *even* degree, $p = 0, 2, 4$.

From the definition (C.2.78), we can read off the rules

$$*(\phi + \psi) = *\phi + *\psi \quad \text{and} \quad *(a\phi) = a*\phi, \quad (\text{C.2.91})$$

for $a \in \mathbb{R}$ and $\phi, \psi \in \Lambda^p V^*$. These linearity properties together with (C.2.84)-(C.2.87) enable one to calculate the Hodge star of any exterior form.

The implementation of these structures in Excalc is simple. By means of the coframe statement, the metric is put in. Excalc provides the operator # as Hodge star.

In electrodynamics the most prominent role of the Hodge star operator is that it maps, up to a dimensionful factor λ , the field strength F into the excitation H , namely $H = \lambda *F$, as we will see in the fifth axiom (D.6.13). Therefore, in Excalc we simply have `excit2 := lam * # farad2`; This spacetime relation is all we need in order to make the Maxwell equations to a complete system. We used this Excalc command already in our Maxwell sample program of Sec. B.5.6.

As a further example, we study the electromagnetic energy-momentum current. Recall that we constructed in (B.5.39) the electromagnetic energy-momentum tensor density in terms of the energy-momentum 3-form. We can now find the corresponding tensor if we take, instead of the diamond operator $\diamond$, the Hodge star operator $*$. Thus,

```
% definitions of o(a) and maxenergy3(a) precede this declaration
pform maxenergy0(a,b)=0;
maxenergy0(-a,b) := #(o(b)^maxenergy3(-a));
```

Or, turning to the η -system of (C.2.88), we can write the program as follows:

```
pform eta0(a,b,c,d)=0,eta1(a,b,c)=1,eta2(a,b)=2,eta3(a)=3,eta4=4$

eta4      := # 1$
eta3(a)   := e(a) _| eta4$
eta2(a,b) := e(b) _| eta3(a)$
eta1(a,b,c) := e(c) _| eta2(a,b)$
eta0(a,b,c,d) := e(d) _| eta1(a,b,c)$
```

We could define η alternatively as `eta4:=o(0)^o(1)^o(2)^o(3)` see (C.2.17). With these tools, the Einstein 3-form $G_\alpha := \frac{1}{2}\eta_{\alpha\beta\gamma} \wedge R^{\beta\gamma}$ now emerges simply as

```
pform einstein3(a)=3;
einstein3(-a):=(1/2)*eta1(-a,-b,-c)^curv2(b,c);
```

Accordingly, exterior calculus and the Excalc package are really of equal power.

⊗ From a metric to the duality operator

Let us assume that a metric is introduced in a 4-dimensional vector space V . Then, the Hodge star map (C.2.79), (C.2.80) is defined for p -forms. Restricting our attention to 2-forms, we find that the Hodge star maps $M^6 = \Lambda^2 V^*$ into itself. The restriction

$$\# = \star \Big|_{\Lambda^2 V^*} \quad (\text{C.2.92})$$

is obviously a duality operator in M^6 . Recalling the definitions of Sec. A.1.11, we can straightforwardly verify that both the closure (A.1.107) and the symmetry (C.2.29) are fulfilled by (C.2.92). The matrix of the duality operator is given by

$$\#_{\rho\sigma}{}^{\alpha\beta} = \overset{\mathbf{g}}{\kappa}_{\rho\sigma}{}^{\alpha\beta} := \eta_{\mu\nu\rho\sigma} g^{\alpha[\mu} g^{\nu]\beta}. \quad (\text{C.2.93})$$

Alternatively, in 6-dimensional notation, it reads

$$\#_I{}^J = \overset{\mathbf{g}}{\kappa}_I{}^J := \begin{pmatrix} \overset{\mathbf{g}}{C} & \overset{\mathbf{g}}{A} \\ \overset{\mathbf{g}}{B} & \overset{\mathbf{g}}{C}^{\text{T}} \end{pmatrix}, \quad (\text{C.2.94})$$

where a straightforward calculation yields the 3×3 blocks

$$\overset{\mathbf{g}}{A}{}^{ab} = \sqrt{-g} (g^{00} g^{ab} - g^{0a} g^{0b}), \quad (\text{C.2.95})$$

$$\overset{\mathbf{g}}{B}{}_{ab} = \frac{1}{4} \sqrt{-g} (g^{ce} g^{df} - g^{de} g^{ef}) \hat{\epsilon}_{acd} \hat{\epsilon}_{bef}, \quad (\text{C.2.96})$$

$$\overset{\mathbf{g}}{C}{}^a{}_b = \frac{\sqrt{-g}}{2} (g^{0c} g^{ad} - g^{ac} g^{0d}) \epsilon_{bcd}. \quad (\text{C.2.97})$$

C.2.9 Manifold with a metric, Levi-Civita connection

The metric on a manifold is introduced pointwise as a smooth scalar product on the tangent spaces. The Levi-Civita (or Riemannian) connection is a unique linear connection with vanishing torsion and covariantly constant metric.

Let X_n be an n -dimensional differentiable manifold. We say that a *metric* is defined on X_n if a metric tensor $\mathbf{g}$ is smoothly assigned to the tangent vector spaces X_x at each point x . In terms of the coframe field,

$$\mathbf{g} = g_{\alpha\beta}(x) \vartheta^\alpha \otimes \vartheta^\beta, \quad \text{where} \quad g_{\alpha\beta}(x) = \mathbf{g}(e_\alpha, e_\beta) \quad (\text{C.2.98})$$

is a smooth tensor field in every local coordinate chart. The manifold with a metric structure defined on it is called a *(pseudo)Riemannian manifold*, denoted $V_n = (X_n, \mathbf{g})$. Usually, a metric (and, correspondingly, a manifold) is called Riemannian if the $g_{\alpha\beta}(x)$ is positive definite for all x . However, in order to simplify formulations, we will omit the “pseudo” and call metrics with a Lorentzian signature Riemannian.

The metric brings a whole bunch of related objects on a manifold V_n . First of all, a metric volume n -form emerges on V_n : For every coframe field $\vartheta^\alpha = (\vartheta^{\hat{1}}, \dots, \vartheta^{\hat{n}})$, it is defined by

$$\eta := \sqrt{-\det g_{\mu\nu}} \vartheta^{\hat{1}} \wedge \dots \wedge \vartheta^{\hat{n}} = \frac{1}{n!} \eta_{\alpha_1 \dots \alpha_n} \vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_n} \quad (\text{C.2.99})$$

$$= \sqrt{-\det g_{kl}} dx^1 \wedge \dots \wedge dx^n = \frac{1}{n!} \eta_{i_1 \dots i_n} dx^{i_1} \wedge \dots \wedge dx^{i_n}. \quad (\text{C.2.100})$$

The *world metric tensor* with components

$$g_{ij}(x) = e_i^\alpha(x) e_j^\beta(x) g_{\alpha\beta}(x) \quad (\text{C.2.101})$$

is defined in every local coordinate chart $\{x^i\}$. The principal difference between $g_{\alpha\beta}$ and g_{ij} is that the former can always be “gauged away” by the suitable choice of the frame field e_α . One can choose an orthonormal frame field, e.g., in which $g_{\alpha\beta}$ has the diagonal form (C.2.8) independent of local coordinates. However, it is impossible in general to choose the coordinates $\{x^i\}$ in such a way that g_{ij} is constant everywhere on V_n .

The Levi-Civita tensor densities in (C.2.99), (C.2.100) are introduced by

$$\begin{aligned} \eta_{\alpha_1 \dots \alpha_n}(x) &= \sqrt{-\det g_{\mu\nu}} \epsilon_{\alpha_1 \dots \alpha_n}, \\ \eta_{i_1 \dots i_n}(x) &= e_{i_1}^{\alpha_1} \dots e_{i_n}^{\alpha_n} \eta_{\alpha_1 \dots \alpha_n} = \sqrt{-\det g_{kl}} \epsilon_{i_1 \dots i_n}, \end{aligned} \quad (\text{C.2.102})$$

with the numerical permutation symbol chosen as $\epsilon_{1\dots n} = +1$. Hats over numerical indices help to distinguish components with respect to local frames from components with respect to coordinate frames.

The next relative of the metric is the Hodge star operator. It is naturally introduced on a V_n pointwise with the help of formulas derived in Sec. C.2.8.

Finally, the most far-reaching and nontrivial consequence of the metric $\mathbf{g}$ is the existence on a V_n of a special covariant differentiation $\tilde{\nabla}$ which is usually called a *Riemannian* or *Levi-Civita connection*. We use tilde to denote this connection and any objects or operators constructed from it. As was shown in Sec. C.1.1, a covariant differentiation is defined on a manifold as soon as in every local chart the connection 1-forms Γ_i^j are given that obey the consistency condition (C.1.11). The Riemannian connection is defined, in each local coordinate system by the *Christoffel symbols* $\tilde{\Gamma}_{ki}^j$:

$$\tilde{\Gamma}_i^j = \tilde{\Gamma}_{ki}^j dx^k, \quad \tilde{\Gamma}_{ki}^j := \frac{1}{2} g^{jl} (\partial_i g_{kl} + \partial_k g_{il} - \partial_l g_{ik}). \quad (\text{C.2.103})$$

The Levi-Civita (or Riemannian) connection $\tilde{\nabla}$ has some special properties that distinguish it from other covariant differentiations. It has *vanishing torsion*. This is trivially seen from (C.1.42) and (C.1.18) if we notice that the Christoffel symbols (C.2.103) are symmetric in its lower indices. Moreover, the covariant exterior derivative of the metric with respect to the Levi-Civita connection vanishes identically:

$$\tilde{D}g_{\alpha\beta} = dg_{\alpha\beta} - \tilde{\Gamma}_\beta^\lambda g_{\alpha\lambda} - \tilde{\Gamma}_\alpha^\lambda g_{\lambda\beta} = dg_{\alpha\beta} - 2\tilde{\Gamma}_{(\alpha\beta)} = 0, \quad (\text{C.2.104})$$

see (C.2.136). A connection for which the covariant derivative of the metric is zero is called *metric compatible*.

C.2.10 $\otimes$ Codifferential and wave operator, also in Excalc

By means of the Hodge star operator we can define the codifferential, which is adjoint to the exterior differential d with respect to the scalar product on exterior forms. This yields directly a wave operator.

Consider the space $\Lambda^p(X)$ of all smooth exterior p -forms on X . The Hodge operator makes it possible to define a natural scalar product on this functional space:

$$(\omega, \varphi) := \int_X \omega \wedge \star \varphi, \quad \text{for all } \omega, \varphi \in \Lambda^p(X). \quad (\text{C.2.105})$$

Then the *codifferential* operator $d^\dagger$ can be introduced as an adjoint to the exterior differential d with respect to the scalar product (C.2.105),

$$(\omega, d^\dagger \varphi) := (d\omega, \varphi). \quad (\text{C.2.106})$$

By construction, the codifferential maps p -forms into $(p-1)$ -forms (contrary to the exterior differential which increases the rank of a form by one). Using the property (C.2.90) of the Hodge operator, one can verify that in an n -dimensional Lorentzian space the *codifferential on p -forms* is given explicitly by

$$d^\dagger = (-1)^{n(p-1)} \star d \star. \quad (\text{C.2.107})$$

The Leibniz rule for d was used at an intermediate step, and the boundary integral (for noncompact manifolds) is vanishing due to the proper behavior of the forms at infinity. Accordingly, in $n=4$ we have $d^\dagger = \star d \star$ for all forms.

In local coordinates, the codifferential of a p -form $\varphi = \frac{1}{p!} \varphi_{i_1 \dots i_p} dx^{i_1} \wedge \dots \wedge dx^{i_p}$ reads:

$$d^\dagger \varphi = \frac{-1}{(p-1)!} \tilde{\nabla}^j \varphi_{j i_1 \dots i_{p-1}} dx^{i_1} \wedge \dots \wedge dx^{i_{p-1}}. \quad (\text{C.2.108})$$

Here $\tilde{\nabla}$ is the covariant differentiation for the Levi-Civita connection (C.2.103). The codifferential is nilpotent,

$$d^\dagger d^\dagger = 0 \quad (\text{C.2.109})$$

which follows directly from (C.2.107), (C.2.90) and the nilpotency property $d^2 = 0$ of the exterior differential (A.2.19). The operator d (resp., $d^\dagger$) increases (resp., decreases) the rank of a form by one. Hence, the combinations $dd^\dagger$ and $d^\dagger d$ both map p -forms into p -forms. However, these operators are not self-adjoint with respect to the scalar product (C.2.105). The symmetrized second-order differential operator

$$\square := d^\dagger d + d d^\dagger \quad (\text{C.2.110})$$

is called the wave operator (or d'Alembertian, also called Laplace–Beltrami operator). It is, by construction, self-adjoint with respect to (C.2.105).

On one occasion, we had to check whether the wave operator, if applied to a certain coframe field, vanishes, i.e., $\square \vartheta^a \stackrel{?}{=} 0$. Excalc could help. After the `coframe` statement specifying the appropriate value for the coframe, we defined a suitable 1-form:

```
pform wavetocoframe1(a)=1$

wavetocoframe1(a) := d(#(d(#o(a)))) + #(d(#(d o(a))));
```

The emerging expression we had to treat with switches and suitable substitutions, but the quite messy computation of the wave operator was given to the machine.

C.2.11 $\otimes$ Nonmetricity

Let a connection and a metric be defined independently on the same spacetime manifold. Then the nonmetricity is a measure of the incompatibility between metric and connection.

Let us consider the general case when on a manifold X_n the metric and connection are defined independently. Such a manifold is denoted $(X_n, \nabla, \mathbf{g})$ and called a *metric-affine* spacetime. Since the metric is a tensor field of type $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$, its covariant differentiation yields a type $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$ tensor field that is called the *non-metricity*:

$$Q(u, v, w) := \mathbf{g}(\nabla_u v, w) + \mathbf{g}(v, \nabla_u w) - u\{\mathbf{g}(v, w)\}, \quad (\text{C.2.111})$$

for all vector fields u, v, w . Nonmetricity measures the extent to which a connection ∇ is incompatible with the metric $\mathbf{g}$. Metric compatibility (also called

metricity) $Q(u, v, w) = 0$ implies the conservation of lengths and angles under parallel transport. A manifold that is endowed with a metric and a metric-compatible connection is said to be a *Riemann–Cartan* manifold (or a U_n). In general, the Riemann–Cartan manifold has a nonvanishing torsion. When the latter is zero, we recover the Riemannian manifold described in Sec. C.2.9.

Like the 2-forms of torsion (C.1.38), (C.1.40) and curvature (C.1.45), (C.1.46), we define the nonmetricity 1-form

$$Q_{\alpha\beta} = Q_{\gamma\alpha\beta} \vartheta^\gamma \quad (\text{C.2.112})$$

by

$$Q_{\gamma\alpha\beta} = Q_{\alpha\beta}(e_\gamma) = Q(e_\gamma, e_\alpha, e_\beta). \quad (\text{C.2.113})$$

Since $u(g_{\alpha\beta}) = dg_{\alpha\beta}(u)$, equation (C.2.111) is equivalent to

$$Q_{\alpha\beta} = -dg_{\alpha\beta} + \Gamma_{\alpha\beta} + \Gamma_{\beta\alpha} = -Dg_{\alpha\beta}, \quad (\text{C.2.114})$$

where $\Gamma_{\alpha\beta} := g_{\beta\gamma}\Gamma_\alpha^\gamma$. If the $g_{\alpha\beta}$ are constants, then it follows from (C.2.114) that $Q_{\alpha\beta} = 2\Gamma_{(\alpha\beta)}$. Hence, in a U_n , where $Q_{\alpha\beta} = 0$, we have antisymmetric connection one-forms

$$\Gamma_{\alpha\beta} = -\Gamma_{\beta\alpha}, \quad (\text{C.2.115})$$

provided the $g_{\alpha\beta}$ are constants, for example, with respect to orthonormal coframe fields.

We shall refer to (C.2.114) as the *0th Cartan structural relation* and shall call the expression obtained as its exterior derivative,

$$DQ_{\alpha\beta} = dQ_{\alpha\beta} - \Gamma_\alpha^\gamma \wedge Q_{\gamma\beta} - \Gamma_\beta^\gamma \wedge Q_{\alpha\gamma} = 2R_{(\alpha\beta)}, \quad (\text{C.2.116})$$

the *0th Bianchi identity*. The proof of (C.2.116) makes use of the Ricci identity (C.1.65).

It is convenient to separate the trace part of the nonmetricity from its traceless piece. Let us define the *Weyl 1-form* (or *Weyl covector*) by

$$Q := \frac{1}{n} Q_{\alpha\beta} g^{\alpha\beta}. \quad (\text{C.2.117})$$

The factor $1/n$ is conventional. Then the nonmetricity is decomposed into its deviator $\mathcal{Q}_{\alpha\beta}$ and its trace according to

$$Q_{\alpha\beta} = \mathcal{Q}_{\alpha\beta} + Q g_{\alpha\beta}. \quad (\text{C.2.118})$$

The trace of the curvature, which is called the *segmental curvature*, can be expressed in terms of the Weyl 1-form:

$$R_\gamma^\gamma = \frac{n}{2} dQ. \quad (\text{C.2.119})$$

The physical importance of the Weyl 1-form is related to the fact that, during parallel transport, the contribution of the Weyl 1-form does *not* affect the light

cone, whereas lengths of non-null vectors are merely scaled with some (path-dependent) factor. A space with $\mathcal{Q}_{\alpha\beta} = 0$ is called a *Weyl–Cartan* manifold Y_n . In this latter case, the position vector (C.1.60) changes according to

$$\Delta r^\alpha = \Delta S \left(T^\alpha - \frac{1}{n} R_\gamma{}^\gamma r^\alpha - R^{[\beta\alpha]} r_\beta \right), \quad (\text{C.2.120})$$

if it is Cartan displaced over a closed loop that encircles the area element ΔS . The first curvature term induces a dilation, while the second one is a pure rotation.

C.2.12 $\otimes$ Post-Riemannian pieces of the connection

An arbitrary linear connection can always be split into the Levi-Civita connection plus a post-Riemannian tensorial piece called the distortion. The latter depends on torsion and nonmetricity. Correspondingly, all the geometric objects and operators can be systematically decomposed into Riemannian and post-Riemannian parts.

“...the question whether this [spacetime] continuum is Euclidean or structured according to the Riemannian scheme or still otherwise is a genuine physical question which has to be answered by experience rather than being a mere convention to be chosen on the basis of expediency.”⁴

The geometrical properties of an arbitrary metric-affine spacetime are described by the 2-forms of curvature $R_\alpha{}^\beta$ and torsion T^α and by the 1-form of nonmetricity $Q_{\alpha\beta}$. Particular values of these fundamental objects specify different geometries which may be realized on a spacetime manifold. Physically, one can think of a number of “phase transitions” that the spacetime geometry undergoes at different energies (or distance scales). Correspondingly, it is convenient to study particular realizations of geometrical structures within the framework of several specific gravitational models. The overview of these models and of the relevant geometries is given in Fig. C.2.2.

The most general gravitational model – metric-affine gravity (MAG) – employs the (L_n, g) geometry in which all three main objects, curvature, torsion, and nonmetricity are nontrivial. Such a geometry could be realized at extremely small distances (high energies) when the hypermomentum current of matter fields plays a central role.

Other gravitational models and the relevant geometries appear as special cases when one or several main geometrical objects are completely or partly “switched off”. The Z_4 geometry is characterized by $T^\alpha = 0$ and was used in

⁴A. Einstein: *Geometrie und Erfahrung* [6], our translation.

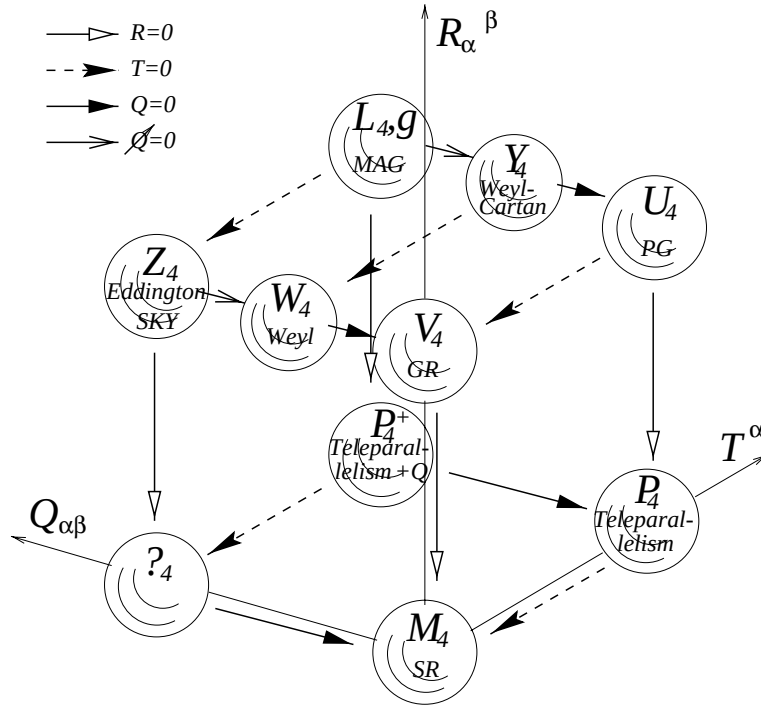


Figure C.2.2: MAGIC cube: Classification of geometries and gravity theories in the three “dimensions” (R, T, Q).

the unified field theory of Eddington and in so-called SKY-gravity (theories of Stephenson–Kilmister–Yang). Switching off the traceless nonmetricity, $Q_{\alpha\beta} = 0$, yields the Weyl–Cartan space Y_4 (with torsion) or standard Weyl theory W_4 (with $T^\alpha = 0$).

Furthermore, switching off the nonmetricity completely, one recovers the Riemann–Cartan geometry U_4 which is the arena of Poincaré gauge (PG) gravity in which the spin current of matter, besides its energy-momentum current, is an additional source of the gravitational field. The Riemannian geometry V_4 (with $Q_{\alpha\beta} = 0$ and $T^\alpha = 0$) describes, via Einstein’s General Relativity (GR), gravitational effects on a macroscopic scale when the energy-momentum current is the only source of gravity. Finally, when the curvature is zero, $R_\alpha^\beta = 0$, one obtains the Weitzenböck space P_4 and the *teleparallelism* theory of gravity (when $Q_{\alpha\beta} = 0$) or a generalized teleparallelism theory in the spacetime with

nontrivial torsion and nonmetricity P_4^+ . The Minkowski spacetime M_4 with vanishing $Q_{\alpha\beta} = 0$, $T^\alpha = 0$, $R_\alpha{}^\beta = 0$ underlies Special Relativity (SR) theory. The relations between the different theories and geometries are given in Fig. C.2.2 by means of arrows of different types that specify which object is switched off.

In a metric-affine space, curvature, torsion, and nonmetricity satisfy the three Bianchi identities (C.2.116), (C.1.67), and (C.1.69):

$$DQ_{\alpha\beta} \equiv 2R_{(\alpha\beta)}, \quad \text{0th Bianchi,} \quad (\text{C.2.121})$$

$$DT^\alpha \equiv R_\beta{}^\alpha \wedge \vartheta^\beta, \quad \text{1st Bianchi,} \quad (\text{C.2.122})$$

$$DR_\alpha{}^\beta \equiv 0, \quad \text{2nd Bianchi.} \quad (\text{C.2.123})$$

In practical calculations, it is important to know exactly the number of geometrical and physical variables and their algebraic properties (e.g., symmetries, orthogonality relations, etc.). These aspects can be clarified with the help of two types of decompositions. A linear connection can always be decomposed into Riemannian and post-Riemannian parts,

$$\Gamma_\beta{}^\alpha = \tilde{\Gamma}_\beta{}^\alpha + N_\beta{}^\alpha, \quad (\text{C.2.124})$$

where the *distortion* 1-form $N_{\alpha\beta}$ is expressed in terms of torsion and nonmetricity as follows:

$$N_{\alpha\beta} = -e_{[\alpha} \lrcorner T_{\beta]} + \frac{1}{2}(e_\alpha \lrcorner e_\beta \lrcorner T_\gamma) \vartheta^\gamma + (e_{[\alpha} \lrcorner Q_{\beta]\gamma}) \vartheta^\gamma + \frac{1}{2}Q_{\alpha\beta}. \quad (\text{C.2.125})$$

The distortion “measures” a deviation of a particular geometry from the purely Riemannian one. As a by-product of the decomposition (C.2.124) we verify that a metric-compatible connection without torsion is unique: It is the Levi-Civita connection. Nonmetricity and torsion can easily be recovered from the distortion, namely

$$Q_{\alpha\beta} = 2N_{(\alpha\beta)}, \quad T^\alpha = N_\beta{}^\alpha \wedge \vartheta^\beta. \quad (\text{C.2.126})$$

If we collect the information we have on the splitting of a connection into Riemannian and non-Riemannian pieces, then, in terms of the metric $g_{\alpha\beta}$, the coframe ϑ^α , the anholonomy C^α , the torsion T^α , and the nonmetricity $Q_{\alpha\beta}$, we have the highly symmetric master formula

$$\begin{aligned} \Gamma_{\alpha\beta} &= \frac{1}{2}dg_{\alpha\beta} + (e_{[\alpha} \lrcorner dg_{\beta]\gamma})\vartheta^\gamma + e_{[\alpha} \lrcorner C_{\beta]} - \frac{1}{2}(e_\alpha \lrcorner e_\beta \lrcorner C_\gamma)\vartheta^\gamma & (V_n) \\ &\quad - e_{[\alpha} \lrcorner T_{\beta]} + \frac{1}{2}(e_\alpha \lrcorner e_\beta \lrcorner T_\gamma)\vartheta^\gamma & (U_n) \\ &\quad + \frac{1}{2}Q_{\alpha\beta} + (e_{[\alpha} \lrcorner Q_{\beta]\gamma})\vartheta^\gamma & (L_n, g). \end{aligned} \quad (\text{C.2.127})$$

The first line of this formula refers to the Riemannian part of the connection. Together with the second line a Riemann–Cartan geometry is encompassed. Only the third line makes the connection a really independent quantity. Note that locally the torsion T_α can be mimicked by the negative of the anholonomy $-C_\alpha$ and the nonmetricity $Q_{\alpha\beta}$ by the exterior derivative $dg_{\alpha\beta}$ of the metric.

In a 4-dimensional metric-affine space, the curvature has 96 components, torsion 24, and nonmetricity 40. In order to make the work with all these variables manageable, one usually decomposes all geometrical quantities into irreducible pieces with respect to the Lorentz group.

C.2.13 $\otimes$ Excalc again

Excalc has a commodity: If a coframe $o(a)$ and a metric g are prescribed, it calculates the Riemannian piece of the connection $\tilde{\Gamma}_\alpha^\beta$ on demand. One just has to issue the command `riemannconx chris1`; Then `chris1` (one could take any other name) is, without further declaration, a 1-form with the index structure `chris1(a,b)`. Since we use Schouten’s conventions in this book, we have to redefine Schrüfer’s `riemannconx` according to `chris1(a,-b):=chris1(-b,a)`;

If you distrust Excalc, you could also compute your own Riemannian connection according to (C.2.127), i.e.,

```
pform anhol2(a)=2,chris1(a,b)=1$

anhol2(a)      := d o(a)$
chris1(-a,-b) := (1/2)*d g(-a,-b)
               +(1/2)*((e(-a)_|(d g(-b,-c)))-(e(-b)_|(d g(-a,-c))))^o(c)
               +(1/2)*( e(-a)_|anhol2(-b) - e(-b)_|anhol2(-a))
               -(1/2)*( e(-a)_|(e(-b)_|anhol2(-c)))^o(c)$
```

But we can assure you that Excalc does its job correctly.

In any case, with `chris1(a,b)` or with `christ1(a,b)` you can equally well compute the distortion 1-form and the nonmetricity 1-form in terms of coframe $o(a)$, metric g , and connection `conn1(a,b)`. The calculation would run as follows:

```
coframe o(0)=...;          % input 1
frame e;
riemannconx chris1; chris1(a,b):=chris1(b,a);
pform conn1(a,b)=1, distort1(a,b)=1, nonmet1(a,b)=1$

conn1(0,0):=...;          % input 2
distort1(a,b):=conn1(a,b)-chris1(a,b);
nonmet1(a,b):=distort1(a,b)+distort1(b,a);
```

and one could continue in this line and compute torsion and curvature according to

```
pform torsion2(a)=2, curv2(a,b)=2;

torsion2(a):=d o(a)+conn1(-b,a)^o(b);
curv2(-a,b):=d conn1(-a,b)-conn1(-a,c)^conn1(-c,b);
```

All other relevant geometrical quantities can be derived therefrom.

Problems

Problem C.1.

Check the geometrical interpretation of torsion given in Fig. C.1.1 by direct calculation using the definition of the parallel transport.

Problem C.2.

Prove (C.1.52) in the infinitesimal case, approximating the curve σ by a small parallelogram.

Problem C.3.

Prove the following relations involving the transposed connection:

1. $T^\alpha = -\widehat{T}^\alpha$, that is, $T^\alpha = -\widehat{D} \vartheta^\alpha$;
2. $\widehat{D} \epsilon^{\alpha\beta\gamma\delta} = 0$;
3. $L_{e_\alpha} \vartheta^\beta = e_\alpha \lrcorner T^\beta$;
4. The covariant Lie derivative of an arbitrary p -form $\Psi = \Psi_{\alpha_1 \dots \alpha_p} \vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_p} / p!$:

$$L_{e_\alpha} \Psi = \frac{1}{p!} \left(\widehat{D}_\alpha \Psi_{\alpha_1 \dots \alpha_p} \right) \vartheta^{\alpha_1} \wedge \dots \wedge \vartheta^{\alpha_p}. \quad (\text{C.2.128})$$

Problem C.4.

1. Find a transformation matrix from an orthonormal basis to the half-null frame in which the metric has the form (C.2.11).
2. Find a transformation matrix from an orthonormal basis to a Newman–Penrose null frame in which the metric has the form (C.2.13).
3. Find a linear transformation $e_\alpha = L_\alpha{}^\beta f_\beta$ that brings the null symmetric Coll–Morales basis f_β back to an orthonormal frame e_α .

Solution:

$$L_\alpha{}^\beta = \frac{1}{2} \begin{pmatrix} 1/\sqrt{3} & \sqrt{2} & 0 & 1 \\ 1/\sqrt{3} & 0 & \sqrt{2} & -1 \\ 1/\sqrt{3} & 0 & -\sqrt{2} & -1 \\ 1/\sqrt{3} & -\sqrt{2} & 0 & 1 \end{pmatrix}. \quad (\text{C.2.129})$$

Check that (C.2.14) is inverse to (C.2.129) up to a Lorentz transformation.

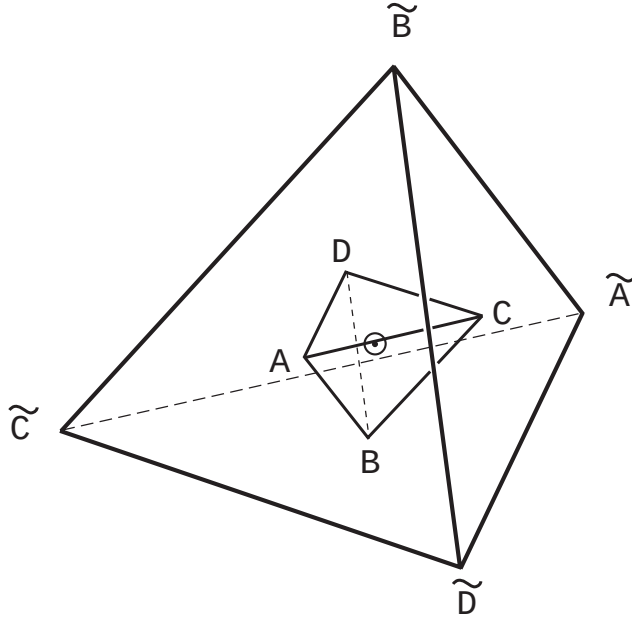


Figure C.2.3: The outer tetrahedron represents a real null symmetric coframe Φ^α . The inner tetrahedron is that of Fig. C.2.1 and it visualizes the corresponding real null symmetric frame f_α . Note that the 3-vectors OA, OB , etc. are perpendicular to the planes $\widetilde{BCD}, \widetilde{CDA}$, etc.

4. Show that the matrix (C.2.129) is represented as a product $L = SR$ of the two matrices:

$$S = \begin{pmatrix} -\sqrt{3} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad R = \frac{1}{2} \begin{pmatrix} -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}. \quad (\text{C.2.130})$$

The matrix S just scales the time coordinate and a short calculation with REDUCE shows that R is an element of $SO(4)$.

5. In analogy to (C.2.14), determine a null symmetric coframe Φ^α and show that it is *not* dual to the corresponding null symmetric frame. The situation is depicted in Fig. C.2.3. Discuss the relation between these two different tetrahedra.

Problem C.5.

1. Prove that for $\phi, \psi \in \Lambda^p V^*$,

$$*\phi \wedge \psi = *\psi \wedge \phi. \quad (\text{C.2.131})$$

2. If $\phi \in \Lambda^p V^*$, show that

$$\vartheta^\alpha \wedge (e_\alpha \lrcorner \phi) = p\phi, \quad (\text{C.2.132})$$

$$*(\phi \wedge \vartheta_\alpha) = e_\alpha \lrcorner * \phi. \quad (\text{C.2.133})$$

3. Prove that for any $\omega \in \Lambda^p V^*$ and $\phi, \psi \in \Lambda^2 V^*$:

$$(e_\alpha \lrcorner \omega) \wedge (e^\alpha \lrcorner * \omega) = 0, \quad (\text{C.2.134})$$

$$(e_\alpha \lrcorner \phi) \wedge (e^\alpha \lrcorner * \psi) = *[(e_\alpha \lrcorner \phi) \wedge (e^\alpha \lrcorner \psi)]; \quad (\text{C.2.135})$$

see [12].

4. Show that in 4-dimensional space

$$(a) \quad \vartheta^\alpha \wedge \eta_\beta = \delta_\beta^\alpha \eta,$$

$$(b) \quad \vartheta^\alpha \wedge \eta_{\beta\gamma} = \delta_\gamma^\alpha \eta_\beta - \delta_\beta^\alpha \eta_\gamma,$$

$$(c) \quad \vartheta^\alpha \wedge \eta_{\beta\gamma\delta} = \delta_\delta^\alpha \eta_{\beta\gamma} + \delta_\gamma^\alpha \eta_{\delta\beta} + \delta_\beta^\alpha \eta_{\gamma\delta},$$

$$(d) \quad \vartheta^\alpha \wedge \eta_{\beta\gamma\delta\mu} = \delta_\mu^\alpha \eta_{\beta\gamma\delta} - \delta_\delta^\alpha \eta_{\beta\gamma\mu} + \delta_\gamma^\alpha \eta_{\beta\delta\mu} - \delta_\beta^\alpha \eta_{\gamma\delta\mu}.$$

Hint: $\vartheta^\alpha \wedge \eta = 0$ because it is a 5-form in a 4-dimensional space.

Problem C.6.

1. Prove that Christoffel symbols define a covariant differentiation by checking the transformation law (C.1.11) for the 1-forms (C.2.103).
2. Show that with respect to an arbitrary local frame field, the Riemannian connection form reads, cf. (C.1.18):

$$\begin{aligned} \tilde{\Gamma}_\alpha{}^\beta &= e_j{}^\beta \tilde{\Gamma}_i{}^j e^i{}_\alpha + e_i{}^\beta d e^i{}_\alpha \\ &= \frac{1}{2} g^{\beta\gamma} \left(dg_{\alpha\gamma} + (e_\alpha \lrcorner dg_{\gamma\lambda} - e_\gamma \lrcorner dg_{\alpha\lambda}) \vartheta^\lambda \right) \\ &\quad + \frac{1}{2} \left(e_\alpha \lrcorner C^\beta - e^\beta \lrcorner C_\alpha - (e_\alpha \lrcorner e^\beta \lrcorner C_\gamma) \vartheta^\gamma \right), \end{aligned} \quad (\text{C.2.136})$$

where $C^\alpha = d\vartheta^\alpha$ is the anholonomy 2-form, see (A.2.35).

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Part D

The Maxwell–Lorentz spacetime relation

So far, the Maxwell equations (B.4.9) and (B.4.10) represent an underdetermined system of partial differential equations of first order for the excitation H and the field strength F . In order to reduce the number of independent variables, we have to set up a universal relation between H and F , which is assumed to be local:

$$H = H(F).$$

We call this the *electromagnetic spacetime relation*. Therefore we can complete electrodynamics, formulated in Part B up to now metric- and connection-free, by introducing a suitable spacetime relation as a *fifth axiom*.

The simplest choice is, of course, a linear relation $H \sim \kappa F$ with the constitutive tensor κ . Eventually, this yields the conventional Maxwell–Lorentz electrodynamics. It is remarkable that this linear relation, if supplemented merely by a *reciprocity* property (basically equivalent to the closure relation $\kappa\kappa = -1$) and by the *symmetry* of κ (symbolically, $\kappa = \kappa^T$), introduces a *light cone* at each point of spacetime. In other words, we are able to *derive* the metric of spacetime, up to a conformal factor, from the existence of a linear κ that is closed and symmetric.⁵

Alternatively, one could simply *assume* the existence of a (pseudo)Riemannian metric g of signature $(+1, -1, -1, -1)$ on the spacetime manifold. In both cases, the Hodge star operator $\star$ is available and ordinary electrodynamics can be recovered via the spacetime relation $H \sim \star F$. Anyone that is not interested in learning about our method of deriving the light cone of spacetime can jump right away to Chap. D.6 where a specific linear spacetime relation is formulated as a fifth axiom.

⁵This type of idea goes back to Toupin [54] and Schönberg [48]; see also Urbantke [56] and Jadczyk [21]. Wang [57] gave a revised presentation of Toupin’s results. A forerunner was Peres [39]; see in this context also the more recent papers by Piron and Moore [42]. For new and recent results, see [36]. It was recognized by Brans [3] that, within general relativity, it is possible to define a duality operator in much the same way as we present below (see (D.4.6)) and that from this duality operator the metric can be recovered. Subsequently numerous authors discussed such structures in the framework of general relativity theory; see, e.g., Capovilla, Jacobson, and Dell [5], ’t Hooft [18], Harnett [14, 15], Obukhov and Tertychniy [37], and the references given there. In Part D, we also use the results of Rubilar and Gross [12, 45].

D.1

A linear relation between H and F

The excitation H is related to the field strength F by means of a universal law. We call it the (electromagnetic) spacetime relation and assume it to be local and linear. Then, the corresponding constitutive tensor κ carries 36 independent components. We decompose κ into three parts, the principal part (20 components), the skewon part (15 components), and the axion part (1 component), and discuss their respective properties.

D.1.1 The constitutive tensor of spacetime

The *electromagnetic spacetime relation*¹ expresses the excitation H in terms of the field strength F . Both are elements of the space of 2-forms $\Lambda^2 X$. However, H is twisted and F is untwisted. Thus one can formulate the spacetime relation as a *local* law

$$H = \kappa(F), \quad (\text{D.1.1})$$

where

$$\kappa : \Lambda^2 X \rightarrow \Lambda^2 X \quad (\text{D.1.2})$$

is an *invertible* operator that maps an untwisted 2-form into a twisted 2-form and vice versa. Local means that the excitation H at some point P of spacetime depends exclusively on the field strength F at the very same point P .

¹Post [43] named it the *constitutive map* including also the constitutive relation for matter (see (E.3.58)). Truesdell & Toupin [55], Toupin [54], and Kovetz [26] use the term *aether relations*.

An important subcase is that of a *linear* law between the 2-forms H and F . Accordingly, the operator (D.1.2) is required to be linear, i.e., for all $a, b \in \Lambda^0 X$ and $\phi, \psi \in \Lambda^2 X$ we have

$$\kappa(a\phi + b\psi) = a\kappa(\phi) + b\kappa(\psi). \quad (\text{D.1.3})$$

For physical applications, it may be useful to present our linear operator κ in a more explicit form. Because of its linearity, it is sufficient to know the action of κ on the basis 2-forms. The corresponding mathematical preliminaries were outlined in Sec. A.1.10. A choice of the natural coframe $\vartheta^i = dx^i$ yields the specific 2-form basis B^I of (A.1.79). The operator κ acts on the 2-form basis $dx^k \wedge dx^l (= B^I$ in the equivalent bivector language) and maps it in twisted 2-forms the latter of which we can again decompose:

$$\kappa(dx^k \wedge dx^l) = \frac{1}{2} \kappa_{ij}{}^{kl} dx^i \wedge dx^j \quad \text{or} \quad \kappa(B^K) = \kappa_I{}^K B^I. \quad (\text{D.1.4})$$

Now, we decompose the 2-forms in (D.1.1):

$$H = \frac{1}{2} H_{ij} dx^i \wedge dx^j \quad \text{or} \quad H = H_I B^I. \quad (\text{D.1.5})$$

Substituting (D.1.5), together with a similar expansion for F , and making use of (D.1.4), we find

$$H_{ij} = \frac{1}{2} \kappa_{ij}{}^{kl} F_{kl} \quad \text{or} \quad H_I = \kappa_I{}^K F_K. \quad (\text{D.1.6})$$

Thus a linear spacetime relation postulates the existence of 6×6 functions $\kappa_{ij}{}^{kl}(x)$ depending on time and space with

$$\kappa_{ij}{}^{kl} = -\kappa_{ji}{}^{kl} = -\kappa_{ij}{}^{lk}. \quad (\text{D.1.7})$$

This twisted tensor field of type $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ is called the *constitutive tensor of spacetime*. This is supposed to remind us that the constitution of spacetime, or rather its constitutive properties relevant for electromagnetism, should be reflected in κ . In the bivector notation,² these 36 independent components are arranged into a 6×6 matrix $\kappa_I{}^K$.

Naively, one could try $\kappa_{ij}{}^{kl} = f \delta_{ij}^{kl}$ as a simple constitutive tensor with a pseudoscalar function f . Then $H = f F$ and Maxwell's equations would transform into $(df) \wedge F = J$ and $dF = 0$. For $f = \text{const}$ and $J \neq 0$, the inhomogeneous equation would be inconsistent — clearly indicating that a spacetime relation $\kappa_{ij}{}^{kl} = f \delta_{ij}^{kl}$ is unphysical.

The choice of the local coordinates in the deduction of (D.1.6) is unimportant. In a different coordinate system, the linear law preserves its form due to the

²Fausser [8] argues that the field strength (E, B) and excitation $(\mathcal{H}, \mathcal{D})$ represent line coordinates in R_6 in the sense of projective geometry (with only points and lines but without metric). Only a linear relation between (E, B) and $(\mathcal{H}, \mathcal{D})$ would then yield a metric geometry.

tensorial transformation properties of $\kappa_{ij}{}^{kl}$. Alternatively, instead of the local coordinates, one may choose an arbitrary (anholonomic) coframe $\vartheta^\alpha = e_i^\alpha dx^i$ and may then decompose the 2-forms H and F with respect to it according to $H = H_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta / 2$ and $F = F_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta / 2$. Then, if we redo the calculations of above, we find

$$H_{\alpha\beta} = \frac{1}{2} \kappa_{\alpha\beta}{}^{\gamma\delta} F_{\gamma\delta} \quad \text{with} \quad \kappa_{\alpha\beta}{}^{\gamma\delta} = e^i{}_\alpha e^j{}_\beta e_k{}^\gamma e_l{}^\delta \kappa_{ij}{}^{kl}. \quad (\text{D.1.8})$$

Here we used also the components of the frame $e_\alpha = e^k{}_\alpha \partial_k$.

As we recall from Sec. A.1.10, the Levi-Civita symbols (A.1.89) and (A.1.91) can be used as a quasi-metric for raising and lowering pairs of indices. We define the constitutive tensor *density* of spacetime,

$$\chi^{ijkl} := \frac{1}{2} \epsilon^{ijmn} \kappa_{mn}{}^{kl} \quad \text{or} \quad \chi^{IK} = \epsilon^{IM} \kappa_M{}^K, \quad (\text{D.1.9})$$

and conversely,

$$\kappa_{ij}{}^{kl} = \frac{1}{2} \hat{\epsilon}_{ijmn} \chi^{mnkl} \quad \text{or} \quad \kappa_I{}^K = \hat{\epsilon}_{IM} \chi^{MK}. \quad (\text{D.1.10})$$

Similar to (D.1.7), we have

$$\chi^{ijkl} = -\chi^{jikl} = -\chi^{ijlk}. \quad (\text{D.1.11})$$

The 36 functions $\chi^{ijkl}(x)$ are equivalent to $\kappa_{ij}{}^{kl}(x)$. Because of the corresponding properties of the Levi-Civita symbol, the χ^{ijkl} represent an (untwisted) tensor density of weight +1. In terms of χ , the spacetime relation reads

$$H_{ij} = \frac{1}{4} \hat{\epsilon}_{ijmn} \chi^{mnkl} F_{kl} \quad \text{or} \quad H_I = \hat{\epsilon}_{IM} \chi^{MK} F_K. \quad (\text{D.1.12})$$

Excitation H and field strength F are measurable quantities. The functions $\kappa_{ij}{}^{kl}$ (or χ^{ijkl}) are “quotients” of H and F . Thus they first of all carry the absolute dimension of $[\kappa] = [\chi] = q^2/h = q/\Phi = (q/t)/(\Phi/t) = \text{current/voltage} = 1/\text{resistance} \stackrel{\text{SI}}{=} 1/\Omega = \text{S}$ (for siemens), and moreover, they are measurable too.

Two *invariants* of κ , a linear and a quadratic one, play a leading role: The twisted scalar

$$\alpha := \frac{1}{12} \kappa_{ij}{}^{ij} = \frac{1}{4!} \hat{\epsilon}_{ijkl} \chi^{ijkl} \quad (\text{D.1.13})$$

and the true scalar

$$\lambda^2 := -\frac{1}{4!} \kappa_{ij}{}^{kl} \kappa_{kl}{}^{ij} = -\frac{1}{96} \hat{\epsilon}_{ijkl} \hat{\epsilon}_{mnpq} \chi^{ijmn} \chi^{pqkl}. \quad (\text{D.1.14})$$

In later applications we will see that it always fulfills $\lambda^2 > 0$. Note that $[\alpha] = [\lambda] = 1/\text{resistance}$. It is as if spacetime carried an intrinsic resistance or the inverse, an intrinsic impedance (commonly called “wave resistance of the vacuum” or “vacuum impedance”).

One could also build up invariants of order p according to the multiplicative pattern of $\kappa_{I_1}{}^{I_2} \kappa_{I_2}{}^{I_3} \dots \kappa_{I_p}{}^{I_1}$, with $p = 1, 2, 3, 4, \dots$, but there doesn't seem to be a need to do so.

D.1.2 Decomposing the constitutive tensor

The tensor $\kappa_{ij}{}^{kl}$ as well as the equivalent tensor density χ^{ijkl} have 36 components. In 6D representation, the latter can be represented by the 6×6 matrix χ^{IK} which can be decomposed into its symmetric and antisymmetric parts, $\chi^{IK} = \chi^{(IK)} + \chi^{[IK]}$. Because of its antisymmetry (or skewsymmetry), we call $\chi^{[IK]}$ the *skewon part* of χ . It is irreducible under the linear group. From the symmetric part $\chi^{(IK)}$ we can still extract the trace $\chi_L{}^L := \epsilon_{LM} \chi^{LM}$, which we call the *axion part* of χ . This yields three irreducible parts,

$$\begin{aligned} \chi^{IK} &= \left(\chi^{(IK)} - \frac{1}{6} \epsilon^{IK} \chi_L{}^L \right) + \chi^{[IK]} + \frac{1}{6} \epsilon^{IK} \chi_L{}^L, \\ 36 &= 20 \quad \oplus \quad 15 \quad \oplus \quad 1. \end{aligned} \quad (\text{D.1.15})$$

Tensor density χ^{ijkl}

It is simple to translate these results into an irreducible decomposition in the 4D representation. We denote the irreducible pieces by $^{(1)}\chi$, $^{(2)}\chi$, $^{(3)}\chi$. The axion and skewon parts read, respectively,

$$^{(3)}\chi^{ijkl} := \chi^{[ijkl]}, \quad ^{(2)}\chi^{ijkl} := \frac{1}{2} (\chi^{ijkl} - \chi^{klji}). \quad (\text{D.1.16})$$

The symmetric tracefree part, which we call the *principal part*, is the rest:

$$^{(1)}\chi^{ijkl} := \chi^{ijkl} - ^{(2)}\chi^{ijkl} - ^{(3)}\chi^{ijkl}. \quad (\text{D.1.17})$$

It is desirable to bring $^{(1)}\chi^{ijkl}$ as well as $^{(3)}\chi^{ijkl}$ into a more compact form. Therefore, we define the analog of $\chi^{(IK)}$:

$$\hat{\chi}^{ijkl} := \frac{1}{2} (\chi^{ijkl} + \chi^{klji}). \quad (\text{D.1.18})$$

Then the axion part can be written as

$$^{(3)}\chi^{ijkl} = \frac{1}{3} (\hat{\chi}^{ijkl} + \hat{\chi}^{iklj} + \hat{\chi}^{iljk}). \quad (\text{D.1.19})$$

For $^{(1)}\chi$, via its definition (D.1.17), we find

$$\begin{aligned} ^{(1)}\chi^{ijkl} &:= \hat{\chi}^{ijkl} - \frac{1}{3} (\hat{\chi}^{ijkl} + \hat{\chi}^{iklj} + \hat{\chi}^{iljk}) \\ &= \frac{1}{3} (2\hat{\chi}^{ijkl} - \hat{\chi}^{iklj} - \hat{\chi}^{iljk}). \end{aligned} \quad (\text{D.1.20})$$

Tensor $\kappa_{ij}{}^{kl}$

Clearly, our irreducible decomposition can also be expressed in terms of the operator κ and its components $\kappa_{ij}{}^{kl}$. The contracted tensor of type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$,

$$\kappa_i{}^k := \kappa_{il}{}^{kl}, \quad (\text{D.1.21})$$

has 16 independent components. The second contraction yields the pseudoscalar function

$$\kappa := \kappa_k{}^k = \kappa_{kl}{}^{kl}. \quad (\text{D.1.22})$$

The traceless piece

$$\not\kappa_i{}^k := \kappa_i{}^k - \frac{1}{4} \kappa \delta_i^k \quad (\text{D.1.23})$$

has 15 independent components. These pieces can now be subtracted from the original constitutive tensor. Then,

$$\kappa_{ij}{}^{kl} = {}^{(1)}\kappa_{ij}{}^{kl} + {}^{(2)}\kappa_{ij}{}^{kl} + {}^{(3)}\kappa_{ij}{}^{kl} \quad (\text{D.1.24})$$

$$= {}^{(1)}\kappa_{ij}{}^{kl} + 2 \not\kappa_{[i}{}^{[k} \delta_{j]}^l] + \frac{1}{6} \kappa \delta_{[i}^k \delta_{j]}^l. \quad (\text{D.1.25})$$

By construction, ${}^{(1)}\kappa_{ij}{}^{kl}$ is the totally traceless part of the constitutive map:

$${}^{(1)}\kappa_{il}{}^{kl} = 0. \quad (\text{D.1.26})$$

Thus, we split κ according to $36 = 20 + 15 + 1$, and the $\begin{smallmatrix} 2 \\ 2 \end{smallmatrix}$ tensor ${}^{(1)}\kappa_{ij}{}^{kl}$ is subject to the 16 constraints (D.1.26) and carries $20 = 36 - 16$ components.

We recall the definition (D.1.9) of χ^{ijkl} . We substitute (D.1.25) into its right-hand side. Then, with

$$\chi^{ijkl} = {}^{(1)}\chi^{ijkl} + {}^{(2)}\chi^{ijkl} + {}^{(3)}\chi^{ijkl}, \quad (\text{D.1.27})$$

we find the translation rules

$${}^{(1)}\chi^{ijkl} = \frac{1}{2} \epsilon^{ijmn} {}^{(1)}\kappa_{mn}{}^{kl}, \quad (\text{D.1.28})$$

$${}^{(2)}\chi^{ijkl} = \frac{1}{2} \epsilon^{ijmn} {}^{(2)}\kappa_{mn}{}^{kl} = -\epsilon^{ijm[k} \not\kappa_m{}^{l]}, \quad (\text{D.1.29})$$

$${}^{(3)}\chi^{ijkl} = \frac{1}{2} \epsilon^{ijmn} {}^{(3)}\kappa_{mn}{}^{kl} = \frac{1}{12} \epsilon^{ijkl} \kappa. \quad (\text{D.1.30})$$

Finally, we can express the spacetime relation as

$$H_{ij} = {}^{(1)}H_{ij} + {}^{(2)}H_{ij} + {}^{(3)}H_{ij}, \quad (\text{D.1.31})$$

with

$${}^{(r)}H_{ij} = {}^{(r)}\kappa_{ij}{}^{kl} F_{kl}, \quad r = 1, 2, 3. \quad (\text{D.1.32})$$

Operator κ

In operator language, the spacetime relation reads

$$H = {}^{(1)}H + {}^{(2)}H + {}^{(3)}H = {}^{(1)}\kappa(F) + {}^{(2)}\kappa(F) + {}^{(3)}\kappa(F). \quad (\text{D.1.33})$$

For every pair of 2-forms ϕ and ψ , the symmetries of the irreducible operators manifest themselves as follows:

$$\psi \wedge {}^{(1)}\kappa(\phi) = \phi \wedge {}^{(1)}\kappa(\psi), \quad (\text{D.1.34})$$

$$\psi \wedge {}^{(2)}\kappa(\phi) = -\phi \wedge {}^{(2)}\kappa(\psi), \quad (\text{D.1.35})$$

$$\psi \wedge {}^{(3)}\kappa(\phi) = \phi \wedge {}^{(3)}\kappa(\psi). \quad (\text{D.1.36})$$

For the axion part (see (D.1.25)), we have

$${}^{(3)}\kappa(\phi) = \alpha \phi, \quad (\text{D.1.37})$$

with a suitable pseudoscalar function α . Synonymously, one also speaks of an *axial* scalar; this is where the name axion part comes from.

D.1.3 $\otimes$ Decomposing energy-momentum and action

Let us go back to our fourth axiom (B.5.7). We substitute (D.1.33) and find that the energy-momenta of the three parts are additive:

$${}^{\text{k}}\Sigma_\alpha = {}^{(1)}{}^{\text{k}}\Sigma_\alpha + {}^{(2)}{}^{\text{k}}\Sigma_\alpha + {}^{(3)}{}^{\text{k}}\Sigma_\alpha. \quad (\text{D.1.38})$$

The ${}^{(3)}\kappa$ part obeys (D.1.37). Therefore, we find

$$\begin{aligned} {}^{(3)}{}^{\text{k}}\Sigma_\alpha &= \frac{1}{2} \left[F \wedge (e_\alpha \lrcorner {}^{(3)}\kappa(F)) - {}^{(3)}\kappa(F) \wedge (e_\alpha \lrcorner F) \right] \\ &= \frac{1}{2} [F \wedge (\alpha e_\alpha \lrcorner F) - \alpha F \wedge (e_\alpha \lrcorner F)] = 0. \end{aligned} \quad (\text{D.1.39})$$

Thus, the axion part drops out from the energy-momentum current:

$${}^{\text{k}}\Sigma_\alpha = {}^{(1)}{}^{\text{k}}\Sigma_\alpha + {}^{(2)}{}^{\text{k}}\Sigma_\alpha. \quad (\text{D.1.40})$$

For a linear spacetime relation $H = \kappa(F)$, the electromagnetic Lagrangian is quadratic in F and has, in accordance with (B.5.90), the form of

$$V = -\frac{1}{2} F \wedge H. \quad (\text{D.1.41})$$

Substitution of (D.1.33) yields

$$V = -\frac{1}{2} \left(F \wedge {}^{(1)}H + F \wedge {}^{(2)}H + F \wedge {}^{(3)}H \right) = {}^{(1)}V + {}^{(2)}V + {}^{(3)}V. \quad (\text{D.1.42})$$

However, because of (D.1.35), for $\psi = \phi = F$, the skewon part of the Lagrangian vanishes:

$${}^{(2)}V = -\frac{1}{2} F \wedge {}^{(2)}H = -\frac{1}{2} F \wedge {}^{(2)}\kappa(F) = 0. \quad (\text{D.1.43})$$

Thus,

$$V = {}^{(1)}V + {}^{(3)}V. \quad (\text{D.1.44})$$

The skewon part drops out of the Lagrangian V . Had we based our considerations exclusively on Lagrangians, we would never have come across the notion of a skewon part. In the light of (D.1.40) and (D.1.44), the principal part ${}^{(1)}\chi$ behaves “normally” whereas the skewon and axion parts are more elusive (see Table D.1.1).

Irreducible part	Contributes to		
	energy-momentum	Lagrangian	TR tensor density
principal ${}^{(1)}\chi$	yes	yes	yes
skewon ${}^{(2)}\chi$ or $\mathcal{S}$	yes	no	yes
axion ${}^{(3)}\chi$ or α	no	yes	no

Table D.1.1: How each irreducible part ${}^{(r)}\chi$ of the constitutive tensor density of spacetime contributes to the electromagnetic energy-momentum 3-form ${}^k\Sigma_\alpha = \frac{1}{2}[F \wedge (e_\alpha \lrcorner H) - H \wedge (e_\alpha \lrcorner F)]$, to the Lagrangian 4-form $V = -\frac{1}{2}H \wedge F$, and to the Tamm-Rubilar tensor density $\mathcal{G}^{ijkl}$ to be defined in (D.2.22). We see that ${}^{(1)}\chi$ is omnipresent and thus indispensable, in accordance with the metric that will finally be derived from it. The skewon piece ${}^{(2)}\chi$ drops out of the Lagrangian because of its dissipative nature and has thus been overlooked in theoretical discussions in the past. The axion part ${}^{(3)}\chi$ is the most elusive one: It contributes to the Lagrangian but neither affects light propagation locally nor carries electromagnetic energy-momentum. So far, there is no experimental evidence for either the axion nor the skewon part nor for more than ten components of the principal part.

D.1.4 Abelian axion field α

Since ${}^{(3)}\chi$ has only one independent component, we introduce in accordance with (D.1.13), the *axion field*,³

$$\alpha(x) = \frac{1}{4!} \hat{\epsilon}_{ijkl} {}^{(3)}\chi^{ijkl}, \quad {}^{(3)}\chi^{ijkl} = \epsilon^{ijkl} \alpha. \quad (\text{D.1.45})$$

It is a twisted 0-form, also called pseudoscalar or axial scalar (see (D.1.37)). As such, it is T and P odd (T stands for time, P for parity). For definiteness, we call it the *Abelian axion*⁴ field, since in our axiomatics it is exclusively related to electrodynamics, the gauge theory of the *Abelian* group⁵ $U(1)$.

It is remarkable that the pseudoscalar axion field α enters here as a quantity that does not interfere at all with the first four axioms of electrodynamics. Already at the pre-metric level, such a field emerges as a not unnatural companion of the electromagnetic field. Hence a possible axion field has a high degree of universality. After all, it arises, in the framework of our axiomatic approach, even *before* the metric field (Einstein's gravitational potential) comes into being.

The axion field $\alpha(x)$ is some kind of universal permittivity/permeability field. One could add a kinetic and a mass term of the α -field to the purely electromagnetic Lagrangian. Then $\alpha(x)$ becomes really propagating and one can associate with it a hypothetical quantum particle, the Abelian axion, with spin = 0 and parity = -1.

Let us collect the principal and the skewon part of κ in

$$\tilde{\kappa} := {}^{(1)}\kappa + {}^{(2)}\kappa. \quad (\text{D.1.46})$$

Then the linear spacetime relation (D.1.1) with (D.1.3) can be written as

$$H = \tilde{\kappa}(F) + \alpha F. \quad (\text{D.1.47})$$

The Maxwell equations in this shorthand notation read

$$d[\tilde{\kappa}(F) + \alpha F] = J, \quad dF = 0. \quad (\text{D.1.48})$$

We can also execute the differentiation in the inhomogeneous equation and substitute the homogeneous one. Then we find for the inhomogeneous Maxwell

³Dicke was seemingly the first to introduce axion terms in a Lagrangian (see [7], p. 51, Eq. (7)). Later Ni [31, 32, 33] discussed in detail an Abelian axion field α in the context of the coupling of electromagnetism to gravity (see also Wilczek [60], de Sabbata & Sivaram [46], and the references given there). Field and Carroll [9] discussed pseudoscalar effects in the early universe, including an Abelian axion and its relation to a possible primordial cosmological helicity; also magnetic helicity, which we addressed earlier in (B.3.17), plays a role therein. One should also compare Haugan & Lämmerzahl [16].

⁴In contrast to the axions related to *non*-Abelian gauge theories, see Peccei and Quinn [38], Weinberg [58], Wilczek and Moody [59, 30], and the reviews in Kolb and Turner [24] and Sikivie [49].

⁵See Huang [19], Chapter III.

equation

$$d\tilde{\kappa}(F) + d\alpha \wedge F = J. \quad (\text{D.1.49})$$

Only the gradient of the axion field enters; that is, an axion field that is constant in time and space does not emerge in the Maxwell equations.

Equation (D.1.49), for $^{(2)}\kappa = 0$, can be derived from the Lagrangian

$$V = -\frac{1}{2} \left[F \wedge {}^{(1)}\kappa(F) + \alpha F \wedge F \right]. \quad (\text{D.1.50})$$

As a look at (B.2.29) shows, the axion part of the Lagrangian reads

$$^{(3)}V = -\frac{1}{2} \alpha F \wedge F = -\alpha E \wedge B \wedge d\sigma, \quad (\text{D.1.51})$$

with the electric field strength 1-form E and the magnetic 2-form B in 3D. This term can also be rewritten as an exact form plus a supplementary term:

$$^{(3)}V = -\frac{1}{2} \alpha F \wedge F = d \left(-\frac{1}{2} \alpha F \wedge A \right) - \frac{1}{2} F \wedge A \wedge d\alpha. \quad (\text{D.1.52})$$

For the special case of $\alpha = \text{const}$, we are left with a pure surface term. Since a surface term doesn't contribute to the field equation, we recover the result mentioned above.

We proved in (D.1.39) that the axion doesn't occur in the energy-momentum current of the electromagnetic field. The axion is, in this energetic sense, a “ghost”: $^{(3)}\kappa_{\Sigma_\alpha} = 0$.

As we mentioned already above, the first four axioms are not touched by the possible existence of the axion field. In particular, charge remains conserved: $dJ = 0$.

Experimentally, the Abelian axion has not been found so far. In particular, it couldn't be traced in ring laser experiments.⁶ Nevertheless, as we shall see below, the axion does not interfere with the light cone structure of spacetime at all. Therefore, this chapter is not yet closed, the Abelian axion remains a serious option for a particle search in experimental high energy physics and in cosmology.

D.1.5 Skewon field $\mathcal{S}_i^j$ and dissipation

Can we simplify $^{(2)}\chi$ in a similar way as $^{(3)}\chi$? Well, it has fifteen independent components. Hence a traceless second rank tensor could fit. That this is correct can be seen from (D.1.29).

⁶See Cooper & Stedman [6] and Stedman [51] for a systematic and extended series of experiments.

Before we start with the actual construction, we remind ourselves of the properties of $^{(2)}\chi$. Because of its irreducibility, it possesses the symmetries

$$^{(2)}\chi^{ijkl} = -^{(2)}\chi^{klij}, \quad ^{(2)}\chi^{[ijkl]} = 0. \quad (\text{D.1.53})$$

Being a piece of χ , the skewon part inherits its antisymmetries in the first and second pair of indices:

$$^{(2)}\chi^{(ij)kl} = 0, \quad ^{(2)}\chi^{ij(kl)} = 0. \quad (\text{D.1.54})$$

We define the *skewon field*⁷ by

$$\mathcal{S}_i{}^j(x) := \frac{1}{4} \hat{\epsilon}_{iklm} ^{(2)}\chi^{klmj}. \quad (\text{D.1.55})$$

Because of (D.1.53)₂, its trace vanishes,

$$\mathcal{S}_n{}^n = \frac{1}{4} \hat{\epsilon}_{nklm} ^{(2)}\chi^{[klmn]} = 0. \quad (\text{D.1.56})$$

In order to indicate the vanishing trace, we added a slash to the S .

⊗ Invert (D.1.55)

We multiply (D.1.55) by ϵ^{inpq} and find

$$\epsilon^{inpq} \mathcal{S}_i{}^j = \frac{1}{4} \epsilon^{inpq} \hat{\epsilon}_{iklm} ^{(2)}\chi^{klmj} = \frac{1}{4} \delta_{klm}^{npq} ^{(2)}\chi^{[klm]j} \quad (\text{D.1.57})$$

or

$$^{(2)}\chi^{[ijk]l} = -\frac{2}{3} \epsilon^{ijkm} \mathcal{S}_m{}^l. \quad (\text{D.1.58})$$

We expand the bracket:

$$^{(2)}\chi^{ijkl} + ^{(2)}\chi^{jkil} + ^{(2)}\chi^{kijl} = -2 \epsilon^{ijkm} \mathcal{S}_m{}^l. \quad (\text{D.1.59})$$

The second term on the left-hand side of this equation, by means of the symmetries (D.1.53)₁ and (D.1.54)₁, can be rewritten as $^{(2)}\chi^{jkil} = -^{(2)}\chi^{iljk} = ^{(2)}\chi^{lijk}$. Thus,

$$^{(2)}\chi^{ijkl} + 2^{(2)}\chi^{(k|ij|l)} = -2 \epsilon^{ijkm} \mathcal{S}_m{}^l, \quad (\text{D.1.60})$$

or because of (D.1.54)₂,

$$^{(2)}\chi^{ijkl} = 2 \epsilon^{ijm[k} \mathcal{S}_m{}^{l]}. \quad (\text{D.1.61})$$

In order to make the symmetry (D.1.53)₁ manifest, we rename the indices

$$^{(2)}\chi^{klij} = 2 \epsilon^{klm[i} \mathcal{S}_m{}^{j]} \quad (\text{D.1.62})$$

⁷The skewon field $\mathcal{S}_i{}^j(x)$ was first introduced by Hehl, Obukhov, and Rubilar [17].

and subtract (D.1.62) from (D.1.61). This yields the final result

$${}^{(2)}\chi^{ijkl} = \epsilon^{ijm[k} \mathcal{S}_m{}^{l]} - \epsilon^{klm[i} \mathcal{S}_m{}^{j]}. \quad (\text{D.1.63})$$

For (D.1.63), all the symmetries (D.1.53) and (D.1.54) can be verified straightforwardly. In (D.1.55), we chose the conventional factor as 1/4 in order to find in (D.1.63) a formula free of inconvenient factors. Thus, (D.1.63) represents the inverse of (D.1.55) and ${}^{(2)}\chi^{ijkl}$ and $\mathcal{S}_i{}^j$ are equivalent, indeed.

⊗ Two identities and a master formula

The two terms on the right-hand side of (D.1.63) can be better understood by means of the following identity: *In four dimensions, every traceless tensor of type $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\mathcal{S}_i{}^j$ fulfills*

$$\epsilon^{ijm[k} \mathcal{S}_m{}^{l]} \equiv -\epsilon^{klm[i} \mathcal{S}_m{}^{j]}. \quad (\text{D.1.64})$$

Let us sketch the proof. In 4D, any object with five completely antisymmetrized indices is zero, $Z_{\dots}^{[ijkl]} \equiv 0$. When four of these five indices belong to the Levi-Civita symbol, we have the identity:

$$\epsilon^{ijmk} Z_{\dots}^l \equiv \epsilon^{l jmk} Z_{\dots}^i + \epsilon^{ilmk} Z_{\dots}^j + \epsilon^{ijlk} Z_{\dots}^m + \epsilon^{ijml} Z_{\dots}^k. \quad (\text{D.1.65})$$

Applying this to the case of $Z = \mathcal{S}_m{}^l$, we find the identity

$$\epsilon^{ijmk} \mathcal{S}_m{}^l \equiv \epsilon^{l jmk} \mathcal{S}_m{}^i + \epsilon^{ilmk} \mathcal{S}_m{}^j + \epsilon^{ijlk} \mathcal{S}_m{}^m + \epsilon^{ijml} \mathcal{S}_m{}^k. \quad (\text{D.1.66})$$

Suppose that $\mathcal{S}_i{}^j$ is a *traceless* tensor, i.e., $\mathcal{S}_m{}^m = 0$. Then a simple rearrangement of the terms in the above identity yields,

$$\epsilon^{ijmk} \mathcal{S}_m{}^l - \epsilon^{ijml} \mathcal{S}_m{}^k \equiv -\epsilon^{klmi} \mathcal{S}_m{}^j + \epsilon^{klmj} \mathcal{S}_m{}^i, \quad \text{qed.} \quad (\text{D.1.67})$$

By means of the identity (D.1.64), it is obvious that (D.1.61) and (D.1.63) are equivalent representations of ${}^{(2)}\chi^{ijkl}$.

A comparison of our results with (D.1.29) and (D.1.30) shows that the skewon and the axion field can be expressed in terms of κ as follows:

$$\mathcal{S}_i{}^j = -\frac{1}{2} \kappa_i{}^j, \quad \alpha = \frac{1}{12} \kappa. \quad (\text{D.1.68})$$

For $\kappa_{ij}{}^{kl}$ one can find the following identity: *In four dimensions, every tensor of type $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$ $\kappa_{ij}{}^{kl}$ that is traceless, $\kappa_{ki}{}^{kj} = 0$, fulfills⁸*

$$\epsilon^{ijmn} \kappa_{mn}{}^{[kl]} = \epsilon^{klmn} \kappa_{mn}{}^{[ij]}. \quad (\text{D.1.69})$$

⁸Incidentally, this property applies in particular to the Weyl curvature tensor $C_{ij}{}^{kl}$ of a 4D Riemannian space. In this case, we recover from (D.1.69) the well known anti-self-duality of the Weyl tensor: $C_{ij}{}^{kl} = \frac{1}{4} \epsilon^{klmn} \hat{\epsilon}_{ijpq} C_{mn}{}^{pq}$.

Let us finally collect the decomposition of the constitutive tensor density of spacetime χ^{ijkl} in the master formula

$$\boxed{\chi^{ijkl} = {}^{(1)}\chi^{ijkl} + \epsilon^{ijm[k} \mathcal{S}_m{}^{l]} - \epsilon^{klm[i} \mathcal{S}_m{}^{j]} + \epsilon^{ijkl} \alpha,} \quad (\text{D.1.70})$$

with

$${}^{(1)}\chi^{ijkl} = {}^{(1)}\chi^{klij}, \quad {}^{(1)}\chi^{[ijkl]} = 0, \quad \mathcal{S}_m{}^m = 0. \quad (\text{D.1.71})$$

We can now forget the details of the derivations. The formulas (D.1.70) and (D.1.71) contain decisive information about the decomposition of the constitutive tensor density of spacetime in terms of the principal part ${}^{(1)}\chi^{ijkl}$ (20 independent components), the skewon field $\mathcal{S}_i{}^j$ (15 components), and the axion field α (1 component).

In terms of $\kappa_{ij}{}^{kl}$, (D.1.70) reads

$$\kappa_{ij}{}^{kl} = {}^{(1)}\kappa_{ij}{}^{kl} + 2\delta_{ij}^{m[k} \mathcal{S}_m{}^{l]} + \delta_{ij}^{kl} \alpha. \quad (\text{D.1.72})$$

This yields for the spacetime relation

$$H_{ij} = \frac{1}{2} {}^{(1)}\kappa_{ij}{}^{kl} F_{kl} + 2\mathcal{S}_{[i}{}^k F_{j]k} + \alpha F_{ij}. \quad (\text{D.1.73})$$

⊗ Properties of $\mathcal{S}_i{}^j$

In contrast to the axion field, the skewon field does contribute to the electromagnetic energy-momentum. In general, ${}^{(2)}\mathbf{k}\Sigma_\alpha \neq 0$ (see (D.1.40)). Therefore the hypothetical skewon field is expected to have more impact on light propagation than the axion field. And this turns out to be true.

What has been said about the axion field should also be stressed here: The high degree of universality of $\mathcal{S}_i{}^j$. The metric g , to be derived from the principal part ${}^{(1)}\chi$, the skewon $\mathcal{S}$ and the axion α come up in a similar context (see (D.1.70)). They just emerge from three irreducible parts of the same quantity. For this reason, the hypothesis of the existence of the skewon field seems reasonable.

However, as we saw in (D.1.43), the skewon part of the Lagrangian vanishes identically: ${}^{(2)}V = 0$. This is the usual argument for discarding ${}^{(2)}\chi$. Since we conventionally assume that all information of a physical system is collected in its Lagrangian, we reject $\mathcal{S}_i{}^j \neq 0$ as being unphysical. Nevertheless, this argument does *not* forbid the existence of $\mathcal{S}_i{}^j \neq 0$. It only implies that V is “insensitive” to $\mathcal{S}_i{}^j$. In other words, if $\mathcal{S}_i{}^j \neq 0$, then not all information about the system is contained in the Lagrangian.

Remember that pre-metric electrodynamics is based on the conservation laws of electric charge and magnetic flux and on an axiom about the (kinematic) electromagnetic energy-momentum current ${}^{\mathbf{k}}\Sigma_\alpha$. No Lagrangian is needed nor assumed. But, of course, the proto-Lagrangian $\Lambda = -F \wedge H/2$ exists anyway and has indeed been introduced in the context of the discussion of ${}^{\mathbf{k}}\Sigma_\alpha$ (see

(B.5.11)). Accordingly, in pre-metric electrodynamics, even when linearity is introduced according to (D.1.1) and (D.1.3), Λ and V have no decisive meaning, and that Λ and V do not depend on $\mathcal{S}_i{}^j$ is interesting to note but no reason for a headache.

This reminds us of a complementary property of the axion α or of ${}^{(3)}\chi$. It features in V (see (D.1.44)), but it drops out of ${}^k\Sigma_\alpha$ (see (D.1.40) and Table D.1.1). Should we be alarmed that the axion doesn't contribute to the electromagnetic energy-momentum current? No, not really. Full-fledged theories of the Abelian axion field can be worked out. By the same token, in linear pre-metric electrodynamics, it is not alarming that $\mathcal{S}_i{}^j$ drops out from the Lagrangian V , and in the future we will take *the possible existence of $\mathcal{S}_i{}^j$ for granted*.

$\otimes$ Dissipation

What is then the possible physical meaning of $\mathcal{S}_i{}^j$? Let us recall the “charge” of (B.5.13) that is defined for every vector field $\xi = \xi^\alpha e_\alpha$ by

$$\mathcal{Q} := \xi^\alpha {}^k\Sigma_\alpha = \frac{1}{2} [F \wedge (\xi \lrcorner H) - H \wedge (\xi \lrcorner F)] . \quad (\text{D.1.74})$$

For its exterior derivative, we found

$$d\mathcal{Q} = (\xi \lrcorner F) \wedge J + \frac{1}{2} (F \wedge \mathcal{L}_\xi H - H \wedge \mathcal{L}_\xi F) \quad (\text{D.1.75})$$

(see (B.5.18)), or in holonomic components (see (B.5.19)),

$$\partial_i \check{\mathcal{Q}}^i = \xi^k F_{kl} \check{\mathcal{J}}^l + \frac{1}{4} (F_{kl} \mathcal{L}_\xi \check{\mathcal{H}}^{kl} - \check{\mathcal{H}}^{kl} \mathcal{L}_\xi F_{kl}) . \quad (\text{D.1.76})$$

Here $\mathcal{L}_\xi$ denotes the Lie derivative along ξ . Now we substitute the linear relation (D.1.12), or $\check{\mathcal{H}}^{kl} = \chi^{klmn} F_{mn}/2$, with $\check{\mathcal{H}}^{kl} = \epsilon^{kl ij} H_{ij}/2$, and find

$$\partial_i \check{\mathcal{Q}}^i = \xi^k F_{kl} \check{\mathcal{J}}^l + \frac{1}{8} [F_{kl} \mathcal{L}_\xi (\chi^{klmn} F_{mn}) - \chi^{klmn} F_{mn} \mathcal{L}_\xi F_{kl}] . \quad (\text{D.1.77})$$

We apply the Leibniz rule of the Lie derivative and rearrange a bit:

$$\partial_i \check{\mathcal{Q}}^i = \xi^k F_{kl} \check{\mathcal{J}}^l + \frac{1}{8} [(\mathcal{L}_\xi \chi^{ijkl}) F_{ij} F_{kl} + (\chi^{ijkl} - \chi^{klij}) F_{ij} \mathcal{L}_\xi F_{kl}] . \quad (\text{D.1.78})$$

We substitute the irreducible pieces of χ^{ijkl} . Then we have

$$\partial_i \check{\mathcal{Q}}^i = \xi^k F_{kl} \check{\mathcal{J}}^l + \frac{1}{8} \mathcal{L}_\xi \left({}^{(1)}\chi^{ijkl} + {}^{(3)}\chi^{ijkl} \right) F_{ij} F_{kl} + \frac{1}{4} {}^{(2)}\chi^{ijkl} F_{ij} \mathcal{L}_\xi F_{kl} . \quad (\text{D.1.79})$$

If the vector ξ is chosen according to $\xi = e_0 = n$, then the charge $\mathcal{Q}$ represents the energy of the electromagnetic field (cf. (B.5.59)). The vanishing of the Lie derivatives $\mathcal{L}_n {}^{(1)}\chi = \mathcal{L}_n {}^{(2)}\chi = 0$ has a clear physical meaning in that these

parts of the constitutive tensor density have the same values on every leaf of the spacetime foliation. In other words, they are constant in “time.” In this case in vacuum, i.e., for $\mathcal{J}^i = 0$, we conclude that the energy $\mathcal{Q}$ is not conserved because of the offending term $^{(2)}\chi F \dot{F}/4 = \mathcal{S}_i{}^j \tilde{\mathcal{F}}^{ik} \dot{F}_{kj}$. Here $\tilde{\mathcal{F}}^{ij} = \epsilon^{ijkl} F_{kl}/2$, and the dot symbolizes the “time” derivative.

Thus $^{(2)}\chi$ or, equivalently, the skewon field $\mathcal{S}_i{}^j$ induces a dissipative term in the energy balance equation carrying a first “time” derivative. This observation is consistent with the dropping out of the skewon field from the Lagrangian since it is well known that dissipative phenomena in general cannot be described within a Lagrangian framework.

It is then our hypothesis that $\mathcal{S}_i{}^j$ represents a field that is *odd under T transformations*. Of course, we must investigate how these *skewons*, as we may call them in a preliminary way, disturb the light cone and whether there exists perhaps merely a viable subclass of the skewons.

D.1.6 Principal part of the constitutive tensor

In the decomposition formula (D.1.70), it is the principal part $^{(1)}\chi^{ijkl}$ that we haven’t discussed so far. It has 20 independent components and we might wonder whether there is, similar to the axion and the skewon parts, the possibility of introducing a simplified version of $^{(1)}\chi^{ijkl}$.

We are not aware of any method of representing $^{(1)}\chi^{ijkl}$ in a useful alternative way. Nevertheless, since a symmetric second rank tensor has 10 independent components, one could try the ansatz

$$^{(1)}\chi^{ijkl} \sim 4 g^{[i[k} h^{l]j]} = g^{ik} h^{lj} - g^{il} h^{kj} - g^{jk} h^{li} + g^{jl} h^{ki}, \quad (\text{D.1.80})$$

with two symmetric fields $g^{ij} = g^{ji}$ and $h^{ij} = h^{ji}$. Our ansatz certainly obeys the algebraic symmetries of $^{(1)}\chi^{ijkl}$ and depends on $10+10$ independent fields. However, (D.1.80) represents only a special case of $^{(1)}\chi^{ijkl}$, as we shall see in Sec. D.4.4, where we construct examples that turn out to be more general.

In spite of this negative outcome of our consideration, we get the idea that, provided $g^{ij} = h^{ij}$, a symmetric second rank tensor may be hidden in $^{(1)}\chi^{ijkl}$. But clearly, one has to find suitable constraints that $^{(1)}\chi^{ijkl}$ has to fulfill in order to be able to get down to this symmetric second rank tensor field.

D.1.7 Six-dimensional representation of the spacetime relation

We already have different versions of the spacetime relation: the operator version (D.1.1) with (D.1.3), the κ -versions in (D.1.6), and the χ -versions in (D.1.12). We now concentrate on the 6D-versions

$$H_I = \kappa_I{}^K F_K = \hat{\epsilon}_{IM} \chi^{MK} F_K. \quad (\text{D.1.81})$$

We decompose excitation and field strength with respect to the B^I cobasis and also into 1 + 3 (see (B.4.6) and (B.4.7)):

$$H = H_I B^I = -\mathcal{H} \wedge d\sigma + \mathcal{D}, \quad (\text{D.1.82})$$

$$F = F_I B^I = E \wedge d\sigma + B. \quad (\text{D.1.83})$$

We recall the (3 + 3)-decomposition (A.1.79) of B^I :

$$B^I = \begin{pmatrix} \beta^a \\ \hat{\epsilon}_b \end{pmatrix} = \begin{pmatrix} \vartheta^0 \wedge \vartheta^a \\ \frac{1}{2} \hat{\epsilon}_{bcd} \vartheta^c \wedge \vartheta^d \end{pmatrix}. \quad (\text{D.1.84})$$

Since every longitudinal (spatial) 1-form can be decomposed with respect to the coframe ϑ^a , whereas every 2-form can be conveniently expanded with respect to the 2-form basis $\hat{\epsilon}_a$, we have ($a, b, \dots = 1, 2, 3$)

$$\mathcal{H} = \mathcal{H}_a \vartheta^a, \quad E = E_a \vartheta^a, \quad (\text{D.1.85})$$

$$\mathcal{D} = \mathcal{D}^b \hat{\epsilon}_b, \quad B = B^b \hat{\epsilon}_b. \quad (\text{D.1.86})$$

We substitute this into (D.1.82), (D.1.83) and find

$$H = H_I B^I = \mathcal{H}_a \beta^a + \mathcal{D}^b \hat{\epsilon}_b, \quad (\text{D.1.87})$$

$$F = F_I B^I = -E_a \beta^a + B^b \hat{\epsilon}_b. \quad (\text{D.1.88})$$

Now we can write H and F as column vectors. Using (D.1.87) and (D.1.88), we can recast (D.1.81) into the form

$$\begin{pmatrix} \mathcal{H}_a \\ \mathcal{D}^a \end{pmatrix} = \begin{pmatrix} \mathcal{C}^b{}_a & \mathcal{B}_{ba} \\ \mathcal{A}^{ba} & \mathcal{D}_b{}^a \end{pmatrix} \begin{pmatrix} -E_b \\ B^b \end{pmatrix}, \quad (\text{D.1.89})$$

with the constitutive tensors

$$\kappa_I{}^K = \begin{pmatrix} \mathcal{C}^b{}_a & \mathcal{B}_{ba} \\ \mathcal{A}^{ba} & \mathcal{D}_b{}^a \end{pmatrix}, \quad \chi^{IK} = \begin{pmatrix} \mathcal{B}_{ab} & \mathcal{D}_a{}^b \\ \mathcal{C}^a{}_b & \mathcal{A}^{ab} \end{pmatrix}. \quad (\text{D.1.90})$$

The new entries here are the constitutive 3×3 matrices $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ introduced by

$$\mathcal{A}^{ba} := \chi^{0a0b}, \quad (\text{D.1.91})$$

$$\mathcal{B}_{ba} := \frac{1}{4} \hat{\epsilon}_{acd} \hat{\epsilon}_{bef} \chi^{cdef}, \quad (\text{D.1.92})$$

$$\mathcal{C}^a{}_b := \frac{1}{2} \hat{\epsilon}_{bcd} \chi^{cd0a}, \quad (\text{D.1.93})$$

$$\mathcal{D}_a{}^b := \frac{1}{2} \hat{\epsilon}_{acd} \chi^{0bcd}. \quad (\text{D.1.94})$$

If we resolve with respect to χ , we find

$$\chi^{0a0b} = \mathcal{A}^{ba}, \quad (\text{D.1.95})$$

$$\chi^{0abc} = \epsilon^{bcd} \mathcal{D}_d{}^a, \quad (\text{D.1.96})$$

$$\chi^{ab0c} = \epsilon^{abd} \mathcal{C}_d{}^c, \quad (\text{D.1.97})$$

$$\chi^{abcd} = \epsilon^{abe} \epsilon^{cdf} \mathcal{B}_{fe}. \quad (\text{D.1.98})$$

The decomposition of χ^{IK} can be found in (D.1.15). In terms of the new matrices, we find straightforwardly

$$^{(1)}\chi^{IK} = \begin{pmatrix} \mathcal{B}_{(ab)} & \frac{1}{2}(\mathcal{P}_a^b + \mathcal{L}^{ba}) \\ \frac{1}{2}(\mathcal{L}_a^b + \mathcal{P}_b^a) & \mathcal{A}^{(ab)} \end{pmatrix}, \quad (\text{D.1.99})$$

$$^{(2)}\chi^{IK} = \begin{pmatrix} \mathcal{B}_{[ab]} & \frac{1}{2}(\mathcal{D}_a^b - \mathcal{C}_b^a) \\ \frac{1}{2}(\mathcal{C}_a^b - \mathcal{D}_b^a) & \mathcal{A}^{[ab]} \end{pmatrix}, \quad (\text{D.1.100})$$

$$^{(3)}\chi^{IK} = \frac{1}{6}(\mathcal{C}_c^c + \mathcal{D}_c^c) \begin{pmatrix} 0_3 & 1_3 \\ 1_3 & 0_3 \end{pmatrix} = \frac{1}{6}(\mathcal{C}_c^c + \mathcal{D}_c^c) \epsilon^{IK}. \quad (\text{D.1.101})$$

By reordering (see (D.1.90)), we can easily find the corresponding irreducible κ_I^K -parts. For a 3D traceless quantity, we used the notation $\mathcal{M}_a^b := M_a^b - M_c^c \delta_a^b/3$. Note that for the corresponding 4D case we use the slash $/$.

Now we would like to express (D.1.100) and (D.1.101) in terms of $\mathcal{S}_i^j$ and α , respectively. We find

$$^{(2)}\chi^{IK} = \begin{pmatrix} -\hat{\epsilon}_{abc}\mathcal{S}_0^c & +\mathcal{S}_a^b - \delta_a^b \mathcal{S}_c^c \\ -\mathcal{S}_b^a + \delta_b^a \mathcal{S}_c^c & \epsilon^{abc}\mathcal{S}_c^0 \end{pmatrix} \quad (\text{D.1.102})$$

and

$$^{(3)}\chi^{IK} = \alpha \begin{pmatrix} 0_3 & 1_3 \\ 1_3 & 0_3 \end{pmatrix} = \alpha \epsilon^{IK}. \quad (\text{D.1.103})$$

The corresponding κ -parts read

$$^{(2)}\kappa_I^K = \begin{pmatrix} -\mathcal{S}_b^a + \delta_b^a \mathcal{S}_c^c & -\hat{\epsilon}_{abc}\mathcal{S}_0^c \\ \epsilon^{abc}\mathcal{S}_c^0 & +\mathcal{S}_a^b - \delta_a^b \mathcal{S}_c^c \end{pmatrix} \quad (\text{D.1.104})$$

and

$$^{(3)}\kappa_I^K = \alpha \begin{pmatrix} 1_3 & 0_3 \\ 0_3 & 1_3 \end{pmatrix} = \alpha \delta_I^K. \quad (\text{D.1.105})$$

Comparing (D.1.101) with (D.1.103) and (D.1.105), we easily recover the axion α (cf. (D.1.68)₂). As for the skewon $\mathcal{S}$ contribution, using our master formula (D.1.70), we find

$$^{(2)}\mathcal{A}^{ba} := ^{(2)}\chi^{0a0b} = \epsilon^{abc}\mathcal{S}_c^0, \quad (\text{D.1.106})$$

$$^{(2)}\mathcal{B}_{ba} := \frac{1}{4}\hat{\epsilon}_{acd}\hat{\epsilon}_{bef} ^{(2)}\chi^{cdef} = -\hat{\epsilon}_{abc}\mathcal{S}_0^c, \quad (\text{D.1.107})$$

$$^{(2)}\mathcal{C}_a^b := \frac{1}{2}\hat{\epsilon}_{bcd} ^{(2)}\chi^{cd0a} = -\mathcal{S}_b^a + \delta_b^a \mathcal{S}_c^c, \quad (\text{D.1.108})$$

$$^{(2)}\mathcal{D}_a^b := \frac{1}{2}\hat{\epsilon}_{acd} ^{(2)}\chi^{0bcd} = \mathcal{S}_a^b - \delta_a^b \mathcal{S}_c^c. \quad (\text{D.1.109})$$

This is what we substituted into (D.1.102) and (D.1.104). For explicit reference, we can also alternatively display the spacetime relations of the skewon and the axion as

$${}^{(2)}\mathcal{D}^a = -\epsilon^{abc}\mathcal{S}_c{}^0 E_b + (-\delta_b^a \mathcal{S}_c{}^c + \mathcal{S}_b{}^a)B^b, \quad (\text{D.1.110})$$

$${}^{(2)}\mathcal{H}_a = (-\delta_a^b \mathcal{S}_c{}^c + \mathcal{S}_a{}^b)E_b - \hat{\epsilon}_{abc}\mathcal{S}_0{}^c B^b, \quad (\text{D.1.111})$$

and

$${}^{(3)}\mathcal{D}^a = +\alpha B^a, \quad (\text{D.1.112})$$

$${}^{(3)}\mathcal{H}_a = -\alpha E_a. \quad (\text{D.1.113})$$

D.1.8 $\otimes$ Special case: Spatially isotropic skewon field

The axion field, as a pseudoscalar function, is isotropic by definition. The skewon field $\mathcal{S}_i{}^j$, as a second rank tensor field, is anisotropic. However, as a specific subcase, it can be 3D isotropic $\mathcal{S}_a{}^b \sim \delta_a^b$. We investigate that case in this section. We hasten to add that the principal part ${}^{(1)}\chi^{ijkl}$ cannot be isotropic, since it has only upper indices and no proportionality with a Kronecker delta can exist.

Consider (D.1.104). If we want 3D (spatial) isotropy, then we have to choose,

$$\mathcal{S}_a{}^b = \frac{s}{2}\delta_a^b, \quad \mathcal{S}_0{}^c = 0, \quad \mathcal{S}_c{}^0 = 0, \quad (\text{D.1.114})$$

with the 3D pseudoscalar function $s = s(x)$. Thus,

$$\mathcal{S}_i{}^j = \frac{s}{2} \begin{pmatrix} -3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (\text{D.1.115})$$

and

$$-\delta_a^b \mathcal{S}_c{}^c + \mathcal{S}_a{}^b = -s\delta_a^b. \quad (\text{D.1.116})$$

Consequently, (D.1.110) and (D.1.111) become⁹

$${}^{(2)}\mathcal{D}^a = -s B^a, \quad {}^{(2)}\mathcal{H}_a = -s E_a. \quad (\text{D.1.117})$$

⁹Nieves and Pal [34, 35] had postulated and discussed such relations within a material medium. Accordingly, the spacetime relations (D.1.110), (D.1.111) are *anisotropic* generalizations of the Nieves and Pal ansatz. The off-diagonal terms with $\mathcal{S}_0{}^a$ and $\mathcal{S}_b{}^0$ lead, respectively, to magnetic and electric Faraday-type effects of the spacetime under consideration; i.e., these terms rotate the polarization of a wave propagating in such a spacetime.

If we substitute (D.1.117) and (D.1.112), (D.1.113) into (D.1.89), we can display the complete spacetime relation as principal part $\oplus$ isotropic skewon field $\oplus$ axion field:

$$\mathcal{D}^a = {}^{(1)}\mathcal{D}^a(E, B) + (-s + \alpha) B^a, \quad (\text{D.1.118})$$

$$\mathcal{H}_a = (-s - \alpha) E_a + {}^{(1)}\mathcal{H}_a(E, B). \quad (\text{D.1.119})$$

Note how the isotropic skewon and the axion act with different signs on E_a and B^a . Thereby they can be distinguished phenomenologically.

D.2

Propagation of electromagnetic waves: Quartic wave surface

We substitute the linear spacetime relation into the vacuum Maxwell equations and determine how electromagnetic waves propagate. We find that the wave covectors lie on quartic surfaces if no constraints are imposed on the constitutive tensor.

D.2.1 Fresnel equation

As soon as the spacetime relation is specified, electrodynamics becomes a predictive theory and one can study various of its physical effects, such as the propagation of electromagnetic disturbances and, in particular, the phenomenon of wave propagation in vacuum.

Often, for such a purpose, one applies the geometrical optics approximation. The field strength is split into a background field plus a wave term of the form $f_{ij} e^{i\Phi}$ where the amplitude f varies slowly compared to the phase Φ . It is assumed that the scale over which the electromagnetic background field varies is much larger than that of the variations of Φ . Then one derives a system of algebraic equations for the amplitude f that ultimately determines the characteristics of the wave solutions in terms of the background field.

An alternative method, which we use because of its mathematical transparency, is Hadamard's theory of weak discontinuities.¹ It yields the same

¹The corresponding theory was developed in detail by Hadamard [13] and Lichnerowicz [28], among others; see also the remarkable paper of Kiehn, Kiehn, and Roberds [23].

physical results. In the theory of partial differential equations, the propagation of disturbances is described by the Hadamard discontinuities of solutions across a characteristic hypersurface S , the wave front. One can locally define S by the equation $\Phi(x^i) = \text{const.}$ The Hadamard discontinuity of any function $\mathcal{F}(x)$ across the hypersurface S is determined as the difference² $[\mathcal{F}](x) := \mathcal{F}(x_+) - \mathcal{F}(x_-)$, where $x_\pm := \lim_{\varepsilon \rightarrow 0} (x \pm \varepsilon)$ are points on the opposite sides of $S \ni x$. We call $[\mathcal{F}](x)$ the *jump* of the function $\mathcal{F}$ at x .

An ordinary electromagnetic wave is a solution of Maxwell's vacuum equations $dH = 0$ and $dF = 0$ for which the *derivatives* of H and F are discontinuous across the wave front hypersurface S . In terms of H and F , we have on the characteristic hypersurface S

$$[H] = 0, \quad [dH] = q \wedge h, \quad (\text{D.2.1})$$

$$[F] = 0, \quad [dF] = q \wedge f. \quad (\text{D.2.2})$$

Here the 2-forms h, f describe the jumps (discontinuities) of the differentials of the electromagnetic field across S , and the wave-covector normal to the front is given by

$$q := d\Phi. \quad (\text{D.2.3})$$

The wave-covector q should not be mixed up with the dimension of charge. Equations (D.2.1) and (D.2.2) represent the Hadamard geometrical compatibility conditions. If we use Maxwell's vacuum equations $dH = 0$ and $dF = 0$, then (D.2.1) and (D.2.2) yield

$$q \wedge h = 0, \quad q \wedge f = 0. \quad (\text{D.2.4})$$

Since we can solve these equations by $h = q \wedge c$ and $f = q \wedge a$, we recognize that these 2-forms fulfill

$$f \wedge h = 0, \quad f \wedge f = 0, \quad h \wedge h = 0; \quad (\text{D.2.5})$$

that is, the first three invariants of the jumps³ h and f vanish (see (B.2.28)–(B.2.33)).

We can say something more about the jump h if we apply the spacetime relation (D.1.47). Provided the components of the linear operator κ (that is, of $\tilde{\kappa}$ and α) are continuous across S , we find by differentiating (D.1.47) and using (D.2.1) and (D.2.2),

$$[dH] = [d\tilde{\kappa}(F)] + \alpha[dF] = q \wedge (\tilde{\kappa}(f) + \alpha f) = q \wedge h. \quad (\text{D.2.6})$$

²In Sec. D.2.1 only, brackets $[\]$ are used to characterize the jump of a function. Elsewhere they denote the dimension of the quantity enclosed by the brackets.

³Incidentally, one can also, somewhat formalistically, determine the energy-momentum of the wave by the replacements ${}^k\Sigma_\alpha \rightarrow {}^k\sigma_\alpha$, $H \rightarrow h$, and $F \rightarrow f$. One finds, ${}^k\sigma_\alpha = q_\alpha \omega$, with the flow 3-form $\omega := q \wedge a \wedge c$.

Accordingly, the jump equations (D.2.4) can be put into the form

$$q \wedge h = q \wedge \tilde{\kappa}(f) = 0, \quad q \wedge f = 0. \quad (\text{D.2.7})$$

Note that the *axion drops out* completely even though it occurs in Maxwell's equations. From the constitutive tensor κ only $\tilde{\kappa} = {}^{(1)}\kappa + {}^{(2)}\kappa$ is left over with $20 + 15$ independent components. If we multiply the two equations of (D.2.7) by q , both vanish identically. Hence only $3 + 3$ equations turn out to be independent. If $\tilde{\kappa}$ is specified, (D.2.7) represents six independent homogeneous algebraic equations for the six independent components of the jump f .

We can solve (D.2.4)₂ by

$$f = q \wedge a. \quad (\text{D.2.8})$$

The 1-form a is only determined up to a gauge transformation

$$a \longrightarrow a + \varphi q, \quad (\text{D.2.9})$$

with φ as an arbitrary scalar function (0-form). We substitute (D.2.8) into (D.2.4)₁:

$$q \wedge \tilde{\kappa}(q \wedge a) = 0. \quad (\text{D.2.10})$$

This is a 3-form. Hence we have four homogeneous algebraic equations for a . However, because of (D.2.9), only three equations are independent.

In order to isolate these three equations, we decompose a with respect to an *arbitrary* coframe $\vartheta^\alpha = e_i^\alpha dx^i$ according to $a = a_\beta \vartheta^\beta$. Here $\alpha, \beta, \dots = 0, 1, 2, 3$. Thus, because of the linearity of the operator $\tilde{\kappa}$, we have

$$q \wedge \tilde{\kappa}(q \wedge \vartheta^{\hat{0}}) a_{\hat{0}} + q \wedge \tilde{\kappa}(q \wedge \vartheta^b) a_b = 0. \quad (\text{D.2.11})$$

“Spatial” anholonomic indices run over $a, b, \dots = 1, 2, 3$. The trick is now to pick up a convenient *specific* coframe. At each point of spacetime, we can always choose to identify the zeroth leg of the coframe with the wave covector:

$$\vartheta^{\hat{0}} = q. \quad (\text{D.2.12})$$

The convenience of this choice is immediately evident from the fact that, with respect to this coframe, we have $a = a_{\hat{0}} q + a_b \vartheta^b$. Correspondingly, the gauge transformation (D.2.9) affects only the zeroth component, $a_{\hat{0}} \rightarrow a_{\hat{0}} + \varphi$, whereas the spatial components a_b are *gauge invariant*. Now, with respect to the coframe chosen, the first term in (D.2.11) vanishes, since $\vartheta^{\hat{0}} \wedge \vartheta^{\hat{0}} = 0$, and we find

$$\vartheta^{\hat{0}} \wedge \tilde{\kappa}(\vartheta^{\hat{0}} \wedge \vartheta^b) a_b = 0. \quad (\text{D.2.13})$$

There are still four equations. But by multiplication (from the right) with ϑ^α , we find a vector-valued 4-form. Again because $\vartheta^{\hat{0}} \wedge \vartheta^{\hat{0}} = 0$, only three equations are left over:

$$\vartheta^{\hat{0}} \wedge \vartheta^a \wedge \tilde{\kappa}(\vartheta^{\hat{0}} \wedge \vartheta^b) a_b = 0. \quad (\text{D.2.14})$$

Our problem is solved! These are three equations for the three spatial components of a , whereas the unphysical gauge-dependent $a_{\hat{0}}$ is successfully expelled.

In order to bring (D.2.14) into a more manageable form, we decompose the operator $\tilde{\kappa}$ according to (D.1.4) and use the 4-volume element $\hat{e}$ as in (A.1.61):

$$\frac{1}{2} \tilde{\kappa}_{cd}^{\hat{0}b} \vartheta^{\hat{0}} \wedge \vartheta^a \wedge \vartheta^c \wedge \vartheta^d a_b = \frac{1}{2} \epsilon^{\hat{0}acd} \tilde{\kappa}_{cd}^{\hat{0}b} a_b \hat{e} = 0. \quad (\text{D.2.15})$$

We recall the definition (D.1.9) of χ and put $\tilde{\chi} := {}^{(1)}\chi + {}^{(2)}\chi$; furthermore, $\hat{e} \neq 0$. Thus,

$$W^{ab} a_b = 0, \quad \text{with} \quad W^{ab} := \tilde{\chi}^{\hat{0}a\hat{0}b}. \quad (\text{D.2.16})$$

The necessary and sufficient condition for the existence of non-trivial solutions for a_b is the vanishing of the determinant $\mathcal{W}$ of the 3×3 matrix W :

$$\mathcal{W} := \det W = \frac{1}{3!} \hat{e}_{abc} \hat{e}_{def} W^{ad} W^{be} W^{cf} = \frac{1}{3!} \hat{e}_{abc} \hat{e}_{def} \tilde{\chi}^{\hat{0}a\hat{0}d} \tilde{\chi}^{\hat{0}b\hat{0}e} \tilde{\chi}^{\hat{0}c\hat{0}f}. \quad (\text{D.2.17})$$

This yields the Fresnel equation $\mathcal{W} = 0$.

However, to begin with, we rewrite $\mathcal{W}$ in a fully 4-dimensional covariant manner. First, we observe that the 3-dimensional Levi-Civita symbol $\hat{e}_{abc}$ is related to the 4-dimensional one by means of $\hat{e}_{abc} \equiv \hat{e}_{\hat{0}abc}$. Then we can extend one $\hat{0}$ -component of the constitutive tensors to a fourth summation index. As a result, we find the identity

$$\hat{e}_{\hat{0}abc} \tilde{\chi}^{\hat{0}a\hat{0}d} \tilde{\chi}^{\hat{0}b\hat{0}e} \tilde{\chi}^{\hat{0}c\hat{0}f} = \hat{e}_{\hat{0}\beta\gamma\delta} \tilde{\chi}^{\hat{0}\beta\hat{0}d} \tilde{\chi}^{\hat{0}\gamma\hat{0}e} \tilde{\chi}^{\hat{0}\delta\hat{0}f} = \frac{1}{2} \hat{e}_{\alpha\beta\gamma\delta} \tilde{\chi}^{\alpha\beta\hat{0}d} \tilde{\chi}^{\hat{0}\gamma\hat{0}e} \tilde{\chi}^{\hat{0}\delta\hat{0}f}, \quad (\text{D.2.18})$$

which holds true due to the (anti)symmetry properties of the Levi-Civita symbol and of the constitutive tensor. This allows us to rewrite (D.2.17) as

$$\mathcal{W} = \frac{1}{3!} \frac{1}{2} \hat{e}_{\alpha\beta\gamma\delta} \hat{e}_{\hat{0}def} \tilde{\chi}^{\alpha\beta\hat{0}d} \tilde{\chi}^{\hat{0}\gamma\hat{0}e} \tilde{\chi}^{\hat{0}\delta\hat{0}f}. \quad (\text{D.2.19})$$

Now we apply the same procedure to the second Levi-Civita symbol and finally obtain

$$\mathcal{W} = \frac{1}{4!} \hat{e}_{\alpha\beta\gamma\delta} \hat{e}_{\lambda\rho\sigma\tau} \tilde{\chi}^{\alpha\beta\hat{0}\rho} \tilde{\chi}^{\hat{0}\gamma\hat{0}\sigma} \tilde{\chi}^{\hat{0}\delta\lambda\tau}. \quad (\text{D.2.20})$$

Due to (D.2.12), the coordinate components of the wave vector are $e_i^{\hat{0}} = q_i$. Therefore, (D.2.20) can, in coordinate components, be rewritten as

$$\mathcal{W} = \frac{\theta^2}{4!} \hat{e}_{mnpq} \hat{e}_{rstu} \tilde{\chi}^{mnr(i} \tilde{\chi}^{jpsk} \tilde{\chi}^{lqtu} q_i q_j q_k q_l, \quad (\text{D.2.21})$$

with $\theta := \det(e_i^\alpha)$. This equation has a tensorial transformation behavior. Thus, it is valid with respect to an arbitrary coframe.

We define the fourth-order Tamm–Rubilar (TR) tensor density of weight +1,

$$\boxed{\mathcal{G}^{ijkl}(\chi) := \frac{1}{4!} \hat{e}_{mnpq} \hat{e}_{rstu} \chi^{mnr(i} \chi^{j|ps|k} \chi^{l)qtu}.} \quad (\text{D.2.22})$$

Here the total symmetrization is extended only over the four indices i, j, k, l , with all the dummy (or dead) summation indices excluded. $\mathcal{G}$ is totally symmetric, $\mathcal{G}^{ijkl}(\chi) = \mathcal{G}^{(ijkl)}(\chi)$. A completely symmetric tensor of rank p in n dimensions has, expressed as binomial coefficient, $\binom{n+p-1}{p} = \binom{n-1+p}{n-1}$ independent components. Thus, $\mathcal{G}$ carries $\binom{4-1+4}{4-1} = 35$ independent components. Note that we defined $\mathcal{G}$ with respect to the total constitutive tensor density χ and not only in terms of $\tilde{\chi}$. However, because $\chi^{ijkl} = \tilde{\chi}^{ijkl} + \alpha \epsilon^{ijkl}$, the total antisymmetry of ϵ yields $\mathcal{G}(\chi) = \mathcal{G}(\tilde{\chi})$. An explicit proof is given below in (D.2.35). This is consistent with the dropping out of the axion field in (D.2.7).

With (D.2.21) and (D.2.22), we find the (extended) Fresnel equation⁴ which is generally covariant in four dimensions:

$$\boxed{\mathcal{G}^{ijkl}(\tilde{\chi}) q_i q_j q_k q_l = 0.} \quad (\text{D.2.23})$$

It is always a *quartic* equation in q_i despite the fact that it was derived from a determinant of a 3×3 matrix quadratic in the wave covectors. Thus, the wave covectors q lie on a quartic Fresnel wave surface, not exactly what we are observing in vacuum at the present epoch of our universe. The rest of Part D is devoted to finding conditions that make the quartic Fresnel equation *factorize* into two quadratic ones and to determining under which circumstances the Fresnel equation turns out to be a perfect square.

D.2.2 $\otimes$ Properties of the Tamm–Rubilar tensor density

The various irreducible pieces of the constitutive tensor density affect wave propagation in different ways. Technically, this can be determined by studying how a certain irreducible piece $^{(a)}\chi$ contributes to the TR-tensor density (D.2.22) and, thereby, to the Fresnel equation (D.2.23), which governs the wave covectors.

We demonstrate here some algebraic properties of the TR-tensor density. The simplest is to check that

$$\mathcal{G}^{ijkl}({}^{(3)}\chi) = 0. \quad (\text{D.2.24})$$

By direct substitution of ${}^{(3)}\chi^{ijkl} = \alpha \epsilon^{ijkl}$ into (D.2.22), we have, before executing the symmetrization,

$$\hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} {}^{(3)}\chi^{mnri} {}^{(3)}\chi^{jpsk} {}^{(3)}\chi^{lqtu} = 4\alpha^3 \delta_{pq}^{ri} \delta_{rs}^{lq} \epsilon^{jpsk} = -16\alpha^3 \epsilon^{jilk}. \quad (\text{D.2.25})$$

⁴The original Fresnel equation described refringence in anisotropic crystals (see [27, 44, 52]). Our extended version, first derived by Obukhov, Fukui, and Rubilar (see [36]) also includes electric and magnetic Faraday rotation, optical activity, and Fizeau–Fresnel phenomena. Tamm [53] studied the relation between fourth- and second-order wave geometry for a special case of a linear constitutive law. He also introduced a “fourth-order metric” of the type of $\mathcal{G}$. A derivation of the 4D formula (D.2.22) was first found by Rubilar [45].

This vanishes upon symmetrization over the indices i, j, k, l . A bit more involved is the proof of

$$\mathcal{G}^{ijkl}({}^{(2)}\chi) = 0. \quad (\text{D.2.26})$$

Using the symmetry properties ${}^{(2)}\chi^{ijkl} = -{}^{(2)}\chi^{jikl} = -{}^{(2)}\chi^{ijlk} - {}^{(2)}\chi^{klij}$, we find

$$\begin{aligned} \hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} {}^{(2)}\chi^{mnri} {}^{(2)}\chi^{jpsk} {}^{(2)}\chi^{lqtu} \\ = -\hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} {}^{(2)}\chi^{rimn} {}^{(2)}\chi^{skjp} {}^{(2)}\chi^{tulq} \\ = -\hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} {}^{(2)}\chi^{tuql} {}^{(2)}\chi^{kspj} {}^{(2)}\chi^{irmn} \\ = -\hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} {}^{(2)}\chi^{mnrl} {}^{(2)}\chi^{kpsj} {}^{(2)}\chi^{iqtu}. \end{aligned} \quad (\text{D.2.27})$$

Upon symmetrization over the indices i, j, k, l , we find $\mathcal{G}({}^{(2)}\chi) = -\mathcal{G}({}^{(2)}\chi)$ or $\mathcal{G}({}^{(2)}\chi) = 0$.

Now we want to turn to “composite” $\mathcal{G}$ ’s like $\mathcal{G}({}^{(1)}\chi + {}^{(2)}\chi + {}^{(3)}\chi)$. As a preparatory step, let us take $\chi = \phi + \psi$. Then, upon substitution into (D.2.22), we have quite generally

$$\begin{aligned} \mathcal{G}^{ijkl}(\chi) &= \mathcal{G}^{ijkl}(\phi) + \mathcal{G}^{ijkl}(\psi) \\ &+ \frac{1}{4!} \left(O_1^{(ijkl)} + O_2^{(ijkl)} + O_3^{(ijkl)} + T_1^{(ijkl)} + T_2^{(ijkl)} + T_3^{(ijkl)} \right). \end{aligned} \quad (\text{D.2.28})$$

Here the mixed terms O_a contain one ψ -factor and the T_a ’s two ψ -factors. These terms read explicitly as follows:

$$O_1^{ijkl}(\phi, \psi, \phi) = \hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} \phi^{mnri} \psi^{jpsk} \phi^{lqtu}, \quad (\text{D.2.29})$$

$$O_2^{ijkl}(\psi, \phi, \phi) = \hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} \psi^{mnri} \phi^{jpsk} \phi^{lqtu}, \quad (\text{D.2.30})$$

$$O_3^{ijkl}(\phi, \phi, \psi) = \hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} \phi^{mnri} \phi^{jpsk} \psi^{lqtu}, \quad (\text{D.2.31})$$

$$T_1^{ijkl}(\psi, \phi, \psi) = \hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} \psi^{mnri} \phi^{jpsk} \psi^{lqtu}, \quad (\text{D.2.32})$$

$$T_2^{ijkl}(\psi, \psi, \phi) = \hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} \psi^{mnri} \psi^{jpsk} \phi^{lqtu}, \quad (\text{D.2.33})$$

$$T_3^{ijkl}(\phi, \psi, \psi) = \hat{\epsilon}_{mnpq} \hat{\epsilon}_{rstu} \phi^{mnri} \psi^{jpsk} \psi^{lqtu}. \quad (\text{D.2.34})$$

We defined the O_a ’s and the T_a ’s without symmetrization and apply symmetrization only afterwards in (D.2.28). Note that such index-stricken expressions can conveniently be manipulated by means of computer algebra in general and the Reduce system in particular.

As already mentioned above, we have the important relation

$$\mathcal{G}^{ijkl}({}^{(1)}\chi + {}^{(2)}\chi + {}^{(3)}\chi) = \mathcal{G}^{ijkl}({}^{(1)}\chi + {}^{(2)}\chi). \quad (\text{D.2.35})$$

For its proof, let us put in formula (D.2.28) $\phi = {}^{(1)}\chi + {}^{(2)}\chi$ and $\psi = {}^{(3)}\chi$. Then we can verify that all the mixed terms (D.2.29)–(D.2.34) vanish. Indeed, using ${}^{(3)}\chi^{ijkl} = \alpha \epsilon^{ijkl}$, we find

$$\begin{aligned} O_1^{ijkl} &= 2\alpha \hat{\epsilon}_{mnpq} \phi^{mnri} \phi^{lqtu} (\delta_r^j \delta_t^p \delta_u^k + \delta_r^p \delta_t^k \delta_u^j + \delta_r^k \delta_t^j \delta_u^p) \\ &= 2\alpha \hat{\epsilon}_{mnpq} (\phi^{mnji} \phi^{lqpk} + \phi^{mnpj} \phi^{lqkj} + \phi^{mnki} \phi^{lqjp}). \end{aligned}$$

This is zero when we impose the symmetrization over i, j, k, l . Similarly, we find

$$\begin{aligned} O_2^{ijkl} &= -2\alpha \hat{\epsilon}_{rstu} (\delta_p^r \delta_q^i - \delta_p^i \delta_q^r) \phi^{jpsk} \phi^{lqtu} \\ &= 2\alpha \hat{\epsilon}_{rstu} (-\phi^{jrsk} \phi^{litu} + \phi^{jisk} \phi^{lrtu}), \end{aligned} \quad (\text{D.2.36})$$

$$\begin{aligned} O_3^{ijkl} &= -2\alpha \hat{\epsilon}_{mnpq} (\delta_r^l \delta_s^q - \delta_r^q \delta_s^l) \phi^{mnri} \phi^{jpsk} \\ &= 2\alpha \hat{\epsilon}_{mnpq} (-\phi^{mnl i} \phi^{jpqk} + \phi^{mnq i} \phi^{jplk}). \end{aligned} \quad (\text{D.2.37})$$

Both expressions vanish when we impose symmetrization over i, j, k, l . The proof that all T 's are equal to zero reduces to the above formulas in which we simply need to replace one of the ϕ factors by the Levi-Civita ϵ . Consequently, (D.2.28) yields $\mathcal{G}(\phi + \psi) = \mathcal{G}(\phi) + \mathcal{G}(\psi) = \mathcal{G}(\phi)$, since $\mathcal{G}(\psi) = \mathcal{G}^{(3)}\chi = 0$, qed. The relation

$$\mathcal{G}^{ijkl(2)}\chi + {}^{(3)}\chi = 0, \quad (\text{D.2.38})$$

is a special case of (D.2.35). Just put ${}^{(1)}\chi = 0$ and use (D.2.26).

Let us pause for a moment with the algebra and let us look back at what we have achieved so far. Because of (D.2.24), (D.2.26), and (D.2.38), neither the axion part ${}^{(2)}\chi$ nor the skewon part ${}^{(3)}\chi$ (alone or together) can provide electromagnetic wave propagation, since the Fresnel equation collapses under those conditions. Therefore the presence of the *principal part* ${}^{(1)}\chi$ of the constitutive tensor density is indispensable for the existence of nontrivial electromagnetic waves and ultimately for the existence of the light cone structure on spacetime. In any case, the TR-tensor density reads

$$\mathcal{G}^{ijkl}(\chi) = \mathcal{G}^{ijkl}({}^{(1)}\chi + {}^{(2)}\chi). \quad (\text{D.2.39})$$

We recover the result that the axion field does not influence the Fresnel wave surfaces at all. Hence a possible axion field can easily remain unnoticed.

But what about the skewon field? In Table D.1.1 we marked already that it does influence the light propagation. Let us show this explicitly. Actually, we can use (D.2.28) to find more exactly the contribution of the skewon to the Fresnel tensor. One can straightforwardly see that

$$\begin{aligned} O_1({}^{(1)}\chi, {}^{(2)}\chi, {}^{(1)}\chi) &= 0, \\ O_2({}^{(2)}\chi, {}^{(1)}\chi, {}^{(1)}\chi) + O_3({}^{(1)}\chi, {}^{(1)}\chi, {}^{(2)}\chi) &= 0, \\ T_2({}^{(2)}\chi, {}^{(2)}\chi, {}^{(1)}\chi) &= T_3({}^{(1)}\chi, {}^{(2)}\chi, {}^{(2)}\chi). \end{aligned} \quad (\text{D.2.40})$$

The proof is analogous to the demonstration of the property $\mathcal{G}^{(2)}\chi = 0$ and is based directly on the skew symmetry ${}^{(2)}\chi^{ijkl} = -{}^{(2)}\chi^{klij}$. As a result, (D.2.28) yields

$$\begin{aligned} \mathcal{G}({}^{(1)}\chi + {}^{(2)}\chi) & \\ &= \mathcal{G}({}^{(1)}\chi) + \frac{1}{4!} [T_1({}^{(2)}\chi, {}^{(1)}\chi, {}^{(2)}\chi) + 2T_2({}^{(2)}\chi, {}^{(2)}\chi, {}^{(1)}\chi)]. \end{aligned} \quad (\text{D.2.41})$$

We substitute here the explicit form of the skewon part (D.1.63) and find

$$\boxed{\mathcal{G}^{ijkl}(\chi) = \mathcal{G}^{ijkl} \left({}^{(1)}\chi \right) + {}^{(1)}\chi^{m(i|n|j} \mathcal{S}_m^k \mathcal{S}_n^l) .} \quad (\text{D.2.42})$$

In fact, this formula subsumes the relations (D.2.24) (put ${}^{(1)}\chi = {}^{(2)}\chi = 0$), (D.2.26) (put ${}^{(1)}\chi = {}^{(3)}\chi = 0$), and (D.2.38) (put ${}^{(1)}\chi = 0$). Also (D.2.35) is contained in it since ${}^{(3)}\chi$ doesn't enter the right-hand side of (D.2.42). In other words, we can forget those relations that we derived earlier and just subscribe to (D.2.42).

Our basic results on light propagation are then that ${}^{(1)}\chi \neq 0$ (otherwise there is no orderly wave propagation), that the axion field α is left arbitrary (it doesn't influence light propagation locally), and that the skewon field $\mathcal{S}_i^j$ makes itself felt via equation (D.2.42).

D.2.3 $\otimes$ Fresnel equation decomposed into time and space

For practical calculations, it is convenient to use a 1 + 3 decomposition of the Fresnel equation (D.2.23). If we put successively

$$\begin{aligned} \mathcal{G}^{0000} &=: M, & \mathcal{G}^{000a} &=: \frac{1}{4}M^a, & \mathcal{G}^{00ab} &=: \frac{1}{6}M^{ab}, \\ \mathcal{G}^{0abc} &=: \frac{1}{4}M^{abc}, & \mathcal{G}^{abcd} &=: M^{abcd}, \end{aligned} \quad (\text{D.2.43})$$

then (D.2.23) reads

$$q_0^4 M + q_0^3 q_a M^a + q_0^2 q_a q_b M^{ab} + q_0 q_a q_b q_c M^{abc} + q_a q_b q_c q_d M^{abcd} = 0. \quad (\text{D.2.44})$$

Thus, we have expressed the 35 independent components of $\mathcal{G}$ in terms of the 35 independent components of the M 's. This comes about as follows: The M 's are totally symmetric expressions in their (spatial) indices. Since totally symmetric expressions of rank p in n dimensions (here $n = 3$) have $\binom{n+p-1}{p} = \binom{n-1+p}{n-1}$ independent components, the M 's altogether carry $1 \oplus 3 \oplus 6 \oplus 10 \oplus 15 = 35$ independent components.

As a comparison with (D.2.22) shows, the components of $\mathcal{G}$ are cubic in χ^{ijkl} . But the components χ^{ijkl} can be, in turn, by means of (D.1.95)–(D.1.98), expressed in terms of the matrices $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$. Upon substitution, we find, after

some algebra:

$$M = \det \mathcal{A}, \quad (\text{D.2.45})$$

$$M^a = -\hat{\epsilon}_{bcd} (\mathcal{A}^{ba} \mathcal{A}^{ce} \mathcal{C}_e^d + \mathcal{A}^{ab} \mathcal{A}^{ec} \mathcal{D}_e^d), \quad (\text{D.2.46})$$

$$\begin{aligned} M^{ab} = & \frac{1}{2} \mathcal{A}^{(ab)} [(\mathcal{C}_d^d)^2 + (\mathcal{D}_c^c)^2 - (\mathcal{C}_d^c + \mathcal{D}_d^c)(\mathcal{C}_c^d + \mathcal{D}_c^d)] \\ & + (\mathcal{C}_c^d + \mathcal{D}_c^d)(\mathcal{A}^{c(a} \mathcal{C}^{b)}_d + \mathcal{D}_d^{(a} \mathcal{A}^{b)c)} - \mathcal{C}_d^d \mathcal{A}^{c(a} \mathcal{C}^{b)}_c \\ & - \mathcal{D}_c^{(a} \mathcal{A}^{b)c} \mathcal{D}_d^d - \mathcal{A}^{dc} \mathcal{C}_c^{(a} \mathcal{D}_d^{b)} \\ & + (\mathcal{A}^{(ab)} \mathcal{A}^{dc} - \mathcal{A}^{d(a} \mathcal{A}^{b)c}) \mathcal{B}_{dc}, \end{aligned} \quad (\text{D.2.47})$$

$$\begin{aligned} M^{abc} = & \epsilon^{de(c} [\mathcal{B}_{df} (\mathcal{A}^{ab)} \mathcal{D}_e^f - \mathcal{D}_e^a \mathcal{A}^{b)f}) + \mathcal{B}_{fd} (\mathcal{A}^{ab)} \mathcal{C}_e^f \\ & - \mathcal{A}^f{}^{(a} \mathcal{C}^{b)}_e) + \mathcal{C}_f^a \mathcal{D}_e^{(b)} \mathcal{D}_d^f + \mathcal{D}_f^a \mathcal{C}_e^{(b)} \mathcal{C}_d^f], \end{aligned} \quad (\text{D.2.48})$$

$$M^{abcd} = \epsilon^{ef(c} \epsilon^{|gh|d} \mathcal{B}_{hf} \left[\frac{1}{2} \mathcal{A}^{ab)} \mathcal{B}_{ge} - \mathcal{C}_e^a \mathcal{D}_g^{b)} \right]. \quad (\text{D.2.49})$$

Actually, the Fresnel equation (D.2.23) is expressed in terms of $\tilde{\chi}$. Accordingly, the M 's, strictly speaking, should be displayed in terms of $\tilde{\mathcal{A}}, \tilde{\mathcal{B}}, \tilde{\mathcal{C}}, \tilde{\mathcal{D}}$. However, because of (D.2.35), the axion terms drop out, and we can just drop the tildes as well.

Results (D.2.45)–(D.2.49) have been checked thoroughly by means of computer algebra.⁵

⁵See Rubilar [45] who used the computer algebra system Maple with its tensor package GrTensor (cf. <http://grtensor.org>).

D.3

First constraint: Electric/magnetic reciprocity

We impose a first constraint on the constitutive tensor of spacetime. The axion-free part of the linear spacetime relation is required to obey electric/magnetic reciprocity. Mathematically, this implies an almost complex structure on spacetime thereby reducing the number of independent components of the axion-free constitutive tensor from 35 to 18.

D.3.1 Reciprocity implies closure

The linear spacetime relation leaves us still with $35 + 1$ independent components of the constitutive tensor κ , even though we recognized in the meantime that the axion field is irrelevant for the propagation of electromagnetic waves. Clearly we need a new method to constrain κ in some way. An obvious choice is to require electric/magnetic reciprocity for the spacetime relation (D.1.1) with (D.1.3). We have discovered electric/magnetic reciprocity

$$H \rightarrow \zeta F, \quad F \rightarrow -\frac{1}{\zeta} H, \quad (\text{D.3.1})$$

in (B.5.21) as a property of the energy-momentum current ${}^k\Sigma_\alpha$ of the electromagnetic field. Why shouldn't we apply it to (D.1.1), too?

However, we should not forget that, according to (D.1.40), the axion field drops out of ${}^k\Sigma_\alpha$ (as it does from the TR-tensor density). Therefore the reciprocity of the energy-momentum current does not affect the axion field at all. In other words, the concept of reciprocity is alien to the axion field. Accordingly,

we apply reciprocity only to the axion-free part of the spacetime relation, that is, to

$$H = {}^{(1)}H + {}^{(2)}H = \tilde{\kappa}(F). \quad (\text{D.3.2})$$

Let us perform an electric/magnetic reciprocity transformation (D.3.1) in (D.3.2) and recall the linearity of $\tilde{\kappa}$:

$$\zeta F = \tilde{\kappa} \left(-\frac{1}{\zeta} H \right) \quad \text{or} \quad \tilde{\kappa}(H) = -\zeta^2 F. \quad (\text{D.3.3})$$

By definition, the reciprocity transformation commutes with the linear operator $\tilde{\kappa}$. Symbolically in the 6D formalism, we can write (D.3.3)₂ as

$$\tilde{\kappa} \tilde{\kappa} = -\zeta^2 \mathbf{1}_6. \quad (\text{D.3.4})$$

Obviously, the spacetime relation is *not* electric/magnetic reciprocal for an arbitrary transformation function ζ (as the energy-momentum current is). The linear operator $\tilde{\kappa}$ is based on the measurable components $\tilde{\kappa}_I{}^K$ (or $\tilde{\kappa}_{ij}{}^{kl}$). If applied twice as in (D.3.4), there must not emerge an arbitrary function. In other words, we can solve (D.3.4) for ζ^2 by taking its trace,

$$\zeta^2 = -\frac{1}{6} \text{Tr}(\tilde{\kappa} \tilde{\kappa}) = -\frac{1}{24} \tilde{\kappa}_{kl}{}^{ij} \tilde{\kappa}_{ij}{}^{kl} =: \tilde{\lambda}^2, \quad (\text{D.3.5})$$

with the quadratic invariant $\tilde{\lambda}^2$ (see (D.1.14)). Thus, we call

$$\tilde{\kappa} \tilde{\kappa} = -\tilde{\lambda}^2 \mathbf{1}_6 \quad (\text{D.3.6})$$

the *closure relation*¹ since applying the operator $\tilde{\kappa}$ twice yields, up to a negative function, the identity $\mathbf{1}_6$ ($= \delta_I^K$ or $= \delta_{ij}^{kl}$ in components). In this sense, the operation closes.

D.3.2 Almost complex structure and dilaton field

If we define a new dimensionless operator by

$$\tilde{\kappa} =: \tilde{\lambda} \mathbf{J}, \quad \text{then} \quad [\tilde{\lambda}] = 1/\text{resistance} \quad (\text{D.3.7})$$

and, because of (D.3.6),

$$\boxed{\mathbf{J} \mathbf{J} = -\mathbf{1}_6}. \quad (\text{D.3.8})$$

As we will see, the minus sign, which originates from (D.3.1)₂, is very decisive: It will eventually yield the Lorentzian signature of the metric of spacetime.

¹Toupin [54], for $\tilde{\lambda}^2 = \text{const}$, just called the closure relation (for vanishing axion and skewon fields) “electric and magnetic reciprocity.”

With the closure relation (D.3.8), the operator $\mathbf{J}$ defines an *almost complex structure* on the space M^6 of 2-forms, as discussed in Sec. A.1.11.

If we apply $\mathbf{J}$ to the 2-form basis (see (D.1.4)), we find

$$\mathbf{J}(E^K) = \frac{1}{\tilde{\lambda}} \tilde{\kappa}(E^K) = \frac{1}{\tilde{\lambda}} \tilde{\kappa}_I{}^K E^I = J_I{}^K E^I. \quad (\text{D.3.9})$$

With $F = F_I B^I$, the spacetime relation can now be written as

$$\boxed{H = \tilde{\lambda} \mathbf{J}(F)} \quad \text{or} \quad H_I = \tilde{\lambda} J_I{}^K F_K, \quad (\text{D.3.10})$$

a remarkably simple result were it not for the symbolic square root $\mathbf{J} = \sqrt{-\mathbf{I}_6}$. However, discussions of the square roots of matrices can be found in the literature.²

The scalar field $\tilde{\lambda}$, as a look at (D.3.5) confirms, can be considered as the absolute value of the axion-free constitutive tensor $\tilde{\kappa}_{ij}{}^{kl}$. We call $\tilde{\lambda}$ the *dilaton field*.³ Besides the axion α , it is the second scalar field possibly related to electrodynamics.

D.3.3 Draw the square root of the 6D negative unit matrix

A comparison with (D.1.90)₁ allows us to express $\mathbf{J}$ in terms of the constitutive functions according to

$$(J_I{}^K) = \frac{1}{\tilde{\lambda}} \begin{pmatrix} \tilde{\mathcal{C}}^b{}_a & \tilde{\mathcal{B}}_{ba} \\ \tilde{\mathcal{A}}^{ba} & \tilde{\mathcal{D}}_b{}^a \end{pmatrix} =: \begin{pmatrix} C^b{}_a & B_{ba} \\ A^{ba} & D_b{}^a \end{pmatrix}. \quad (\text{D.3.11})$$

Because of (D.3.8), the 3×3 blocks A, B, C, D are constrained by

$$C^c{}_a C^b{}_c + B_{ca} A^{bc} = -\delta_a^b, \quad (\text{D.3.12})$$

$$C^c{}_a B_{bc} + B_{ca} D_b{}^c = 0, \quad (\text{D.3.13})$$

$$A^{ca} C^b{}_c + D_c{}^a A^{bc} = 0, \quad (\text{D.3.14})$$

$$A^{ac} B_{bc} + D_c{}^a D_b{}^c = -\delta_b^a. \quad (\text{D.3.15})$$

²For the square root of such a negative unit matrix (see Gantmacher [11], p.214 et seq.).

³Among the first to consider a scalar field coupled to electromagnetism (and to gravity) have been Jordan [22] and Brans & Dicke [4]. The related notion of a “dilaton field” emerged seemingly during the late 1960s as a field coupled to the dila[t]ation current, the Noether current of scale (or dila[t]ation) transformations. Isham, Salam, and Strathdee [20] used this name. They considered the dilaton as a Goldstone boson related to the breaking of scale invariance. The dilaton also features in the low energy limit of string theory. There it is usually written as $\zeta(x) = e^{-b\phi(x)}$, with a constant b and the dilation field $\phi(x)$. For more recent literature, see, e.g., Bekenstein [2] and Sandvik et al. [47].

We have, for instance, $C^c{}_a C^b{}_c + B^{ca} A_{bc} = C^c{}_a C^b{}_c + A_{bc} B^{ca}$. Then we can put our scheme in matrix form

$$C^2 + AB = -\mathbf{1}_3, \quad (\text{D.3.16})$$

$$BC + DB = 0, \quad (\text{D.3.17})$$

$$CA + AD = 0, \quad (\text{D.3.18})$$

$$BA + D^2 = -\mathbf{1}_3. \quad (\text{D.3.19})$$

We are able to solve this closure relation. Assume that $\det B \neq 0$. Consider (D.3.17). Then we can make an ansatz for the matrix C , namely

$$C = B^{-1}K. \quad (\text{D.3.20})$$

We substitute this into (D.3.17) and find

$$D = -KB^{-1}. \quad (\text{D.3.21})$$

Next, we straightforwardly solve (D.3.16) with respect to A :

$$A = -B^{-1} - (B^{-1}K)^2 B^{-1}. \quad (\text{D.3.22})$$

Now we turn to (D.3.18). Equations (D.3.20) and (D.3.22) yield,

$$CA = -B^{-1}KB^{-1} - (B^{-1}K)^3 B^{-1}, \quad (\text{D.3.23})$$

$$AD = B^{-1}KB^{-1} + (B^{-1}K)^3 B^{-1}. \quad (\text{D.3.24})$$

Thus, we conclude that (D.3.18) is satisfied. Finally, (D.3.19) is left for consideration. From (D.3.22) and (D.3.21), we obtain:

$$BA = -1 - (KB^{-1})^2, \quad (\text{D.3.25})$$

$$D^2 = (KB^{-1})^2. \quad (\text{D.3.26})$$

Thus, in view of (D.3.25) and (D.3.26), equation (D.3.19) is also fulfilled.

Summing up, we have derived the general solution of the closure equations (D.3.16)–(D.3.19). They read,

$$A = -B^{-1} - (B^{-1}K)^2 B^{-1}, \quad (\text{D.3.27})$$

$$C = B^{-1}K, \quad (\text{D.3.28})$$

$$D = -KB^{-1}, \quad (\text{D.3.29})$$

or if we put this in 6×6 matrix form,

$$\mathbf{J} = \begin{pmatrix} C & B \\ A & D \end{pmatrix} = \begin{pmatrix} B^{-1}K & B \\ -[1 + (B^{-1}K)^2] B^{-1} & -KB^{-1} \end{pmatrix}. \quad (\text{D.3.30})$$

By squaring $\mathbf{J}$, one can directly verify the closure relation (D.3.8). The *two arbitrary* 3×3 matrices B and K parametrize the solution, which thus has $2 \times 9 = 18$ independent components.

Clearly, it would be possible to substitute K via $K = BC$, but the matrix K turns out to be particularly useful in Chap. D.4. If we measure the elements of the 3×3 matrices $\tilde{\mathcal{A}}, \tilde{\mathcal{B}}, \tilde{\mathcal{C}}, \tilde{\mathcal{D}}$ of $\tilde{\kappa}$ and thereby, according to (D.3.11), also those of A, B, C, D , then closure is only guaranteed provided the relations (D.3.27)–(D.3.29) are fulfilled. Therefore, closure has a well-defined operational meaning.

If we assume that $\det A \neq 0$, then an analogous derivation leads to

$$\mathbf{J} = \begin{pmatrix} C & B \\ A & D \end{pmatrix} = \begin{pmatrix} \hat{K}A^{-1} & -A^{-1}[1 + (\hat{K}A^{-1})^2] \\ A & -A^{-1}\hat{K} \end{pmatrix}. \quad (\text{D.3.31})$$

Before turning to the different subcases, a word to the physics of all of this appears to be in order. We know from experiments in vacuum that $\mathcal{D} \sim \varepsilon_0 E$ and $\mathcal{H} \sim (1/\mu_0) B$. If we compare this with the spacetime relation (D.1.89), then we recognize that ε_0 is related to the 3×3 matrix $\tilde{\mathcal{A}}$ and μ_0 to the 3×3 matrix $\tilde{\mathcal{B}}$:

$$H = \tilde{\lambda} \mathbf{J}(F) \quad \text{or} \quad \begin{cases} \mathcal{H}_a = -\tilde{\mathcal{C}}^b{}_a E_b + \tilde{\mathcal{B}}_{ba} B^b, \\ \mathcal{D}^a = -\tilde{\mathcal{A}}^{ba} E_b + \tilde{\mathcal{D}}_b{}^a B^b. \end{cases} \quad (\text{D.3.32})$$

Therefore it is safe to assume that $\det A \neq 0$ and $\det B \neq 0$. From this practical point of view the other subcases don't seem to be of much interest.

We could take the matrices A, C, D of (D.3.27)–(D.3.29), multiply them by $\tilde{\lambda}$, e.g., $\mathcal{A} = \tilde{\lambda} A$, substitute them into the M 's of (D.2.45)–(D.2.49) of the Fresnel equation, and then discuss the corresponding consequences. However, we get a much more decisive restructuring of the Fresnel equation if we require, in addition to the reciprocity relation, the vanishing of the skewon field to which we now turn.

D.4

Second constraint: Vanishing skewon field. Emergence of the light cone

We found an almost complex structure on spacetime by linearity and electric/magnetic reciprocity. Now we impose a second constraint on the constitutive tensor of spacetime by removing its skewon part. Thereby we reduce the number of independent components of the constitutive tensor to 9 plus 1 components. One goes for the axion field, whereas the 9 components mentioned reduce the quartic wave surface to a quadratic one that spans the light cone at each point of spacetime.

D.4.1 Lagrangian and symmetry

The removal of the skewon field $\mathcal{S}$ is equivalent to ${}^{(2)}\kappa = 0$. Then κ and $\tilde{\kappa}$ are symmetric. Accordingly, we can alternatively require the operator $\tilde{\kappa}$ to be *symmetric*,

$$\tilde{\kappa}(\phi) \wedge \psi = \phi \wedge \tilde{\kappa}(\psi), \quad (\text{D.4.1})$$

for arbitrary twisted or untwisted 2-forms ϕ and ψ . As a consequence, $\mathbf{J}$ and κ are symmetric, too. The axion part ${}^{(3)}\kappa = \kappa - \tilde{\kappa}$ is not touched by this constraint since it is symmetric by definition.

This can be motivated in that we usually assume that a Lagrange 4-form exists for a fundamental theory, as we discussed in Sec. B.5.4. If we do this in the context of our linear spacetime relation then, because $H = -\partial V/\partial F$, the Lagrangian must be quadratic in F , that is, $V = -F \wedge H/2$. As we saw in (D.1.44), the skewon part drops out in such a quadratic Lagrangian (cf. Table D.1.1). Hence, if we want all information on the field equation to be contained

in the Lagrangian, then we have to abandon a possible skewon contribution. A strict Lagrangian point of view — including a lack of dissipation — strangles the skewon.

In the language of M^6 space, the 6×6 matrix $\tilde{\chi}^{IK}$ is now symmetric and, as a comparison with (D.1.90) shows, it can be represented with the help of 3×3 matrices $\tilde{\mathcal{A}}, \tilde{\mathcal{B}}, \tilde{\mathcal{C}}$ as

$$\boxed{\tilde{\chi}^{IK} = \tilde{\chi}^{KI}} = \begin{pmatrix} \tilde{\mathcal{B}} & \tilde{\mathcal{C}}^T \\ \tilde{\mathcal{C}} & \tilde{\mathcal{A}} \end{pmatrix}, \quad \tilde{\mathcal{A}} = \tilde{\mathcal{A}}^T, \quad \tilde{\mathcal{B}} = \tilde{\mathcal{B}}^T. \quad (\text{D.4.2})$$

Equivalently, we have (see (D.1.90)),

$$\tilde{\kappa}_I{}^K = \begin{pmatrix} \tilde{\mathcal{C}} & \tilde{\mathcal{B}} \\ \tilde{\mathcal{A}} & \tilde{\mathcal{C}}^T \end{pmatrix}. \quad (\text{D.4.3})$$

Note that we have $\mathcal{D} = \mathcal{C}^T$ in this case. Accordingly, the spacetime relation (D.1.47) with the symmetry (D.4.1) now reads

$$H = \tilde{\kappa}(F) + \alpha F, \quad \text{with} \quad \tilde{\kappa}(F) \wedge F = F \wedge \tilde{\kappa}(F), \quad (\text{D.4.4})$$

or

$$H = {}^{(1)}\kappa(F) + \alpha F. \quad (\text{D.4.5})$$

D.4.2 Duality operator

We know from Sec. C.2.4 that, in the framework of the almost complex structure of (D.3.7), the symmetry (D.4.1) introduces a duality operator. Therefore, we define a duality operator according to

$$\boxed{\# := \mathbf{J} = \frac{1}{\lambda} \tilde{\kappa}, \quad \text{with} \quad \mathbf{J}(\phi) \wedge \psi = \phi \wedge \mathbf{J}(\psi).} \quad (\text{D.4.6})$$

Then, we also have self-adjointness with respect to the 6-metric,

$$\varepsilon(\varphi, \# \omega) = \varepsilon(\# \varphi, \omega). \quad (\text{D.4.7})$$

Alternatively, we can also write (D.4.6) as

$$\# = \mathbf{J} = \frac{1}{\lambda} {}^{(1)}\kappa. \quad (\text{D.4.8})$$

Thus, electric/magnetic reciprocity and symmetry of the linear ansatz eventually lead to the spacetime relation

$$H = \tilde{\lambda} \# F + \alpha F. \quad (\text{D.4.9})$$

We can consider the duality operator $\#$ also from another point of view. It is our desire to describe eventually empty spacetime with such a linear ansatz. Therefore we have to reduce the number of independent components of $\tilde{\kappa}_{ij}{}^{kl}$ somehow. The only constants with even parity are the generalized Kronecker deltas. Recognizing that in the framework of electrodynamics in matter a linear ansatz with $\tilde{\kappa}_{ij}{}^{kl}$ can describe anisotropic media, we need a condition in order to guarantee isotropy. A “square” of $\tilde{\kappa}$ does the job,

$$\frac{1}{2} \tilde{\kappa}_{ij}{}^{mn} \tilde{\kappa}_{mn}{}^{kl} = -\tilde{\lambda}^2 \delta_{ij}^{kl}, \quad (\text{D.4.10})$$

with δ_{rs}^{ij} as a generalized Kronecker delta. Alternatively, (D.4.10) can also be written as

$$\frac{1}{2} \#_{ij}{}^{mn} \#_{mn}{}^{kl} = -\delta_{ij}^{kl}, \quad (\text{D.4.11})$$

which is, for symmetric $\tilde{\kappa}$, equation (D.3.8) in another disguise. However, in contrast to requiring reciprocity of the spacetime relation, this consideration does not yield the minus sign in (D.4.10) and (D.4.11) which, in turn, induces the Lorentz signature of the spacetime metric.

D.4.3 Algebraic solution of the closure and symmetry relations

In addition to the almost complex structure $\mathbf{J}$, we arrived at the symmetry of the constitutive matrix (D.4.2). As a consequence, the constraints (D.3.16)–(D.3.19), following from the closure relation, pick up the additional properties

$$A = A^T, \quad B = B^T, \quad D = C^T. \quad (\text{D.4.12})$$

Then they reduce to

$$C^b{}_c C^c{}_a + A^{bc} B_{ca} = -\delta_a^b \quad \text{or} \quad C^2 + AB = -\mathbf{1}_3, \quad (\text{D.4.13})$$

$$C^c{}_{(a} B_{b)c} = 0 \quad \text{or} \quad BC + C^T B = 0, \quad (\text{D.4.14})$$

$$C^{(a}{}_c A^{b)c} = 0 \quad \text{or} \quad CA + AC^T = 0. \quad (\text{D.4.15})$$

Naturally, we would like to resolve these algebraic constraints.

Preliminary analysis

Being a solution of the system (D.4.13)–(D.4.15), the matrix C has very specific properties. First of all, because $\text{Tr } \tilde{\kappa} = 0$, we find from (D.4.3), $\text{Tr } C + \text{Tr } C^T = 0$. Thus, $\text{Tr } C = 0$. More generally, the traces of all odd powers of the matrix C are zero:

$$\text{Tr } C = \text{Tr}(C^3) = \text{Tr}(C^5) = \cdots = 0. \quad (\text{D.4.16})$$

Indeed, multiplying (D.4.13) by C from the right and taking the trace, we find $\text{Tr}(ABC) + \text{Tr}(C^3) = 0$. On the other hand, if we transpose (D.4.13) and multiply the result by C^T , then the trace yields $\text{Tr}(BAC^T) + \text{Tr}(C^3) = 0$. The sum of the two last equations reads $\text{Tr}(A(BC + C^T B)) + 2\text{Tr}(C^3) = 0$. In view of (D.4.14), we then conclude that $\text{Tr}(C^3) = 0$. The same line of argument yields generalizations to the higher odd powers.

It follows from (D.4.16) that the matrix C is always degenerate,

$$\det C = 0. \quad (\text{D.4.17})$$

Indeed, recall that the determinant of an arbitrary 3×3 matrix $N (= N_b{}^a)$ reads

$$\begin{aligned} \det N &= \frac{1}{6} \hat{\epsilon}_{abc} \epsilon^{a'b'c'} N_{a'}{}^a N_{b'}{}^b N_{c'}{}^c \\ &= \frac{1}{6} [(\text{Tr} N)^3 - 3\text{Tr} N \text{Tr}(N^2) + 2\text{Tr}(N^3)]. \end{aligned} \quad (\text{D.4.18})$$

Because of (D.4.16), we can immediately read off (D.4.17).

Let us now analyse the determinants of A and B . When the term C^2 is moved from the left-hand side of (D.4.13) to the right-hand side, a direct computation of the determinant yields:

$$\det A \det B = -\det(1 + C^2) = -\left(1 + \frac{1}{2} \text{Tr}(C^2)\right)^2. \quad (\text{D.4.19})$$

We used the formula (D.4.18) and the properties (D.4.16) and (D.4.17) to evaluate the right-hand side.

Accordingly, the matrices A and B cannot be both positive definite. Moreover, when at least one of them is degenerate, we find that necessarily $\text{Tr}(C^2) = -2$.

General regular solution

Let us consider the case when both or at least one of A and B are regular matrices, i.e., $\det A \neq 0$, $\det B \neq 0$. The general solution has been given in (D.3.30) and (D.3.31). Together with the symmetries (D.4.12), the general solution of (D.4.12)–(D.4.14) can be presented in one of the following two equivalent forms.

B-representation:

$$A = -[1 + (B^{-1}K)^2] B^{-1}, \quad (\text{D.4.20})$$

$$C = B^{-1}K, \quad (\text{D.4.21})$$

where $K = -K^T$. An arbitrary *symmetric* matrix B and an arbitrary *anti-symmetric* matrix K describe the $6 + 3 = 9$ degrees of freedom of the general solution.

A-representation:

$$B = -A^{-1} \left[1 + (\hat{K}A^{-1})^2 \right], \quad (\text{D.4.22})$$

$$C = \hat{K}A^{-1}, \quad (\text{D.4.23})$$

where $\hat{K} = -\hat{K}^T$. In this case, the $6 + 3 = 9$ degrees of freedom of the general solution are encoded in the matrices A (symmetric) and $\hat{K}$ (antisymmetric).

The transition between the two representations is established by means of the relation

$$\hat{K} = B^{-1}KA. \quad (\text{D.4.24})$$

One can readily check that (D.4.20), (D.4.21) and (D.4.22), (D.4.23) are really the solutions of the closure and symmetry relations. Indeed, since the matrix B is nondegenerate, we find that the ansatz $C = B^{-1}K$ solves (D.4.14) provided $K + K^T = 0$. Then, from (D.4.13), we obtain the matrix A in the form (D.4.20). Finally, from (D.4.20), (D.4.21) we get

$$CA = -B^{-1}KB^{-1} - B^{-1}KB^{-1}KB^{-1}KB^{-1}. \quad (\text{D.4.25})$$

The right-hand side is obviously antisymmetric (i.e., the sign is changed under transposition). Hence equation (D.4.15) is satisfied identically.

If, instead, we start from a nondegenerate A , then the analogous ansatz $C = \hat{K}A^{-1}$ solves (D.4.15), whereas B , because of (D.4.13), is found to be in the form of (D.4.22). This time, equation (D.4.14) is fulfilled because of (D.4.22) and (D.4.23).

In the case in which both A and B are nondegenerate, the formulas (D.4.20), (D.4.21) and (D.4.22), (D.4.23) are merely two alternative representations of the same solution. By using (D.4.24), one can recast (D.4.20) and (D.4.21) into (D.4.22) and (D.4.23), and vice versa. However, if $\det A = 0$ and $\det B \neq 0$, then the B-representation (D.4.20), (D.4.21) can be used for the solution of the problem. In the opposite case, i.e., for $\det A \neq 0$ and $\det B = 0$, we turn to the A-representation (D.4.22) and (D.4.23). In these cases the equivalence of both sets, mediated by (D.4.24), is removed. The totally degenerate case is treated in the next subsection.

It is useful to write the regular solution explicitly in components. As usual, we denote the components of the matrices as $B = B_{ab}$, $A = A^{ab}$, $C = C^a{}_b$, and the components of the *inverse* matrices as $(B^{-1}) = B^{ab}$ and $(A^{-1}) = A_{ab}$. We introduce the antisymmetric matrices by $K = K_{ab}$ and $\hat{K} = \hat{K}^{ab}$. Then the component version of the B-representation (D.4.20), (D.4.21) reads:

$$\begin{aligned} A^{ab} &= -B^{ab} - B^{am}K_{mc}B^{cd}K_{dn}B^{nb} \\ &= -B^{ab} + \frac{1}{\det B} (k^2 B^{ab} - k^a k^b), \end{aligned} \quad (\text{D.4.26})$$

$$\begin{aligned} C^a{}_b &= B^{ac}K_{cb} \\ &= B^{ac}\hat{\epsilon}_{cbd}k^d = \frac{1}{\det B}\epsilon^{acd}B_{cb}k_d. \end{aligned} \quad (\text{D.4.27})$$

Here we introduced $k^a := \frac{1}{2}\epsilon^{abc}K_{bc}$ and $k_a := B_{ab}k^b$; moreover, $k^2 := k_a k^a$. Analogously, the A -representation (D.4.22), (D.4.23) reads:

$$\begin{aligned} B_{ab} &= -A_{ab} - A_{am}\hat{K}^{mc}A_{cd}\hat{K}^{dn}A_{nb} \\ &= -A_{ab} + \frac{1}{\det A}(\hat{k}^2 A_{ab} - \hat{k}_a \hat{k}_b), \end{aligned} \quad (\text{D.4.28})$$

$$\begin{aligned} C^a{}_b &= \hat{K}^{ac}A_{cb} \\ &= A_{bc}\epsilon^{acd}\hat{k}_d = \frac{1}{\det A}\hat{\epsilon}_{cbd}A^{ac}\hat{k}^d, \end{aligned} \quad (\text{D.4.29})$$

where $\hat{k}_a := \frac{1}{2}\hat{\epsilon}_{abc}\hat{K}^{bc}$, $\hat{k}^a := A^{ab}\hat{k}_b$, and $\hat{k}^2 := \hat{k}_a \hat{k}^a$.

⊗ Degenerate solution

Besides the regular case outlined above, the closure and symmetry relations also admit a degenerate case when all the matrices are singular, i.e.,

$$\det A = \det B = 0. \quad (\text{D.4.30})$$

Recall that we always have $\det C = 0$ (see (D.4.17)). As mentioned in the context of (D.3.32), from a physical point of view this case is not too interesting. However, for the sake of completeness, we study it in this subsection.

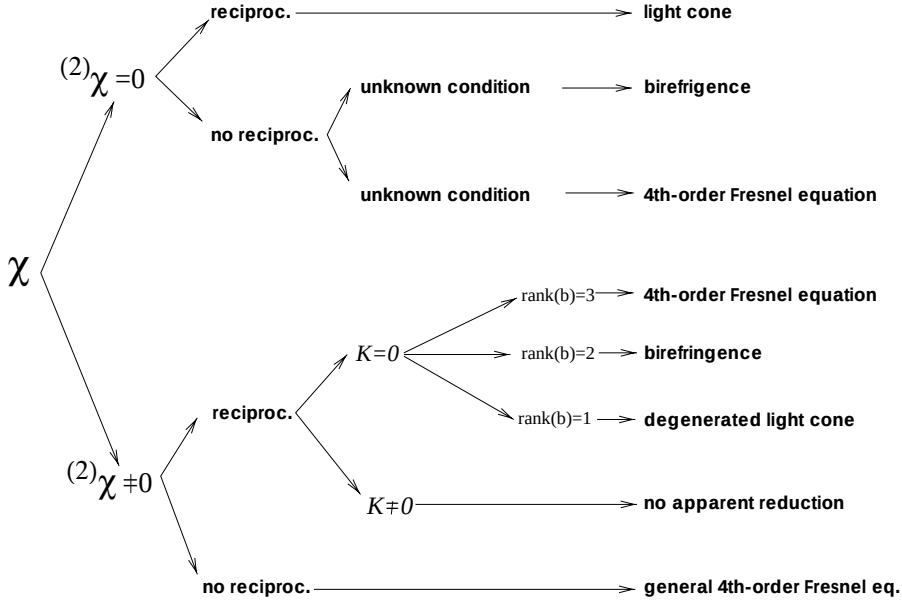
We do not give a detailed analysis of the degenerate case because, in a certain sense to be explained below, it reduces to the regular solution. Nevertheless, let us outline the main steps that yield the explicit construction of the degenerate solution. The basic tool for this is the use of the “gauge” freedom of the system (D.4.13)–(D.4.15), which is obviously invariant under the action of the general linear group $GL(3, \mathbb{R}) \ni L_b{}^a$ defined by

$$\begin{aligned} A^{ab} &\longrightarrow L_c{}^a L_d{}^b A^{cd}, \\ B_{ab} &\longrightarrow (L^{-1})_a{}^c (L^{-1})_b{}^d B_{cd}, \\ C^a{}_b &\longrightarrow L_c{}^a (L^{-1})_b{}^d C^c{}_d. \end{aligned} \quad (\text{D.4.31})$$

This transformation does not change the determinants of the matrices, and hence by means of (D.4.31), the degenerate solutions are mapped again into degenerate ones. We can use the freedom (D.4.31) in order to simplify the construction of the singular solutions.

The vanishing of a determinant $\det B = 0$ means that the algebraic rank of the matrix B is less than three (and the same for A). A rather lengthy analysis then shows that the system (D.4.13)–(D.4.15) does not have real solutions when the matrices A or B have rank 2. As a result, we have to admit that both A

Figure D.4.1: In this tree graph, we represent the different subcases of the linear spacetime relation and the implications for the light cone structure. Note that the axion part $^{(3)}\chi$ drops out and doesn't play a role here whereas the principal part is always nonvanishing, $^{(1)}\chi \neq 0$. The skewon part $^{(2)}\chi$ is of decisive importance. The 3×3 matrix K of (D.3.30) describes electric-magnetic mixing terms, as can be seen from (D.3.32). If $K = 0$, then we have simply $\mathcal{D} \sim E$ and $\mathcal{H} \sim B$. The 3×3 matrix b , in (D.3.30) denoted by B , since no mixing with the magnetic field was possible, represents a part of $^{(1)}\chi \neq 0$. It can be of rank 1, 2, or 3.



and B carry rank 1. Then direct inspection shows that the degenerate matrices A and B can be represented in the general form

$$A^{ab} = -v^a v^b, \quad B_{ab} = u_a u_b. \quad (\text{D.4.32})$$

Here v^a and u_a are arbitrary 3D vectors and covectors. We substitute (D.4.32) into (D.4.13). This yields, for the square of the C matrix, $C^a_c C^c_b = -\delta^a_b + (v^c u_c) v^a u_b$. Taking into account the constraint $\text{Tr}(C^2) = -2$, which arises from (D.4.19), we find the general structure of C^2 as

$$C^a_c C^c_b = -\delta^a_b + v^a u_b, \quad \text{with} \quad v^c u_c = 1. \quad (\text{D.4.33})$$

If we multiply (D.4.33)₁ with u_a and v^b , respectively, we find that u_a and v^b are eigenvectors of C^2 with eigenvalues zero; this is necessary for the validity of the equations (D.4.15) and (D.4.14).

It remains to find the matrix C as the square root of $(D.4.33)_1$. Although this is a rather tedious task, one can solve it with the help of the linear transformations (D.4.31). It is always possible to use (D.4.31) and to bring the column v^a and the row u_a into the specific form

$${}^{\circ}v^a = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad {}^{\circ}u_a = (0, 0, 1). \quad (D.4.34)$$

Then (D.4.33) can be solved explicitly and yields

$${}^{\circ}C^a{}_b = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad (D.4.35)$$

Summarizing, the general degenerate solution is given by the matrices A, B of (D.4.32), with $v^a = L_b{}^a {}^{\circ}v^b$, $u_a = (L^{-1})_a{}^b {}^{\circ}u_b$, and the matrix $C^a{}_b = L_c{}^a (L^{-1})_b{}^d {}^{\circ}C^c{}_d$. An arbitrary matrix $L_b{}^a \in GL(3, \mathbb{R})$ embodies the nine degrees of freedom of this solution.

D.4.4 From a quartic wave surface to the light cone

Having taken care of closure and symmetry in the last section and having determined the explicit form of the matrix $\mathbf{J}$, we can come back to the Fresnel equation (D.2.44) and its M coefficients (D.2.45)–(D.2.49). The latter can now be calculated. The regular and degenerate solutions should be considered separately. Let us begin with the regular case.

Regular case

Starting from (D.4.26)–(D.4.27) or from (D.4.28)–(D.4.29), direct calculation yields for the coefficients of the Fresnel equation (D.2.45)–(D.2.49):

$$M = -\frac{1}{\det B} \left(1 - \frac{k^2}{\det B}\right)^2 \quad (\text{D.4.36})$$

$$= \det A, \quad (\text{D.4.37})$$

$$M^a = \frac{1}{\det B} 4k^a \left(1 - \frac{k^2}{\det B}\right) \quad (\text{D.4.38})$$

$$= 4\widehat{k}^a, \quad (\text{D.4.39})$$

$$M^{ab} = -\frac{1}{\det B} 4k^a k^b + 2B^{ab} \left(1 - \frac{k^2}{\det B}\right) \quad (\text{D.4.40})$$

$$= -2A^{ab} + \frac{6}{\det A} \widehat{k}^a \widehat{k}^b, \quad (\text{D.4.41})$$

$$M^{abc} = -4 B^{(ab} k^{c)} \quad (\text{D.4.42})$$

$$= \frac{4}{\det A} \left(-A^{(ab} \widehat{k}^{c)} + \frac{1}{\det A} \widehat{k}^a \widehat{k}^b \widehat{k}^c \right), \quad (\text{D.4.43})$$

$$M^{abcd} = -(\det B) B^{(ab} B^{cd)} \quad (\text{D.4.44})$$

$$= \frac{1}{\det A} \left(A^{(ab} A^{cd)} - \frac{2A^{(ab} \widehat{k}^c \widehat{k}^d)}{\det A} + \frac{\widehat{k}^a \widehat{k}^b \widehat{k}^c \widehat{k}^d}{\det A} \right). \quad (\text{D.4.45})$$

Here every M is described by two lines, the first displaying the expression in the B -representation and the second in the A -representation. Substituting all this into the general Fresnel equation (D.2.44), we find

$$\begin{aligned} & -\frac{1}{\det B} \left[q_0^2 \left(1 - \frac{k^2}{\det B}\right) - 2q_0 q_a k^a - q_a q_b (\det B) B^{ab} \right]^2 \\ &= \frac{1}{\det A} \left[q_0^2 \det A + 2q_0 q_a \widehat{k}^a - q_a q_b \left(A^{ab} - \frac{\widehat{k}^a \widehat{k}^b}{\det A} \right) \right]^2 \\ &= - (q_i q_j g^{ij})^2 = 0. \end{aligned} \quad (\text{D.4.46})$$

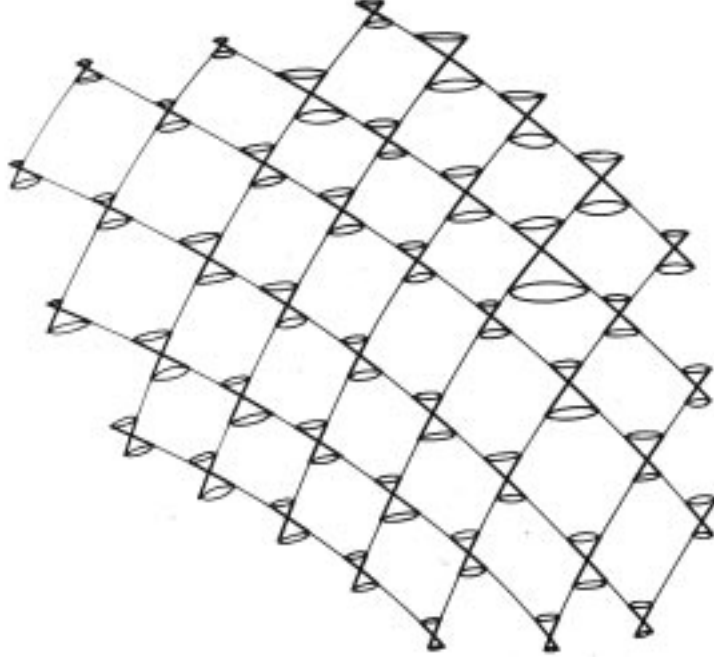


Figure D.4.2: Null cones fitted together to form a conformal manifold (see Pirani and Schild [41]).

Here g^{ij} is a symmetric tensor field. Let, for definiteness, $\det B > 0$. Hence, $\det A < 0$. Then from (D.4.46) we read off its components as

$$g^{ij} = \frac{1}{\sqrt{\det B}} \left(\frac{1 - (\det B)^{-1} B_{cd} k^c k^d}{-k^a} \mid \frac{-k^b}{-(\det B) B^{ab}} \right) \quad (\text{D.4.47})$$

$$= \frac{1}{\sqrt{-\det A}} \left(\frac{\det A}{\widehat{k}^a} \mid \frac{\widehat{k}^b}{-A^{ab} + (\det A)^{-1} \widehat{k}^a \widehat{k}^b} \right). \quad (\text{D.4.48})$$

One can prove that this tensor has Lorentz signature. Hence it can be understood as the *metric of spacetime*. For a flow diagram indicating the different constraints that led to the light cone, see Fig. D.4.1.

Thus, from our general analysis, we indeed recover the null or light cone structure $q_i q^i = q_i q_j g^{ij} = 0$ for the propagation of electromagnetic waves: Provided the linear spacetime relation satisfies reciprocity and symmetry, the quartic wave surface in (D.2.44) reduces to two coinciding *quadratic* wave surfaces, that is, to the light cone for the metric g^{ij} . A visualization of the corresponding conformal manifold is given in Fig. D.4.2.

⊗ Degenerate case

The same conclusion is also true for the degenerate case. Substituting (D.4.32) into (D.2.46)–(D.2.49) and using (D.4.33)–(D.4.35), we find

$$M = 0, \quad M^a = 0, \quad (\text{D.4.49})$$

$$M^{ab} = -4v^a v^b, \quad (\text{D.4.50})$$

$$M^{abc} = 4u_e C^{(a}_f v^b \epsilon^{c)ef}, \quad (\text{D.4.51})$$

$$M^{abcd} = -u_e C^{(a}_f \epsilon^{b|ef} u_g C^{c|}_h \epsilon^{d)gh}. \quad (\text{D.4.52})$$

Inserting this into (D.2.44), we obtain

$$-[-2q_0 q_a v^a + q_a q_b u_e C^a_f \epsilon^{bef}]^2 = -(q_i q_j g^{ij})^2 = 0. \quad (\text{D.4.53})$$

This time the tensor field g^{ij} is described by

$$g^{ij} = \left(\begin{array}{c|c} 0 & -v^b \\ \hline -v^a & u_c C^{(a}_d \epsilon^{b)cd} \end{array} \right). \quad (\text{D.4.54})$$

This tensor is nondegenerate and it has Lorentzian signature.

⊗ Is reciprocity also necessary for the light cone?

Reciprocity is sufficient for the light cone to exist. Is it also a necessary condition for the reduction of the quartic geometry (D.2.44) to the light cone? Such a conjecture is substantiated by some particular examples, but its general proof is still missing. For the special case of vanishing C , that is, for $C = 0$, we are also able to prove necessity.

Here we do *not* assume closure of $\tilde{\kappa}$, but we keep its symmetry. Putting $C^a_b = 0$, we find from (D.2.46)–(D.2.49) that $M^a = 0$ and $M^{abc} = 0$, whereas

$$M^{ab} = B_{cd}(A^{ab} A^{cd} - A^{ac} A^{bd}), \quad (\text{D.4.55})$$

$$M^{abcd} = (\det B) A^{(ab} B^{cd)}. \quad (\text{D.4.56})$$

Consequently, (D.2.44) reduces to

$$\det A q_0^4 + q_0^2 \gamma + \det B \alpha \beta = 0, \quad (\text{D.4.57})$$

where $\alpha := A^{ab} q_a q_b$, $\beta := B^{ab} q_a q_b$, and $\gamma := M^{ab} q_a q_b$. Assuming that (D.4.57) describes a light cone, one concludes that the roots for q_0^2 should coincide. Thus necessarily

$$\gamma^2 = 4 \det A \det B \alpha \beta. \quad (\text{D.4.58})$$

Let us write $(\det A \det B) = s |\det A \det B|$, with $s = \text{sign}(\det A \det B)$. Then (D.4.58) yields

$$2\sqrt{|\det A \det B|} \frac{\alpha}{\gamma} = s \lambda, \quad 2\sqrt{|\det A \det B|} \frac{\beta}{\gamma} = \frac{1}{\lambda}, \quad (\text{D.4.59})$$

where λ is an arbitrary scalar factor. We recall the definitions of α, β, γ and find

$$A^{ab} = s \lambda^2 B^{ab}. \quad (\text{D.4.60})$$

Consequently, $M = \det A = s \lambda^6 / \det B$ and $M^{ab} = 2 \lambda^4 B^{ab}$. Thus, (D.4.57) simplifies to

$$\frac{s \lambda^2}{\det B} (\lambda^2 q_0^2 + s q_a q_b B^{ab} \det B)^2 = 0. \quad (\text{D.4.61})$$

For $s = -1$, we immediately recognize that the quadratic form in (D.4.61) can have a signature of either $(+ - - -)$ or $(+ + + -)$. Similarly, for $s = 1$, the signature is either $(+ + + +)$ or $(+ + - -)$. Therefore, the Fresnel equation describes the correct (hyperbolic) light *cone* structure only in the case $s = -1$.

Finally, one can verify that the above solutions satisfy

$$\tilde{\kappa}_{ij}{}^{mn} \tilde{\kappa}_{mn}{}^{kl} = s \lambda^2 \delta_{ij}^{kl}, \quad (\text{D.4.62})$$

which reproduces the closure relation (D.3.4) for $s = -1$.

D.5

Extracting the metric by an alternative method

We discussed wave propagation in linear pre-metric electrodynamics and found that reciprocity and symmetry enabled us to obtain the metric of spacetime from the spacetime relation.

Here we present an alternative construction of the Lorentzian metric from the constitutive tensor of spacetime. As above, the crucial point is a well-known mathematical fact. Here it is the one-to-one correspondence between the duality operators $\#$ and the conformal classes of the metrics of spacetime. Hence, with the linearity condition (D.1.6), together with reciprocity or closure (D.3.4) and symmetry (D.4.2), we can construct, up to a conformal factor, the metric of spacetime from the components of the duality operator $\#$. In this sense, the metric is a concept derived from the electrodynamic spacetime relation.

At the center of this derivation is the triplet of (anti)self-dual 2-forms, which provides the basis of the (anti)self-dual subspace of the complexified space of all 2-forms $M^6(\mathbb{C})$.

D.5.1 $\otimes$ Triplet of self-dual 2-forms and metric

In Sec. C.2.5 we saw that the basis of the self-dual 2-forms can be described either by (C.2.45) or by (C.2.46). These triplets are linearly dependent, and one can use any of them for the actual computation. For example, one can take as the fundamental triplet $S^{(a)} = \overset{(s)}{\beta}^a$, with $G^{ab} = A^{ab}/4$. Alternatively, we can work with $S^{(a)} = B^{ab} \overset{(s)}{\epsilon}_b$, with $G^{ab} = B^{ab}/4$. Both choices yield equivalent

results if A and B are non degenerate. For definiteness, let us choose the second option. Then from (C.2.46), in the B -representation, we have explicitly

$$\begin{aligned} S^{(a)} &= \frac{1}{2} S_{ij}^{(a)} dx^i \wedge dx^j = B^{ab} \epsilon_b^{(s)} \\ &= \frac{1}{2} \left(-i dx^0 \wedge dx^a + i(\det B)^{-1} k_b dx^b \wedge dx^a \right. \\ &\quad \left. + \frac{1}{2} B^{ab} \hat{\epsilon}_{bcd} dx^c \wedge dx^d \right). \end{aligned} \quad (\text{D.5.1})$$

Here $x^0 = \sigma$ and x^a are the spatial coordinates, with $a, b, c, \dots = 1, 2, 3$.

A useful calculational tool is provided by a set of *1-forms* $S_i^{(a)} := \partial_i \lrcorner S^{(a)} = S_{ij}^{(a)} dx^j$. With their help, we can read off the contractions needed in the Schönberg–Urbantke formulas (C.2.73), (C.2.74) from the exterior products

$$\begin{aligned} S_i^{(a)} \wedge S^{(b)} \wedge S_j^{(c)} &= \frac{1}{2} S_{ik}^{(a)} S_{lm}^{(b)} S_{nj}^{(c)} dx^k \wedge dx^l \wedge dx^m \wedge dx^n \\ &= \frac{1}{2} \epsilon^{klmn} S_{ik}^{(a)} S_{lm}^{(b)} S_{nj}^{(c)} \text{Vol}. \end{aligned} \quad (\text{D.5.2})$$

Here, as usual, $\text{Vol} = dx^0 \wedge dx^1 \wedge dx^2 \wedge dx^3$.

From (D.5.1) we have

$$S_0^{(a)} = -\frac{i}{2} dx^a, \quad (\text{D.5.3})$$

$$\begin{aligned} S_b^{(a)} &= -\frac{i}{2} \left(-dx^0 \delta_b^a + i B^{ac} \hat{\epsilon}_{cbd} dx^d \right. \\ &\quad \left. + (\det B)^{-1} k_c dx^c \delta_b^a - (\det B)^{-1} k_b dx^a \right). \end{aligned} \quad (\text{D.5.4})$$

By a rather lengthy calculation we find

$$G_{ab} S^{(a)} \wedge S^{(b)} = -6i \text{Vol}, \quad (\text{D.5.5})$$

$$\hat{\epsilon}_{abc} S_0^{(a)} \wedge S^{(b)} \wedge S_0^{(c)} = -\frac{3i}{4} \text{Vol}, \quad (\text{D.5.6})$$

$$\hat{\epsilon}_{abc} S_n^{(a)} \wedge S^{(b)} \wedge S_0^{(c)} = \frac{3i}{4} (\det B)^{-1} k_n \text{Vol}, \quad (\text{D.5.7})$$

$$\begin{aligned} \hat{\epsilon}_{abc} S_m^{(a)} \wedge S^{(b)} \wedge S_n^{(c)} &= -\frac{3i}{4} \left(-(\det B)^{-1} B_{mn} \right. \\ &\quad \left. + (\det B)^{-2} k_m k_n \right) \text{Vol}. \end{aligned} \quad (\text{D.5.8})$$

We use (D.5.5)–(D.5.8) together with (D.5.2). Then, the Schönberg–Urbantke formulas (C.2.73), (C.2.74), for the components of the spacetime metric yield

$$g_{ij} = \frac{1}{\sqrt{\det B}} \left(\begin{array}{c|c} \det B & -k_b \\ \hline -k_a & -B_{ab} + (\det B)^{-1} k_a k_b \end{array} \right). \quad (\text{D.5.9})$$

One can find the determinant of this expression and verify that $(i\sqrt{\det g}) = \sqrt{-\det g} = 1$ in accordance with (C.2.74).

If instead of the B -representation (D.4.26), (D.4.27) of the solution of the closure and symmetry relations, we start from the A -representation (D.4.28), (D.4.29), then the Schönberg–Urbantke formulas yield an alternative form of the spacetime metric,

$$g_{ij} = \frac{1}{\sqrt{-\det A}} \left(\frac{1 - (\det A)^{-1} A^{cd} \widehat{k}_c \widehat{k}_d}{-\widehat{k}_a} \middle| \frac{-\widehat{k}_b}{-(\det A) A_{ab}} \right). \quad (\text{D.5.10})$$

The triplet of the self-dual 2-forms $S^{(a)}$ is defined up to an arbitrary scalar factor: By multiplying them with an arbitrary (in general complex) function $h(x)$, one preserves the completeness condition (C.2.52), (C.2.53). Correspondingly, the determinant of the metric is rescaled by a factor h^4 , whereas the metric itself is rescaled by a factor h . In other words, the whole procedure defines a *conformal class* of metrics rather than a metric itself. Clearly, one can always choose the conformal factor h so as to eliminate the first factor in (D.5.9).

For the metric (D.5.9), the line element of spacetime reads explicitly:

$$ds^2 = \frac{1}{\sqrt{\det B}} \left[\det B \left(d\sigma - \frac{k_a dx^a}{\det B} \right)^2 - B_{ab} dx^a dx^b \right]. \quad (\text{D.5.11})$$

However, as just discussed, a conformal factor is irrelevant. Therefore we may limit ourselves to the line element

$$ds'^2 = \det B \left(d\sigma - \frac{k_a dx^a}{\det B} \right)^2 - B_{ab} dx^a dx^b. \quad (\text{D.5.12})$$

If we consider the 3×3 matrix B , we can distinguish four different cases with the signatures $(+++)$, $(-++)$, $(--++)$, and $(---)$, respectively. Let us demonstrate that all these cases lead to a Lorentzian signature.

For $(+++)$, the determinant is positive and the expression $B_{ab} dx^a dx^b$ is positive definite. Thus, we can read off the Lorentzian signature immediately: σ is the time coordinate, the x^a 's the three spatial coordinates.

For $(-++)$, the determinant is negative and the expression $B_{ab} dx^a dx^b$ is indefinite. Therefore the square of one dx coordinate differential, say $(dx^1)^2$, carries a positive sign and can be identified as the time coordinate, whereas σ can be identified as the space coordinate.

For $(--++)$, the determinant is positive again and the expression $-B_{ab} dx^a dx^b$ is indefinite with signature $(++-)$. Therefore x^3 is the time coordinate in this case.

For $(---)$, the determinant is negative and the expression $-B_{ab} dx^a dx^b$ is positive definite. In other words, this corresponds to the case $(+++)$ with an overall sign change.

Accordingly, the structure in (D.5.12) is rather robust and the quantity k_a doesn't influence the Lorentzian signature of (D.5.12).

It is quite satisfactory to note that the Schönberg–Urbantke formalism produces the same Lorentzian metric that we recovered earlier in Sec. D.4.4 when discussing the reduction of the fourth-order wave surface to the (second-order) light cone. In that approach, since the wave vector is a 1-form, we obtained the contravariant components (D.4.47) and (D.4.48) of the metric, which are just the inverses of (D.5.9) and (D.5.10), respectively. However, our new “reduction” method of Sec. D.4.4 also produces a spacetime metric for the degenerate solution of the closure relation (see (D.4.54)). The Schönberg–Urbantke formulas are inapplicable to the degenerate solutions. In this sense, the reduction method is farther reaching.

D.5.2 Maxwell–Lorentz spacetime relation and Minkowski spacetime

Let us take a particular example for the spacetime relation. We assume that we are in a suitable specific frame such that we measure

$$\begin{aligned} H_{01} &= \frac{1}{\mu_0} F_{23}, & H_{02} &= \frac{1}{\mu_0} F_{31}, & H_{03} &= \frac{1}{\mu_0} F_{12}, \\ H_{23} &= -\varepsilon_0 F_{01}, & H_{31} &= -\varepsilon_0 F_{02}, & H_{12} &= -\varepsilon_0 F_{03}. \end{aligned} \quad (\text{D.5.13})$$

This set is known as the *Maxwell–Lorentz* spacetime relation in the frame chosen. The components of the constitutive tensor of spacetime in (D.1.6), for the example discussed, are independent of the spacetime coordinates and are given by (D.5.13) in terms of a pair of basic constants ε_0, μ_0 , with $(a, b, c = 1, 2, 3)$

$$[\varepsilon_0] = \left[\frac{H_{ab}}{F_{0c}} \right] = \left[\frac{H/l^2}{F/(tl)} \right] = \frac{q^2 l}{h} \frac{\text{SI}}{t} \frac{\text{A m}}{\text{V s}} = \frac{1}{\Omega} \frac{\text{m}}{\text{s}}, \quad (\text{D.5.14})$$

$$[\mu_0] = \left[\frac{F_{ab}}{H_{0c}} \right] = \left[\frac{F/l^2}{H/(tl)} \right] = \frac{h}{q^2} \frac{l}{t} \frac{\text{SI}}{t} \frac{\text{V m}}{\text{A s}} = \Omega \frac{\text{m}}{\text{s}}. \quad (\text{D.5.15})$$

We can read off $\kappa_{ij}{}^{kl}$ from (D.5.13). Then a direct computation shows that the quadratic invariant (D.3.5) is equal to $\tilde{\lambda}^2 = \varepsilon_0/\mu_0$, with $[\tilde{\lambda}] = q^2/h \stackrel{\text{SI}}{=} 1/\Omega$. Consequently, the duality operator $\#$ of (D.4.6), as defined by the spacetime relation (D.5.13), is given by the 6×6 matrix

$$\#_I^K = \begin{pmatrix} 0 & -\frac{1}{c} \mathbf{1}_3 \\ c \mathbf{1}_3 & 0 \end{pmatrix} \quad (\text{D.5.16})$$

(cf. (D.1.6)). Here we denote $c := 1/\sqrt{\varepsilon_0\mu_0}$, with $[c] = \text{velocity}$. Thus we have the matrices $A = -B^{-1}$, $B = c \mathbf{1}_3$, whereas $C = 0$. One then immediately finds

the metric (D.5.9), (D.5.10) in the form:

$$g_{ij} = \frac{1}{\sqrt{c}} \begin{pmatrix} c^2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (\text{D.5.17})$$

This is, up to the factor $1/\sqrt{c}$, the standard Minkowski metric in orthonormal coordinates. If we include $1/\sqrt{c}$, then we find the Minkowski metric density of weight $-1/2$.

D.5.3 Hodge star operator and isotropy

The inverse of (D.5.9) is given by

$$g^{ij} = \frac{1}{\sqrt{\det B}} \left(\frac{1 - (\det B)^{-1} k^2}{-k^a} \middle| \frac{-k^b}{-(\det B) B^{ab}} \right). \quad (\text{D.5.18})$$

With the help of (D.5.9) and (D.5.18), we can define the corresponding Hodge star operator * attached to the metric extracted by means of the Schönberg–Urbantke formalism. Its action on a 2-form, say F , is described by (C.2.92) or, explicitly, by

$$^*F_{ij} := \frac{\sqrt{-g}}{2} \hat{\epsilon}_{ijkl} g^{km} g^{ln} F_{mn}. \quad (\text{D.5.19})$$

Such a Hodge duality operator (see (C.2.33)) has the constitutive tensor of spacetime $\overset{g}{\kappa}_{ij}{}^{kl} = \hat{\epsilon}_{ijmn} \overset{g}{\chi}{}^{mnkl}/2$ with

$$\overset{g}{\chi}{}^{ijkl} = \sqrt{-g} (g^{ik} g^{jl} - g^{jk} g^{il}). \quad (\text{D.5.20})$$

Note that $\overset{g}{\chi}{}^{ijkl}$ is invariant under conformal transformations $g_{ij} \rightarrow e^{\lambda(x)} g_{ij}$; therefore, only nine of the possible ten components of the metric can ever enter $\overset{g}{\chi}{}^{ijkl}$.

We can compare $\overset{g}{\chi}{}^{ijkl}$ with the original $\tilde{\chi}{}^{ijkl}$ of the linear spacetime relation. For this purpose, we have to substitute the metric components (D.5.9) into the A, B, C blocks (C.2.95)–(C.2.97) of the Hodge duality matrix (C.2.94). Then inspection of (D.4.26)–(D.4.29) demonstrates the exact coincidence

$$\overset{g}{A}{}^{ab} = A^{ab}, \quad \overset{g}{B}_{ab} = B_{ab}, \quad \overset{g}{C}{}^a{}_b = C^a{}_b. \quad (\text{D.5.21})$$

Summing up, it turns out that $\overset{g}{\chi}{}^{IJ} = \tilde{\chi}{}^{IJ}$; i.e., *the metric extracted allows us to write the original duality operator $^\#$ as the Hodge star operator associated with that metric.* Therefore, the original linear constitutive tensor (D.1.70) for a vanishing skewon field can then finally be written as

$$\chi^{ijkl} = \tilde{\lambda}(x) \sqrt{-g} (g^{ik} g^{jl} - g^{jk} g^{il}) + \alpha(x) \epsilon^{ijkl} \quad (\text{D.5.22})$$

(see also (D.4.9)), with the dilaton field $\tilde{\lambda}$ and the axion field α .

Given a metric, we can define the notion of *local isotropy*. Let $T^{i_1 \dots i_p}$ be the contravariant coordinate components of a tensor field and $T^{\alpha_1 \dots \alpha_p} := e_{i_1}^{\alpha_1} \dots e_{i_p}^{\alpha_p} T^{i_1 \dots i_p}$ its frame components with respect to an orthonormal frame $e_\alpha = e^i_\alpha \partial_i$. A tensor is said to be locally isotropic at a given point if its frame components are invariant under a Lorentz rotation of the orthonormal frame. Similar considerations extend to tensor densities.

There are only two geometrical objects that are numerically invariant under (local) Lorentz transformations: the Minkowski metric $\eta^{\alpha\beta} = \text{diag}(+1, -1, -1, -1)$ and the Levi-Civita tensor density $\epsilon^{\alpha\beta\gamma\delta}$. Thus

$$\mathcal{T}^{\alpha\beta\gamma\delta} = \phi(x) \sqrt{-g} (g^{\alpha\gamma} g^{\beta\delta} - g^{\beta\gamma} g^{\alpha\delta}) + \varphi(x) \epsilon^{\alpha\beta\gamma\delta} \quad (\text{D.5.23})$$

is the most general locally isotropic contravariant fourth-rank tensor density of weight +1 with the symmetries $\mathcal{T}^{\alpha\beta\gamma\delta} = -\mathcal{T}^{\beta\alpha\gamma\delta} = -\mathcal{T}^{\alpha\beta\delta\gamma} = \mathcal{T}^{\gamma\delta\alpha\beta}$. Here ϕ and φ are scalar and pseudoscalar fields, respectively.

Accordingly, in view of (D.5.22) we have proved that the constitutive tensor density χ^{ijkl} with the constraints of reciprocity and symmetry is *locally isotropic with respect to the metric* (D.5.9).

D.5.4 $\otimes$ Covariance properties

The discussion above was confined to a fixed coordinate system. However, one may ask: What happens when the coordinates are changed or, more generally, when a local (co)frame is transformed? Up to now, we considered only a holonomic coframe $\vartheta^\alpha = \delta_i^\alpha dx^i$, but in physically important cases one often needs to go to nonholonomic coframes. In this section we study the covariance properties of the Schönberg–Urbantke construction.

The behavior of the metric (C.2.73), (C.2.74) under coordinate and frame transformations is by no means obvious. Although the fundamental completeness relation (C.2.52), (C.2.53) looks covariant, one should recall that the index ^(a) of the self-dual S -forms comes, in a noncovariant manner, from the 3+3 splitting of the basis B^I . A transformation of the spacetime coframe (A.1.92) acts on both types of indices in the coefficients $S_{\alpha\beta}^{(a)}$ that enter the Schönberg–Urbantke formulas (C.2.73), (C.2.74). Thus, the determination of the new components of the metric with respect to transformed frame becomes a nontrivial problem.

For the sake of generality, we consider an arbitrary linear transformation (A.1.92) of the coframe that includes the holonomic coordinate transformation as a particular case. With respect to the original coframe $\vartheta^\alpha = \delta_i^\alpha dx^i$, the duality operator is described by the components of the spacetime matrix $\#_{\alpha\beta}{}^{\mu\nu} = \delta_\alpha^i \delta_\beta^j \delta_k^\mu \delta_l^\nu \#_{ij}{}^{kl}$. Accordingly, the components of the extracted spacetime metric read $g_{\alpha\beta} = \delta_\alpha^i \delta_\beta^j g_{ij}$.

Because of the tensorial nature of the definition of the duality operator (D.1.46), a linear transformation (A.1.92) of the basis, $\vartheta^\alpha = L_{\alpha'}{}^\alpha \vartheta^{\alpha'}$, yields

the corresponding transformation of the duality components,

$$\#_{\alpha'\beta'}{}^{\mu'\nu'} = L_{\alpha'}{}^\alpha L_{\beta'}{}^\beta L_{\mu'}{}^\mu L_{\nu'}{}^\nu \#_{\alpha\beta}{}^{\mu\nu}. \quad (\text{D.5.24})$$

Recall that $L_{\alpha'}{}^{\alpha'}$ is the inverse of $L_{\alpha'}{}^\alpha$.

We now demonstrate that the Schönberg–Urbantke construction is completely covariant and that the extracted metric (C.2.73) transforms as

$$g'_{\alpha'\beta'} = (\det L)^{-\frac{1}{2}} L_{\alpha'}{}^\alpha L_{\beta'}{}^\beta g_{\alpha\beta} \quad (\text{D.5.25})$$

under the linear transformation (A.1.92), (D.5.24).

The proof is technically simple and straightforward, but it is somewhat lengthy. We have prepared the necessary tools in Sec. A.1.10. Combining now the matrix equations (A.1.95), (A.1.98), and (C.2.37), we find that, with respect to the new coframe, the duality operator

$$\# \begin{pmatrix} \beta' \\ \hat{\epsilon}' \end{pmatrix} = \begin{pmatrix} C' & A' \\ B' & C'^T \end{pmatrix} \begin{pmatrix} \beta' \\ \hat{\epsilon}' \end{pmatrix} \quad (\text{D.5.26})$$

is described by the new matrix components

$$\begin{pmatrix} C' & A' \\ B' & C'^T \end{pmatrix} = \frac{1}{\det L} \begin{pmatrix} Q^T & W^T \\ Z^T & P^T \end{pmatrix} \begin{pmatrix} C & A \\ B & C^T \end{pmatrix} \begin{pmatrix} P & W \\ Z & Q \end{pmatrix}. \quad (\text{D.5.27})$$

This is the direct matrix remake of the tensorial version (D.5.24). The matrices P, Q, W, Z are described in (A.1.96) and (A.1.97). Explicitly, (D.5.27) yields

$$A' = (\det L)^{-1} (Q^T A Q + W^T B W + Q^T C W + W^T C^T Q), \quad (\text{D.5.28})$$

$$B' = (\det L)^{-1} (P^T B P + Z^T A Z + Z^T C P + P^T C^T Z), \quad (\text{D.5.29})$$

$$C' = (\det L)^{-1} (Q^T C P + W^T C^T Z + Q^T A Z + W^T B P). \quad (\text{D.5.30})$$

It is convenient to consider the three subcases of the linear transformation (A.1.101)–(A.1.103) because their product (A.1.100) describes a general linear transformation.

For $L = L_1$, we have (A.1.104) with only the matrix $W^{ab} = \epsilon^{abc} U_c$ being nontrivial. Then (D.5.28)–(D.5.30) yield

$$\begin{aligned} A' &= A + W^T B W + C W + W^T C^T, \\ B' &= B, \\ C' &= C + W^T B. \end{aligned} \quad (\text{D.5.31})$$

Comparing this with the B -representation (D.4.20) and (D.4.21), we see that the transformation (D.5.31) means a mere shift of the antisymmetric matrix:

$$K' = K + B W^T B, \quad \text{or equivalently, } k'_a = k_a - (\det B) U_a. \quad (\text{D.5.32})$$

Substituting this into (D.5.9), we find the transformed metric:

$$g'_{\alpha\beta} = g_{\alpha\beta} + \frac{1}{\sqrt{\det B}} \left(\begin{array}{c|c} 0 & (\det B) U_b \\ \hline (\det B) U_a & -k_a U_b - k_b U_a + (\det B) U_a U_b \end{array} \right). \quad (\text{D.5.33})$$

Comparing with (A.1.101), we get a particular case of (D.5.25):

$$g'_{\alpha'\beta'} = L_{\alpha'}^{\alpha} L_{\beta'}^{\beta} g_{\alpha\beta}, \quad \text{with} \quad L = L_1. \quad (\text{D.5.34})$$

When $L = L_2$, the matrix $Z_{ab} = \hat{\epsilon}_{abc} V^c$ describes the case (A.1.105). Then, from (D.5.28)–(D.5.30), we find:

$$\begin{aligned} A' &= A, \\ B' &= B + Z^T A Z + Z^T C + C^T Z, \\ C' &= C + AZ. \end{aligned} \quad (\text{D.5.35})$$

Contrary to the first case, it is now more convenient to proceed in the A -representation. From (D.4.22) and (D.4.23), we then see that the transformation (D.5.35) means a mere shift of the antisymmetric matrix

$$\hat{K}' = \hat{K} + AZA, \quad \text{or equivalently,} \quad \hat{k}'_a = \hat{k}_a - (\det A) A_{ab} V^b. \quad (\text{D.5.36})$$

Substituting this into (D.5.10), we obtain the transformed metric:

$$g'_{\alpha\beta} = g_{\alpha\beta} + \frac{1}{\sqrt{\det A}} \left(\begin{array}{c|c} 2\hat{k}_c V^c - (\det A) A_{cd} V^c V^d & (\det B) A_{bd} V^d \\ \hline (\det A) A_{ac} V^c & 0 \end{array} \right). \quad (\text{D.5.37})$$

We compare with (A.1.102) and recover a subcase of (D.5.25),

$$g'_{\alpha'\beta'} = L_{\alpha'}^{\alpha} L_{\beta'}^{\beta} g_{\alpha\beta}, \quad \text{with} \quad L = L_2. \quad (\text{D.5.38})$$

Finally, for $L = L_3$ we have (A.1.106). Then (D.5.28)–(D.5.30) reduce to

$$\begin{aligned} A' &= \frac{\det \Lambda}{\Lambda_0^0} \Lambda^{-1} A (\Lambda^{-1})^T, \\ B' &= \frac{\Lambda_0^0}{\det \Lambda} \Lambda^T B \Lambda, \\ C' &= \Lambda^{-1} C \Lambda. \end{aligned} \quad (\text{D.5.39})$$

For the antisymmetric matrix K , this yields:

$$K' = \frac{\Lambda_0^0}{\det \Lambda} \Lambda^T K \Lambda, \quad \text{or equivalently,} \quad k'^a = \Lambda_0^0 (\Lambda^{-1})^a_b k^b. \quad (\text{D.5.40})$$

As we saw above, the analysis of the case (A.1.104) was easier in the B -representation, whereas the A -representation was more suitable for the treatment of the case (A.1.105). However, the last case (A.1.106), (D.5.39) looks the

same in both pictures. For definiteness, let us choose the B -representation. A new and nontrivial feature of the present case is that the transformation L_3 is not unimodular, $\det L_3 = \Lambda_0^0 \det \Lambda \neq 1$. Recall that $\det L_1 = \det L_2 = 1$. As a consequence, one should carefully study the behavior of the determinant of the metric defined by the second Schönberg–Urbantke formula (C.2.74).

From (A.1.98) we have the transformation of the self-dual basis (C.2.46):

$$\epsilon_a^{(s)'} = \frac{1}{\det \Lambda} \Lambda_a^b \epsilon_b^{(s)}. \quad (\text{D.5.41})$$

Hence, for the S -forms (D.5.1), we find

$$S'^{(a)} = \frac{1}{\Lambda_0^0} (\Lambda^{-1})_b^a S^{(b)}, \quad (\text{D.5.42})$$

and consequently,

$$B'_{ab} S'^{(a)} \wedge S'^{(b)} = \frac{1}{\det L_3} B_{ab} S^{(a)} \wedge S^{(b)}. \quad (\text{D.5.43})$$

Using this in (C.2.74), one finally proves the invariance of the determinant:

$$\sqrt{\det g'} = \sqrt{\det g}. \quad (\text{D.5.44})$$

Taking this result into account when substituting (D.5.39) and (D.5.40) into (D.5.9), one obtains the transformed metric

$$g'_{\alpha\beta} = \frac{1}{\sqrt{\det B \det L_3}} \left(\frac{(\Lambda_0^0)^2 \det B}{-\Lambda_0^0 \Lambda_a^c k_c} \left| \frac{-\Lambda_0^0 \Lambda_b^d k_d}{\Lambda_a^c \Lambda_b^d [-B_{cd} + (\det B)^{-1} k_c k_d]} \right. \right). \quad (\text{D.5.45})$$

Thus, the metric transforms as a tensor *density*,

$$g'_{\alpha'\beta'} = \frac{1}{\sqrt{\det L}} L_{\alpha'}^\alpha L_{\beta'}^\beta g_{\alpha\beta} \quad \text{with} \quad L = L_3. \quad (\text{D.5.46})$$

This transformation law is completely consistent with the invariance of the determinant (D.5.44).

Turning now to the case of a general linear transformation, one can use the factorization (A.1.100) and perform the three transformations (A.1.101)–(A.1.103) one after another. This yields subsequently (D.5.34), (D.5.38), and (D.5.46). The final result is then represented in (D.5.25) with an arbitrary transformation matrix L .

Reducing the degenerate case to the regular one

Using the formalism above, we can show that the degenerate solutions of the closure relation, A, B, C , can always be transformed into the regular configurations. Indeed, let us take the degenerate solution (D.4.32), (D.4.34), (D.4.35),

which is explicitly given by the three matrices

$$\overset{\circ}{A} = - \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \overset{\circ}{B} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \overset{\circ}{C} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{D.5.47})$$

Consider a simple transformation of the coframe $\vartheta^\alpha = L_{\alpha'}^\alpha \vartheta^{\alpha'}$ by means of the $L = L_1$ matrix (A.1.101) with $U_a = (0, 0, 1)$. This induces the transformation of the 2-form basis (A.1.95) where the matrix W of (A.1.104) reads explicitly

$$W^{ab} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}. \quad (\text{D.5.48})$$

The other matrices are $P = Q = \mathbf{I}_3$ and $Z = 0$. Then, using (D.5.31), we find the transformed spacetime relation matrices as

$$\overset{\circ}{A}' = - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \overset{\circ}{B}' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \overset{\circ}{C}' = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{D.5.49})$$

With respect to the transformed basis, as we immediately see, the constitutive matrices become *nondegenerate*, $\det \overset{\circ}{A}' \neq 0$, and we can use the A -representation to describe this configuration and to construct the corresponding metric of spacetime.

Evidently, the general degenerate solution can also be reduced to the regular case. Then the linear transformation above, with $L = L_1$, should be supplemented by the appropriate $GL(3, \mathbb{R})$ transformation (D.4.31). We thus conclude that the separate treatment of the degenerate case is in fact not necessary: The degeneracy (D.4.30) merely reflects an unfortunate choice of the frame that can be easily removed by a linear transformation.

D.6

Fifth axiom: Maxwell–Lorentz spacetime relation

D.6.1 The axiom

Summarizing the content of Part D, we can now give a clear formulation of the fifth axiom. In Maxwell–Lorentz electrodynamics, the 2-forms of the electromagnetic excitation H and the field strength F are related by the universal, *local, linear* spacetime relation

$$H = \kappa(F) . \quad (\text{D.6.1})$$

The linear operator

$$\kappa(a\phi + b\psi) = a\kappa(\phi) + b\kappa(\psi) \quad (\text{D.6.2})$$

(here a, b are functions and ϕ, ψ 2-forms) can be represented in its spacetime components such that (D.6.1) reads

$$H_{ij} = \frac{1}{2} \kappa_{ij}{}^{kl} F_{kl} = \frac{1}{4} \hat{\epsilon}_{ijmn} \chi^{mnkl} F_{kl} . \quad (\text{D.6.3})$$

Under the linear group, the operator κ can be decomposed irreducibly into three parts,

$$\kappa = {}^{(1)}\kappa + {}^{(2)}\kappa + {}^{(3)}\kappa = \tilde{\kappa} + {}^{(3)}\kappa , \quad \text{with} \quad {}^{(3)}\kappa = \alpha \mathbf{1} . \quad (\text{D.6.4})$$

Here α is the axion field. In terms of components, the corresponding spacetime relation reads

$$H_{ij} = \frac{1}{2} {}^{(1)}\kappa_{ij}{}^{kl} F_{kl} + 2 \mathcal{J}_{[i}{}^k F_{j]k} + \alpha F_{ij} , \quad (\text{D.6.5})$$

with the principal part ⁽¹⁾ κ with twenty and the (traceless) skewon field $\mathcal{S}$ with fifteen independent components.

We demand *electric/magnetic reciprocity* of the spacetime relation $H = \tilde{\kappa}(F)$ and find the closure property

$$\tilde{\kappa} \tilde{\kappa} = -\tilde{\lambda}^2 \mathbf{1}_6 \quad \text{or} \quad \tilde{\kappa}_{ij}{}^{mn} \tilde{\kappa}_{mn}{}^{kl} = -\tilde{\lambda}^2 \delta_{ij}^{kl}, \quad (\text{D.6.6})$$

with $\tilde{\lambda}^2 = -\text{Tr}(\tilde{\kappa} \tilde{\kappa})/6$. We define $\mathbf{J} = \tilde{\kappa}/\tilde{\lambda}$. Then, at this stage, the spacetime relation has the form¹

$$H = \tilde{\lambda} \mathbf{J}(F) + \alpha F, \quad \text{with} \quad \mathbf{J} \mathbf{J} = -\mathbf{1}. \quad (\text{D.6.7})$$

Furthermore, we demand the vanishing of the skewon field $\mathcal{S}$, i.e., the *symmetry* of the operator κ or, equivalently, of $\tilde{\kappa}$,

$$\tilde{\kappa}(\phi) \wedge \psi = \phi \wedge \tilde{\kappa}(\psi) \quad \text{or} \quad \epsilon^{ijmn} \tilde{\kappa}_{mn}{}^{kl} = \epsilon^{klmn} \tilde{\kappa}_{mn}{}^{ij}. \quad (\text{D.6.8})$$

Then $\mathbf{J}$ becomes a duality operator, the Hodge star operator $\star$ that defines a metric tensor field up to an undetermined (conformal) factor:²

$$H = \tilde{\lambda} \star F + \alpha F. \quad (\text{D.6.9})$$

Accordingly, linearity, reciprocity, and symmetry provide a unique *light cone* structure for the propagation of electromagnetic waves (see Fig. D.4.2). As a result, the spacetime metric with the correct *Lorentzian signature* is, up to a conformal factor, reconstructed from the constitutive tensor $\kappa_{ij}{}^{kl}$ of spacetime.

Maxwell–Lorentzian electrodynamics is specified, among the models defined in (D.6.9), by a constant universal dilation factor

$$\tilde{\lambda} = \left[-\frac{1}{6} \text{Tr}(\tilde{\kappa} \tilde{\kappa}) \right]^{\frac{1}{2}} = \lambda_0 = \text{const} \quad (\text{no dilaton}). \quad (\text{D.6.10})$$

The actual value of this universal constant³ characterizing the vacuum has to be taken from experiment, basically from a Weber–Kohlrausch type of experiment relating electric to magnetic effects. In the SI system, its value is

$$\lambda_0 \stackrel{\text{SI}}{=} \sqrt{\frac{\varepsilon_0}{\mu_0}} \approx \frac{1}{377 \, \Omega}, \quad (\text{D.6.11})$$

¹With the understanding that H and F also live in M_6 , we could write (D.6.7)₁ symbolically as $H = \tilde{\lambda} \sqrt{-\mathbf{I}_6} F + \alpha F$. Isn't that a remarkably simple expression after having only applied linearity and reciprocity?

²Sowa [50] has proposed a “nonlinear Maxwell theory” with $H = \tilde{\lambda} \star F$, i.e., with $\alpha = 0$. His dilaton — he calls it the filling factor, following the terminology of the QHE (see our equation (B.4.54)) — obeys a nonlinear wave equation $\square f + a \star (F \wedge \star F) f = \nu f$ with the constants a and ν . For vacuum, $J = 0$, Sowa found interesting vortex-like solutions of his theory.

³Up-to-date information about the fundamental constants in nature (including the electric ones), their measurement, and their SI values has been provided by Flowers & Petley [10] and by Mohr & Taylor [29].

with the *electric constant* $\varepsilon_0 \approx 8.854 \times 10^{-12} \text{ s}/(\Omega \text{ m})$ (permittivity of vacuum) and the *magnetic constant* $\mu_0 = 4\pi \times 10^{-7} \Omega \text{ s}/\text{m}$ (permeability of vacuum).

So much about the vacuum impedance of the spacetime relation (D.6.9). What are we going to do about the axion α in the same equation? Well, we remove it by putting

$$\alpha = \frac{1}{6} \text{Tr } \kappa = 0 \quad (\text{no axion}). \quad (\text{D.6.12})$$

Then, making use of the extracted metric and of the corresponding Hodge star operator $*$, the *Maxwell–Lorentz spacetime relation* can be written as

$$\boxed{H = \lambda_0 *F} \quad (\text{fifth axiom}), \quad (\text{D.6.13})$$

or in spacetime components, as

$$H_{ij} = \frac{\lambda_0}{4} \hat{\epsilon}_{ijmn} \sqrt{-g} (g^{mk} g^{nl} - g^{nk} g^{ml}) F_{mn} = \frac{\lambda_0}{2} \hat{\epsilon}_{ij}{}^{kl} \sqrt{-g} F_{kl}. \quad (\text{D.6.14})$$

An alternative formulation is (raising the indices of F_{ij})

$$\check{\mathcal{H}}^{ij} = \lambda_0 \sqrt{-g} F^{ij}, \quad \text{with} \quad \check{\mathcal{H}}^{ij} = \frac{1}{2} \epsilon^{ijkl} H_{kl}. \quad (\text{D.6.15})$$

Accordingly, the experimentally well-established Maxwell–Lorentz electrodynamics is distinguished from other models by linearity, reciprocity, symmetry (no skewon), no dilaton, and no axion.

D.6.2 On the Poincaré and the Lorentz groups

The light cone is invariant under the 15-parameter group of local conformal transformations.⁴ The 10-parameter Poincaré group (inhomogeneous Lorentz group) is a subgroup of the conformal group. In our considerations, we never recover the rigid (global) Poincaré group unless the metric is Minkowskian.

In our whole axiomatics in Part B and Part D we did not use the Poincaré group nor its 6-parameter Lorentz subgroup at all. In the conventional approach to electrodynamics, the Poincaré group is an essential ingredient for cooking up the formalism of electrodynamics. In the general covariant approach, with electric charge and magnetic flux conservation as its basis, which we followed, the metric is distilled from a linear spacetime relation with reciprocity and symmetry as additives. The metric, and thus the gravitational potential, is a derived concept. The metric gets its meaning from electrodynamics; it is not a fundamental field.

⁴See Barut and Rączka [1].

D.6.3 $\otimes$ Extensions. Dissolving Lorentz invariance?

We formulated our deductions of the Maxwell–Lorentz spacetime relation already in such a way that the possible generalizations of the Maxwell–Lorentz framework are apparent: Have a dilaton, have an axion, have a skewon, either one of them, some of them, or all of them. But all of these possible generalizations to dilaton electrodynamics, axion(-dilaton) electrodynamics, and so on take place in the framework of the *linear* spacetime relation (D.6.1) cum (D.6.2).

This linearity gave birth to the axion, the skewon, and an unstructured principal part ${}^{(1)}\chi^{ijkl} = {}^{(1)}\chi^{klij}$. The subsequent assumption of reciprocity and symmetry created the metric g of spacetime. Had we given up linearity, that is, had we allowed nonlinearity at a fundamental level in the spacetime relation (D.6.1), then the metric and their siblings axion and skewon would be irretrievably lost. In fact, we are not aware of any attempt to construct a nonlinear electrodynamic model at this fundamental level, and we see no way of doing so.

However, if we take the metric for granted, i.e., fundamental linearity, reciprocity, and symmetry, then, with the help of the metric, one can construct nonlocal and nonlinear models that, nevertheless, are based on fundamental linearity à la (D.6.2). Presupposing a metric of spacetime implies a linear spacetime relation willy-nilly. To fix this linear relation later by some decorative nonlinear and nonlocal terms doesn't change the underlying rationale. For this reason, we discuss these nonlocal and nonlinear models, which respect the linear spacetime relation in the sense described, in Chap. E.2.

Let us come back to the possible extensions of the Maxwell–Lorentz electrodynamics by means of introducing dilaton, axion, and/or skewon fields. There are two distinct classes: Those that respect the light cone (one could call them the harmless extensions) and those that don't. To the former belong axion, dilaton, and axion-dilaton electrodynamics, to the latter skewon electrodynamics (that is, with $\mathcal{S} \neq 0$) with possible admixtures of axion and/or dilaton fields.

As soon as one mixes a skewon field into the spacetime relation (see (D.6.5)), one cannot recover the light cone any longer and the Lorentz group is dissolved. Several such attempts are discussed in the literature.⁵ If one wants to look for possible new physics violating Lorentz invariance, then the assumption of a skewon appears to be the most natural possibility. As a first attempt one could try to simplify (D.6.5) by brute force to

$$H_{ij} = \lambda_0 \hat{\epsilon}_{ij}{}^{kl} \sqrt{-g} F_{kl} + 2 \mathcal{S}_{[i}{}^k F_{j]k} + \alpha F_{ij} \quad (\text{D.6.16})$$

and then investigate the influence of $\mathcal{S}$ on light propagation.

⁵See, for instance, Kostelecký [25].

Moreover, recently, there was some observational evidence from astronomy that Sommerfeld's fine structure constant may depend on (cosmic) time:

$$\alpha_f = \frac{e^2}{4\pi\epsilon_0 c \hbar} \approx \frac{1}{137}, \quad \alpha_f = \alpha_f(t) = ? \quad (\text{D.6.17})$$

(e = elementary charge, $\hbar$ = reduced Planck constant). Even though these observations did not find independent support, the posed question may be interesting for future. Provided that we want to uphold the pre-metric Maxwellian structure (B.4.1), then, in the light of (B.4.2), only $\epsilon_0 c$ could be time dependent: The speed of light would not be necessarily constant.⁶ We saw that $\epsilon_0 c$ codifies a response of spacetime to the propagation of electromagnetic disturbances. Dimensionwise, we have $[\epsilon_0 c] = 1/\text{resistance}$, that is, it represents an impedance of spacetime ("vacuum"). It is conceivable that such a "vacuum" impedance picks up some time dependence or, more generally, becomes a field dependig on time and space. Such an approach would dismantle local Lorentz invariance and thereby general relativity too.

⁶This is the main result of a recent article of Peres [40]. Here we partly follow his arguments.

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Part E

Electrodynamics in vacuum and in matter

After the spacetime relation is added as fifth axiom, the formulation of electrodynamics, as a predictive theory, is complete.

In Part E, we present the basic aspects of standard Maxwell–Lorentz electrodynamics in vacuum and outline the possible generalizations of the theory when a metric is prescribed and the spacetime relation becomes nonlocal and/or nonlinear.

The electromagnetic spacetime relation is universal. It is determined by the electric and magnetic properties of spacetime itself and is intimately related to the metric structure of spacetime. Additionally, we have to take into account the electric and magnetic properties of the material continuum as well as its state of motion. This leads to the necessity of introducing the last, namely the sixth, axiom, which expresses the splitting of the current into two parts, one referring to the bound charge and the other to the free charge. Thereby a constitutive relation emerges. Throughout this part, we consistently distinguish between the spacetime relation and the constitutive relation with particular attention paid to the fundamental theoretical and experimental aspects of the electrodynamics of moving bodies.

E.1

Standard Maxwell–Lorentz theory in vacuum

We describe standard Maxwell–Lorentz electrodynamics based on the fifth axiom that we put forward in Chap.D.6. The field equations as well as the action are formulated in a generally covariant manner on an arbitrary curved spacetime. For a given spacetime metric, we introduce a special foliation adapted to the metric structure. This gives rise to the effective permittivity and permeability functions of the vacuum which are expressed in terms of the components of the metric. Furthermore, we use the metric in order to prove the symmetry of the electromagnetic energy-momentum.

E.1.1 Maxwell–Lorentz equations, impedance of the vacuum

When the *Maxwell–Lorentz spacetime relation* (D.6.13) is substituted into the Maxwell equations (B.4.9), (B.4.10), we find the Maxwell–Lorentz equations

$$d^*F = J/\lambda_0, \quad dF = 0 \quad (\text{E.1.1})$$

of standard electrodynamics. According to (D.6.11), the characteristic *impedance* (or wave resistance) of vacuum $1/\lambda_0$ is fixed by experiment.¹

¹In quantum field theory, the electromagnetic coupling constant is given by the dimensionless fine structure constant $\alpha_f := e^2/(2\epsilon_0 h c) = 1/137.036$, with e = elementary charge and h = Planck's constant (see Pohl [28, p. 299]). We can express λ_0 in terms of α_f and find $\lambda_0 = e^2/(2\alpha_f h)$.

The impedance $1/\lambda_0$ represents a fundamental constant that describes the basic electromagnetic property of spacetime; that is, one can interpret spacetime as a special type of medium (sometimes called vacuum, or aether, in the old terminology). In this sense, one can understand (D.6.13) as the constitutive relation for spacetime itself. The Maxwell–Lorentz spacetime relation (D.6.13) is universal. It is equally valid in Minkowski, Riemannian, and post-Riemannian spacetimes.

The *electric constant* ε_0 and the *magnetic constant* μ_0 (also called permittivity of vacuum and permeability of vacuum, respectively) determine the universal constant of nature

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}, \quad (\text{E.1.2})$$

which yields the vacuum velocity of light.

The homogeneous Maxwell equation can be solved by $F = dA$. Using this and (E.1.2), we can recast the inhomogeneous Maxwell equation (E.1.1)₁ in the form

$$\varepsilon_0 d^* F = \varepsilon_0 d^* dA = \frac{1}{c} J. \quad (\text{E.1.3})$$

Recalling the definitions of the codifferential (C.2.107), $d^\dagger := *d^*$, and of the wave operator (d'Alembertian) (C.2.110), we can rewrite (E.1.3) as

$$(\square - d d^\dagger) A = \frac{1}{\varepsilon_0 c} *J. \quad (\text{E.1.4})$$

Because of gauge invariance, one can impose the Lorenz condition² $d^\dagger A = 0$. Then we find a wave equation for the electromagnetic potential 1-form:

$$\square A = \frac{1}{\varepsilon_0 c} *J, \quad \text{with} \quad d^\dagger A = 0. \quad (\text{E.1.5})$$

In components, the left-hand side reads:

$$\square A = \left(-\tilde{\nabla}^k \tilde{\nabla}_k A_i + \widetilde{\text{Ric}}_i{}^k A_k \right) dx^i. \quad (\text{E.1.6})$$

Here $\text{Ric}_{ij} := R_{kij}{}^k$ is the Ricci tensor, see (C.1.51). The tilde denotes the covariant differentiation and the geometric objects defined by the Levi-Civita connection (C.2.103).

Also F obeys a wave equation. We take the Hodge star of (E.1.3) and differentiate it. We substitute $dF = 0$ and find

$$\square F = \frac{1}{\varepsilon_0 c} d^* J. \quad (\text{E.1.7})$$

²Ludvig Valentin *Lorenz* (Danish physicist, 1829–1891): Lorenz condition; Hendrik Antoon *Lorentz* (Dutch physicist, 1853–1928): Lorentz transformation, Maxwell–Lorentz electrodynamics. The Lorentz–Lorenz formula in molecular polarizability.

The left-hand side of this equation, in terms of components, can be determined by substituting (C.2.110) and (C.2.107):

$$\square F = \frac{1}{2} \left(-\tilde{\nabla}^k \tilde{\nabla}_k F_{ij} - 2 \widetilde{\text{Ric}}_{[i}{}^k F_{j]k} + \tilde{R}^{kl}{}_{ij} F_{kl} \right) dx^i \wedge dx^j. \quad (\text{E.1.8})$$

Accordingly, curvature dependent terms surface in a natural way, both in (E.1.6) and in (E.1.8).

E.1.2 Action

According to (B.5.84), the excitation can be expressed in terms of the electromagnetic Lagrangian V by

$$H = -\frac{\partial V}{\partial F}. \quad (\text{E.1.9})$$

Because of the Maxwell–Lorentz spacetime relation (D.6.13), the excitation H is linear and homogeneous in F . Therefore the action V is homogeneous in F of degree 2. Then by Euler’s theorem for homogeneous functions, we have

$$F \wedge \frac{\partial V}{\partial F} = 2V, \quad (\text{E.1.10})$$

or with (E.1.9) and (D.6.13),

$$V = -\frac{1}{2} F \wedge H = -\frac{\lambda_0}{2} F \wedge \star F = -\frac{1}{2} \sqrt{\frac{\varepsilon_0}{\mu_0}} F \wedge \star F. \quad (\text{E.1.11})$$

This is the twisted Lagrangian 4-form of the electromagnetic field à la Maxwell–Lorentz.

E.1.3 Foliation of a spacetime carrying a metric. Effective permeabilities

In the previous sections, the standard Maxwell–Lorentz theory was presented in 4-dimensional form. In order to visualize the separate electric and magnetic pieces, we have to use the $(1+3)$ -decomposition technique.

In the presence of a metric, it becomes necessary to further specialize the vector field n , which is our basic tool in a $(1+3)$ -decomposition. Before we introduced the metric, all possible vectors n described the same spacetime foliation without really distinguishing between “time” and “space” (see Fig. B.1.3), since the very notions of *timelike* and *spacelike* vectors and subspaces were absent. Now we choose the three functions n^a in such a way that

$$\mathbf{g}(n, \partial_a) = g_{(\sigma)a} + g_{ab} n^b = 0, \quad (\text{E.1.12})$$

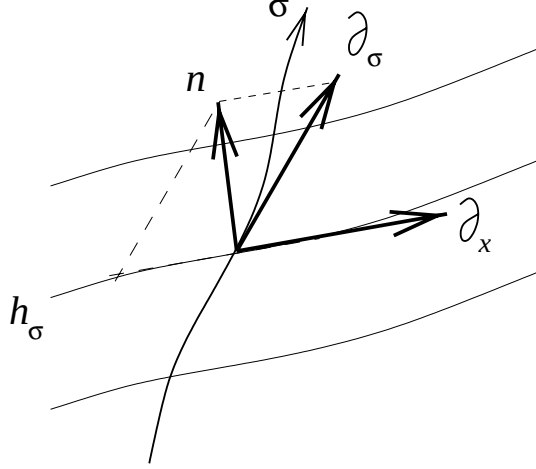


Figure E.1.1: The vector field n adapted to the $(1+3)$ -foliation on a manifold with metric: The vector n is orthogonal to the hypersurface h_σ (compare with Fig. B.1.3).

where $g_{ab} := \mathbf{g}(\partial_a, \partial_b)$, $g_{(\sigma)a} := \mathbf{g}(\partial_\sigma, \partial_a)$, and $g_{(\sigma)(\sigma)} := \mathbf{g}(\partial_\sigma, \partial_\sigma)$. Provided

$$\mathbf{g}(n, n) = g_{(\sigma)(\sigma)} - g_{ab} n^a n^b = N^2 > 0, \quad (\text{E.1.13})$$

the vector field n is timelike, and thus we can indeed consider σ as a local time coordinate. The condition (E.1.12) guarantees that the folia h_σ of constant σ are orthogonal to n (see Fig. E.1.1). Thus they are really 3-dimensional spacelike hypersurfaces. The metric

$$^{(3)}\mathbf{g}(\partial_a, \partial_b) := -g_{ab} \quad (\text{E.1.14})$$

is evidently a *positive* definite Riemannian metric on h_σ . We denote by $\underline{\cdot}$ the Hodge star operator defined in terms of the 3-dimensional metric (E.1.14).

Applying the general definitions (C.2.85) to our foliation compatible coframe (B.1.33), we find relations between the 4-dimensional and the 3-dimensional star operators:

$$*(d\sigma \wedge \underline{\Psi}^{(p)}) = (-1)^p \frac{1}{N} \underline{\Psi}^{(p)}, \quad (\text{E.1.15})$$

$$\underline{\Psi}^{(p)} = N d\sigma \wedge \underline{\Psi}^{(p)}. \quad (\text{E.1.16})$$

Here $\underline{\Psi}^{(p)}$ is an arbitrary *transversal* (i.e., purely *spatial*) p -form. Note that $\underline{\cdot}^* = 1$ for all forms.

We substitute the $(1 + 3)$ decompositions (B.1.39) and (B.2.9) into (D.6.13),

$$\underline{H} = \lambda_0 \underline{(*F)}, \quad {}^\perp H = \lambda_0 {}^\perp (*F). \quad (\text{E.1.17})$$

Then the 3-dimensional version of the Maxwell–Lorentz spacetime relation reads

$$\mathcal{D} = \varepsilon_g \varepsilon_0 {}^*E \quad \text{and} \quad B = \mu_g \mu_0 {}^*\mathcal{H}, \quad (\text{E.1.18})$$

where we introduced the *effective* electric and magnetic permeabilities

$$\varepsilon_g = \mu_g = \frac{c}{N}, \quad (\text{E.1.19})$$

see (D.6.13) and (E.1.2). In general, these quantities are functions of the coordinates since N , according to (E.1.13), is determined by the geometry of spacetime. Thus the *gravitational field acts*, via its potential — the metric — *on spacetime and makes it look like a medium with nontrivial polarization properties*. In particular, the propagation of light, described by the Maxwell equations, is affected by these refractive properties of curved spacetime. In flat Minkowski space in the frame chosen, $N = c$, and hence, $\varepsilon_g = \mu_g = 1$.

E.1.4 Electromagnetic energy and momentum

If the spacetime metric g is given, then there exists a unique torsion-free and metric-compatible Levi-Civita connection $\tilde{\Gamma}_\alpha{}^\beta$, see (C.2.103), (C.2.136). Consider the conservation law (B.5.52) of the energy-momentum. In a Riemannian space, the covariant Lie derivative $\tilde{\mathcal{L}}_\xi = \tilde{D} \xi \lrcorner + \xi \lrcorner \tilde{D}$ commutes with the Hodge operator, $\tilde{\mathcal{L}}_\xi {}^* = {}^* \tilde{\mathcal{L}}_\xi$. Thus (B.5.53) straightforwardly yields

$$\hat{X}_\alpha = \frac{\lambda_0}{2} \left({}^*F \wedge \tilde{\mathcal{L}}_{e_\alpha} F - F \wedge \tilde{\mathcal{L}}_{e_\alpha} {}^*F \right) = 0. \quad (\text{E.1.20})$$

Therefore in *general relativity* (GR), with the Maxwell–Lorentz spacetime relation, (B.5.52) simply reduces to

$$\tilde{D} {}^k \Sigma_\alpha = (e_\alpha \lrcorner F) \wedge J. \quad (\text{E.1.21})$$

The energy-momentum current (B.5.7) now reads

$${}^k \Sigma_\alpha = \frac{\lambda_0}{2} [F \wedge (e_\alpha \lrcorner {}^*F) - (e_\alpha \lrcorner F) \wedge {}^*F]. \quad (\text{E.1.22})$$

In the absence of sources ($J = 0$), we find the energy-momentum law

$$\tilde{D} {}^k \Sigma_\alpha = 0. \quad (\text{E.1.23})$$

In the *flat* Minkowski spacetime of SR, we can *globally* choose coordinates in such a way that $\tilde{\Gamma}_\alpha{}^\beta = 0$. Thus $\tilde{D} = d$ and $d {}^k \Sigma_\alpha = 0$.

As we already know from (B.5.20), the current (E.1.22) is *traceless* ($\vartheta^\alpha \wedge {}^k\Sigma_\alpha = 0$). Moreover, we can now use the metric and also prove its symmetry. We multiply (E.1.22) by $\vartheta_\beta := g_{\beta\gamma} \vartheta^\gamma$ and antisymmetrize:

$$\begin{aligned} \frac{4}{\lambda_0} \vartheta_{[\beta} \wedge \Sigma_{\alpha]} &= \vartheta_\beta \wedge F \wedge (e_\alpha \lrcorner {}^*F) - \vartheta_\beta \wedge (e_\alpha \lrcorner F) \wedge {}^*F \\ &\quad - \vartheta_\alpha \wedge F \wedge (e_\beta \lrcorner {}^*F) + \vartheta_\alpha \wedge (e_\beta \lrcorner F) \wedge {}^*F. \end{aligned} \quad (\text{E.1.24})$$

Because of (C.2.133) and (C.2.131), the first term on the right-hand side can be rewritten,

$$\begin{aligned} \vartheta_\beta \wedge F \wedge (e_\alpha \lrcorner {}^*F) &= \vartheta_\alpha \wedge F \wedge {}^*(\vartheta_\beta \wedge F) \\ &= \vartheta_\alpha \wedge F \wedge (e_\beta \lrcorner {}^*F); \end{aligned} \quad (\text{E.1.25})$$

i.e., it is compensated by the third term. We apply the analogous technique to the second term. Because ${}^{**}F = -F$, we have

$$\begin{aligned} \vartheta_\beta \wedge {}^*({}^*F \wedge \vartheta_\alpha) \wedge {}^*F &= -{}^*({}^*F \wedge \vartheta_\alpha) \wedge {}^*F \wedge \vartheta_\beta \\ &= -{}^*({}^*F \wedge \vartheta_\beta) \wedge {}^*F \wedge \vartheta_\alpha \\ &= -\vartheta_\alpha \wedge (e_\beta \lrcorner F) \wedge {}^*F. \end{aligned} \quad (\text{E.1.26})$$

In other words, the second term is compensated by the fourth one and we have

$$\vartheta_{[\beta} \wedge {}^k\Sigma_{\alpha]} = 0. \quad (\text{E.1.27})$$

Alternatively, we can work with the energy-momentum tensor. We decompose the 3-form ${}^k\Sigma_\alpha$ with respect to the η -basis. This is now possible since a metric is available. Because $\vartheta^\alpha \wedge \eta_\gamma = \delta^\alpha_\gamma \eta$, we find

$${}^k\Sigma_\alpha =: {}^kT_{\alpha}{}^\beta \eta_\beta \quad \text{or} \quad {}^kT_{\alpha\beta} = -{}^*(\vartheta_\beta \wedge {}^k\Sigma_\alpha), \quad (\text{E.1.28})$$

see (B.5.37),(B.5.38). Thus,

$${}^kT_{\alpha\beta} = {}^k\mathcal{T}_{\alpha\beta} / \sqrt{-g}. \quad (\text{E.1.29})$$

Its tracelessness $T_\gamma{}^\gamma = 0$ has already been established in equation (B.5.20). Its symmetry

$${}^kT_{[\alpha\beta]} = 0 \quad (\text{E.1.30})$$

can be either read off from (E.1.28) and (E.1.27) or directly from (B.5.49) with $\tilde{\mathcal{H}}^{ij} \sim F^{ij}$. A manifestly symmetric version of the energy-momentum tensor can be derived from (E.1.28)₂ and (E.1.22):

$${}^kT_{\alpha\beta} = -\lambda_0 {}^*\left[(e_\alpha \lrcorner F) \wedge (e_\beta \lrcorner F) + \frac{1}{2} g_{\alpha\beta} ({}^*F \wedge F) \right]. \quad (\text{E.1.31})$$

Thus ${}^kT_{\alpha\beta}$ is a traceless symmetric tensor(-valued 0-form) with nine independent components. Its symmetry is sometimes called a *bastard symmetry* since it

interrelates two indices of totally different origin, as can be seen from (E.1.27). Without using a metric, it cannot even be formulated, see (B.5.38).

It is a reflection of the symmetry of ${}^kT_{\alpha\beta}$ that the energy flux density 2-form s of (B.5.62) and the momentum density 3-form p_a of (B.5.63) are closely inter-related. Indeed, using the spacetime relation (E.1.18) in the definition (B.5.63), we obtain:

$$p_a = -\varepsilon_g \varepsilon_0 B \wedge (e_a \lrcorner {}^*E) = -\varepsilon_g \mu_g \varepsilon_0 \mu_0 {}^*\mathcal{H} \wedge {}^*(E \wedge dx_a) = \frac{\varepsilon_g \mu_g}{c^2} dx_a \wedge E \wedge \mathcal{H}. \quad (\text{E.1.32})$$

Here we used the identities (C.2.133), (C.2.131) and (E.1.2). Taking into account (E.1.19) and comparing with (B.5.62), we finally find

$$p_a = \frac{1}{N^2} dx_a \wedge s. \quad (\text{E.1.33})$$

In a Minkowski space, we have $N = c$. This is the electromagnetic version of the relativistic formula $m = \frac{1}{c^2} E$ in a field-theoretical disguise.

The energy density (B.5.61) becomes explicitly positive when we substitute the spacetime relation (E.1.18) in it:

$$u = \frac{\varepsilon_g}{2} \left(\varepsilon_0 E \wedge {}^*E + \frac{1}{\mu_0} B \wedge {}^*B \right). \quad (\text{E.1.34})$$

E.2

Electromagnetic spacetime relations beyond locality and linearity

Here we outline the structure of electrodynamics that arises when we keep the first four axioms but generalize the fifth one. Particular examples of spacetime relations are described that are either nonlocal or nonlinear. In these models the existence of a metric of spacetime is presupposed.

E.2.1 Keeping the first four axioms fixed

The *Lamb shift* in the spectrum of a hydrogen atom and the *Casimir force* between two uncharged conducting plates attest to the possibility of polarizing spacetime electromagnetically. We have electromagnetic “vacuum polarization”. These effects are to be described in the framework of quantum electrodynamics. However, since the Lamb shift and the Casimir effect are low-energy effects involving slowly varying electromagnetic fields, they can be described quasi-classically in the framework of classical electrodynamics with an altered spacetime relation. Thus the *linear* Maxwell–Lorentz relation has to be substituted by the *nonlinear* Heisenberg–Euler spacetime relation (see Sec.E.2.3). However, the nonlinearity turns out to be only a correction to the linear Maxwell–Lorentz spacetime relation. And the existence of a metric has to be *assumed* before one can formulate the Heisenberg–Euler spacetime relation. In this sense, the nonlinearity is not a fundamental one. The same is valid for all the models to be described below.

It should be understood that in our axiomatic setup, when only the Maxwell–Lorentz relation, the fifth axiom, is generalized, the basic structure of electro-

dynamics is untouched; that is, the first four axioms stand firm. If we turn the argument around: The limits of the Maxwell–Lorentz spacetime relation become visible. The fifth axiom is built on shakier grounds than the first four axioms.

Obviously, besides nonlinearity in the spacetime relation, the *nonlocality* can be and has been explored. This is further away from present-day experiments but it may be unavoidable in the end. We turn first to a discussion of nonlocality.

E.2.2 $\otimes$ Mashhoon

A spacetime has *dispersion* properties when the electromagnetic fields produce noninstantaneous polarization and magnetization effects. The most general *linear* spacetime relation is then given, in the comoving system, by means of the Volterra integral

$$H_{ij}(\tau, \xi) = \frac{1}{2} \int d\tau' K_{ij}{}^{kl}(\tau, \tau') F_{kl}(\tau', \xi). \quad (\text{E.2.1})$$

The coefficients of the kernel $K_{ij}{}^{kl}(\tau, \tau')$ are called the response functions. We expect the *metric to be involved* in their setup. Their form is defined by the intrinsic properties of spacetime itself.

Mashhoon has proposed a physically very interesting example of such a nonlocal electrodynamics in which nonlocality is induced by the *noninertial* dynamics of observers; that is, it results from the observers' accelerations. In this case, instead of decomposing the electromagnetic field with respect to $dx^i \wedge dx^j$, one should do it with respect to the coframe $\vartheta^\alpha = e_i^\alpha dx^i$ of a noninertial observer:

$$H = \frac{1}{2} H_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta, \quad F = \frac{1}{2} F_{\alpha\beta} \vartheta^\alpha \wedge \vartheta^\beta. \quad (\text{E.2.2})$$

The spacetime relation is then replaced by¹

$$H_{\alpha\beta}(\tau, \xi) = \frac{1}{2} \int d\tau' K_{\alpha\beta}{}^{\gamma\delta}(\tau, \tau') F_{\gamma\delta}(\tau', \xi). \quad (\text{E.2.3})$$

The response kernel in (E.2.3) is now defined by the acceleration and rotation of the observer's reference system. It is a constitutive law for the vacuum as viewed from a non-inertial frame of reference.

Mashhoon originally imposed the additional assumption that the kernel is of a *convolution* type, i.e., $K_{\alpha\beta}{}^{\gamma\delta}(\tau, \tau') = K_{\alpha\beta}{}^{\gamma\delta}(\tau - \tau')$. This is called “dynamic memory”.² Then the kernel can be uniquely determined by means of the

¹See Mashhoon [22, 23].

²The notions “dynamic” and “kinematic” memory were introduced and extensively discussed by Chicone and Mashhoon [8, 9].

Volterra technique. It turned out, however, that the integral in (E.2.3) diverges in the convolution case under certain physically reasonable conditions.

Preserving the main ideas of Mashhoon’s approach, one can abandon the convolution condition and assume that, apart from the local term proportional to the delta function $\delta(\tau - \tau')$, the kernel $K_{\alpha\beta}{}^{\gamma\delta}(\tau, \tau')$ depends only on the value of τ' . This is the case of “kinematic memory”. Let u be the observer’s 4-velocity and $\Gamma_{\alpha}{}^{\beta}$ the linear connection 1-form of the underlying spacetime. Then the general form of the kernel can be worked out explicitly:³

$$K_{\alpha\beta}{}^{\gamma\delta}(\tau, \tau') = \frac{1}{2} \epsilon_{\alpha\beta}{}^{\mu[\delta} \left(\delta_{\mu}^{\gamma]} \delta(\tau - \tau') - u \lrcorner \Gamma_{\mu}{}^{\gamma]}(\tau') \right). \quad (\text{E.2.4})$$

The influence of noninertiality is manifest in the presence of the connection 1-form. Clearly a metric is here assumed beforehand. The kernel (E.2.4) coincides with the original Mashhoon kernel in the case of constant acceleration and rotation, but in general the two kernels are different. The kernel (E.2.4) yields convergent integrals and is thus a reasonable choice. Only direct observations could establish the true form of the nonlocal spacetime relation. However, such nonlocal effects have not been confirmed experimentally as yet.

E.2.3 Heisenberg–Euler

Quantum electrodynamical vacuum corrections to the Maxwell–Lorentz theory can be accounted for by an effective *nonlinear* spacetime relation derived by Heisenberg and Euler [15]. To first order in the fine structure constant $\alpha_f = e^2/4\pi\epsilon_0\hbar c \approx 1/137$, it is given by⁴

$$H = \sqrt{\frac{\epsilon_0}{\mu_0}} \left\{ \left[1 + \frac{8\alpha_f}{45 B_k^2} \star (F \wedge \star F) \right] \star F + \frac{14\alpha_f}{45 B_k^2} \star (F \wedge F) F \right\}, \quad (\text{E.2.5})$$

where the magnetic field strength⁵ $B_k = m^2 c^2 / e\hbar \approx 4.4 \times 10^9 \text{ T}$, with the mass of the electron m . Again, post-Riemannian structures don’t interfere here. This theory is a valid physical theory.⁶

According to (E.2.5), the vacuum is treated as a specific type of medium the polarizability and magnetizability properties of which are determined by

³See Hehl and Obukhov [14] and Muench et al. [25].

⁴See, for instance, Itzykson and Zuber [18] or Heyl and Hernquist [16].

⁵Let us check the dimensional consistency of (E.2.5). We abbreviate the dimensions of length and magnetic flux by ℓ and Φ , respectively. If we assume that the line element ds^2 has dimension $[ds^2] = \ell^2$, then the metric volume element η has $[\eta] = \ell^4$. Accordingly, since $\star\eta = 1$, $[\star]_4 = 1/\ell^4$. As we indicated with the subscript 4, the latter formula is only valid if applied to a 4-form. For a 2-form, we have $[F] = [\star F]$, that is, $[\star]_2 = 1$. Hence in (E.2.5), we find $[\star(F \wedge \star F)] = [F][\star F]/\ell^4 = (\phi/\ell^2)^2 \stackrel{\text{SI}}{=} (\text{Wb}/\text{m}^2)^2 = \text{T}^2$.

⁶Recently, Brodin et al. [7] proposed a possible *direct* measurement of the Heisenberg–Euler effect.

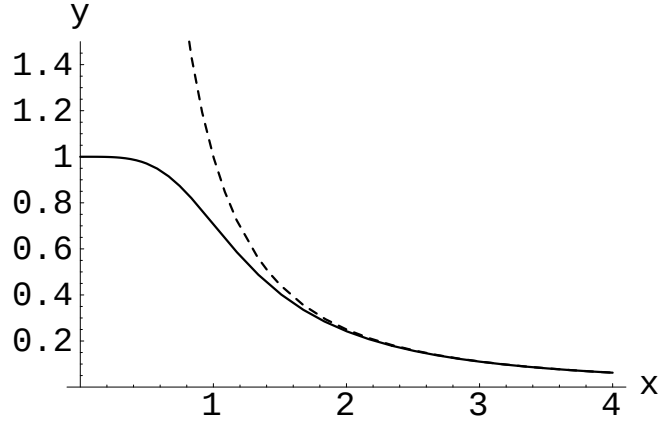


Figure E.2.1: Spherically symmetric electric field of a point charge in the Born–Infeld (solid line) and in Maxwell–Lorentz theory (dashed line). On the axes we have dimensionless variables $x := r/r_0$ and $y := E/E_e$.

the “clouds” of virtual charges surrounding the real currents and charges. As already explained, the first four axioms remain untouched by (E.2.5).

E.2.4 $\otimes$ Born–Infeld

The nonlinear Born–Infeld theory represents a *classical* generalization of the Maxwell–Lorentz theory for accommodating stable solutions for the description of “electrons”. Its spacetime relation reads⁷

$$H = \sqrt{\frac{\varepsilon_0}{\mu_0}} \frac{{}^*F + \frac{1}{2f_e^2} {}^*(F \wedge F) F}{\sqrt{1 - \frac{1}{f_e^2} {}^*(F \wedge {}^*F) - \frac{1}{4f_e^4} [{}^*(F \wedge F)]^2}}. \quad (\text{E.2.6})$$

Because of the nonlinearity, the field of a point charge, for example, turns out to be finite at $r = 0$, in contrast to the well-known $1/r^2$ singularity of the Coulomb field in Maxwell–Lorentz electrodynamics (see Fig.E.2.1). The dimensional parameter $f_e := E_e/c$ is defined by the so-called maximal field strength achieved in the Coulomb configuration of an electron:⁸ $E_e := e/4\pi\varepsilon_0 r_0^2$, where $r_0 := \xi r_e$, with the classical electron radius $r_e := e^2/4\pi\varepsilon_0 mc^2 = \alpha_f \hbar/mc$ and a numerical constant $\xi \approx 1$. Explicitly, we have $E_e \approx 1.8 \times 10^{20}$ V/m and $f_e = E_e/c \approx 6.4 \times 10^{11}$ T.

The spacetime relation (E.2.6) — like (E.2.5) — leads to a nonlinear equation for the dynamical evolution of the field strength F . As a consequence, the

⁷See Born and Infeld [4], Sommerfeld [33], and Gibbons and Rasheed [11].

⁸In quantum string theory, the Born–Infeld spacetime relation arises as an effective model with $f_e = 1/2\pi\alpha'$, where α' is the inverse string tension constant.

characteristic surface, the light cone, depends on the field strength, and the superposition principle for the electromagnetic field doesn't hold any longer.

E.2.5 $\otimes$ Plebański

Both (E.2.6) and (E.2.5) are special cases of Plebański's more general nonlinear electrodynamics [27]. Let the quadratic invariants of the electromagnetic field strength be denoted by

$$\mathcal{I}_1 := \frac{1}{2} {}^* (F \wedge {}^* F) \quad \text{and} \quad \mathcal{I}_2 := \frac{1}{2} {}^* (F \wedge F), \quad (\text{E.2.7})$$

where $\mathcal{I}_1$ is an untwisted and $\mathcal{I}_2$ a twisted scalar (the Hodge operator is twisted). The relations to our 4-forms of Part B and their $(1+3)$ -decompositions, see (B.2.28) and (B.2.29), are given by

$$\begin{aligned} \mathcal{I}_1 &= \frac{1}{2\lambda_0} {}^* I_1 = -\frac{1}{2} {}^* (B \wedge {}^* B - (E/c) \wedge {}^* (E/c)) \\ &= -\frac{1}{2} \left(\vec{B}^2 - (\vec{E}/c)^2 \right), \end{aligned} \quad (\text{E.2.8})$$

$$\mathcal{I}_2 = \frac{1}{2} {}^* I_2 = {}^* (B \wedge E/c) = (\vec{B} \cdot \vec{E}/c). \quad (\text{E.2.9})$$

Then Plebański postulated a nonlinear electrodynamics with the spacetime relation⁹

$$H = U(\mathcal{I}_1, \mathcal{I}_2) {}^* F + V(\mathcal{I}_1, \mathcal{I}_2) F, \quad (\text{E.2.10})$$

where U and V are functions of the two invariants. Note that in the Born–Infeld case U and V depend on both invariants whereas in the Heisenberg–Euler case we have $U_{\text{HE}}(\mathcal{I}_1)$ and $V_{\text{HE}}(\mathcal{I}_2)$. Nevertheless, in both cases U is required as well as V . And in both cases, see (E.2.6) and (E.2.5), U is an even function and V an odd one so as to preserve parity invariance. If one chooses $V(\mathcal{I}_1, \mathcal{I}_2)$ to be an even function, then parity violating terms would emerge, a case that is not visible in experiment.

⁹Strictly, Plebański assumed a Lagrangian that yields the Maxwell equations together with the *structural relations* $F = u(\mathcal{I}_1, \mathcal{I}_2) {}^* H + v(\mathcal{I}_1, \mathcal{I}_2) H$. The latter law, apart from singular cases, is equivalent to (E.2.10).

E.3

Electrodynamics in matter, constitutive law

When considering electromagnetic phenomena inside a material medium with nontrivial electric and magnetic properties, we have to distinguish between bound and free charges and the corresponding currents. This leads to the sixth axiom, which describes the splitting of the electromagnetic current and introduces the polarization and magnetization forms. The Maxwell equations are then formulated in terms of the external excitations (which correspond to the free charges and currents). The electric and magnetic susceptibilities of matter are defined by linear constitutive relations. The fundamental physical consequence of the current-splitting axiom is the corresponding split of the energy-momentum current, which turns out to be the sum of free and bound (material) energy-momenta. In order to illustrate the physical contents of the latter, we study the experiments of Walker & Walker and James who measured a ponderomotive force that affects a matter with nontrivial electric and magnetic properties placed in an external electromagnetic field.

E.3.1 Splitting of the current: Sixth axiom

In this chapter we present a consistent *microscopic* approach to the electrodynamics of continuous media.¹ Besides the field strength F , the excitation H is

¹In a great number of texts on electrodynamics, the electric and magnetic properties of media are described following the *macroscopic averaging* scheme of Lorentz [21]. However, this formalism has a number of serious limitations; see the relevant criticism of Hirst [17], for

a microscopic field in its own right, as we have shown in our axiomatic discussion in Part B. The total current density is the sum of the two contributions originating “from the inside” of the medium (bound or material charge) and “from the outside” (free or external charge):

$$J = J^{\text{mat}} + J^{\text{ext}} \quad (\text{sixth axiom a}). \quad (\text{E.3.1})$$

Accordingly, the bound electric current inside matter is denoted by *mat* and the external current by *ext*. The same notational scheme is also applied to the excitation H , so we have H^{mat} and H^{ext} .

Bound charges and bound currents are inherent characteristics of matter determined by the medium itself. They only emerge *inside* the medium. In contrast, free charges and free currents in general appear outside and inside matter. They can be prepared for a specific purpose by a suitable experimental arrangement. We can, for instance, prepare a beam of charged particles, described by J^{ext} , and scatter them at the medium, or we could study the reaction of a medium in response to a prescribed configuration of charges and currents, J^{ext} .

We assume that the charge bound by matter fulfills the usual charge conservation law separately:

$$dJ^{\text{mat}} = 0 \quad (\text{sixth axiom b}). \quad (\text{E.3.2})$$

We call (E.3.1) and (E.3.2) the *sixth axiom*. It specifies the properties of the classical material medium. In view of the relation $dJ = 0$, resulting from the first axiom, the assumption (E.3.2) means that there is no physical exchange (or conversion) between the bound and free charges. The sixth axiom certainly does not exhaust all possible types of material media, but it is valid for a sufficiently wide class of media.

Mathematically, (E.3.2) has the same form as $dJ = 0$. As a consequence, we can repeat the arguments of Sec. B.1.3 and find the corresponding excitation H^{mat} as a “potential” for the bound current:

$$J^{\text{mat}} = dH^{\text{mat}}. \quad (\text{E.3.3})$$

The $(1 + 3)$ -decomposition, following the pattern of (B.1.39), yields

$$H^{\text{mat}} = {}^\perp H^{\text{mat}} + \underline{H}^{\text{mat}} = -\mathcal{H}^{\text{mat}} \wedge d\sigma + \mathcal{D}^{\text{mat}}. \quad (\text{E.3.4})$$

The conventional names for these newly introduced excitations are *polarization* 2-form P and *magnetization* 1-form M , i.e.,

$$\mathcal{D}^{\text{mat}} \equiv -P, \quad \mathcal{H}^{\text{mat}} \equiv M. \quad (\text{E.3.5})$$

The minus sign is conventional. Thus, in analogy to the inhomogeneous Maxwell equations (B.1.44), (B.1.45), we find

$$-\underline{d}P = \rho^{\text{mat}}, \quad \underline{d}M + \dot{P} = j^{\text{mat}}. \quad (\text{E.3.6})$$

example. An appropriate modern presentation of the microscopic approach to this subject has been given in the textbook of Kovetz [20].

The identifications (E.3.5) are only true up to an exact form. However, if we require $\mathcal{D}^{\text{mat}} = 0$ for $E = 0$ and $\mathcal{H}^{\text{mat}} = 0$ for $B = 0$, as we do in (E.3.56), uniqueness is guaranteed.

E.3.2 Maxwell's equations in matter

The Maxwell equations (B.4.9),

$$dH = J \quad \text{or} \quad \begin{cases} \underline{d}\mathcal{D} = \rho, \\ \underline{d}\mathcal{H} - \dot{\mathcal{D}} = j, \end{cases} \quad (\text{E.3.7})$$

are *linear* partial differential equations of first order. Therefore it is useful to define the *external excitation*

$$\mathcal{H} := H - H^{\text{mat}} \quad \text{or} \quad \begin{cases} \mathfrak{D} := \mathcal{D} - \mathcal{D}^{\text{mat}} = \mathcal{D} + P \\ \mathfrak{H} := \mathcal{H} - \mathcal{H}^{\text{mat}} = \mathcal{H} - M \end{cases}. \quad (\text{E.3.8})$$

The external excitation $\mathcal{H} = (\mathfrak{H}, \mathfrak{D})$ can be understood as an auxiliary quantity. We differentiate (E.3.8) and eliminate dH and dH^{mat} by (E.3.7) and (E.3.3), respectively. Then, using (E.3.1), the *inhomogeneous* Maxwell equation for matter finally reads

$$d\mathcal{H} = J^{\text{ext}}, \quad (\text{E.3.9})$$

or in $(1+3)$ -decomposed form,

$$\underline{d}\mathfrak{D} = \rho^{\text{ext}}, \quad (\text{E.3.10})$$

$$\underline{d}\mathfrak{H} - \dot{\mathfrak{D}} = j^{\text{ext}}. \quad (\text{E.3.11})$$

From (E.3.8) and the universal spacetime relation (E.1.18) we obtain

$$\mathfrak{D} = \varepsilon_g \varepsilon_0 \, {}^*E + P(E, B), \quad (\text{E.3.12})$$

$$\mathfrak{H} = \frac{1}{\mu_g \mu_0} \, {}^*B - M(B, E). \quad (\text{E.3.13})$$

The polarization $P(E, B)$ is a functional of the electromagnetic field strengths E and B . In general, it can depend also on the temperature T and possibly on other thermodynamic variables specifying the material continuum under consideration; similar remarks apply to the magnetization $M(B, E)$. The system (E.3.10), (E.3.11) looks similar to the Maxwell equations (E.3.7). However, the former equations refer only to the external fields and sources. We stress that the *homogeneous* Maxwell equation *remains valid* in its original form.

E.3.3 Energy-momentum currents in matter

In a medium, the total electric current J is the sum (E.3.1) of the external or free charge J^{ext} and the material or bound charge J^{mat} . When both of these

currents are nontrivial, the (ponderomotive) force acting on the medium is the sum of the two Lorentz forces,

$$f_\alpha = (e_\alpha \lrcorner F) \wedge J = f_\alpha^{\text{ext}} + f_\alpha^{\text{mat}}. \quad (\text{E.3.14})$$

Here the Lorentz force density $f_\alpha^{\text{ext}} = (e_\alpha \lrcorner F) \wedge J^{\text{ext}}$ describes the influence of the electric and magnetic fields on the external (free) current J^{ext} and the Lorentz force density $f_\alpha^{\text{mat}} = (e_\alpha \lrcorner F) \wedge J^{\text{mat}}$ specifies the action on the material (bound) charges.

Recalling the (1 + 3)-analysis of Sec. B.2.2, we can write now the total longitudinal force as the sum $f_\alpha = d\sigma \wedge ({}^f k_\alpha + {}^b k_\alpha)$ with

$${}^f k_{\hat{0}} = -j^{\text{ext}} \wedge E, \quad {}^f k_a = \rho^{\text{ext}} \wedge (e_a \lrcorner E) + j^{\text{ext}} \wedge (e_a \lrcorner B), \quad (\text{E.3.15})$$

$${}^b k_{\hat{0}} = -j^{\text{mat}} \wedge E, \quad {}^b k_a = \rho^{\text{mat}} \wedge (e_a \lrcorner E) + j^{\text{mat}} \wedge (e_a \lrcorner B). \quad (\text{E.3.16})$$

Starting from the *left* equality in (E.3.14), we find that the energy-momentum current of the electromagnetic field in matter has the form (B.5.7). We may call (B.5.7) the total energy-momentum current in a material medium. Upon substitution of the Maxwell–Lorentz spacetime relation (D.6.13), it reads²

$${}^k \Sigma_\alpha = \frac{\lambda_0}{2} [F \wedge (e_\alpha \lrcorner {}^* F) - {}^* F \wedge (e_\alpha \lrcorner F)] \quad (\text{E.3.17})$$

(see (E.1.22)). Looking at the *right* equality in (E.3.14), we observe that this total energy-momentum current is the sum of the two separate contributions on the right-hand side:

$${}^k \Sigma_\alpha = {}^f \Sigma_\alpha + {}^b \Sigma_\alpha. \quad (\text{E.3.18})$$

We can study these two contributions separately because there is, as we assumed in Sec. E.3.1, no physical mixing between the free and the bound currents; in particular, they are conserved separately.

The corresponding analysis gives rise to two energy-momentum currents that are specified below. In actual observations in media, we are inspecting how the electric and magnetic fields act on the *external or free charges and currents*. By using the inhomogeneous Maxwell equations in matter, namely (E.3.9) or (E.3.10), (E.3.11), and repeating the derivations of Sec. B.5.3, we obtain

$$f_\alpha^{\text{ext}} = (e_\alpha \lrcorner F) \wedge J^{\text{ext}} = d {}^f \Sigma_\alpha + {}^f X_\alpha. \quad (\text{E.3.19})$$

Here the “*free-charge*” *energy-momentum* 3-form of the electromagnetic field and the supplementary term are, respectively, given by

$${}^f \Sigma_\alpha := \frac{1}{2} [F \wedge (e_\alpha \lrcorner \mathbb{H}) - \mathbb{H} \wedge (e_\alpha \lrcorner F)], \quad (\text{E.3.20})$$

$${}^f X_\alpha := -\frac{1}{2} (F \wedge \mathcal{L}_{e_\alpha} \mathbb{H} - \mathbb{H} \wedge \mathcal{L}_{e_\alpha} F). \quad (\text{E.3.21})$$

²This energy-momentum current was proposed in [29] and [26].

This energy-momentum describes the action of the electromagnetic field on the free charges (hence the notation where the superscript “f” stands for “free”).

First, let us analyze the supplementary term. In $(1+3)$ -decomposed form, we have $F = E \wedge d\sigma + B$ and $\mathcal{H} = -\mathfrak{H} \wedge d\sigma + \mathfrak{D}$. The Lie derivatives of these 2-forms can be easily computed,

$$\begin{aligned}\mathcal{L}_{e_0} F &= \dot{E} \wedge d\sigma + \dot{B}, \\ \mathcal{L}_{e_a} F &= (\underline{\mathcal{L}}_{e_a} E) \wedge d\sigma + \underline{\mathcal{L}}_{e_a} B,\end{aligned}\tag{E.3.22}$$

and similarly,

$$\begin{aligned}\mathcal{L}_{e_0} \mathcal{H} &= -\dot{\mathfrak{H}} \wedge d\sigma + \dot{\mathfrak{D}}, \\ \mathcal{L}_{e_a} \mathcal{H} &= -(\underline{\mathcal{L}}_{e_a} \mathfrak{H}) \wedge d\sigma + \underline{\mathcal{L}}_{e_a} \mathfrak{D}.\end{aligned}\tag{E.3.23}$$

Here $\underline{\mathcal{L}}_{e_a} := \underline{d}e_a \lrcorner + e_a \lrcorner \underline{d}$ is the purely spatial Lie derivative. Substituting (E.3.22), (E.3.23) into (E.3.21), we find

$${}^f X_0 = \frac{1}{2} d\sigma \wedge [\mathfrak{H} \wedge \dot{B} - \dot{\mathfrak{H}} \wedge B + E \wedge \dot{\mathfrak{D}} - \dot{E} \wedge \mathfrak{D}],\tag{E.3.24}$$

$${}^f X_a = \frac{1}{2} d\sigma \wedge [\mathfrak{H} \wedge \underline{\mathcal{L}}_{e_a} B - \underline{\mathcal{L}}_{e_a} \mathfrak{H} \wedge B + E \wedge \underline{\mathcal{L}}_{e_a} \mathfrak{D} - \underline{\mathcal{L}}_{e_a} E \wedge \mathfrak{D}].\tag{E.3.25}$$

The structure of the free-charge energy-momentum is revealed via the standard $(1+3)$ -decomposition:

$${}^f \Sigma_0 = {}^f u - d\sigma \wedge {}^f s,\tag{E.3.26}$$

$${}^f \Sigma_a = -{}^f p_a - d\sigma \wedge {}^f S_a.\tag{E.3.27}$$

Here, in complete analogy with (B.5.59), (B.5.60) and (B.5.61)–(B.5.64), we introduced the *energy* density 3-form

$${}^f u := \frac{1}{2} (E \wedge \mathfrak{D} + B \wedge \mathfrak{H}),\tag{E.3.28}$$

the *energy flux* density (or Poynting) 2-form

$${}^f s := E \wedge \mathfrak{H},\tag{E.3.29}$$

the *momentum* density 3-form

$${}^f p_a := -B \wedge (e_a \lrcorner \mathfrak{D}),\tag{E.3.30}$$

and the *stress* (or momentum flux density) 2-form of the electromagnetic field

$${}^f S_a := \frac{1}{2} [(e_a \lrcorner E) \wedge \mathfrak{D} - (e_a \lrcorner \mathfrak{D}) \wedge E + (e_a \lrcorner \mathfrak{H}) \wedge B - (e_a \lrcorner B) \wedge \mathfrak{H}].\tag{E.3.31}$$

In the absence of free charges and currents, we have the balance equations for the electromagnetic field energy and momentum $d^f \Sigma_\alpha + {}^f X_\alpha = 0$. In the $(1+3)$ -decomposed form this reads, analogously to (B.5.71), (B.5.72):

$${}^f \dot{u} + \underline{d}^f s + ({}^f X_{\hat{0}})_\perp = 0, \quad (\text{E.3.32})$$

$$-{}^f \dot{p}_a + \underline{d}^f S_a + ({}^f X_a)_\perp = 0. \quad (\text{E.3.33})$$

The Minkowski energy-momentum (E.3.20) is associated with free charges and has no relation to the forces that can arise in a medium due to its polarization and magnetization properties. It is, however, indispensable for analyzing electromagnetic wave phenomena in matter.

Let us now turn our attention back to the forces $f_\alpha^{\text{mat}} = (e_\alpha \lrcorner F) \wedge J^{\text{mat}}$ acting on the bound charges. In complete analogy to the derivation of (E.3.19)–(E.3.21), we find

$$f_\alpha^{\text{mat}} = (e_\alpha \lrcorner F) \wedge J^{\text{mat}} = d^b \Sigma_\alpha + {}^b X_\alpha. \quad (\text{E.3.34})$$

Here we introduce a new *material energy-momentum* 3-form of the electromagnetic field and the corresponding supplementary term:

$${}^b \Sigma_\alpha := \frac{1}{2} [F \wedge (e_\alpha \lrcorner H^{\text{mat}}) - H^{\text{mat}} \wedge (e_\alpha \lrcorner F)], \quad (\text{E.3.35})$$

$${}^b X_\alpha := -\frac{1}{2} (F \wedge \mathcal{L}_{e_\alpha} H^{\text{mat}} - H^{\text{mat}} \wedge \mathcal{L}_{e_\alpha} F). \quad (\text{E.3.36})$$

The material energy-momentum describes the action of the electromagnetic field on the bound charges (hence the notation where superscript “b” stands for “bound”). In $(1+3)$ -decomposed form, we have $H^{\text{mat}} = -(M \wedge d\sigma + P)$ and, as usual, $F = E \wedge d\sigma + B$. Thus the energy-momentum (E.3.35) is ultimately expressed in terms of the polarization P and magnetization M forms (E.3.5). When there are no free charges and currents, the energy-momentum (B.5.7) reduces to (E.3.35), and the latter should be used for the computation of the forces acting on dielectric and magnetic matter.

It is straightforward to perform, as in (E.3.22)–(E.3.25), a derivation of the supplementary term. The result reads

$${}^b X_\alpha = -{}^f X_\alpha + X_\alpha. \quad (\text{E.3.37})$$

Here X_α is given by (B.5.8). This relation is valid for *all* constitutive laws and spacetime relations.

The $(1+3)$ -decomposition yields a structure similar to (E.3.26), (E.3.27):

$${}^b \Sigma_{\hat{0}} = {}^b u - d\sigma \wedge {}^b s, \quad (\text{E.3.38})$$

$${}^b \Sigma_a = -{}^b p_a - d\sigma \wedge {}^b S_a. \quad (\text{E.3.39})$$

Here, in complete analogy with (B.5.59), (B.5.60) and (B.5.61)–(B.5.64), we introduced the *bound-charge energy* density 3-form

$${}^b u := \frac{1}{2} (B \wedge M - E \wedge P), \quad (\text{E.3.40})$$

the *bound-charge energy flux* density 2-form

$${}^b s := E \wedge M, \quad (\text{E.3.41})$$

the *bound-charge momentum* density 3-form

$${}^b p_a := B \wedge (e_a \lrcorner P), \quad (\text{E.3.42})$$

and the *bound-charge stress* (or momentum flux density) 2-form of the electromagnetic field

$${}^b S_a := \frac{1}{2} [(e_a \lrcorner P) \wedge E - (e_a \lrcorner E) \wedge P + (e_a \lrcorner M) \wedge B - (e_a \lrcorner B) \wedge M]. \quad (\text{E.3.43})$$

The (1 + 3)-decomposed balance equations for the bound-charge energy-momentum are analogous to (B.5.71), (B.5.72) and (E.3.32), (E.3.33). Writing the purely longitudinal force 4-form as $f_\alpha^{\text{mat}} = d\sigma \wedge {}^b k_\alpha$, we find:

$${}^b k_0 = {}^b \dot{u} + \underline{d} {}^b s + ({}^b X_{\hat{0}})_\perp, \quad (\text{E.3.44})$$

$${}^b k_a = -{}^b \dot{p}_a + \underline{d} {}^b S_a + ({}^b X_a)_\perp. \quad (\text{E.3.45})$$

In absence of the free charges, the integral of the 3-form of the force density (E.3.45) over the 3-dimensional domain Ω^{mat} , occupied by a material body or a medium, yields the total 3-force acting on the body or the medium,

$$K_a = \int_{\Omega^{\text{mat}}} {}^b k_a. \quad (\text{E.3.46})$$

⊗ Abraham versus Minkowski

The discussion on the correct energy-momentum current in macroscopic electrodynamics is quite old.³ Nevertheless, up to now this question was not settled theoretically,⁴ and not even by experiments was it possible to make a definite and decisive choice for electromagnetic energy and momentum in material media. Here we recall the definitions of the Minkowski and Abraham energy-momenta and compare them with our energy-momentum current.

The Minkowski energy-momentum is defined by (E.3.20), (E.3.26)–(E.3.27). Its properties are described in this section above. The corresponding force density reads:

$${}^M k_a = -{}^f \dot{p}_a + \underline{d} {}^f S_a. \quad (\text{E.3.47})$$

When the free charges are absent, the Minkowski force density, in view of (E.3.33), reduces merely to ${}^M k_a = -({}^f X_a)_\perp$.

³The beginning of this dispute goes back to Abraham [1] who raised some objections against Minkowski's work [24]; for a good review see Brevik [6].

⁴See [2], for example.

In Maxwell–Lorentz electrodynamics, the presence of the spacetime metric allows to discuss the symmetry of the energy-momentum tensor. The latter is defined by ${}^f\Sigma_\alpha =: {}^fT_\alpha{}^\beta \eta_\beta$ (see (E.1.28)). Let us study this question in (flat) Minkowski spacetime for the case of a medium that is at rest in the laboratory frame. Using the definition (E.3.20), we find for the off-diagonal components:

$${}^fT_0{}^a = -\eta^{abc} \mathfrak{H}_b E_c, \quad {}^fT_a{}^0 = \frac{\varepsilon\mu}{c^2} \eta_{abc} \mathfrak{H}^b E^c. \quad (\text{E.3.48})$$

If the upper index is lowered by means of the spacetime metric, ${}^fT_i{}^k g_{kj}$, we find indeed that

$${}^fT_i{}^k g_{kj} \neq {}^fT_j{}^k g_{ki}, \quad (\text{E.3.49})$$

since ${}^fT_0{}^b g_{ba} = \eta_{abc} \mathfrak{H}^b E^c$, whereas ${}^fT_a{}^0 g_{00} = \varepsilon\mu \eta_{abc} \mathfrak{H}^b E^c$. The extra factor is the square of the refractive index $n^2 = \varepsilon\mu$ of matter.

Moreover, the Minkowski stress (E.3.31) is also nonsymmetric: One can easily prove that $\underline{dx}_b \wedge {}^fS_a \neq \underline{dx}_a \wedge {}^fS_b$ in general. However, for an *isotropic* medium with the constitutive law (E.3.58) the stress becomes symmetric.

It is, nevertheless, interesting to observe that the Minkowski energy-momentum tensor is symmetric provided we use the *optical metric* (to be specified in (E.4.33) below) for the lowering of the upper index:

$${}^fT_i{}^k g_{kj}^{\text{opt}} = {}^fT_j{}^k g_{ki}^{\text{opt}}. \quad (\text{E.3.50})$$

This can easily be verified by means of (E.4.35). The *generalized symmetry* (E.3.50) also holds true in an arbitrarily moving medium. This fact highlights the fundamental position that the optical metric occupies in the Maxwell–Lorentz theory. Thus, it is evident why the Minkowski energy-momentum turns out to be the most useful one for the discussion of optical phenomena in material media.

The symmetric Abraham energy-momentum was proposed as a replacement of the nonsymmetric Minkowski energy-momentum. Contrary to a widely held view, there does not exist a derivation of the Abraham current ${}^A\Sigma_\alpha$ from first principles. In contrast, the Minkowski current emerges naturally from the Lorentz force axiom. The (1+3)-decomposed definition of the Abraham energy-momentum form reads:

$${}^A\Sigma_{\hat{0}} = {}^f u - d\sigma \wedge {}^f s, \quad (\text{E.3.51})$$

$${}^A\Sigma_a = -\frac{1}{c^2} \underline{dx}_a \wedge {}^f s - d\sigma \wedge {}^A S_a. \quad (\text{E.3.52})$$

Here the Abraham stress 2-form is introduced by

$$\begin{aligned} {}^A S_a := & \frac{1}{4} [(e_a \lrcorner E) \wedge \mathfrak{D} + (e_a \lrcorner {}^* \mathfrak{D}) \wedge {}^* E - (e_a \lrcorner \mathfrak{D}) \wedge E \\ & - (e_a \lrcorner {}^* E) \wedge {}^* \mathfrak{D} + (e_a \lrcorner \mathfrak{H}) \wedge B + (e_a \lrcorner {}^* B) \wedge {}^* \mathfrak{H} \\ & - (e_a \lrcorner B) \wedge \mathfrak{H} - (e_a \lrcorner {}^* \mathfrak{H}) \wedge {}^* B]. \end{aligned} \quad (\text{E.3.53})$$

One can straightforwardly verify that $\underline{dx}_b \wedge {}^A S_a = \underline{dx}_a \wedge {}^A S_b$. The corresponding energy-momentum tensor, introduced by means of ${}^A \Sigma_\alpha =: {}^A T_\alpha{}^\beta \eta_\beta$, is explicitly symmetric:

$${}^A T_i{}^k g_{kj} = {}^A T_j{}^k g_{ki}. \quad (\text{E.3.54})$$

The electromagnetic Abraham force density is then given by

$${}^A k_a = -\frac{1}{c^2} \underline{dx}_a \wedge {}^f \dot{s} + \underline{d} {}^A S_a. \quad (\text{E.3.55})$$

Within the framework of our axiomatic approach, the Abraham energy-momentum appears to be an unnatural object. As we established, the energy-momentum current is well defined (both in vacuum and in matter) already in the pre-metric formulation of electrodynamics. The fact that for the introduction of the Abraham energy-momentum one necessarily needs the metric makes ${}^A \Sigma_\alpha$ far less fundamental than the Minkowski energy-momentum current. Below we demonstrate that such an object also lacks substance from an experimental point of view.

E.3.4 Linear constitutive law

“It should be needless to remark that while from the mathematical standpoint a constitutive equation is a postulate or a definition, the first guide is physical experience, perhaps fortified by experimental data.”

C. Truesdell and R.A. Toupin (1960)

In the simplest case of a *homogeneous isotropic medium at rest* with nontrivial polarizational/magnetizational properties, we have the linear constitutive laws

$$P = \varepsilon_g \varepsilon_0 \chi_E {}^*E, \quad M = \frac{1}{\mu_g \mu_0} \chi_B {}^*B, \quad (\text{E.3.56})$$

with the electric and magnetic susceptibilities (χ_E, χ_B) . If we introduce the material constants

$$\varepsilon := 1 + \chi_E, \quad \mu := \frac{1}{1 - \chi_B}, \quad (\text{E.3.57})$$

we can rewrite the constitutive laws (E.3.56) as

$$\mathfrak{D} = \varepsilon \varepsilon_g \varepsilon_0 {}^*E \quad \text{and} \quad B = \mu \mu_g \mu_0 {}^*H. \quad (\text{E.3.58})$$

In curved spacetime, the quantities $(\varepsilon \varepsilon_g)$ and $(\mu \mu_g)$, in general, are functions of coordinates, but in flat Minkowski spacetime they are usually constant. However, $\varepsilon \neq \mu$, contrary to the effective gravitational permeabilities (E.1.19). In the general case, their values are determined by the electric and magnetic polarizability of a material medium.

A medium characterized by (E.3.58) is called *simple*. For a conductive medium, one usually adds one more constitutive relation that specifies the dependence of the free-charge current on the electric field, namely, Ohm's law:

$$j^{\text{ext}} = \sigma \lrcorner E. \quad (\text{E.3.59})$$

Here σ is the conductivity of the simple medium.

The constitutive law (E.3.58) can be written in an alternative way by using the foliation projectors explicitly:

$$\underline{H} = \varepsilon \lambda_0 (\star F), \quad \mu \lrcorner H = \lambda_0 \lrcorner (\star F). \quad (\text{E.3.60})$$

This form is particularly convenient for the discussion of the transition to vacuum. Then $\varepsilon = \mu = 1$ and (E.3.60) reduces immediately to the universal space-time relation (D.6.13).

For *anisotropic* media the constitutive laws (E.3.58), (E.3.59) are further generalized by replacing ε, μ, σ by the linear operators $\boldsymbol{\varepsilon}, \boldsymbol{\mu}, \boldsymbol{\sigma}$ acting on the spaces of transversal 2- and 1-forms. The easiest way to formulate these constitutive laws explicitly is to use the 3-vector components of the electromagnetic field strengths and excitations that were introduced earlier in (D.1.87), (D.1.88). In this description, the linear operators are just 3×3 matrices $\boldsymbol{\varepsilon} = \varepsilon^{ab}$ and $\boldsymbol{\mu} = \mu_{ab}$. One can write explicitly

$$\begin{pmatrix} \mathfrak{H}_a \\ \mathfrak{D}^a \end{pmatrix} = \lambda_0 \begin{pmatrix} 0 & \frac{N}{\sqrt{g}} \mu_{ab} \\ -\frac{\sqrt{g}}{N} \varepsilon^{ab} & 0 \end{pmatrix} \begin{pmatrix} -E_b \\ B^b \end{pmatrix}. \quad (\text{E.3.61})$$

In general, the matrices ε^{ab} and μ_{ab} depend on the spacetime coordinates. The contribution of the gravitational field is included in the metric-dependent factors N and $\sqrt{g} = \sqrt{{}^{(3)}g}$. In Minkowski spacetime, we have $N = c$, $\sqrt{g} = 1$.

After specifying the material constitutive relation, we can make the structure of the energy-momentum simpler and more transparent. In particular, in Minkowski spacetime for matter with the general linear constitutive law (E.3.61), we find for the supplementary force term (E.3.21):

$${}^f X_\alpha = \frac{1}{2} \text{Vol} \left[\varepsilon_0 (\partial_\alpha \varepsilon^{ab}) E_a E_b - \frac{1}{\mu_0} (\partial_\alpha \mu_{ab}) B^a B^b \right], \quad (\text{E.3.62})$$

where Vol is the 4-form of the spacetime volume. Thus ${}^f X_\alpha$ vanishes for homogeneous media with constant electric and magnetic permeabilities. Furthermore, within Maxwell–Lorentz electrodynamics with a general linear constitutive relation, we recover (E.3.62) with the opposite sign for the supplementary bound charge force (E.3.37).

The Abraham energy-momentum is also simplified, and its comparison with the Minkowski energy-momentum current becomes straightforward. For *isotropic* media with the constitutive law (E.3.58), the Abraham stress (E.3.53) coincides

with that of Minkowski (E.3.31), ${}^A S_a = {}^f S_a$. As a result, the comparison of (E.3.47) and (E.3.55) yields the relation:

$${}^A k_a = {}^M k_a + \Delta k_a. \quad (\text{E.3.63})$$

The expression

$$\Delta k_a := {}^f \dot{p}_a - c^{-2} \underline{dx}_a \wedge {}^f \dot{s} \quad (\text{E.3.64})$$

is usually called the Abraham term. Substituting (E.3.58) into the definition of the Minkowski momentum density (E.3.30) and using the identities (C.2.133) and (C.2.131), we find:

$${}^f p_a = -\varepsilon \varepsilon_0 B \wedge (e_a \lrcorner {}^* E) = -\varepsilon \mu \varepsilon_0 \mu_0 {}^* \mathfrak{H} \wedge {}^*(E \wedge \underline{dx}_a) = \frac{\varepsilon \mu}{c^2} \underline{dx}_a \wedge E \wedge \mathfrak{H}. \quad (\text{E.3.65})$$

Accordingly, the Abraham term (E.3.64) reads:

$$\Delta k_a := \frac{\varepsilon \mu - 1}{c^2} \underline{dx}_a \wedge {}^f \dot{s} = \frac{\varepsilon \mu - 1}{c^2} \underline{dx}_a \wedge \left(\dot{E} \wedge \mathfrak{H} + E \wedge \dot{\mathfrak{H}} \right). \quad (\text{E.3.66})$$

E.3.5 $\otimes$ Experiment of Walker & Walker

Let us consider an explicit example that shows how the energy-momentum current (E.3.18), (E.3.20) and (E.3.35) works. For concreteness, we analyze the experiment of Walker & Walker who measured the force acting on a dielectric disk placed in a vertical magnetic field B , as shown in Fig. E.3.1. The time-dependent magnetic field was synchronized with the alternating voltage applied to the inner and outer cylindrical surfaces of the disk at radius ρ_1 and ρ_2 , respectively, thus creating the electric field along the radial direction. The experiment revealed the torque around the vertical z -axis. We derive this torque by using the bound-charge energy-momentum current.

We have the Minkowski spacetime geometry. In cylindrical coordinates (ρ, φ, z) , the torque density around the z -axis is given by the product $\rho {}^b k_\varphi$. Hence the total torque is the integral

$$N^z = \int_{\text{disk}} \rho {}^b k_\varphi. \quad (\text{E.3.67})$$

Assuming the harmonically oscillating electric and magnetic fields, we find for the excitations inside the disk

$$\mathfrak{D} = \lambda_0 n \sin(\omega t) dz \wedge \left[a_1 J_1\left(\frac{\rho \omega n}{c}\right) d\rho - \frac{a_2}{\rho} \cos\left(\frac{z \omega n}{c}\right) \rho d\varphi \right], \quad (\text{E.3.68})$$

$$\mathfrak{H} = \frac{\cos(\omega t)}{\mu_0} \left[a_1 J_0\left(\frac{\rho \omega n}{c}\right) dz - \frac{a_2}{\rho} \sin\left(\frac{z \omega n}{c}\right) \rho d\varphi \right]. \quad (\text{E.3.69})$$

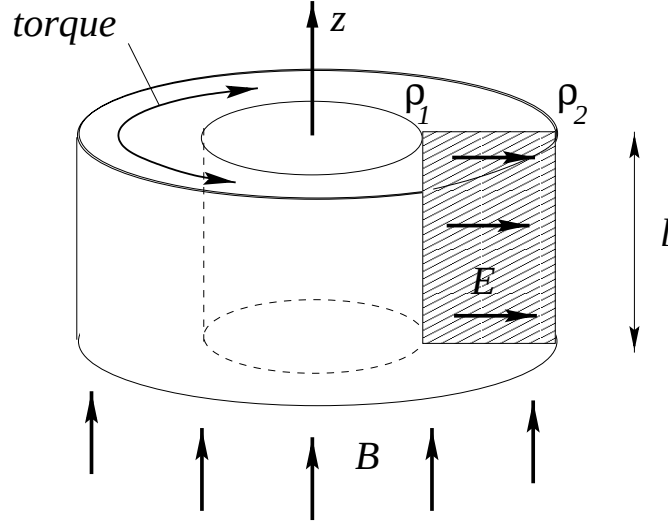


Figure E.3.1: Experiment of Walker & Walker. A dielectric disk, with inner radius ρ_1 and outer radius ρ_2 , is placed in a vertical magnetic field B between the poles of an electromagnet.

Here $\lambda_0 = \sqrt{\varepsilon_0/\mu_0}$ and $n := \sqrt{\varepsilon}$ is the refractive index of the medium. The disk consists of nonmagnetic dielectric material with $\mu = 1$. Here ω is the oscillation frequency and J_0, J_1 are Bessel functions. The two integration constants

$$a_1 = B_0, \quad a_2 = \frac{n U_0}{c \log(\rho_2/\rho_1)}, \quad (\text{E.3.70})$$

determine, respectively, the magnitude B_0 of the oscillating magnetic field and the amplitude U_0 of the voltage applied between the inner ($\rho = \rho_1$) and the outer ($\rho = \rho_2$) cylindrical surfaces of the disk, $\Delta U = U_0 \sin(\omega t)$. The field strength forms look similar:

$$E = \sin(\omega t) \frac{c}{n} \left[a_1 J_1 \left(\frac{\rho \omega n}{c} \right) \rho d\varphi + \frac{a_2}{\rho} \cos \left(\frac{z \omega n}{c} \right) d\rho \right], \quad (\text{E.3.71})$$

$$B = \cos(\omega t) d\rho \wedge \left[a_1 J_0 \left(\frac{\rho \omega n}{c} \right) \rho d\varphi + \frac{a_2}{\rho} \sin \left(\frac{z \omega n}{c} \right) dz \right]. \quad (\text{E.3.72})$$

One can verify by substitution that the electromagnetic field (E.3.68)–(E.3.72) represents an exact solution of the Maxwell equations plus the constitutive relations (E.3.58).

In the actual Walker & Walker experiment⁵, the disk, made of barium titanate with $\varepsilon = 3340$, has $l \approx 2$ cm height and the internal and external radius

⁵See Walker and Walker [36].

$\rho_1 \approx 0.4$ cm and $\rho_2 \approx 2.6$ cm, respectively. The oscillation frequency is rather low, namely $\omega = 60$ Hz. Correspondingly, one can verify that everywhere in the disk we have

$$\frac{\rho\omega n}{c} \sim \frac{z\omega n}{c} \sim 10^{-7} \ll 1. \quad (\text{E.3.73})$$

Then the field strengths read approximately

$$E = \sin(\omega t) \left[\frac{1}{2} a_1 \omega \rho^2 d\varphi + \frac{ca_2}{n\rho} d\rho \right], \quad (\text{E.3.74})$$

$$B = \cos(\omega t) \left[a_1 d\rho \wedge \rho d\varphi - \frac{na_2}{c\rho} \omega z dz \wedge d\rho \right]. \quad (\text{E.3.75})$$

Using the constitutive relations (E.3.56), we find the polarization 2-form

$$P = \varepsilon_0 \chi_E \sin(\omega t) \left[\frac{1}{2} a_1 \omega \rho dz \wedge d\rho + \frac{ca_2}{n} d\varphi \wedge dz \right]. \quad (\text{E.3.76})$$

The 1-form of magnetization is vanishing, $M = 0$, since $\mu = 1$. For the computation of the torque around the vertical axis, we need only the azimuthal components of the momentum (E.3.42) and of the stress form (E.3.43). Note that $e_\varphi = \frac{1}{\rho} \partial_\varphi$. A simple calculation yields:

$${}^b p_\varphi = \varepsilon_0 \chi_E \sin(\omega t) \cos(\omega t) \frac{ca_1 a_2}{n\rho} d\rho \wedge \rho d\varphi \wedge dz, \quad (\text{E.3.77})$$

$$\underline{d} {}^b S_\varphi = -\varepsilon_0 \chi_E \sin^2(\omega t) \frac{c\omega a_1 a_2}{n\rho} d\rho \wedge \rho d\varphi \wedge dz. \quad (\text{E.3.78})$$

As a result of the cylindrical symmetry, the permittivity and permeability do not depend on the angular coordinate and, accordingly, (E.3.62) yields ${}^b X_\varphi = -{}^f X_\varphi = 0$. Substituting all this into (E.3.45) and subsequently computing the integral (E.3.67), we find the torque

$$N^z = -\varepsilon_0(\varepsilon - 1) \pi l (\rho_2^2 - \rho_1^2) \frac{B_0 U_0 \omega}{\log(\rho_2/\rho_1)} \cos^2(\omega t). \quad (\text{E.3.79})$$

We already substituted the values of the integration constants (E.3.70); l is the height of the disk, whereas ρ_1 and ρ_2 are its inner and outer radius, as shown on Fig. E.3.1. Formula (E.3.79) has been verified experimentally by Walker & Walker.

It is rather curious that this fact was considered an argument in favor of the Abraham energy-momentum tensor⁶ that was introduced in the previous section. A certain formal coincidence is taking place, indeed, in the following sense: Recall that our starting point for deriving the bound-charge energy-momentum

⁶See, e.g., [36] and [2].

was the Lorentz force (E.3.34). Quite generally, for the 3-force density (E.3.34), we have from (E.3.3)–(E.3.6):

$$\begin{aligned} f_a^{\text{mat}} &= d\sigma \wedge (-B \wedge e_a \lrcorner j^{\text{mat}} + E \wedge e_a \lrcorner \rho^{\text{mat}}) \\ &= -d\sigma \wedge \left(B \wedge e_a \lrcorner \dot{P} \right. \\ &\quad \left. + B \wedge e_a \lrcorner \underline{d}M + E \wedge e_a \lrcorner \underline{d}P \right). \end{aligned} \quad (\text{E.3.80})$$

In the Walker & Walker experiment, we have $M = 0$. By differentiating (E.3.76), one can prove that $\underline{d}P = 0$. Thus the last line in (E.3.80) vanishes. Accordingly, the force density reduces to

$$-B \wedge e_a \lrcorner \dot{P} = -\mu\mu_0\chi_E\varepsilon_0 {}^*\mathfrak{H} \wedge (e_a \lrcorner {}^*\dot{E}) = \frac{\mu(\varepsilon - 1)}{c^2} \underline{dx}_a \wedge \dot{E} \wedge \mathfrak{H}, \quad (\text{E.3.81})$$

which resembles the Abraham term (E.3.66). However, our derivation was not based on the Abraham energy-momentum, and moreover, the argument of the symmetry of the energy-momentum tensor is absolutely irrelevant. As one can see, our bound-charge energy-momentum is manifestly asymmetric since the energy flux (E.3.41) is plainly zero whereas the momentum (E.3.42) is nonvanishing.

E.3.6 $\otimes$ Experiment of James

As another example we consider the experiment of James⁷ which is in many respects very similar to the one of Walker & Walker. James also placed a disk into crossed electric and magnetic fields (see Fig. E.3.2). The small cylinders were made of a composition of nickel-zinc ferrite with $\mu = 16$ or 43 and $\varepsilon \approx 7$. Like in the experiment of the Walkers, the *radial* electric field was created by means of an oscillating voltage $\Delta U = U_0 \sin(\omega_u t)$ applied between the inner and the outer cylindrical surface of the disk. However, instead of an axial magnetic field, an *azimuthal* magnetic field was produced inside matter by an alternating $I = I_0 \sin(\omega_i t)$ electric current in a conducting wire placed along the axis of the disk. The frequencies ω_i and ω_u are different (in the Walker & Walker experiment, electric and magnetic fields were oscillating with the same frequency). The resulting configuration for the excitations inside the disk is

$$\mathfrak{D} = \varepsilon\varepsilon_0 d\varphi \wedge dz \left[\frac{a_2 c}{n} \sin(\omega_u t) - a_3 \cos(\omega_i t) z\omega_i \right], \quad (\text{E.3.82})$$

$$\mathfrak{H} = \frac{1}{\mu\mu_0} d\varphi \left[-\frac{a_2 n}{c} \cos(\omega_u t) z\omega_u + a_3 \sin(\omega_i t) \right]. \quad (\text{E.3.83})$$

⁷The results of his observations are only described in his thesis and his short paper [19]. One can find, though, a somewhat detailed discussion in [5].

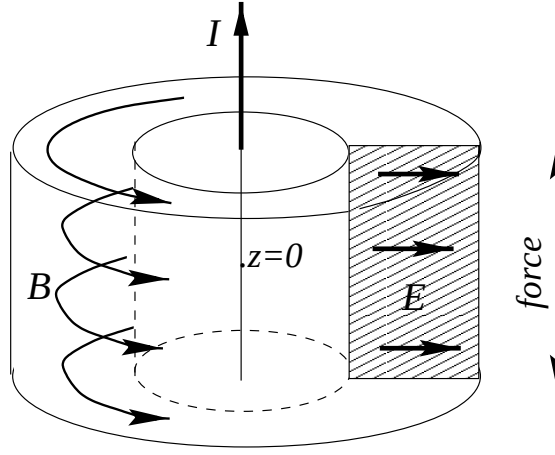


Figure E.3.2: Experiment of James. A ferrite disk is placed in an azimuthal magnetic field B created by an electric current flowing along the z -axis. A voltage, applied between the inner and outer cylindrical surfaces, creates the radial electric field E .

Here $n^2 = \varepsilon\mu$ and $\mu_0 c^2 = 1/\varepsilon_0$. Correspondingly, the field strength forms are:

$$E = \frac{1}{\rho} d\rho \left[\frac{a_2 c}{n} \sin(\omega_u t) - a_3 \cos(\omega_i t) z \omega_i \right], \quad (\text{E.3.84})$$

$$B = \frac{1}{\rho} dz \wedge d\rho \left[-\frac{a_2 n}{c} \cos(\omega_u t) z \omega_u + a_3 \sin(\omega_i t) \right]. \quad (\text{E.3.85})$$

Like in (E.3.74)–(E.3.75), we limit ourselves again to an approximation: The formulas (E.3.82)–(E.3.85) hold true for

$$\frac{z \omega_u n}{c} \sim \frac{z \omega_i n}{c} \sim 10^{-5} \ll 1 \quad (\text{E.3.86})$$

that is fulfilled everywhere inside the disk. The length (height) l of the disk, as well as the inner and outer radii ρ_1 and ρ_2 , are of order 1–3 cm, whereas the frequencies ω_i and ω_u were varied in the course of the experiment between 10 and 30 kHz.

We can verify that (E.3.82)–(E.3.85) represent a solution of the Maxwell equations. The integration constants are determined by the magnitudes of the electric current I_0 and the voltage U_0 :

$$a_2 = \frac{n U_0}{c \log(\rho_2/\rho_1)}, \quad a_3 = \mu \mu_0 \frac{I_0}{2\pi}. \quad (\text{E.3.87})$$

James measured the force acting along the z -axis of the disk in the crossed electric and magnetic fields (E.3.82)–(E.3.85). Let us derive the theoretical value

of this force by using the general expression for the force density (E.3.14). There are no free charges and currents inside matter, $\rho^{\text{ext}} = 0$ and $j^{\text{ext}} = 0$. As a result, the corresponding part (E.3.15) vanishes and we are left with a contribution of the bound charge and current (E.3.16).

Using the constitutive relations (E.3.56), we find the polarization 2-form and the magnetization 1-form:

$$P = \chi_E \varepsilon_0 d\varphi \wedge dz \left[\frac{a_2 c}{n} \sin(\omega_u t) - a_3 \cos(\omega_i t) z \omega_i \right], \quad (\text{E.3.88})$$

$$M = \frac{\chi_B}{\mu_0} d\varphi \left[-\frac{a_2 n}{c} \cos(\omega_u t) z \omega_u + a_3 \sin(\omega_i t) \right]. \quad (\text{E.3.89})$$

Bound charge and current densities are obtained by exterior differentiation according to (E.3.6). Since $\underline{d}P \equiv 0$, we find vanishing charge $\rho^{\text{mat}} = 0$. The bound current is nontrivial:

$$j^{\text{mat}} = \dot{P} + \underline{d}M = (\varepsilon\mu - 1)\varepsilon_0 d\varphi \wedge dz \left[\frac{\omega_u a_2 c}{n} \cos(\omega_u t) + a_3 \omega_i^2 z \sin(\omega_i t) \right]. \quad (\text{E.3.90})$$

Inserting this together with (E.3.85) into (E.3.16), we find the force density:

$$\begin{aligned} {}^b k_z &= j^{\text{mat}} \wedge (e_z \lrcorner B) = (\varepsilon\mu - 1)\varepsilon_0 \frac{1}{\rho} d\rho \wedge d\varphi \wedge dz \left[-\omega_u^2 a_2^2 \cos^2(\omega_u t) z \right. \\ &\quad \left. + \omega_i^2 a_3^2 \sin^2(\omega_i t) z + \frac{\omega_u a_2 a_3 c}{n} \sin(\omega_i t) \cos(\omega_u t) \right]. \end{aligned} \quad (\text{E.3.91})$$

The total force is computed as the integral (E.3.46) over the disk:

$$K_z = \int_{\text{disk}} {}^b k_a = \frac{(\mu\varepsilon - 1)\mu l U_0 I_0}{c^2} \omega_u \sin(\omega_i t) \cos(\omega_u t). \quad (\text{E.3.92})$$

We substituted (E.3.87) here. According to James, we choose $\omega_i = \omega_u \pm \omega_0$ with ω_0 the mechanical resonance frequency of the disk. Then, specializing to the component that varies with the frequency ω_0 , we find the final expression for the force

$$K_z = \pm \frac{(\mu\varepsilon - 1)\mu l U_0 I_0}{2c^2} \omega_u \sin(\omega_0 t). \quad (\text{E.3.93})$$

This theoretical prediction was actually verified in the experiment of James.

For comparison, let us derive the alternative theoretical predictions based on the Minkowski and Abraham energy-momentum currents. In the absence of free charges in matter, the Minkowski force density (E.3.47) reduces to ${}^M k_a = -({}^f X_a)_\perp$. Since the permittivity has the constant value of $\varepsilon \neq 1$ inside the body, i.e., for $-l/2 < z < l/2$, and drops to $\varepsilon = 1$ outside of that interval, the derivative of such a stepwise function reads $\partial_z \varepsilon(z) = (\varepsilon - 1) [\delta(z + l/2) - \delta(z - l/2)]$. A similar relation holds for the derivative of the permeability function: $\partial_z \mu(z) = (\mu - 1) [\delta(z + l/2) - \delta(z - l/2)]$. Correspondingly, from (E.3.62) we find for the

Minkowski force

$$\begin{aligned}
 {}^M K_z &= - \int_{\text{disk}} ({}^f X_z)_\perp = - \frac{1}{2} \int_{\text{disk}} [\varepsilon_0 E \wedge {}^* E \partial_z \varepsilon + \mu_0 \mathfrak{H} \wedge {}^* \mathfrak{H} \partial_z \mu] \\
 &= - \frac{l U_0 I_0}{c^2} \left[\mu(\varepsilon - 1) \omega_i \sin(\omega_u t) \cos(\omega_i t) \right. \\
 &\quad \left. + \varepsilon(\mu - 1) \omega_u \sin(\omega_i t) \cos(\omega_u t) \right]. \quad (\text{E.3.94})
 \end{aligned}$$

Let us now evaluate the Abraham term. Substituting (E.3.84) and (E.3.83) into (E.3.29), we find the energy flux density 2-form:

$$\begin{aligned}
 {}^f s &= \frac{1}{\mu \mu_0 \rho} d\rho \wedge d\varphi \left[\frac{a_2 a_3 c}{n} \sin(\omega_i t) \sin(\omega_u t) \right. \\
 &\quad \left. - \omega_u a_2^2 \sin(\omega_u t) \cos(\omega_u t) z - \omega_i a_3^2 \sin(\omega_i t) \cos(\omega_i t) z \right]. \quad (\text{E.3.95})
 \end{aligned}$$

Differentiating this and using (E.3.66), we obtain the additional contribution to the total force:

$$\begin{aligned}
 \Delta K_z &= \int_{\text{disk}} \Delta k_z = \int_{\text{disk}} \frac{\mu \varepsilon - 1}{c^2} dz \wedge {}^f s \\
 &= \frac{(\mu \varepsilon - 1) l U_0 I_0}{c^2} [\omega_i \sin(\omega_u t) \cos(\omega_i t) + \omega_u \sin(\omega_i t) \cos(\omega_u t)]. \quad (\text{E.3.96})
 \end{aligned}$$

In James' experiment, we put $\omega_i = \omega_u \pm \omega_0$ and select only the component varying with the mechanical resonance frequency ω_0 of the body. Then (E.3.63), (E.3.94), and (E.3.96) yield the Minkowski and the Abraham forces, respectively:

$${}^M K_z = \frac{l U_0 I_0}{2c^2} [\mu(\varepsilon - 1) \omega_0 \mp (\mu - \varepsilon) \omega_u] \sin(\omega_0 t), \quad (\text{E.3.97})$$

$${}^A K_z = {}^M K_z + \Delta K_z = \frac{l U_0 I_0}{2c^2} [(1 - \mu) \omega_0 \mp (\mu - \varepsilon) \omega_u] \sin(\omega_0 t). \quad (\text{E.3.98})$$

All the theoretical expressions for the electromagnetic force look similar: compare (E.3.93) with (E.3.97) and (E.3.98). However, the crucial difference is revealed when we take into account that James measured not the force itself but a “reduced force” defined as the mean value

$$K_z^{\text{red}} := \frac{1}{2} [K_z(\omega_u, \omega_i = \omega_u + \omega_0) + K_z(\omega_u, \omega_i = \omega_u - \omega_0)]. \quad (\text{E.3.99})$$

With high accuracy, James observed the *vanishing* of the reduced force in his experiment, $K_z^{\text{red}} = 0$. This observation is in complete agreement with the theoretical derivation (E.3.93) based on our energy-momentum current (E.3.17), whereas both, the expressions of Minkowski (E.3.97) and of Abraham (E.3.98), clearly contradict this experiment.

E.4

Electrodynamics of moving continua

The adapted spacetime foliation, which is associated with the metric structure, determines what can be considered as a laboratory frame of reference in spacetime. When a material medium moves with respect to such a laboratory foliation, its 4-vector field of the average velocity of matter induces another $(1 + 3)$ -splitting of the spacetime that we call the material foliation. We establish the relation between the components of the electromagnetic field with respect to the two foliations and derive the Minkowski constitutive law for arbitrarily moving matter. The motion of the material medium results, in particular, in the “dragging of light” phenomenon that tells us the electromagnetic waves in a moving medium propagate along the light cone of the so-called optical metric rather than along the cone of the original spacetime metric. Furthermore, we determine the generation of an electromagnetic field inside a moving medium. We discuss the two corresponding experiments of Röntgen and Wilson & Wilson who discovered this important physical effect. Finally, we discuss Maxwell–Lorentz electrodynamics as seen by a noninertial observer.

E.4.1 Laboratory and material foliation

The electric and magnetic parts of current, excitation, and field strength are only determined with respect to a certain foliation of spacetime. In Sec. B.1.4 we assumed the existence of a foliation specified by a formal “time” parameter σ and a vector field n . We know how to project all the physical and geometrical objects into transversal and longitudinal parts by using the coordinate-free $(1 +$

3)-decomposition technique of Sec. B.1.4. This original spacetime foliation is called the *laboratory foliation*.

Moving *macroscopic* matter, by means of its own velocity, defines another (1+3)-splitting of spacetime that is different from the original foliation discussed above. Here we describe this *material foliation* and its interrelationship with the laboratory foliation.

Let us denote a 3-dimensional matter-filled domain by V . Mathematically, we start with a 3-dimensional arithmetic space R^3 equipped with the coordinates ξ^a , where $a = 1, 2, 3$. We consider a smooth mapping $x_{(0)} : R^3 \rightarrow V \in X_4$ into spacetime that defines a 3-dimensional domain (hypersurface) V representing the initial distribution of matter. The coordinates ξ^a (known as the *Lagrange* coordinates in continuum mechanics) serve as labels that denote the elements of the material medium.

Given the initial configuration V of matter, we parametrize the dynamics of the medium by the coordinate τ , which is defined as the proper time measured along an element's world line from the original hypersurface V . The resulting local coordinates (τ, ξ^a) are usually called the normalized comoving coordinates. The motion of matter is thus described by the functions $x^i(\tau, \xi^a)$, and we subsequently define the (mean) velocity 4-vector field by

$$u := \partial_\tau = \left(\frac{dx^i}{d\tau} \right)_{\xi^a = \text{const}} \partial_i. \quad (\text{E.4.1})$$

By construction, this vector field is timelike and it is normalized according to

$$\mathbf{g}(u, u) = c^2. \quad (\text{E.4.2})$$

Evidently, a family of observers comoving with matter is characterized by the same timelike congruence $x^i(\tau, \xi^a)$. They are making physical (in particular, electro-dynamical) measurements in their local reference frames drifting with the material motion. By the *hypothesis of locality* it is assumed that the instruments in the comoving frame are not affected in an appreciable way by the local acceleration they experience. They measure the same as if they were in a suitable comoving instantaneous *inertial* frame.

After these preliminaries, we are ready to find the relation between the two foliations. Technically, the crucial point is to express the *laboratory* coframe $(d\sigma, \underline{dx}^a)$ in terms of the coframe adapted to the *material* foliation. Recall that according to our conventions formulated in Sec. B.1.4, $\underline{dx}^a = dx^a - n^a d\sigma$ is the transversal projection of the spatial coframe.

The motion of a medium uniquely determines the (1 + 3)-decomposition of spacetime through a *material foliation* that is obtained by replacing $n, d\sigma$ by $u, d\tau$. Note that the proper time differential is $d\tau = c^{-2} u_i dx^i$. Thus evidently

$$u \lrcorner d\tau = 1. \quad (\text{E.4.3})$$

We consider an arbitrary motion of matter. The velocity field u is arbitrary, and one does *not* assume that the laboratory and moving reference systems are related by a Lorentz transformation.

The technique of the $(1+3)$ -splitting is similar to that described in Sec. B.1.4 for the laboratory foliation. Namely, following the pattern of (B.1.22) and (B.1.23), one defines the decompositions with respect to the material foliation: For any p -form Ψ we denote the part longitudinal to the velocity vector u by

$${}^\perp\Psi := d\tau \wedge \Psi_\perp, \quad \Psi_\perp := u \lrcorner \Psi, \quad (\text{E.4.4})$$

and the part transversal to the velocity u by

$$\underline{\Psi} := u \lrcorner (d\tau \wedge \Psi) = (1 - {}^\perp)\Psi, \quad u \lrcorner \underline{\Psi} \equiv 0. \quad (\text{E.4.5})$$

Please note that the projectors are denoted now differently (${}^\perp$ and $\underline{}$) in order to distinguish them from the corresponding projectors (${}^\perp$ and $\underline{}$) of the laboratory foliation.

With the spacetime metric introduced on the X_4 by means of the Maxwell–Lorentz spacetime relation, we assume that the laboratory foliation is consistent with the metric structure in the sense outlined in Sec. E.1.3. In particular, taking into account (E.1.12) and (E.1.13), one finds the line element with respect to the laboratory foliation coframe:

$$ds^2 = N^2 d\sigma^2 + g_{ab} \underline{dx}^a \underline{dx}^b = N^2 d\sigma^2 - {}^{(3)}g_{ab} \underline{dx}^a \underline{dx}^b. \quad (\text{E.4.6})$$

Now it is straightforward to find the relation between the two foliations. Technically, by using (E.4.4) and (E.4.5), one just needs to $(1+3)$ -decompose the basis 1-forms of the laboratory coframe $(d\sigma, \underline{dx}^a)$ with respect to the material foliation. Taking into account that, in local coordinates, $n = \partial_\sigma + n^a \partial_a$ and similarly $u = u^{(\sigma)} \partial_\sigma + u^a \partial_a$, the result, in a convenient matrix form, reads:

$$\begin{pmatrix} d\sigma \\ \underline{dx}^a \end{pmatrix} = \begin{pmatrix} \gamma c/N & v_b/(cN) \\ \gamma v^a & \delta_b^a \end{pmatrix} \begin{pmatrix} d\tau \\ \underline{dx}^b \end{pmatrix}. \quad (\text{E.4.7})$$

Here we introduced for the *relative velocity* 3-vector the notation

$$v^a := \frac{c}{N} \left(\frac{u^a}{u^{(\sigma)}} - n^a \right); \quad (\text{E.4.8})$$

furthermore

$$\gamma := \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (\text{E.4.9})$$

Observe that (E.4.7) is *not* a Lorentz transformation since it relates two frames that are both *noninertial* in general.

As usual, the spatial indices are raised and lowered by the 3-space metric of (E.1.14), ${}^{(3)}g_{ab} := -g_{ab}$. In particular, we have explicitly $v_a = {}^{(3)}g_{ab} v^b$ and $v^2 := v_a v^a$. By means of the normalization (E.4.2), we can express the zeroth

Table E.4.1: Two foliations

	laboratory frame	material frame
vector field	n	u
“time”	σ	τ
longitudinal	${}^\perp\Psi$	${}^{\perp}\Psi$
transversal	$\underline{\Psi}$	$\underline{\Psi}$
time coframe	$d\sigma$	$d\tau$
3D coframe	$\underline{dx}^a$	$\underline{dx}^a$
4D line element of spacetime	$ds^2 = N^2 d\sigma^2$ $- {}^{(3)}g_{ab} \underline{dx}^a \underline{dx}^b$	$ds^2 = c^2 d\tau^2$ $- ({}^{(3)}g_{ab} - \frac{1}{c^2} v_a v_b) \underline{dx}^a \underline{dx}^b$

(time) component of the velocity as $u^{(\sigma)} = \gamma(c/N)$. Hence the explicit form of the matter 4-velocity reads:

$$u = u^{(\sigma)} \partial_\sigma + u^a \partial_a = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \left(\frac{c}{N} e_{\hat{0}} + v^a e_a \right). \quad (\text{E.4.10})$$

Here $(e_{\hat{0}}, e_a)$ is the frame dual to the adapted laboratory coframe $(d\sigma, \underline{dx}^a)$, i.e., $e_{\hat{0}} = n, e_a = \partial_a$.

When the relative 3-velocity is zero $v^a = 0$, the material and the laboratory foliations coincide because the corresponding foliation 1-forms turn out to be proportional to each other $u = (c/N) n$.

Substituting (E.4.7) into (E.4.6), we find for the line element in terms of the new variables

$$ds^2 = c^2 d\tau^2 - \hat{g}_{ab} \underline{dx}^a \underline{dx}^b, \quad \text{where} \quad \hat{g}_{ab} = {}^{(3)}g_{ab} - \frac{1}{c^2} v_a v_b. \quad (\text{E.4.11})$$

Comparing this with (E.4.6), we recognize that the transition from a laboratory frame to a moving material frame changes the form of the line element from (E.4.6) to (E.4.11). Consequently, this transition corresponds to a linear homogeneous transformation that is anholonomic in general. It is not a Lorentz transformation which, by definition, preserves the form of the metric coefficients.

The metric $\hat{g}_{ab}$ of the material foliation has the inverse

$$\hat{g}^{ab} = {}^{(3)}g^{ab} + \frac{\gamma^2}{c^2} v^a v^b. \quad (\text{E.4.12})$$

For its determinant one finds $(\det \widehat{g}_{ab}) = (\det g_{ab}) \gamma^{-2}$.

E.4.2 Electromagnetic field in laboratory and material frames

Let us consider the case of a simple medium with homogeneous and isotropic electric and magnetic properties. The constitutive law (E.3.60) for such a *medium at rest with respect to the laboratory frame* has to be understood as a result of a laboratory foliation. A moving medium is naturally at rest with respect to its own material foliation. Consequently, the constitutive law for such a simple medium reads

$$\underline{\mathcal{H}} = \varepsilon \lambda_0 (*F), \quad \mu^{-1} \underline{\mathcal{H}} = \lambda_0^{-1} (*F). \quad (\text{E.4.13})$$

How does the constitutive law look as seen from the original laboratory frame? For this purpose we will use the results of Sec. D.5.4 and the relations between the two foliations established in the previous section.

To begin with, recall of how the excitation and the field strength 2-forms decompose with respect to the *laboratory frame*,

$$\underline{\mathcal{H}} = -\mathfrak{H} \wedge d\sigma + \mathfrak{D}, \quad F = E \wedge d\sigma + B, \quad (\text{E.4.14})$$

and analogously, with respect to the *material frame*:

$$\underline{\mathcal{H}} = -\mathfrak{H}' \wedge d\tau + \mathfrak{D}', \quad F = E' \wedge d\tau + B'. \quad (\text{E.4.15})$$

Clearly, we preserve the same symbols $\underline{\mathcal{H}}$ and F on the left-hand sides of (E.4.14) and (E.4.15) because these are just the same physical objects. In contrast, the right-hand sides are of course different, hence we use primes. The constitutive law (E.4.13), according to the results of the previous section, can be rewritten as

$$\mathfrak{D}' = \varepsilon \varepsilon_0 \widehat{*} E', \quad \mathfrak{H}' = \frac{1}{\mu \mu_0} \widehat{*} B'. \quad (\text{E.4.16})$$

Here the Hodge star $\widehat{*}$ corresponds to the metric $\widehat{g}_{ab}$ of the *material* foliation. (Please do not mix it up with the Hodge star $*$ defined by the 3-space metric $^{(3)}g_{ab}$ of the laboratory foliation.) Now, (E.4.16) can be presented in the equivalent matrix form

$$\begin{pmatrix} \mathfrak{H}'_a \\ \mathfrak{D}'^a \end{pmatrix} = \lambda \begin{pmatrix} 0 & \frac{c}{n} \frac{\gamma}{\sqrt{g}} \widehat{g}_{ab} \\ -\frac{n}{c} \frac{\sqrt{g}}{\gamma} \widehat{g}^{ab} & 0 \end{pmatrix} \begin{pmatrix} -E'_b \\ B'^b \end{pmatrix}. \quad (\text{E.4.17})$$

The components of the constitutive matrices read explicitly

$$A'^{ab} = -\frac{n}{c} \frac{\sqrt{g}}{\gamma} \widehat{g}^{ab}, \quad B'_{ab} = \frac{c}{n} \frac{\gamma}{\sqrt{g}} \widehat{g}_{ab}, \quad C'^a{}_b = 0, \quad (\text{E.4.18})$$

$$\text{with } \lambda = \sqrt{\frac{\varepsilon \varepsilon_0}{\mu \mu_0}}, \quad n := \sqrt{\mu \varepsilon}. \quad (\text{E.4.19})$$

In order to find the constitutive law in the laboratory frame, we have to perform some very straightforward manipulations in matrix algebra along the lines described in Sec. D.5.4. Given is the linear transformation of the coframes (E.4.7). The corresponding transformation of the 2-form basis (A.1.95) turns out to be

$$\begin{aligned} P^a{}_b &= \frac{\gamma c}{N} \left(\delta_b^a - \frac{1}{c^2} v^a v_b \right), & Q_b{}^a &= \delta_b^a, \\ Z_{ab} &= -\gamma \hat{\epsilon}_{abc} v^c, & W^{ab} &= \frac{1}{Nc} \epsilon^{abc} v_c. \end{aligned} \quad (\text{E.4.20})$$

We use these results in (D.5.27)–(D.5.30). Then, after a lengthy matrix computation, we obtain from (E.4.18) the constitutive matrices in the laboratory foliation:

$$A^{ab} = \frac{1}{1 - \frac{v^2}{c^2}} \frac{\sqrt{{}^{(3)}g}}{N} \left[{}^{(3)}g^{ab} \left(\frac{v^2}{c^2 n} - n \right) + \frac{1}{c^2} v^a v^b \left(n - \frac{1}{n} \right) \right], \quad (\text{E.4.21})$$

$$B_{ab} = \frac{1}{1 - \frac{v^2}{c^2}} \frac{N}{\sqrt{{}^{(3)}g}} \left[{}^{(3)}g_{ab} \left(\frac{1}{n} - \frac{v^2 n}{c^2} \right) + \frac{1}{c^2} v_a v_b \left(n - \frac{1}{n} \right) \right], \quad (\text{E.4.22})$$

$$C^a{}_b = \frac{1}{1 - \frac{v^2}{c^2}} \left(n - \frac{1}{n} \right) {}^{(3)}\eta^{ac}{}_b \frac{v_c}{c}. \quad (\text{E.4.23})$$

The resulting constitutive law

$$\begin{pmatrix} \mathfrak{H}_a \\ \mathfrak{D}^a \end{pmatrix} = \lambda \begin{pmatrix} C^b{}_a & B_{ab} \\ A^{ab} & C^a{}_b \end{pmatrix} \begin{pmatrix} -E_b \\ B^b \end{pmatrix} \quad (\text{E.4.24})$$

can be presented in terms of exterior forms as:

$$\begin{aligned} \mathfrak{H} &= \frac{\gamma^2}{\mu_0 \mu_g} \left[{}^*B \left(\frac{1}{\mu} - \varepsilon \frac{v^2}{c^2} \right) + \frac{1}{c^2} v {}^*(v \wedge B) \left(\varepsilon - \frac{1}{\mu} \right) \right] \\ &\quad + \gamma^2 \varepsilon_0 {}^*(v \wedge E) \left(\varepsilon - \frac{1}{\mu} \right), \end{aligned} \quad (\text{E.4.25})$$

$$\begin{aligned} \mathfrak{D} &= \gamma^2 \varepsilon_0 \varepsilon_g \left[{}^*E \left(\varepsilon - \frac{v^2}{\mu c^2} \right) - \frac{1}{c^2} {}^*v {}^*(v \wedge {}^*E) \left(\varepsilon - \frac{1}{\mu} \right) \right] \\ &\quad + \gamma^2 \varepsilon_0 v \wedge {}^*B \left(\varepsilon - \frac{1}{\mu} \right). \end{aligned} \quad (\text{E.4.26})$$

Here we introduced the 3-velocity 1-form

$$v := v_a dx^a. \quad (\text{E.4.27})$$

The 3-velocity vector is decomposed according to $v^a e_a$. If we lower the index v^a by means of the 3-metric ${}^{(3)}g_{ab}$, we find the covariant components v_a of the

3-velocity that enter (E.4.27). Direct inspection shows that the constitutive law (E.4.25), (E.4.26) of above can alternatively be recast into the pair of equations:

$$\mathfrak{D} + \frac{\mu_g}{c^2} v \wedge \mathfrak{H} = \varepsilon \varepsilon_0 (\varepsilon_g {}^*E + v \wedge {}^*B), \quad (\text{E.4.28})$$

$$B - \frac{\varepsilon_g}{c^2} v \wedge E = \mu \mu_0 (\mu_g {}^*\mathfrak{H} - v \wedge {}^*\mathfrak{D}). \quad (\text{E.4.29})$$

These are the famous Minkowski relations for the electromagnetic field in a moving medium.¹ Originally, the constitutive relations (E.4.28), (E.4.29) were derived by Minkowski with the help of the Lorentz transformations for the case of a flat spacetime and a uniformly moving medium. We stress, however, that the Lorentz group never entered the scene in our derivation above. This demonstrates (contrary to the traditional view) that the role and the value of the Lorentz invariance in electrodynamics should not be overestimated.

The constitutive law (E.4.25), (E.4.26) or, equivalently, (E.4.28), (E.4.29) describes a moving simple medium on an *arbitrary curved background*. The influence of the spacetime geometry is manifest in ε_g, μ_g and in ${}^{(3)}g_{ab}$, which enters the Hodge star operator. In flat Minkowski spacetime in Cartesian coordinates, these quantities reduce to $\varepsilon_g = \mu_g = 1, {}^{(3)}g_{ab} = \delta_{ab}$.

The physical sources of the electric and magnetic excitations $\mathfrak{D}$ and $\mathfrak{H}$ are free charges and currents. Recalling the definitions (E.3.8) and (E.3.5), we can find the polarization P and the magnetization M that have bound charges and currents as their sources. A direct substitution of (E.4.25) and (E.4.26) into (E.3.8) yields:

$$\begin{aligned} P = & \gamma^2 \varepsilon_0 \varepsilon_g \left\{ \chi_E \left[{}^*E - \frac{1}{c^2} {}^*v {}^*(v \wedge {}^*E) + \frac{1}{\mu_g} v \wedge {}^*B \right] \right. \\ & \left. + \chi_B \left[\frac{v^2}{c^2} {}^*E - \frac{1}{c^2} {}^*v {}^*(v \wedge {}^*E) + \frac{1}{\mu_g} v \wedge {}^*B \right] \right\}, \end{aligned} \quad (\text{E.4.30})$$

$$\begin{aligned} M = & \frac{\gamma^2}{\mu_0 \mu_g} \left\{ \chi_B \left[{}^*B - \frac{1}{c^2} v {}^*(v \wedge B) - \frac{\varepsilon_g}{c^2} {}^*(v \wedge E) \right] \right. \\ & \left. + \chi_E \left[\frac{v^2}{c^2} {}^*B - \frac{1}{c^2} v {}^*(v \wedge B) - \frac{\varepsilon_g}{c^2} {}^*(v \wedge E) \right] \right\}. \end{aligned} \quad (\text{E.4.31})$$

Here χ_E and χ_B are the electric and magnetic susceptibilities (E.3.57). When the matter is at rest, i.e., $v = 0$, the equations (E.4.30), (E.4.31) reduce to the rest frame relations (E.3.56).

¹See the discussions of various aspects of the electrodynamics of moving media in [3, 6, 20, 30, 31, 35].

E.4.3 Optical metric from the constitutive law

A direct check shows that the constitutive matrices (E.4.21)–(E.4.23) satisfy the closure relation (D.3.4), (D.4.13), (D.4.14). Consequently, a metric of Lorentzian signature is induced by the constitutive law (E.4.24).

The general reconstruction of a metric from a linear constitutive law is given by (D.5.9). Starting from (E.4.18), we immediately find the induced metric in the material foliation:

$$g_{ij}^{\text{opt}} = \sqrt{\frac{n}{c} \frac{\gamma}{\sqrt{g}}} \left(\frac{\frac{c^2}{n^2}}{0} \middle| \frac{0}{-\widehat{g}_{ab}} \right). \quad (\text{E.4.32})$$

Making use of relation (E.4.7) between the foliations and the covariance properties proven in Sec. D.5.4, we find the explicit form of the induced metric in the laboratory foliation:

$$g_{ij}^{\text{opt}} = \left(\frac{\varepsilon\mu}{-\det g} \right)^{\frac{1}{4}} \left\{ g_{ij} - \left(1 - \frac{1}{\varepsilon\mu} \right) \frac{u_i u_j}{c^2} \right\}. \quad (\text{E.4.33})$$

Here g_{ij} are the components of the metric tensor of spacetime and u_i are the covariant components of the 4-velocity of matter (E.4.10). Note that $g^{ij} u_i u_j = c^2$, as usual. The contravariant induced metric reads:

$$g^{\text{opt } ij} = \left(\frac{-\det g}{\varepsilon\mu} \right)^{\frac{1}{4}} \left\{ g^{ij} - (1 - \varepsilon\mu) \frac{u^i u^j}{c^2} \right\}. \quad (\text{E.4.34})$$

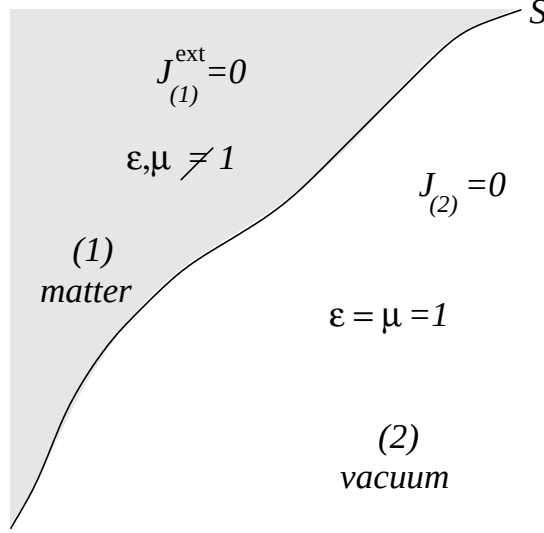
Such an induced metric g_{ij}^{opt} is usually called the *optical metric* in order to distinguish it from the *true* spacetime metric g_{ij} . It describes the “dragging of the aether” (“Mitführung des Äthers”²). The adjective “optical” expresses the fact that all the optical effects in moving matter are determined by the Fresnel equation (D.2.44), which reduces to the equation for the light cone determined by the metric (E.4.33) in the present case.

The nontrivial polarization/magnetization properties of matter are manifestly present even when the medium in the laboratory frame is at rest. Let us consider Minkowski spacetime with $g_{ij} = \text{diag}(c^2, -1, -1, -1)$, for example, and a medium at rest in it. Then $u = \partial_t$ or, in components, $u^i = (1, 0, 0, 0)$. We substitute this into (E.4.33) and find the optical metric

$$g_{ij}^{\text{opt}} = \sqrt{\frac{n}{c}} \left(\frac{\frac{c^2}{n^2}}{0} \middle| \frac{0}{-\delta_{ab}} \right). \quad (\text{E.4.35})$$

Evidently, the velocity of light c is replaced by c/n , with n as the refractive index of the dielectric and magnetic media.

²See Gordon [12] who first introduced the optical metric.

Figure E.4.1: Two regions divided by a surface S .

E.4.4 Electromagnetic field generated in moving continua

Let us consider an explicit example that demonstrates the power of the generally covariant constitutive law. For simplicity, we study electrodynamics in flat Minkowski spacetime in which the laboratory frame is determined by the usual time coordinate t and the Cartesian spatial coordinates $\vec{x}$. Then the metric has the components $N = c$, ${}^{(3)}g_{ab} = \delta_{ab}$. Correspondingly, $\epsilon_g = \mu_g = 1$. The motion of matter is, as shown in Sec. E.4.1, described by the material foliation as specified by the relative velocity v^a .

Next, let a surface S be the border between the two regions, the first of which [labeled as (1)] is filled *with matter* having nontrivial magnetic and electric properties with $\mu \neq 1$, $\epsilon \neq 1$. In the second region [labeled as (2)], matter is absent, and hence, $\mu = \epsilon = 1$.

We assume that the matter in the first region does not contain any free (i.e., external) charges and currents and that the motion of the medium is *stationary*. Then all the variables are independent of time.

Consider the case when the second (matter-free) region contains *constant* magnetic and/or electric fields, i.e., the components $B_{(2)}^a$ and $E_a^{(2)}$ of the field

strengths forms

$$\begin{aligned} B_{(2)} &= B^1 dx^2 \wedge dx^3 + B^2 dx^3 \wedge dx^1 + B^3 dx^1 \wedge dx^2, \\ E_{(2)} &= E_1 dx^1 + E_2 dx^2 + E_3 dx^3, \end{aligned} \quad (\text{E.4.36})$$

do not depend on $t, \vec{x}$. Then, the constitutive law (E.4.24)–(E.4.26) yields that the components $\mathfrak{H}_a^{(2)}$ and $\mathfrak{D}_{(2)}^a$ of the magnetic and electric excitations forms

$$\begin{aligned} \mathfrak{D}_{(2)} &= \varepsilon_0 (E^1 dx^2 \wedge dx^3 + E^2 dx^3 \wedge dx^1 + E^3 dx^1 \wedge dx^2), \\ \mathfrak{H}_{(2)} &= \frac{1}{\mu_0} (B_1 dx^1 + B_2 dx^2 + B_3 dx^3), \end{aligned} \quad (\text{E.4.37})$$

are also constant in space and time. The spatial indices are raised and lowered by the spatial metric ${}^{(3)}g_{ab} = \delta_{ab}$. These assumptions evidently guarantee that both the homogeneous and inhomogeneous Maxwell equations, $dF = 0$, $d\mathcal{H} = J$, are satisfied for the trivial sources $J = 0$ ($\rho = 0$ and $j = 0$) in the second region.

Let us now verify that the motion of matter generates *nontrivial* electric and magnetic fields in the first region. In order to find their configurations, it is necessary to use the constitutive law (E.4.24)–(E.4.26) and the boundary conditions at the surface S . Recalling the jump conditions on the separating surface (B.4.25), (B.4.26) and (B.4.27), (B.4.28), we find, in the absence of free charges and currents:

$$\tau_A \lrcorner \mathfrak{H}_{(1)} \Big|_S = \tau_A \lrcorner \mathfrak{H}_{(2)} \Big|_S, \quad \nu \wedge \mathfrak{D}_{(1)} \Big|_S = \nu \wedge \mathfrak{D}_{(2)} \Big|_S, \quad (\text{E.4.38})$$

$$\tau_A \lrcorner E_{(1)} \Big|_S = \tau_A \lrcorner E_{(2)} \Big|_S, \quad \nu \wedge B_{(1)} \Big|_S = \nu \wedge B_{(2)} \Big|_S. \quad (\text{E.4.39})$$

Since the matter is confined to the first region, we conclude that the 3-velocity vector on the boundary surface S has only two tangential components:

$$v^a \partial_a \Big|_S = v^A \tau_A, \quad A = 1, 2. \quad (\text{E.4.40})$$

Let us assume that the two tangential vectors are mutually orthogonal and have unit length (which is always possible to achieve by the suitable choice of the variables ξ^A parametrizing the boundary surface).

The solution of the Maxwell equations $dF = 0$ and $d\mathcal{H} = 0$ in the second region is uniquely defined by the continuity conditions (E.4.38), (E.4.39). Let us write them down explicitly. Applying $\tau_A \lrcorner$ to (E.4.25) and $\nu \wedge$ to (E.4.26),

we find:

$$\begin{aligned}\tau_A \lrcorner \mathfrak{H} &= \frac{\gamma^2}{\mu_0 \mu_g} \left[\left(\frac{1}{\mu} - \varepsilon \frac{v^2}{c^2} \right) \delta_A^B + \left(\varepsilon - \frac{1}{\mu} \right) \frac{1}{c^2} v_A v^B \right] \tau_B \lrcorner \mathfrak{B} \\ &\quad + \gamma^2 \varepsilon_0 \left(\varepsilon - \frac{1}{\mu} \right) \epsilon_{AB} v^B \mathfrak{E}(\nu \wedge \mathfrak{E}),\end{aligned}\quad (\text{E.4.41})$$

$$\begin{aligned}\nu \wedge \mathfrak{D} &= \gamma^2 \varepsilon_0 \varepsilon_g \left(\varepsilon - \frac{v^2}{\mu c^2} \right) \nu \wedge \mathfrak{E} \\ &\quad + \gamma^2 \varepsilon_0 \left(\varepsilon - \frac{1}{\mu} \right) \epsilon^{AB} v_A \mathfrak{E}(\tau_B \lrcorner \mathfrak{B}).\end{aligned}\quad (\text{E.4.42})$$

Here $\epsilon_{AB} = -\epsilon_{BA}$, with $\epsilon_{12} = 1$ (and the same for ϵ^{AB}). These equations should be taken on the boundary surface S . A simple but rather lengthy calculation yields the inverse relations:

$$\begin{aligned}\tau_A \lrcorner \mathfrak{B} &= \gamma^2 \mu_0 \mu_g \left[\left(\mu - \frac{v^2}{\varepsilon c^2} \right) \delta_A^B - \left(\mu - \frac{1}{\varepsilon} \right) \frac{1}{c^2} v_A v^B \right] \tau_B \lrcorner \mathfrak{H} \\ &\quad - \gamma^2 \mu_0 \left(\mu - \frac{1}{\varepsilon} \right) \epsilon_{AB} v^B \mathfrak{D}(\nu \wedge \mathfrak{D}),\end{aligned}\quad (\text{E.4.43})$$

$$\begin{aligned}\nu \wedge \mathfrak{E} &= \frac{\gamma^2}{\varepsilon_0 \varepsilon_g} \left(\frac{1}{\varepsilon} - \mu \frac{v^2}{c^2} \right) \nu \wedge \mathfrak{D} \\ &\quad - \gamma^2 \mu_0 \left(\mu - \frac{1}{\varepsilon} \right) \epsilon^{AB} v_A \mathfrak{D}(\tau_B \lrcorner \mathfrak{H}).\end{aligned}\quad (\text{E.4.44})$$

The three equations (E.4.43), (E.4.44) taken on S , together with the three equations (E.4.39) specify all six components of the electric and magnetic field strengths E and B on the boundary S in terms of the constant values of the field strengths (E.4.36) in the matter free region.

The standard way of finding the static electromagnetic field in region 1 is as follows. The $[(1+3)\text{-decomposed}]$ homogeneous Maxwell equations $dE_{(1)} = 0, dB_{(1)} = 0$ are solved by $E_{(1)} = d\varphi, B_{(1)} = d\mathcal{A}$. Substituting this, by using the constitutive law (E.4.25), (E.4.26), into the inhomogeneous Maxwell equations $d\mathfrak{H}_{(1)} = 0, d\mathfrak{D}_{(1)} = 0$, we obtain the four second-order differential equations for the four independent components of the electromagnetic potential $\varphi(x), \mathcal{A}(x)$. The unique solution of the resulting partial differential system is determined by the boundary conditions (E.4.39) and (E.4.43), (E.4.44). In the general case, this is a highly nontrivial problem. However, there are two special cases of physical importance for which the solution is straightforward. They describe the experiments of Röntgen and the Wilsons with moving dielectric bodies.

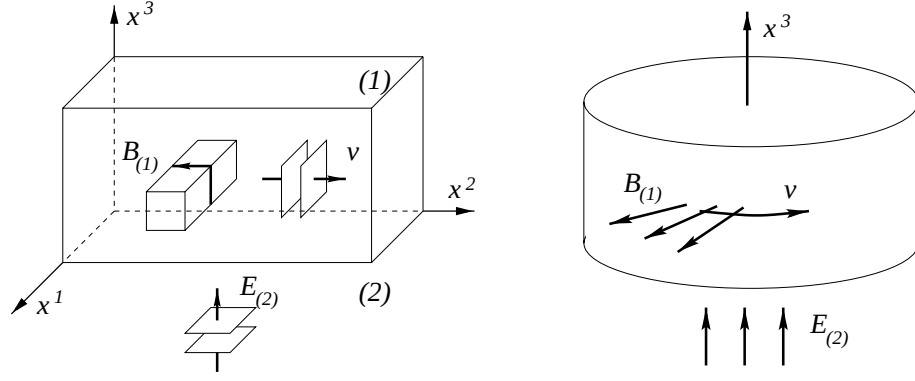


Figure E.4.2: Experiment of Röntgen.

E.4.5 The experiments of Röntgen and Wilson & Wilson

In both cases, we choose the boundary as a plane $S = \{x^3 = 0\}$, so that the tangential vectors and the normal 1-form are

$$\tau_1 = \partial_1, \quad \tau_2 = \partial_2; \quad \nu = dx^3. \quad (\text{E.4.45})$$

We assume that the upper half-space (corresponding to $x^3 > 0$) is filled with matter moving with the horizontal velocity

$$v = v_1 dx^1 + v_2 dx^2. \quad (\text{E.4.46})$$

Röntgen experiment

Let us consider the case when the magnetic field is absent in the matter-free region, whereas the *electric field* is directed towards the boundary:

$$B_{(2)} = 0, \quad E_{(2)} = E_3 dx^3. \quad (\text{E.4.47})$$

Then, from (E.4.37), we have

$$\mathfrak{H}_{(2)} = 0, \quad \mathfrak{D}_{(2)} = \varepsilon_0 E^3 dx^1 \wedge dx^2. \quad (\text{E.4.48})$$

It is straightforward to verify that, for uniform motion (with constant v), the forms

$$B_{(1)} = \frac{1}{1 - \frac{v^2}{c^2}} \left(\frac{1}{\varepsilon} - \mu \right) \frac{1}{c^2} v \wedge E_3 dx^3, \quad (\text{E.4.49})$$

$$E_{(1)} = \frac{1}{1 - \frac{v^2}{c^2}} \left(\frac{1}{\varepsilon} - \mu \frac{v^2}{c^2} \right) E_3 dx^3, \quad (\text{E.4.50})$$

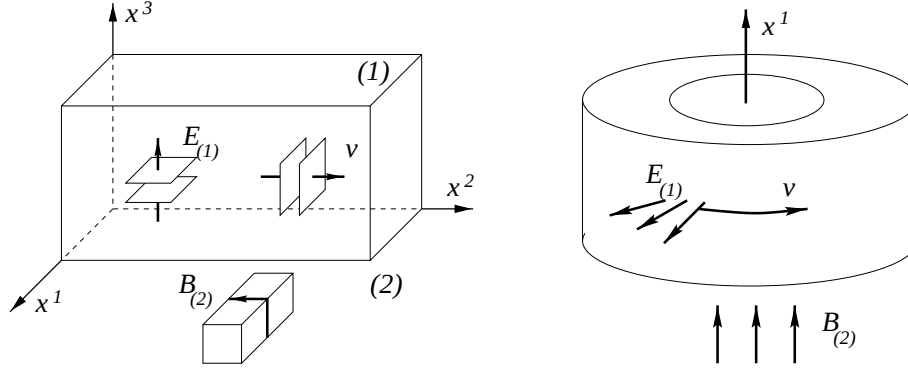


Figure E.4.3: Experiment of Wilson & Wilson.

describe a solution of the Maxwell equations satisfying the boundary conditions (E.4.39) and (E.4.43), (E.4.44). Using the constitutive law (E.4.25), (E.4.26), we find from (E.4.49) and (E.4.50) the corresponding excitation forms:

$$\mathfrak{H}_{(1)} = 0, \quad \mathfrak{D}_{(1)} = \varepsilon_0 E^3 dx^1 \wedge dx^2. \quad (\text{E.4.51})$$

This situation is described in the left part of Figure E.4.2: A magnetic field is generated along the x^1 axis by the motion of matter along the x^2 axis.

In order to simplify the derivations, we have studied here the case of the uniform translational motion of matter. However, in the actual experiment of Röntgen³ in 1888 he observed this effect for a rotating dielectric disk, as shown schematically in the right part of Figure E.4.2. One can immediately see that in the nonrelativistic approximation (neglecting terms with v^2/c^2), the formulas (E.4.49), (E.4.50) describe a solution of the Maxwell equations provided $dv = 0$. This includes, in particular, the case of the slow uniform rotation of a small disk. The magnetic field generated along the radial direction can be detected by means of a magnetic needle, for example.

Wilson and Wilson experiment

In the “dual” case, the electric field is absent in the matter-free region whereas a magnetic field is pointing along the boundary:

$$B_{(2)} = B^1 dx^2 \wedge dx^3 + B^2 dx^3 \wedge dx^1, \quad E_{(2)} = 0. \quad (\text{E.4.52})$$

Then, from (E.4.37), we find

$$\mathfrak{H}_{(2)} = \frac{1}{\mu_0} (B_1 dx^1 + B_2 dx^2), \quad \mathfrak{D}_{(2)} = 0. \quad (\text{E.4.53})$$

³See Röntgen [32] and the later thorough experimental study of Eichenwald [10].

A solution of the Maxwell equations satisfying the boundary conditions (E.4.39), (E.4.43), (E.4.44) is straightforwardly obtained for uniform motion of matter:

$$E_{(1)} = \frac{1}{1 - \frac{v^2}{c^2}} \left(\frac{1}{\varepsilon} - \mu \right) (v^1 B^2 - v^2 B^1) dx^3, \quad (\text{E.4.54})$$

$$B_{(1)} = \mu (B^1 dx^2 \wedge dx^3 + B^2 dx^3 \wedge dx^1) + \frac{1}{1 - \frac{v^2}{c^2}} \left(\frac{1}{\varepsilon} - \mu \right) \frac{1}{c^2} v \wedge (v^1 B^2 - v^2 B^1) dx^3. \quad (\text{E.4.55})$$

This situation is depicted in the left part of Figure E.4.3. There, without restricting generality, we have chosen the velocity along x^2 and the magnetic field $B_{(2)}$ along x^1 . Then the electric field generated is directed along the x^3 axis.

The electric and magnetic excitations in matter are obtained from the constitutive law (E.4.25), (E.4.26) which, for (E.4.54) and (E.4.55), yields

$$\mathfrak{H}_{(1)} = \frac{1}{\mu_0} (B_1 dx^1 + B_2 dx^2), \quad \mathfrak{D}_{(1)} = 0. \quad (\text{E.4.56})$$

Like the experiment of Röntgen, the experiment of Wilson & Wilson⁴ was actually performed for rotating matter, not for the uniform translational motion as described above. The true scheme of the experiment is given on the right side of Figure E.4.3. In fact, the rotating cylinder is formally obtained from the left figure by identifying $x^1 = z$, $x^2 = \phi$, $x^3 = \rho$ with the standard cylindrical coordinates (polar angle ϕ , radius ρ). Usually, one should be careful if one uses curvilinear coordinates in which the components of the metric are nonconstant. However, the use of exterior calculus makes all computations transparent and simple. We leave it as an exercise to the reader to verify that the Maxwell equations yield the following exact solution for the cylindrical configuration of the Wilsons experiment:

$$\mathfrak{H}_{(2)} = \frac{1}{\mu_0} B dz, \quad \mathfrak{D}_{(2)} = 0, \quad (\text{E.4.57})$$

$$B_{(2)} = B d\rho \wedge \rho d\phi, \quad E_{(2)} = 0, \quad (\text{E.4.58})$$

$$\mathfrak{H}_{(1)} = \frac{1}{\mu_0} B dz, \quad \mathfrak{D}_{(1)} = 0, \quad (\text{E.4.59})$$

$$B_{(1)} = \frac{1}{1 - \frac{(\omega\rho)^2}{c^2}} \left(\mu - \frac{(\omega\rho)^2}{\varepsilon c^2} \right) B d\rho \wedge \rho d\phi, \quad (\text{E.4.60})$$

$$E_{(1)} = \frac{1}{1 - \frac{(\omega\rho)^2}{c^2}} \left(\frac{1}{\varepsilon} - \mu \right) \omega\rho B d\rho. \quad (\text{E.4.61})$$

The boundary conditions (E.4.39), (E.4.43), (E.4.44) are satisfied for (E.4.57)–(E.4.61). Note that now $\nu = d\rho$, $\tau_1 = \partial_z$, $\tau_2 = \partial_\phi$. The velocity 1-form reads

$$v = \omega\rho^2 d\phi, \quad (\text{E.4.62})$$

⁴See Wilson & Wilson [37].

with constant angular velocity ω . The radial electric field (E.4.61) that is generated in the rotating cylinder can be detected by measuring the voltage between the inner and the outer surfaces of the cylinder.

One may wonder what physical source is behind the electric and magnetic fields that are generated in moving matter. After all, we have assumed that there are no free charges and currents inside region 1. However, we have bound charges and currents therein described by the polarization and magnetization (E.4.30) and (E.4.31). Substituting (E.4.60), (E.4.61) into (E.4.30), (E.4.31), we find:

$$P = \frac{\varepsilon_0}{1 - \frac{(\omega\rho)^2}{c^2}} \left(\mu - \frac{1}{\varepsilon} \right) \omega \rho^2 B d\phi \wedge dz, \quad (\text{E.4.63})$$

$$M = \frac{\varepsilon_0}{1 - \frac{(\omega\rho)^2}{c^2}} \left[\mu - 1 + \frac{(\omega\rho)^2}{c^2} \left(1 - \frac{1}{\varepsilon} \right) \right] c^2 B dz. \quad (\text{E.4.64})$$

From the definitions of these quantities in (E.3.5), we obtain, merely by taking the exterior differential, the charge and current densities:

$$\rho^{\text{mat}} = \frac{2\varepsilon_0}{\left(1 - \frac{(\omega\rho)^2}{c^2}\right)^2} \left(\frac{1}{\varepsilon} - \mu \right) \omega B d\rho \wedge \rho d\phi \wedge dz, \quad (\text{E.4.65})$$

$$j^{\text{mat}} = \frac{2\varepsilon_0}{\left(1 - \frac{(\omega\rho)^2}{c^2}\right)^2} \left(\frac{1}{\varepsilon} - \mu \right) \omega^2 \rho B dz \wedge d\rho. \quad (\text{E.4.66})$$

It is these charge and current densities that generate the nontrivial electric and magnetic fields in the rotating cylinder of the experiment of the Wilsons. The bound current and charge density (E.4.65), (E.4.66) satisfy the relation

$$j^{\text{mat}} = (*\rho^{\text{mat}})^*v. \quad (\text{E.4.67})$$

E.4.6 $\otimes$ Noninertial “rotating coordinates”

How is the Maxwell–Lorentz electrodynamics seen by a noninertial observer? We need a procedure of two steps for the installation of such an observer. In this section the first step is done by introducing suitable noninertial coordinates.

We assume the absence of the gravitational field. Then spacetime is Minkowskian and a global Cartesian coordinate system t, x^a (with $a = 1, 2, 3$) can be introduced that spans the inertial (reference) frame. The line element of spacetime reads

$$ds^2 = c^2 dt^2 - \delta_{ab} dx^a dx^b. \quad (\text{E.4.68})$$

As usual, the electromagnetic excitation and the field strength are given by

$$\mathcal{H} = -\mathfrak{H} \wedge dt + \mathfrak{D}, \quad F = E \wedge dt + B. \quad (\text{E.4.69})$$

Assuming matter to be at rest in the inertial frame, we have the constitutive law

$$\mathfrak{H} = \frac{1}{\mu\mu_0} {}^*B, \quad \mathfrak{D} = \varepsilon\varepsilon_0 {}^*E. \quad (\text{E.4.70})$$

Equivalently, we have $\lambda_0 = \sqrt{\frac{\varepsilon\varepsilon_0}{\mu\mu_0}}$ and the constitutive matrices

$$A^{ab} = -\frac{n}{c} \delta^{ab}, \quad B_{ab} = \frac{c}{n} \delta_{ab}, \quad C^a{}_b = 0. \quad (\text{E.4.71})$$

The corresponding optical metric is given by (E.4.35).

Now we want to introduce noninertial “rotating coordinates” (t', x'^a) by

$$t = t', \quad x^a = L_b{}^a x'^b, \quad (\text{E.4.72})$$

with the 3×3 matrix

$$L_b{}^a = n^a n_b + (\delta_b^a - n^a n_b) \cos \varphi + \hat{\epsilon}^a{}_{cb} n^c \sin \varphi. \quad (\text{E.4.73})$$

The matrix defines a rotation of an angle $\varphi = \varphi(t)$ around the direction specified by the constant unit vector $\vec{n} = n^a$, with $\delta_{ab} n^a n^b = 1$. The Latin (spatial) indices are raised and lowered by means of the Euclidean metric δ_{ab} and δ^{ab} ($\hat{\epsilon}^a{}_{cb} = \delta^{ad} \hat{\epsilon}_{dc b}$, for example). We put “rotating coordinates” in quotes since it is strictly speaking the natural frame (dt', dx'^a) attached to the coordinates (t', x'^a) that is rotating with respect to the Cartesian frame.

The electromagnetic 2-forms H and F are independent of coordinates. However, their *components* are different in different coordinate systems. In exterior calculus it is easy to find the components of forms: one only needs to substitute the original natural coframe (dt, dx^a) by the transformed one. A straightforward calculation, using (E.4.73), yields

$$\begin{pmatrix} dt \\ dx^a \end{pmatrix} = \left(\frac{1}{L_c{}^a [\vec{\omega} \times \vec{x}']^c} \middle| \frac{0}{L_b{}^a} \right) \begin{pmatrix} dt' \\ dx'^b \end{pmatrix}. \quad (\text{E.4.74})$$

Here the angular velocity 3-vector is defined by

$$\vec{\omega} := \omega \vec{n}, \quad \omega := \frac{d\varphi}{dt}. \quad (\text{E.4.75})$$

Substituting these differentials into (E.4.68), we find the line element in rotating coordinates:

$$\begin{aligned} ds^2 &= c^2 (dt')^2 \left[1 + (\vec{\omega} \cdot \vec{x}'/c)^2 - (\vec{\omega} \cdot \vec{\omega}/c^2)(\vec{x}' \cdot \vec{x}') \right] \\ &\quad - 2dt' d\vec{x}' \cdot [\vec{\omega} \times \vec{x}'] - d\vec{x}' \cdot d\vec{x}'. \end{aligned} \quad (\text{E.4.76})$$

The electromagnetic excitation and the field strength are, as usual, decomposed with respect to the rotating frame:

$$\mathcal{H} = -\mathfrak{H}' \wedge dt + \mathfrak{D}', \quad F = E' \wedge dt + B'. \quad (\text{E.4.77})$$

Note that $dt = dt'$. The constitutive matrices are derived from (E.4.71) by means of the transformation (D.5.28)–(D.5.30). Given (E.4.74), we find the matrices (A.1.96), (A.1.97) as

$$P^a{}_b = L^a{}_b, \quad Q_b{}^a = (L^{-1})^a{}_b, \quad W^{ab} = 0, \quad Z_{ab} = (L^{-1})^c{}_a \hat{e}_{bcd} v^d. \quad (\text{E.4.78})$$

Hereafter we use the abbreviation

$$\vec{v} := [\vec{\omega} \times \vec{x}']. \quad (\text{E.4.79})$$

We substitute (E.4.78) into (D.5.28)–(D.5.30) and find the constitutive matrices in the rotating natural frame as

$$A'^{ab} = \frac{1}{c} [-n \delta^{ab}], \quad (\text{E.4.80})$$

$$B'_{ab} = c \left[\frac{1}{n} \delta_{ab} + n \left(-\delta_{ab} \frac{v^2}{c^2} + \frac{1}{c^2} v_a v_b \right) \right], \quad (\text{E.4.81})$$

$$C'^a{}_b = n \delta^{ac} \hat{e}_{cbd} \frac{v^d}{c}, \quad (\text{E.4.82})$$

with $n = \sqrt{\mu\epsilon}$. These matrices satisfy the algebraic closure relation (D.4.13), (D.4.14). Thus they define an induced spacetime metric. The latter is obtained from (D.5.25) by using (E.4.35) and (E.4.74):

$$g'^{\text{opt}}_{ij} = \sqrt{\frac{n}{c}} \left(\begin{array}{c|c} \frac{c^2}{n^2} - v^2 & -v_b \\ \hline -v_a & -\delta_{ab} \end{array} \right). \quad (\text{E.4.83})$$

E.4.7 Rotating observer

However, the behavior of fields with respect to a rotating frame is usually of minor physical interest to us. The observer rather measures all physical quantities with respect to a local frame ϑ^α that is anholonomic in general, i.e., $d\vartheta^\alpha \neq 0$. The observer is, in fact, *comoving* with that frame and the components of excitation and field strength should be determined with respect to ϑ^α . Consequently, the observer's 4-velocity vector reads

$$e_{\hat{0}} = e^i_{\hat{0}} \partial_i = \gamma \partial_{t'}. \quad (\text{E.4.84})$$

The “Lorentz” factor

$$\gamma = \frac{1}{\sqrt{1 - \vec{v}^2/c^2}}, \quad (\text{E.4.85})$$

is determined for the metric (E.4.76) by the normalization condition $\mathbf{g}(e_{\hat{0}}, e_{\hat{0}}) = e^i_{\hat{0}} e^j_{\hat{0}} g'_{ij} = c^2$. Note that $\vec{v}^2 = \delta_{ab} v^a v^b$.

The observer's rotating frame e_α , with (E.4.84) and

$$e_{\hat{a}} = \partial_{x'^a} + \frac{\gamma^2}{c^2} v_a \partial_{t'}, \quad (\text{E.4.86})$$

is dual to the corresponding coframe ϑ^α with

$$\vartheta^{\hat{0}} = \frac{1}{\gamma} dt' - \frac{\gamma}{c^2} \vec{v} \cdot d\vec{x}', \quad \vartheta^{\hat{a}} = dx'^a. \quad (\text{E.4.87})$$

Expressed in terms of this coframe, the metric (E.4.76) reads

$$ds^2 = c^2 (\vartheta^{\hat{0}})^2 - \left(\delta_{ab} + \frac{\gamma^2}{c^2} v_a v_b \right) \vartheta^{\hat{a}} \otimes \vartheta^{\hat{b}}. \quad (\text{E.4.88})$$

Combining (E.4.74) with (E.4.87), we obtain the transformation from the inertial (dt, dx^a) coframe to the noninertial $(\vartheta^{\hat{0}}, \vartheta^{\hat{a}})$ frame as follows:

$$\begin{pmatrix} dt \\ dx^a \end{pmatrix} = \left(\frac{\gamma}{\gamma L_c^a v^c} \middle| \frac{\frac{\gamma^2}{c^2} v_b}{L_c^a \left(\delta_b^c + \frac{\gamma^2}{c^2} v^c v_b \right)} \right) \begin{pmatrix} \vartheta^{\hat{0}} \\ \vartheta^{\hat{b}} \end{pmatrix}. \quad (\text{E.4.89})$$

With respect to ϑ^α , the electromagnetic excitation and field strength read

$$\mathcal{H} = -\mathfrak{H}' \wedge \vartheta^{\hat{0}} + \mathfrak{D}', \quad F = E' \wedge \vartheta^{\hat{0}} + B'. \quad (\text{E.4.90})$$

Using (E.4.89) in the transformation formulas (D.5.28)–(D.5.30), the constitutive law in the frame of a rotating observer turns out to be defined by the constitutive matrices

$$A'^{ab} = \frac{\gamma}{c} \left[n \left(-\delta^{ab} + \frac{1}{c^2} v^a v^b \right) \right], \quad (\text{E.4.91})$$

$$B'_{ab} = \frac{c}{\gamma} \left[\frac{1}{n} \left(\delta_{ab} + \frac{\gamma^2}{c^2} v_a v_b \right) + n \left(-\delta_{ab} \frac{v^2}{c^2} + \frac{1}{c^2} v_a v_b \right) \right], \quad (\text{E.4.92})$$

$$C'^a{}_b = n \delta^{ac} \hat{\epsilon}_{cbd} \frac{v^d}{c}. \quad (\text{E.4.93})$$

The matrices (E.4.91)–(E.4.93) satisfy the algebraic closure relation (D.4.13), (D.4.14). The corresponding optical metric is obtained from (D.5.25), (E.4.35), and (E.4.89) as

$$g_{ij}^{\text{opt}} = \sqrt{\frac{n}{c\gamma}} \left(\frac{\frac{c^2}{n^2} - v^2}{-\gamma v_a} \middle| \frac{-\gamma v_b}{-\left(\delta_{ab} + \frac{\gamma^2}{c^2} v_a v_b \right)} \right). \quad (\text{E.4.94})$$

In exterior calculus, the constitutive law (E.4.91)–(E.4.93) in the rotating frame reads

$$\mathfrak{H}' = \frac{1}{\mu_0} \left(\frac{1}{\mu} - \varepsilon \frac{v^2}{c^2} \right) {}^{*'}B' + \varepsilon \varepsilon_0 \left[v' {}^{*'}(v' \wedge B') - {}^{*'}(v' \wedge E') \right], \quad (\text{E.4.95})$$

$$\mathfrak{D}' = \varepsilon \varepsilon_0 \left({}^{*'}E' - v' \wedge {}^{*'}B' \right). \quad (\text{E.4.96})$$

Here ${}^{*'}$ denotes the Hodge operator with respect to the corresponding 3-space metric $\delta_{ab} + \frac{\gamma^2}{c^2} v_a v_b$ (see (E.4.88)), and we introduced the velocity 1-form

$$v' := \gamma \delta_{ab} v^a dx'^b. \quad (\text{E.4.97})$$

E.4.8 Accelerating observer

Let us now analyze the case of pure acceleration. It is quite similar to the pure rotation that was considered in Sec. E.4.6. More concretely, we study the motion in a fixed spatial direction with an acceleration 3-vector parametrized as

$$\vec{a} = a \vec{n}, \quad \text{or} \quad a^b = a n^b. \quad (\text{E.4.98})$$

Here $a^2 := \vec{a} \cdot \vec{a}$ is the magnitude of acceleration, and the unit vector $\vec{n}$, with $\vec{n} \cdot \vec{n} = n^b n_b = 1$, gives its direction in space. Recall that we are in the Minkowski spacetime (E.4.68). The accelerating coordinates (t', x'^a) can be obtained from the Cartesian ones (t, x^a) by means of the transformation

$$t = \frac{1}{c} \sinh \phi n_a x'^a + \int_{t'}^{t'} d\tau \cosh \phi(\tau), \quad (\text{E.4.99})$$

$$x^a = \mathcal{K}_b{}^a x'^b + c n^a \int_{t'}^{t'} d\tau \sinh \phi(\tau). \quad (\text{E.4.100})$$

Here we introduce the 3×3 matrix

$$\mathcal{K}_b{}^a = (\delta_b^a - n^a n_b) + n^a n_b \cosh \phi. \quad (\text{E.4.101})$$

The scalar function $\phi(t')$ determines the magnitude of the acceleration by

$$a(t') = c \frac{d\phi}{dt'}. \quad (\text{E.4.102})$$

Differentiating (E.4.99), (E.4.100), we obtain the transformation from the inertial coframe to the accelerating one:

$$\begin{pmatrix} dt \\ dx^a \end{pmatrix} = \left(\begin{array}{c|c} \left(1 + \frac{\vec{a} \cdot \vec{x}'}{c^2}\right) \cosh \phi & \frac{n_b}{c} \sinh \phi \\ \hline \left(1 + \frac{\vec{a} \cdot \vec{x}'}{c^2}\right) c n^a \sinh \phi & \mathcal{K}_b{}^a \end{array} \right) \begin{pmatrix} dt' \\ dx'^b \end{pmatrix}. \quad (\text{E.4.103})$$

Substituting (E.4.103) into (E.4.68), we find the metric in accelerating coordinates:

$$ds^2 = c^2 \left(1 + \vec{a} \cdot \vec{x}' / c^2\right)^2 (dt')^2 - d\vec{x}' \cdot d\vec{x}'. \quad (\text{E.4.104})$$

This is one of the possible forms of the well-known Rindler spacetime.

It is straightforward to construct the local frame of a noninertial observer that is comoving with the accelerating coordinate system. With respect to the original Cartesian coordinates, it reads:

$$e_{\hat{0}} = \frac{1}{c} u, \quad e_{\hat{a}} = \frac{n_a}{c} \sinh \phi \partial_t + \mathcal{K}_a{}^b \partial_{x^b}. \quad (\text{E.4.105})$$

Here

$$u = \cosh \phi \partial_t + c n^a \sinh \phi \partial_{x^a} \quad (\text{E.4.106})$$

is the observer's velocity 4-vector field, which satisfies $\mathbf{g}(u, u) = c^2$. Clearly, the vectors of the basis (E.4.105), in the sense of the Minkowski 4-metric (E.4.68), are mutually orthogonal and normalized:

$$\mathbf{g}(e_{\hat{0}}, e_{\hat{0}}) = 1, \quad \mathbf{g}(e_{\hat{0}}, e_{\hat{a}}) = 0, \quad \mathbf{g}(e_{\hat{a}}, e_{\hat{b}}) = -\delta_{ab}. \quad (\text{E.4.107})$$

Because of (E.4.103), the coordinate bases are related by

$$\begin{pmatrix} \partial_t \\ \partial_{x^a} \end{pmatrix} = \begin{pmatrix} \cosh \phi / \left(1 + \frac{\vec{a} \cdot \vec{x}'}{c^2}\right) & -c n^b \sinh \phi \\ -\frac{n_a}{c} \sinh \phi / \left(1 + \frac{\vec{a} \cdot \vec{x}'}{c^2}\right) & \mathcal{K}_a{}^b \end{pmatrix} \begin{pmatrix} \partial_{t'} \\ \partial_{x'^b} \end{pmatrix}. \quad (\text{E.4.108})$$

Thus, the accelerated frame (E.4.105), with respect to the *accelerating* coordinate system, is described by the simple expressions

$$e_{\hat{0}} = \frac{1}{c(1 + \vec{a} \cdot \vec{x}/c^2)} \partial_{t'}, \quad e_{\hat{a}} = \partial_{x'^a}. \quad (\text{E.4.109})$$

According to the definition of the covariant differentiation, see (C.1.15), one has $\nabla_u e_\alpha = \Gamma_\alpha{}^\beta(u) e_\beta$. This enables us to compute the proper time derivatives for the frame (E.4.105), (E.4.109):

$$\dot{u} := \nabla_u u = \frac{a^b}{1 + \vec{a} \cdot \vec{x}/c^2} e_{\hat{b}}, \quad (\text{E.4.110})$$

$$\dot{e}_{\hat{b}} := \nabla_u e_{\hat{b}} = \frac{a_b/c^2}{1 + \vec{a} \cdot \vec{x}/c^2} u. \quad (\text{E.4.111})$$

This means that the frame (E.4.105) is Fermi–Walker transported along the observer's world line.

The coframe dual to the frame (E.4.105), (E.4.109) reads, with respect to the accelerating coordinate system,

$$\vartheta^{\hat{0}} = (1 + \vec{a} \cdot \vec{x}/c^2) c dt', \quad \vartheta^{\hat{a}} = dx'^a. \quad (\text{E.4.112})$$

E.4.9 The proper reference frame of the noninertial observer (“noninertial frame”)

The line elements (E.4.76) and (E.4.104) of spacetime represent the Minkowski space in rotating and accelerating coordinate systems, respectively. Both are particular cases of the line element

$$\begin{aligned} ds^2 = & c^2 (dt')^2 \left[\left(1 + \vec{a} \cdot \vec{x}'/c^2\right)^2 + (\vec{\omega} \cdot \vec{x}'/c)^2 - (\vec{\omega} \cdot \vec{\omega}/c^2)(\vec{x}' \cdot \vec{x}') \right] \\ & - 2 dt' d\vec{x}' \cdot [\vec{\omega} \times \vec{x}'] - d\vec{x}' \cdot d\vec{x}'. \end{aligned} \quad (\text{E.4.113})$$

Here the 3-vectors of acceleration $\vec{a}$ ($= a^b$) and of angular velocity $\vec{\omega}$ ($= \omega^c$) can be arbitrary functions of time t' . The line element (E.4.113) reduces to the diagonal form

$$ds^2 = \vartheta^{\hat{0}} \otimes \vartheta^{\hat{0}} - \vartheta^{\hat{1}} \otimes \vartheta^{\hat{1}} - \vartheta^{\hat{2}} \otimes \vartheta^{\hat{2}} - \vartheta^{\hat{3}} \otimes \vartheta^{\hat{3}} \quad (\text{E.4.114})$$

in terms of the noninertial orthonormal coframe:⁵

$$\vartheta^{\hat{0}} = (1 + \vec{a} \cdot \vec{x}/c^2) c dt', \quad (\text{E.4.115})$$

$$\vartheta^{\hat{a}} = dx'^a + [\vec{\omega} \times \vec{x}']^a dt'. \quad (\text{E.4.116})$$

The corresponding basis vectors of the (dual) frame are

$$e_{\hat{0}} = \frac{1}{c(1 + \vec{a} \cdot \vec{x}'/c^2)} (\partial_{t'} - [\vec{\omega} \times \vec{x}']^a \partial_{x'^a}), \quad (\text{E.4.117})$$

$$e_{\hat{a}} = \partial_{x'^a}. \quad (\text{E.4.118})$$

With respect to the local frame chosen, the components Γ_{α}^{β} of the Levi-Civita connection read:

$$\Gamma_{\hat{0}}^{\hat{b}} = \Gamma_{\hat{b}}^{\hat{0}} = \frac{(a^b/c^2)}{1 + \vec{a} \cdot \vec{x}'/c^2} \vartheta^{\hat{0}} = (a^b/c) dt', \quad (\text{E.4.119})$$

$$\Gamma_{\hat{a}}^{\hat{b}} = \frac{\hat{\epsilon}_{abc} (\omega^c/c)}{1 + \vec{a} \cdot \vec{x}'/c^2} \vartheta^{\hat{0}} = \hat{\epsilon}_{abc} \omega^c dt'. \quad (\text{E.4.120})$$

We can readily check that $d\Gamma_{\alpha}^{\beta} = 0$ and that the exterior products of the connection 1-forms are zero. As a result, the Riemannian curvature 2-form of the metric (E.4.113) vanishes, $\tilde{R}_{\alpha}^{\beta} = 0$. Thus, indeed, we are in a flat spacetime as seen by a noninertial observer moving with acceleration $\vec{a}$ and angular velocity $\vec{\omega}$.

After these geometrical preliminaries, we can address the problem of how a noninertial observer (accelerating and/or rotating) sees the electrodynamical effects in his proper reference frame (E.4.117), (E.4.118). In order to apply the results of the previous sections, let us specialize either to the case of pure rotation or of pure acceleration.

To begin with, we note that the constitutive relation has its usual form (E.4.70), (E.4.71) in the inertial Cartesian coordinate system (E.4.68). Putting $\vec{a} = 0$, we find from (E.4.115), (E.4.116) the proper coframe of a rotating observer:

$$\vartheta^{\hat{0}} = dt', \quad \vartheta^{\hat{a}} = dx'^a + [\vec{\omega} \times \vec{x}']^a dt'. \quad (\text{E.4.121})$$

Combining this with (E.4.74), we find the transformation of the inertial coframe to the noninertial (rotating) one,

$$\begin{pmatrix} dt \\ dx^a \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & L_b^a \end{pmatrix} \begin{pmatrix} \vartheta^{\hat{0}} \\ \vartheta^{\hat{b}} \end{pmatrix}. \quad (\text{E.4.122})$$

⁵See Hehl & Ni [13] who discussed the Dirac equation in this accelerating and rotating coframe.

Correspondingly, substituting this into the transformation (D.5.28)–(D.5.30), we immediately find from (E.4.70) the constitutive law in the rotating observer's frame:

$$\mathfrak{H}'_{\hat{a}} = \frac{1}{\mu\mu_0} \delta_{ab} B^{\hat{b}}, \quad \mathfrak{D}'^{\hat{a}} = \varepsilon\varepsilon_0 \delta^{ab} E'_{\hat{b}}. \quad (\text{E.4.123})$$

The same result holds true for an accelerating observer. If we put $\vec{\omega} = 0$ in (E.4.115), (E.4.116), we arrive at the coframe (E.4.112). Combined with (E.4.103), this yields the transformation of the inertial coframe to the noninertial (accelerating) one:

$$\begin{pmatrix} dt \\ dx^a \end{pmatrix} = \begin{pmatrix} \cosh \phi & \frac{n_b}{c} \sinh \phi \\ c n^a \sinh \phi & \mathcal{K}_b^a \end{pmatrix} \begin{pmatrix} \vartheta^{\hat{0}} \\ \vartheta^{\hat{b}} \end{pmatrix}. \quad (\text{E.4.124})$$

When we use this in (D.5.28)–(D.5.30), the final constitutive law again turns out to be (E.4.123).

Summing up, despite the fact that the proper coframe (E.4.115), (E.4.116) is *noninertial*, the constitutive relation remains in this coframe formally the same as in the inertial coordinate system.⁶

E.4.10 Universality of the Maxwell–Lorentz spacetime relation

The use of foliations and of exterior calculus for the description of the reference frames enables us to establish the universality of the Maxwell–Lorentz spacetime relation.

Let us put $\mu = \varepsilon = 1$ (hence $n = 1$) into the formulas above. Physically, this corresponds to a transformation from one frame (σ -foliation) to another frame (τ -foliation) that moves with an arbitrary velocity u relative to the first one. Then the relation (E.4.13) reduces to

$$\underline{\mathcal{H}} = \lambda_0 \, ({}^*F), \quad {}^{\perp}\mathcal{H} = \lambda_0 \, {}^{\perp}({}^*F). \quad (\text{E.4.125})$$

On the other hand, from (E.4.21)–(E.4.23), we find for the constitutive matrices:

$$A^{ab} = -\frac{\sqrt{{}^{(3)}g}}{N} {}^{(3)}g^{ab}, \quad B_{ab} = \frac{N}{\sqrt{{}^{(3)}g}} {}^{(3)}g_{ab}, \quad C^a{}_b = 0. \quad (\text{E.4.126})$$

Equivalently, from (E.4.24), we read off that

$$\mathfrak{H}_a = \sqrt{\frac{\varepsilon_0}{\mu_0}} N B_a / \sqrt{g}, \quad \mathfrak{D}^a / \sqrt{g} = \sqrt{\frac{\varepsilon_0}{\mu_0}} \frac{1}{N} E^a, \quad (\text{E.4.127})$$

⁶This shows that it is misleading to associate the “Cartesian form” of a constitutive relation with inertial frames of reference. Kovetz [20], for example, following Truesdell & Toupin [34], takes (E.4.123) as a sort of definition of inertial frames.

or returning to exterior forms,

$$\mathfrak{D} = \varepsilon_0 \varepsilon_g \, {}^*E, \quad \mathfrak{H} = \frac{1}{\mu_0 \mu_g} \, {}^*B. \quad (\text{E.4.128})$$

This is nothing but a $(1 + 3)$ -decomposition with respect to the laboratory foliation:

$$\underline{\underline{H}} = \lambda_0 \, \underline{({}^*F)}, \quad {}^\perp H = \lambda_0 \, {}^\perp ({}^*F). \quad (\text{E.4.129})$$

Comparing (E.4.125) with (E.4.129), we arrive at the conclusion that (E.4.125) and (E.4.129) are just different “projections” of the generally valid Maxwell–Lorentz spacetime relation $\underline{\underline{H}} = \lambda_0 \, {}^*F$. Since here ${}^{\text{mat}}H = 0$, we have

$$H = \lambda_0 \, {}^*F. \quad (\text{E.4.130})$$

In this form, the Maxwell–Lorentz spacetime relation is valid *always and everywhere*. Neither the choice of coordinates nor the choice of a specific reference frame (foliation) play any role. Consequently, our fifth axiom has a universal physical meaning.



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⊗ Outlook

Electrodynamics describes only one out of *four* interactions in nature. And classical electrodynamics covers only the nonquantum aspects of the electromagnetic field. Therefore electrodynamics is related to the other fields of knowledge in physics in a multitude of different ways.

In this tour d'horizon, we look out in different directions for neighboring disciplines of classical electrodynamics and indicate how they are related to classical electrodynamics. We first explore the *classical* domain, namely the relation of electrodynamics to gravity and topology. Subsequently, we turn to *quasi-classical* sources of the electromagnetic field in the context of superconductivity (Ginzburg–Landau theory) and first quantized Dirac theory. This allows us to construct the electric current 3-form in terms of a complex-valued classical *scalar* and *spinor* field, respectively. The Maxwell equations remain untouched otherwise.

The *quantum* domain represents new territory, which we only name: the Quantum Hall Effect (Jain's composite fermion picture fits nicely into our approach to electrodynamics), the quantum version of electrodynamics, that is, quantum electrodynamics (QED), and the unified electroweak Glashow–Salam–Weinberg model. The latter unifies the electromagnetic with the weak interaction.

How does gravity affect electrodynamics?

The only other known interaction, besides the electromagnetic one, that can be described by means of the classical field concept, is the gravitational interaction. Einstein's theory of gravity, general relativity (GR),⁷ describes the gravitational field successfully in the macrophysical domain. In GR, spacetime is a 4-dimensional Riemannian manifold with a metric g of Lorentzian signature. The metric is the gravitational potential. The curvature 2-form $\tilde{R}_\alpha{}^\beta$, subsuming up to second derivatives of the metric, represents the tidal forces of the gravitational field. Einstein's field equation, with respect to an arbitrary coframe ϑ^α , reads⁸

$$\frac{1}{2} \eta_{\alpha\beta\gamma} \wedge \tilde{R}^{\beta\gamma} + \Lambda \eta_\alpha = \frac{8\pi G}{c^3} \overset{\text{Mat}}{\sigma}_\alpha . \quad (\text{O.1})$$

The tilde labels Riemannian objects, G is Newton's gravitational constant, and Λ the cosmological constant, which we neglect in our future considerations. The source on the right-hand side is a *symmetric* energy-momentum current of "matter,"

$$\vartheta_{[\alpha} \wedge \overset{\text{Mat}}{\sigma}_{\beta]} = 0 , \quad (\text{O.2})$$

embodying all nongravitational contributions to energy.

Electrodynamics fits smoothly into this picture. The Maxwell equations remain absolutely the same,

$$dH = J , \quad dF = 0 . \quad (\text{O.3})$$

After all, the first four axioms of electrodynamics don't depend on the metric. Consequently, gravity leaves them unaffected. However, the Hodge star $*$ in the spacetime relation, the fifth axiom,

$$H = \lambda_0 *F , \quad (\text{O.4})$$

"feels" the dynamical metric g fulfilling the Einstein equation. Hence it is via the Maxwell–Lorentz spacetime relation that gravity makes itself felt.

The electromagnetic field, in the framework of GR, belongs to the matter side,

$$\overset{\text{Mat}}{\sigma}_\alpha = \overset{\text{mat}}{\sigma}_\alpha + \overset{\text{Max}}{\Sigma}_\alpha . \quad (\text{O.5})$$

In other words, the electromagnetic field enters the gravity scene explicitly via its energy-momentum current $\overset{\text{Max}}{\Sigma}_\alpha$. Since $\vartheta_{[\alpha} \wedge \overset{\text{Max}}{\Sigma}_{\beta]} = 0$, this is also possible in a smooth way.

These are the basics of the gravito-electromagnetic complex. Let us illustrate it by an example.

⁷Compare with Landau & Lifshitz [21], for example. Einstein's Princeton lectures [7] still give a good idea of the underlying principles and some of the main results of GR. Frankel [8] has written a little book on GR underlining its geometrical character and, in particular, developing it in terms of exterior calculus.

⁸Recall $\eta_{\alpha\beta\gamma} = *(\vartheta_\alpha \wedge \vartheta_\beta \wedge \vartheta_\gamma)$, $\eta_\alpha = *\vartheta_\alpha$.

Reissner–Nordström solution

We consider a point source of mass m and charge Q and study its gravitational and electromagnetic fields in the electro-vacuum, that is, outside the mass m and the charge Q . The Einstein equation in this case reads

$$\eta_{\alpha\beta\gamma} \wedge \tilde{R}^{\beta\gamma} = \frac{8\pi G}{c^3} \lambda_0 [F \wedge (e_\alpha \lrcorner {}^*F) - {}^*F \wedge (e_\alpha \lrcorner F)] . \quad (\text{O.6})$$

For the electromagnetic equations one should compare (O.3) and (O.4).

To solve such a problem, we take a *spherically symmetric coframe* and express it in polar coordinates r, θ, ϕ :

$$\vartheta^0 = f c dt, \quad \vartheta^1 = \frac{1}{f} dr, \quad \vartheta^2 = r d\theta, \quad \vartheta^3 = r \sin \theta d\phi. \quad (\text{O.7})$$

It contains the 0-form $f = f(r)$ and is assumed to be orthonormal; i.e., the metric reads

$$ds^2 = o_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = f^2 c^2 dt^2 - \frac{1}{f^2} dr^2 - r^2 (d\theta^2 + \sin^2 \theta d\phi^2) . \quad (\text{O.8})$$

Let us first turn to the electromagnetic case *without gravity* (pure electric case). Then we have a Minkowski spacetime with $f = 1$. The spherically symmetric electromagnetic field is then described by the Coulomb ansatz:

$$A = -\frac{qc}{r} dt, \quad (\text{O.9})$$

$$F = dA = \frac{qc}{r^2} dr \wedge dt = \frac{q}{r^2} \vartheta^1 \wedge \vartheta^0. \quad (\text{O.10})$$

Here q is a constant. The homogeneous Maxwell equation $dF = 0$ as well as the inhomogeneous equation for the vacuum case $d{}^*F = 0$ are both fulfilled. The energy-momentum current can be determined by substituting (O.10) into (E.1.22):

$$\overset{\text{Max}}{\Sigma}_\alpha = \lambda_0 \frac{q^2}{2r^4} \left(\delta_\alpha^0 \eta_0 + \delta_\alpha^1 \eta_1 - \delta_\alpha^2 \eta_2 - \delta_\alpha^3 \eta_3 \right). \quad (\text{O.11})$$

Clearly, since $f = 1$, Einstein's equation (O.1) is *not* fulfilled: The geometric left-hand side vanishes and does not counterbalance the nontrivial right-hand side (O.11).

The integration constant q is related to the total charge Q of the source which, by means of (B.1.1), (B.1.45), and the Stokes theorem, turns out to be

$$Q = \int_{\Omega_3} \rho = \int_{\partial\Omega_3} \mathcal{D}, \quad (\text{O.12})$$

where the 3-dimensional domain Ω_3 encloses the source. From (O.10) and (O.4) we find the excitation

$$H = \mathcal{D} = \lambda_0 \frac{q}{r^2} \vartheta^2 \wedge \vartheta^3 = \lambda_0 q \sin \theta d\theta \wedge d\phi. \quad (\text{O.13})$$

If integrated according to (O.12), we have

$$Q = 4\pi\lambda_0 q. \quad (\text{O.14})$$

We recall (D.6.11) and (E.1.2), that is, $\lambda_0 = \sqrt{\varepsilon_0/\mu_0}$ and $c = 1/\sqrt{\varepsilon_0\mu_0}$. Then we find the standard SI form of the Coulomb potential (O.9):

$$A = -\frac{Q}{4\pi\varepsilon_0 r} dt. \quad (\text{O.15})$$

Let us now turn to the gravitational case for an electrically *uncharged* sphere (pure gravity case). Then, as is known from GR, we have the Schwarzschild solution with

$$f^2 = 1 - \frac{2Gm}{c^2 r}. \quad (\text{O.16})$$

Here m is the mass of the source. It is an easy exercise in computer algebra to derive this solution by substituting the spherically symmetric frame (O.7) into the vacuum Einstein equation $\frac{1}{2}\eta_{\alpha\beta\gamma}\wedge\tilde{R}^{\beta\gamma}=0$ and by subsequently integration the emerging second-order ordinary differential equation (ODE).

GR is a nonlinear field theory. Nevertheless, if we now treat the *combined case* with electromagnetic *and* gravitational fields, we can sort of superimpose the single solutions because of our coordinate and frame invariant presentation of electrodynamics. We now have $f \neq 1$, but we still keep the ansatz for the Coulomb field (O.9). The form of the field strength (O.10) remains the same in terms of the orthonormal coframe (O.7). Also the energy-momentum current (O.11) does not change. Hence we can write down the Einstein field equation (O.6) with an explicitly known right-hand side. For the unknown function f^2 we can make the ansatz $f^2 = 1 - 2Gm/c^2 r + U(r)$. For $U = 0$, we recover the Schwarzschild case. If we substitute this into the left-hand side of (O.6), then we find (also most conveniently by means of computer algebra) an ordinary differential equation of second-order for $U(r)$ that can be easily solved. The result reads:

$$f^2 = 1 - \frac{2G}{c^2 r} \left(m - \frac{Q^2}{8\pi\varepsilon_0 c^2 r} \right). \quad (\text{O.17})$$

This, together with the orthonormal coframe (O.7) and the electric potential (O.15), represents the *Reissner–Nordström* solution of GR for a massive charged “particle.”

The electromagnetic field of the Reissner–Nordström solution has the same innocent appearance as that of a point charge in *flat* Minkowski spacetime. It is clear, however, that all relevant geometric objects, coframe, metric, connection, curvature, “feel” — via the 0-form f — the presence of the electric charge.

If the charge satisfies the inequality

$$\frac{Q^2}{4\pi\varepsilon_0} \leq Gm^2, \quad (\text{O.18})$$

then the spacetime metric (O.8) has a horizon that corresponds to the zeros of the function (O.17). However, as it is clearly seen from (O.9), (O.10), and especially from (O.11), the electromagnetic field is regular everywhere except at the origin. The emerging geometry describes a charged *black hole*. When the charge is so large that (O.18) becomes invalid, then the solution is no black hole but rather describes a bare (or naked) singularity.

These results on spherical symmetry can be straightforwardly generalized to gauge theories of gravity with *post*-Riemannian pieces in the linear connection (see [26, 13]).

Rotating source: Kerr–Newman solution

When a source is rotating, its electromagnetic and gravitational fields are no longer spherically symmetric. Instead, the Reissner–Nordström geometry discussed above is replaced by the *axially symmetric* configuration described by the coframe⁹

$$\begin{aligned}\vartheta^{\hat{0}} &= \sqrt{\frac{\Delta}{\Sigma}} (cdt - a \sin^2 \theta d\phi), \\ \vartheta^{\hat{1}} &= \sqrt{\frac{\Sigma}{\Delta}} dr, \\ \vartheta^{\hat{2}} &= \sqrt{\Sigma} d\theta, \\ \vartheta^{\hat{3}} &= \frac{\sin \theta}{\sqrt{\Sigma}} [-acdt + (r^2 + a^2) d\phi],\end{aligned}\tag{O.19}$$

where $\Delta = \Delta(r)$, $\Sigma = \Sigma(r, \theta)$, and a is a constant. The latter is directly related to the angular momentum of the source.

The electromagnetic potential 1-form reads

$$A = A_{\hat{0}} \vartheta^{\hat{0}},\tag{O.20}$$

with $A_{\hat{0}} = A_{\hat{0}}(r, \theta)$. Substituting the ansatz (O.19), (O.20) into the Einstein–Maxwell field equations (O.6), (O.3), and (O.4), one finds:

$$\Delta = r^2 + a^2 - \frac{2Gmr}{c^2} + \frac{GQ^2}{4\pi\epsilon_0 c^4},\tag{O.21}$$

$$\Sigma = r^2 + a^2 \cos^2 \theta,\tag{O.22}$$

$$A_{\hat{0}} = -\frac{Q}{4\pi\epsilon_0} \frac{r}{\sqrt{\Delta\Sigma}}.\tag{O.23}$$

⁹ A Reduce–Excalc program for such a coframe can be found in Sec. C.2.2 (including the cosmological constant for `ffsqr` ≠ 1). Puntigam et al. [27] derived the electrically charged Kerr metric, the Kerr–Newman metric, by means of Excalc from the (uncharged) Kerr metric in a manner analogous to the derivation of the Reissner–Nordström metric from the Schwarzschild metric shown above.

Accordingly, the electromagnetic field strength reads:

$$F = dA = \frac{Q}{4\pi\epsilon_0\Sigma^2} \left((a^2 \cos^2 \theta - r^2) \vartheta^{\hat{0}} \wedge \vartheta^{\hat{1}} + \frac{2a^2 r \sin \theta \cos \theta}{\sqrt{\Delta}} \vartheta^{\hat{0}} \wedge \vartheta^{\hat{2}} \right). \quad (\text{O.24})$$

We denote $\rho^2 := (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta$ and introduce the 4-vector field n of the adapted spacetime foliation by

$$n = \partial_t + n^\phi \partial_\phi, \quad \text{with} \quad n^\phi = \frac{2aGmr}{c\rho^2}. \quad (\text{O.25})$$

Then we can write the metric of spacetime in the standard form:

$$ds^2 = o_{\alpha\beta} \vartheta^\alpha \otimes \vartheta^\beta = N^2 dt^2 - {}^{(3)}g_{ab} \underline{dx}^a \underline{dx}^b. \quad (\text{O.26})$$

Here, making use of (O.19) and (O.21), (O.22), we have

$$N^2 = \frac{\Delta \Sigma c^2}{\rho^2}, \quad {}^{(3)}g_{ab} = \begin{pmatrix} \frac{\Sigma}{\Delta} & 0 & 0 \\ 0 & \Sigma & 0 \\ 0 & 0 & \frac{\sin^2 \theta \rho^2}{\Sigma} \end{pmatrix}, \quad (\text{O.27})$$

whereas the transversal spatial 3-coframe reads, as usual,

$$\underline{dr} = dr, \quad \underline{d\theta} = d\theta, \quad \underline{d\phi} = d\phi - n^\phi dt. \quad (\text{O.28})$$

For large distances, we find from (O.19), (O.21), and (O.22) the asymptotic line element of spacetime as

$$\begin{aligned} ds^2 &= \left[1 - \frac{2Gm}{c^2 r} + \frac{GQ^2}{4\pi\epsilon_0 c^4 r^2} + \mathcal{O}(r^{-3}) \right] c^2 dt^2 \\ &\quad - a \sin^2 \theta \left[\frac{4Gm}{c^2 r} - \frac{GQ^2}{2\pi\epsilon_0 c^4 r^2} + \mathcal{O}(r^{-3}) \right] c dt d\phi \\ &\quad - \left[1 + \frac{2Gm}{c^2 r} - \frac{GQ^2}{4\pi\epsilon_0 c^4 r^2} + \mathcal{O}(r^{-3}) \right] dr^2 \\ &\quad - r^2 \left(1 + \frac{a^2}{r^2} \right) [d\theta^2 + \sin^2 \theta d\phi^2 (1 + \mathcal{O}(r^{-3}))]. \end{aligned} \quad (\text{O.29})$$

All components of the spacetime metric g_{ij} depend only on the two coordinates r, θ . Accordingly, the Lie derivatives of the metric with respect to the tangent vector fields ∂_t and ∂_ϕ vanish, $\mathcal{L}_\xi g_{ij} = 0$. These are Killing vectors of the metric. Thus the Kerr–Newman metric possesses the two Killing vectors

$$\overset{(t)}{\xi} = \partial_t \quad \text{and} \quad \overset{(\phi)}{\xi} = \partial_\phi. \quad (\text{O.30})$$

In GR, knowledge of the Killing vectors provides important information about the gravitating system. In particular, for a compact source, the total mass M

and the total angular momentum L can be given in terms of the Killing vectors by means of the so-called Komar formulas:¹⁰

$$M = \frac{c}{8\pi G} \int_{S_\infty} {}^\star(d^{(t)}k), \quad L = -\frac{c^3}{16\pi G} \int_{S_\infty} {}^\star(d^{(\phi)}k). \quad (\text{O.31})$$

The integrals are taken over the spatial boundary of a sphere with infinite radius. Furthermore, we used the canonical map (C.2.3) to define the 1-forms

$${}^{(t)}k = \tilde{\mathbf{g}}({}^{(t)}\xi), \quad {}^{(\phi)}k = \tilde{\mathbf{g}}({}^{(\phi)}\xi) \quad (\text{O.32})$$

in terms of the Killing vector fields.

It is sufficient to use in the Komar expressions of (O.31) the asymptotic formula (O.29). Then, for the Kerr–Newman metric, we can prove

$$M = m, \quad L = mca. \quad (\text{O.33})$$

This explains the physical meaning of the parameters m and a in the Kerr–Newman solution. If we put $a = 0$, we fall back to the Reissner–Nordström solution.

Electrodynamics outside black holes and neutron stars

Neutron stars and black holes arise from the gravitational collapse of ordinary matter. The gravitational effects become very strong near such objects, and GR is necessary for the description of the corresponding spacetime geometry. Normally, the total electric charge of the collapsing matter vanishes. Then we are left, in general, with the Kerr metric, which can be obtained from (O.19)–(O.29) by putting $Q = 0$.

Near the surface of a neutron star and outside a black hole, one can expect many interesting electrodynamical effects. To describe them, we need to solve Maxwell's equations in a prescribed Kerr metric.

It is amazingly simple to find the exact solution of the Maxwell equations in the Kerr geometry. The crucial points are: (i) the Kerr geometry describes a (matter-free) vacuum spacetime and (ii) there exist the two Killing vector fields (O.30). It is straightforward to prove that every Killing vector ξ defines, in a vacuum spacetime, a harmonic 1-form $k = \tilde{\mathbf{g}}(\xi)$ that satisfies $\square k = 0$ and $d^\dagger k = 0$. Recalling the Maxwell equations in the form of the wave equation (E.1.4), we immediately find that the ansatz

$$A = -\frac{B_0}{2} \left(\frac{2a}{c} {}^{(t)}k + {}^{(\phi)}k \right) \quad (\text{O.34})$$

¹⁰Compare, e.g., Heusler [15].

yields an exact solution of the Maxwell equations on the background of the Kerr metric. Here B_0 is constant and the coefficient in the first term is chosen in accordance with (O.31) and (O.33) in order to guarantee a totally vanishing charge. Substituting (O.32) into (O.34), we find explicitly

$$A = aB_0 \frac{Gm}{c^2} (1 + \cos^2 \theta) \frac{r}{\sqrt{\Delta \Sigma}} \vartheta^{\hat{0}} + \frac{B_0}{2} (r^2 + a^2) \sin^2 \theta d\phi. \quad (\text{O.35})$$

The physical interpretation is straightforward: The 1-form potential (O.35) is a kind of superposition of the Coulomb-type electric piece (the first term on the right-hand side, cf. (O.20), (O.23)) and the asymptotically homogeneous magnetic piece (the second term). With respect to the coordinate foliation (for which $n = \partial_t$), the electromagnetic field strength reads $F = dA = E \wedge dt + B$, with

$$E = - \frac{B_0 G M a}{c \Sigma^2} [(1 + \cos^2 \theta) (r^2 - a^2 \cos^2 \theta) dr + 2 (r^2 - a^2) \sin \theta \cos \theta d\theta], \quad (\text{O.36})$$

$$B = \frac{B_0 G M a}{c^2 \Sigma^2} [(1 + \cos^2 \theta) (r^2 - a^2 \cos^2 \theta) \sin^2 \theta dr - 2r \sin \theta \cos \theta (2r^2 \cos^2 \theta + a^2 + a^2 \cos^4 \theta) d\theta] \wedge d\phi + B_0 [r \sin^2 \theta dr + (r^2 + a^2) \sin \theta \cos \theta d\theta] \wedge d\phi. \quad (\text{O.37})$$

For large distances, the last line in (O.37) dominates, yielding asymptotically the homogeneous constant magnetic field directed along the z -axis,

$$F = B_0 dx \wedge dy + \mathcal{O}(1/r^2). \quad (\text{O.38})$$

Here we performed the usual transformation from spherical polar coordinates (r, θ, ϕ) to Cartesian rectangular coordinates (x, y, z) .

The electric field vanishes for $a = 0$, i.e., for a non-rotating black hole. We can draw a direct parallel to the Wilson & Wilson experiment where the magnetic field induces an electric field inside the rotating body. The spacetime of a rotating Kerr geometry acts similarly and induces an electric field around the black hole.

In the membrane approach,¹¹ the physics outside a rotating black hole is described in terms of the horizon understood as a conducting membrane with surface charge and current density, as well as with surface resistivity. In particular, it is possible to develop a mechanism of extracting (rotational) energy from a black hole by means of external magnetic fields.

In realistic situations, one has a plasma around neutron stars and black holes. Therefore plasma physics¹² or, more specifically, magnetohydrodynamics has to be applied.

¹¹See, e.g., Straumann [28] and the literature cited therein.

¹²See, e.g., Cap [3], Hora [16], and also Knoepfel [19].

Force-free electrodynamics

Near a black hole or a neutron star, the energy-momentum of the electromagnetic field heavily dominates the energy-momentum of matter and, thus, approximately force-free fields can naturally emerge in the plasma of electrons and positrons.¹³ In Sec. B.2.2, we have defined such electromagnetic fields by the condition of the vanishing of the Lorentz force (B.2.21). Using the space-time relation (O.4), we can now develop a more substantial analysis of such a situation. The force-free condition now reads:

$$(e_\alpha \lrcorner F) \wedge d^*F = 0. \quad (\text{O.39})$$

To begin with, let us recall the *sourceless* solution considered above. The magnetic field (O.37) has the evident structure

$$B = d\Psi \wedge d\phi, \quad (\text{O.40})$$

with

$$\Psi = B_0 \sin^2 \theta \left(\frac{r^2 + a^2}{2} - \frac{GMa^2 r}{c^2 \Sigma} (1 + \cos^2 \theta) \right). \quad (\text{O.41})$$

Thus, the magnetic field is manifestly axially symmetric, that is, its Lie derivative with respect to the vector field ∂_ϕ vanishes:

$$\mathcal{L}_{\partial_\phi} B = d(\partial_\phi \lrcorner B) + \partial_\phi \lrcorner dB = -dd\Psi \equiv 0. \quad (\text{O.42})$$

Here we used the Maxwell equation $dB = 0$, which is identically fulfilled for any function Ψ that does not depend on ϕ , i.e., $\Psi = \Psi(r, \theta)$.

Returning to the problem under consideration (*with* nontrivial plasma source), we also demand that the magnetic field be axially symmetric. As a more general structure of the field we then expect

$$B = d\Psi \wedge d\phi + \Phi dr \wedge d\theta. \quad (\text{O.43})$$

Clearly, this 2-form satisfies the Maxwell equation $dB = 0$ for every $\Phi = \Phi(r, \theta)$, which also guarantees that the axial symmetry condition (O.42) is fulfilled. Since the 2-form $dr \wedge d\theta$ is proportional to the Hodge dual ${}^\pm d\phi$, it is more convenient to rewrite the last term in the magnetic field (O.43) as $\Phi dr \wedge d\theta = (I/N) {}^\pm d\phi$, with a new function $I = I(r, \theta)$.

Now we need also an ansatz for the electric field strength 1-form E . Taking into account the axial symmetry, a natural assumption reads

$$E = v \lrcorner B, \quad (\text{O.44})$$

where the vector field

$$v = \Omega \partial_\phi \quad (\text{O.45})$$

¹³For recent developments see, e.g., Komissarov [20].

can be interpreted as the vorticity of the magnetic field lines. The function Ω does not depend on the angular coordinate ϕ . Furthermore, substituting (O.43) into (O.44), we find

$$E = -\Omega d\Psi. \quad (\text{O.46})$$

The Maxwell equation $dE = 0$ is fulfilled if $\Omega = \Omega(\Psi)$.

Combining (O.43) and (O.46), we find as general ansatz for the electromagnetic field strength 2-form

$$F = -\Omega d\Psi \wedge dt + d\Psi \wedge d\phi + \frac{I}{N} \underline{*d\phi}. \quad (\text{O.47})$$

This must be inserted into the “force-free condition” (O.39), which will be evaluated with respect to the natural (or coordinate) frame $e_\alpha = \delta_\alpha^i \partial_i$. Direct inspection shows that equation (O.39) is identically fulfilled for ∂_t and ∂_ϕ provided $I = I(\Psi)$. Substituting (O.47) into (O.39) for ∂_r and ∂_θ , we find a nontrivial differential equation

$$\underline{d}(\alpha \underline{*d\Psi}) + \beta d\Psi \wedge \underline{*d\Psi} + \gamma \underline{*I} \frac{dI}{d\Psi} = 0. \quad (\text{O.48})$$

Here the functions $\alpha = \alpha(r, \theta)$, $\beta = \beta(r, \theta)$, $\gamma = \gamma(r, \theta)$ are constructed from Ω and from the components of the spacetime metric according to

$$\alpha = \frac{N}{({}^{(3)}g_{33})} - \frac{(\Omega - n^\phi)^2}{N}, \quad \beta = \frac{(\Omega - n^\phi)}{N} \frac{d\Omega}{d\Psi}, \quad \gamma = \frac{1}{({}^{(3)}g_{33}N)}. \quad (\text{O.49})$$

Recall that $({}^{(3)}g_{33}) = \sin^2\theta \rho^2/\Sigma$. Equation (O.48) is called the *Grad-Shafranov equation* and, with the given functions $\Omega = \Omega(\Psi)$ and $I = I(\Psi)$, the solution Ψ of (O.48) describes completely the force-free electromagnetic field configuration. The corresponding distribution of the charge and current density is derived from the Maxwell equation $J = dH$: Explicitly, the charge density 3-form reads

$$\rho = -\lambda_0 \underline{d} \left[\frac{(\Omega - n^\phi)}{N} \underline{*d\Psi} \right], \quad (\text{O.50})$$

whereas the electric current 2-form turns out to be

$$j = \lambda_0 \left\{ -\partial_\phi \lrcorner \left[\underline{d} \left(\frac{N}{({}^{(3)}g_{33})} \underline{*d\Psi} \right) + \frac{(\Omega - n^\phi)}{N} dn^\phi \wedge \underline{*d\Psi} \right] + dI \wedge \underline{d\phi} \right\}. \quad (\text{O.51})$$

With these few examples, we wanted to demonstrate that the electrodynamic formalism, which we developed in this book, is ideally suited for “marrying” electrodynamics to general relativity; this includes interesting astrophysical scenarios. There seems to be no better way than the one outlined above for the investigation of the mutual influence of the electromagnetic and the gravitational field on each other.

Remarks on topology and electrodynamics

In our book, we did not discuss genuine topological aspects of electrodynamics. However, topology can play a very important and nontrivial role in electrodynamics and in magnetohydrodynamics.¹⁴

A word of caution is in order: One should carefully distinguish the physical situations in which an underlying spacetime (or space¹⁵) has a complicated topology from the case in which the electromagnetic *field configuration* is topologically nontrivial.¹⁶ Usually the decisive role is played by the *pure gauge* contribution to the electromagnetic potential 1-form.

A manifest example is given by the force-free magnetic fields that approximately describe the twisted flux tubes in the models of solar prominence (sheets of luminous gas emanating from the Sun's surface).¹⁷ Recall equation (B.2.23); which determines a force-free magnetic field. It is identically fulfilled when

$$\underline{d}\mathcal{H} = \alpha B. \quad (\text{O.52})$$

Indeed, since B is a transversal 2-form (i.e., living in three spatial dimensions), we have $B \wedge B \equiv 0$. Consequently, $B \wedge e_a \lrcorner B = 0$. Thus (O.52) solves (B.2.23) for any function $\alpha = \alpha(x)$.

Note that the ansatz (O.52) can be used even in the metric-free formulation of electrodynamics. In Maxwell–Lorentz electrodynamics, (O.52) further reduces to

$$\underline{d}^*B = \alpha B. \quad (\text{O.53})$$

Incidentally, this is the field equation of the so-called “topologically massive electrodynamics” in three dimensions¹⁸ provided α is a constant.

Specializing to axially symmetric configurations in Minkowski spacetime, we can easily find a solution of (O.53) for any choice of α . For example, in cylindrical coordinates (ρ, ϕ, z) , for $\alpha(\rho) = 2/[a(1 + \rho^2/a^2)]$, we obtain

$$B = \frac{B_0 \rho}{1 + \frac{\rho^2}{a^2}} d\rho \wedge \left(d\phi - \frac{1}{a} dz \right). \quad (\text{O.54})$$

This solution describes a uniformly twisted flux tube. The constant parameter a determines the twist. Although $\underline{d}\alpha \neq 0$, $\underline{d}\alpha \wedge B = 0$, which guarantees the consistency of the solution.

It is straightforward to read off from (O.54) the potential 1-form

$$\mathcal{A} = \frac{B_0 a^2}{2} \log \left(1 + \frac{\rho^2}{a^2} \right) \left(d\phi - \frac{1}{a} dz \right) + d\chi, \quad (\text{O.55})$$

¹⁴See the reviews of Moffatt and Marsh [24, 22, 23].

¹⁵See the article of Frankel [9] and the experimental investigations of Tsebro and Omel'yanovskii [30].

¹⁶This could be true for the ball-like solutions of the vacuum Maxwell equations of Chubykalo and Espinoza [5]; see also the literature cited there.

¹⁷This is discussed in Marsh [23], for example.

¹⁸See Deser, Jackiw, and Templeton [6].

where χ is an arbitrary gauge function.

There exist various topological numbers (or invariants) that evaluate the topological complexity of an electric and magnetic field configuration. The magnetic helicity density in (B.3.17) provides an explicit example of such a number. Defined by the integral

$$h := \int_V \mathcal{A} \wedge B, \quad (\text{O.56})$$

the magnetic helicity h measures the “linkage” of the magnetic field lines. One can establish a direct relation of h to the classical Hopf invariant, which classifies the maps of a 3-sphere onto a 2-sphere, and can interpret it in terms of the Gauss linking number.

Turning again to the twisted flux-tube solution above, we see from (O.55) and (O.54) that in the exterior product of $\mathcal{A}$ with B only the contribution from the pure gauge survives: $\mathcal{A} \wedge B = \underline{d}\chi \wedge B = \underline{d}(\chi B)$. As a result, a nontrivial value of the helicity (O.56) can only be obtained by assuming a toroidal topology of space. This can be achieved by gluing two 2-dimensional cross sections at some values z_1 and z_2 of the third coordinate and, moreover, by assigning a nontrivial jump $\delta\chi = \chi(z_2) - \chi(z_1)$ to the gauge function.

There is a close relation between magnetic interaction energy and helicity. For the *force-free* magnetic field, we have explicitly,

$$E_{\text{mag}} = \int \mathcal{A} \wedge j = \int \alpha \mathcal{A} \wedge B, \quad (\text{O.57})$$

where we used the Oersted–Ampère law $j = \underline{d}\mathcal{H}$ and the ansatz (O.52). When α is constant and $\dot{\mathcal{D}}$ can be neglected, then the energy is proportional to the magnetic helicity.¹⁹

Another manifestation of topology in electrodynamical systems is more of a quantum nature: the Aharonov–Bohm effect.²⁰ If the wave function of an electron surrounds an isolated magnetic flux such that it cannot penetrate it, then the electron wave resides in a region free of a magnetic field B (but not free of the covector potential A). Nevertheless, the phase of the wave function is influenced by the enclosed magnetic flux. By interferometric experiments this flux can be measured.

We shall now turn to the quasi-classical fields that built up the electric current 3-form.

¹⁹More on the electrodynamics in multiply connected domains can be found in Marsh [22, 23]. The corresponding effects underline the physical importance of the electromagnetic potential 1-form A .

²⁰See the theoretical discussion by Aharonov and Bohm [1] and the first experimental findings by Chambers [4]. A recent evaluation has been given by Nambu [25].

Superconductivity: Remarks on Ginzburg–Landau theory

Generalizing the classical Maxwell–London theory of superconductivity of 1935, Ginzburg and Landau achieved in 1950, by introducing a complex 3D scalar “order parameter” ψ , a quasi-classical static description of superconductivity for $T = 0$.²¹ The field ψ describes the Cooper pairs (two bound electrons in a superconductor), the density of which is $|\psi|^2$. The Ginzburg–Landau model is specified by the 3D energy functional

$$E = \int_{\Omega_3} \left[\frac{1}{2\mu_0} d\mathcal{A} \wedge {}^*d\mathcal{A} + \frac{1}{2m} (\hbar d\bar{\psi} + iq\mathcal{A}\bar{\psi}) \wedge {}^*(\hbar d\psi - iq\mathcal{A}\psi) + U(|\psi|^2) \eta \right], \quad (\text{O.58})$$

with $q = 2e$ as the charge of a Cooper pair, $\hbar$ the Planck constant, and $U(x) = -\beta x + \frac{1}{2}\gamma x^2$. Here $\bar{\psi}$ denotes the complex conjugate field. The positive quantities m, β, γ are the parameters of the theory. By varying (O.58) with respect to the fields $\mathcal{A}$ and ψ , one finds the field equations, namely the Oersted–Ampère law with the superconducting current

$$j = \frac{i\hbar q}{2m} (\bar{\psi} {}^*d\psi - \psi {}^*d\bar{\psi}) + \frac{q^2}{m} |\psi|^2 {}^*\mathcal{A} \quad (\text{O.59})$$

and the Ginzburg–Landau equation.

This nonrelativistic theory describes the static properties of superconductors very well. It represents a good approximation to the exact theory of superconductivity for temperatures close to $T = 0$. One consequence of this theory is the Abrikosov lattice of magnetic flux lines, which we displayed in Fig. B.3.2.

Classical (first quantized) Dirac field

In classical electrodynamics, the electric current 3-form J is phenomenologically specified; it cannot be resolved any further. In the Ginzburg–Landau theory, we expressed the superconducting current (O.59) in terms of the 3D scalar order parameter ψ . Quite generally, we know, however, that electric charge is carried by the fundamental particles, namely by leptons and quarks. They are both fermions of spin $\hbar/2$. The electron, a lepton, is responsible for many everyday effects. Its field can be described by means of a Dirac spinor, a half-integer representation of the Lorentz group. Thus, spinors can only be introduced after the 5th axiom has been adopted. Then a metric is available and the Lorentz group surfaces.

Strictly, the second quantized Dirac theory governs the behavior of the *electron* and its interaction with the photon. However, if the energies involved in an

²¹See Tinkham [29] and Zirnbauser [31], for example.

experiment are much smaller than the mass of the electron, then the electron can be approximately viewed as a classical matter wave, i.e., as a first quantized Dirac wave function. An electron microscope and its resolution may well be described in such a manner; similarly, an electron interferometer for sensing rotation (Sagnac type of effect) doesn't need a more refined description.

Let us then assume that the Dirac matrices, referred to an orthonormal coframe, are given by

$$\gamma_{(\alpha} \gamma_{\beta)} = o_{\alpha\beta} . \quad (\text{O.60})$$

Here $o_{\alpha\beta} = \text{diag}(+1, -1, -1, -1)$. Then we can introduce the Dirac-algebra-valued 1-form $\gamma := \gamma_\alpha \vartheta^\alpha$. The Dirac equation, minimally coupled to the electromagnetic field, reads

$$i\hbar \star \gamma \wedge D \Psi + \star mc \Psi = 0, \quad \text{with} \quad D = d + i \frac{e}{\hbar} A . \quad (\text{O.61})$$

The Dirac adjoint is $\bar{\Psi} := \Psi^\dagger \gamma^0$. The Dirac equation can be derived from the Lagrangian 4-form

$$L_D = \frac{i}{2} \hbar (\bar{\Psi} \star \gamma \wedge D \Psi + \overline{D \Psi} \wedge \star \gamma \Psi) + \star mc \bar{\Psi} \Psi . \quad (\text{O.62})$$

The conserved electric current turns out to be

$$J := \frac{\delta L_D}{\delta A} = -ie \bar{\Psi} \star \gamma \Psi, \quad dJ = 0 . \quad (\text{O.63})$$

The energy-momentum current of the Dirac field, according to the Lagrange-Noether procedure developed in Sec. B.5.5, reads

$$\Sigma_\alpha = \frac{\delta L_D}{\delta \vartheta^\alpha} = \frac{i\hbar}{2} (\bar{\Psi} \star \gamma D_\alpha \Psi - D_\alpha \bar{\Psi} \star \gamma \Psi) , \quad D \Sigma_\alpha = 0 . \quad (\text{O.64})$$

Here $D_\alpha := e_\alpha \lrcorner D$. Therefore the self-consistent Dirac-Maxwell system reads

$$d \star F = J, \quad dF = 0 \quad (\text{O.65})$$

$$J = -ie \bar{\Psi} \star \gamma \Psi, \quad i\hbar \star \gamma \wedge D \Psi + \star mc \Psi = 0 . \quad (\text{O.66})$$

With *gravity*, one must generalize the spinor covariant derivative, which now should read

$$D = d + i \frac{e}{\hbar} A + \frac{i}{4} \hat{\sigma}_{\alpha\beta} \tilde{\Gamma}^{\alpha\beta}, \quad (\text{O.67})$$

where, as usual, $\hat{\sigma}^{\alpha\beta} := i\gamma^{[\alpha} \gamma^{\beta]}$, and $\tilde{\Gamma}^{\alpha\beta}$ is the Levi-Civita connection. In order to be able to formulate Einstein's gravitational field equation, one must symmetrize the energy-momentum current,

$$\overset{D}{\sigma}_\alpha = \overset{D}{\Sigma}_\alpha - D\mu_\alpha, \quad \text{with} \quad \overset{D}{\tau}_{\alpha\beta} = \vartheta_{[\alpha} \wedge \mu_{\beta]}, \quad (\text{O.68})$$

where $\tau_{\alpha\beta}^{\text{D}} = (\partial L_{\text{D}}/D\Psi)(i\hat{\sigma}_{\alpha\beta}/4)\Psi$ is the canonical spin current 3-form of the Dirac field. Then the complete Einstein–Dirac–Maxwell system, together with (E.1.22), (O.68), (O.65), and (O.66), reads

$$\frac{1}{2}\eta_{\alpha\beta\gamma}\wedge\tilde{R}^{\beta\gamma}=\frac{8\pi G}{c^3}\left(\overset{\text{Max}}{\Sigma}_{\alpha}+\overset{\text{D}}{\sigma}_{\alpha}\right). \quad (\text{O.69})$$

An alternative Dirac coupling to gravity can be achieved via the *Einstein–Cartan theory* of gravity.²² We allow, generalizing the Levi-Civita connection of GR, for a metric compatible connection $\Gamma^{\alpha\beta}=-\Gamma^{\beta\alpha}$ carrying a torsion piece. Then the spin current of the Dirac field can be defined according to

$$\overset{\text{D}}{\tau}_{\alpha\beta}:=-\frac{\delta\overset{\text{D}}{L}}{\delta\Gamma^{\alpha\beta}}=\frac{\hbar}{4}\vartheta_{\alpha}\wedge\vartheta_{\beta}\wedge\overline{\Psi}\gamma_5\Psi, \quad D\overset{\text{D}}{\tau}_{\alpha\beta}+\vartheta_{[\alpha}\wedge\overset{\text{D}}{\Sigma}_{\beta]}=0, \quad (\text{O.70})$$

where $\gamma_5:=-i\eta_{\alpha\beta\mu\nu}\gamma^{\alpha}\gamma^{\beta}\gamma^{\mu}\gamma^{\nu}/4!$, and for the gravitational sector of the Einstein–Cartan–Dirac–Maxwell system, we find

$$\frac{1}{2}\eta_{\alpha\beta\gamma}\wedge R^{\beta\gamma}=\frac{8\pi G}{c^3}\left(\overset{\text{Max}}{\Sigma}_{\alpha}+\overset{\text{D}}{\Sigma}_{\alpha}\right), \quad (\text{O.71})$$

$$\frac{1}{2}\eta_{\alpha\beta\gamma}\wedge T^{\gamma}=\frac{8\pi G}{c^3}\overset{\text{D}}{\tau}_{\alpha\beta}. \quad (\text{O.72})$$

This is a viable alternative to the Einstein–Dirac–Maxwell system. It looks very symmetric. Note that the electromagnetic field (since it is massless) doesn’t carry dynamical spin.

If we take either the Einstein–Dirac–Maxwell or the Einstein–Cartan–Dirac–Maxwell system, in any case we can study the gravitational properties of the Dirac electron.²³ And in both cases we recognize that there is no need to change electrodynamics. It is insensitive to the gravitational model applied. Via the 5th axiom the Hodge star corresponds to a dynamic metric obeying the Einstein or the Einstein–Cartan equations.

On the quantum Hall effect and the composite fermion

In Sec. B.4.5 we presented a classical description of the quantum Hall effect (QHE). However, a true understanding of QHE is only possible when the quantum aspects of the phenomenon are carefully studied.

²²See Blagojević [2] or Gronwald et al. [11] and references given there.

²³See [14] for a discussion of the gravitational moments of the Dirac electron.

We have no tools in our book to pursue this goal. What one has to do is to quantize the Lagrangian (B.4.72) by, for instance, path integral methods. In other words, QHE is described within the framework of an effective topological quantum field theory with a Chern–Simons action. The explanation of the quantization of the Hall conductance σ_H is the ultimate goal achieved in these studies. The best thing we can do is to direct the interested reader to the corresponding literature.²⁴

Incidentally, if one describes QHE for low lying Landau levels, then Jain’s concept of a *composite fermion* is very helpful: It consists of one *electron* and an even number of magnetic *fluxoids* that are attached to it.²⁵ Isn’t that a very clear additional indication of what the fundamental quantities are in electrodynamics? Namely, electric charge (see first axiom) and magnetic flux (see third axiom).

On quantum electrodynamics

The approach in our book is essentially classical. This is a legitimate assumption within the well known and rather wide limits of the idealized representation of electric charges and currents by means of particles and continuous media and of electromagnetic radiation by means of classical waves of electric and magnetic fields.

However, it is experimentally well established that an electron, under certain conditions, may display wave properties, whereas electromagnetic radiation must sometimes be described in terms of particles, i.e., photons. These facts are taken into account by quantum electrodynamics (QED).

The mathematical framework of QED is the scheme of second quantization. In this scheme, the electromagnetic (Maxwell) field A and the electron (Dirac) field ψ are replaced by field operators acting in the Hilbert space of all quantum states. Physical processes are then described in terms of creation, annihilation, and propagation of the quanta of the electromagnetic and the spinor fields. The photon γ is massless and has spin 1 (better: helicity 1), in units of $\hbar$, whereas the electron is massive and has spin $1/2$.

In QED, the electromagnetic potential 1-form A plays a fundamental role, representing the “generalized coordinate” with the excitation H being its canonically conjugated “momentum.” The gauge symmetry $A \rightarrow A + d\chi$ moves to center stage: One can consistently build electrodynamics on the basis of the principle of local gauge invariance. The underlying symmetry group is the Abelian $U(1)$ and A emerges naturally as the $U(1)$ -gauge potential, which is geometrically interpreted as the connection on the principal $U(1)$ -bundle of spacetime.

²⁴See Fröhlich and Pedrini [10], e.g., and the literature cited therein.

²⁵See Jain [17, 18] and, in this general context, also Nambu [25].

The gauge freedom is responsible for the masslessness of a photon and, thus, ultimately for the long range character of the electromagnetic interaction.

QED describes correctly all quantum phenomena involving electrons and photons. It is experimentally very well verified; in fact, it is the most precisely tested physical theory. The most precise experiments are the proof of the predicted value of the anomalous magnetic moment of the electron and the observation of the relativistic shift of energy levels in atoms.

The success of QED is to a great extent related to the smallness of its coupling constant $\alpha_f = e^2/(4\pi\epsilon_0\hbar c)$, which allows one to effectively use a perturbation approach.

On electroweak unification

Since the 1960s, considerable progress has been achieved in the construction of unified theories of physical interactions. In particular, QED is found to be naturally unified with the weak interaction.

A typical example of a quantum process governed by the weak interaction is the radioactive decay of the neutron into a proton, an electron, and an antineutrino. The modern understanding of weak forces is based on the gauge approach. In the Glashow–Salam–Weinberg model of the electroweak interaction, the fundamental symmetry group $SU(2) \times U(1)$ gives rise to four gauge potentials as mediators of the electroweak interaction: the intermediate vector boson W (its positively or negatively charged versions $W^\pm$), the neutral intermediate vector boson Z , and the photon γ . The spin 1 particles W and Z become massive via the mechanism of spontaneous symmetry breaking of the gauge group, whereas γ corresponds to the unbroken exact subgroup and remains massless.

In the standard model of elementary particle physics, the group $SU(2) \times U(1)$ is enlarged to $SU(3) \times SU(2) \times U(1)$ and the resulting theory represents a unification of the electroweak and the strong forces. The color group $SU(3)$ brings to life the eight additional gauge potential 1-forms A^a , $a = 1, \dots, 8$, which are called the gluon potentials. In the standard model, one describes the interaction of quarks (fermionic constituents of the strongly interacting particles, the hadrons) and leptons (electron, muon, tauon, and their neutrinos) by means of the gluons A^a and the intermediate gauge bosons W , Z , and γ . And classical electrodynamics is encapsulated in the γ .

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